From Group Theory to Representation Theory

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Contents

Chapter 1

Rudiments of Group Theory

1 Review

Group Axioms

A group consists of a non-empty set G and a binary operation on G such that the following axioms hold:

- $\forall g, h, k \in G, (gh)k = g(hk)$ (associativity)
- $\exists e \in G$ s.t. $eg = ge = g \ \forall g \in G$ (identity)
- $\forall g \in G, \exists g^{-1} \in G$ s.t. $gg^{-1} = g^{-1}g = e$ (inverses)

Properties of Groups

- If $xy = yx$, the commutator $[x, y] = xyx^{-1}y^{-1} = e$
- If $xy = yx \ \forall x, y \in G$, then G is an **abelian** group.
- A $x \in G$ is of finite order if $\exists n \in \mathbb{N}$ s.t. $x^n = e$
- $|G|$, the number of elements in G is defined to be the order of a finite group G. Every element of a finite group is of finite order
- There are infinite groups that has every element having a finite order, but there are also infinite groups in which the identity is the only emenet of finite order (torsion-free groups such as $\mathbb{Z}$)

Properties of Subgroups

- H is a **subgroup** of G, denoted $H \leq G$, if $H \subseteq G$ and H is a group under the same operation as G. In particular, $e \in H$ and $h^{-1} \in H \ \forall h \in H$
- Every subgroup of a finite group is finite, but an infinite group always has both finite and infinite subgroups
- Every subgroup of an abelian group is abelian, but a non-abelian group can have both abelian and non-abelian subgroups

Proposition 1.1. *If $H, K \leq G$, then $H \cap K \leq G$. More generally, $H_i \leq G \Rightarrow \bigcap H_i \leq G$*

Proof. Suppose $H_i \leq G \ \forall i$
Since each $H_i \leq G$, we have $e \in H_i \ \forall i$, so $e \in \bigcap H_i$ and $\bigcap H_i \neq \emptyset$. For $x, y \in \bigcap H_i$, we have $x, y \in H_i \ \forall i$, so by closure of each H_i, $xy \in H_i \ \forall i$, giving $xy \in \bigcap H_i$. Similarly, $x^{-1} \in H_i \ \forall i$, so $x^{-1} \in \bigcap H_i$.
$\therefore \bigcap H_i \leq G$ $\square$

Theorem (Lagrange's Theorem)**.** *Let G be a* ***finite*** *group, and let $H \leq G$. Then $|H| \mid |G|$.*

Proof. Let $x \in G$. Define $f_x \colon H \to xH$ by $f_x(h) = xh$. If $f_x(h_1) = f_x(h_2)$, then $xh_1 = xh_2$, and multiplying on the left by x^{-1} gives $h_1 = h_2$. Also, every element of xH has the form xh for some $h \in H$, so f_x is surjective. Thus f_x is a bijection, and $|xH| = |H|$.
We claim that any two sets of the form xH are either equal or disjoint. Suppose that $xH \cap yH \neq \emptyset$. Then $xh_1 = yh_2$ for some $h_1, h_2 \in H$, so $x = y(h_2h_1^{-1})$. Since $h_2h_1^{-1} \in H$, we have $x = yh$ for some $h \in H$, and so $xH = yhH = yH$. Thus the distinct sets of the form xH partition G.
Since G is finite, there are finitely many such sets, say r. Each has cardinality $|H|$, so $|G| = r|H|$.
$\therefore |H| \mid |G|$. ☐

If $X \subseteq G, \langle X \rangle \coloneqq \bigcap\{H \leq G \mid X \subseteq H\}$. By Proposition 1.1, $\langle X \rangle \leq G$, and it is the smallest subgroup of G containing X. We call $\langle X \rangle$ the **subgroup generated by X**, and we say that X **generates** $\langle X \rangle$.

Proposition 1.2. *Let $X \subseteq G$. Then $\langle X \rangle = \{e\} \cup \{x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} \mid r \geq 1,\ x_i \in X,\ \epsilon_i = \pm 1,\ 1 \leq i \leq r\}$.*

Proof. Let $S = \{e\} \cup \{x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} \mid r \geq 1,\ x_i \in X,\ \epsilon_i = \pm 1,\ 1 \leq i \leq r\}$.
We first show that $S \leq G$. Clearly $e \in S$. If $a = e$ or $b = e$, then $ab \in S$. Otherwise, if $a = x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} \in S$ and $b = y_1^{\delta_1} \cdots y_s^{\delta_s} \in S$, then $ab = x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} y_1^{\delta_1} \cdots y_s^{\delta_s} \in S$. Also $e^{-1} = e$, and if $a = x_1^{\epsilon_1} \cdots x_r^{\epsilon_r}$, then $a^{-1} = x_r^{-\epsilon_r} \cdots x_1^{-\epsilon_1} \in S$. Hence $S \leq G$.
Let $\mathcal{H}_X = \{H \leq G \mid X \subseteq H\}$. By Proposition 1.1, $\langle X \rangle = \bigcap_{H \in \mathcal{H}_X} H \leq G$. Since $X \subseteq S$, we have $S \in \mathcal{H}_X$, and so $\langle X \rangle = \bigcap_{H \in \mathcal{H}_X} H \subseteq S$.
Conversely, let $H \in \mathcal{H}_X$. Since $x_i \in X \subseteq H$, we have $x_i^{\epsilon_i} \in H$, so $x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} \in H$. Also $e \in H$, hence $S \subseteq H$. Thus $S \subseteq H$ for all $H \in \mathcal{H}_X$, so $S \subseteq \bigcap_{H \in \mathcal{H}_X} H = \langle X \rangle$.
$\therefore S = \langle X \rangle$. ☐

Cyclic Groups

- A group G is **cyclic** if $G = \langle g \rangle$ for some $g \in G$. By Proposition 1.2, $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$ s.t. $|\langle g \rangle| = n$ and $o(g) = n$
- All cyclic groups are abelian
- If g is not of finite order, then $\langle g \rangle$ is a torsion-free infinite abelian group, and $\langle g \rangle \cong \mathbb{Z}$
- If g is of finite order, then $\langle g \rangle \cong \mathbb{Z}/n\mathbb{Z}$
- If $g \in G$ and $o(g) = n \in \mathbb{Z}$, $\langle g \rangle \leq G$, so by Lagrange's Theorem, $n \mid |G|$. Thus, the order of an element of a finite group must divide the order of that group

If $X, Y \subseteq G, XY \coloneqq \{xy \mid x \in X, y \in Y\} \subseteq G$. Also, $X^{-1} \coloneqq \{x^{-1} \mid x \in X\} \subseteq G$. If $H \neq \emptyset$, then $H \leq G \iff HH = H$ and $H^{-1} = H$

Proposition 1.3. *Let $H, K \leq G$. Then $HK \leq G \iff HK = KH$*

Proof. ($\Rightarrow$) Suppose $HK \leq G$. Since HK is a subgroup, $(HK)^{-1} = HK$. Also $H^{-1} = H$ and $K^{-1} = K$, since $H, K \leq G$. Hence $(HK)^{-1} = K^{-1}H^{-1} = KH$. Therefore $HK = (HK)^{-1} = KH$.
($\Leftarrow$) Suppose $HK = KH$. Since $e \in H \cap K$, we have $e = ee \in HK$. Let $h_1k_1, h_2k_2 \in HK$. Since $k_1h_2 \in KH = HK$, write $k_1h_2 = h_3k_3$ for some $h_3 \in H,\ k_3 \in K$. Then $(h_1k_1)(h_2k_2) = h_1(k_1h_2)k_2 = h_1h_3k_3k_2 \in HK$. Also $(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$ for all $h \in H,\ k \in K$.
$\therefore HK \leq G$. ☐

Theorem 1.4. *Let $G = \langle g \rangle$ be a cyclic group of order n*

- *For each $d \mid n$, there is exactly one subgroup of order d, $\langle g^{\frac{n}{d}} \rangle$*
- *If $d, e \mid n$, then $\langle g^{\frac{n}{d}} \rangle \cap \langle g^{\frac{n}{e}} \rangle = \langle g^{\frac{n}{\gcd(d,e)}} \rangle$*
- *If $d, e \mid n$, then $\langle g^{\frac{n}{d}} \rangle \langle g^{\frac{n}{e}} \rangle = \langle g^{\frac{n}{\operatorname{lcm}(d,e)}} \rangle$*

Proof. First note that if $m \mid n$, then $o(g^m) = \frac{n}{m}$. Indeed, $(g^m)^{\frac{n}{m}} = g^n = e$, and if $(g^m)^t = e$, then $n \mid mt$, so $\frac{n}{m} \mid t$.
Let $d \mid n$. Taking $m = \frac{n}{d}$, we get $o(g^{\frac{n}{d}}) = d$, so $\langle g^{\frac{n}{d}} \rangle$ is a subgroup of order d.
For uniqueness, let $H \leq G$ with $|H| = d$, and let $a = \min\{m \in \mathbb{N} \mid g^m \in H\}$. This set is non-empty since $g^n = e \in H$. If $g^b \in H$, write $b = qa + r$, $0 \leq r < a$. Then $g^r = g^b(g^a)^{-q} \in H$, so by minimality of

a, $r = 0$. Hence $a \mid b$, so $H \subseteq \langle g^a \rangle$. Since $g^a \in H$, $\langle g^a \rangle \subseteq H$, and hence $H = \langle g^a \rangle$. Applying the same division argument to $n = qa + r$ gives $a \mid n$, so $|H| = |\langle g^a \rangle| = \frac{n}{a} = d$. Thus $a = \frac{n}{d}$, and $H = \langle g^{\frac{n}{d}} \rangle$.
Now let $d, e \mid n$. An element $g^b \in \langle g^{\frac{n}{d}} \rangle \cap \langle g^{\frac{n}{e}} \rangle$ iff $\frac{n}{d} \mid b$ and $\frac{n}{e} \mid b$, iff $\operatorname{lcm}(\frac{n}{d}, \frac{n}{e}) = \frac{n}{\gcd(d,e)}$ divides b. Hence $\langle g^{\frac{n}{d}} \rangle \cap \langle g^{\frac{n}{e}} \rangle = \langle g^{\frac{n}{\gcd(d,e)}} \rangle$.
Finally, $\langle g^{\frac{n}{d}} \rangle \langle g^{\frac{n}{e}} \rangle = \{g^{\frac{n}{d}r + \frac{n}{e}s} \mid r, s \in \mathbb{Z}\}$. Since $\frac{n}{d}\mathbb{Z} + \frac{n}{e}\mathbb{Z} = \gcd(\frac{n}{d}, \frac{n}{e})\mathbb{Z} = \frac{n}{\operatorname{lcm}(d,e)}\mathbb{Z}$, we get $\langle g^{\frac{n}{d}} \rangle \langle g^{\frac{n}{e}} \rangle = \langle g^{\frac{n}{\operatorname{lcm}(d,e)}} \rangle$.
$\therefore$ all three assertions hold. □

Cosets

- If $H \leq G$ and $x \in G, xH$ is called a left coset of H in G and Hx is called a right coset of H in g
- There is a bijective correspondence between left and right cosets of H in G, sending xH to its inverse $(xH)^{-1} = Hx^{-1}$
- Any two cosets of are equal or disjoint, with $xH = yH \iff y^{-1}x \in H$
- A $x \in G$ lies in exacty one coset of H, xH
- $\forall x \in G, \exists$ a bijective correspondence between H and xH
- We define the index of H in G, $|G : H|$ to be the number of cosets of H in G
- The cosets of H in G partition G into $|G : H|$ disjoint sets of cardinality $|H|$, and we have $|G| = |G : H||H|$ (Lagrange's Theorem)
- All subgroups of a finite group are of finite index, while subgroups of an infinite group may be of finite or infinite index

Theorem 1.5. *Let $G = \langle g \rangle$ be an infinite cyclic group*

- *For each $d \in \mathbb{N}, \exists$ exactly one subgroup of G of index d, $\langle g^d \rangle$. Furthermore, every non-trivial subgroup of G is of finite index*
- *The intersection of the subgroups of indices d and e is the subgroup of index* $\operatorname{lcm}(d, e)$
- *The product of subgroups of indices d and e is the subgroup of index* $\gcd(d, e)$

Proof. As in the proof of Theorem 1.4, let $1 \neq H \leq G$, and choose $a = \min\{m \in \mathbb{N} \mid g^m \in H\}$. This set is non-empty since H contains some $g^b \neq e$, and then either $b > 0$ or $g^{-b} \in H$ with $-b > 0$. If $g^c \in H$, write $c = qa + r$, $0 \leq r < a$. Then $g^r = g^c(g^a)^{-q} \in H$, so minimality of a gives $r = 0$. Hence $a \mid c$, so $H = \langle g^a \rangle$.
For $d \in \mathbb{N}$, the cosets of $\langle g^d \rangle$ are $g^i \langle g^d \rangle$, $0 \leq i < d$. Hence $|G : \langle g^d \rangle| = d$. By the previous paragraph, every non-trivial subgroup is $\langle g^a \rangle$ for some $a \in \mathbb{N}$, so it has finite index a, and $\langle g^d \rangle$ is the unique subgroup of index d.
If $d, e \in \mathbb{N}$, then $g^b \in \langle g^d \rangle \cap \langle g^e \rangle \iff d \mid b$ and $e \mid b \iff \operatorname{lcm}(d, e) \mid b$. Thus $\langle g^d \rangle \cap \langle g^e \rangle = \langle g^{\operatorname{lcm}(d,e)} \rangle$, which has index $\operatorname{lcm}(d, e)$.
Since G is abelian, $\langle g^d \rangle \langle g^e \rangle \leq G$ by Proposition 1.3. Moreover $\langle g^d \rangle \langle g^e \rangle = \{g^{dr+es} \mid r, s \in \mathbb{Z}\} = \langle g^{\gcd(d,e)} \rangle$, since $d\mathbb{Z} + e\mathbb{Z} = \gcd(d, e)\mathbb{Z}$. Hence the product has index $\gcd(d, e)$.
$\therefore$ all three assertions hold. □

Theorem 1.6. *If $K \leq H \leq G$, then $|G : K| = |G : H||H : K|$*

Proof. By the coset-counting identity in Lagrange's Theorem, the coefficient of $|H|$ in $|G|$ is $|G : H|$, and the coefficient of $|K|$ in $|H|$ is $|H : K|$. Hence $|G| = |G : H||H|$ and $|H| = |H : K||K|$.
Substituting, $|G| = |G : H||H : K||K|$. But applying the same identity directly to $K \leq G$ gives $|G| = |G : K||K|$.
$\therefore |G : K| = |G : H||H : K|$. □

Let $H \leq G$, and let $\mathcal{I}$ be an indexing set that is in bijective correspondence with G/H. $T = \{t_i \mid i \in \mathcal{I}\} \subseteq G$ is said to be a (*left*) *transversal* for H (or a set of (*left*) *coset representatives* of H in G) if the sets t_iH are precisely the distinct cosets of H in G

Normal Subgroups

- N is a *normal subgroup* of G if $xN = Nx\ \forall x \in G$, or equivalently if $xNx^{-1} \subseteq N$ for all $x \in G$
- If G is abelian, then every subgroup of G is normal.
- If {e} and G are the only normal subgroups of G, then we say that G is *simple*.
- If $H \trianglelefteq G$ and $K \trianglelefteq H$, then it is not necessarily true that $K \trianglelefteq G$. However, it is clearly true that if $K \trianglelefteq G$ and $K \leq H \leq G$, then $K \trianglelefteq H$.

Proposition 1.7. *Let $H, K \leq G$. If $K \trianglelefteq G$, then $HK \leq G$ and $H \cap K \trianglelefteq H$; if also $H \trianglelefteq G$, then $HK \trianglelefteq G$ and $H \cap K \trianglelefteq G$.*

Proof. Let $H, K \leq G$. Suppose that $K \trianglelefteq G$.
Then we have $gK = Kg\ \forall g \in G$. Since $H \leq G$, we have $hK = Kh\ \forall h \in H$.
Take $x \in HK$, so $x = hk$ for some $h \in H, k \in K$. Since $hK = Kh, \exists$ some $h' \in H$ such that $x = kh'$, so we have $HK \subseteq KH$. By a similar argument, we have $KH \subseteq HK$, so we conclude $HK = KH$. By Proposition 1.3, we have $HK \leq G$.
Take $y \in H \cap K$ and $h \in H$. Since $y \in H$, we have $hyh^{-1} \in H$. Since $y \in K$ and $K \trianglelefteq G$, we have $hKh^{-1} \subseteq K \Rightarrow hyh^{-1} \in K$. It follows that $hyh^{-1} \in H \cap K$, so $H \cap K \trianglelefteq H$.
Suppose also that $H \trianglelefteq G$. Take $hk \in HK$ and $g \in G$. Then $g(hk)g^{-1} = (ghg^{-1})(gkg^{-1})$, where $ghg^{-1} \in H$ since $H \trianglelefteq G$ and $gkg^{-1} \in K$ since $K \trianglelefteq G$. Hence $g(hk)g^{-1} \in HK$, so $gHKg^{-1} \subseteq HK \Rightarrow HK \trianglelefteq G$.
Take $y \in H \cap K$ and $g \in G$. Since $H \trianglelefteq G$, $gyg^{-1} \in H$, and since $K \trianglelefteq G$, $gyg^{-1} \in K$. So $gyg^{-1} \in H \cap K \Rightarrow H \cap K \trianglelefteq G$. □

Proposition 1.8. *Any subgroup of index 2 is normal*

Proof. Let $H \leq G$ and suppose that $|G : H| = 2$
Then there are 2 left cosets, H and G - H
It follows that $x \in H \iff xH = H = Hx$, and $x \notin H \iff xH = G - H = Hx$
$\therefore H \trianglelefteq G$ □

Theorem 1.9. *If $N \trianglelefteq G$, then G/N forms the quotient group under the operation $(xN)(yN) = (xy)N$*

Proof. First, we show that the operation is well-defined. Suppose $xN = x'N$ and $yN = y'N$. Then $x' = xn_1$ and $y' = yn_2$ for some $n_1, n_2 \in N$. Since $N \trianglelefteq G$, $y^{-1}n_1y \in N$, so $n_1y = y(y^{-1}n_1y)$. Hence $x'y' = xn_1yn_2 = xy(y^{-1}n_1y)n_2 \in xyN$, and so $x'y'N = xyN$. Thus $(xN)(yN) = (xy)N$ is well-defined.
Associativity follows from associativity in G, since $((xN)(yN))(zN) = ((xy)z)N$ and $(xN)((yN)(zN)) = (x(yz))N$.
The identity is $N = eN$, since $N(xN) = xN = (xN)N$. Also, $(xN)(x^{-1}N) = N = (x^{-1}N)(xN)$, so $x^{-1}N$ is the inverse of xN.
$\therefore G/N$ is a group under the operation $(xN)(yN) = (xy)N$. □

The identity element of G/N is N, and the inverse is $x^{-1}N$. If G is abelian, then G/N is also abelian

Conjugation

- The conjugate of an element x by g is defined as gxg^{-1}
- x and y are conjugate if $\exists g \in G$ s.t. $y = gxg^{-1}$
- No 2 distinct elements of an abelian group can be conjugate
- $N \trianglelefteq G \iff \forall n \in N, gng^{-1} \in N$

Permutations

- A permutation of a set X is a bijective set map from X to X. The set of permutations of X forms a group under composition
- If $X = \{1, \ldots, n\}$, then this groups is called the symmetric group, S_n
- S_n is a finite group of order $n!$

- $\rho \in S_n$ is called a r-cycle if $\exists$ distinct integers $1 \leq a_1, \dots, a_r \leq n$ s.t. $\rho(a_i) = a_{i+1} \forall\ 1 \leq i \leq r, \rho(a_r) = a_1, \rho(b) = b\ \forall 1 \leq b \leq n$ which is not equal to some a_i
- Two cycles are disjoint if there is no number that is moved by both cycles
- Every element of S_n can be written as a product of disjoint cycles
- The cycle structure refers to the partion of n consisting of the lengths of the cycles that appear in a disjoint cycle decomposition of ρ

Proposition 1.10. *Let $n \in \mathbb{N}$. Then two elements of S_n are conjugate iff they have the same cycle structure*

Proof. Let $\alpha, \beta \in S_n$.
($\Rightarrow$) Suppose that α and β are conjugate, so $\beta = \sigma\alpha\sigma^{-1}$ for some $\sigma \in S_n$. We show that $\sigma\alpha\sigma^{-1}$ is the permutation with the same cycle structure as α obtained by applying σ to the symbols in α. Let π be this permutation.
Suppose that α fixes some i, so $\sigma(i)$ resides in a 1-cycle of π, and π fixes $\sigma(i)$. Then $\sigma\alpha\sigma^{-1}(\sigma(i)) = \sigma\alpha(i) = \sigma(i)$, so $\sigma\alpha\sigma^{-1}$ fixes $\sigma(i)$ as well.
Suppose that α moves i, and let $\alpha(i) = j$. Take the complete factorization of α to be

$$\alpha = \gamma_1\gamma_2 \cdots (\cdots i\ j \cdots) \cdots \gamma_t$$

Take $\sigma(i) = k$ and $\sigma(j) = l$, so $\pi : k \mapsto l$. Since $\sigma\alpha\sigma^{-1} : k \mapsto i \mapsto j \mapsto l$, we have $\sigma\alpha\sigma^{-1}(k) = \pi(k)$. So π and $\sigma\alpha\sigma^{-1}$ agree on all symbols of the form $k = \sigma(i)$. Since σ is a surjection, it follows that $\pi = \sigma\alpha\sigma^{-1} = \beta$, so α and β have the same cycle structure.
($\Leftarrow$) Suppose that α and β have the same cycle structure. Place the complete factorization of α over that of β so that cycles of the same length correspond, and let $\gamma \in S_n$ be the function sending the top to the bottom. For example, if

$$\alpha = \gamma_1\gamma_2 \cdots (\cdots i\ j \cdots) \cdots \gamma_t$$
$$\beta = \delta_1\delta_2 \cdots (\cdots k\ l \cdots) \cdots \delta_t$$

then $\gamma(i) = k, \gamma(j) = l$, etc. γ is a permutation, since every i between 1 and n occurs exactly once in a complete factorization. By the forward direction, $\gamma\alpha\gamma^{-1} = \beta$, so α and β are conjugate. □

- A **transposition** in S_n is a 2-cycle
- Every element of S_n can be written as a (not necessarily disjoint) product of transpositions in many ways. However, any two such expressions use the same number, modulo 2, of transpositions
- A permutation is **even** (resp. **odd**) if it is a product of an even (resp. odd) number of transpositions; every permutation is exactly one of these. Since an r-cycle is a product of $r-1$ transpositions, an r-cycle is even iff r is odd
- The subset of S_n consisting of all even permutations is a subgroup of index 2, hence is normal in S_n by Proposition 1.8; this is the **alternating group** of degree n, denoted A_n

Homomorphisms

- Let G and H be groups. A **homomorphism** is a map $\varphi \colon G \to H$ satisfying $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in G$
- If φ is a homomorphism, then $\varphi(1) = 1$ and $\varphi(x^{-1}) = \varphi(x)^{-1}$ for any $x \in G$
- The **trivial homomorphism** from G to H sends every element of G to the identity of H
- φ is a **monomorphism** if injective, an **epimorphism** if surjective, and an **isomorphism** if bijective. If φ is an isomorphism, then so is $\varphi^{-1} : H \to G$
- A homomorphism $\varphi : G \to G$ is an **endomorphism** of G; a bijective endomorphism is an **automorphism**
- If there is an isomorphism $\varphi : G \to H$, we say G and H are **isomorphic** and write $G \cong H$. Isomorphism is an equivalence relation on groups, so we can speak of the *isomorphism class* of a group.

Standard examples of homomorphisms and isomorphisms:

- Let $G = \langle g \rangle$ and $H = \langle h \rangle$ be cyclic groups of order n. The map $\varphi : G \to H$ by $\varphi(g^a) = h^a$ is an isomorphism. Hence any two finite cyclic groups of the same order are isomorphic.
- Let G be a group, $H \le G$, $g \in G$. The **conjugate** of H by g is $gHg^{-1} = \{ghg^{-1} \mid h \in H\}$, and $gHg^{-1} \le G$. The map $\varphi : H \to gHg^{-1}$ by $\varphi(h) = ghg^{-1}$ is an isomorphism, so conjugate subgroups are isomorphic. (However, isomorphic subgroups need not be conjugate)
- Let $X = \{x_1, \ldots, x_n\}$ and let S_X be the group of permutations of X. The map $\varphi : S_n \to S_X$ by $\varphi(\rho)(x_i) = x_{\rho(i)}$ is an isomorphism
- Let G be a group, $N \trianglelefteq G$. The **natural map** $\eta : G \to G/N$ defined by $\eta(x) = xN$ is an epimorphism

Kernel and image:

- If $\varphi : G \to H$ is a homomorphism, the **kernel** of φ is $\ker\varphi = \{g \in G \mid \varphi(g) = 1\}$, and the **image** is $\operatorname{im}\varphi = \{\varphi(g) \mid g \in G\}$
- We use $\varphi(K) = \{\varphi(k) \mid k \in K\}$ for the image of $K \le G$
- E.g., if $N \trianglelefteq G$ and $\eta : G \to G/N$ is the natural map, then $\ker\eta = N$ and $\eta(K) = KN/N$ for any $K \le G$. ($\eta(K) = K/N$ if K contains N.)

Proposition 1.11. *Let G and H be groups, and let $\varphi : G \to H$ be a homomorphism. Then* $\ker\varphi \trianglelefteq G$, *and $\varphi(K) \le H$ for any $K \le G$.*

Proof. First, $\varphi(1) = 1$, so $1 \in \ker\varphi$. If $x, y \in \ker\varphi$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = 1$, so $xy^{-1} \in \ker\varphi$. Hence $\ker\varphi \le G$.
Let $x \in \ker\varphi$ and $g \in G$. Then $\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g^{-1}) = \varphi(g)\varphi(g)^{-1} = 1$. Thus $gxg^{-1} \in \ker\varphi$, so $\ker\varphi \trianglelefteq G$.
Now let $K \le G$. Since $1 \in K$, we have $1 = \varphi(1) \in \varphi(K)$. If $a, b \in \varphi(K)$, then $a = \varphi(k_1)$ and $b = \varphi(k_2)$ for some $k_1, k_2 \in K$. Hence $ab^{-1} = \varphi(k_1)\varphi(k_2)^{-1} = \varphi(k_1k_2^{-1}) \in \varphi(K)$.
$\therefore \varphi(K) \le H$. □

Theorem (Fundamental Theorem on Homomorphisms)**.** *If G and H are groups and $\varphi : G \to H$ is a homomorphism, then there is an isomorphism $\psi : G/K \to \varphi(G)$ such that $\varphi = \psi \circ \eta$, where $K = \ker\varphi$ and $\eta : G \to G/K$ is the natural map; moreover, the map ψ is uniquely determined.*

Proof. Define $\psi\colon G/K \to \varphi(G)$ by $\psi(xK) = \varphi(x)$. This is well-defined: if $xK = yK$, then $y^{-1}x \in K$, so $1 = \varphi(y^{-1}x) = \varphi(y)^{-1}\varphi(x)$, and hence $\varphi(x) = \varphi(y)$.
The map ψ is a homomorphism, since $\psi((xK)(yK)) = \psi(xyK) = \varphi(xy) = \varphi(x)\varphi(y) = \psi(xK)\psi(yK)$.
It is surjective by definition of $\varphi(G)$: if $h \in \varphi(G)$, then $h = \varphi(x) = \psi(xK)$ for some $x \in G$. It is injective since $\psi(xK) = 1 \Rightarrow \varphi(x) = 1 \Rightarrow x \in K \Rightarrow xK = K$. Hence ψ is an isomorphism.
For all $x \in G$, $(\psi \circ \eta)(x) = \psi(xK) = \varphi(x)$, so $\varphi = \psi \circ \eta$.
Finally, suppose $\theta\colon G/K \to \varphi(G)$ also satisfies $\varphi = \theta \circ \eta$. For any $xK \in G/K$, $\theta(xK) = \theta(\eta(x)) = \varphi(x) = \psi(xK)$. Thus $\theta = \psi$, so ψ is uniquely determined. □

Theorem (First Isomorphism Theorem)**.** *Let G be a group. If $N \trianglelefteq G$ and $H \le G$, then $HN/N \cong H/(H \cap N)$.*

Proof. By Proposition 1.7, $HN \le G$ and $H \cap N \trianglelefteq H$. Since $N \trianglelefteq G$ and $HN \le G$, we also have $N \trianglelefteq HN$, so HN/N is a group.
Let $\eta\colon G \to G/N$ be the natural map $\eta(g) = gN$, and let $\varphi = \eta|_H\colon H \to G/N$. Then φ is a homomorphism, and $\ker\varphi = \{h \in H \mid hN = N\} = H \cap N$.
For the natural map, $\eta(K) = KN/N$ for any $K \le G$. Taking $K = H$, we get $\varphi(H) = \eta(H) = HN/N$, so we may regard φ as a homomorphism $H \to HN/N$.
By the Fundamental Theorem on Homomorphisms, $H/\ker\varphi \cong \varphi(H)$. Thus $H/(H \cap N) \cong HN/N$.
$\therefore HN/N \cong H/(H \cap N)$. □

Theorem (Correspondence Theorem)**.** *Let G and H be groups, and let $\varphi : G \to H$ be an epimorphism having kernel N. Then there is a bijective correspondence given by φ between the set of subgroups of G that contain N and the set of subgroups of H. If K is a subgroup of G containing N, then this correspondence sends K to $\varphi(K)$; if L is a subgroup of H, then the subgroup of G sent to L under this correspondence is $\varphi^{-1}(L) = \{x \in G \mid \varphi(x) \in L\}$. Moreover, if K_1 and K_2 are subgroups of G containing N, then:*

- *$K_2 \leq K_1$ iff $\varphi(K_2) \leq \varphi(K_1)$, and in this case we have $|K_1 : K_2| = |\varphi(K_1) : \varphi(K_2)|$.*
- *$K_2 \trianglelefteq K_1$ iff $\varphi(K_2) \trianglelefteq \varphi(K_1)$, and in this case the map from K_1/K_2 to $\varphi(K_1)/\varphi(K_2)$ sending xK_2 to $\varphi(x)\varphi(K_2)$ is an isomorphism.*

Proof. Let $\mathcal{S} = \{K \leq G \mid N \leq K\}$ and $\mathcal{T} = \{L \leq H\}$. For $K \in \mathcal{S}$, Proposition 1.11 gives $\varphi(K) \leq H$. If $L \in \mathcal{T}$, then $\varphi^{-1}(L) \leq G$ by the subgroup test, and $N \leq \varphi^{-1}(L)$, so $\varphi^{-1}(L) \in \mathcal{S}$.
Define $\Phi\colon \mathcal{S} \to \mathcal{T}$ by $\Phi(K) = \varphi(K)$, and define $\Psi\colon \mathcal{T} \to \mathcal{S}$ by $\Psi(L) = \varphi^{-1}(L)$. Since φ is surjective, $\Phi(\Psi(L)) = \varphi(\varphi^{-1}(L)) = L$. If $K \in \mathcal{S}$, then $K \subseteq \Psi(\Phi(K))$. Conversely, if $x \in \Psi(\Phi(K))$, then $\varphi(x) = \varphi(k)$ for some $k \in K$, so $k^{-1}x \in \ker\varphi = N \leq K$, and hence $x \in K$. Thus $\Psi(\Phi(K)) = K$. Hence Φ and Ψ are inverse bijections.
Let $K_1, K_2 \in \mathcal{S}$. If $K_2 \leq K_1$, then $\varphi(K_2) \leq \varphi(K_1)$. Conversely, if $\varphi(K_2) \leq \varphi(K_1)$, then $K_2 = \Psi(\Phi(K_2)) \leq \Psi(\Phi(K_1)) = K_1$.
Now suppose $K_2 \leq K_1$. Define $\alpha\colon K_1/K_2 \to \varphi(K_1)/\varphi(K_2)$ by $\alpha(xK_2) = \varphi(x)\varphi(K_2)$. If $xK_2 = yK_2$, then $y^{-1}x \in K_2$, so $\varphi(y)^{-1}\varphi(x) \in \varphi(K_2)$, and $\alpha(xK_2) = \alpha(yK_2)$. Thus α is well-defined. It is surjective by definition. If $\alpha(xK_2) = \alpha(yK_2)$, then $\varphi(y)^{-1}\varphi(x) = \varphi(k)$ for some $k \in K_2$. Hence $k^{-1}y^{-1}x \in N \leq K_2$, so $y^{-1}x \in K_2$, and $xK_2 = yK_2$. Thus α is bijective, so $|K_1 : K_2| = |\varphi(K_1) : \varphi(K_2)|$.
If $K_2 \trianglelefteq K_1$, then for $x \in K_1$, $k \in K_2$, we have $\varphi(x)\varphi(k)\varphi(x)^{-1} = \varphi(xkx^{-1}) \in \varphi(K_2)$, so $\varphi(K_2) \trianglelefteq \varphi(K_1)$.
Conversely, suppose $\varphi(K_2) \trianglelefteq \varphi(K_1)$. If $x \in K_1$, $k \in K_2$, then $\varphi(xkx^{-1}) \in \varphi(K_2)$, so $xkx^{-1} \in \Psi(\Phi(K_2)) = K_2$. Hence $K_2 \trianglelefteq K_1$.
In this case, α is a homomorphism, since $\alpha((xK_2)(yK_2)) = \alpha(xyK_2) = \varphi(xy)\varphi(K_2) = \varphi(x)\varphi(y)\varphi(K_2) = \alpha(xK_2)\alpha(yK_2)$. Since α is already bijective, it is an isomorphism.
$\therefore$ the correspondence has all the stated properties. □

As a special case of the Correspondence Theorem, if G is a group and $N \trianglelefteq G$, then every subgroup of G/N has the form K/N for some subgroup $K \leq G$ with $N \leq K$. Here we take φ to be the natural map $G \to G/N$.

Theorem (Second Isomorphism Theorem)**.** *Let H and K be normal subgroups of a group G. If H contains K, then $G/H \cong (G/K)/(H/K)$.*

Proof. Let $\eta\colon G \to G/K$ be the natural map. Then $\ker\eta = K$. Since $K \leq H \leq G$, the Correspondence Theorem applies to $H \leq G$.
Since $H \trianglelefteq G$, applying the normality and quotient-isomorphism parts with $K_1 = G$ and $K_2 = H$ gives $\eta(H) \trianglelefteq \eta(G)$ and $G/H \cong \eta(G)/\eta(H)$.
Now $\eta(G) = G/K$ and $\eta(H) = H/K$. Hence $G/H \cong (G/K)/(H/K)$.
$\therefore G/H \cong (G/K)/(H/K)$. □

Exercises

Exercise 1.1. Prove, or complete the sketched proof of, each result in this section.

Solution. The sketched proofs are completed above in:

- Proposition 1.1, Lagrange's Theorem, Proposition 1.2, and Proposition 1.3
- Theorem 1.4, Theorem 1.5, and Theorem 1.6
- Proposition 1.7, Proposition 1.8, Theorem 1.9, Proposition 1.10, and Proposition 1.11
- Fundamental Theorem on Homomorphisms, First Isomorphism Theorem, Correspondence Theorem, and Second Isomorphism Theorem

□

Exercise 1.2. We say that a group G has *exponent* e if e is the smallest positive integer such that $x^e = 1$ for every $x \in G$. Show that if G has exponent 2, then G is abelian. For what integers e is a group having exponent e necessarily abelian?

Solution. First suppose that G has exponent 2. Let $x, y \in G$. Then $x^2 = y^2 = (xy)^2 = 1$, so $xyxy = 1$. Multiplying on the left by x and on the right by y, and using $x^2 = y^2 = 1$, gives $yx = xy$. Since x, y were arbitrary, G is abelian.
If $e = 1$, then $x = 1$ for all $x \in G$, so $G = \{1\}$, which is abelian. We claim that these are the only possibilities.
Let $e \geq 3$. If e is even, take the dihedral group

$$D_e = \langle r, s \mid r^e = s^2 = 1,\ srs^{-1} = r^{-1} \rangle.$$

This group is non-abelian, since $srs^{-1} = r^{-1} \neq r$. Its rotations have order dividing e, and its reflections have order 2. Since e is even, the exponent is $\operatorname{lcm}(e, 2) = e$.
Now suppose that $e \geq 3$ is odd. Let

$$U_e = \left\{ M(a,b,c) = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \;\middle|\; a, b, c \in \mathbb{Z}/e\mathbb{Z} \right\},$$

under matrix multiplication, with entries computed in $\mathbb{Z}/e\mathbb{Z}$. Direct multiplication gives

$$M(a,b,c)M(a',b',c') = M(a + a', b + b', c + c' + ab').$$

Hence U_e is closed under multiplication. The identity is $M(0,0,0)$, and

$$M(a,b,c)^{-1} = M(-a, -b, ab - c).$$

Associativity follows from associativity of matrix multiplication, so U_e is a group. It is non-abelian, since

$$M(1,0,0)M(0,1,0) = M(1,1,1), \qquad M(0,1,0)M(1,0,0) = M(1,1,0).$$

For $n \in \mathbb{N}$, induction using the multiplication formula gives

$$M(a,b,c)^n = M\left(na, nb, nc + \binom{n}{2}ab\right).$$

Taking $n = e$, we get

$$M(a,b,c)^e = M\left(0, 0, \binom{e}{2}ab\right).$$

Since e is odd, $\binom{e}{2} = e(e-1)/2$ is divisible by e, so $M(a,b,c)^e = M(0,0,0)$ for every $M(a,b,c) \in U_e$. Thus every element has order dividing e. However $M(1,0,0)^n = M(n,0,0)$, so $M(1,0,0)$ has order e. Hence U_e is a non-abelian group of exponent e.
$\therefore$ a group of exponent e is necessarily abelian iff $e = 1$ or $e = 2$. $\square$

Exercise 1.3. Let G be a finite group, and suppose that the map $\varphi : G \to G$ defined by $\varphi(x) = x^3$ for $x \in G$ is a homomorphism. Show that if 3 does not divide $|G|$, then G must be abelian. (See [2] for a generalization.)

Solution. Let $x, y \in G$. Since φ is a homomorphism, we have

$$x^3y^3 = \varphi(x)\varphi(y) = \varphi(xy) = (xy)^3 = xyxyxy.$$

Cancelling x on the left and y on the right gives

$$(yx)^2 = x^2y^2.$$

Swapping x and y, we also have

$$(xy)^2 = y^2x^2.$$

Using this in the original equation,

$$x^3y^3 = (xy)^3 = (xy)(xy)^2 = xy(y^2x^2) = xy^3x^2.$$

Cancelling x on the left gives
$$y^3x^2 = x^2y^3.$$
Hence every cube commutes with every square.
Now let $z \in \ker\varphi$. Then $z^3 = 1$, so $o(z) \mid 3$. Then we also have $o(z) \mid |G|$. Since $3 \nmid |G|$, we must have $o(z) = 1$, and hence $z = 1$. Thus $\ker\varphi = \{1\}$, so φ is injective. Since G is finite, φ is also surjective. Therefore every element of G is a cube.
Let $g \in G$. Since every element of G is a cube, write $g = y^3$ for some $y \in G$. Then for every $x \in G$, the relation above gives
$$gx^2 = y^3x^2 = x^2y^3 = x^2g.$$
Thus every square is central. In particular, x^2 and y^2 commute for all $x, y \in G$. Since $(xy)^2 = y^2x^2$, it follows that
$$(xy)^2 = x^2y^2.$$
Hence $xyxy = xxyy$. Cancelling x on the left and y on the right gives $yx = xy$. Since x, y were arbitrary, G is abelian. □

Exercise 1.4. Let g be an element of a group G, and suppose that $|G| = mn$ where m and n are coprime. Show that there are unique elements x and y of G such that $xy = g = yx$ and $x^m = 1 = y^n$. (In the case where m is a power of some prime p, we call x the p-part of g and y the p'-part of g; more generally, if π is a set of primes which includes all prime divisors of m and no prime divisors of n, then x and y are called the π-part and π'-part, respectively, of g.)

Solution. Since $\gcd(m, n) = 1$, choose $a, b \in \mathbb{Z}$ such that $am + bn = 1$. Since $\langle g\rangle \le G$, Lagrange's Theorem gives $o(g) = |\langle g\rangle| \mid |G| = mn$, so $g^{mn} = 1$. Define
$$x = g^{bn}, \qquad y = g^{am}.$$
Then x and y commute, since both are powers of g. Moreover,
$$xy = g^{bn}g^{am} = g^{am+bn} = g,$$
and similarly $yx = g$. Also,
$$x^m = g^{bmn} = (g^{mn})^b = 1, \qquad y^n = g^{amn} = (g^{mn})^a = 1.$$
Thus such elements $x, y \in G$ exist.
We now show that they are unique. Suppose that $x', y' \in G$ also satisfy
$$x'y' = g = y'x', \qquad (x')^m = 1, \qquad (y')^n = 1.$$
Since x' and y' commute, we may take powers of $x'y'$ termwise. Thus
$$g^{bn} = (x'y')^{bn} = (x')^{bn}(y')^{bn}.$$
But $(y')^{bn} = ((y')^n)^b = 1$, so
$$g^{bn} = (x')^{bn}.$$
Since $bn = 1 - am$, we get
$$(x')^{bn} = (x')^{1-am} = x'((x')^m)^{-a} = x'.$$
Therefore $x' = g^{bn} = x$. Similarly,
$$g^{am} = (x'y')^{am} = (x')^{am}(y')^{am} = (y')^{am},$$
and since $am = 1 - bn$,
$$(y')^{am} = (y')^{1-bn} = y'((y')^n)^{-b} = y'.$$
Therefore $y' = g^{am} = y$. Hence the required elements x and y exist and are unique.

□

Exercise 1.5. Let r, s, and t be positive integers greater than 1. Show that there is a finite group G having elements x and y such that x has order r, y has order s, and xy has order t.

Solution. First approach: a linear construction. Let

$$N = 2rst.$$

Choose a prime $p \nmid N$. By taking a suitable power $q = p^d$, we may arrange that

$$q \equiv 1 \pmod N,$$

since p is invertible modulo N. Thus $2r, 2s, 2t$ all divide $q-1$. Since $\mathbb{F}_q^\times$ is cyclic, we may choose elements $u, v, w \in \mathbb{F}_q^\times$ of orders $2r, 2s, 2t$, respectively. Define

$$A = \begin{pmatrix} u & 1 \\ 0 & u^{-1} \end{pmatrix}, \qquad B = \begin{pmatrix} v & 0 \\ \lambda & v^{-1} \end{pmatrix},$$

where

$$\lambda = w + w^{-1} - uv - u^{-1}v^{-1}.$$

Then $A, B \in \mathrm{SL}(2, \mathbb{F}_q)$. The characteristic polynomial of A is

$$X^2 - (u + u^{-1})X + 1,$$

with distinct roots u, u^{-1}, since u has order $2r > 2$. Hence A is conjugate over $\mathbb{F}_q$ to $\mathrm{diag}(u, u^{-1})$, and so A has order $2r$. Similarly, B has characteristic polynomial

$$X^2 - (v + v^{-1})X + 1$$

with distinct roots v, v^{-1}, and therefore B has order $2s$. Finally,

$$AB = \begin{pmatrix} uv + \lambda & v^{-1} \\ u^{-1}\lambda & u^{-1}v^{-1} \end{pmatrix},$$

so

$$\mathrm{tr}(AB) = uv + \lambda + u^{-1}v^{-1} = w + w^{-1}.$$

Since $\det(AB) = 1$, the characteristic polynomial of AB is

$$X^2 - (w + w^{-1})X + 1,$$

with distinct roots w, w^{-1}. Thus AB has order $2t$.

Now pass to the finite quotient

$$G = \mathrm{SL}(2, \mathbb{F}_q)/\{\pm I\}.$$

The element $-I$ is the unique element of order 2 in $\mathrm{SL}(2, \mathbb{F}_q)$: indeed, if $C^2 = I$, then C is diagonalizable with eigenvalues in $\{\pm 1\}$, and the condition $\det C = 1$ forces either $C = I$ or $C = -I$. Hence, if an element $g \in \mathrm{SL}(2, \mathbb{F}_q)$ has order $2n$, its image modulo $\{\pm I\}$ has order n. For g^n is the unique element of order 2, hence $g^n = -I$, so the image has order dividing n; and if a smaller positive power became trivial in the quotient, then $g^m = \pm I$ for some $m < n$. The case $g^m = I$ contradicts $o(g) = 2n$, while $g^m = -I = g^n$ gives $g^{n-m} = I$, again impossible. Therefore the images x, y of A, B in G have orders r, s, respectively, and xy, the image of AB, has order t.

Second approach: finite quotients of triangle groups. We use one standard fact about a particular group given by generators and relations. For $a, b, c > 1$, the triangle group $\Delta(a, b, c)$ is the abstract group

$$\Delta(a, b, c) = \langle U, V \mid U^a = 1,\ V^b = 1,\ (UV)^c = 1 \rangle$$

generated by two formal elements U and V, subject to the relations $U^a = 1$, $V^b = 1$, and $(UV)^c = 1$. The word "triangle" is only the standard name for this presentation; no geometry is needed here. Equivalently, $\Delta(a, b, c)$ is the universal group with these three relations: whenever a group H has elements $\alpha, \beta \in H$ satisfying

$$\alpha^a = 1, \qquad \beta^b = 1, \qquad (\alpha\beta)^c = 1,$$

there is a homomorphism $\Delta(a, b, c) \to H$ sending U to α and V to β.

The standard fact we use is that U, V, UV have exact orders a, b, c, respectively, in $\Delta(a, b, c)$, and that $\Delta(a, b, c)$ is residually finite. Here residually finite means that if $g \neq 1$, then there is a homomorphism

from $\Delta(a,b,c)$ to some finite group in which the image of g is not 1.
Now apply this fact with $a = r$, $b = s$, and $c = t$. Put

$$\Gamma = \langle X, Y \mid X^r = 1,\ Y^s = 1,\ (XY)^t = 1\rangle.$$

In Γ, the elements X, Y, XY have exact orders r, s, t. Hence every element of the finite set

$$\mathcal{S} = \{X^i \mid 1 \le i < r\} \cup \{Y^j \mid 1 \le j < s\} \cup \{(XY)^k \mid 1 \le k < t\}$$

is non-identity. By residual finiteness, for each $g \in \mathcal{S}$, there is a finite group Q_g and a homomorphism

$$\varphi_g \colon \Gamma \to Q_g$$

such that $\varphi_g(g) \ne 1$. Now define

$$Q = \prod_{g\in\mathcal{S}} Q_g, \qquad \Phi\colon \Gamma \to Q, \qquad \Phi(h) = \big(\varphi_g(h)\big)_{g\in\mathcal{S}}.$$

Let $G = \Phi(\Gamma) \le Q$, and set

$$x = \Phi(X), \qquad y = \Phi(Y).$$

Since Q is finite, G is finite. Also, since $X^r = 1$, $Y^s = 1$, and $(XY)^t = 1$ in Γ, we have

$$x^r = 1, \qquad y^s = 1, \qquad (xy)^t = 1.$$

Thus $o(x) \mid r$, $o(y) \mid s$, and $o(xy) \mid t$.
It remains to show that no smaller positive powers become trivial in this finite quotient. Suppose $1 \le i < r$. Since $X^i \in \mathcal{S}$, the X^i-coordinate of $\Phi(X^i)$ is $\varphi_{X^i}(X^i)$, which is not 1. Hence $\Phi(X^i) \ne 1$, so $x^i \ne 1$. Therefore $o(x) = r$.
Similarly, if $1 \le j < s$, then the Y^j-coordinate of $\Phi(Y^j)$ is $\varphi_{Y^j}(Y^j) \ne 1$, so $y^j \ne 1$. Hence $o(y) = s$.
Finally, if $1 \le k < t$, then the $(XY)^k$-coordinate of $\Phi((XY)^k)$ is $\varphi_{(XY)^k}((XY)^k) \ne 1$, so $(xy)^k \ne 1$. Hence $o(xy) = t$.

Third approach: Miller's permutation construction. We now give Miller's construction in modern notation. Miller's "substitutions" are permutations, and his "circular substitutions" are cycles. The proof is based on the following explicit overlap calculation. Let two cycles share a consecutive block

$$a_1, a_2, \dots, a_c$$

and have no other letters in common. If $c = 2h+1$ is odd, arrange the two cycles as

$$S_1 = (a_1\ \cdots\ a_{2j}\ a_{2j+1}\ \cdots\ a_{2h+1}\ b_1\ \cdots\ b_p),$$

$$S_2 = (a_{2j+1}\ \cdots\ a_1\ a_{2j+2}\ \cdots\ a_{2h+1}\ d_1\ \cdots\ d_q).$$

Tracing the product gives

$$S_1S_2 = (a_{2j+1}\ a_{2j+3}\ \cdots\ a_{2h+1}\ b_1\ \cdots\ b_p\ a_{2j+2}\ a_{2j+4}\ \cdots\ a_{2h}\ d_1\ \cdots\ d_q),$$

so S_1S_2 is one cycle and the omitted common letters are exactly

$$a_1, \dots, a_{2j}.$$

Thus an odd overlap lets us omit any prescribed even number of common letters. Similarly, if $c = 2h$ is even and the same calculation is made with the first omitted block $a_1, \dots, a_{2j-1}$, then the product is one cycle omitting exactly $2j-1$ common letters. Thus an even overlap lets us omit any prescribed odd number of common letters.
If the required parity is opposite to the parity supplied by the two-cycle calculation, Miller uses one cycle together with a permutation of the form

(one transposition)(one disjoint cycle).

The same trace shows that this changes the parity of the number of omitted common letters. This is harmless whenever the corresponding order is even, since the least common multiple of 2 and an even cycle length is still the required even order. When both relevant orders are odd, no transposition is inserted; instead, one uses two adjacent odd cycles and performs the same overlap twice. The first overlap changes the parity, and the second restores the required odd order. This is the only extra bookkeeping in Miller's all-odd case.

We now use this calculation to build permutations L, M, N of prescribed orders $\ell, m, n > 1$ with $LM = N$. Since

$$LM = N \quad \Longleftrightarrow \quad MN^{-1} = L^{-1} \quad \Longleftrightarrow \quad N^{-1}L = M^{-1},$$

we may construct the triple after arranging the desired orders as $\ell \leq m \leq n$, and then translate back to the original order. Write

$$n = am + b, \qquad 0 \leq b < m.$$

Take M to be a product of a disjoint m-cycles,

$$M = (a_1 \, \cdots \, a_m)(a_{m+1} \, \cdots \, a_{2m}) \cdots (a_{(a-1)m+1} \, \cdots \, a_{am}).$$

The permutation L is chosen so that its cycles join these m-cycles end-to-end. If $a = 1$, this is exactly the overlap calculation above, since $m \leq n < m + \ell$. For instance, when ℓ is even and $a > 1$, the joining part may be written schematically as

$$\begin{aligned} L = {} & (a_1 \, a_{am+1})(a_2 \, a_{am+2}) \cdots (a_b \, a_{am+b}) \\ & \cdot (a_m \, a_{m+1})(a_{2m} \, a_{2m+1}) \cdots \\ & \cdot (a_{(a-1)m} \, a_{(a-1)m+1} \, \cdots \, a_{(a-1)m+\ell-1}). \end{aligned}$$

where the first row is omitted if $b = 0$. Here the displayed transpositions perform the ordinary joins, and the final ℓ-cycle makes $o(L) = \ell$. A direct trace explains the construction: starting at a_1, the product LM first follows the first m-cycle of M; when it reaches the end of that cycle, the next factor of L sends it to the beginning of the next m-cycle; the same process repeats until the trace reaches the final block. At the final block the overlap calculation above omits exactly $m - b$ common letters, so the resulting product has precisely

$$am + b = n$$

letters. Hence LM is an n-cycle.

If ℓ is odd, the same threading is done with ℓ-cycles rather than transpositions. The only issue is parity of the number of omitted common letters in the final overlap. When m is even, the transposition-plus-cycle variant above supplies the missing parity while keeping $o(M) = m$. When both ℓ and m are odd, two neighboring odd overlaps are used: the first overlap changes the parity of the omitted block and the second overlap joins the next m-cycle back into the same long trace. Since every component of L is then an ℓ-cycle and every component of M is an m-cycle, the orders remain

$$o(L) = \ell, \qquad o(M) = m.$$

The same trace as before shows that the product is still a single n-cycle. Thus in all cases we have constructed permutations L, M, N such that

$$o(L) = \ell, \qquad o(M) = m, \qquad o(N) = n, \qquad LM = N.$$

We now return to r, s, t. If t is the largest of the three orders, take $\ell = r$, $m = s$, and $n = t$, and set $x = L$, $y = M$. If r is the largest, take $\ell = s$, $m = t$, and $n = r$. Then $LM = N$, so $N^{-1}L = M^{-1}$; setting $x = N^{-1}$ and $y = L$ gives $o(x) = r$, $o(y) = s$, and $o(xy) = t$. If s is the largest, take $\ell = t$, $m = r$, and $n = s$. Then $LM = N$, so $MN^{-1} = L^{-1}$; setting $x = M$ and $y = N^{-1}$ gives $o(x) = r$, $o(y) = s$, and $o(xy) = t$. In all cases the permutations act on a finite set Ω. Let

$$G = \langle x, y \rangle \leq \mathrm{Sym}(\Omega).$$

Then G is finite, and by construction $o(x) = r$, $o(y) = s$, and $o(xy) = t$. $\therefore$ the required finite group exists. $\square$

Exercise 1.6. Let X and Y be subsets of a group G. Are $\langle X \rangle \cap \langle Y \rangle$ and $\langle X \cap Y \rangle$ necessarily equal? Are $\langle \langle X \rangle \cup \langle Y \rangle \rangle$ and $\langle X \cup Y \rangle$ necessarily equal?

Solution. The first pair need not be equal. Let $G = C_3 = \langle g \rangle$, $X = \{1, g\}$, and $Y = \{1, g^2\}$. Then $\langle X \rangle = G$, since $g \in X$, and $\langle Y \rangle = G$, since g^2 also generates C_3. Hence

$$\langle X \rangle \cap \langle Y \rangle = G.$$

However, $X \cap Y = \{1\}$, so

$$\langle X \cap Y \rangle = \{1\}.$$

Thus $\langle X \rangle \cap \langle Y \rangle$ and $\langle X \cap Y \rangle$ are not necessarily equal.
The second pair is always equal. Since $X \subseteq \langle X \rangle$ and $Y \subseteq \langle Y \rangle$, we have

$$X \cup Y \subseteq \langle X \rangle \cup \langle Y \rangle.$$

Therefore

$$\langle X \cup Y \rangle \leq \langle \langle X \rangle \cup \langle Y \rangle \rangle.$$

Conversely, since $X \subseteq X \cup Y$ and $Y \subseteq X \cup Y$, we have

$$\langle X \rangle \leq \langle X \cup Y \rangle, \qquad \langle Y \rangle \leq \langle X \cup Y \rangle.$$

Hence

$$\langle X \rangle \cup \langle Y \rangle \subseteq \langle X \cup Y \rangle.$$

Since $\langle X \cup Y \rangle$ is a subgroup containing $\langle X \rangle \cup \langle Y \rangle$, it contains the subgroup generated by that union, so

$$\langle \langle X \rangle \cup \langle Y \rangle \rangle \leq \langle X \cup Y \rangle.$$

$\therefore \langle \langle X \rangle \cup \langle Y \rangle \rangle = \langle X \cup Y \rangle.$ □

Exercise 1.7. Let G be a finite group and let $H \leq G$. Show that there is a subset T of G which is simultaneously a left transversal for H and a right transversal for H.

Solution. We use the following elementary matching fact. Suppose $A_1, \ldots, A_n$ are finite sets of possible choices. If, for every $I \subseteq \{1, \ldots, n\}$,

$$\left| \bigcup_{i \in I} A_i \right| \geq |I|,$$

then we can choose distinct representatives $a_i \in A_i$.
Let

$$L_1, \ldots, L_n$$

be the distinct left cosets of H, and let

$$R_1, \ldots, R_n$$

be the distinct right cosets of H. There are the same number n of each, since both left and right cosets partition G into subsets of size $|H|$. For each i, define

$$A_i = \{ j \in \{1, \ldots, n\} \mid L_i \cap R_j \neq \emptyset \}.$$

Thus A_i is the set of right cosets which the left coset L_i meets. We check the matching condition. Choose any m left cosets, say those indexed by I, where $|I| = m$. Suppose these m left cosets meet q right cosets. Since distinct left cosets are disjoint,

$$\left| \bigcup_{i \in I} L_i \right| = m|H|.$$

This union is contained in the union of the q right cosets it meets, whose total size is $q|H|$. Hence

$$m|H| \leq q|H|,$$

so $m \leq q$. Therefore any m left cosets meet at least m right cosets. In terms of the sets A_i, this says exactly that

$$\left| \bigcup_{i \in I} A_i \right| \geq |I|.$$

By the matching fact, choose distinct $j_1, \ldots, j_n$ such that $j_i \in A_i$ for every i. Since $j_i \in A_i$,

$$L_i \cap R_{j_i} \neq \emptyset,$$

so choose

$$t_i \in L_i \cap R_{j_i}.$$

Set

$$T = \{t_1, \ldots, t_n\}.$$

Because $t_i \in L_i$, the set T contains one element from each left coset. Since the L_i are disjoint, it contains exactly one element from each left coset, so T is a left transversal for H.
Because the indices $j_1, \ldots, j_n$ are distinct, the right cosets $R_{j_1}, \ldots, R_{j_n}$ are all distinct. There are n of them, and there are only n right cosets in total, so they are all the right cosets. Since $t_i \in R_{j_i}$, the set T also contains exactly one element from each right coset. Hence T is a right transversal for H.
$\therefore$ there exists a subset $T \subseteq G$ which is simultaneously a left transversal and a right transversal for H. $\square$

Exercise 1.8. Suppose that $\mathcal{C}$ is a family of subsets of a group G which forms a partition of G, and suppose further that $gC \in \mathcal{C}$ for any $g \in G$ and $C \in \mathcal{C}$. (Recall that a *partition* of a set S is a collection $\mathcal{S}$ of subsets of S with the property that every element of S lies in exactly one member of $\mathcal{S}$.) Show that $\mathcal{C}$ is the set of cosets of some subgroup of G.

Solution. Since $\mathcal{C}$ is a partition of G, there is a unique member $H \in \mathcal{C}$ such that $1 \in H$. We first show that $H \leq G$.
Let $h \in H$. By the hypothesis, $hH \in \mathcal{C}$. Also, since $1 \in H$, we have $h = h1 \in hH$. Thus both H and hH are members of $\mathcal{C}$ containing h. Since $\mathcal{C}$ is a partition, we must have

$$hH = H.$$

Hence for any $h, k \in H$, we have $hk \in hH = H$, so H is closed under multiplication. Also, since $1 \in H = hH$, there exists $u \in H$ such that $hu = 1$. Thus $u = h^{-1}$, so $h^{-1} \in H$. Therefore $H \leq G$.
Now let $C \in \mathcal{C}$. Choose $g \in C$. By the hypothesis, $gH \in \mathcal{C}$, and since $1 \in H$, we have $g \in gH$. Thus both C and gH are members of $\mathcal{C}$ containing g. Since $\mathcal{C}$ is a partition, $C = gH$. Therefore every member of $\mathcal{C}$ is a left coset of H. Conversely, for every $g \in G$, the set gH belongs to $\mathcal{C}$ by the hypothesis.
$\therefore \mathcal{C} = \{gH \mid g \in G\}$, the set of left cosets of the subgroup H. $\square$

Exercise 1.9. Suppose that $\mathcal{C}$ is a family of subsets of a group G which forms a partition of G, and suppose further that $XY \in \mathcal{C}$ for any $X, Y \in \mathcal{C}$. Show that exactly one of the sets belonging to $\mathcal{C}$ is a subgroup of G, that this subgroup is normal in G, and that $\mathcal{C}$ consists of its cosets.

Solution. Since $\mathcal{C}$ is a partition of G, there is a unique member $H \in \mathcal{C}$ such that $1 \in H$. We first show that $H \leq G$.
By the hypothesis, $HH \in \mathcal{C}$. Since $1 \in H$, we have $1 = 1 \cdot 1 \in HH$. Thus H and HH are both members of $\mathcal{C}$ containing 1, so $HH = H$. Hence H is closed under multiplication.
Now let $h \in H$. Let $K \in \mathcal{C}$ be the unique member containing h^{-1}. By the hypothesis, $HK \in \mathcal{C}$. Since $h \in H$ and $h^{-1} \in K$, we have $1 = hh^{-1} \in HK$. Thus HK and H are both members of $\mathcal{C}$ containing 1, so $HK = H$. Since $1 \in H$ and $h^{-1} \in K$, it follows that $h^{-1} = 1h^{-1} \in HK = H$. Therefore $H \leq G$.
We now show that every member of $\mathcal{C}$ is a left coset of H. Let $X \in \mathcal{C}$, and choose $x \in X$. By the hypothesis, $XH \in \mathcal{C}$. Since $1 \in H$, we have $x = x1 \in XH$. Thus XH and X are both members of $\mathcal{C}$ containing x, so $XH = X$. Hence $xH \subseteq XH = X$.
For the reverse inclusion, let $X' \in \mathcal{C}$ be the unique member containing x^{-1}. By the hypothesis, $X'X \in \mathcal{C}$. Since $x^{-1}x = 1 \in X'X$, we have $X'X = H$. If $y \in X$, then $x^{-1}y \in X'X = H$, and hence

$$y = x(x^{-1}y) \in xH.$$

Thus $X \subseteq xH$. Therefore $X = xH$, and so every member of $\mathcal{C}$ is a left coset of H. Conversely, if $g \in G$, and $X \in \mathcal{C}$ contains g, then $X = gH$. Hence

$$\mathcal{C} = \{gH \mid g \in G\}.$$

The subgroup H is the only member of $\mathcal{C}$ which is a subgroup, since it is the only member containing 1. It remains to show that H is normal. Let $g \in G$. Since $gH, g^{-1}H \in \mathcal{C}$, the product $(gH)(g^{-1}H)$ is a member of $\mathcal{C}$. It contains 1, since $g \in gH$, $g^{-1} \in g^{-1}H$, and $gg^{-1} = 1$. Thus

$$(gH)(g^{-1}H) = H.$$

Now let $h \in H$. Since $gh \in gH$ and $g^{-1} \in g^{-1}H$, we have

$$ghg^{-1} \in (gH)(g^{-1}H) = H.$$

Thus $gHg^{-1} \subseteq H$. Replacing g by g^{-1}, we also have $g^{-1}Hg \subseteq H$, and conjugating this inclusion by g gives $H \subseteq gHg^{-1}$. Hence $gHg^{-1} = H$ for all $g \in G$. Therefore $H \trianglelefteq G$.
$\therefore$ exactly one member of $\mathcal{C}$ is a subgroup, this subgroup is normal in G, and $\mathcal{C}$ consists of its cosets. □

Exercise 1.10. Prove the following generalization of Proposition 1.8: If G is a finite group and $H \leq G$ is such that $|G : H|$ is equal to the smallest prime divisor of $|G|$, then $H \trianglelefteq G$.

Solution. Let $p = |G : H|$. By hypothesis, p is the smallest prime divisor of $|G|$. Let

$$\Omega = \{xH \mid x \in G\}$$

be the set of left cosets of H. Then $|\Omega| = p$. Define

$$\varphi\colon G \to S_\Omega$$

by

$$\varphi(g)(xH) = gxH$$

for all $g, x \in G$. This is well-defined: if $xH = yH$, then $gxH = gyH$. For each $g \in G$, the map $\varphi(g)$ is a permutation of Ω, with inverse $\varphi(g^{-1})$. Also,

$$\varphi(g_1g_2)(xH) = g_1g_2xH = \varphi(g_1)(g_2xH) = \big(\varphi(g_1)\varphi(g_2)\big)(xH),$$

so φ is a homomorphism.
Let $K = \ker\varphi$. Then $K \trianglelefteq G$. We claim that $K \leq H$. If $k \in K$, then $\varphi(k)$ fixes every left coset, so in particular

$$H = \varphi(k)(H) = kH.$$

Thus $k \in H$, and hence $K \leq H$.
By the Fundamental Theorem on Homomorphisms, $G/K \cong \operatorname{im}\varphi \leq S_\Omega$. Hence

$$|G : K| = |\operatorname{im}\varphi| \mid |S_\Omega| = p!.$$

Also, by Lagrange's Theorem, $|G : K| \mid |G|$. Therefore every prime divisor of $|G : K|$ is a prime divisor of $|G|$, and hence is at least p. On the other hand, since $|G : K| \mid p!$, every prime divisor of $|G : K|$ is at most p. Thus the only possible prime divisor of $|G : K|$ is p.
Since $K \leq H \leq G$, we have

$$|G : K| = |G : H|\,|H : K| = p|H : K|.$$

Thus $p \mid |G : K|$. Since the only possible prime divisor of $|G : K|$ is p, we may write $|G : K| = p^a$ for some $a \geq 1$. But $|G : K| \mid p!$, and $p^2 \nmid p!$, so $a = 1$. Hence

$$|G : K| = p.$$

It follows that $p|H : K| = p$, so $|H : K| = 1$. Therefore $H = K$. Since $K \trianglelefteq G$, we conclude that $H \trianglelefteq G$. □

Further Exercises

If $K \trianglelefteq H \leq G$, then H/K is called a *section* of G. We say that two sections H_1/K_1 and H_2/K_2 of G are *incident* if every coset of K_1 in H_1 has non-empty intersection with exactly one coset of K_2 in H_2, and vice versa. (In other words, two sections are incident if the relation of non-empty intersection gives a bijective correspondence between their elements.)

Exercise 1.11. Show that incident sections are isomorphic.

Solution. Let H_1/K_1 and H_2/K_2 be incident sections of G. Define

$$\varphi\colon H_1/K_1 \to H_2/K_2$$

by letting $\varphi(xK_1)$ be the unique coset yK_2 such that

$$xK_1 \cap yK_2 \neq \varnothing.$$

This is well-defined by the definition of incidence. We first show that φ is bijective. If $\varphi(xK_1) = \varphi(x'K_1) = yK_2$, then yK_2 has non-empty intersection with both xK_1 and $x'K_1$. Since incidence also says that each coset of K_2 in H_2 has non-empty intersection with exactly one coset of K_1 in H_1, we have $xK_1 = x'K_1$. Hence φ is injective. If $yK_2 \in H_2/K_2$, then, again by incidence, yK_2 has non-empty intersection with some coset $xK_1 \in H_1/K_1$, and then $\varphi(xK_1) = yK_2$. Hence φ is surjective.

It remains to show that φ is a homomorphism. Suppose

$$\varphi(xK_1) = yK_2 \quad \text{and} \quad \varphi(x'K_1) = y'K_2.$$

Choose

$$a \in xK_1 \cap yK_2 \quad \text{and} \quad b \in x'K_1 \cap y'K_2.$$

Then $a = xk$ and $b = x'k'$ for some $k, k' \in K_1$. Hence

$$ab = xkx'k' = xx'(x'^{-1}kx')k' \in xx'K_1,$$

because $K_1 \trianglelefteq H_1$ and $x' \in H_1$. Similarly, writing $a = y\ell$ and $b = y'\ell'$ with $\ell, \ell' \in K_2$, we get

$$ab = y\ell y'\ell' = yy'(y'^{-1}\ell y')\ell' \in yy'K_2,$$

because $K_2 \trianglelefteq H_2$ and $y' \in H_2$. Therefore

$$xx'K_1 \cap yy'K_2 \neq \varnothing.$$

By the uniqueness in the definition of incidence, this implies

$$\varphi(xx'K_1) = yy'K_2.$$

On the other hand,

$$\varphi(xK_1)\varphi(x'K_1) = (yK_2)(y'K_2) = yy'K_2.$$

Thus

$$\varphi((xK_1)(x'K_1)) = \varphi(xK_1)\varphi(x'K_1).$$

Therefore φ is a bijective homomorphism, so $H_1/K_1 \cong H_2/K_2$. □

Exercise 1.12. (cont.) Suppose that $N \trianglelefteq G$ and $H \leq G$. Show that HN/N and $H/(H \cap N)$ are incident. (Exercises 1.11 and 1.12 provide an alternate proof of the First Isomorphism Theorem.)

Solution. Let $K = H \cap N$. By Proposition 1.7, $HN \leq G$ and $K \trianglelefteq H$. Since $N \trianglelefteq G$ and $HN \leq G$, we also have $N \trianglelefteq HN$, so HN/N and H/K are sections of G. We show that they are incident.
First let $xN \in HN/N$. Since $x \in HN$, we may write $x = hn$ for some $h \in H$ and $n \in N$. Hence

$$xN = hnN = hN.$$

Since $1 \in K$, we have $h \in hN \cap hK$, so $hN \cap hK \neq \varnothing$. Thus every coset of N in HN has non-empty intersection with at least one coset of K in H. We now prove uniqueness. Suppose that

$$hN \cap h'K \neq \varnothing$$

for some $h' \in H$. Choose $z \in hN \cap h'K$. Then $z = hn_1 = h'k$ for some $n_1 \in N$ and $k \in K$, and hence

$$h^{-1}h' = n_1k^{-1}.$$

Since $n_1 \in N$ and $k \in K \leq N$, we have $n_1k^{-1} \in N$. Since also $h, h' \in H$, we have $h^{-1}h' \in H$. Hence $h^{-1}h' \in H \cap N = K$, so $hK = h'K$. Therefore each coset of N in HN intersects exactly one coset of K in H.

Conversely, let $hK \in H/K$. Since $h \in hK \cap hN$, the coset hK has non-empty intersection with at least one coset of N in HN. Suppose that $hK \cap xN \neq \varnothing$ for some $xN \in HN/N$. As above, we may write $xN = h'N$ for some $h' \in H$. Choose $z \in hK \cap h'N$. Then $z = hk = h'n_2$ for some $k \in K$ and $n_2 \in N$, and hence

$$h^{-1}h' = kn_2^{-1}.$$

Since $k \in K \leq N$ and $n_2 \in N$, we get $h^{-1}h' \in N$. Therefore $h'N = hN$. Thus each coset of K in H intersects exactly one coset of N in HN.
$\therefore HN/N$ and $H/(H \cap N)$ are incident. $\square$

If L/M is a section of G and $H \leq G$, then the *projection* of H on L/M is the subset of L/M consisting of those cosets of M in L which contain elements of H.

Exercise 1.13. (cont.) Show that the projection of H on L/M is the subgroup $(L \cap H)M/M$ of L/M.

Solution. Let

$$P = \{lM \in L/M \mid lM \cap H \neq \varnothing\}$$

be the projection of H on L/M. Since L/M is a section of G, we have $M \trianglelefteq L$. Let $\eta\colon L \to L/M$ be the natural map. By Proposition 1.11,

$$\eta(L \cap H) = (L \cap H)M/M$$

is a subgroup of L/M. It remains to show that $P = \eta(L \cap H)$.
First let $lM \in P$. Then $lM \cap H \neq \varnothing$, so we may choose $h \in lM \cap H$. Since $h \in lM \subseteq L$, and also $h \in H$, we have $h \in L \cap H$. Moreover $h \in lM$, so $hM = lM$. Hence

$$lM = hM \in \eta(L \cap H).$$

Thus $P \subseteq \eta(L \cap H)$.
Conversely, let $xM \in \eta(L \cap H)$. Then $xM = hM$ for some $h \in L \cap H$. Since $h \in H$ and $h \in hM = xM$, we have $xM \cap H \neq \varnothing$. Hence $xM \in P$. Thus $\eta(L \cap H) \subseteq P$.
$\therefore P = \eta(L \cap H) = (L \cap H)M/M$, so the projection of H on L/M is the subgroup $(L \cap H)M/M$ of L/M. $\square$

Let H_1/K_1 and H_2/K_2 be sections of a group G.

Exercise 1.14. (cont.) Show that the projection of K_2 on H_1/K_1 is a normal subgroup of the projection of H_2 on H_1/K_1. The quotient group obtained thereby is called the projection of H_2/K_2 on H_1/K_1.

Solution. By Exercise 1.13, the projection of K_2 on H_1/K_1 is

$$(H_1 \cap K_2)K_1/K_1,$$

and the projection of H_2 on H_1/K_1 is

$$(H_1 \cap H_2)K_1/K_1.$$

We first show that $H_1 \cap K_2 \trianglelefteq H_1 \cap H_2$. Let $x \in H_1 \cap K_2$ and $y \in H_1 \cap H_2$. Since $x, y \in H_1$, we have $yxy^{-1} \in H_1$. Since $x \in K_2$, $y \in H_2$, and $K_2 \trianglelefteq H_2$, we also have $yxy^{-1} \in K_2$. Hence

$$yxy^{-1} \in H_1 \cap K_2.$$

Therefore $H_1 \cap K_2 \trianglelefteq H_1 \cap H_2$.
Now take

$$xK_1 \in (H_1 \cap K_2)K_1/K_1 \quad \text{and} \quad yK_1 \in (H_1 \cap H_2)K_1/K_1,$$

where $x \in H_1 \cap K_2$ and $y \in H_1 \cap H_2$. Then

$$(yK_1)(xK_1)(yK_1)^{-1} = (yxy^{-1})K_1.$$

By the first paragraph, $yxy^{-1} \in H_1 \cap K_2$, so

$$(yxy^{-1})K_1 \in (H_1 \cap K_2)K_1/K_1.$$

Thus

$$(H_1 \cap K_2)K_1/K_1 \trianglelefteq (H_1 \cap H_2)K_1/K_1.$$

$\therefore$ the projection of K_2 on H_1/K_1 is a normal subgroup of the projection of H_2 on H_1/K_1. $\square$

Exercise 1.15. (cont.) Show that the projection of H_1/K_1 on H_2/K_2 and the projection of H_2/K_2 on H_1/K_1 are incident. Deduce that, if $H_1, H_2 \leq G$, $K_1 \trianglelefteq H_1$, and $K_2 \trianglelefteq H_2$, then

$$(H_1 \cap H_2)K_1/(H_1 \cap K_2)K_1 \cong (H_1 \cap H_2)K_2/(K_1 \cap H_2)K_2.$$

Solution. Let

$$A = H_1 \cap H_2, \quad B_1 = (H_1 \cap K_2)K_1, \quad B_2 = (K_1 \cap H_2)K_2.$$

By Exercise 1.13 and Exercise 1.14, the projection of H_2/K_2 on H_1/K_1 is

$$AK_1/B_1,$$

and the projection of H_1/K_1 on H_2/K_2 is

$$AK_2/B_2.$$

Every coset of B_1 in AK_1 has the form aB_1 for some $a \in A$, since $K_1 \leq B_1$. Similarly, every coset of B_2 in AK_2 has the form aB_2 for some $a \in A$, since $K_2 \leq B_2$. For each $a \in A$, the two cosets aB_1 and aB_2 have non-empty intersection, since both contain a.
We now prove uniqueness. Suppose that

$$aB_1 \cap bB_2 \neq \varnothing$$

for some $a, b \in A$. Choose $z \in aB_1 \cap bB_2$. Then $z = as = bt$ for some $s \in B_1$ and $t \in B_2$. Write

$$s = ck, \quad c \in H_1 \cap K_2, \quad k \in K_1,$$

and

$$t = d\ell, \quad d \in K_1 \cap H_2, \quad \ell \in K_2.$$

From $ack = bd\ell$, we get

$$k = c^{-1}a^{-1}bd\ell.$$

Since $a, b \in A \leq H_2$, $c \in K_2 \leq H_2$, $d \in H_2$, and $\ell \in K_2 \leq H_2$, it follows that $k \in H_2$. Thus $k \in K_1 \cap H_2$. Since $c \in K_2$ and $K_2 \trianglelefteq H_2$, we have

$$s = ck = k(k^{-1}ck) \in (K_1 \cap H_2)K_2 = B_2.$$

Since $t \in B_2$, the equality $as = bt$ gives

$$b^{-1}a = ts^{-1} \in B_2.$$

Hence $aB_2 = bB_2$. Therefore each coset of B_1 in AK_1 intersects at most one coset of B_2 in AK_2.
Conversely, suppose that

$$aB_2 \cap bB_1 \neq \varnothing$$

for some $a, b \in A$. Choose $z \in aB_2 \cap bB_1$. Then $z = at = bs$ with $t \in B_2$ and $s \in B_1$. Write

$$t = d\ell, \quad d \in K_1 \cap H_2, \quad \ell \in K_2,$$

and

$$s = ck, \quad c \in H_1 \cap K_2, \quad k \in K_1.$$

From $ad\ell = bck$, we get

$$\ell = d^{-1}a^{-1}bck.$$

Since $a, b \in A \leq H_1$, $d \in K_1 \leq H_1$, $c \in H_1$, and $k \in K_1 \leq H_1$, it follows that $\ell \in H_1$. Thus $\ell \in H_1 \cap K_2$. Since $d \in K_1$ and $K_1 \trianglelefteq H_1$, we have

$$t = d\ell = \ell(\ell^{-1}d\ell) \in (H_1 \cap K_2)K_1 = B_1.$$

Since $s \in B_1$, the equality $at = bs$ gives

$$b^{-1}a = st^{-1} \in B_1.$$

Hence $aB_1 = bB_1$. Therefore each coset of B_2 in AK_2 intersects at most one coset of B_1 in AK_1.
Thus the projections AK_1/B_1 and AK_2/B_2 are incident. □

Theorem (Third Isomorphism Theorem)**.** *Let $H_1, H_2 \leq G$, let $K_1 \trianglelefteq H_1$, and let $K_2 \trianglelefteq H_2$. Then*

$$(H_1 \cap H_2)K_1/(H_1 \cap K_2)K_1 \cong (H_1 \cap H_2)K_2/(K_1 \cap H_2)K_2.$$

Proof. By Exercise 1.15, the sections

$$(H_1 \cap H_2)K_1/(H_1 \cap K_2)K_1 \quad \text{and} \quad (H_1 \cap H_2)K_2/(K_1 \cap H_2)K_2$$

are incident. By Exercise 1.11, incident sections are isomorphic. Hence

$$(H_1 \cap H_2)K_1/(H_1 \cap K_2)K_1 \cong (H_1 \cap H_2)K_2/(K_1 \cap H_2)K_2.$$

□

(This result is also called the fourth isomorphism theorem, or Zassenhaus' lemma — after its discoverer, who proved it as a student at the age of 21 — or even the butterfly lemma. The last name refers to the shape of the diagram showing the inclusion relations between the many subgroups involved.)

2 Automorphisms

The Automorphism Group

- The set of automorphisms of a group G is denoted $\mathrm{Aut}(G)$.
- If $\varphi, \rho \in \mathrm{Aut}(G)$, then $\varphi \circ \rho \in \mathrm{Aut}(G)$, so composition is a binary operation on $\mathrm{Aut}(G)$.
- Under composition, $\mathrm{Aut}(G)$ is a group: identity is the trivial automorphism, and the inverse of φ is its set-map inverse φ^{-1}.
- Call $\mathrm{Aut}(G)$ with this operation the *automorphism group* of G; we may write $\varphi\rho$ for $\varphi \circ \rho$.

Inner Automorphisms, $\mathrm{Inn}(G)$, and the Center

- Each $g \in G$ defines a conjugation homomorphism $\varphi_g : G \to G$ by $\varphi_g(x) = gxg^{-1}$.
- Each φ_g is an automorphism: for $x \in G$, $x = \varphi_g(g^{-1}xg)$, and if $\varphi_g(x) = \varphi_g(y)$ then $x = y$ by cancellation. These are the *inner automorphisms* of G.
- $\varphi_g\varphi_h = \varphi_{gh}$ for $g, h \in G$, since $g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1}$. Hence there is a homomorphism $G \to \mathrm{Aut}(G)$ sending g to φ_g.
- Its image is the *inner automorphism group* of G, denoted $\mathrm{Inn}(G)$; its kernel is the *center* of G, denoted $Z(G)$. Observe

$$Z(G) = \{g \in G \mid \varphi_g(x) = x \text{ for all } x \in G\} = \{g \in G \mid gx = xg \text{ for all } x \in G\},$$

  so $Z(G)$ consists of those elements of G which commute with every element of G.
- Clearly G is abelian iff $Z(G) = G$.

Outer Automorphisms, $\mathrm{Out}(G)$

- If $\sigma \in \mathrm{Aut}(G)$ and $\varphi_g \in \mathrm{Inn}(G)$, then $\sigma\varphi_g\sigma^{-1} = \varphi_{\sigma(g)}$. Hence $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$.
- The quotient $\mathrm{Aut}(G)/\mathrm{Inn}(G)$ is the *outer automorphism group* of G, denoted $\mathrm{Out}(G)$.
- "Outer automorphism" refers to automorphisms of G which are not inner and which thus have non-trivial image in $\mathrm{Out}(G)$.
- If G is abelian, then all non-trivial automorphisms of G are outer in this sense, since $\mathrm{Inn}(G) = 1$.

Automorphism Groups of Cyclic Groups

- Let $G = \langle x \rangle \cong \mathbb{Z}$, and let φ be an automorphism of G. Then $\varphi(x)$ must generate G; the only generators are x and x^{-1}. So φ either fixes each element or sends each to its inverse, and $\mathrm{Aut}(G) \cong \mathbb{Z}_2$.
- Let $n \in \mathbb{N}$ and $G = \langle x \rangle \cong \mathbb{Z}_n$. Suppose φ is an endomorphism of G. Then $\varphi(x) = x^m$ for some $0 \le m < n$, so φ sends every element to its mth power. Hence G has exactly n endomorphisms, namely the mth power maps σ_m for $0 \le m < n$.

Proposition 2.1. *Let $G = \langle x \rangle \cong \mathbb{Z}_n$ for $n \in \mathbb{N}$, and for each $0 \le m < n$ let σ_m be the endomorphism of G sending x to x^m. Then $\mathrm{Aut}(G)$ consists precisely of those σ_m for which $m \ne 0$ and $\gcd(m, n) = 1$. Furthermore, $\mathrm{Aut}(G)$ is abelian and is isomorphic with the group $(\mathbb{Z}/n\mathbb{Z})^\times$ of units of the ring $\mathbb{Z}/n\mathbb{Z}$.*

Proof. The map σ_0 has trivial image, so it is not an automorphism. Let $1 \le m < n$ and consider σ_m. Suppose first that $\gcd(m, n) = 1$. Then there exist $a, b \in \mathbb{Z}$ such that $am + bn = 1$, and hence

$$\sigma_m(x^a) = x^{am} = x^{1-bn} = x(x^n)^{-b} = x.$$

Thus $x \in \operatorname{im}\sigma_m$, so σ_m is surjective. Since G is finite, σ_m is also injective, and therefore $\sigma_m \in \mathrm{Aut}(G)$. Conversely, suppose that $\sigma_m \in \mathrm{Aut}(G)$. Since σ_m is surjective, there is some $a \in \mathbb{Z}$ such that $x = \sigma_m(x^a) = x^{am}$. Hence $x^{am-1} = 1$, so $am - 1 = bn$ for some $b \in \mathbb{Z}$. It follows that any common divisor

of m and n divides 1, and therefore $\gcd(m,n)=1$. This proves the first assertion.
Now let $1 \le m_1, m_2 < n$ with $\gcd(m_1,n)=\gcd(m_2,n)=1$. If t is chosen with $1 \le t < n$ and $m_1m_2 \equiv t \pmod{n}$, then

$$(\sigma_{m_1}\sigma_{m_2})(x) = \sigma_{m_1}(x^{m_2}) = x^{m_1m_2} = x^t.$$

Hence $\sigma_{m_1}\sigma_{m_2} = \sigma_t = \sigma_{m_2}\sigma_{m_1}$, so $\mathrm{Aut}(G)$ is abelian. Finally, the map

$$\mathrm{Aut}(G) \to (\mathbb{Z}/n\mathbb{Z})^\times, \qquad \sigma_m \mapsto m + n\mathbb{Z}$$

is a bijection by the first assertion, and the computation above shows that it preserves multiplication. Hence $\mathrm{Aut}(G) \cong (\mathbb{Z}/n\mathbb{Z})^\times$. $\square$

The Totient and Primitive Roots

- The *totient* of $n \in \mathbb{N}$ is the number of positive integers less than n and coprime to n (the value at n of the Euler phi-function).
- If $n = p_1^{a_1} \cdots p_r^{a_r}$ with the p_i distinct primes, then the totient of n is $(p_1^{a_1} - p_1^{a_1-1}) \cdots (p_r^{a_r} - p_r^{a_r-1})$. To see this, note that m is coprime to n iff no p_i divides m. Among $1, \ldots, n$, exactly n/p_i integers are divisible by p_i, exactly $n/(p_ip_j)$ are divisible by both p_i and p_j, and so on. Hence inclusion-exclusion gives
$$\varphi(n) = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right) = \prod_{i=1}^{r} \left(p_i^{a_i} - p_i^{a_i-1}\right).$$
- By Proposition 2.1, the order of $\mathrm{Aut}(\mathbb{Z}_n)$ is the totient of n. In particular, $|\mathrm{Aut}(\mathbb{Z}_p)| = p-1$ when p is prime.

Proposition 2.2. *Let p be a prime. Then* $\mathrm{Aut}(\mathbb{Z}_p) \cong \mathbb{Z}_{p-1}$.

Proof. Let $F = \mathbb{F}_p$. By Proposition 2.1, $\mathrm{Aut}(\mathbb{Z}_p)$ is isomorphic with $F^\times = (\mathbb{Z}/p\mathbb{Z})^\times$. Thus it suffices to show that $F^\times$ is cyclic of order $p-1$.
For each divisor d of $p-1$, define

$$f_d = |\{x \in F^\times \mid o(x) = d\}|, \qquad z_d = |\{x \in \mathbb{Z}_{p-1} \mid o(x) = d\}|.$$

We first show that $f_d \le z_d$ for every $d \mid (p-1)$. If $x \in F^\times$ has order dividing d, then $x^d = 1$, so x is a root of $T^d - 1 \in F[T]$. Since this polynomial has degree d, it has at most d roots in F. Now suppose $F^\times$ has an element x of order d. Then the d elements of $\langle x \rangle$ are all roots of $T^d - 1$, so they are all the roots. Hence every element of $F^\times$ of order d lies in $\langle x \rangle \cong \mathbb{Z}_d$. Therefore either $f_d = 0$, or f_d is the number of elements of order d in a cyclic group of order d. By Theorem 1.4, the cyclic group $\mathbb{Z}_{p-1}$ has a unique subgroup of order d; every element of order d generates a subgroup of order d, so all elements of order d in $\mathbb{Z}_{p-1}$ lie in this unique subgroup. This subgroup is cyclic of order d, and hence $f_d \le z_d$.
Since every element of $F^\times$ has order dividing $|F^\times| = p-1$, and every element of $\mathbb{Z}_{p-1}$ has order dividing $p-1$, we have

$$\sum_{d|(p-1)} f_d = |F^\times| = p-1 = |\mathbb{Z}_{p-1}| = \sum_{d|(p-1)} z_d.$$

Since $f_d \le z_d$ for every $d \mid (p-1)$ and the sums are equal, we must have $f_d = z_d$ for every $d \mid (p-1)$. In particular, $f_{p-1} = z_{p-1}$. But $\mathbb{Z}_{p-1}$ has an element of order $p-1$, so $z_{p-1} > 0$, and hence $f_{p-1} > 0$. Therefore $F^\times$ has an element of order $p-1$, so $F^\times \cong \mathbb{Z}_{p-1}$.
$\therefore \mathrm{Aut}(\mathbb{Z}_p) \cong \mathbb{Z}_{p-1}$. $\square$

Primitive Roots Modulo n

- Let $G = \langle x \rangle \cong \mathbb{Z}_n$, $1 \le m \le n$, $\gcd(m,n) = 1$. Induction gives $(\sigma_m)^k(x) = x^{m^k}$.
- The order of σ_m is the least positive k such that $x^{m^k} = x$, i.e. the smallest $k \in \mathbb{N}$ with $m^k \equiv 1 \pmod{n}$.
- If the order of σ_m equals the totient of n, we call m a *primitive root modulo n*.
- $\mathrm{Aut}(\mathbb{Z}_n)$ is cyclic iff there is a primitive root modulo n.

- For composite n, the structure of $\mathrm{Aut}(\mathbb{Z}_n)$ belongs to number theory more than group theory.

Theorem 2.3. *For $n > 1$, $\mathrm{Aut}(\mathbb{Z}_n)$ is cyclic iff $n = 2$ or 4, or $n = p^k$ or $2p^k$ for some odd prime p and some $k \in \mathbb{N}$.*

Proof. By Proposition 2.1,

$$\mathrm{Aut}(\mathbb{Z}_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$

Under this isomorphism, σ_m corresponds to the unit $m + n\mathbb{Z}$, and the order of σ_m is the least $k \in \mathbb{N}$ such that $m^k \equiv 1 \pmod{n}$. Hence $\mathrm{Aut}(\mathbb{Z}_n)$ is cyclic iff there is a primitive root modulo n. It remains to prove the number-theoretic classification of those $n > 1$ which have primitive roots.

We use Euler's theorem in the following form: if $\gcd(a, q) = 1$, then $a^{\varphi(q)} \equiv 1 \pmod{q}$. First suppose $q = 2^s$ with $s \geq 3$. If a is odd, then $a^2 \equiv 1 \pmod{8}$. We claim that

$$a^{2^{s-2}} \equiv 1 \pmod{2^s}$$

for all $s \geq 3$. The case $s = 3$ is the congruence just observed. If $a^{2^{s-2}} = 1 + b2^s$, then

$$a^{2^{s-1}} = (1 + b2^s)^2 \equiv 1 \pmod{2^{s+1}},$$

so the claim follows by induction. But $\varphi(2^s) = 2^{s-1}$, so every odd residue modulo 2^s has order at most $2^{s-2} < \varphi(2^s)$. Thus 2^s has no primitive root for $s \geq 3$.

Next suppose $\gcd(u, v) = 1$ with $u, v > 2$. Let a be coprime to uv, and put

$$h = \mathrm{lcm}(\varphi(u), \varphi(v)).$$

Since $u, v > 2$, both $\varphi(u)$ and $\varphi(v)$ are even (the reduced residue classes pair as b and $-b$), so

$$h \leq \frac{\varphi(u)\varphi(v)}{2} = \frac{\varphi(uv)}{2}.$$

Euler's theorem gives $a^h \equiv 1 \pmod{u}$ and $a^h \equiv 1 \pmod{v}$; since $\gcd(u, v) = 1$, this implies $a^h \equiv 1 \pmod{uv}$. Hence no unit modulo uv can have order $\varphi(uv)$, and uv has no primitive root.

These two obstructions leave only 2, 4, odd prime powers, and twice odd prime powers. Indeed, if

$$n = 2^a p_1^{e_1} \cdots p_t^{e_t}$$

is the prime factorization of $n > 1$, then two distinct odd prime factors are ruled out by the previous paragraph, as is $2^a p^e$ with $a \geq 2$ and p odd. The pure powers 2^a with $a \geq 3$ were ruled out in the paragraph before that. Therefore any $n > 1$ with a primitive root must be one of 2, 4, p^e, or $2p^e$ with p odd.

It remains to see that all numbers in this list actually have primitive roots. Clearly 1 is a primitive root modulo 2, and 3 is a primitive root modulo 4. Now let p be an odd prime. By Proposition 2.2, there is a primitive root r modulo p. We may choose such an r with

$$r^{p-1} \not\equiv 1 \pmod{p^2}.$$

For if a chosen primitive root r satisfies $r^{p-1} \equiv 1 \pmod{p^2}$, then $r + p$ is still primitive modulo p, and

$$(r + p)^{p-1} \equiv r^{p-1} + (p - 1)pr^{p-2} \equiv 1 - pr^{p-2} \not\equiv 1 \pmod{p^2}.$$

We claim that this r is primitive modulo p^e for every $e \in \mathbb{N}$. For $e = 1$ this is how r was chosen. For $e \geq 2$, we first show by induction that

$$r^{p^{e-2}(p-1)} \not\equiv 1 \pmod{p^e}.$$

The case $e = 2$ is our choice of r. If the assertion holds for e, then Euler's theorem modulo p^{e-1} gives

$$r^{p^{e-2}(p-1)} = 1 + cp^{e-1}$$

for some integer c not divisible by p. Raising both sides to the pth power gives

$$r^{p^{e-1}(p-1)} = (1 + cp^{e-1})^p \equiv 1 + cp^e \not\equiv 1 \pmod{p^{e+1}},$$

proving the induction step. Now let d be the order of r modulo p^e. Then $d \mid \varphi(p^e) = p^{e-1}(p-1)$, and reducing modulo p shows $p-1 \mid d$. Hence $d = p^j(p-1)$ for some $0 \le j \le e-1$. If $j < e-1$, then $d \mid p^{e-2}(p-1)$, contradicting the congruence just proved. Thus $d = p^{e-1}(p-1) = \varphi(p^e)$, so r is primitive modulo p^e.
Finally, let r be primitive modulo p^e. Replacing r by $r + p^e$ if necessary, we may assume that r is odd; this does not change its residue modulo p^e. Let d be the order of r modulo $2p^e$. Since $r^d \equiv 1 \pmod{2p^e}$ implies $r^d \equiv 1 \pmod{p^e}$, we have $\varphi(p^e) \mid d$. Also $d \mid \varphi(2p^e) = \varphi(p^e)$. Therefore $d = \varphi(2p^e)$, and r is primitive modulo $2p^e$.
We have proved that primitive roots exist exactly for $n = 2, 4, p^e$, and $2p^e$ with p odd. Therefore $\mathrm{Aut}(\mathbb{Z}_n)$ is cyclic exactly for the same values of n. □

The number-theoretic primitive-root classification used in this proof follows [9, Section 8.3].

Fixed and Characteristic Subgroups

- Let $\varphi \in \mathrm{Aut}(G)$ and $H \le G$. We say H is *fixed by* φ if $\varphi(H) = H$; in this case the restriction is an automorphism of H.
- If $L \le \mathrm{Aut}(G)$, H is *fixed by* L if H is fixed by every $\varphi \in L$.
- H is normal in G iff H is fixed by $\mathrm{Inn}(G)$.
- H is a *characteristic subgroup* of G (or H *characteristic* in G) if H is fixed by $\mathrm{Aut}(G)$. (Some authors write $H \operatorname{char} G$.)
- Example: $Z(G)$ is always characteristic. For if $x \in Z(G)$ and $\varphi \in \mathrm{Aut}(G)$, then $\varphi(x)y = \varphi(x\varphi^{-1}(y)) = \varphi(\varphi^{-1}(y)x) = y\varphi(x)$ for any $y \in G$, showing $\varphi(x) \in Z(G)$.
- Characteristic subgroups are normal, but the converse fails. In fact, an infinite abelian group need not have any non-trivial proper characteristic subgroups (Exercise 2.5).

Lemma 2.4. *If K is a characteristic subgroup of H and H is a characteristic subgroup of G, then K is a characteristic subgroup of G.*

Proof. Let $\varphi \in \mathrm{Aut}(G)$. We need to show that $\varphi(K) = K$. Since H is characteristic in G, we have $\varphi(H) = H$. Hence the restriction $\varphi|_H : H \to H$ is an automorphism of H: it is still a homomorphism, it is injective because φ is injective, and it is surjective onto H because $\varphi(H) = H$.
Now K is characteristic in H, so every automorphism of H fixes K. Applying this to $\varphi|_H$, we obtain $(\varphi|_H)(K) = K$. But $K \le H$, so $(\varphi|_H)(K) = \varphi(K)$. Therefore $\varphi(K) = K$. Since $\varphi \in \mathrm{Aut}(G)$ was arbitrary, K is characteristic in G. □

The Derived Group

- Normality is not transitive: if $N \trianglelefteq G$, the restriction to N of an element of $\mathrm{Inn}(G)$ lies in $\mathrm{Aut}(N)$ but need not lie in $\mathrm{Inn}(N)$.
- For $x, y \in G$ the *commutator* is $[x, y] = xyx^{-1}y^{-1}$.
- The *derived group* G' of G is the subgroup generated by all commutators: $G' = \langle\{[x, y] \mid x, y \in G\}\rangle$.
- G is abelian iff $G' = 1$.
- If $H \le G$, then $H' \le G'$.
- In general, G' contains more than just the commutators of elements of G.
- Since $[x, y]^{-1} = (xyx^{-1}y^{-1})^{-1} = yxy^{-1}x^{-1} = [y, x]$, by Proposition 1.2 an arbitrary element of G' is a finite product of commutators of elements of G.

Lemma 2.5. *Let G be a group. Then G' is characteristic in G.*

Proof. Let $\varphi \in \mathrm{Aut}(G)$. For any $x, y \in G$, we have

$$\varphi([x,y]) = \varphi(xyx^{-1}y^{-1}) = \varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1} = [\varphi(x), \varphi(y)].$$

Thus φ sends each commutator of elements of G to another commutator of elements of G. Now let $g \in G'$. By Proposition 1.2 and the preceding observation about inverses of commutators, we may write

$$g = [x_1, y_1] \cdots [x_r, y_r]$$

for some $x_i, y_i \in G$. Applying φ, we obtain

$$\varphi(g) = [\varphi(x_1), \varphi(y_1)] \cdots [\varphi(x_r), \varphi(y_r)] \in G'.$$

Hence $\varphi(G') \leq G'$. Applying the same argument to φ^{-1} gives $\varphi^{-1}(G') \leq G'$, and therefore $G' \leq \varphi(G')$. Thus $\varphi(G') = G'$. Since φ was arbitrary, G' is characteristic in G. □

Proposition 2.6. *Let G be a group, and let $N \trianglelefteq G$. Then G/N is abelian iff $G' \leq N$.*

Proof. By Lemma 2.5, G' is characteristic in G, and hence $G' \trianglelefteq G$. We first observe that G/G' is abelian. Indeed, for any $x, y \in G$,

$$[xG', yG'] = [x,y]G' = G',$$

since $[x,y] \in G'$. Thus every commutator in G/G' is trivial, so G/G' is abelian.
Suppose first that $G' \leq N$. Since $G' \leq N$, the set N/G' is a subgroup of G/G'. Moreover, it is normal in G/G': if $g \in G$ and $n \in N$, then

$$(gG')(nG')(gG')^{-1} = gng^{-1}G' \in N/G',$$

because $N \trianglelefteq G$. Thus $N/G' \trianglelefteq G/G'$, and by Second Isomorphism Theorem,

$$G/N \cong (G/G')/(N/G').$$

Since G/G' is abelian, every quotient of G/G' is abelian. Hence G/N is abelian.
Conversely, suppose that G/N is abelian. For any $x, y \in G$, the commutator of xN and yN is trivial, so

$$N = [xN, yN] = [x,y]N.$$

Therefore $[x,y] \in N$ for all $x, y \in G$. Since N contains every commutator of elements of G, it contains the subgroup generated by all such commutators. Hence $G' \leq N$. □

External Direct Products

- Let $n \in \mathbb{N}$ and let $G_1, \ldots, G_n$ be groups. The Cartesian product $G_1 \times \cdots \times G_n$ becomes a group under componentwise multiplication $(g_1, \ldots, g_n)(g_1', \ldots, g_n') = (g_1g_1', \ldots, g_ng_n')$. The identity is $(1, \ldots, 1)$ and inverses are $(g_1^{-1}, \ldots, g_n^{-1})$.
- This is the (*external*) *direct product* of $G_1, \ldots, G_n$.
- Order of factors is irrelevant: $G_1 \times \cdots \times G_n \cong G_{\rho(1)} \times \cdots \times G_{\rho(n)}$ for any $\rho \in S_n$.
- For $G = G_1 \times \cdots \times G_n$:
  - Each $H_i = \{(1, \ldots, g_i, \ldots, 1) \mid g_i \in G_i\}$ is a normal subgroup of G isomorphic with G_i. Moreover G/H_i is isomorphic with the direct product of the remaining G_j.
  - Every $g \in G$ has a unique expression $g = h_1 \cdots h_n$ with $h_i \in H_i$; if $g = (g_1, \ldots, g_n)$ then $h_i = (1, \ldots, g_i, \ldots, 1)$.
  - If the G_i are finite, then $|G| = |G_1| \cdots |G_n|$.

Internal Direct Products

- Suppose G has subgroups $H_1, \ldots, H_n$ such that:

  1. $H_i \trianglelefteq G$ for each $1 \leq i \leq n$.
  2. Every $g \in G$ has a unique expression $g = h_1 \cdots h_n$ where $h_i \in H_i$.

  Conditions (1) and (2) imply:

  3. $G = H_1 \cdots H_n$.
  4. $H_i \cap H_1 \cdots H_{i-1} H_{i+1} \cdots H_n = 1$ for each i.
  5. If $i \neq j$, then elements of H_i commute with elements of H_j.
  6. If $g = h_1 \cdots h_n$ and $g' = h_1' \cdots h_n'$ with $h_i, h_i' \in H_i$, then $gg' = (h_1 h_1') \cdots (h_n h_n')$.

- Under these conditions, there is a unique isomorphism from G to the external direct product $H_1 \times \cdots \times H_n$ sending H_i to $1 \times \cdots \times H_i \times \cdots \times 1$. Call G the (*internal*) *direct product* of $H_1, \ldots, H_n$ and write $G = H_1 \times \cdots \times H_n$ (slight abuse of notation).

- If (1) holds, then (2) holds iff both (3) and (4) hold; so to check a direct product, it suffices to verify either (1) and (2) or (1), (3), and (4). (For $n = 2$, (4) reduces to $H_1 \cap H_2 = 1$.)

Lemma 2.7. *Let G be a group having normal subgroups H and K such that $G = HK$. Then $G/H \cap K \cong H/H \cap K \times K/H \cap K$.*

Proof. Put $M = H \cap K$. By Proposition 1.7, we have $M \trianglelefteq G$, and hence G/M, H/M, and K/M are quotient groups. By Correspondence Theorem, the subgroups H/M and K/M are normal in G/M.
We first compute their intersection. Suppose that $xM \in (H/M) \cap (K/M)$. Since $xM \in H/M$, there is some $h \in H$ such that $xM = hM$, and because $M \leq H$ this implies $x \in H$. Similarly, $x \in K$. Thus $x \in H \cap K = M$, and so $xM = M$. Therefore

$$(H/M) \cap (K/M) = 1.$$

Next let $gM \in G/M$. Since $G = HK$, we may write $g = hk$ with $h \in H$ and $k \in K$. Then

$$gM = hkM = (hM)(kM),$$

so $(G/M) = (H/M)(K/M)$. Hence G/M is the internal direct product of H/M and K/M, and therefore is isomorphic with the corresponding external direct product. Thus

$$G/(H \cap K) \cong H/(H \cap K) \times K/(H \cap K).$$

□

Lemma 2.8. *Let $n \in \mathbb{N}$, and write $n = p_1^{a_1} \cdots p_r^{a_r}$ where the p_i are distinct primes and the a_i are positive integers. Then we have $\mathbb{Z}_n \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_r^{a_r}}$.*

Proof. If $n = 1$, there is nothing to prove, so assume $n > 1$. For each i, let

$$P_i = \langle x_i \rangle \cong \mathbb{Z}_{p_i^{a_i}},$$

and put $P = P_1 \times \cdots \times P_r$. Then

$$|P| = |P_1| \cdots |P_r| = p_1^{a_1} \cdots p_r^{a_r} = n.$$

Let $x = (x_1, \ldots, x_r) \in P$. For $k \in \mathbb{N}$, we have

$$x^k = (1, \ldots, 1)$$

iff $x_i^k = 1$ for every i, iff $p_i^{a_i} \mid k$ for every i. Since the prime powers $p_i^{a_i}$ are pairwise coprime, this is equivalent to $n \mid k$. Hence $o(x) = n$.
Therefore $\langle x \rangle$ has n elements and lies in P, which also has n elements. Thus $\langle x \rangle = P$, so P is cyclic of order n. It follows that $P \cong \mathbb{Z}_n$, and substituting the groups $P_i \cong \mathbb{Z}_{p_i^{a_i}}$ gives

$$\mathbb{Z}_n \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_r^{a_r}}.$$

□

Corollary 2.9. *If* $\gcd(a, b) = 1$*, then* $\mathbb{Z}_{ab} \cong \mathbb{Z}_a \times \mathbb{Z}_b$*.*

Proof. If $a = 1$ or $b = 1$, the assertion is immediate because $\mathbb{Z}_1$ is the trivial group. Thus assume $a, b > 1$. Write

$$a = p_1^{\alpha_1} \cdots p_s^{\alpha_s}, \qquad b = q_1^{\beta_1} \cdots q_t^{\beta_t}$$

as prime-power factorizations. Since $\gcd(a, b) = 1$, the primes $p_1, \ldots, p_s$ are distinct from the primes $q_1, \ldots, q_t$. Hence

$$ab = p_1^{\alpha_1} \cdots p_s^{\alpha_s} q_1^{\beta_1} \cdots q_t^{\beta_t}$$

is the prime-power factorization of ab. Applying Lemma 2.8 to ab gives

$$\mathbb{Z}_{ab} \cong \mathbb{Z}_{p_1^{\alpha_1}} \times \cdots \times \mathbb{Z}_{p_s^{\alpha_s}} \times \mathbb{Z}_{q_1^{\beta_1}} \times \cdots \times \mathbb{Z}_{q_t^{\beta_t}}.$$

Applying Lemma 2.8 separately to a and to b, and regrouping the factors, gives

$$\mathbb{Z}_{ab} \cong \mathbb{Z}_a \times \mathbb{Z}_b.$$

□

Proposition 2.10. *Suppose that a finite group G is the direct product of its subgroups $H_1, \ldots, H_n$, where the orders $|H_i|$ are pairwise coprime. Then any subgroup L of G is the direct product of $L \cap H_1, \ldots, L \cap H_n$.*

Proof. We first prove the case $n = 2$. Write $G = H \times K$, where $\gcd(|H|, |K|) = 1$, and let $L \leq G$. Put

$$A = L \cap H, \qquad B = L \cap K.$$

Since $H, K \trianglelefteq G$, we have $A, B \trianglelefteq L$. Also $A \cap B \leq H \cap K = 1$, so $A \cap B = 1$. It remains to show that $L = AB$.

Let $g \in L$. Since $G = H \times K$, we may write $g = hk$ with $h \in H$ and $k \in K$, and h and k commute. By Lagrange's Theorem, $o(h) \mid |H|$ and $o(k) \mid |K|$, so $o(h)$ and $o(k)$ are coprime. Put $a = o(h)$ and $b = o(k)$. If $(hk)^m = 1$, then, since h and k commute,

$$h^m = k^{-m} \in H \cap K = 1.$$

Hence $a \mid m$ and $b \mid m$, and because $\gcd(a, b) = 1$, we have $ab \mid m$. Conversely, $(hk)^{ab} = 1$, so $o(hk) = ab$. Now $\langle hk \rangle \leq \langle h \rangle \langle k \rangle$. The subgroup $\langle h \rangle \langle k \rangle$ is the internal direct product of $\langle h \rangle$ and $\langle k \rangle$, so it has order ab. Since $\langle hk \rangle$ also has order ab, we have

$$\langle hk \rangle = \langle h \rangle \langle k \rangle.$$

Thus $h, k \in \langle hk \rangle = \langle g \rangle \leq L$. Therefore $h \in A$ and $k \in B$, and so $g = hk \in AB$. This proves $L \leq AB$, while $AB \leq L$ is immediate. Hence $L = AB$. Since A and B are normal in L, have trivial intersection, and commute with one another, $L = A \times B$.

We now argue by induction on n. The case $n = 1$ is immediate, and the case $n = 2$ has just been proved. Suppose $n > 2$ and that the result is known for direct products with $n - 1$ factors. Let

$$M = H_1 \times \cdots \times H_{n-1}.$$

Then $G = M \times H_n$, and pairwise coprimeness of the orders $|H_i|$ gives $\gcd(|M|, |H_n|) = 1$. Applying the $n = 2$ case to $G = M \times H_n$ gives

$$L = (L \cap M) \times (L \cap H_n).$$

Inside M, the subgroup $L \cap M$ is a subgroup of the direct product $H_1 \times \cdots \times H_{n-1}$, whose factor orders are still pairwise coprime. By the induction hypothesis,

$$L \cap M = (L \cap H_1) \times \cdots \times (L \cap H_{n-1}).$$

Substituting this into the preceding display gives

$$L = (L \cap H_1) \times \cdots \times (L \cap H_n),$$

as required. □

Internal Semidirect Products

- Suppose G has subgroup H and normal subgroup N with $G = NH$ and $N \cap H = 1$. Then G is the (*internal*) *semidirect product* of N by H, written $G = N \rtimes H$. (Notation common but not standard; alternatives include $N \rtimes H$, $H \ltimes N$, no notation.)
- If in addition $H \trianglelefteq G$, then G is the direct product of N and H.
- Example: $G = S_3$, $N = A_3$, $H = \langle(1\,2)\rangle$. Then $G = N \rtimes H$, but H is not normal, so G is not the direct product.
- Properties of $G = N \rtimes H$:
  - $H \cong H/(N \cap H) \cong NH/N = G/N$ by First Isomorphism Theorem; if G is finite, $|G| = |N||G : N| = |N||H|$.
  - Each $x \in G$ has a unique expression $x = nh$ with $n \in N$, $h \in H$.
  - If $x = n_1h_1$, $y = n_2h_2$, then $xy = n_1h_1n_2h_1^{-1} \cdot h_1h_2 = n'h'$ with $n' = n_1h_1n_2h_1^{-1} \in N$ and $h' = h_1h_2 \in H$.
  - For $h \in H$, conjugation by h maps N to N, so $\varphi_h : N \to N$, $\varphi_h(n) = hnh^{-1}$, is an automorphism of N, and $\varphi_h \circ \varphi_{h'} = \varphi_{hh'}$. Hence the *conjugation homomorphism* $\varphi : H \to \mathrm{Aut}(N)$, $\varphi(h) = \varphi_h$.
  - $(n_1h_1)(n_2h_2) = n_1\varphi(h_1)(n_2) \cdot h_1h_2$, so the multiplication of G is determined by the operations of N and H together with φ.
  - If φ is trivial, then $hnh^{-1} = n$ for all $n \in N$, $h \in H$, so $H \trianglelefteq G$ and $G = N \times H$. Conversely, if $G = N \times H$, then H commutes with N, so φ is trivial.
  - If φ is non-trivial, G is non-abelian: there exist $h \in H$, $n \in N$ with $hnh^{-1} = \varphi(h)(n) \neq n$.

External Semidirect Products

- Given groups N, H and a homomorphism $\varphi : H \to \mathrm{Aut}(N)$, define a binary operation on $N \times H$ by
$$(n_1, h_1)(n_2, h_2) = (n_1\varphi(h_1)(n_2), h_1h_2).$$
This makes $N \times H$ a group with identity $(1, 1)$ and inverse $(n, h)^{-1} = (\varphi(h^{-1})(n^{-1}), h^{-1})$.
- Call this the (*external*) *semidirect product* of N by H corresponding to φ, denoted $G = N \rtimes_\varphi H$.
- This generally differs from the direct product unless φ is trivial.
- $\mathcal{N} = N \times 1$ and $\mathcal{H} = 1 \times H$ in G are isomorphic copies of N, H respectively. Computation shows $\mathcal{N} \trianglelefteq G$, $G = \mathcal{N}\mathcal{H}$, $\mathcal{N} \cap \mathcal{H} = 1$, so G is the internal semidirect product of $\mathcal{N}$ by $\mathcal{H}$.
- Identifying $\mathcal{N}$ with N and $\mathcal{H}$ with H, the conjugation homomorphism from $\mathcal{H}$ to $\mathrm{Aut}(\mathcal{N})$ corresponds to the original φ.
- Write $N \rtimes_\varphi H = \{nh \mid n \in N, h \in H\}$ with multiplication $(n_1h_1)(n_2h_2) = n_1\varphi(h_1)(n_2) \cdot h_1h_2$. Note $hnh^{-1} = \varphi(h)(n)$.
- Distinct $\varphi, \psi : H \to \mathrm{Aut}(N)$ need not yield isomorphic semidirect products. The next two propositions are partial converses.

Proposition 2.11. *Let H be a cyclic group and let N be an arbitrary group. If φ and ψ are monomorphisms from H to* $\mathrm{Aut}(N)$ *such that $\varphi(H) = \psi(H)$, then we have $N \rtimes_\varphi H \cong N \rtimes_\psi H$.*

Proof. Write $H = \langle x\rangle$. Since $\varphi(H) = \psi(H)$, the automorphisms $\varphi(x)$ and $\psi(x)$ generate the same cyclic subgroup of $\mathrm{Aut}(N)$. Hence there exist $a, b \in \mathbb{Z}$ such that

$$\varphi(x)^a = \psi(x), \qquad \psi(x)^b = \varphi(x).$$

It follows that, for every $h \in H$,

$$\varphi(h^a) = \psi(h), \qquad \psi(h^b) = \varphi(h).$$

Indeed, if $h = x^m$, then $\varphi(h^a) = \varphi(x^{am}) = \varphi(x)^{am} = \psi(x)^m = \psi(h)$, and the second equality is similar. Define

$$\tau\colon N \rtimes_\psi H \to N \rtimes_\varphi H, \qquad \tau(nh) = nh^a.$$

We show that τ is a homomorphism. Let $n_1, n_2 \in N$ and $h_1, h_2 \in H$. Multiplication in $N \rtimes_\psi H$ gives

$$(n_1h_1)(n_2h_2) = n_1\psi(h_1)(n_2)h_1h_2.$$

Since H is cyclic, it is abelian, so $(h_1h_2)^a = h_1^ah_2^a$. Therefore

$$\tau((n_1h_1)(n_2h_2)) = n_1\psi(h_1)(n_2)(h_1h_2)^a = n_1\varphi(h_1^a)(n_2)h_1^ah_2^a.$$

But this is exactly the product of $n_1h_1^a$ and $n_2h_2^a$ in $N \rtimes_\varphi H$, so

$$\tau((n_1h_1)(n_2h_2)) = \tau(n_1h_1)\tau(n_2h_2).$$

Thus τ is a homomorphism. Similarly, the map

$$\lambda\colon N \rtimes_\varphi H \to N \rtimes_\psi H, \qquad \lambda(nh) = nh^b$$

is a homomorphism.
It remains to check that these two homomorphisms are inverse to one another. Since

$$\varphi(x) = \psi(x)^b = (\varphi(x)^a)^b = \varphi(x^{ab}),$$

and φ is injective, we have $x = x^{ab}$. Hence $h = h^{ab}$ for every $h \in H$. Similarly, using the injectivity of ψ, we get $h = h^{ba}$ for every $h \in H$. Therefore

$$(\tau \circ \lambda)(nh) = \tau(nh^b) = nh^{ba} = nh, \qquad (\lambda \circ \tau)(nh) = \lambda(nh^a) = nh^{ab} = nh.$$

Thus τ and λ are mutually inverse isomorphisms, and so $N \rtimes_\varphi H \cong N \rtimes_\psi H$. □

Proposition 2.12. *Let N and H be groups, let $\psi : H \to \mathrm{Aut}(N)$ be a homomorphism, and let $f \in \mathrm{Aut}(N)$. If $\hat{f}$ is the inner automorphism of $\mathrm{Aut}(N)$ induced by f, then $N \rtimes_{\hat{f}\circ\psi} H \cong N \rtimes_\psi H$.*

Proof. By definition, the inner automorphism $\hat{f}$ of $\mathrm{Aut}(N)$ sends $\alpha \in \mathrm{Aut}(N)$ to $f \circ \alpha \circ f^{-1}$. Define

$$\theta\colon N \rtimes_\psi H \to N \rtimes_{\hat{f}\circ\psi} H, \qquad \theta(nh) = f(n)h.$$

We show that θ is a homomorphism. Let $n_1, n_2 \in N$ and $h_1, h_2 \in H$. Multiplication in $N \rtimes_\psi H$ gives

$$(n_1h_1)(n_2h_2) = n_1\psi(h_1)(n_2)h_1h_2.$$

Hence

$$\theta((n_1h_1)(n_2h_2)) = f(n_1\psi(h_1)(n_2))h_1h_2 = f(n_1)f(\psi(h_1)(n_2))h_1h_2.$$

On the other hand, multiplication in $N \rtimes_{\hat{f}\circ\psi} H$ gives

$$\theta(n_1h_1)\theta(n_2h_2) = (f(n_1)h_1)(f(n_2)h_2) = f(n_1)(\hat{f} \circ \psi)(h_1)(f(n_2))h_1h_2.$$

Since

$$(\hat{f} \circ \psi)(h_1)(f(n_2)) = (f \circ \psi(h_1) \circ f^{-1})(f(n_2)) = f(\psi(h_1)(n_2)),$$

the two expressions are equal. Thus θ is a homomorphism.
Finally, the map

$$N \rtimes_{\hat{f}\circ\psi} H \to N \rtimes_\psi H, \qquad nh \mapsto f^{-1}(n)h$$

is inverse to θ. Therefore θ is an isomorphism, and so $N \rtimes_{\hat{f}\circ\psi} H \cong N \rtimes_\psi H$. □

Dihedral Groups

- Let $N \cong \mathbb{Z}_n$ ($n \in \mathbb{N}$), $H \cong \mathbb{Z}_2$, $\varphi : H \to \mathrm{Aut}(N)$ send the generator of H to the automorphism sending each element of N to its inverse (so $\varphi(h) = \sigma_{n-1}$).
- The group $N \rtimes_\varphi H$ is the *dihedral group* of order $2n$, denoted D_{2n}. (Some authors write D_n.)
- D_{2n} is non-abelian iff $n > 2$; when $n = 2$, φ is trivial and $D_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.
- Geometrically D_{2n} is the symmetry group of a regular n-gon: the generator of N is rotation by $2\pi/n$, the generator of H is reflection through a fixed axis.
- Also: the *infinite dihedral group* $D_\infty = N \rtimes_\varphi H$ where $N \cong \mathbb{Z}$ and H, φ as above.

Proposition 2.13. *Let s and t be elements of order 2 in a group G. (Such elements are called* involutions.*) Then $\langle s, t\rangle$ is a dihedral group; in particular, $\langle s, t\rangle = \langle st\rangle \rtimes \langle s\rangle$.*

Proof. Put

$$L = \langle s, t\rangle, \qquad r = st, \qquad N = \langle r\rangle, \qquad H = \langle s\rangle.$$

Since s and t have order 2, we have $s^{-1} = s$ and $t^{-1} = t$. Therefore

$$srs^{-1} = s(st)s = ts = (st)^{-1} = r^{-1}.$$

Similarly,

$$trt^{-1} = t(st)t = ts = r^{-1}.$$

Thus both generators s and t of L conjugate r to r^{-1}. Hence both generators normalize $N = \langle r\rangle$, and so

$$N \trianglelefteq L.$$

Also $H = \langle s\rangle \le L$. We next show that $L = NH$. Since $r = st \in N$ and $s \in H$, we have

$$t = sr \in HN.$$

But $N \trianglelefteq L$, so $HN = NH$. Hence both s and t lie in NH, and therefore

$$L = \langle s, t\rangle \le NH.$$

The reverse inclusion is immediate from $N, H \le L$, so $L = NH$.
It remains to check that $N \cap H = 1$. Since $H = \langle s\rangle$ and s has order 2, the only possible non-identity element of H is s. Suppose, for contradiction, that $s \in N$. Then s is a power of r. Since $t = sr$, it follows that $t \in N$ as well. Hence $L = \langle s, t\rangle \le N$, so $L = N$ is cyclic. But a cyclic group has at most one element of order 2, so $s = t$. Then $r = st = 1$, and hence $N = \langle r\rangle = 1$, contradicting $s \in N$. Therefore $s \notin N$, and so $N \cap H = 1$.
We have shown that

$$L = NH, \qquad N \trianglelefteq L, \qquad N \cap H = 1.$$

Hence L is the internal semidirect product $N \rtimes H$. Finally, the conjugation action of H on N sends the generator r of N to r^{-1}, so it sends every element of N to its inverse. This is precisely the action used in the definition of a dihedral group. Thus $\langle s, t\rangle$ is dihedral; more explicitly, if $r = st$ has finite order n, then $L \cong D_{2n}$, while if r has infinite order, then $L \cong D_\infty$. □

Reference note. Reference [7] is Aschbacher's *Finite Group Theory*; the local reference copy is the second edition, and Section 45 is titled "Involutions in finite groups." The point of the reference is that the small fact proved above becomes a major tool once one studies finite simple groups. In a finite group, two involutions x and y generate a dihedral group whose rotation subgroup is $\langle xy\rangle$. This means that the product xy controls the subgroup $\langle x, y\rangle$, and the elementary geometry of dihedral groups gives information about conjugacy among involutions.
Aschbacher begins Section 45 by using this exact observation. If $D = \langle x, y\rangle$ and $n = o(xy)$, then D has order $2n$. When n is odd, all involutions of D are conjugate in D, so x and y are already conjugate inside the small subgroup they generate. When n is even, there is a unique involution in the rotation subgroup $\langle xy\rangle$, and this central involution gives an additional conjugacy possibility. Thus the parity of $o(xy)$ tells us how involutions can fuse, meaning how they may become conjugate in the larger group.

This is important because involutions are unavoidable in non-abelian finite simple groups: by the Feit–Thompson Odd Order Theorem, every non-abelian finite simple group has even order, and therefore has involutions. Section 45 uses dihedral subgroups generated by pairs of involutions to count and control such elements. Two central results in that section are the Brauer–Fowler Theorem and the Thompson Order Formula. The Brauer–Fowler Theorem says, roughly, that fixing the centralizer of an involution leaves only finitely many possible finite simple groups. This suggests that finite simple groups can be attacked by studying the centralizers of their involutions.

The Thompson Order Formula is more explicit. When a finite group has at least two conjugacy classes of involutions, it expresses the order of the group in terms of centralizers of involutions and the fusion of involutions inside those centralizers. In one form, if $x_1, \ldots, x_k$ represent the conjugacy classes of involutions and n_i counts certain ordered pairs of involutions whose product-generated subgroup contains x_i, then

$$|G| = |C_G(x_1)|\,|C_G(x_2)| \sum_{i=1}^{k} \frac{n_i}{|C_G(x_i)|}.$$

The important point is not the formula itself at this stage, but the method behind it: because pairs of involutions generate dihedral groups, their products and conjugacy relations can be counted very precisely. Thus Proposition 2.13 is not just a cute exercise. It is one of the basic mechanisms behind the local analysis of finite simple groups, where one tries to understand a group by studying centralizers and normalizers of special subgroups.

Exercises

Exercise 2.1. Let H be a subgroup of a cyclic group G. Show that every automorphism of H is the restriction to H of an automorphism of G.

Solution. Let $G = \langle g \rangle$ and let $H \leq G$. Since every subgroup of a cyclic group is cyclic, we may write $H = \langle g^k \rangle$ for some integer k.

First suppose that G is infinite cyclic. If $H = 1$, then the only automorphism of H is the restriction of the identity automorphism of G. Otherwise $H = \langle g^k \rangle$ is infinite cyclic, and every automorphism of H sends the generator g^k either to g^k or to g^{-k}. These two automorphisms are respectively the restrictions of the automorphisms of G given by $g \mapsto g$ and $g \mapsto g^{-1}$. Hence the claim holds when G is infinite cyclic.

Now suppose that G is finite of order n, and put $|H| = m$. Let $\psi \in \mathrm{Aut}(H)$. Since g^k generates H, we have

$$\psi(g^k) = (g^k)^a$$

for some integer a with $\gcd(a, m) = 1$. We want to choose an automorphism $\varphi \in \mathrm{Aut}(G)$ whose restriction to H is ψ. By Proposition 2.1, an automorphism of G has the form $\varphi(g) = g^b$ for some integer b with $\gcd(b, n) = 1$. For such a map,

$$\varphi(g^k) = \varphi(g)^k = (g^b)^k = g^{bk} = (g^k)^b.$$

Thus φ will agree with ψ on H provided $b \equiv a \pmod{m}$, since g^k has order m.

It remains to choose b so that $b \equiv a \pmod{m}$ and $\gcd(b, n) = 1$. Let q be the product of the distinct prime divisors of n which do not divide m; if there are no such primes, take $q = 1$. Then $\gcd(m, q) = 1$, so by the Chinese Remainder Theorem there is an integer b such that

$$b \equiv a \pmod{m}, \qquad b \equiv 1 \pmod{q}.$$

If p is a prime divisor of n and $p \mid m$, then $p \nmid a$, since $\gcd(a, m) = 1$, and hence $p \nmid b$ because $b \equiv a \pmod{m}$. If $p \mid n$ but $p \nmid m$, then $p \mid q$, so $b \equiv 1 \pmod{p}$, and again $p \nmid b$. Therefore no prime divisor of n divides b, so $\gcd(b, n) = 1$.

Define $\varphi(g) = g^b$. Then $\varphi \in \mathrm{Aut}(G)$ by Proposition 2.1, and

$$\varphi(g^k) = (g^k)^b = (g^k)^a = \psi(g^k).$$

Since g^k generates H, it follows that $\varphi|_H = \psi$. Thus every automorphism of H is the restriction to H of an automorphism of G. □

Exercise 2.2. Show that $\mathrm{Aut}(\mathbb{Z}_8) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

Solution. By Proposition 2.1,
$$\mathrm{Aut}(\mathbb{Z}_8) \cong (\mathbb{Z}/8\mathbb{Z})^\times.$$
The units modulo 8 are precisely the residue classes represented by integers coprime to 8, so
$$(\mathbb{Z}/8\mathbb{Z})^\times = \{1+8\mathbb{Z}, 3+8\mathbb{Z}, 5+8\mathbb{Z}, 7+8\mathbb{Z}\}.$$
Each non-identity element has order 2, since
$$3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod 8.$$
Thus $(\mathbb{Z}/8\mathbb{Z})^\times$ has four elements and every non-identity element has order 2. Therefore it is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$, and hence
$$\mathrm{Aut}(\mathbb{Z}_8) \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$
□

Exercise 2.3. Show that $\mathrm{Aut}(\mathbb{Z}_{p^2}) \cong \mathbb{Z}_{p^2-p}$ for p a prime. (HINT: Let m be a primitive root modulo p, and show that either m or $m+p$ be a primitive root modulo p^2.)

Solution. If $p = 2$, then Proposition 2.1 gives
$$\mathrm{Aut}(\mathbb{Z}_4) \cong (\mathbb{Z}/4\mathbb{Z})^\times = \{1+4\mathbb{Z}, 3+4\mathbb{Z}\} \cong \mathbb{Z}_2,$$
and this is $\mathbb{Z}_{p^2-p}$. Hence we may assume that p is odd.
By Proposition 2.1,
$$\mathrm{Aut}(\mathbb{Z}_{p^2}) \cong (\mathbb{Z}/p^2\mathbb{Z})^\times.$$
The latter group has order $\varphi(p^2) = p^2 - p = p(p-1)$. It remains to show that it has an element of order $p(p-1)$. By Proposition 2.2, there is a primitive root m modulo p, so m has order $p-1$ modulo p. Consider m as a unit modulo p^2. If d is its order modulo p^2, then $d \mid p(p-1)$ by Euler's theorem. Also, reducing $m^d \equiv 1 \pmod{p^2}$ modulo p gives $m^d \equiv 1 \pmod p$, so $p-1 \mid d$. Therefore d is either $p-1$ or $p(p-1)$. If $d = p(p-1)$, then m is already a primitive root modulo p^2, so $(\mathbb{Z}/p^2\mathbb{Z})^\times$ is cyclic of order $p(p-1)$.
It remains to handle the case $d = p-1$. Then
$$m^{p-1} \equiv 1 \pmod{p^2}.$$
We claim that $m+p$ has order $p(p-1)$ modulo p^2. Since $m+p \equiv m \pmod p$, the order of $m+p$ modulo p^2 is again divisible by $p-1$ and divides $p(p-1)$. Thus it is enough to show that $(m+p)^{p-1} \not\equiv 1 \pmod{p^2}$. By the binomial theorem,
$$(m+p)^{p-1} \equiv m^{p-1} + (p-1)pm^{p-2} \pmod{p^2},$$
since all remaining terms contain a factor p^2. Using $m^{p-1} \equiv 1 \pmod{p^2}$, we obtain
$$(m+p)^{p-1} \equiv 1 + p(p-1)m^{p-2} \pmod{p^2}.$$
Since $p \nmid m$ and $p \nmid (p-1)$, the integer $(p-1)m^{p-2}$ is not divisible by p. Hence $p(p-1)m^{p-2}$ is divisible by p but not by p^2, and therefore
$$(m+p)^{p-1} \not\equiv 1 \pmod{p^2}.$$
Thus $m+p$ has order $p(p-1)$ modulo p^2. In either case, $(\mathbb{Z}/p^2\mathbb{Z})^\times$ is cyclic of order $p(p-1) = p^2 - p$. Therefore
$$\mathrm{Aut}(\mathbb{Z}_{p^2}) \cong \mathbb{Z}_{p^2-p}.$$
□

Exercise 2.4. Show that if $H \trianglelefteq G$, then any characteristic subgroup of H is normal in G.

Solution. Let N be a characteristic subgroup of H. We want to show that $N \trianglelefteq G$. Let $g \in G$ be arbitrary, and consider the inner automorphism

$$\varphi\colon G \to G, \qquad \varphi(x) = gxg^{-1}.$$

Since $H \trianglelefteq G$, we have

$$\varphi(H) = gHg^{-1} = H.$$

Hence the restriction $\varphi|_H\colon H \to H$ is an automorphism of H. Since N is characteristic in H, every automorphism of H fixes N, so

$$\varphi|_H(N) = N.$$

But $\varphi|_H(N) = gNg^{-1}$. Therefore $gNg^{-1} = N$. Since $g \in G$ was arbitrary, $N \trianglelefteq G$. □

Exercise 2.5. Let F be a field, and consider F as a group under its additive operation. Show that F has no non-trivial proper characteristic subgroups.

Solution. The subgroups 0 and F are characteristic, so it remains to show that no other characteristic subgroups exist. Let H be a characteristic subgroup of the additive group of F, and suppose that H is non-trivial. Then there is some $h \in H$ with $h \neq 0$.

For each $a \in F^\times$, define

$$\varphi_a\colon F \to F, \qquad \varphi_a(f) = af.$$

Since

$$\varphi_a(f_1 + f_2) = a(f_1 + f_2) = af_1 + af_2 = \varphi_a(f_1) + \varphi_a(f_2),$$

and $\varphi_{a^{-1}}$ is inverse to φ_a, the map φ_a is an automorphism of the additive group of F. Since H is characteristic, we have $\varphi_a(H) = H$ for every $a \in F^\times$.

Now let $f \in F$. If $f = 0$, then $f \in H$ because H is a subgroup. If $f \neq 0$, then $a = fh^{-1}$ lies in $F^\times$, and hence

$$f = (fh^{-1})h = \varphi_a(h) \in H.$$

Thus every element of F lies in H, so $H = F$. Therefore the only characteristic subgroups of the additive group of F are 0 and F, and hence F has no non-trivial proper characteristic subgroups. □

Exercise 2.6. Verify the claim made on page 19 that if (1) holds, then (2) holds iff (3) and (4) hold.

Solution. Assume throughout that (1) holds, so $H_i \trianglelefteq G$ for every $1 \leq i \leq n$. Suppose first that (2) holds. Then every $g \in G$ has an expression $g = h_1 \cdots h_n$ with $h_i \in H_i$, so $G \subseteq H_1 \cdots H_n$. Conversely, since each $H_i \leq G$, every product $h_1 \cdots h_n$ lies in G. Hence $H_1 \cdots H_n \subseteq G$, and therefore

$$G = H_1 \cdots H_n.$$

Thus (3) holds.

To prove (4), fix i and take

$$x \in H_i \cap H_1 \cdots H_{i-1}H_{i+1} \cdots H_n.$$

Since $x \in H_i$, we have one expression

$$x = 1 \cdots 1 \cdot x \cdot 1 \cdots 1,$$

where x appears in the ith position. Since $x \in H_1 \cdots H_{i-1}H_{i+1} \cdots H_n$, we also have an expression

$$x = h_1 \cdots h_{i-1} \cdot 1 \cdot h_{i+1} \cdots h_n$$

with $h_j \in H_j$ for $j \neq i$. By uniqueness in (2), the ith factors in these two expressions are equal, so $x = 1$. Hence

$$H_i \cap H_1 \cdots H_{i-1}H_{i+1} \cdots H_n = 1$$

for every i, proving (4).

Conversely, suppose that (3) and (4) hold. By (3), every $g \in G$ has at least one expression $g = h_1 \cdots h_n$ with $h_i \in H_i$. It remains to show that this expression is unique. We first observe that elements from different factors commute. Let $i \neq j$, and take $h_i \in H_i$, $h_j \in H_j$. Since $H_j \trianglelefteq G$, we have $h_i h_j h_i^{-1} \in H_j$, and hence

$$[h_i, h_j] = (h_i h_j h_i^{-1})h_j^{-1} \in H_j.$$

Similarly, since $H_i \trianglelefteq G$, we have $h_j h_i^{-1} h_j^{-1} \in H_i$, and hence

$$[h_i, h_j] = h_i(h_j h_i^{-1} h_j^{-1}) \in H_i.$$

Therefore $[h_i, h_j] \in H_i \cap H_j$. But $H_j \leq H_1 \cdots H_{i-1} H_{i+1} \cdots H_n$, so (4) gives $H_i \cap H_j = 1$. Thus $[h_i, h_j] = 1$, and hence h_i and h_j commute.
Now suppose

$$g = h_1 \cdots h_n = h_1' \cdots h_n', \qquad h_i, h_i' \in H_i.$$

Fix i. Since elements from different factors commute, we may write

$$g = h_i k_i = h_i' k_i',$$

where

$$k_i \in H_1 \cdots H_{i-1} H_{i+1} \cdots H_n, \qquad k_i' \in H_1 \cdots H_{i-1} H_{i+1} \cdots H_n.$$

Since the factors in $H_1 \cdots H_{i-1} H_{i+1} \cdots H_n$ commute with one another, this product is closed under multiplication and inverses. Also, from $h_i k_i = h_i' k_i'$ we get

$$h_i'^{-1} h_i = k_i' k_i^{-1}.$$

The left hand side lies in H_i, and the right hand side lies in $H_1 \cdots H_{i-1} H_{i+1} \cdots H_n$. Therefore, by (4),

$$h_i'^{-1} h_i = 1.$$

Hence $h_i = h_i'$. Since this holds for every i, the expression of g is unique. Thus (2) holds. □

Exercise 2.7. Verify the claim made on page 19 that (1) and (2) together imply (3) through (6).

Solution. Assume that (1) and (2) hold. By Exercise 2.6, conditions (1) and (2) imply (3) and (4). It remains to prove (5) and (6).
To prove (5), let $i \neq j$, and take $h_i \in H_i$, $h_j \in H_j$. Since $H_j \trianglelefteq G$, we have $h_i h_j h_i^{-1} \in H_j$, and hence

$$[h_i, h_j] = (h_i h_j h_i^{-1}) h_j^{-1} \in H_j.$$

Similarly, since $H_i \trianglelefteq G$, we have $h_j h_i^{-1} h_j^{-1} \in H_i$, and hence

$$[h_i, h_j] = h_i(h_j h_i^{-1} h_j^{-1}) \in H_i.$$

Thus $[h_i, h_j] \in H_i \cap H_j$. But $H_j \leq H_1 \cdots H_{i-1} H_{i+1} \cdots H_n$, so (4) gives $H_i \cap H_j = 1$. Hence $[h_i, h_j] = 1$, and therefore h_i and h_j commute. Since $i \neq j$ was arbitrary, elements of distinct factors commute.
Finally, let

$$g = h_1 \cdots h_n, \qquad g' = h_1' \cdots h_n',$$

where $h_i, h_i' \in H_i$ for each i. By (5), elements from distinct factors commute, so we may move each h_i' left across all factors h_j with $j \neq i$. Therefore

$$gg' = h_1 \cdots h_n h_1' \cdots h_n' = (h_1 h_1') \cdots (h_n h_n').$$

This proves (6), and hence (1) and (2) together imply (3) through (6). □

Exercise 2.8. Let G_1 and G_2 be groups, and let $H \leq G_1 \times G_2$. Define

$$P_1 = \{x \in G_1 \mid (x, y) \in H \text{ for some } y \in G_2\},$$

$$I_1 = \{x \in G_1 \mid (x, 1) \in H\},$$

and analogously define subsets P_2 and I_2 of G_2.

(a) Show for $i = 1, 2$ that $P_i \leq G_i$ and $I_i \trianglelefteq P_i$.

(b) Show that there is a unique isomorphism from P_1/I_1 to P_2/I_2 sending xI_1 to yI_2, where y is any element of G_2 such that $(x, y) \in H$.

(c) Prove *Goursat's theorem*: There is a bijective correspondence between subgroups of $G_1 \times G_2$ and triples (S_1, S_2, φ), where S_i is a section of G_i $(i = 1, 2)$ and $\varphi : S_1 \to S_2$ is an isomorphism. (Recall that a section of a group G is a group L/M, where $M \trianglelefteq L \leq G$.)

Solution.

(a) Fix $i \in \{1, 2\}$, and let j be the other index. First we show that $P_i \leq G_i$. Since $(1, 1) \in H$, we have $1 \in P_i$, so P_i is non-empty. Let $a, b \in P_i$. By definition of P_i, there are elements $a_j, b_j \in G_j$ such that

$$(a_1, a_2), (b_1, b_2) \in H, \qquad a_i = a, \quad b_i = b.$$

Since $H \leq G_1 \times G_2$, we have

$$(a_1, a_2)(b_1, b_2)^{-1} = (a_1 b_1^{-1}, a_2 b_2^{-1}) \in H.$$

The ith coordinate of this element is ab^{-1}, so $ab^{-1} \in P_i$. Hence $P_i \leq G_i$.
Next we show that $I_i \trianglelefteq P_i$. First, $I_i \leq P_i$: if $x \in I_i$, then the element with x in the ith coordinate and 1 in the jth coordinate lies in H, so $x \in P_i$; and the same subgroup test as above, with the jth coordinate always equal to 1, shows that $I_i \leq G_i$.
It remains to prove normality in P_i. Let $a \in P_i$ and $x \in I_i$. Since $a \in P_i$, there is some $a_j \in G_j$ such that, putting $a_i = a$, we have

$$(a_1, a_2) \in H.$$

Since $x \in I_i$, putting $x_i = x$ and $x_j = 1$, we have

$$(x_1, x_2) \in H.$$

Since H is a subgroup,

$$(a_1, a_2)(x_1, x_2)(a_1, a_2)^{-1} = (a_1 x_1 a_1^{-1}, a_2 x_2 a_2^{-1}) \in H.$$

The ith coordinate of this element is axa^{-1}, and the jth coordinate is $a_j 1 a_j^{-1} = 1$. Hence $axa^{-1} \in I_i$. Therefore $I_i \trianglelefteq P_i$.

(b) Define

$$\varphi\colon P_1/I_1 \to P_2/I_2$$

by setting $\varphi(xI_1) = yI_2$, where $y \in G_2$ is chosen so that $(x, y) \in H$. We first check that this is well-defined. Suppose $(x, y), (x, y') \in H$. Since H is a subgroup,

$$(x, y)^{-1}(x, y') = (1, y^{-1}y') \in H.$$

Hence $y^{-1}y' \in I_2$, so $yI_2 = y'I_2$. Thus the choice of y does not matter.
Now suppose $xI_1 = x'I_1$, and choose $y, y' \in G_2$ such that $(x, y), (x', y') \in H$. Since $xI_1 = x'I_1$, we have $x^{-1}x' \in I_1$, so

$$(x^{-1}x', 1) \in H.$$

Also,

$$(x, y)^{-1}(x', y') = (x^{-1}x', y^{-1}y') \in H.$$

Hence

$$(x^{-1}x', 1)^{-1}(x^{-1}x', y^{-1}y') = (1, y^{-1}y') \in H.$$

Therefore $y^{-1}y' \in I_2$, so $yI_2 = y'I_2$. Thus φ is well-defined on cosets.
Next, φ is a homomorphism. Let $(x, y), (x', y') \in H$. Then $(xx', yy') \in H$, and hence

$$\varphi(xI_1 \cdot x'I_1) = \varphi(xx'I_1) = yy'I_2 = yI_2 \cdot y'I_2 = \varphi(xI_1)\varphi(x'I_1).$$

Define

$$\psi\colon P_2/I_2 \to P_1/I_1$$

by setting $\psi(yI_2) = xI_1$, where $x \in G_1$ is chosen so that $(x, y) \in H$. By the same argument with the two coordinates interchanged, ψ is a well-defined homomorphism. Moreover, if $(x, y) \in H$, then

$$\psi(\varphi(xI_1)) = \psi(yI_2) = xI_1, \qquad \varphi(\psi(yI_2)) = \varphi(xI_1) = yI_2.$$

Thus $\psi \circ \varphi$ and $\varphi \circ \psi$ are the identity maps, so φ is a bijection. Therefore φ is an isomorphism. Finally, suppose $\varphi' \colon P_1/I_1 \to P_2/I_2$ is another function satisfying the same rule: whenever $(x, y) \in H$, we have $\varphi'(xI_1) = yI_2$. Every coset of P_1/I_1 has the form xI_1 with $x \in P_1$, and by definition of P_1 there is some $y \in G_2$ such that $(x, y) \in H$. Hence

$$\varphi'(xI_1) = yI_2 = \varphi(xI_1).$$

Therefore $\varphi' = \varphi$, proving uniqueness.

(c) Let

$$A = \{H \mid H \leq G_1 \times G_2\},$$

and let

$$B = \{(S_1, S_2, \varphi) \mid S_i \text{ is a section of } G_i,\ \varphi \colon S_1 \to S_2 \text{ is an isomorphism}\}.$$

Define

$$\theta \colon A \to B$$

as follows. Given $H \in A$, let P_i and I_i be the subgroups defined above. By (a), $I_i \trianglelefteq P_i \leq G_i$, so P_i/I_i is a section of G_i. By (b), there is a unique isomorphism

$$\varphi_H \colon P_1/I_1 \to P_2/I_2$$

such that $\varphi_H(xI_1) = yI_2$ whenever $(x, y) \in H$. Thus we define

$$\theta(H) = (P_1/I_1, P_2/I_2, \varphi_H).$$

Define

$$\theta' \colon B \to A$$

as follows. Let $(S_1, S_2, \varphi) \in B$. Since S_i is a section of G_i, there are subgroups $M_i \trianglelefteq L_i \leq G_i$ such that

$$S_i = L_i/M_i.$$

Define

$$H_\varphi = \{(x, y) \in L_1 \times L_2 \mid \varphi(xM_1) = yM_2\}.$$

Let $(x, y), (x', y') \in H_\varphi$. Since φ is an isomorphism,

$$\varphi(xM_1) = yM_2, \qquad \varphi(x'M_1) = y'M_2,$$

and hence

$$\varphi(xx'^{-1}M_1) = \varphi(xM_1)\varphi(x'M_1)^{-1} = (yM_2)(y'M_2)^{-1} = yy'^{-1}M_2.$$

Therefore $(xx'^{-1}, yy'^{-1}) \in H_\varphi$, so

$$H_\varphi \leq L_1 \times L_2 \leq G_1 \times G_2.$$

Thus $H_\varphi \in A$, and we define

$$\theta'(S_1, S_2, \varphi) = H_\varphi.$$

We now show that θ and θ' are inverse maps. First let $H \in A$, and form $\theta(H) = (P_1/I_1, P_2/I_2, \varphi_H)$. Then

$$\theta'(\theta(H)) = \{(x, y) \in P_1 \times P_2 \mid \varphi_H(xI_1) = yI_2\}.$$

If $(x, y) \in H$, then $\varphi_H(xI_1) = yI_2$, so $(x, y) \in \theta'(\theta(H))$. Conversely, suppose $(x, y) \in \theta'(\theta(H))$. Since $x \in P_1$, there is some $z \in G_2$ such that $(x, z) \in H$. Then $\varphi_H(xI_1) = zI_2$, while $\varphi_H(xI_1) = yI_2$ by assumption. Hence $zI_2 = yI_2$, so $z^{-1}y \in I_2$ and therefore $(1, z^{-1}y) \in H$. Thus

$$(x, z)(1, z^{-1}y) = (x, y) \in H.$$

Hence $\theta'(\theta(H)) = H$.

Conversely, let $(S_1, S_2, \varphi) \in B$, and write $S_i = L_i/M_i$ with $M_i \trianglelefteq L_i \leq G_i$. Form $H_\varphi = \theta'(S_1, S_2, \varphi)$. We compute the subgroups associated to H_φ. If $(x, y) \in H_\varphi$, then $x \in L_1$, so $P_1 \leq L_1$. Conversely, if $x \in L_1$, then $\varphi(xM_1) = yM_2$ for some $y \in L_2$, and so $(x, y) \in H_\varphi$. Thus $P_1 = L_1$. By a similar

argument, using the surjectivity of φ, we have $P_2 = L_2$.
For I_1, we have

$$x \in I_1 \iff (x, 1) \in H_\varphi \iff \varphi(xM_1) = M_2 \iff xM_1 = M_1 \iff x \in M_1,$$

so $I_1 = M_1$. Similarly, $I_2 = M_2$. Therefore the first two entries recovered by θ are

$$P_1/I_1 = L_1/M_1 = S_1, \qquad P_2/I_2 = L_2/M_2 = S_2.$$

Finally, the isomorphism recovered by (b) sends xM_1 to yM_2 whenever $(x, y) \in H_\varphi$. By the definition of H_φ, this means exactly that $yM_2 = \varphi(xM_1)$. Hence, by the uniqueness in (b), the recovered isomorphism is φ. Thus $\theta(\theta'(S_1, S_2, \varphi)) = (S_1, S_2, \varphi)$. Therefore θ and θ' are inverse maps, so θ is a bijection.

□

Exercise 2.9. (cont.) Use Exercise 2.8 to give a different proof of Proposition 2.10.

Solution. Suppose that

$$G = H_1 \times \cdots \times H_n$$

where the orders $|H_i|$ are pairwise coprime. We want to show that any subgroup $L \leq G$ is the direct product of $L \cap H_1, \ldots, L \cap H_n$. We prove this by induction on n.
First suppose $n = 2$, and write $G = H \times K$ with $\gcd(|H|, |K|) = 1$. Let $L \leq H \times K$. By Exercise 2.8, the subgroup L corresponds to a triple

$$(P_H/I_H, P_K/I_K, \varphi),$$

where P_H/I_H is a section of H, P_K/I_K is a section of K, and

$$\varphi\colon P_H/I_H \to P_K/I_K$$

is an isomorphism. Hence

$$|P_H/I_H| = |P_K/I_K|.$$

Since $I_H \trianglelefteq P_H \leq H$, we have $|P_H/I_H| \mid |P_H|$ and $|P_H| \mid |H|$, so $|P_H/I_H| \mid |H|$. Similarly, $|P_K/I_K| \mid |K|$. But $\gcd(|H|, |K|) = 1$, and the two quotient groups have the same order, so

$$|P_H/I_H| = |P_K/I_K| = 1.$$

Therefore P_H/I_H and P_K/I_K are trivial, and hence $P_H = I_H$ and $P_K = I_K$. By the reconstruction in Exercise 2.8, we have

$$L = \{(h, k) \in P_H \times P_K \mid \varphi(hI_H) = kI_K\}.$$

Since both quotient groups are trivial, the condition $\varphi(hI_H) = kI_K$ is automatic, so

$$L = P_H \times P_K.$$

Also $I_H = L \cap H$ and $I_K = L \cap K$, so $P_H = L \cap H$ and $P_K = L \cap K$. Thus

$$L = (L \cap H) \times (L \cap K),$$

proving the case $n = 2$.
Now suppose $n > 2$ and that the result is known for direct products with $n - 1$ factors. Put

$$M = H_1 \times \cdots \times H_{n-1}.$$

Then $G = M \times H_n$. Since the orders $|H_i|$ are pairwise coprime, we have $\gcd(|M|, |H_n|) = 1$. Applying the $n = 2$ case to $G = M \times H_n$ gives

$$L = (L \cap M) \times (L \cap H_n).$$

Now $L \cap M$ is a subgroup of $M = H_1 \times \cdots \times H_{n-1}$, and the orders $|H_1|, \ldots, |H_{n-1}|$ are still pairwise coprime. By the induction hypothesis,

$$L \cap M = (L \cap H_1) \times \cdots \times (L \cap H_{n-1}).$$

Substituting this into the preceding expression gives

$$L = (L \cap H_1) \times \cdots \times (L \cap H_n).$$

This proves the result for all n. □

Exercise 2.10. Let $G = N \rtimes H$, and suppose that $N \leq K \leq G$. Show that $K = N \rtimes (K \cap H)$.

Solution. We check the internal semidirect product conditions inside K. Since $G = N \rtimes H$, we have $N \trianglelefteq G$, $G = NH$, and $N \cap H = 1$. Since $N \trianglelefteq G$ and $N \leq K \leq G$, it follows that $N \trianglelefteq K$. Also, since K and H are subgroups of G, their intersection $K \cap H$ is a subgroup of K.
Next,

$$N \cap (K \cap H) = K \cap (N \cap H) = K \cap 1 = 1.$$

It remains to show that $K = N(K \cap H)$. First, since $N \leq K$ and $K \cap H \leq K$, and since K is a subgroup, we have

$$N(K \cap H) \leq K.$$

Conversely, take $k \in K$. Since $G = N \rtimes H$, there are $n \in N$ and $h \in H$ such that

$$k = nh.$$

Since $N \leq K$, we have $n \in K$. Also $k \in K$, so

$$h = n^{-1}k \in K.$$

Since $h \in H$, we have $h \in K \cap H$. Therefore

$$k = nh \in N(K \cap H).$$

Hence $K \leq N(K \cap H)$, and so $K = N(K \cap H)$. Thus K is the internal semidirect product

$$K = N \rtimes (K \cap H).$$

□

Further Exercises

Let N and H be groups. An *extension* of N by H is a group E along with a monomorphism $i : N \to E$ and an epimorphism $\pi : E \to H$ such that $i(N) = \ker \pi$ (so that N imbeds in E as a normal subgroup, with the quotient group being isomorphic with H). We shall usually refer to an extension (E, i, π) simply by the group E; however, the nature of the maps i and π are important in distinguishing between extensions. We identify N with its image under i, and H with the quotient of E by N. As an example, let $\varphi : H \to \mathrm{Aut}(N)$ be a homomorphism; then the semidirect product $N \rtimes_\varphi H$ is an extension of N by H in an obvious way, taking i to be the inclusion map sending $n \in N$ to $(n, 1)$ and π to be the projection map sending (n, h) to h.

Exercise 2.11. We say that an extension E of N by H is a *split extension* if there is a homomorphism $t : H \to E$ (called a *splitting map* for the extension) such that $\pi \circ t$ is the identity map on H, in which case $t(H)$ will be a transversal for N in E. Show that E is a split extension iff it is a semidirect product of N by H.

Solution. ($\Rightarrow$) Suppose that E is a split extension of N by H. Thus there are homomorphisms

$$i \colon N \to E, \qquad \pi \colon E \to H, \qquad t \colon H \to E$$

such that i is injective, π is surjective, $i(N) = \ker \pi$, and $\pi \circ t = \mathrm{id}_H$. Following the convention above, identify N with $i(N)$. Then

$$N = \ker \pi \trianglelefteq E.$$

Also $t(H) \leq E$, since t is a homomorphism. We show that

$$E = N \rtimes t(H).$$

First let $x \in N \cap t(H)$. Since $x \in t(H)$, we may write $x = t(h)$ for some $h \in H$. Since $x \in N = \ker \pi$, we have

$$1 = \pi(x) = \pi(t(h)) = (\pi \circ t)(h) = h.$$

Hence $x = t(1) = 1$, so $N \cap t(H) = 1$.
Next let $e \in E$, and put $h = \pi(e)$. Then

$$\pi(et(h)^{-1}) = \pi(e)\pi(t(h))^{-1} = h(\pi \circ t)(h)^{-1} = hh^{-1} = 1.$$

Therefore $et(h)^{-1} \in \ker \pi = N$. Say $et(h)^{-1} = n$ with $n \in N$. Then

$$e = nt(h) \in Nt(H),$$

so $E \le Nt(H)$. Since $N \le E$ and $t(H) \le E$, the reverse inclusion is clear, and hence $E = Nt(H)$. Thus $E = N \rtimes t(H)$. Finally, $t(H) \cong H$ because $\pi \circ t = \mathrm{id}_H$ forces t to be injective. Therefore E is a semidirect product of N by H.
($\Leftarrow$) Conversely, suppose that E is a semidirect product of N by H. Thus, after identifying N and H with their copies inside E, we have

$$E = N \rtimes H.$$

Hence $N \trianglelefteq E$, $H \le E$, $E = NH$, and $N \cap H = 1$. Every element $e \in E$ has a unique expression $e = nh$ with $n \in N$ and $h \in H$. Define

$$i\colon N \to E, \qquad i(n) = n,$$

and

$$\pi\colon E \to H, \qquad \pi(nh) = h.$$

The map i is the inclusion map, so it is injective. We check that π is a homomorphism. Let $e_1 = n_1h_1$ and $e_2 = n_2h_2$. Since $N \trianglelefteq E$, we have $h_1n_2h_1^{-1} \in N$, and hence

$$e_1e_2 = n_1h_1n_2h_2 = n_1(h_1n_2h_1^{-1})h_1h_2.$$

Therefore the H-part of e_1e_2 is h_1h_2, so

$$\pi(e_1e_2) = h_1h_2 = \pi(e_1)\pi(e_2).$$

Thus π is a homomorphism. It is surjective because $\pi(h) = h$ for all $h \in H$, and

$$\ker \pi = \{nh \in E \mid h = 1\} = N.$$

Hence E is an extension of N by H. Now define

$$t\colon H \to E, \qquad t(h) = h,$$

the inclusion of H into E. Then t is a homomorphism, and for every $h \in H$,

$$(\pi \circ t)(h) = \pi(h) = h.$$

Therefore $\pi \circ t = \mathrm{id}_H$, so the extension splits. □

Exercise 2.12. (cont.) Let Q be the quaternion group of order 8. (We can consider Q as the set $\{\pm1, \pm i, \pm j, \pm k\}$ with multiplication given by the rules $i^2 = j^2 = k^2 = -1$ and $ij = k = -ji$.) Show that Q can be realized as a non-trivial extension in four ways—thrice as an extension of $\mathbb{Z}_4$ by $\mathbb{Z}_2$, and once as an extension of $\mathbb{Z}_2$ by $\mathbb{Z}_2 \times \mathbb{Z}_2$—but that none of these extensions is split. (In other words, Q cannot be written non-trivially as a semidirect product.)

Solution. From Exercise 2.11, an extension of N by H splits exactly when Q can be written as

$$Q = N \rtimes t(H)$$

for some subgroup $t(H) \le Q$ with $t(H) \cong H$ and $N \cap t(H) = 1$. We first exhibit the four extensions. For the extensions of $\mathbb{Z}_4$ by $\mathbb{Z}_2$, take

$$N = \langle i\rangle, \qquad N = \langle j\rangle, \qquad N = \langle k\rangle.$$

Each of these subgroups has order 4, since

$$\langle i\rangle = \{1, i, -1, -i\},$$

and similarly for j and k. Hence each is isomorphic to $\mathbb{Z}_4$. Also each has index 2 in Q, so each is normal in Q. Thus for each of these choices of N,

$$Q/N \cong \mathbb{Z}_2.$$

Therefore Q is an extension of $\mathbb{Z}_4$ by $\mathbb{Z}_2$ in three ways.
For the extension of $\mathbb{Z}_2$ by $\mathbb{Z}_2 \times \mathbb{Z}_2$, take

$$N = Z(Q) = \{1, -1\}.$$

Indeed, -1 commutes with every element of Q, while i, j, k do not commute with one another, for example $ij = k$ but $ji = -k$. The same is true for $-i, -j, -k$, so $Z(Q) = \{1, -1\}$. Thus $N \cong \mathbb{Z}_2$ and $N \trianglelefteq Q$. The quotient Q/N has four cosets:

$$N, \qquad iN, \qquad jN, \qquad kN.$$

Moreover,

$$(iN)^2 = i^2 N = (-1)N = N,$$

and the same argument gives $(jN)^2 = N$ and $(kN)^2 = N$. Hence every non-identity element of Q/N has order 2. Therefore

$$Q/N \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Thus Q is an extension of $\mathbb{Z}_2$ by $\mathbb{Z}_2 \times \mathbb{Z}_2$.
It remains to show that none of these extensions splits. The only element of order 2 in Q is -1: the elements $\pm i, \pm j, \pm k$ all have order 4, and 1 has order 1.
First consider one of the three extensions with

$$N \in \{\langle i\rangle, \langle j\rangle, \langle k\rangle\}.$$

If the extension split, then by Exercise 2.11 there would be a subgroup

$$t(H) \cong \mathbb{Z}_2$$

such that $N \cap t(H) = 1$. But the only subgroup of Q isomorphic to $\mathbb{Z}_2$ is

$$\langle -1\rangle = \{1, -1\}.$$

Since $-1 \in N$ for each of $N = \langle i\rangle, \langle j\rangle, \langle k\rangle$, we would have

$$N \cap t(H) = \{1, -1\} \neq 1,$$

a contradiction. Hence none of the three extensions of $\mathbb{Z}_4$ by $\mathbb{Z}_2$ splits.
Now consider the extension with $N = Z(Q) = \{1, -1\}$. If this extension split, then there would be a subgroup

$$t(H) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$$

of Q. But $\mathbb{Z}_2 \times \mathbb{Z}_2$ has three non-identity elements, all of order 2, while Q has only one element of order 2, namely -1. Thus Q has no subgroup isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. This is again a contradiction. Therefore this extension also does not split.
Hence Q has the four stated non-trivial extensions, but none is split. Equivalently, Q cannot be written non-trivially as a semidirect product. □

If E is an extension of N by H, then we cannot expect to find a homomorphism $t : H \to E$ such that $t(H)$ will be a transversal for N in E, for if such a t existed then E would be split. However, since $H \cong E/N$, we can always find a set map $t : H \to E$ whose image is a transversal for N; such a map is called a *section* of the extension. Moreover, we can always choose t so that $t(1) = 1$, in which case we say that t is *normalized*. (We use normalized sections instead of arbitrary sections in order to keep the notational complexity to a minimum.)

Exercise 2.13. (cont.) Let t be a normalized section of an extension E. Let $\Psi : E \to \mathrm{Aut}(E)$ be the homomorphism sending an element of E to the corresponding inner automorphism of E. We shall, for $x \in E$, regard $\Psi(x)$ as being an automorphism of N, which is possible since $N \trianglelefteq E$. Define set maps $f : H \times H \to N$ and $\varphi : H \to \mathrm{Aut}(N)$ by

$$f(\alpha, \beta) = t(\alpha)t(\beta)t(\alpha\beta)^{-1},$$

$$\varphi(\alpha) = \Psi(t(\alpha)).$$

We call (f, φ) the *factor pair* arising from t. Show that (f, φ) has the following properties:

1 $f(\alpha, 1) = f(1, \alpha) = 1$ for every $\alpha \in H$, and $\varphi(1)$ is the identity in $\mathrm{Aut}(N)$.

2 $\varphi(\alpha)\varphi(\beta) = \Psi(f(\alpha, \beta))\varphi(\alpha\beta)$ for $\alpha, \beta \in H$.

3 $f(\alpha, \beta)f(\alpha\beta, \gamma) = \varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma)$ for $\alpha, \beta, \gamma \in H$.

Solution. Let $\pi\colon E \to H$ be the epimorphism of the extension, and identify N with $\ker\pi$. Since t is a section, we have

$$\pi(t(\alpha)) = \alpha$$

for every $\alpha \in H$. First note that

$$\pi(f(\alpha, \beta)) = \pi(t(\alpha)t(\beta)t(\alpha\beta)^{-1}) = \alpha\beta(\alpha\beta)^{-1} = 1,$$

so $f(\alpha, \beta) \in \ker\pi = N$. Also, since $N \trianglelefteq E$, conjugation by any element of E restricts to an automorphism of N. Thus $\varphi(\alpha) = \Psi(t(\alpha))$ is indeed an element of $\mathrm{Aut}(N)$.

1 Since t is normalized, $t(1) = 1$. Hence

$$f(\alpha, 1) = t(\alpha)t(1)t(\alpha \cdot 1)^{-1} = t(\alpha)t(\alpha)^{-1} = 1,$$

and similarly

$$f(1, \alpha) = t(1)t(\alpha)t(1 \cdot \alpha)^{-1} = t(\alpha)t(\alpha)^{-1} = 1.$$

Moreover,

$$\varphi(1) = \Psi(t(1)) = \Psi(1).$$

For every $n \in N$, we have

$$\Psi(1)(n) = 1n1^{-1} = n,$$

so $\varphi(1) = \mathrm{id}_N$.

2 By the definition of f, we have

$$t(\alpha)t(\beta) = f(\alpha, \beta)t(\alpha\beta).$$

Since Ψ is a homomorphism,

$$\varphi(\alpha)\varphi(\beta) = \Psi(t(\alpha))\Psi(t(\beta)) = \Psi(t(\alpha)t(\beta)).$$

Using the displayed identity, this becomes

$$\Psi(t(\alpha)t(\beta)) = \Psi(f(\alpha, \beta)t(\alpha\beta)) = \Psi(f(\alpha, \beta))\Psi(t(\alpha\beta)).$$

Therefore

$$\varphi(\alpha)\varphi(\beta) = \Psi(f(\alpha, \beta))\varphi(\alpha\beta).$$

3 Again using

$$t(\alpha)t(\beta) = f(\alpha, \beta)t(\alpha\beta),$$

we compute

$$(t(\alpha)t(\beta))t(\gamma) = f(\alpha, \beta)t(\alpha\beta)t(\gamma) = f(\alpha, \beta)f(\alpha\beta, \gamma)t(\alpha\beta\gamma).$$

On the other hand,

$$t(\alpha)(t(\beta)t(\gamma)) = t(\alpha)f(\beta, \gamma)t(\beta\gamma).$$

Since

$$\varphi(\alpha)(n) = t(\alpha)nt(\alpha)^{-1}$$

for every $n \in N$, we have

$$t(\alpha)f(\beta, \gamma) = \varphi(\alpha)(f(\beta, \gamma))t(\alpha).$$

Thus

$$t(\alpha)(t(\beta)t(\gamma)) = \varphi(\alpha)(f(\beta, \gamma))t(\alpha)t(\beta\gamma) = \varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma)t(\alpha\beta\gamma).$$

By associativity in E,

$$(t(\alpha)t(\beta))t(\gamma) = t(\alpha)(t(\beta)t(\gamma)).$$

Comparing the two expressions and cancelling $t(\alpha\beta\gamma)$ on the right, we obtain

$$f(\alpha, \beta)f(\alpha\beta, \gamma) = \varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma).$$

□

Exercise 2.14. (cont.) Just as we were able to externalize the notion of semidirect product, so should we be able to externalize the notion of extension; that is, given groups N and H and appropriate additional data, we should be able to construct an extension of N by H. Using Exercise 2.13 as a guide, formulate such an external construction and prove that it works.

Solution. Suppose that we are given groups N and H, together with set maps

$$f\colon H \times H \to N, \qquad \varphi\colon H \to \mathrm{Aut}(N),$$

satisfying the three identities from Exercise 2.13. Thus, for all $\alpha, \beta, \gamma \in H$,

$$f(\alpha, 1) = f(1, \alpha) = 1, \qquad \varphi(1) = \mathrm{id}_N,$$

$$\varphi(\alpha)\varphi(\beta) = \Psi(f(\alpha, \beta))\varphi(\alpha\beta),$$

and

$$f(\alpha, \beta)f(\alpha\beta, \gamma) = \varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma),$$

where, for $a \in N$, $\Psi(a)$ denotes conjugation by a on N. We construct a group E whose underlying set is

$$E = N \times H.$$

If $(n, \alpha), (m, \beta) \in N \times H$, where $n, m \in N$ and $\alpha, \beta \in H$, define

$$(n, \alpha)(m, \beta) = \big(n\varphi(\alpha)(m)f(\alpha, \beta), \alpha\beta\big).$$

This is a well-defined element of $N \times H$, since $\varphi(\alpha)(m) \in N$, $f(\alpha, \beta) \in N$, and $\alpha\beta \in H$.
We first check that this multiplication makes E into a group. The identity element is $(1, 1)$. Indeed, for $n \in N$ and $\alpha \in H$,

$$(1, 1)(n, \alpha) = \big(\varphi(1)(n)f(1, \alpha), \alpha\big) = (n, \alpha),$$

and

$$(n, \alpha)(1, 1) = \big(n\varphi(\alpha)(1)f(\alpha, 1), \alpha\big) = (n, \alpha).$$

Next we prove associativity. Let

$$(n, \alpha), \quad (m, \beta), \quad (\ell, \gamma) \in N \times H,$$

where $n, m, \ell \in N$ and $\alpha, \beta, \gamma \in H$. First,

$$((n, \alpha)(m, \beta))(\ell, \gamma)$$

has first coordinate

$$n\varphi(\alpha)(m)f(\alpha, \beta)\varphi(\alpha\beta)(\ell)f(\alpha\beta, \gamma),$$

and second coordinate $\alpha\beta\gamma$. On the other hand,

$$(n, \alpha)((m, \beta)(\ell, \gamma))$$

has first coordinate

$$n\varphi(\alpha)\big(m\varphi(\beta)(\ell)f(\beta, \gamma)\big)f(\alpha, \beta\gamma).$$

Since $\varphi(\alpha)$ is an automorphism of N, this is

$$n\varphi(\alpha)(m)\varphi(\alpha)\varphi(\beta)(\ell)\varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma).$$

By the second identity,

$$\varphi(\alpha)\varphi(\beta)(\ell) = f(\alpha, \beta)\varphi(\alpha\beta)(\ell)f(\alpha, \beta)^{-1}.$$

Therefore the first coordinate of $(n, \alpha)((m, \beta)(\ell, \gamma))$ is

$$n\varphi(\alpha)(m)f(\alpha, \beta)\varphi(\alpha\beta)(\ell)f(\alpha, \beta)^{-1}\varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma).$$

By the third identity,

$$f(\alpha, \beta)^{-1}\varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma) = f(\alpha\beta, \gamma),$$

so this becomes
$$n\varphi(\alpha)(m)f(\alpha,\beta)\varphi(\alpha\beta)(\ell)f(\alpha\beta,\gamma).$$
This agrees with the first coordinate of $((n,\alpha)(m,\beta))(\ell,\gamma)$, and the second coordinates are both $\alpha\beta\gamma$. Hence multiplication is associative.
Now we find inverses. For $(n,\alpha)\in N\times H$, put
$$(n,\alpha)^{-1}=\big(f(\alpha^{-1},\alpha)^{-1}\varphi(\alpha^{-1})(n^{-1}),\alpha^{-1}\big).$$
Let
$$p=f(\alpha^{-1},\alpha)^{-1}\varphi(\alpha^{-1})(n^{-1}).$$
Then
$$(p,\alpha^{-1})(n,\alpha)=\big(p\varphi(\alpha^{-1})(n)f(\alpha^{-1},\alpha),1\big)=(1,1).$$
It remains only to check the product in the other order. Taking $(\alpha,\beta,\gamma)=(\alpha,\alpha^{-1},\alpha)$ in the third identity gives
$$f(\alpha,\alpha^{-1})=\varphi(\alpha)(f(\alpha^{-1},\alpha)).$$
Also, taking $(\alpha,\beta)=(\alpha,\alpha^{-1})$ in the second identity gives
$$\varphi(\alpha)\varphi(\alpha^{-1})=\Psi(f(\alpha,\alpha^{-1})).$$
Hence
$$\begin{aligned}(n,\alpha)(p,\alpha^{-1})&=\big(n\varphi(\alpha)(p)f(\alpha,\alpha^{-1}),1\big)\\&=\big(n\varphi(\alpha)(f(\alpha^{-1},\alpha)^{-1})\varphi(\alpha)\varphi(\alpha^{-1})(n^{-1})f(\alpha,\alpha^{-1}),1\big)\\&=\big(nf(\alpha,\alpha^{-1})^{-1}f(\alpha,\alpha^{-1})n^{-1}f(\alpha,\alpha^{-1})^{-1}f(\alpha,\alpha^{-1}),1\big)\\&=(1,1).\end{aligned}$$
Thus every element has an inverse, and so $E=N\times H$ is a group under the multiplication above.
Finally, define
$$i\colon N\to E,\qquad i(n)=(n,1),$$
and
$$\pi\colon E\to H,\qquad \pi(n,\alpha)=\alpha.$$
The map i is injective, and for $n,m\in N$,
$$i(n)i(m)=(n,1)(m,1)=(nm,1)=i(nm),$$
so i is a monomorphism. Also π is a homomorphism, since
$$\pi((n,\alpha)(m,\beta))=\pi\big(n\varphi(\alpha)(m)f(\alpha,\beta),\alpha\beta\big)=\alpha\beta=\pi(n,\alpha)\pi(m,\beta).$$
It is surjective because $\pi(1,\alpha)=\alpha$ for every $\alpha\in H$. Finally,
$$\ker\pi=\{(n,\alpha)\in E\mid\alpha=1\}=\{(n,1)\mid n\in N\}=i(N).$$
Therefore (E,i,π) is an extension of N by H. □

We shall return to these ideas in the further exercises to Section 9.

3 Group Actions

Defining group actions

- A (*left*) *action* of G on a set X is a map $G \times X \to X$, the image of (g, x) denoted gx, satisfying $1x = x$ for every $x \in X$ and $(g_1g_2)x = g_1(g_2x)$ for all $g_1, g_2 \in G$, $x \in X$.
- If G acts on X, we say X is a G-set; for any $H \leq G$, X is also an H-set by restriction.
- For $H \leq G$, the coset space G/H is a G-set under left multiplication: $(g, xH) \mapsto gxH$.

Proposition 3.1. *There is a natural bijective correspondence between the set of actions of G on a set X and the set of homomorphisms from G to S_X.*

Proof. Suppose first that G acts on X. For each $g \in G$, define a function

$$\sigma_g : X \to X, \qquad \sigma_g(x) = gx.$$

We claim that σ_g is a permutation of X. Indeed, $\sigma_{g^{-1}}$ is its inverse, since for every $x \in X$,

$$\sigma_{g^{-1}}(\sigma_g(x)) = g^{-1}(gx) = (g^{-1}g)x = 1x = x$$

and similarly $\sigma_g(\sigma_{g^{-1}}(x)) = x$. Hence $\sigma_g \in S_X$. Now define

$$\Phi : G \to S_X, \qquad \Phi(g) = \sigma_g.$$

For $g_1, g_2 \in G$ and $x \in X$, the action law gives

$$\Phi(g_1g_2)(x) = \sigma_{g_1g_2}(x) = (g_1g_2)x = g_1(g_2x) = \sigma_{g_1}(\sigma_{g_2}(x)).$$

Thus $\Phi(g_1g_2) = \Phi(g_1)\Phi(g_2)$, so Φ is a homomorphism. Conversely, suppose that $\Phi : G \to S_X$ is a homomorphism. Define a rule $G \times X \to X$ by

$$gx = \Phi(g)(x).$$

This rule lands in X because each $\Phi(g)$ is a permutation of X. Also,

$$1x = \Phi(1)(x) = x,$$

since $\Phi(1)$ is the identity permutation of X. For $g_1, g_2 \in G$ and $x \in X$,

$$(g_1g_2)x = \Phi(g_1g_2)(x) = (\Phi(g_1)\Phi(g_2))(x) = \Phi(g_1)(\Phi(g_2)(x)) = g_1(g_2x).$$

Therefore this rule is an action of G on X. Finally, these two constructions undo one another. Starting from an action, the homomorphism sends g to the permutation $x \mapsto gx$, so reconstructing the action gives the same rule. Starting from a homomorphism Φ, the action is defined by $gx = \Phi(g)(x)$, so the permutation attached to g is exactly $\Phi(g)$. Hence the correspondence is bijective. $\square$

Faithful actions and Cayley's Theorem

- If $H < G$ with $|G : H| = n$, then the action of G on G/H gives a non-trivial homomorphism $G \to S_n$ via Proposition 3.1; if G is simple, this homomorphism must be injective.
- The action of G on X is *faithful* (or G acts *faithfully*) if the corresponding homomorphism $G \to S_X$ is injective; equivalently, the only $g \in G$ satisfying $gx = x$ for every $x \in X$ is the identity.
- If G acts faithfully on X, we call G a *permutation group* on X, since G is then imbedded isomorphically in S_X.

Theorem (Cayley's Theorem)**.** *G is isomorphic with a subgroup of S_G; in particular, if G is finite with $|G| = n$, then G is isomorphic with a subgroup of S_n.*

Proof. Let G act on itself by left multiplication. In other words, for $g, x \in G$, define

$$gx$$

to be the product of g and x in the group G. This is an action: the identity satisfies $1x = x$, and associativity in G gives

$$(g_1g_2)x = g_1(g_2x)$$

for all $g_1, g_2, x \in G$. By Proposition 3.1, this action gives a homomorphism

$$\Phi\colon G \to S_G.$$

Explicitly, $\Phi(g)$ is the permutation of G given by left multiplication by g:

$$\Phi(g)(x) = gx.$$

We now show that Φ is injective. Suppose $g \in \ker\Phi$. Then $\Phi(g)$ is the identity permutation of G, so

$$gx = x$$

for every $x \in G$. Taking $x = 1$, we get $g1 = 1$, and hence $g = 1$. Thus $\ker\Phi = 1$, so Φ is injective. Therefore $G \cong \operatorname{im}\Phi$, and $\operatorname{im}\Phi \leq S_G$. Hence G is isomorphic with a subgroup of S_G. Finally, if G is finite with $|G| = n$, then S_G is isomorphic with S_n, since both are symmetric groups on sets with n elements. Thus G is isomorphic with a subgroup of S_n. □

G-set homomorphisms and orbits

- If X, Y are G-sets, $\varphi : X \to Y$ is a *G-set homomorphism* if it commutes with the actions: $\varphi(gx) = g\varphi(x)$ for all $g \in G, x \in X$. A bijective such φ is a *G-set isomorphism*; we then write $X \cong Y$.
- For each $x \in X$, the *orbit* of x is $Gx = \{gx \mid g \in G\} \subseteq X$, and the *stabilizer* of x is $G_x = \{g \in G \mid gx = x\} \leq G$.
- Gx is itself a G-set under the induced action; a subset of X is a G-set under the induced action iff it is a union of orbits.

Lemma 3.2. *If X is a G-set, then $G_{gx} = gG_xg^{-1}$ for any $g \in G$ and $x \in X$.*

Proof. Let $u \in G$. We show that $u \in G_{gx}$ iff $u \in gG_xg^{-1}$. By definition, $u \in G_{gx}$ means that u fixes the element gx, so

$$u \in G_{gx} \iff u(gx) = gx.$$

Using the action law, this is equivalent to

$$(ug)x = gx.$$

Now act on both sides by g^{-1}. Again using the action law, we get

$$(g^{-1}ug)x = x.$$

This means exactly that $g^{-1}ug \in G_x$. Equivalently, $u \in gG_xg^{-1}$. Therefore $G_{gx} = gG_xg^{-1}$. □

Transitive actions and the orbit decomposition

- The action of G on X is *transitive* (or G acts *transitively* on X) if there is some $x \in X$ such that $Gx = X$, equivalently, if for any $x_1, x_2 \in X$ there exists some $g \in G$ such that $gx_1 = x_2$.
- A subset of X is a transitive G-set under the induced action iff it consists of a single orbit.
- For $H \leq G$, the action of G on G/H is transitive: $(yx^{-1})xH = yH$.

Proposition 3.3. *Any G-set has a unique partition consisting of transitive G-sets, namely its partition into orbits.*

Proof. We first show that the orbits form a partition of X. Every $x \in X$ lies in its own orbit Gx, since $x = 1x$. Now let $x, y \in X$, and suppose that $Gx \cap Gy \neq \varnothing$. Then there are $g_1, g_2 \in G$ such that

$$g_1x = g_2y.$$

Hence

$$y = (g_2^{-1}g_1)x,$$

so $y \in Gx$. It follows that $Gy \subseteq Gx$, since for any $u \in G$,

$$uy = u((g_2^{-1}g_1)x) = (ug_2^{-1}g_1)x \in Gx.$$

By symmetry, we also have $Gx \subseteq Gy$. Therefore $Gx = Gy$. Thus any two orbits are either equal or disjoint, so the orbits partition X. Each orbit is a transitive G-set. Indeed, if $y, z \in Gx$, then $y = g_1x$ and $z = g_2x$ for some $g_1, g_2 \in G$, and

$$(g_2g_1^{-1})y = (g_2g_1^{-1})(g_1x) = g_2x = z.$$

Hence any element of the orbit can be moved to any other element of the same orbit. Finally, we prove uniqueness. Suppose X has some partition into transitive G-sets, and let P be one piece of this partition. Choose some $x \in P$. Since P is itself a G-set, it is closed under the action of G. Thus

$$Gx \subseteq P.$$

On the other hand, since P is transitive, every $y \in P$ can be reached from x. That is, for every $y \in P$, there is some $g \in G$ such that $gx = y$. Hence every $y \in P$ lies in Gx, so

$$P \subseteq Gx.$$

Therefore $P = Gx$. Since every transitive piece P is an orbit, the only possible partition into transitive G-sets is the partition into orbits. □

Classification of transitive G-sets

- Since any G-set is a union of transitive G-sets, classifying G-sets reduces to classifying transitive G-sets.
- Coset spaces are examples of transitive G-sets; we now show every transitive G-set is isomorphic to one.

Proposition 3.4. *If X is a transitive G-set, then $X \cong G/G_x$ as G-sets for any $x \in X$.*

Proof. Fix $x \in X$. Define

$$\varphi\colon G/G_x \to X, \qquad \varphi(gG_x) = gx.$$

We first check that φ is well-defined. Suppose $gG_x = hG_x$. Then $h^{-1}g \in G_x$, so $(h^{-1}g)x = x$. Acting by h on both sides gives

$$gx = hx.$$

Hence $\varphi(gG_x) = \varphi(hG_x)$. We next show that φ is injective. Suppose $\varphi(gG_x) = \varphi(hG_x)$. Then $gx = hx$, so acting by h^{-1} on both sides gives

$$(h^{-1}g)x = x.$$

Thus $h^{-1}g \in G_x$, and therefore $gG_x = hG_x$. Hence φ is injective. Since X is transitive, for every $y \in X$ there exists some $g \in G$ such that $gx = y$. Thus $y = \varphi(gG_x)$, so φ is surjective. Finally, we check that φ is a G-set homomorphism. For $u, g \in G$,

$$\varphi(u(gG_x)) = \varphi(ugG_x) = (ug)x = u(gx) = u\varphi(gG_x).$$

Therefore φ is a bijective G-set homomorphism, and hence $X \cong G/G_x$ as G-sets. □

Corollary 3.5 (Orbit-Stabilizer)**.** *Let X be a G-set. Then $Gx \cong G/G_x$ as G-sets for any $x \in X$; in particular, if G is finite, then $|Gx| = |G : G_x|$.*

Proof. The orbit Gx is a transitive G-set under the induced action. The stabilizer of x for this restricted action is still G_x. Hence, by Proposition 3.4,

$$Gx \cong G/G_x$$

as G-sets. If G is finite, then the number of left cosets of G_x in G is $|G : G_x|$. Since Gx is in bijection with G/G_x, we have

$$|Gx| = |G/G_x| = |G : G_x|.$$

$\square$

When are transitive G-sets isomorphic?

- Having decomposed G-sets into transitive ones and identified each with a coset space, the remaining question is when two transitive G-sets are isomorphic.

Lemma 3.6. *Let $\varphi : X \to Y$ be a homomorphism of G-sets, and let $x \in X$. Then $G_x \leq G_{\varphi(x)}$, and if φ is an isomorphism, then $G_x = G_{\varphi(x)}$.*

Proof. Let $g \in G_x$. Then $gx = x$. Since φ is a G-set homomorphism, we have

$$g\varphi(x) = \varphi(gx) = \varphi(x).$$

Thus $g \in G_{\varphi(x)}$, and therefore $G_x \leq G_{\varphi(x)}$. Now suppose that φ is an isomorphism. Then $\varphi^{-1} \colon Y \to X$ is also a G-set homomorphism. Applying the first part to φ^{-1} and the element $\varphi(x) \in Y$, we obtain

$$G_{\varphi(x)} \leq G_{\varphi^{-1}(\varphi(x))} = G_x.$$

Together with the inclusion $G_x \leq G_{\varphi(x)}$, this gives $G_x = G_{\varphi(x)}$. $\square$

Proposition 3.7. *If H and K are subgroups of G, then the G-sets G/H and G/K are isomorphic iff H and K are conjugate in G.*

Proof. The stabilizer of the coset $H \in G/H$ is exactly H. Since every coset of G/H has the form gH for some $g \in G$, Lemma 3.2 shows that the set of stabilizers arising from the G-set G/H is precisely the set of conjugates of H. Similarly, the set of stabilizers arising from G/K is precisely the set of conjugates of K. Suppose first that $\varphi \colon G/H \to G/K$ is a G-set isomorphism. By Lemma 3.6,

$$H = G_H = G_{\varphi(H)}.$$

Now $\varphi(H)$ is some coset $gK \in G/K$, and by Lemma 3.2 its stabilizer is gKg^{-1}. Hence

$$H = G_{\varphi(H)} = G_{gK} = gKg^{-1}.$$

Thus H and K are conjugate in G. Conversely, suppose that $H = gKg^{-1}$ for some $g \in G$. Then H is the stabilizer of the coset $gK \in G/K$. Since G/K is transitive, Proposition 3.4 gives

$$G/K \cong G/H$$

as G-sets. Therefore G/H and G/K are isomorphic as G-sets. $\square$

Doubly transitive actions and maximality

- Let X be a G-set. We say X is *doubly transitive* (and G acts *doubly transitively* on X) if whenever (x_1, x_2) and (y_1, y_2) are elements of $X \times X$ with $x_1 \neq x_2$ and $y_1 \neq y_2$, there exists some $g \in G$ such that $gx_1 = y_1$ and $gx_2 = y_2$.
- The natural action of S_n on $\{1, \ldots, n\}$ for $n \geq 2$ is doubly transitive. Any doubly transitive G-set is transitive.
- A proper subgroup H of G is *maximal* if there is no proper subgroup of G that properly contains H. Any subgroup of prime index is necessarily maximal by Theorem 1.6.

Proposition 3.8. *Let G be a group, let X be a doubly transitive G-set, and let $x \in X$. Then G_x is a maximal subgroup of G.*

Proof. By Proposition 3.4, we have $X \cong G/G_x$ as G-sets. Thus we may work with the doubly transitive G-set G/G_x. Suppose, for a contradiction, that G_x is not maximal. Then there exists a subgroup K such that

$$G_x < K < G.$$

Choose $g \in G$ with $g \notin K$, and choose $k \in K$ with $k \notin G_x$. Since $k \notin G_x$, the cosets G_x and kG_x are distinct. Since $g \notin K$ and $G_x \leq K$, the cosets G_x and gG_x are also distinct. Because G/G_x is doubly transitive, there exists some $u \in G$ such that

$$uG_x = G_x \quad \text{and} \quad u(kG_x) = gG_x.$$

The first equality gives $u \in G_x$, and hence $u \in K$. Since $k \in K$, we have $uk \in K$. The second equality says

$$ukG_x = gG_x,$$

so $g^{-1}uk \in G_x \leq K$. Since both uk and $g^{-1}uk$ lie in K, it follows that

$$g = uk(g^{-1}uk)^{-1} \in K,$$

contradicting our choice of $g \notin K$. Therefore no subgroup lies strictly between G_x and G, and so G_x is maximal. □

Blocks and primitivity

- A non-empty subset B of a transitive G-set X is a *block* if B and $gB = \{gx \mid x \in B\}$ are either equal or disjoint for every $g \in G$.
- X itself is always a block, and any one-element subset of X is always a block; the transitive G-set X is *primitive* if these are the only blocks.

Proposition 3.9. *There is a natural bijective correspondence between the set of blocks of a transitive G-set X which contain a given element x and the set of subgroups of G which contain G_x.*

Proof. Let B be a block of X containing x, and define

$$H_B = \{g \in G \mid gx \in B\}.$$

We first show that H_B is a subgroup of G. Since $x \in B$, we have $1x = x \in B$, so $1 \in H_B$. Now let $g, g' \in H_B$. Then $gx \in B$ and $g'x \in B$. Since both x and gx lie in B, the sets B and gB meet, and therefore $gB = B$ because B is a block. Hence

$$(gg')x = g(g'x) \in gB = B,$$

so $gg' \in H_B$. Similarly, if $g \in H_B$, then $gx \in B$. Since $g^{-1}(gx) = x \in B$, the sets $g^{-1}B$ and B meet, and therefore $g^{-1}B = B$. Thus $g^{-1}x \in B$, so $g^{-1} \in H_B$. Therefore $H_B \leq G$. Also, if $u \in G_x$, then $ux = x \in B$, so $u \in H_B$. Hence $G_x \leq H_B$.
Thus every block B containing x gives a subgroup H_B containing G_x. We now show that this assignment is injective. Suppose B and B' are distinct blocks containing x. Choose $y \in B'$ with $y \notin B$. Since X is transitive, there exists some $g \in G$ such that $gx = y$. Then $g \in H_{B'}$, but $g \notin H_B$. Hence $H_B \neq H_{B'}$.
Conversely, let H be a subgroup of G containing G_x, and define

$$B_H = Hx = \{hx \mid h \in H\}.$$

Clearly $x \in B_H$, so B_H is non-empty. We claim that B_H is a block. If $g \in H$, then $gB_H = B_H$. Now suppose that $gB_H \cap B_H \neq \varnothing$. Then for some $h_1, h_2 \in H$,

$$gh_1x = h_2x.$$

Hence

$$(h_2^{-1}gh_1)x = x,$$

so $h_2^{-1}gh_1 \in G_x \leq H$. Since $h_1, h_2 \in H$, we get

$$g = h_2(h_2^{-1}gh_1)h_1^{-1} \in H.$$

Therefore, whenever gB_H meets B_H, we must have $g \in H$, and hence $gB_H = B_H$. If gB_H does not meet B_H, then it is disjoint from B_H. Thus B_H is a block.
Finally, we check that this construction gives back the subgroup H. By definition,

$$H_{B_H} = \{g \in G \mid gx \in B_H\}.$$

If $g \in H$, then $gx \in Hx = B_H$, so $g \in H_{B_H}$. Thus $H \leq H_{B_H}$. Conversely, if $g \in H_{B_H}$, then $gx = hx$ for some $h \in H$. Therefore

$$(h^{-1}g)x = x,$$

so $h^{-1}g \in G_x \leq H$. Since $h \in H$, we get $g \in H$. Hence $H_{B_H} = H$, proving that every subgroup containing G_x arises from a unique block containing x. □

Corollary 3.10. *Let X be a transitive G-set. Then X is primitive iff G_x is a maximal subgroup of G for every $x \in X$.*

Proof. Suppose first that X is primitive, and fix $x \in X$. By Proposition 3.9, the blocks of X which contain x correspond exactly to the subgroups of G which contain G_x. Since X is primitive, the only blocks containing x are $\{x\}$ and X. These correspond to the subgroups G_x and G, respectively. Hence there is no subgroup H with

$$G_x < H < G.$$

Therefore G_x is maximal in G.
Conversely, suppose that G_x is maximal in G for every $x \in X$. Let B be a block of X, and choose some $x \in B$. Again by Proposition 3.9, the block B corresponds to a subgroup H_B satisfying

$$G_x \leq H_B \leq G.$$

Since G_x is maximal, either $H_B = G_x$ or $H_B = G$. If $H_B = G_x$, then the corresponding block is $\{x\}$; if $H_B = G$, then the corresponding block is X. Thus every block of X is either a one-element subset or all of X. Hence X is primitive. □

Consequences for primitivity

- If X is a transitive G-set, all stabilizers are conjugate by Lemma 3.2; so if G_x is maximal for some x, it is maximal for every x, and Corollary 3.10 can be modified accordingly.

Corollary 3.11. *Any doubly transitive G-set is primitive.*

Proof. This follows from Proposition 3.8 and Corollary 3.10. Indeed, if X is doubly transitive, then Proposition 3.8 shows that G_x is maximal in G for every $x \in X$. Hence, by Corollary 3.10, the G-set X is primitive. □

Counting via group actions

Proposition 3.12. *If G is finite and $H, K \leq G$, then*

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proof. Let $X = G/K$, and consider X as an H-set by restricting the usual left multiplication action of G on G/K. The orbit of the coset $K \in G/K$ under this action is $\{hK \mid h \in H\}$. The union of the cosets in this orbit is exactly

$$\bigcup_{h \in H} hK = HK.$$

Since distinct left cosets of K are disjoint and each has $|K|$ elements, we have

$$|HK| = |\{hK \mid h \in H\}|\,|K|.$$

The stabilizer of K under the action of H is

$$H_K = \{h \in H \mid hK = K\}.$$

But $hK = K$ iff $h \in K$, and we already have $h \in H$. Hence

$$H_K = H \cap K.$$

By Corollary 3.5, this orbit has size

$$|\{hK \mid h \in H\}| = |H : H \cap K| = \frac{|H|}{|H \cap K|}.$$

Therefore

$$|HK| = |\{hK \mid h \in H\}|\,|K| = \frac{|H|}{|H \cap K|}|K| = \frac{|H||K|}{|H \cap K|}.$$

□

Centralizers and conjugacy classes

- For $x \in G$, the *centralizer* of x in G is $C_G(x) = \{g \in G \mid gx = xg\} \leq G$.
- For $S \subseteq G$, $C_G(S) = \{g \in G \mid gx = xg \text{ for all } x \in S\} = \bigcap_{x \in S} C_G(x)$, called the centralizer of S in G.
- $Z(G) = C_G(G)$, and $x \in Z(G)$ iff $C_G(x) = G$.
- The *conjugacy class* of $x \in G$ is $\{gxg^{-1} \mid g \in G\}$. By Proposition 1.10, the conjugacy class of $\rho \in S_n$ consists of all elements of S_n having the same cycle structure as ρ.
- More explicitly, if $\rho \in S_n$ has m_i cycles of length i for each $i \geq 1$, then the size of its conjugacy class in S_n is
$$\frac{n!}{\prod_{i \geq 1} i^{m_i} m_i!}.$$
To see the denominator, first arrange the n symbols in an ordered list and then cut this list into m_i blocks of length i for each i. This gives $n!$ arrangements. However, each i-cycle can be written in i cyclic rotations, and these rotations define the same cycle; this gives a factor of i^{m_i}. Also, the m_i cycles of the same length may be permuted among themselves without changing the resulting disjoint cycle decomposition; this gives a factor of $m_i!$. Dividing by all these repetitions gives the formula [24, p. 47].

Proposition 3.13. *The conjugacy classes of G form a partition of G, and if G is finite then an element $x \in G$ has $|G : C_G(x)|$ conjugates in G.*

Proof. Let G act on itself by conjugation: for $g, x \in G$, define

$$g \cdot x = gxg^{-1}.$$

This is a left action, since $1 \cdot x = x$ and

$$g_1 \cdot (g_2 \cdot x) = g_1(g_2 x g_2^{-1})g_1^{-1} = (g_1 g_2)x(g_1 g_2)^{-1} = (g_1 g_2) \cdot x.$$

The orbit of $x \in G$ under this action is

$$\{gxg^{-1} \mid g \in G\},$$

which is exactly the conjugacy class of x. Hence the first assertion follows from Proposition 3.3, since the orbits of any group action form a partition. If G is finite, then Corollary 3.5 says that the size of the orbit of x is the index of the stabilizer of x. The stabilizer is

$$G_x = \{g \in G \mid gxg^{-1} = x\}.$$

But $gxg^{-1} = x$ iff $gx = xg$, so

$$G_x = C_G(x).$$

Therefore the conjugacy class of x has size $|G : C_G(x)|$. □

Normalizers and conjugates of subgroups

- A subgroup of G is normal iff it is a union of conjugacy classes; such a union is in fact disjoint.
- The order of a normal subgroup of a finite group G must be a sum of orders of conjugacy classes of G.
- For $H \leq G$, the *normalizer* of H in G is $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$. We have $N_G(H) \leq G$ and $H \trianglelefteq N_G(H)$; $N_G(H)$ is the largest subgroup of G in which H is normal, so $N_G(H) = G$ iff $H \trianglelefteq G$.

Proposition 3.14. *A subgroup H of a finite group G has exactly $|G : N_G(H)|$ conjugates in G. In particular, the number of conjugates of H in G divides $|G : H|$ and is equal to 1 iff $H \trianglelefteq G$.*

Proof. Let $\mathcal{P}(G)$ be the set of subsets of G, and let G act on $\mathcal{P}(G)$ by conjugation:

$$g \cdot S = gSg^{-1}$$

for $g \in G$ and $S \subseteq G$. This is a left action, since $1S1^{-1} = S$ and

$$g_1 \cdot (g_2 \cdot S) = g_1(g_2Sg_2^{-1})g_1^{-1} = (g_1g_2)S(g_1g_2)^{-1} = (g_1g_2) \cdot S.$$

The orbit of H under this action is

$$\{gHg^{-1} \mid g \in G\},$$

which is precisely the set of conjugates of H in G. The stabilizer of H is

$$\{g \in G \mid gHg^{-1} = H\} = N_G(H).$$

Therefore, by Corollary 3.5, the number of conjugates of H in G is

$$|G : N_G(H)|.$$

Since $H \leq N_G(H) \leq G$, we have

$$|G : H| = |G : N_G(H)|\,|N_G(H) : H|.$$

Hence the number of conjugates of H divides $|G : H|$. Finally, H has exactly one conjugate iff $|G : N_G(H)| = 1$, iff $N_G(H) = G$. This is equivalent to $gHg^{-1} = H$ for every $g \in G$, which is exactly the condition that $H \trianglelefteq G$. ☐

Double cosets

- Let G be a group and let H, K be subgroups of G. A *double coset* of H and K in G is a set $HxK = \{hxk \mid h \in H, k \in K\}$ for some $x \in G$.
- Any two double cosets are either disjoint or equal, as is the case with ordinary cosets.

Proposition 3.15. *If G is finite and $H, K \leq G$, then for any $x \in G$ we have*

$$|HxK| = \frac{|H||K|}{|H \cap xKx^{-1}|} = \frac{|H||K|}{|x^{-1}Hx \cap K|}.$$

Proof. Consider G/K as an H-set by restricting the usual left multiplication action. The orbit of $xK \in G/K$ under this action is

$$\{hxK \mid h \in H\}.$$

The union in G of the cosets in this orbit is exactly HxK. Since distinct left cosets of K are disjoint and each has $|K|$ elements, we have

$$|HxK| = |\{hxK \mid h \in H\}|\,|K|.$$

The stabilizer of xK under the action of H is

$$\{h \in H \mid hxK = xK\}.$$

Now

$$hxK = xK \iff x^{-1}hx \in K \iff h \in xKx^{-1}.$$

Thus the stabilizer is $H \cap xKx^{-1}$. By Corollary 3.5,

$$|\{hxK \mid h \in H\}| = |H : H \cap xKx^{-1}| = \frac{|H|}{|H \cap xKx^{-1}|}.$$

Therefore

$$|HxK| = \frac{|H|}{|H \cap xKx^{-1}|}|K| = \frac{|H||K|}{|H \cap xKx^{-1}|}.$$

Finally, conjugating the denominator by x^{-1} gives

$$x^{-1}(H \cap xKx^{-1})x = x^{-1}Hx \cap K.$$

Conjugation is a bijection, so these two intersections have the same order. Hence

$$\frac{|H||K|}{|H \cap xKx^{-1}|} = \frac{|H||K|}{|x^{-1}Hx \cap K|}.$$

□

Exercises

Exercise 3.1. Show that a finite simple group whose order is at least $r!$ cannot have a proper subgroup of index r.

Solution. Suppose, for a contradiction, that $H < G$ and $|G : H| = r$. Let G act on the coset space G/H by left multiplication. Since G/H has r elements, Proposition 3.1 gives a homomorphism

$$\Phi \colon G \to S_{G/H} \cong S_r.$$

This is the same permutation-representation idea as in Cayley's Theorem, with the coset space G/H in place of G itself. Now $\ker \Phi \trianglelefteq G$. Since G is simple, either $\ker \Phi = G$ or $\ker \Phi = 1$. If $\ker \Phi = G$, then every $g \in G$ fixes every coset. In particular,

$$gH = H$$

for every $g \in G$. Hence every $g \in G$ lies in H, so $G = H$, contradicting that H is proper. Therefore $\ker \Phi \neq G$, and so $\ker \Phi = 1$. Thus Φ is injective, so G is isomorphic to a subgroup of S_r. Hence

$$|G| \leq |S_r| = r!.$$

By hypothesis $|G| \geq r!$, so $|G| = r!$. Therefore $\Phi(G)$ is a subgroup of S_r with the same order as S_r, and hence $\Phi(G) = S_r$. Thus $G \cong S_r$. For $r \geq 3$, this is impossible because $A_r \trianglelefteq S_r$ is a non-trivial proper normal subgroup. Hence no such subgroup H exists. As stated, the exercise also needs the condition $r \geq 3$: for $r = 2$, the group C_2 is simple, has order $2 = 2!$, and has the proper subgroup 1 of index 2. □

Exercise 3.2. Show that a group G acts doubly transitively on a set X iff G_x acts transitively on $X - \{x\}$ for every $x \in X$. (Here we must assume that X has more than two elements.)

Solution. Suppose first that G acts doubly transitively on X, and fix $x \in X$. We show that G_x acts transitively on $X - \{x\}$. First, G_x preserves $X - \{x\}$. Indeed, if $g \in G_x$ and $y \neq x$, then $gy \neq x$; otherwise $gy = x$, and applying g^{-1} gives $y = g^{-1}x = x$, a contradiction. Now let $y, z \in X - \{x\}$. Then (x, y) and (x, z) are ordered pairs of distinct elements of X. Since G acts doubly transitively on X, there exists $g \in G$ such that

$$gx = x \quad \text{and} \quad gy = z.$$

The first equality says that $g \in G_x$, and the second says that this element of G_x sends y to z. Hence G_x acts transitively on $X - \{x\}$. Conversely, suppose that G_x acts transitively on $X - \{x\}$ for every $x \in X$. Let (x_1, x_2) and (y_1, y_2) be ordered pairs of distinct elements of X. We must find some $g \in G$ such that $gx_1 = y_1$ and $gx_2 = y_2$. We first find an element $a \in G$ such that $ax_1 = y_1$. If $x_1 = y_1$, take $a = 1$. If $x_1 \neq y_1$, choose

$$w \in X - \{x_1, y_1\},$$

which is possible since X has more than two elements. By hypothesis, G_w acts transitively on $X - \{w\}$. Since $x_1, y_1 \in X - \{w\}$, there exists $a \in G_w$ such that

$$ax_1 = y_1.$$

Thus, in either case, we have some $a \in G$ with $ax_1 = y_1$. Since a acts as a bijection on X and $x_2 \neq x_1$, we have $ax_2 \neq ax_1 = y_1$. Hence $ax_2 \in X - \{y_1\}$. Also $y_2 \in X - \{y_1\}$, since $y_2 \neq y_1$. By hypothesis, G_{y_1} acts transitively on $X - \{y_1\}$, so there exists $b \in G_{y_1}$ such that

$$b(ax_2) = y_2.$$

Since $b \in G_{y_1}$, we also have $by_1 = y_1$. Now set $g = ba$. Then

$$gx_1 = b(ax_1) = by_1 = y_1$$

and

$$gx_2 = b(ax_2) = y_2.$$

Therefore G acts doubly transitively on X. □

Exercise 3.3. Show directly from the definitions (that is, without reference to Propositions 3.8 and 3.9) that a doubly transitive G-set is primitive.

Solution. Let B be a block of X. We show that B is either a one-element subset of X or all of X. If B has exactly one element, then B is one of the trivial blocks. Suppose instead that B has at least two elements, and choose distinct elements $x, y \in B$. We will show that $B = X$. Let $z \in X$. If $z = x$, then $z \in B$. So suppose that $z \neq x$. Since $x \neq y$ and $x \neq z$, the ordered pairs

$$(x, y) \quad \text{and} \quad (x, z)$$

are pairs of distinct elements of X. Since G acts doubly transitively on X, there exists $g \in G$ such that

$$gx = x \quad \text{and} \quad gy = z.$$

Now $x \in B$ and $gx = x \in B$, so gB meets B. Since B is a block, this forces $gB = B$. But $y \in B$, so $gy \in gB$. Therefore

$$z = gy \in gB = B.$$

Hence every $z \in X$ lies in B, so $B = X$. Thus every block of X is either a one-element subset or all of X. Therefore X is primitive. □

Exercise 3.4. Let G be the subgroup of S_5 generated by $(1\ 2\ 3\ 4\ 5)$, and let G act on $X = \{1, 2, 3, 4, 5\}$ in the canonical way. Show that this action is primitive, but not doubly transitive.

Solution. Let $\sigma = (1\ 2\ 3\ 4\ 5)$, so $G = \langle \sigma \rangle$ has order 5. First we show that the action is primitive. The action is transitive, since powers of σ move any element of X to any other element of X. Fix $x \in X$. No non-identity power of σ fixes x, since σ^k moves x by k steps around the 5-cycle for $1 \leq k \leq 4$. Hence

$$G_x = 1.$$

Since G has prime order, its only subgroups are 1 and G. Therefore $G_x = 1$ is maximal in G. By Corollary 3.10, the action of G on X is primitive. We now show that the action is not doubly transitive. If it were doubly transitive, then there would exist some $g \in G$ sending the ordered pair $(1, 2)$ to the ordered pair $(1, 3)$; that is,

$$g1 = 1 \quad \text{and} \quad g2 = 3.$$

But the only element of G which fixes 1 is the identity, and the identity sends 2 to 2, not to 3. Thus no such g exists. Therefore the action is not doubly transitive. □

Exercise 3.5. Let $N \trianglelefteq G$, and let $y \in N$. Show that the conjugacy class of y in G is a union of conjugacy classes of the group N. Show further that there is a bijective correspondence between the conjugacy classes of N which comprise the conjugacy class of y in G and the cosets of $NC_G(y)$ in G.

Solution. Let
$$\mathcal{C} = \{gyg^{-1} \mid g \in G\}$$
be the conjugacy class of y in G. Since $N \trianglelefteq G$ and $y \in N$, every element gyg^{-1} lies in N. Hence $\mathcal{C} \subseteq N$. Now let $z \in \mathcal{C}$. By definition of $\mathcal{C}$, there is some $g \in G$ such that
$$z = gyg^{-1}.$$
If $n \in N$, then
$$nzn^{-1} = n(gyg^{-1})n^{-1} = (ng)y(ng)^{-1}.$$
Since $ng \in G$, this is again an element of $\mathcal{C}$. Therefore, whenever $\mathcal{C}$ contains one element z, it contains every N-conjugate of z. Thus $\mathcal{C}$ is a union of conjugacy classes of N. For the second assertion, set $C = C_G(y)$. Since $N \trianglelefteq G$, the product NC is a subgroup of G. We define a map from the left cosets of NC in G to the N-conjugacy classes contained in $\mathcal{C}$ by
$$\Psi\colon G/NC \to \{N\text{-conjugacy classes contained in } \mathcal{C}\}, \qquad \Psi(gNC) = \{n(gyg^{-1})n^{-1} \mid n \in N\}.$$
In words, the coset gNC is sent to the N-conjugacy class of gyg^{-1}. We first check that this is well-defined. Suppose $hNC = gNC$. Then $h = gmc$ for some $m \in N$ and $c \in C$. Since $c \in C_G(y)$, we have $cyc^{-1} = y$. Hence
$$hyh^{-1} = gmcyc^{-1}m^{-1}g^{-1} = gmym^{-1}g^{-1}.$$
But
$$gmym^{-1}g^{-1} = (gmg^{-1})(gyg^{-1})(gmg^{-1})^{-1}.$$
Since $N \trianglelefteq G$, we have $gmg^{-1} \in N$. Therefore hyh^{-1} is N-conjugate to gyg^{-1}. Thus hNC and gNC give the same N-conjugacy class, so Ψ is well-defined. The map Ψ is onto. Indeed, take any N-conjugacy class contained in $\mathcal{C}$. It contains some element gyg^{-1} with $g \in G$, so it is exactly the image of the coset gNC. Finally, we show that Ψ is one-to-one. Suppose
$$\Psi(gNC) = \Psi(hNC).$$
Then gyg^{-1} and hyh^{-1} are N-conjugate, so there exists $n \in N$ such that
$$hyh^{-1} = n(gyg^{-1})n^{-1} = (ng)y(ng)^{-1}.$$
This means that h and ng conjugate y to the same element. Equivalently,
$$((ng)^{-1}h)y((ng)^{-1}h)^{-1} = y.$$
Hence $(ng)^{-1}h \in C_G(y) = C$. So $h = ngc$ for some $c \in C$. Since N is normal in G, we have $g^{-1}ng \in N$, and therefore
$$h = ngc = g(g^{-1}ng)c \in gNC.$$
Thus $hNC = gNC$. Therefore Ψ is injective. Hence the N-conjugacy classes which make up the G-conjugacy class of y are in bijective correspondence with the cosets of $NC_G(y)$ in G. □

Exercise 3.6. Let G be a finite group, and let r be the number of conjugacy classes of G. Show that $|\{(a,b) \in G \times G \mid ab = ba\}| = r|G|$.

Solution. We count the same set in a slower way. Fix $a \in G$. The elements $b \in G$ such that $ab = ba$ are exactly the elements of the centralizer $C_G(a)$. Hence the number of commuting pairs is
$$\sum_{a \in G} |C_G(a)|.$$
We now group this sum by conjugacy classes. Let K be one conjugacy class of G, and choose a representative $x \in K$. If $a \in K$, then $a = gxg^{-1}$ for some $g \in G$. Conjugation by g gives a bijection
$$C_G(x) \to C_G(a), \qquad u \mapsto gug^{-1},$$
so $|C_G(a)| = |C_G(x)|$. Thus, as a runs through the elements of K, the number $|C_G(a)|$ is always the same number $|C_G(x)|$. Therefore the part of the sum coming from this one conjugacy class is
$$\sum_{a \in K} |C_G(a)| = \underbrace{|C_G(x)| + \cdots + |C_G(x)|}_{|K| \text{ times}} = |K|\,|C_G(x)|.$$

By Proposition 3.13,

$$|K| = |G : C_G(x)|.$$

Since G is finite, this means

$$|K|\,|C_G(x)| = |G : C_G(x)|\,|C_G(x)| = |G|.$$

Hence each conjugacy class contributes exactly $|G|$ to the total sum. Since there are r conjugacy classes, we get

$$|\{(a, b) \in G \times G \mid ab = ba\}| = r|G|.$$

□

Exercise 3.7. Show that $C_G(gxg^{-1}) = gC_G(x)g^{-1}$ for any elements g and x of a group G.

Solution. Let G act on itself by conjugation, so that $u \cdot x = uxu^{-1}$ for $u, x \in G$. Under this action, the stabilizer of x is

$$G_x = \{u \in G \mid uxu^{-1} = x\}.$$

But $uxu^{-1} = x$ iff $ux = xu$, so

$$G_x = C_G(x).$$

Also, the image of x under g is

$$g \cdot x = gxg^{-1}.$$

Therefore Lemma 3.2, applied to this conjugation action, gives

$$C_G(gxg^{-1}) = G_{g\cdot x} = gG_xg^{-1} = gC_G(x)g^{-1}.$$

□

Exercise 3.8. Let $n \geq 5$, and assume that A_n is simple. (A proof of this fact is outlined in Exercise 7.8.) Use Exercise 3.1 to show that S_n has no proper subgroup of index less than n other than A_n.

Solution. Let $H < S_n$ be a proper subgroup with

$$|S_n : H| < n.$$

We show that $H = A_n$. Suppose, for a contradiction, that $H \neq A_n$, and set

$$K = H \cap A_n.$$

We first check that K is a proper subgroup of A_n. If $K = A_n$, then $A_n \leq H$. Since $H \neq A_n$, the subgroup H must contain some odd permutation. But then H contains all even permutations and at least one odd permutation. Multiplying that odd permutation by the elements of A_n gives all odd permutations, so $H = S_n$, contradicting that H is proper. Hence

$$K < A_n.$$

We now show that K has index less than n in A_n. There are two cases. First suppose that $H \leq A_n$. Then $K = H$, and

$$|S_n : H| = |S_n : A_n|\,|A_n : H| = 2|A_n : K|.$$

Therefore $|A_n : K| < n$. Now suppose that $H \not\leq A_n$. Then H contains an odd permutation, so

$$HA_n = S_n.$$

By Proposition 3.12,

$$|S_n| = |HA_n| = \frac{|H||A_n|}{|H \cap A_n|} = \frac{|H||A_n|}{|K|}.$$

Dividing by $|H|$ gives

$$|S_n : H| = |A_n : K|.$$

Hence again $|A_n : K| < n$. Thus, in either case, K is a proper subgroup of A_n of index

$$s = |A_n : K| < n.$$

Since $s < n$, we have $s \leq n-1$, and hence

$$s! \leq (n-1)! \leq \frac{n!}{2} = |A_n|$$

because $n \geq 5$. If $s = 2$, then K is a subgroup of index 2 in A_n, hence normal in A_n, contradicting the simplicity of A_n. If $s \geq 3$, then Exercise 3.1 applies to the simple group A_n and rules out a proper subgroup of index s. This contradiction shows that our assumption $H \neq A_n$ was false. Therefore the only proper subgroup of S_n of index less than n is A_n. □

Further Exercises

Let X be a transitive G-set. A *system of imprimitivity* of X is a partition of X which is permuted by the action of G. Note that a system of imprimitivity of a G-set is itself a G-set.

Exercise 3.9. Show that there is a bijective correspondence between the set of blocks which contain a given element of X and the set of systems of imprimitivity of X. (Observe that when X is finite, this implies that any two elements of X lie in the same number of blocks.)

Solution. Fix $x \in X$. We construct maps in both directions. First let B be a block of X with $x \in B$. Define

$$\mathcal{P}_B = \{gB \mid g \in G\}.$$

We show that $\mathcal{P}_B$ is a system of imprimitivity of X. The sets in $\mathcal{P}_B$ are either equal or disjoint. Indeed, suppose

$$gB \cap hB \neq \varnothing.$$

Applying g^{-1} gives

$$B \cap g^{-1}hB \neq \varnothing.$$

Since B is a block, this forces $B = g^{-1}hB$. Multiplying by g, we get $gB = hB$. The sets in $\mathcal{P}_B$ also cover X. Let $y \in X$. Since X is transitive, there exists $g \in G$ such that $gx = y$. Since $x \in B$, we have $y = gx \in gB$. Hence every element of X lies in some member of $\mathcal{P}_B$. Thus $\mathcal{P}_B$ is a partition of X. Moreover, G permutes the pieces, because if $u \in G$, then

$$u(gB) = (ug)B,$$

which is again an element of $\mathcal{P}_B$. Hence $\mathcal{P}_B$ is a system of imprimitivity. Conversely, let $\mathcal{P}$ be a system of imprimitivity of X. Since $\mathcal{P}$ is a partition, there is a unique piece $B \in \mathcal{P}$ such that $x \in B$. We claim that B is a block. Let $g \in G$. Since $\mathcal{P}$ is permuted by G, the set gB is also a piece of $\mathcal{P}$. Since $\mathcal{P}$ is a partition, the two pieces B and gB are either equal or disjoint. Therefore B is a block containing x. These two constructions undo one another. Starting with a block B containing x, the system $\mathcal{P}_B$ contains the piece $B = 1B$. Since $\mathcal{P}_B$ is a partition and $x \in B$, this is the unique piece of $\mathcal{P}_B$ containing x. Thus passing from B to $\mathcal{P}_B$ and then taking the piece containing x returns the original block B. Conversely, start with a system of imprimitivity $\mathcal{P}$, and let B be the unique piece of $\mathcal{P}$ containing x. We claim that the system obtained from this block is the original system:

$$\mathcal{P} = \{gB \mid g \in G\}.$$

Since $\mathcal{P}$ is permuted by G, every set gB is a piece of $\mathcal{P}$. Conversely, let $P \in \mathcal{P}$. Choose $y \in P$. Since X is transitive, there exists $g \in G$ such that $gx = y$. Now $x \in B$, so $y = gx \in gB$. Also gB is a piece of $\mathcal{P}$. Thus P and gB are two pieces of the partition $\mathcal{P}$ containing y, so they must be equal. Hence $P = gB$, and every piece of $\mathcal{P}$ is a translate of B. Therefore the blocks containing x are in bijective correspondence with the systems of imprimitivity of X. If X is finite, then for each chosen element $x \in X$, the blocks containing x are in bijection with the same set of systems of imprimitivity. Hence the number of blocks containing x is independent of x, so any two elements of X lie in the same number of blocks. □

Exercise 3.10. Suppose that the G-set Y is an epimorphic image of a G-set X. Show that there is a G-set isomorphism between Y and some system of imprimitivity of X.

Solution. Since Y is an epimorphic image of X, there is a surjective G-set homomorphism

$$\varphi \colon X \to Y.$$

For each $y \in Y$, let

$$X_y = \varphi^{-1}(y) = \{x \in X \mid \varphi(x) = y\}.$$

Since φ is surjective, each X_y is non-empty. The fibers of φ form a partition of X, so set

$$\mathcal{P} = \{X_y \mid y \in Y\}.$$

We show that $\mathcal{P}$ is a system of imprimitivity. Let $g \in G$ and $y \in Y$. We claim that

$$gX_y = X_{gy}.$$

Indeed, if $x \in X_y$, then $\varphi(x) = y$, and since φ is a G-set homomorphism,

$$\varphi(gx) = g\varphi(x) = gy.$$

Thus $gx \in X_{gy}$, so $gX_y \subseteq X_{gy}$. Conversely, if $z \in X_{gy}$, then $\varphi(z) = gy$. Since $g^{-1}z \in X$, we have

$$\varphi(g^{-1}z) = g^{-1}\varphi(z) = g^{-1}(gy) = y.$$

Hence $g^{-1}z \in X_y$, and therefore $z \in gX_y$. Thus $X_{gy} \subseteq gX_y$, proving $gX_y = X_{gy}$. Therefore the action of G permutes the pieces of $\mathcal{P}$, so $\mathcal{P}$ is a system of imprimitivity of X. Now define

$$\theta\colon Y \to \mathcal{P}, \qquad \theta(y) = X_y.$$

This map is surjective by definition of $\mathcal{P}$. It is injective because distinct elements of Y have disjoint fibers: if $X_y = X_{y'}$, then this common fiber is non-empty, so choose x in it. Then $\varphi(x) = y$ and $\varphi(x) = y'$, hence $y = y'$. Finally, θ is a G-set homomorphism. For $g \in G$ and $y \in Y$,

$$\theta(gy) = X_{gy} = gX_y = g\theta(y).$$

Therefore θ is a G-set isomorphism between Y and the system of imprimitivity $\mathcal{P}$. □

If X is a G-set, we use $[X]$ to denote the isomorphism class of X.

Exercise 3.11. Let G be a finite group, and let $S(G)$ be the set of isomorphism classes of finite G-sets. Show that we can define sum and product operations on $S(G)$ by $[X]+[Y] = [X \cup Y]$ and $[X][Y] = [X \times Y]$.

Solution. We must check that these operations do not depend on the chosen representatives of the isomorphism classes. Suppose $X \cong X'$ and $Y \cong Y'$ as G-sets. Choose G-set isomorphisms

$$\alpha\colon X \to X' \quad \text{and} \quad \beta\colon Y \to Y'.$$

First consider the sum operation. We regard $X \cup Y$ as the disjoint union of X and Y, with G acting on each piece in the usual way. Define

$$\alpha \cup \beta\colon X \cup Y \to X' \cup Y'$$

by

$$(\alpha \cup \beta)(z) = \begin{cases} \alpha(z), & z \in X, \\ \beta(z), & z \in Y. \end{cases}$$

Since α and β are bijections, $\alpha \cup \beta$ is a bijection. It also commutes with the G-action, because if $z \in X$, then

$$(\alpha \cup \beta)(gz) = \alpha(gz) = g\alpha(z) = g(\alpha \cup \beta)(z),$$

and the same calculation using β works when $z \in Y$. Thus $X \cup Y \cong X' \cup Y'$ as G-sets, so $[X]+[Y] = [X \cup Y]$ is well-defined. Now consider the product operation. The product $X \times Y$ is a G-set by the diagonal action

$$g(x, y) = (gx, gy).$$

Define

$$\alpha \times \beta\colon X \times Y \to X' \times Y', \qquad (\alpha \times \beta)(x, y) = (\alpha(x), \beta(y)).$$

Since α and β are bijections, $\alpha \times \beta$ is a bijection. Also,

$$(\alpha \times \beta)(g(x, y)) = (\alpha \times \beta)(gx, gy) = (\alpha(gx), \beta(gy)) = (g\alpha(x), g\beta(y)) = g(\alpha(x), \beta(y)).$$

Therefore $\alpha \times \beta$ is a G-set isomorphism, so $X \times Y \cong X' \times Y'$ as G-sets. Hence $[X][Y] = [X \times Y]$ is well-defined. □

Exercise 3.12. (cont.) Let $B(G)$ be the set obtained from $S(G)$ by adjoining formal additive inverses of isomorphism classes, in the same way that $\mathbb{Z}$ is obtained from $\mathbb{N}$ by adjoining the additive inverses of positive integers. (The additive identity here will be the isomorphism class of the empty set.) Show that the operations defined above on $S(G)$ extend to give a commutative ring structure on $B(G)$. This ring $B(G)$ is called the *Burnside ring* of G.

Solution. By Exercise 3.11, the operations

$$[X] + [Y] = [X \cup Y], \qquad [X][Y] = [X \times Y]$$

are well-defined on isomorphism classes of finite G-sets. The set $B(G)$ is obtained by formally adjoining additive inverses, so every element of $B(G)$ may be represented as a formal difference

$$[X] - [Y],$$

where X and Y are finite G-sets. Addition is then defined by collecting the positive and negative parts:

$$([X] - [Y]) + ([U] - [V]) = [X \cup U] - [Y \cup V].$$

This gives an abelian group under addition. The zero element is $[\varnothing]$, and the additive inverse of $[X]-[Y]$ is $[Y] - [X]$, since

$$([X] - [Y]) + ([Y] - [X]) = [X \cup Y] - [Y \cup X] = [\varnothing]$$

in the formal group obtained from $S(G)$. We extend multiplication by the distributive rule:

$$([X] - [Y])([U] - [V]) = [X \times U] - [X \times V] - [Y \times U] + [Y \times V].$$

This agrees with the original product on $S(G)$ when $Y = V = \varnothing$. The usual bijections

$$X \cup Y \cong Y \cup X, \qquad (X \cup Y) \cup Z \cong X \cup (Y \cup Z)$$

are G-set isomorphisms, so addition is commutative and associative. Similarly,

$$X \times Y \cong Y \times X, \qquad (X \times Y) \times Z \cong X \times (Y \times Z)$$

as G-sets, so multiplication is commutative and associative. The multiplicative identity is the class $[\{*\}]$ of the one-point G-set, where $g* = *$ for every $g \in G$, because

$$X \times \{*\} \cong X$$

as G-sets. Finally, Cartesian product distributes over disjoint union:

$$X \times (Y \cup Z) \cong (X \times Y) \cup (X \times Z)$$

as G-sets. Hence multiplication distributes over addition in $B(G)$. Therefore the operations on $S(G)$ extend to make $B(G)$ a commutative ring. □

Exercise 3.13. (cont.) Show that any element of $B(G)$ can be written uniquely as a $\mathbb{Z}$-linear combination of isomorphism classes of transitive G-sets.

Solution. Let $\mathcal{T}$ be the set of isomorphism classes of transitive finite G-sets. Let X be a finite G-set. By Proposition 3.3, X has a unique decomposition into orbits:

$$X = O_1 \cup \cdots \cup O_r.$$

Each orbit O_i is a transitive G-set, and hence

$$[X] = [O_1] + \cdots + [O_r]$$

in $S(G)$. If several of the orbits are isomorphic as G-sets, we collect their isomorphism classes together. Thus

$$[X] = n_1[T_1] + \cdots + n_k[T_k],$$

where the T_i are pairwise non-isomorphic transitive G-sets and $n_i \in \mathbb{N}$ records the number of orbits of X isomorphic to T_i. This expression is unique in $S(G)$, because the coefficient n_i is exactly the number of orbits of X whose isomorphism class is $[T_i]$. Therefore $S(G)$ is the free commutative monoid on the set $\mathcal{T}$. Now let $\alpha \in B(G)$. By construction of $B(G)$, we may write

$$\alpha = [X] - [Y]$$

for finite G-sets X and Y. Decompose both X and Y into transitive orbits:

$$[X] = n_1[T_1] + \cdots + n_k[T_k], \qquad [Y] = m_1[T_1] + \cdots + m_k[T_k],$$

after enlarging the list of T_i if necessary. Then

$$\alpha = [X] - [Y] = (n_1 - m_1)[T_1] + \cdots + (n_k - m_k)[T_k].$$

Hence every element of $B(G)$ is a $\mathbb{Z}$-linear combination of isomorphism classes of transitive G-sets. Since $S(G)$ is the free commutative monoid on $\mathcal{T}$, adjoining formal additive inverses turns it into the free abelian group on $\mathcal{T}$. Thus an element of $B(G)$ is determined by its integer coefficient attached to each transitive isomorphism class. Therefore no non-trivial $\mathbb{Z}$-linear relation among distinct transitive isomorphism classes is possible, and the expression above is unique. □

Exercise 3.14. (cont.) Let $H \leq G$. Show that there is a unique ring homomorphism from $B(G)$ to $\mathbb{Z}$ which sends an isomorphism class $[X]$ to the number of elements of X fixed by H.

Solution. For a finite G-set X, define

$$X^H = \{x \in X \mid hx = x \text{ for every } h \in H\}.$$

We first check that the rule $[X] \mapsto |X^H|$ is well-defined on isomorphism classes. Suppose that $X \cong Y$ as G-sets, and let

$$f\colon X \to Y$$

be a G-set isomorphism. If $x \in X^H$, then for every $h \in H$,

$$hf(x) = f(hx) = f(x),$$

so $f(x) \in Y^H$. Applying the same argument to f^{-1} shows that f restricts to a bijection $X^H \to Y^H$. Hence $|X^H| = |Y^H|$. Define

$$\varphi_H([X]) = |X^H|$$

on $S(G)$. This map respects addition, because

$$(X \cup Y)^H = X^H \cup Y^H$$

as a disjoint union. Hence

$$\varphi_H([X] + [Y]) = |(X \cup Y)^H| = |X^H| + |Y^H| = \varphi_H([X]) + \varphi_H([Y]).$$

It also respects multiplication, because an element $(x, y) \in X \times Y$ is fixed by every element of H iff both x and y are fixed by every element of H. Thus

$$(X \times Y)^H = X^H \times Y^H,$$

and so

$$\varphi_H([X][Y]) = |(X \times Y)^H| = |X^H|\,|Y^H| = \varphi_H([X])\varphi_H([Y]).$$

The multiplicative identity $[\{*\}]$ is sent to 1, since the one point of $\{*\}$ is fixed by every element of H. Since $B(G)$ is obtained from $S(G)$ by adjoining formal additive inverses, the additive map φ_H extends uniquely to $B(G)$ by

$$\varphi_H([X] - [Y]) = |X^H| - |Y^H|.$$

The multiplication calculation above shows that this extension is multiplicative, so it is a ring homomorphism $B(G) \to \mathbb{Z}$. Finally, this homomorphism is unique. If $\psi\colon B(G) \to \mathbb{Z}$ is any ring homomorphism satisfying $\psi([X]) = |X^H|$ for every finite G-set X, then for every formal difference $[X] - [Y] \in B(G)$,

$$\psi([X] - [Y]) = \psi([X]) - \psi([Y]) = |X^H| - |Y^H|.$$

Thus ψ must agree with φ_H on every element of $B(G)$, proving uniqueness. □

Exercise 3.15. (cont.) Show that any non-zero homomorphism from $B(G)$ to $\mathbb{Z}$ arises from the construction in Exercise 3.14 and that the intersection of the kernels of all such homomorphisms is zero.

Solution. By Proposition 3.4, every transitive finite G-set is isomorphic to G/K for some subgroup $K \leq G$. Here G/K means the set of left cosets of K in G, with G acting by left multiplication; it is not being used as a quotient group. Thus, by Exercise 3.13, the elements $[G/K]$, with K running over representatives of conjugacy classes of subgroups, generate $B(G)$ additively.
Let $\psi\colon B(G) \to \mathbb{Z}$ be a non-zero ring homomorphism. Since $[G/G]$ is the multiplicative identity of $B(G)$, we have

$$\psi([G/G])^2 = \psi([G/G]^2) = \psi([G/G]).$$

The only integers satisfying $n^2 = n$ are 0 and 1. Hence $\psi([G/G])$ is either 0 or 1. If it were 0, then for every $\alpha \in B(G)$,

$$\psi(\alpha) = \psi([G/G]\alpha) = \psi([G/G])\psi(\alpha) = 0,$$

contrary to the assumption that ψ is non-zero. Thus $\psi([G/G]) = 1$. Choose a subgroup $H \leq G$ with $\psi([G/H]) \neq 0$ and with $|H|$ minimal among all such subgroups. We claim that $\psi = \varphi_H$, where φ_H is the fixed-point homomorphism from Exercise 3.14. Let $K \leq G$. We decompose the product $G/H \times G/K$ into G-orbits using Proposition 3.3, where G acts diagonally:

$$u(aH, bK) = (uaH, ubK).$$

Since the first coordinate of an element of $G/H \times G/K$ has the form aH, acting by a^{-1} sends (aH, bK) to

$$(H, a^{-1}bK).$$

Hence every G-orbit has a representative of the form (H, gK). We now compute the stabilizer of this representative. An element $u \in G$ fixes (H, gK) iff

$$u(H, gK) = (H, gK),$$

which is equivalent to the two conditions

$$uH = H \qquad \text{and} \qquad ugK = gK.$$

The first condition says $u \in H$. The second condition says $g^{-1}ug \in K$, or equivalently $u \in gKg^{-1}$. Therefore the stabilizer is

$$H \cap gKg^{-1}.$$

By Proposition 3.4, the orbit of (H, gK) is therefore isomorphic to

$$G/(H \cap gKg^{-1}).$$

Combining the orbit decomposition from Proposition 3.3 with the orbit-stabilizer identification from Proposition 3.4, we get

$$[G/H][G/K] = \sum [G/(H \cap gKg^{-1})],$$

where the sum runs over representatives of the H-orbits on G/K. Applying ψ gives

$$\psi([G/H])\psi([G/K]) = \sum \psi([G/(H \cap gKg^{-1})]).$$

If $H \cap gKg^{-1}$ is a proper subgroup of H, then its order is smaller than $|H|$, so its term is zero by the minimal choice of H. The only remaining terms occur when

$$H \cap gKg^{-1} = H,$$

equivalently when $H \leq gKg^{-1}$. This is exactly the condition that the coset gK is fixed by every element of H. Indeed, gK is fixed by H iff for every $h \in H$,

$$hgK = gK.$$

Multiplying on the left by g^{-1}, this is equivalent to

$$g^{-1}hgK = K,$$

which holds iff $g^{-1}hg \in K$. Thus gK is fixed by every element of H iff $g^{-1}Hg \leq K$, equivalently iff $H \leq gKg^{-1}$. Hence the number of remaining terms is $|(G/K)^H|$, and the equality above becomes

$$\psi([G/H])\psi([G/K]) = |(G/K)^H|\psi([G/H]).$$

Since $\psi([G/H]) \neq 0$, we may cancel it in $\mathbb{Z}$, obtaining

$$\psi([G/K]) = |(G/K)^H| = \varphi_H([G/K]).$$

Since the classes $[G/K]$ generate $B(G)$ additively, this proves that $\psi = \varphi_H$. Thus every non-zero homomorphism $B(G) \to \mathbb{Z}$ arises from the construction in Exercise 3.14.
It remains to prove that the intersection of the kernels is zero. Let

$$\alpha = \sum_K a_K[G/K] \in B(G),$$

where K runs over one representative from each conjugacy class of subgroups of G. Suppose that $\alpha \neq 0$. Choose K with $a_K \neq 0$ and with $|K|$ maximal among all subgroups appearing with non-zero coefficient. We evaluate at the fixed-point homomorphism φ_K. For another subgroup L, the set $(G/L)^K$ is non-empty iff some coset gL is fixed by K, which is equivalent to

$$g^{-1}Kg \leq L.$$

Thus, if $(G/L)^K$ is non-empty, then $|K| \leq |L|$. If also $a_L \neq 0$, maximality of $|K|$ gives $|L| \leq |K|$, and hence $|L| = |K|$. Therefore $g^{-1}Kg = L$, so L is conjugate to K. Since we chose one representative from each conjugacy class, this forces $L = K$. Hence all non-zero coefficient terms except the K-term make no contribution to $\varphi_K(\alpha)$, and

$$\varphi_K(\alpha) = a_K|(G/K)^K|.$$

The coset $K \in G/K$ is fixed by every element of K, so $|(G/K)^K| > 0$. Since $a_K \neq 0$, we have $\varphi_K(\alpha) \neq 0$. Therefore α is not in the kernel of φ_K. Thus no non-zero element of $B(G)$ lies in the kernels of all fixed-point homomorphisms, and so the intersection of all these kernels is zero. □

An element e of a ring R is called an *idempotent* if $e^2 = e$.

Exercise 3.16. (cont.) Show that if G has no self-normalizing proper subgroup, then $B(G)$ has no idempotents other than the additive and multiplicative identities. (A subgroup H of G is called *self-normalizing* if $N_G(H) = H$. We shall see in Section 11 that there is an important class of groups, namely the nilpotent groups, that have no self-normalizing proper subgroups.)

Solution. Let $e \in B(G)$ be an idempotent, so $e^2 = e$. For every subgroup $H \leq G$, let

$$\varphi_H \colon B(G) \to \mathbb{Z}$$

be the fixed-point homomorphism defined by

$$\varphi_H([X]) = |X^H|$$

for every finite G-set X. Since φ_H is a ring homomorphism,

$$\varphi_H(e)^2 = \varphi_H(e^2) = \varphi_H(e).$$

Hence $\varphi_H(e)$ is an integer satisfying $n^2 = n$, so

$$\varphi_H(e) \in \{0, 1\}.$$

We next show that these values do not change when we enlarge a subgroup inside its normalizer. Suppose that

$$H < K \leq N_G(H)$$

and that $|K : H| = p$ is prime. Since $K \leq N_G(H)$, the quotient group K/H acts on X^H for every finite G-set X. The fixed points of this action are exactly X^K. Since K/H has prime order p, Corollary 3.5 shows that each orbit of K/H on X^H has size either 1 or p. The orbits of size 1 are precisely the points of X^K, while every other orbit has size p. Hence, for some integer $r \geq 0$,

$$|X^H| = |X^K| + rp.$$

Therefore
$$|X^H| \equiv |X^K| \pmod p.$$
By additivity, the same congruence holds for every element of $B(G)$; in particular,
$$\varphi_H(e) \equiv \varphi_K(e) \pmod p.$$
Since both $\varphi_H(e)$ and $\varphi_K(e)$ are either 0 or 1, this congruence forces
$$\varphi_H(e) = \varphi_K(e).$$
Now let $H < G$ be a proper subgroup. By hypothesis, H is not self-normalizing, so
$$H < N_G(H).$$
Thus $N_G(H)/H$ is a non-trivial finite group. Choose a subgroup K/H of prime order in $N_G(H)/H$. Then
$$H < K \leq N_G(H),$$
and the previous paragraph gives $\varphi_H(e) = \varphi_K(e)$. If $K < G$, we repeat the same argument with K in place of H. Since G is finite, this process must eventually reach G. Hence
$$\varphi_H(e) = \varphi_G(e)$$
for every subgroup $H \leq G$.
Therefore all fixed-point homomorphisms take the same value on e, and this common value is either 0 or 1. If it is 0, then e lies in the kernel of every φ_H, so Exercise 3.15 gives $e = 0$. If it is 1, then for every $H \leq G$,
$$\varphi_H(e - [G/G]) = \varphi_H(e) - \varphi_H([G/G]) = 1 - 1 = 0,$$
since G/G is the one-point G-set. Again by Exercise 3.15, we get
$$e - [G/G] = 0.$$
Thus $e = [G/G]$, the multiplicative identity of $B(G)$. Therefore the only idempotents in $B(G)$ are 0 and 1. □

Chapter 2

The General Linear Group

4 Basic Structure

Setup and notation

- F a field, $n \in \mathbb{N}$. $\mathcal{M}_n(F)$ denotes the set of all $n \times n$ matrices with entries in F; we write $M = (m_{ij})$ with $m_{ij} \in F$ the (i, j)-entry (row i, column j).
- The *general linear group* $\mathrm{GL}(n, F)$ is the subset of $\mathcal{M}_n(F)$ of invertible matrices, equivalently those with non-zero determinant. It forms a group under matrix multiplication; I denotes the identity.
- More generally, for a finite-dimensional F-vector space V, $\mathrm{GL}(V)$ is the group of invertible linear transformations of V under composition. Taking $V = F^n$ recovers $\mathrm{GL}(n, F)$ up to obvious isomorphism, and any n-dimensional F-vector space is isomorphic to F^n, so we lose nothing by working with $\mathrm{GL}(n, F)$.
- If F is finite, then $|F| = p^a$ for a prime p and $a \in \mathbb{N}$ (Moore's theorem, 1893). For q a prime power we write $\mathrm{GL}(n, q)$ for $\mathrm{GL}(n, F)$ where F is the unique field of order q.

Proposition 4.1. *Let E be a finite abelian group of exponent p, where p is prime. Then* $\mathrm{Aut}(E) \cong \mathrm{GL}(n, p)$*, where $n \in \mathbb{N}$ is such that $|E| = p^n$.*

(Recall that the exponent of a group is the least common multiple of the orders of its elements.)

Proof. Let $K = \mathbb{Z}/p\mathbb{Z}$. We give E the structure of a K-vector space. Since E is abelian, we may define vector addition by

$$x + y = xy$$

for $x, y \in E$, with the identity element of E serving as the zero vector. If $a \in K$, choose an integer representative r of a, and define scalar multiplication by

$$ax = x^r.$$

This is well-defined: if $r \equiv s \pmod p$, then $r - s = mp$ for some $m \in \mathbb{Z}$, and since E has exponent p we have $x^{r-s} = (x^p)^m = e$, so $x^r = x^s$.

The vector-space axioms follow from the group law in E. For instance, if $a \in K$ is represented by r, then

$$a(x + y) = (xy)^r = x^r y^r = ax + ay,$$

where we use that E is abelian. The remaining compatibility laws are checked similarly from the identities $x^{r+s} = x^r x^s$, $x^{rs} = (x^s)^r$, and $x^1 = x$. Thus E is a vector space over K.

Now every group endomorphism of E is automatically K-linear. Indeed, if $\varphi \in \mathrm{End}(E)$ and $a \in K$ is represented by r, then

$$\varphi(x + y) = \varphi(xy) = \varphi(x)\varphi(y) = \varphi(x) + \varphi(y)$$

and

$$\varphi(ax) = \varphi(x^r) = \varphi(x)^r = a\varphi(x).$$

Conversely, every K-linear transformation of E is a homomorphism of the underlying abelian group, because it preserves vector addition, which is precisely the group operation on E. Hence the automorphisms of E as a group are exactly the invertible K-linear transformations of E.
If $\dim_K E = n$, then $|E| = |K|^n = p^n$. Therefore

$$\mathrm{Aut}(E) \cong \mathrm{GL}(E) \cong \mathrm{GL}(n, p),$$

as required. $\square$

Elementary abelian p-groups

- If E is as in Proposition 4.1 with basis $\{x_1, \dots, x_n\}$ over the field of p elements, then as groups $E = \langle x_1 \rangle \times \cdots \times \langle x_n \rangle$ with each $\langle x_i \rangle$ cyclic of order p.
- Hence any finite abelian group of prime exponent p is isomorphic with a direct product of copies of $\mathbb{Z}_p$. Such groups are called *elementary abelian p-groups*.
- The *rank* of an elementary abelian p-group E is n, where $|E| = p^n$.
- In particular $\mathrm{Aut}(\mathbb{Z}_p) \cong \mathrm{GL}(1, p) \cong (\mathbb{Z}/p\mathbb{Z})^\times$ by Proposition 4.1; this was previously established in Proposition 2.1.

Proposition 4.2. *Let $n \in \mathbb{N}$, and let q be a prime power. Then*

$$|\mathrm{GL}(n, q)| = \prod_{k=1}^{n} (q^n - q^{k-1}) = q^{\frac{n(n-1)}{2}} (q^n - 1) \cdots (q - 1).$$

Proof. Let F be the field of order q. An $n \times n$ matrix over F is invertible if and only if its rows form a linearly independent subset of F^n. We therefore count the number of ways to choose the rows of such a matrix.

The first row may be any non-zero vector of F^n, so there are $q^n - 1$ choices. Suppose now that $1 < k \leq n$ and that the first $k - 1$ rows have already been chosen to be linearly independent. Their span is a $(k-1)$-dimensional subspace of F^n, and hence contains q^{k-1} vectors. The kth row must avoid this span, and so there are

$$q^n - q^{k-1}$$

choices for it.

Multiplying these independent choices gives

$$|\mathrm{GL}(n, q)| = \prod_{k=1}^{n} (q^n - q^{k-1}).$$

Finally, since

$$q^n - q^{k-1} = q^{k-1}(q^{n-k+1} - 1),$$

we obtain

$$\prod_{k=1}^{n} (q^n - q^{k-1}) = q^{0+1+\cdots+(n-1)}(q^n - 1)(q^{n-1} - 1) \cdots (q - 1) = q^{\frac{n(n-1)}{2}} (q^n - 1) \cdots (q - 1).$$

$\square$

Fixing notation for the section

- We now fix a field F and some $n \in \mathbb{N}$, and write G instead of $\mathrm{GL}(n, F)$.
- For $M = (m_{ij}) \in \mathcal{M}_n(F)$: the *main diagonal* of M consists of the entries m_{ii} for $1 \leq i \leq n$.
- M is *diagonal* if its only non-zero entries lie on the main diagonal.
- M is *upper triangular* if all entries below the main diagonal are zero, i.e. $m_{ij} = 0$ whenever $i > j$.

Proposition 4.3. *The set B consisting of all invertible upper triangular matrices is a subgroup of G, called the* standard Borel subgroup.

Proof. The identity matrix lies in B. If $M = (m_{ij})$ and $N = (n_{ij})$ are upper triangular, then for $i > j$ the (i, j)-entry of MN is

$$\sum_{k=1}^{n} m_{ik} n_{kj}.$$

If $k < i$, then $m_{ik} = 0$, while if $k \geq i > j$, then $n_{kj} = 0$. Hence every term in the sum is zero, so MN is again upper triangular. Since M and N are invertible, MN is also invertible; hence $MN \in B$.
It remains to show that inverses of elements of B again lie in B. Let $M = (m_{ij}) \in B$, and let $N = (n_{ij}) = M^{-1}$. Since M is upper triangular, its determinant is the product of its diagonal entries. As M is invertible, this determinant is non-zero, and hence each m_{ii} is non-zero.
We show that $n_{ij} = 0$ whenever $i > j$. Since $MN = I$, for each i and j we have

$$\sum_{k=1}^{n} m_{ik} n_{kj} = \delta_{ij}.$$

Fix j. Taking $i = n$ and $j < n$ gives $m_{nn} n_{nj} = 0$, and hence $n_{nj} = 0$. Now argue upward from the bottom row. Suppose that for some i we already know $n_{kj} = 0$ whenever $k > i$ and $j < k$. If $j < i$, then the (i, j)-entry of $MN = I$ gives

$$0 = \sum_{k=1}^{n} m_{ik} n_{kj} = m_{ii} n_{ij} + \sum_{k=i+1}^{n} m_{ik} n_{kj}.$$

The sum on the right is zero by the induction hypothesis, so $m_{ii} n_{ij} = 0$. Since $m_{ii} \neq 0$, it follows that $n_{ij} = 0$. Thus N is upper triangular.
Therefore B contains the identity, is closed under multiplication, and is closed under inverses. Hence $B \leq G$. □

More generally, a *Borel subgroup* of G is any conjugate of the standard Borel subgroup B.

Permutation matrices

- A *permutation matrix* is a matrix in which every row and column has a unique non-zero entry, and all non-zero entries equal 1.
- The identity matrix is a permutation matrix; every permutation matrix is obtained from I by switching columns (or rows).
- Every permutation matrix is orthogonal, so its inverse (= transpose) is again a permutation matrix. Hence all $n \times n$ permutation matrices lie in G.

Proposition 4.4. *The set W consisting of all permutation matrices is a subgroup of G, called the* Weyl subgroup*.*

Proof. We already know that the identity matrix lies in W, and that the inverse of a permutation matrix is again a permutation matrix. It remains to check closure under multiplication. Let $M = (m_{ij})$ and $N = (n_{ij})$ be permutation matrices, and set $P = MN = (p_{ij})$. Then

$$p_{ij} = \sum_{k=1}^{n} m_{ik} n_{kj}.$$

Fix a row i. Since M is a permutation matrix, there is a unique k_i such that $m_{ik_i} = 1$, and all other entries in row i of M are zero. Hence

$$p_{ij} = n_{k_i j}$$

for every j. The k_ith row of N has exactly one non-zero entry, and that entry is 1, so the ith row of P has exactly one non-zero entry, also equal to 1.
Similarly, fix a column j. Since N is a permutation matrix, there is a unique l_j such that $n_{l_j j} = 1$, and all other entries in column j of N are zero. Hence

$$p_{ij} = m_{il_j}$$

for every i. The l_jth column of M has exactly one non-zero entry, and that entry is 1, so the jth column of P has exactly one non-zero entry, also equal to 1. Thus P is a permutation matrix. Therefore W is closed under multiplication, and so $W \leq G$. □

Action on the standard basis

- Let $V_n(F)$ denote the vector space of n-dimensional column vectors with entries in F, and let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be the standard basis.
- Multiplying $\mathbf{v}_i$ on the left by an $n \times n$ matrix M produces the ith column of M; we say that M *sends* $\mathbf{v}_i$ to the ith column of M.

Proposition 4.5. $W \cong S_n$.

Proof. Let $X = \{1, \dots, n\}$. For each $w \in W$ and each $i \in X$, the ith column of w is some standard basis vector $\mathbf{v}_k$. Define
$$\varphi(w)(i) = k$$
where $w\mathbf{v}_i = \mathbf{v}_k$. Since w is a permutation matrix, each standard basis vector appears exactly once among the columns of w. Hence $\varphi(w)$ is a bijection $X \to X$, so $\varphi(w) \in S_n$. This gives a map $\varphi\colon W \to S_n$.
If $w, u \in W$, then for each $i \in X$ we have
$$u\mathbf{v}_i = \mathbf{v}_{\varphi(u)(i)}$$
and therefore
$$wu\mathbf{v}_i = w\mathbf{v}_{\varphi(u)(i)} = \mathbf{v}_{\varphi(w)(\varphi(u)(i))}.$$
Thus $\varphi(wu) = \varphi(w) \circ \varphi(u)$, and φ is a homomorphism.
If $\varphi(w) = \varphi(u)$, then $w\mathbf{v}_i = u\mathbf{v}_i$ for every i, so w and u have the same columns. Hence $w = u$, and φ is injective. Conversely, if $\sigma \in S_n$, let w_σ be the matrix whose ith column is $\mathbf{v}_{\sigma(i)}$ for each i. Then w_σ is a permutation matrix and $\varphi(w_\sigma) = \sigma$, so φ is surjective. Therefore $W \cong S_n$. □

Permutation matrices as permutations

- We often implicitly regard a permutation matrix w as the element of S_n sending i to j, where $\mathbf{v}_j$ is the ith column of w.
- For example, the matrix
$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$
corresponds to $(1\ 3)(2\ 4) \in S_4$, and so if w is this matrix then we may write $w(1) = 3$, and so forth.
- Useful observation: if $w(i) = j$, then for any $M \in \mathcal{M}_n(F)$, the jth row of wM equals the ith row of M, and the ith column of Mw equals the jth column of M.

Transvections

- For distinct $1 \le i, j \le n$ and $\alpha \in F$, define $X_{ij}(\alpha)$ to be the $n \times n$ matrix whose (k, l)-entry equals α if $(k, l) = (i, j)$ and equals δ_{kl} for all other (k, l).
- For example, $X_{23}(\alpha) \in \mathcal{M}_3(F)$ is the matrix
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{pmatrix}.$$
- These matrices $X_{ij}(\alpha)$, and their conjugates by elements of G, are called *transvections*.

Lemma 4.6. *Let $\alpha, \beta \in F$, and let i and j be distinct.*

(i) $X_{ij}(\alpha)$ has determinant 1 and hence lies in G.

(ii) If $\alpha \neq 0$, then $X_{ij}(\alpha) \in B$ iff $i < j$.

(iii) $X_{ij}(\alpha)X_{ij}(\beta) = X_{ij}(\alpha + \beta)$, and hence $X_{ij}(\alpha)^{-1} = X_{ij}(-\alpha)$.

(iv) $[X_{ij}(\alpha), X_{jk}(\beta)] = X_{ik}(\alpha\beta)$ *whenever* i, j, k *are distinct.*

(v) If $w \in W$*, then* $wX_{ij}(\alpha)w^{-1} = X_{w(i)\,w(j)}(\alpha)$*.*

(vi) $X_{ij}(\alpha)$ *sends* $\mathbf{v}_j$ *to* $\mathbf{v}_j + \alpha\mathbf{v}_i$ *and fixes* $\mathbf{v}_k$ *whenever* $k \neq j$*.*

(vii) If $M \in \mathcal{M}_n(F)$*, then the* i*th row of* $X_{ij}(\alpha)M$ *is equal to the sum of the* i*th row of* M *and* α *times the* j*th row of* M*, and for* $k \neq i$ *the* k*th row of* $X_{ij}(\alpha)M$ *is equal to the* k*th row of* M*.*

Proof. For this proof, write E_{rs} for the matrix with 1 in the (r,s)-entry and 0 everywhere else. Thus

$$X_{ij}(\alpha) = I + \alpha E_{ij}.$$

For (i), if $i < j$, then $X_{ij}(\alpha)$ is upper triangular with every diagonal entry equal to 1; if $i > j$, then it is lower triangular with every diagonal entry equal to 1. In either case its determinant is 1, and hence $X_{ij}(\alpha) \in G$.
For (ii), assume $\alpha \neq 0$. If $i < j$, then the only possible non-zero off-diagonal entry of $X_{ij}(\alpha)$ lies above the main diagonal, so $X_{ij}(\alpha) \in B$. If $i > j$, then the (i,j)-entry lies below the main diagonal and is non-zero, so $X_{ij}(\alpha)$ is not upper triangular. Hence $X_{ij}(\alpha) \in B$ iff $i < j$.
For (iii), since $i \neq j$ we have $E_{ij}^2 = 0$. Therefore

$$X_{ij}(\alpha)X_{ij}(\beta) = (I + \alpha E_{ij})(I + \beta E_{ij}) = I + (\alpha + \beta)E_{ij} = X_{ij}(\alpha + \beta).$$

Taking $\beta = -\alpha$ gives $X_{ij}(\alpha)X_{ij}(-\alpha) = I$, and hence $X_{ij}(\alpha)^{-1} = X_{ij}(-\alpha)$.
For (iv), suppose that i, j, k are distinct. Using the convention $[x, y] = xyx^{-1}y^{-1}$ and part (iii), we compute

$$\begin{aligned}
[X_{ij}(\alpha), X_{jk}(\beta)] &= (I + \alpha E_{ij})(I + \beta E_{jk})(I - \alpha E_{ij})(I - \beta E_{jk}) \\
&= (I + \alpha E_{ij} + \beta E_{jk} + \alpha\beta E_{ik})(I - \alpha E_{ij})(I - \beta E_{jk}) \\
&= (I + \beta E_{jk} + \alpha\beta E_{ik})(I - \beta E_{jk}) \\
&= I + \alpha\beta E_{ik} \\
&= X_{ik}(\alpha\beta).
\end{aligned}$$

Here we used $E_{ij}E_{jk} = E_{ik}$, together with the fact that the other products of these matrix units which occur in the calculation vanish because i, j, k are distinct.
For (v), let $w \in W$. Since we are regarding w as the corresponding permutation of $\{1, \ldots, n\}$, we have $w\mathbf{v}_r = \mathbf{v}_{w(r)}$ for every r. If $r = w(j)$, then

$$wX_{ij}(\alpha)w^{-1}\mathbf{v}_r = wX_{ij}(\alpha)\mathbf{v}_j = w(\mathbf{v}_j + \alpha\mathbf{v}_i) = \mathbf{v}_{w(j)} + \alpha\mathbf{v}_{w(i)}.$$

If $r \neq w(j)$, then $w^{-1}(r) \neq j$, so $X_{ij}(\alpha)$ fixes $\mathbf{v}_{w^{-1}(r)}$, and hence $wX_{ij}(\alpha)w^{-1}$ fixes $\mathbf{v}_r$. Thus $wX_{ij}(\alpha)w^{-1}$ sends $\mathbf{v}_{w(j)}$ to $\mathbf{v}_{w(j)} + \alpha\mathbf{v}_{w(i)}$ and fixes all other standard basis vectors. Therefore

$$wX_{ij}(\alpha)w^{-1} = X_{w(i)\,w(j)}(\alpha).$$

For (vi), the jth column of $X_{ij}(\alpha)$ is $\mathbf{v}_j + \alpha\mathbf{v}_i$, while its kth column is $\mathbf{v}_k$ whenever $k \neq j$. Since multiplying $\mathbf{v}_k$ on the left by a matrix gives the kth column of that matrix, the asserted action on the standard basis follows.
For (vii), let $M = (m_{rs}) \in \mathcal{M}_n(F)$, and write $X_{ij}(\alpha)M = (p_{rs})$. Then

$$p_{rs} = \sum_{t=1}^{n} (X_{ij}(\alpha))_{rt} m_{ts}.$$

If $r = i$, this gives $p_{is} = m_{is} + \alpha m_{js}$ for every s, so the ith row of $X_{ij}(\alpha)M$ is the ith row of M plus α times the jth row of M. If $r \neq i$, then the rth row of $X_{ij}(\alpha)$ is the rth row of the identity matrix, so $p_{rs} = m_{rs}$ for every s. Hence every row other than the ith row is unchanged. □

Root subgroups

- For distinct i and j, define $X_{ij} = \{X_{ij}(\alpha) \mid \alpha \in F\}$; by parts (i) and (iii) of Lemma 4.6, this is a subgroup of G.

- The subgroups X_{ij} are called *root subgroups* of G. (Terminology from the theory of Lie algebras.)

- These subgroups feed into the Bruhat Decomposition Theorem, the main result of the section.

Lemma 4.7. *Let $M \in G$. Then there is a product b of upper triangular transvections such that the following property holds: For each $1 \le i \le n$, bM has exactly one row, say the k_ith row, whose entries in the first $i-1$ columns are zero and which has a non-zero entry in the ith column.*

Proof. Let $M = (m_{ij}) \in G$. Since M is invertible, its first column is not the zero column. Choose k_1 maximal such that $m_{k_1,1} \neq 0$; equivalently, $m_{i,1} = 0$ for all $i > k_1$. For example, if $n = 5$ and $k_1 = 3$, then M has the schematic form

$$M = \begin{pmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & * & * & * & * \end{pmatrix},$$

where the symbol $*$ denotes an entry whose value is not relevant to the argument.
We now premultiply by transvections of the form $X_{ik_1}(\alpha)$ with $i < k_1$, choosing $\alpha = -m_{i,1}m_{k_1,1}^{-1}$ in order to clear the $(i,1)$-entry. By Lemma 4.6(vii), this replaces row i by (row i) + α(row k_1); since row k_1 has first entry $m_{k_1,1}$, the new first entry in row i is 0. Each such transvection is upper triangular because $i < k_1$. Thus we obtain a matrix $M' = (m'_{ij})$ whose only non-zero entry in the first column lies in row k_1. In the example above,

$$M' = \begin{pmatrix} 0 & * & * & * & * \\ 0 & * & * & * & * \\ * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & * & * & * & * \end{pmatrix}.$$

All transvections used so far lie in B, and the resulting matrix is still invertible.
Since M' is invertible, its first two columns are linearly independent. Therefore the second column of M' has a non-zero entry in some row other than the k_1th row. Choose $k_2 \neq k_1$ maximal such that $m'_{k_2,2} \neq 0$ and $m'_{i,2} = 0$ for all $i > k_2$ with $i \neq k_1$. We premultiply M' by transvections of the form $X_{ik_2}(\alpha)$, where $i < k_2$ and $i \neq k_1$, so as to clear all entries in the second column except those in rows k_1 and k_2. These transvections are again upper triangular, and they do not disturb the first column because row k_2 has zero first entry.
We continue in this way. Suppose that the rows $k_1, \dots, k_{r-1}$ have already been chosen, and that after the preceding operations the first $r-1$ columns have the required shape. Since the matrix is still invertible, the first r columns are linearly independent, so the rth column must have a non-zero entry in some row not among $k_1, \dots, k_{r-1}$. Choose k_r maximal among such rows. Then premultiply by transvections $X_{ik_r}(\alpha)$ with $i < k_r$ and $i \notin \{k_1, \dots, k_{r-1}\}$ to clear the other entries in the rth column among the rows whose first $r-1$ entries are zero. Because row k_r has zero entries in the first $r-1$ columns, these operations preserve all earlier columns.
After n stages we obtain a matrix bM, where b is a product of upper triangular transvections, with the desired property. For example, if $n = 5$ and

$$(k_1, k_2, k_3, k_4, k_5) = (3, 5, 4, 1, 2),$$

then the final shape is

$$bM = \begin{pmatrix} 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * \\ * & * & * & * & * \\ 0 & 0 & * & * & * \\ 0 & * & * & * & * \end{pmatrix}.$$

Of course, the numbers $k_1, \dots, k_n$ are just $1, \dots, n$ in some order. Thus for each $1 \le i \le n$, the matrix bM has exactly one row, namely row k_i, whose entries in the first $i-1$ columns are zero and whose entry in the ith column is non-zero. □

Theorem (Bruhat Decomposition Theorem)**.** *$G = BWB$.*

Proof. Let $M \in G$. By Lemma 4.7, there is a product b of upper triangular transvections such that, for each $1 \le i \le n$, the matrix bM has exactly one row, say the k_ith row, whose entries in the first $i-1$ columns are zero and whose entry in the ith column is non-zero. Since each upper triangular transvection

lies in B, and since $B \leq G$ by Proposition 4.3, we have $b \in B$.
Let $w \in W$ be the permutation matrix whose k_ith column is $\mathbf{v}_i$ for each i. Equivalently, $w(k_i) = i$. By the row-permutation observation above, the ith row of wbM is equal to the k_ith row of bM. Hence the entries of the ith row of wbM in columns $1, \ldots, i-1$ are all zero. This holds for every i, so wbM is upper triangular. Since w, b, and M are invertible, wbM is also invertible; therefore $wbM \in B$.
It follows that

$$M = b^{-1}w^{-1}(wbM) \in BWB.$$

Thus $G \subseteq BWB$. The reverse inclusion is immediate from $B \leq G$ and $W \leq G$, so $G = BWB$. □

We have now expressed G as a union of the double cosets BwB, as w ranges through W. We will now show that this union is disjoint. Again we first need a lemma.

Lemma 4.8. *If $w_1, w_2 \in W$ and $b \in B$ are such that $w_1 b w_2 \in B$, then $w_2 = w_1^{-1}$.*

Proof. Fix $j \in \{1, \ldots, n\}$. Choose i such that $w_1(i) = j$, and then choose k such that $w_2(k) = i$. With our convention for permutation matrices, $w_1(i) = j$ means that left multiplication by w_1 sends row i to row j. Thus the jth row of $w_1 b$ is the ith row of b. Also, since $w_2(k) = i$, the kth column of $(w_1 b)w_2$ is the ith column of $w_1 b$. Therefore the (j,k)-entry of $w_1 b w_2$ is the (i,i)-entry of b.
Since $b \in B$, its diagonal entries are non-zero, so this (j,k)-entry of $w_1 b w_2$ is non-zero. But $w_1 b w_2 \in B$ is upper triangular, and hence a non-zero (j,k)-entry can occur only when $j \leq k$. By construction,

$$k = w_2^{-1}(i) = w_2^{-1}w_1^{-1}(j).$$

Thus the permutation $\sigma = w_2^{-1}w_1^{-1}$ satisfies $\sigma(j) \geq j$ for every j. Summing these inequalities over all j gives

$$\sum_{j=1}^{n} \sigma(j) \geq \sum_{j=1}^{n} j.$$

The two sums are equal because σ is a permutation of $\{1, \ldots, n\}$, so every inequality $\sigma(j) \geq j$ must in fact be an equality. Hence $\sigma(j) = j$ for every j, and so $\sigma = 1$. Therefore $w_2^{-1}w_1^{-1} = 1$, which is equivalent to $w_2 = w_1^{-1}$. □

Corollary 4.9. *If w and w' are distinct elements of W, then BwB and $Bw'B$ are disjoint; consequently, G is the disjoint union as w runs through W of the $n!$ double cosets BwB.*

Proof. Suppose first that $BwB \cap Bw'B$ is non-empty. Since double cosets are either equal or disjoint, this gives $BwB = Bw'B$. In particular, $w' \in BwB$, so there exist $b, b' \in B$ such that

$$w' = bwb'.$$

Rearranging, we obtain

$$w^{-1}b^{-1}w' = b' \in B.$$

Applying Lemma 4.8 with $w_1 = w^{-1}$, $b = b^{-1}$, and $w_2 = w'$, we get

$$w' = (w^{-1})^{-1} = w.$$

Thus if $w \neq w'$, then BwB and $Bw'B$ are disjoint.
By Bruhat Decomposition Theorem, every element of G lies in BWB, so

$$G = \bigcup_{w \in W} BwB.$$

The disjointness just proved shows that this is a disjoint union. Finally, by Proposition 4.5, we have $W \cong S_n$, and hence $|W| = n!$. Therefore G is the disjoint union of the $n!$ double cosets BwB. □

The following somewhat technical observation, which we shall need in the next section, is an immediate consequence of what we have done.

Corollary 4.10. *Let $M \in G$, and let w be the unique element of W such that $M \in BwB$. Then w sends $\mathbf{v}_i$ to $\mathbf{v}_{k_i}$ for each i, where the numbers k_i are as defined in the statement of Lemma 4.7. In particular, if M sends $\mathbf{v}_1$ to $\alpha_1\mathbf{v}_1 + \cdots + \alpha_k\mathbf{v}_k$ where $\alpha_k \neq 0$, then w sends $\mathbf{v}_1$ to $\mathbf{v}_k$.*

Proof. In the proof of Bruhat Decomposition Theorem, Lemma 4.7 gives row indices $k_1, \dots, k_n$ and an element $b \in B$ such that bM has the prescribed row-echelon form. The permutation matrix used there is the matrix $p \in W$ satisfying

$$p(k_i) = i$$

for every i, or equivalently, the k_ith column of p is $\mathbf{v}_i$. With this choice, left multiplication by p moves row k_i of bM to row i, and hence $pbM \in B$. It follows that

$$M \in Bp^{-1}B.$$

By the uniqueness of the Bruhat double coset from Corollary 4.9, the unique element $w \in W$ such that $M \in BwB$ is $w = p^{-1}$. Therefore

$$w(i) = p^{-1}(i) = k_i,$$

and so $w\mathbf{v}_i = \mathbf{v}_{k_i}$ for each i.
For the final assertion, $M\mathbf{v}_1$ is the first column of M. If

$$M\mathbf{v}_1 = \alpha_1\mathbf{v}_1 + \cdots + \alpha_k\mathbf{v}_k$$

with $\alpha_k \neq 0$ and no later terms, then k is precisely the lowest row in which the first column of M has a non-zero entry. Thus the index selected in Lemma 4.7 for the first column is $k_1 = k$, and the result above gives $w\mathbf{v}_1 = \mathbf{v}_k$. □

We close this section with a result giving a smaller generating set for G than that given by the Bruhat decomposition. Once again, we start with a lemma.

Lemma 4.11. *Let $b \in B$. Then there exists a product t of transvections such that tb is a diagonal matrix having the same main diagonal entries as b.*

Proof. Write $b = (b_{ij})$. Since $b \in B$, it is upper triangular and invertible, so each diagonal entry b_{ii} is non-zero. We shall clear the entries above the main diagonal by premultiplying by transvections, beginning at the far right. This direction is important: operations used to clear column j will add multiples of row j to rows above it, and if we have already treated columns to the right of j, then row j has zero entries in those right-hand columns except at its own diagonal position where appropriate, so the lower-right block is not disturbed.
For example, in the 3×3 case, clearing the $(1,2)$-entry first would give

$$X_{12}(-ad_2^{-1}) \begin{pmatrix} d_1 & a & c \\ 0 & d_2 & e \\ 0 & 0 & d_3 \end{pmatrix} = \begin{pmatrix} d_1 & 0 & c - ad_2^{-1}e \\ 0 & d_2 & e \\ 0 & 0 & d_3 \end{pmatrix},$$

so an operation aimed at column 2 can change an entry in column 3. This is precisely what the right-to-left procedure avoids. By contrast, within one fixed column the relevant transvections may be applied in any order: they add the same pivot row to different rows, and they leave that pivot row unchanged.
Start with column n. For each $i < n$, premultiply by

$$X_{in}(-b_{in}b_{nn}^{-1}).$$

By Lemma 4.6(vii), this adds $-b_{in}b_{nn}^{-1}$ times row n to row i, and therefore clears the (i,n)-entry. These operations do not change any diagonal entry: row n has zeros in columns $1, \dots, n-1$, so adding a multiple of row n to row $i < n$ does not affect the (i,i)-entry, and row n itself is unchanged.
Now move to column $n-1$. Using transvections of the form $X_{i,n-1}(\alpha)$ with $i < n-1$, and choosing α so as to cancel the current $(i, n-1)$-entry, we clear all entries above the $(n-1, n-1)$-entry. These operations do not alter the diagonal entries, and they do not reintroduce entries in column n, because after the previous stage row $n-1$ has zero in column n.
Continuing leftward in this way, at the stage for column j we use transvections $X_{ij}(\alpha)$ with $i < j$ to clear the entries above the (j,j)-entry. The lower-right block formed by columns and rows $j+1, \dots, n$ has already been put in diagonal form, and it is not disturbed because row j has zeros in those columns. The diagonal entries are preserved throughout.
Let t be the product of all the transvections used, in the order in which they are applied. Then tb has every off-diagonal entry equal to zero and has the same main diagonal entries as b. Hence tb is the required diagonal matrix. □

Theorem 4.12. *G is generated by the set consisting of all invertible diagonal matrices and all transvections.*

Proof. Let H be the subgroup of G generated by all invertible diagonal matrices and all transvections. By Bruhat Decomposition Theorem, it is enough to prove that $B \leq H$ and $W \leq H$.
First let $b \in B$. By Lemma 4.11, there is a product t of transvections such that $tb = d$, where d is a diagonal matrix with the same main diagonal entries as b. Since b is invertible, d is an invertible diagonal matrix. Thus $t \in H$ and $d \in H$, so

$$b = t^{-1}d \in H.$$

Hence $B \leq H$.
It remains to show that $W \leq H$. By Proposition 4.5, the group W is generated by the permutation matrices corresponding to transpositions. Let $i \neq j$. Consider

$$X_{ji}(1)X_{ij}(-1)X_{ji}(1).$$

This product only acts non-trivially on the two-dimensional subspace spanned by $\mathbf{v}_i$ and $\mathbf{v}_j$. With respect to the ordered basis $(\mathbf{v}_i, \mathbf{v}_j)$, its action is represented by

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Thus, equivalently by Lemma 4.6(vi), this product sends $\mathbf{v}_i$ to $\mathbf{v}_j$, sends $\mathbf{v}_j$ to $-\mathbf{v}_i$, and fixes every $\mathbf{v}_k$ with $k \notin \{i, j\}$. Now premultiply by the diagonal matrix whose (i,i)-entry is -1 and whose other diagonal entries are 1. The resulting matrix sends $\mathbf{v}_i$ to $\mathbf{v}_j$, sends $\mathbf{v}_j$ to $\mathbf{v}_i$, and fixes all other standard basis vectors, so it is exactly the permutation matrix for the transposition $(i\ j)$. In characteristic 2 this argument still works, since then $-1 = 1$.
Therefore each transposition matrix lies in H. Since every permutation is a product of transpositions, every permutation matrix is a product of such transposition matrices. Hence $W \leq H$.
Now Bruhat Decomposition Theorem gives

$$G = BWB \leq H.$$

The reverse inclusion $H \leq G$ is immediate from the definition of H, so $G = H$. □

Exercises

Exercise 4.1. Show that $\mathrm{GL}(2, 2) \cong S_3$.

Solution. Let $E = \mathbb{Z}_2 \times \mathbb{Z}_2$. Then E is abelian, has exponent 2, and $|E| = 4 = 2^2$. Hence Proposition 4.1 gives

$$\mathrm{Aut}(E) \cong \mathrm{GL}(2, 2).$$

It remains to show that $\mathrm{Aut}(E) \cong S_3$. Write

$$E = \{e, a, b, c\},$$

where e is the identity and a, b, c are the three non-identity elements. Since E has exponent 2, each of a, b, c has order 2. Also, the product of two distinct elements among a, b, c is the third one.
Every automorphism of E fixes e and permutes the set $X = \{a, b, c\}$. Therefore we get a homomorphism

$$\theta\colon \mathrm{Aut}(E) \to S_X \cong S_3, \qquad \theta(\varphi) = \varphi|_X.$$

If $\varphi, \psi \in \mathrm{Aut}(E)$ and $x \in X$, then $\psi(x) \in X$, and so

$$\theta(\varphi \circ \psi)(x) = (\varphi \circ \psi)(x) = \varphi(\psi(x)) = \theta(\varphi)(\theta(\psi)(x)).$$

Hence $\theta(\varphi \circ \psi) = \theta(\varphi) \circ \theta(\psi)$, so θ is a homomorphism.
If $\theta(\varphi)$ is the identity permutation of X, then φ fixes a, b, c and also fixes e, so φ is the identity automorphism of E. Hence θ is injective.
By Proposition 4.2, we have

$$|\mathrm{GL}(2, 2)| = (2^2 - 1)(2^2 - 2) = 6.$$

Therefore $|\operatorname{Aut}(E)| = 6$. Also $|S_X| = 3! = 6$. Since θ is an injective homomorphism between finite groups of the same order, θ is an isomorphism. Thus

$$\operatorname{Aut}(E) \cong S_X \cong S_3.$$

Combining this with $\operatorname{Aut}(E) \cong \mathrm{GL}(2,2)$ gives $\mathrm{GL}(2,2) \cong S_3$. $\square$

Exercise 4.2. (cont.) Construct a monomorphism $\varphi\colon \mathrm{GL}(1,4) \to \mathrm{GL}(2,2)$ that corresponds to the inclusion of A_3 in S_3. (Recall that the field $\mathbb{F}_4$ of 4 elements can be written as $\{0, 1, \alpha, \alpha^2\}$, where $\alpha + 1 = \alpha^2$ and $\lambda + \lambda = 0$ for all $\lambda \in \mathbb{F}_4$.) Show further that the extension of φ to a map from $\mathbb{F}_4$ to $\mathcal{M}_2(\mathbb{F}_2)$, where $\mathbb{F}_2 = \{0, 1\}$ is the field of 2 elements, is a monomorphism of rings.

Solution. We regard $\mathbb{F}_4$ as a 2-dimensional vector space over $\mathbb{F}_2$ with ordered basis

$$\mathcal{B} = (1, \alpha).$$

For each $\lambda \in \mathbb{F}_4$, let

$$m_\lambda\colon \mathbb{F}_4 \to \mathbb{F}_4, \qquad m_\lambda(x) = \lambda x$$

be multiplication by λ. We define

$$\widetilde{\varphi}\colon \mathbb{F}_4 \to \mathcal{M}_2(\mathbb{F}_2), \qquad \widetilde{\varphi}(\lambda) = [m_\lambda]_{\mathcal{B}}^{\mathcal{B}}.$$

Thus m_λ is the linear map, and $\widetilde{\varphi}(\lambda)$ is the matrix of that map in the basis $\mathcal{B}$.
We compute this map on the elements of $\mathbb{F}_4$. Multiplication by 0 is the zero map, so

$$\widetilde{\varphi}(0) = [m_0]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Multiplication by 1 is the identity map, so

$$\widetilde{\varphi}(1) = [m_1]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Next, since $\alpha \cdot 1 = \alpha$ and $\alpha \cdot \alpha = \alpha^2 = 1 + \alpha$, the first column of $\widetilde{\varphi}(\alpha)$ is the coordinate column of α, and the second column is the coordinate column of $1 + \alpha$. Hence

$$\widetilde{\varphi}(\alpha) = [m_\alpha]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Finally, since $\alpha^2 \cdot 1 = \alpha^2 = 1 + \alpha$ and $\alpha^2 \cdot \alpha = \alpha^3 = 1$, we get

$$\widetilde{\varphi}(\alpha^2) = [m_{\alpha^2}]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Restricting $\widetilde{\varphi}$ to the non-zero elements $\mathbb{F}_4^\times = \{1, \alpha, \alpha^2\} = \mathrm{GL}(1,4)$ gives

$$\varphi\colon \mathrm{GL}(1,4) \to \mathrm{GL}(2,2), \qquad \varphi(\lambda) = [m_\lambda]_{\mathcal{B}}^{\mathcal{B}}.$$

This is a group homomorphism because, for $\lambda, \mu \in \mathbb{F}_4^\times$,

$$m_{\lambda\mu} = m_\lambda \circ m_\mu,$$

and taking matrices in the basis $\mathcal{B}$ gives

$$\varphi(\lambda\mu) = [m_{\lambda\mu}]_{\mathcal{B}}^{\mathcal{B}} = [m_\lambda]_{\mathcal{B}}^{\mathcal{B}}[m_\mu]_{\mathcal{B}}^{\mathcal{B}} = \varphi(\lambda)\varphi(\mu).$$

It is injective because if $\varphi(\lambda) = \varphi(\mu)$, then $[m_\lambda]_{\mathcal{B}}^{\mathcal{B}} = [m_\mu]_{\mathcal{B}}^{\mathcal{B}}$, so $m_\lambda = m_\mu$. Applying both maps to 1 gives $\lambda = \mu$. Therefore φ is a monomorphism.
It remains to connect this with the inclusion $A_3 \leq S_3$. Since φ is injective and $\mathrm{GL}(1,4) = \mathbb{F}_4^\times$ has order 3, the subgroup $\operatorname{im}\varphi$ has order 3. Under the isomorphism $\mathrm{GL}(2,2) \cong S_3$ from Exercise 4.1, this image corresponds to a subgroup of S_3 of order 3. The only such subgroup is

$$\{e, (1\ 2\ 3), (1\ 3\ 2)\} = A_3.$$

Thus φ corresponds to the usual inclusion $A_3 \leq S_3$. Finally, the map $\widetilde{\varphi}\colon \mathbb{F}_4 \to \mathcal{M}_2(\mathbb{F}_2)$ is a ring homomorphism. Indeed, for $\lambda, \mu, x \in \mathbb{F}_4$,

$$m_{\lambda+\mu}(x) = (\lambda+\mu)x = \lambda x + \mu x = (m_\lambda + m_\mu)(x),$$

so $m_{\lambda+\mu} = m_\lambda + m_\mu$. Taking matrices in the basis $\mathcal{B}$ gives

$$\widetilde{\varphi}(\lambda+\mu) = [m_{\lambda+\mu}]_{\mathcal{B}}^{\mathcal{B}} = [m_\lambda]_{\mathcal{B}}^{\mathcal{B}} + [m_\mu]_{\mathcal{B}}^{\mathcal{B}} = \widetilde{\varphi}(\lambda) + \widetilde{\varphi}(\mu).$$

Similarly,

$$m_{\lambda\mu} = m_\lambda \circ m_\mu,$$

so

$$\widetilde{\varphi}(\lambda\mu) = [m_{\lambda\mu}]_{\mathcal{B}}^{\mathcal{B}} = [m_\lambda]_{\mathcal{B}}^{\mathcal{B}}[m_\mu]_{\mathcal{B}}^{\mathcal{B}} = \widetilde{\varphi}(\lambda)\widetilde{\varphi}(\mu),$$

and $\widetilde{\varphi}(1) = I$. If $\widetilde{\varphi}(\lambda) = \widetilde{\varphi}(\mu)$, then $[m_\lambda]_{\mathcal{B}}^{\mathcal{B}} = [m_\mu]_{\mathcal{B}}^{\mathcal{B}}$, so $m_\lambda = m_\mu$. Applying both maps to 1 gives $\lambda = \mu$. Hence $\widetilde{\varphi}$ is injective, and so it is a monomorphism of rings. □

Exercise 4.3. (cont.) Construct an explicit monomorphism from $\mathrm{GL}(n,4)$ to $\mathrm{GL}(2n,2)$ for any $n \in \mathbb{N}$.

Solution. Let $\mathcal{B} = (1, \alpha)$ be the ordered $\mathbb{F}_2$-basis of $\mathbb{F}_4$ used in Exercise 4.2. Write

$$\rho(\lambda) = [m_\lambda]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_2(\mathbb{F}_2)$$

for the matrix of multiplication by λ on $\mathbb{F}_4$. By Exercise 4.2, the map $\rho\colon \mathbb{F}_4 \to \mathcal{M}_2(\mathbb{F}_2)$ is an injective ring homomorphism.
Now let $M = (\lambda_{ij}) \in \mathrm{GL}(n,4)$. Define $\Phi(M)$ to be the $2n \times 2n$ block matrix over $\mathbb{F}_2$ whose (i,j)-block is $\rho(\lambda_{ij})$:

$$\Phi(M) = \begin{pmatrix} \rho(\lambda_{11}) & \rho(\lambda_{12}) & \cdots & \rho(\lambda_{1n}) \\ \rho(\lambda_{21}) & \rho(\lambda_{22}) & \cdots & \rho(\lambda_{2n}) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(\lambda_{n1}) & \rho(\lambda_{n2}) & \cdots & \rho(\lambda_{nn}) \end{pmatrix}.$$

This is the same as replacing each entry of M by its 2×2 multiplication matrix $[m_{\lambda_{ij}}]_{\mathcal{B}}^{\mathcal{B}}$.
We claim that

$$\Phi\colon \mathrm{GL}(n,4) \to \mathrm{GL}(2n,2)$$

is a monomorphism. First, if $M = (\lambda_{ij})$ and $N = (\mu_{ij})$ are in $\mathrm{GL}(n,4)$, then the (i,j)-block of $\Phi(M)\Phi(N)$ is

$$\sum_{k=1}^{n} \rho(\lambda_{ik})\rho(\mu_{kj}).$$

Since ρ is a ring homomorphism, this equals

$$\sum_{k=1}^{n} \rho(\lambda_{ik}\mu_{kj}) = \rho\left(\sum_{k=1}^{n} \lambda_{ik}\mu_{kj}\right),$$

which is exactly the (i,j)-block of $\Phi(MN)$. Hence $\Phi(MN) = \Phi(M)\Phi(N)$.
If $M \in \mathrm{GL}(n,4)$, let M^{-1} be its inverse over $\mathbb{F}_4$. Then

$$\Phi(M)\Phi(M^{-1}) = \Phi(I_n) = I_{2n},$$

so $\Phi(M)$ is invertible. Hence $\Phi(M) \in \mathrm{GL}(2n,2)$.
Finally, suppose $\Phi(M) = \Phi(N)$. Then every corresponding 2×2 block is equal, so $\rho(\lambda_{ij}) = \rho(\mu_{ij})$ for all i, j. Since ρ is injective, $\lambda_{ij} = \mu_{ij}$ for all i, j, and therefore $M = N$. Thus Φ is injective.
Therefore Φ is an explicit monomorphism from $\mathrm{GL}(n,4)$ to $\mathrm{GL}(2n,2)$. □

Exercise 4.4. (cont.) More generally, show for any $n \in \mathbb{N}$ and any prime power q that $\mathrm{GL}(2n,q)$ has a subgroup that is isomorphic with $\mathrm{GL}(n,q^2)$. Will $\mathrm{GL}(n,q^m)$ always have a subgroup isomorphic with $\mathrm{GL}(mn,q)$ for any $m, n \in \mathbb{N}$ and any prime power q?

Solution. Let $\mathbb{F}_{q^2}$ be the field with q^2 elements. Since $\mathbb{F}_{q^2}$ is a 2-dimensional vector space over $\mathbb{F}_q$, choose an ordered $\mathbb{F}_q$-basis

$$\mathcal{B} = (\beta_1, \beta_2)$$

of $\mathbb{F}_{q^2}$. For each $\lambda \in \mathbb{F}_{q^2}$, let

$$m_\lambda \colon \mathbb{F}_{q^2} \to \mathbb{F}_{q^2}, \qquad m_\lambda(x) = \lambda x,$$

and define

$$\rho(\lambda) = [m_\lambda]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_2(\mathbb{F}_q).$$

This is an injective ring homomorphism. Indeed, for $\lambda, \mu \in \mathbb{F}_{q^2}$ and $x \in \mathbb{F}_{q^2}$, we have

$$m_{\lambda+\mu}(x) = (\lambda + \mu)x = \lambda x + \mu x = (m_\lambda + m_\mu)(x),$$

so $\rho(\lambda + \mu) = \rho(\lambda) + \rho(\mu)$. Also

$$m_{\lambda\mu}(x) = (\lambda\mu)x = \lambda(\mu x) = (m_\lambda \circ m_\mu)(x),$$

so $\rho(\lambda\mu) = \rho(\lambda)\rho(\mu)$. Finally, if $\rho(\lambda) = \rho(\mu)$, then $m_\lambda = m_\mu$, so evaluating both maps at 1 gives $\lambda = \mu$. Thus ρ is injective.
Now let $M = (\lambda_{ij}) \in \mathrm{GL}(n, q^2)$. Define $\Phi(M)$ to be the $2n \times 2n$ block matrix over $\mathbb{F}_q$ whose (i,j)-block is $\rho(\lambda_{ij})$:

$$\Phi(M) = \begin{pmatrix} \rho(\lambda_{11}) & \rho(\lambda_{12}) & \cdots & \rho(\lambda_{1n}) \\ \rho(\lambda_{21}) & \rho(\lambda_{22}) & \cdots & \rho(\lambda_{2n}) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(\lambda_{n1}) & \rho(\lambda_{n2}) & \cdots & \rho(\lambda_{nn}) \end{pmatrix}.$$

Let $N = (\mu_{ij}) \in \mathrm{GL}(n, q^2)$. The (i,j)-block of $\Phi(M)\Phi(N)$ is

$$\sum_{k=1}^{n} \rho(\lambda_{ik})\rho(\mu_{kj}) = \sum_{k=1}^{n} \rho(\lambda_{ik}\mu_{kj}) = \rho\left(\sum_{k=1}^{n} \lambda_{ik}\mu_{kj}\right),$$

which is exactly the (i,j)-block of $\Phi(MN)$. Hence

$$\Phi(MN) = \Phi(M)\Phi(N)$$

for $M, N \in \mathrm{GL}(n, q^2)$. If $M \in \mathrm{GL}(n, q^2)$, then

$$\Phi(M)\Phi(M^{-1}) = \Phi(I_n) = I_{2n},$$

so $\Phi(M) \in \mathrm{GL}(2n, q)$. Also, if $\Phi(M) = \Phi(N)$, then $\rho(\lambda_{ij}) = \rho(\mu_{ij})$ for all i, j, and since ρ is injective, $\lambda_{ij} = \mu_{ij}$ for all i, j. Hence $M = N$. Therefore Φ is an injective homomorphism

$$\Phi \colon \mathrm{GL}(n, q^2) \to \mathrm{GL}(2n, q).$$

Thus $\mathrm{GL}(2n, q)$ has a subgroup isomorphic with $\mathrm{GL}(n, q^2)$.
The proposed reverse-looking statement is not always true. Take $n = 1$, $m = 2$, and $q = 2$. Then it would say that $\mathrm{GL}(1, 4)$ has a subgroup isomorphic with $\mathrm{GL}(2, 2)$. But

$$|\mathrm{GL}(1, 4)| = |\mathbb{F}_4^\times| = 3,$$

whereas by Proposition 4.2,

$$|\mathrm{GL}(2, 2)| = (2^2 - 1)(2^2 - 2) = 6.$$

A group of order 3 cannot have a subgroup of order 6. Therefore $\mathrm{GL}(n, q^m)$ does not always have a subgroup isomorphic with $\mathrm{GL}(mn, q)$. □

Exercise 4.5. Show that $\mathrm{GL}(4, 2) \cong A_8$.

Solution. Let $G = \mathrm{GL}(4, 2)$. We first record the plan. By Proposition 4.2,

$$|G| = (2^4 - 1)(2^4 - 2)(2^4 - 2^2)(2^4 - 2^3) = 15 \cdot 14 \cdot 12 \cdot 8 = 20160.$$

Also

$$|A_8| = \frac{8!}{2} = 20160.$$

Thus it is enough to construct an embedding

$$G \hookrightarrow S_8,$$

because the image will then have order $8!/2$, hence will be an index 2 subgroup of S_8. The unique index 2 subgroup of S_8 is A_8.

The hard part is finding the eight objects on which G should act. We shall find them by first constructing a subgroup $H \leq G$ with $H \cong A_7$; then the eight objects will be the left cosets of H in G.

Step 1: Constructing fifteen Fano planes.

Let $\Omega = \{1, \dots, 7\}$. A *Fano plane structure* on Ω is a set of seven 3-element subsets of Ω, called lines, such that any two points of Ω lie in exactly one line. For example, take

$$\mathcal{P}_0 = \{\{1,2,3\}, \{1,4,5\}, \{1,6,7\}, \{2,4,6\}, \{2,5,7\}, \{3,4,7\}, \{3,5,6\}\}.$$

We may identify the points $1, \dots, 7$ with the seven non-zero vectors of $\mathbb{F}_2^3$ by

point	1	2	3	4	5	6	7
vector	e_1	e_2	e_1+e_2	e_3	e_1+e_3	e_2+e_3	$e_1+e_2+e_3$.

With this identification, the lines of $\mathcal{P}_0$ are exactly the triples

$$\{u, v, u+v\}$$

where u and v are distinct non-zero vectors. Therefore every invertible linear map of $\mathbb{F}_2^3$ permutes the seven points and preserves the lines, giving

$$\mathrm{GL}(3,2) \leq \mathrm{Aut}(\mathcal{P}_0).$$

Conversely, every automorphism of $\mathcal{P}_0$ is determined by the images of three non-collinear points, for instance $1, 2, 4$. These images are again non-collinear, hence form a basis of $\mathbb{F}_2^3$, and so determine a unique invertible linear map. Thus

$$\mathrm{Aut}(\mathcal{P}_0) \cong \mathrm{GL}(3,2).$$

By Proposition 4.2,

$$|\mathrm{Aut}(\mathcal{P}_0)| = |\mathrm{GL}(3,2)| = (2^3-1)(2^3-2)(2^3-2^2) = 168.$$

Moreover $\mathrm{Aut}(\mathcal{P}_0) \leq A_7$. To see this, note that by Theorem 4.12, $\mathrm{GL}(3,2)$ is generated by transvections, since the only invertible diagonal matrix over $\mathbb{F}_2$ is the identity. A transvection of $\mathbb{F}_2^3$ fixes three non-zero vectors and interchanges the remaining four non-zero vectors in two pairs. Hence it is an even permutation of the seven non-zero vectors. So every element of $\mathrm{Aut}(\mathcal{P}_0)$ is an even permutation.

Let

$$\mathcal{P} = A_7 \mathcal{P}_0$$

be the orbit of $\mathcal{P}_0$ under the natural action of A_7 on Ω. Since the stabilizer of $\mathcal{P}_0$ in A_7 is $\mathrm{Aut}(\mathcal{P}_0)$, we have

$$|\mathcal{P}| = \frac{|A_7|}{|\mathrm{Aut}(\mathcal{P}_0)|} = \frac{7!/2}{168} = 15.$$

So $\mathcal{P}$ is a set of fifteen Fano plane structures on the same seven-point set Ω.

We shall need one elementary fact about these fifteen Fano planes: any two distinct elements of $\mathcal{P}$ have exactly one common line. It is enough to check this with one of the two planes equal to $\mathcal{P}_0$. Let $\ell = \{1,2,3\}$ and let $\tau = (1\ 2\ 3)$. Then $\tau\mathcal{P}_0$ and $\tau^2\mathcal{P}_0$ are the two Fano planes

$$\tau\mathcal{P}_0 = \{\{1,2,3\}, \{2,4,5\}, \{2,6,7\}, \{3,4,6\}, \{3,5,7\}, \{1,4,7\}, \{1,5,6\}\},$$
$$\tau^2\mathcal{P}_0 = \{\{1,2,3\}, \{3,4,5\}, \{3,6,7\}, \{1,4,6\}, \{1,5,7\}, \{2,4,7\}, \{2,5,6\}\}.$$

Each has exactly the line ℓ in common with $\mathcal{P}_0$. Since $\mathrm{Aut}(\mathcal{P}_0)$ is transitive on the seven lines of $\mathcal{P}_0$, the same statement holds for every line of $\mathcal{P}_0$. Thus the two rotations of each of the seven lines give fourteen distinct elements of $\mathcal{P}$ different from $\mathcal{P}_0$. Since $|\mathcal{P}| = 15$, these are all the other elements of $\mathcal{P}$. Therefore

every element of $\mathcal{P} - \{\mathcal{P}_0\}$ has exactly one line in common with $\mathcal{P}_0$. Applying an element of A_7 gives the same conclusion for every pair of distinct elements of $\mathcal{P}$.
It follows also that if $\mathcal{Q} \in \mathcal{P}$ and ℓ is a line of $\mathcal{Q}$, then exactly three elements of $\mathcal{P}$ contain ℓ: namely $\mathcal{Q}$ and the two rotations of $\mathcal{Q}$ around ℓ.
Step 2: Turning the fifteen planes into a vector space.
We now put a vector space structure over $\mathbb{F}_2$ on

$$V = \mathcal{P} \cup \{0\}.$$

The zero element is the new symbol 0. For $\mathcal{Q} \in \mathcal{P}$, define

$$\mathcal{Q} + 0 = 0 + \mathcal{Q} = \mathcal{Q}, \qquad \mathcal{Q} + \mathcal{Q} = 0.$$

If $\mathcal{Q}_1, \mathcal{Q}_2 \in \mathcal{P}$ are distinct, let ℓ be their unique common line. There is a unique third element $\mathcal{Q}_3 \in \mathcal{P}$ which contains ℓ. Define

$$\mathcal{Q}_1 + \mathcal{Q}_2 = \mathcal{Q}_3.$$

This addition is commutative, has identity 0, and every element has order 2. We check associativity. If one of the three entries is 0, or if two of the three entries are equal, associativity follows directly from the definition. If $\mathcal{Q}_1, \mathcal{Q}_2, \mathcal{Q}_3$ are distinct and have one common line, then

$$\mathcal{Q}_1 + \mathcal{Q}_2 = \mathcal{Q}_3, \qquad \mathcal{Q}_2 + \mathcal{Q}_3 = \mathcal{Q}_1,$$

and hence

$$(\mathcal{Q}_1 + \mathcal{Q}_2) + \mathcal{Q}_3 = \mathcal{Q}_3 + \mathcal{Q}_3 = 0 = \mathcal{Q}_1 + \mathcal{Q}_1 = \mathcal{Q}_1 + (\mathcal{Q}_2 + \mathcal{Q}_3).$$

It remains to check the case where $\mathcal{Q}_1, \mathcal{Q}_2, \mathcal{Q}_3$ are distinct and do not all share one common line. There are $15 \cdot 14 \cdot 12 = 2520$ ordered triples of this type: after choosing $\mathcal{Q}_1$ and $\mathcal{Q}_2$, the third entry can be any element of $\mathcal{P}$ except the three planes containing the common line of $\mathcal{Q}_1$ and $\mathcal{Q}_2$. We show that A_7 is transitive on these triples by looking at one example. Take $\mathcal{P}_0$ as above and set

$$\mathcal{Q}_1 = \{\{1,2,3\},\{1,4,6\},\{1,5,7\},\{2,4,7\},\{2,5,6\},\{3,4,5\},\{3,6,7\}\},$$
$$\mathcal{Q}_2 = \{\{1,2,4\},\{1,3,5\},\{1,6,7\},\{2,3,7\},\{2,5,6\},\{3,4,6\},\{4,5,7\}\}.$$

The three pairwise common lines are

$$\mathcal{P}_0 \cap \mathcal{Q}_1 = \{\{1,2,3\}\}, \qquad \mathcal{P}_0 \cap \mathcal{Q}_2 = \{\{1,6,7\}\}, \qquad \mathcal{Q}_1 \cap \mathcal{Q}_2 = \{\{2,5,6\}\}.$$

Any element of A_7 which fixes the ordered triple $(\mathcal{P}_0, \mathcal{Q}_1, \mathcal{Q}_2)$ must fix these three lines setwise. It therefore fixes their pairwise intersections 1, 2, and 6; then it fixes the remaining point on each of the three lines, namely 3, 7, and 5; and finally it fixes the last point 4. Thus the stabilizer of this ordered triple is trivial. Its orbit has size $|A_7| = 2520$, which is the total number of triples in this case. Hence every such triple is in this one orbit, so it is enough to check associativity for the displayed example. Direct use of the addition rule gives

$$\mathcal{P}_0 + \mathcal{Q}_1 = \{\{1,2,3\},\{1,4,7\},\{1,5,6\},\{2,4,5\},\{2,6,7\},\{3,4,6\},\{3,5,7\}\},$$
$$\mathcal{Q}_1 + \mathcal{Q}_2 = \{\{1,2,7\},\{1,3,6\},\{1,4,5\},\{2,3,4\},\{2,5,6\},\{3,5,7\},\{4,6,7\}\},$$

and both $(\mathcal{P}_0 + \mathcal{Q}_1) + \mathcal{Q}_2$ and $\mathcal{P}_0 + (\mathcal{Q}_1 + \mathcal{Q}_2)$ are equal to

$$\{\{1,2,6\},\{1,3,7\},\{1,4,5\},\{2,3,5\},\{2,4,7\},\{3,4,6\},\{5,6,7\}\}.$$

Thus V is an elementary abelian group of order 16, hence a 4-dimensional vector space over $\mathbb{F}_2$.
Step 3: Getting A_7 inside $\mathrm{GL}(4,2)$.
The action of A_7 on $\mathcal{P}$ preserves the addition on V, since the definition of addition only uses common lines. Therefore we obtain a homomorphism

$$A_7 \to \mathrm{GL}(V).$$

We now show this homomorphism is injective. We use the elementary fact that A_7 is simple. Here is a proof. Let $1 \neq N \trianglelefteq A_7$, and choose a non-identity element of N moving as few letters as possible. The possible non-identity cycle types in A_7 are represented by

$$(1\ 2\ 3), \quad (1\ 2)(3\ 4), \quad (1\ 2\ 3\ 4\ 5), \quad (1\ 2\ 3\ 4\ 5\ 6\ 7), \quad (1\ 2\ 3)(4\ 5\ 6), \quad (1\ 2\ 3\ 4)(5\ 6), \quad (1\ 2\ 3)(4\ 5)(6\ 7).$$

If the chosen element is not a 3-cycle, then after conjugating in A_7 and replacing by its inverse if necessary, one of the following commutator calculations applies:

$$\begin{aligned}
[(1\ 2)(3\ 4), (1\ 2\ 5)] &= (1\ 2\ 5),\\
[(1\ 2\ 3\ 4\ 5), (1\ 2\ 3)] &= (1\ 4\ 2),\\
[(1\ 2\ 3\ 4\ 5\ 6\ 7), (1\ 2\ 3)] &= (1\ 4\ 2),\\
[(1\ 2\ 3)(4\ 5\ 6), (1\ 2\ 4)] &= (1\ 4\ 3\ 5\ 2),\\
[(1\ 2\ 3\ 4)(5\ 6), (1\ 2\ 3)] &= (1\ 4\ 2),\\
[(1\ 2\ 3)(4\ 5)(6\ 7), (1\ 2\ 4)] &= (1\ 4\ 3\ 5\ 2).
\end{aligned}$$

Each commutator lies in N because N is normal in A_7, and each displayed commutator is non-trivial and moves fewer letters than the original element. This contradicts the minimal choice unless N contains a 3-cycle. Since all 3-cycles are conjugate in A_7, N contains every 3-cycle. Finally, 3-cycles generate A_7: every even permutation is a product of pairs of transpositions, and a product of two transpositions is either a 3-cycle or, when the transpositions are disjoint,

$$(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d).$$

It follows that $N = A_7$. Thus A_7 is simple.

The kernel of the action of A_7 on V is a normal subgroup of A_7. The action is not trivial because the orbit $\mathcal{P} = A_7\mathcal{P}_0$ has 15 elements. Hence the kernel is not all of A_7, and by simplicity it must be trivial. Thus

$$A_7 \hookrightarrow \mathrm{GL}(V).$$

Since $\dim_{\mathbb{F}_2} V = 4$, we have $\mathrm{GL}(V) \cong \mathrm{GL}(4, 2)$. Therefore $G = \mathrm{GL}(4, 2)$ contains a subgroup

$$H \cong A_7.$$

Step 4: Acting on the eight cosets of H.

Since

$$|H| = |A_7| = \frac{7!}{2} = 2520,$$

we have

$$|G : H| = \frac{20160}{2520} = 8.$$

Thus G acts by left multiplication on the set G/H of eight left cosets of H. This gives a homomorphism

$$\varphi\colon G \to S_8.$$

We claim that $\ker\varphi = 1$. The kernel of the action is contained in H, because the identity coset H is fixed precisely by elements of H. Also $\ker\varphi \trianglelefteq G$, so $\ker\varphi \trianglelefteq H$. Since $H \cong A_7$ is simple, $\ker\varphi$ is either 1 or H.

It remains to rule out $\ker\varphi = H$. If $\ker\varphi = H$, then $H \trianglelefteq G$ and G/H has order 8. But G is generated by transvections: over $\mathbb{F}_2$ every invertible diagonal matrix is the identity, so Theorem 4.12 says that G is generated by transvections. Also, since $4 \geq 3$, every transvection is a commutator. Indeed, for distinct i, j, choose k distinct from both i and j, and Lemma 4.6(iv) gives

$$[X_{ik}(1), X_{kj}(1)] = X_{ij}(1).$$

Hence $G = G'$, so every quotient of G is perfect. This is impossible for a quotient of order 8. Indeed, if a group Q has order 8, then either Q has an element of order 4 or 8, in which case it has a cyclic subgroup of index 2 and therefore a non-trivial abelian quotient, or every non-identity element of Q has order 2, in which case Q is abelian because $(ab)^2 = 1$ implies $ab = ba$. Thus no group of order 8 is perfect. Therefore $\ker\varphi \neq H$, and so $\ker\varphi = 1$.

Therefore φ embeds G into S_8. Its image has order $20160 = 8!/2$, so it is an index 2 subgroup of S_8. Finally, the unique index 2 subgroup of S_8 is A_8. Indeed, an index 2 subgroup is normal, so it is the kernel of a non-trivial homomorphism $S_8 \to C_2$; since transpositions generate S_8 and are all conjugate, this kernel consists exactly of the even permutations. Hence

$$\varphi(G) = A_8.$$

Thus

$$\mathrm{GL}(4, 2) \cong A_8.$$

□

Exercise 4.6. Let β be a non-trivial automorphism of the field F. Use β to construct an outer automorphism of $\mathrm{GL}(n, F)$. Do all outer automorphisms of $\mathrm{GL}(n, F)$ arise in this way?

Solution. For $A = (a_{ij}) \in \mathrm{GL}(n, F)$, define

$$\Phi_\beta(A) = (\beta(a_{ij})).$$

That is, Φ_β applies β to every entry of the matrix. Since β preserves addition and multiplication, the (i, j)-entry of $\Phi_\beta(AB)$ is

$$\beta\left(\sum_{k=1}^{n} a_{ik}b_{kj}\right) = \sum_{k=1}^{n} \beta(a_{ik})\beta(b_{kj}),$$

which is exactly the (i, j)-entry of $\Phi_\beta(A)\Phi_\beta(B)$. Hence

$$\Phi_\beta(AB) = \Phi_\beta(A)\Phi_\beta(B).$$

If $A \in \mathrm{GL}(n, F)$, then A^{-1} exists and

$$\Phi_\beta(A)\Phi_\beta(A^{-1}) = \Phi_\beta(AA^{-1}) = I,$$

so $\Phi_\beta(A)$ is again invertible. Thus Φ_β is a homomorphism $\mathrm{GL}(n, F) \to \mathrm{GL}(n, F)$. Its inverse is obtained by applying β^{-1} entry by entry, and hence Φ_β is an automorphism of $\mathrm{GL}(n, F)$.
We now show that it is outer. Suppose, for contradiction, that Φ_β is inner. Then there exists $P \in \mathrm{GL}(n, F)$ such that

$$\Phi_\beta(A) = PAP^{-1}$$

for every $A \in \mathrm{GL}(n, F)$. Since β is non-trivial and fixes 0 and 1, choose $\lambda \in F^\times$ such that $\beta(\lambda) \neq \lambda$. Taking $A = \lambda I$, we get on one hand

$$\Phi_\beta(\lambda I) = \beta(\lambda)I.$$

On the other hand, if Φ_β were conjugation by P, then

$$\Phi_\beta(\lambda I) = P(\lambda I)P^{-1} = \lambda PP^{-1} = \lambda I.$$

Thus $\beta(\lambda)I = \lambda I$, forcing $\beta(\lambda) = \lambda$, a contradiction. Therefore Φ_β is not inner, and hence it is outer.
Not all outer automorphisms arise in this way. For $n \geq 2$, define

$$\tau(A) = (A^{-1})^T.$$

Then

$$\tau(AB) = ((AB)^{-1})^T = (B^{-1}A^{-1})^T = (A^{-1})^T(B^{-1})^T = \tau(A)\tau(B),$$

so τ is an automorphism of $\mathrm{GL}(n, F)$. It is outer by the same scalar-matrix test: choose $\lambda \in F^\times$ with $\lambda^{-1} \neq \lambda$; such a λ exists under the present hypothesis, since otherwise every non-zero element of F would satisfy $x^2 = 1$, forcing $|F^\times| \leq 2$ and leaving no non-trivial field automorphism. Then $\tau(\lambda I) = \lambda^{-1}I$, while every inner automorphism fixes λI. Finally, τ is not obtained from a field automorphism, because a field automorphism γ sends

$$X_{ij}(\alpha) \mapsto X_{ij}(\gamma(\alpha)),$$

keeping the same matrix position (i, j), whereas

$$\tau(X_{ij}(\alpha)) = (X_{ij}(\alpha)^{-1})^T = X_{ji}(-\alpha),$$

which swaps the position from (i, j) to (j, i). Therefore field automorphisms give outer automorphisms of $\mathrm{GL}(n, F)$, but they do not account for all of them. □

Further Exercises

Exercise 4.7. A matrix is said to be *monomial* if each row and column has exactly one non-zero entry. Let N be the subset of $G = \mathrm{GL}(n, F)$ consisting of all monomial matrices. Show that $N \leq G$, that $T = B \cap N$ is the subgroup of G consisting of all diagonal matrices, that $N = N_G(T)$, and that $N = T \rtimes W$.

Solution. Let $\mathcal{D}$ denote the subgroup of G consisting of all invertible diagonal matrices. We first show that

$$N = \mathcal{D}W.$$

Let $m = (m_{ij}) \in N$. In each row i of m, there is exactly one non-zero entry. Let that entry occur in column $\sigma(i)$. Since each column also has exactly one non-zero entry, σ is a permutation of $\{1, \ldots, n\}$. Let w be the permutation matrix whose non-zero entries occur in the same positions as the non-zero entries of m, and let

$$D = \operatorname{diag}(m_{1,\sigma(1)}, \ldots, m_{n,\sigma(n)}).$$

Then $D \in \mathcal{D}$, because all these entries are non-zero, and left multiplication by D scales row i of w by $m_{i,\sigma(i)}$. Hence $m = Dw$, so $m \in \mathcal{D}W$. Thus $N \subseteq \mathcal{D}W$. Conversely, if $D \in \mathcal{D}$ and $w \in W$, then Dw is obtained from the permutation matrix w by scaling each row by a non-zero scalar. Therefore every row and column still has exactly one non-zero entry, so $Dw \in N$. Hence $\mathcal{D}W \subseteq N$, and therefore $N = \mathcal{D}W$.
We now prove that $N \leq G$. The identity matrix is monomial, so N is non-empty. Let $m, l \in N$. Write

$$m = Dw, \qquad l = Eu,$$

where $D, E \in \mathcal{D}$ and $w, u \in W$. Then

$$ml = DwEu = D(wEw^{-1})(wu).$$

The matrix wEw^{-1} is diagonal, because conjugating a diagonal matrix by a permutation matrix only permutes the diagonal entries. Also $wu \in W$. Hence ml is diagonal times permutation, so $ml \in N$. Similarly,

$$m^{-1} = (Dw)^{-1} = w^{-1}D^{-1} = (w^{-1}D^{-1}w)w^{-1}.$$

Again $w^{-1}D^{-1}w$ is diagonal and $w^{-1} \in W$, so $m^{-1} \in N$. Therefore $N \leq G$.
Next we identify $B \cap N$. Let $x \in B \cap N$. Since $x \in N$, the non-zero entries of x have the form of a permutation: in row i, the unique non-zero entry occurs in column $\sigma(i)$, where σ is a permutation of $\{1, \ldots, n\}$. Since $x \in B$, all entries below the diagonal are zero, so this non-zero entry cannot lie below the diagonal. Hence

$$\sigma(i) \geq i$$

for every i. But σ is a permutation, so

$$\sum_{i=1}^{n} \sigma(i) = \sum_{i=1}^{n} i.$$

Since each $\sigma(i) \geq i$, equality of the sums forces $\sigma(i) = i$ for every i. Thus the only non-zero entries of x are diagonal entries, so x is diagonal. Therefore $B \cap N \subseteq \mathcal{D}$. Conversely, every invertible diagonal matrix is both upper triangular and monomial, so $\mathcal{D} \subseteq B \cap N$. Hence

$$T = B \cap N = \mathcal{D}.$$

In particular, T is the subgroup of G consisting of all invertible diagonal matrices.
We now prove that $N = N_G(T)$. For this assertion we assume $F \neq \mathbb{F}_2$; otherwise, if $n > 1$, then $T = \{I\}$ and $N_G(T) = G$, so the displayed equality is false. First let $m \in N$. Write $m = Dw$ with $D \in T$ and $w \in W$. If $t \in T$, then

$$mtm^{-1} = Dwtw^{-1}D^{-1}.$$

The matrix wtw^{-1} is diagonal, hence lies in T. Since diagonal matrices commute, we get

$$Dwtw^{-1}D^{-1} = wtw^{-1} \in T.$$

Thus $mTm^{-1} \subseteq T$. Applying the same argument to $m^{-1} \in N$ gives $T \subseteq mTm^{-1}$, and hence $mTm^{-1} = T$. Therefore $m \in N_G(T)$, so $N \subseteq N_G(T)$.
Conversely, let $g \in N_G(T)$. We prove that g is monomial. Let

$$V = F^n$$

with standard basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Every matrix $A \in G$ acts on V by left multiplication:

$$L_A \colon V \to V, \qquad L_A(x) = Ax.$$

In particular, $g\mathbf{v}_i = L_g(\mathbf{v}_i)$ is exactly the ith column of the matrix g. Since $F \neq \mathbb{F}_2$, choose $\lambda \in F^\times$ with $\lambda \neq 1$. For each i, define

$$t_i = \operatorname{diag}(1, \dots, 1, \lambda, 1, \dots, 1),$$

where λ is in the ith diagonal position. Then $t_i \in T$. As a linear transformation, L_{t_i} satisfies

$$L_{t_i}(\mathbf{v}_i) = \lambda \mathbf{v}_i,$$

and, for $k \neq i$,

$$L_{t_i}(\mathbf{v}_k) = \mathbf{v}_k.$$

Thus the λ-eigenspace of L_{t_i} is exactly

$$E_\lambda(L_{t_i}) = F\mathbf{v}_i = \{\alpha \mathbf{v}_i \mid \alpha \in F\}.$$

Indeed, if $x = \sum_{k=1}^n a_k \mathbf{v}_k$ and $t_i x = \lambda x$, then for every $k \neq i$ we have $a_k = \lambda a_k$, so $(\lambda - 1)a_k = 0$, and hence $a_k = 0$. Thus only the $\mathbf{v}_i$-coordinate can remain.
Because $g \in N_G(T)$ and $t_i \in T$, the conjugate

$$s_i = g t_i g^{-1}$$

also lies in T. Hence s_i is diagonal. Now compute the action of s_i on $g\mathbf{v}_i$:

$$s_i(g\mathbf{v}_i) = (g t_i g^{-1})(g\mathbf{v}_i) = g t_i \mathbf{v}_i = g(\lambda \mathbf{v}_i) = \lambda g \mathbf{v}_i.$$

So $g\mathbf{v}_i$ is a λ-eigenvector of the diagonal matrix s_i. More precisely, conjugation transports eigenspaces:

$$E_\lambda(L_{s_i}) = g(E_\lambda(L_{t_i})) = g(F\mathbf{v}_i).$$

Since s_i is diagonal, its λ-eigenspace is spanned by those standard basis vectors whose diagonal entries in s_i are equal to λ. Since s_i is conjugate to t_i, this eigenspace has the same dimension as $E_\lambda(L_{t_i})$, namely dimension 1. Therefore there is some index j such that

$$E_\lambda(L_{s_i}) = F\mathbf{v}_j.$$

In particular,

$$g\mathbf{v}_i \in F\mathbf{v}_j.$$

Thus there is some $\alpha_i \in F^\times$ such that

$$g\mathbf{v}_i = \alpha_i \mathbf{v}_j.$$

But $g\mathbf{v}_i$ is the ith column of g, so the ith column of g has exactly one non-zero entry. Since this holds for every i, every column of g has exactly one non-zero entry. Finally, g is invertible, so its columns are linearly independent. Hence two different columns cannot be non-zero scalar multiples of the same standard basis vector. Therefore the non-zero entries occur in different rows, and every row also has exactly one non-zero entry. Thus g is monomial, so $g \in N$. Hence $N_G(T) \subseteq N$. Combining both inclusions gives

$$N = N_G(T).$$

It remains to prove the semidirect product statement. We already proved $N = TW$, since $T = \mathcal{D}$. Also $T \trianglelefteq N$: if $m = Dw \in N$ and $t \in T$, then the same computation as above gives

$$mtm^{-1} = wtw^{-1} \in T.$$

Finally,

$$T \cap W = \{I\}.$$

Indeed, if $x \in T \cap W$, then x is diagonal, say

$$x = \operatorname{diag}(\mu_1, \dots, \mu_n),$$

and x is also a permutation matrix. Since x is diagonal, the only possible non-zero entry in row i is the diagonal entry μ_i. Since x is a permutation matrix, the non-zero entry in each row is equal to 1. Hence $\mu_i = 1$ for every i, and so $x = I$. The reverse inclusion $I \in T \cap W$ is immediate. Therefore $T \cap W = \{I\}$.
Since $N = TW$, $T \trianglelefteq N$, and $T \cap W = \{I\}$, we conclude that

$$N = T \rtimes W.$$

□

Let G be a group. Suppose that G has subgroups B and N of G satisfying the following conditions:

- G is generated by B and N.
- $T = B \cap N$ is a normal subgroup of N.
- $W = N/T$ is generated by a finite set S of involutions (elements of order 2). In addition, if for each $w \in W$ we choose some $\dot{w} \in N$ such that $\dot{w}T = w$, then we must have
  - $\dot{s}B\dot{w} \subseteq B\dot{w}B \cup B\dot{s}\dot{w}B$ for any $s \in S$ and $w \in W$,
  - $\dot{s}B\dot{s} \nsubseteq B$ for any $s \in S$.

In this case we say that B and N form a *BN-pair* of G, or that (G, B, N, S) is a *Tits system* (after Jacques Tits). We call B the *Borel subgroup* of G, and $W = N/B \cap N$ the *Weyl group* associated with the Tits system. The *rank* of the Tits system is defined to be $|S|$.

Exercise 4.8. (cont.) Let $G = \mathrm{GL}(n, F)$, let B be the standard Borel subgroup of G, let N be the subgroup of G consisting of all monomial matrices, and let $T = B \cap N$. By Exercise 4.7 above, we know that we can regard $W = N/T$ as being imbedded in G as the group of permutation matrices. (By making this identification, we can replace $\dot{w}$ by w in the above formulas.) Let S be the subset of W consisting of those permutation matrices that are obtained from the identity matrix by switching two adjacent columns. (In other words, if we identify W with S_n as in Proposition 4.5, then S corresponds to the set $\{(1\ 2), (2\ 3), \ldots, (n-1\ n)\}$.) Show that (G, B, N, S) is a Tits system of rank $n-1$.

Solution. We check the defining conditions one at a time. To keep the quotient notation separate from the matrix notation, write $s_i \in S \leq N/T$ for the generator corresponding to the adjacent transposition $(i\ i+1)$, and write $w_i \in W \leq G$ for its permutation-matrix representative. Thus, in the notation of the definition above, we choose

$$\dot{s_i} = w_i.$$

Similarly, for an element $w \in W$, we use the same symbol w for its permutation-matrix representative in G; in the dotted notation this means $\dot{w} = w$. With these choices, the condition

$$\dot{s}B\dot{w} \subseteq B\dot{w}B \cup B\dot{s}\dot{w}B$$

becomes

$$w_iBw \subseteq BwB \cup Bw_iwB.$$

We now verify the defining conditions using these representatives.
First, by the Bruhat Decomposition Theorem,

$$G = BWB.$$

Since every permutation matrix is monomial, we have $W \leq N$. Hence

$$G = BWB \subseteq \langle B, N\rangle.$$

The reverse inclusion is automatic because $B \leq G$ and $N \leq G$. Therefore

$$G = \langle B, N\rangle.$$

Second, Exercise 4.7 gives $T \trianglelefteq N$.
Third, Exercise 4.7 gives $N = T \rtimes W$. Thus the quotient N/T is naturally isomorphic to W: the map

$$W \to N/T, \qquad w \mapsto wT$$

is surjective because every element of N has the form tw with $t \in T$ and $w \in W$, and it is injective because its kernel is $T \cap W = \{I\}$. By Proposition 4.5, $W \cong S_n$. Write

$$S = \{s_1, \ldots, s_{n-1}\},$$

where s_i is the generator of N/T corresponding to the adjacent transposition

$$(i\ i+1).$$

The representative $w_i \in W$ switches the adjacent basis vectors $\mathbf{v}_i$ and $\mathbf{v}_{i+1}$. Then $w_i^2 = I$, so s_i has order 2 in N/T. The elements s_i generate N/T, because their representatives w_i generate the permutation-matrix subgroup W: by repeatedly swapping neighbouring entries, we can move any chosen entry to any chosen position, and hence build any permutation matrix. Therefore N/T is generated by the finite set S of involutions. Since $|S| = n-1$, the rank will be $n-1$ once the remaining two conditions are checked.
It remains to verify the two double-coset conditions. If $n = 1$, then $S = \varnothing$ and these conditions are vacuous, so assume $n \geq 2$. Fix $s_i \in S$, and let $w_i \in W$ be its adjacent permutation-matrix representative. Write

$$X_i(\alpha) = X_{i,i+1}(\alpha).$$

We first prove that

$$w_i B w \subseteq BwB \cup Bw_i wB$$

for all $w \in W$. Let $b = (b_{rs}) \in B$. Since b is invertible upper triangular, $b_{ii} \neq 0$. Set

$$\alpha = b_{i,i+1} b_{ii}^{-1}, \qquad b_0 = bX_i(-\alpha).$$

Right multiplication by $X_i(-\alpha)$ subtracts α times column i from column $i+1$. Hence the $(i, i+1)$-entry of b_0 is

$$b_{i,i+1} - \alpha b_{ii} = 0.$$

Thus $b = b_0 X_i(\alpha)$, with $b_0 \in B$. Moreover $w_i b_0 w_i \in B$. Indeed, conjugating by w_i only swaps the adjacent rows and columns i and $i+1$; the only possible below-diagonal entry which could appear is the old $(i, i+1)$-entry of b_0, and this entry is zero. Hence $b_0 \in B \cap w_i B w_i$.
Now

$$w_i b w = w_i b_0 X_i(\alpha) w = (w_i b_0 w_i)(w_i X_i(\alpha) w).$$

Since $w_i b_0 w_i \in B$, it is enough to show that $w_i X_i(\alpha) w$ lies in $BwB \cup Bw_i wB$. Put

$$p = w^{-1}(i), \qquad q = w^{-1}(i+1).$$

By Lemma 4.6(v),

$$X_i(\alpha) w = w X_{pq}(\alpha).$$

If $p < q$, then $X_{pq}(\alpha) \in B$, and therefore

$$w_i X_i(\alpha) w = w_i w X_{pq}(\alpha) \in Bw_i wB.$$

Now suppose that $p > q$. If $\alpha = 0$, then $w_i X_i(0) w = w_i w \in Bw_i wB$. If $\alpha \neq 0$, then in the rows and columns $i, i+1$ we have the 2×2 calculation

$$w_i X_i(\alpha) = \begin{pmatrix} 0 & 1 \\ 1 & \alpha \end{pmatrix} = \begin{pmatrix} -\alpha^{-1} & 1 \\ 0 & \alpha \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha^{-1} & 1 \end{pmatrix}.$$

Thus $w_i X_i(\alpha) = b_1 X_{i+1,i}(\alpha^{-1})$ for some $b_1 \in B$. Therefore

$$w_i X_i(\alpha) w = b_1 X_{i+1,i}(\alpha^{-1}) w = b_1 w X_{qp}(\alpha^{-1}).$$

Since $p > q$, we have $q < p$, so $X_{qp}(\alpha^{-1}) \in B$. Hence

$$w_i X_i(\alpha) w \in BwB.$$

In all cases, $w_i X_i(\alpha) w \in BwB \cup Bw_i wB$, and so

$$w_i B w \subseteq BwB \cup Bw_i wB.$$

Finally, we verify that the representative w_i satisfies $w_i B w_i \not\subseteq B$. Take $X_i(1) = X_{i,i+1}(1) \in B$. Since $w_i = w_i^{-1}$, Lemma 4.6(v) gives

$$w_i X_i(1) w_i = X_{i+1,i}(1).$$

This matrix has a non-zero entry below the diagonal, so it is not in B. Hence $w_i B w_i \not\subseteq B$.
All the defining conditions of a Tits system are now verified. Therefore (G, B, N, S) is a Tits system. Since S consists of the $n-1$ adjacent transposition generators, its rank is $n-1$. □

Exercise 4.9. (cont.) Let G be a group with a BN-pair. Verify that, for $w \in W$, the set $B\dot{w}B$ is independent of the choice of $\dot{w} \in N$ such that $\dot{w}T = w$. Show that we have a Bruhat decomposition

$$G = \bigcup_{w \in W} BwB$$

in which the union is disjoint, where BwB is taken to mean $B\dot{w}B$ for any $\dot{w} \in N$ with $\dot{w}T = w$.

Solution. Let

$$T = B \cap N, \qquad W = N/T.$$

For each $w \in W$, choose a representative $\dot{w} \in N$ with $\dot{w}T = w$, and write

$$D_w = B\dot{w}B.$$

We will prove that D_w is well-defined and that

$$G = \coprod_{w \in W} D_w,$$

where $\coprod$ means that the union is disjoint.

Step 1: D_w does not depend on the representative. Suppose that $n, n' \in N$ both represent the same element $w \in W$. This means

$$nT = n'T.$$

Hence $n' = nt$ for some $t \in T$. Since $T = B \cap N$, we have $t \in B$. Therefore

$$Bn'B = BntB = BnB,$$

because the final B already contains t. Thus $B\dot{w}B$ depends only on w, not on the chosen representative $\dot{w}$.

Step 2: multiplying by a simple double coset has only two possible outcomes. Let $s \in S$ and $w \in W$. We claim that

$$D_s D_w \subseteq D_w \cup D_{sw}.$$

Indeed,

$$D_s D_w = B\dot{s}B\dot{w}B = B(\dot{s}B\dot{w})B.$$

By the BN-pair axiom,

$$\dot{s}B\dot{w} \subseteq B\dot{w}B \cup B\dot{s}\dot{w}B.$$

The element $\dot{s}\dot{w}$ represents $sw \in W$, so $B\dot{s}\dot{w}B = D_{sw}$. Multiplying the inclusion above on the left and right by B gives

$$D_s D_w \subseteq D_w \cup D_{sw}.$$

Also,

$$D_{sw} = B\dot{s}\dot{w}B \subseteq B\dot{s}B\dot{w}B = D_s D_w.$$

So the useful statement is

$$D_{sw} \subseteq D_s D_w \subseteq D_w \cup D_{sw}.$$

Step 3: the union covers G. Let

$$X = \bigcup_{w \in W} D_w.$$

We show that $X = G$. First,

$$B = D_1 \subseteq X.$$

Also, if $b \in B$, then

$$bD_w \subseteq D_w,$$

because $D_w = B\dot{w}B$ already has B on the left. Hence X is closed under left multiplication by elements of B.

Now take $s \in S$. If $x \in D_w$, then

$$\dot{s}x \in \dot{s}D_w = \dot{s}B\dot{w}B.$$

By Step 2,

$$\dot{s}B\dot{w}B \subseteq D_w \cup D_{sw} \subseteq X.$$

Thus X is closed under left multiplication by each chosen representative $\dot{s}$.
We now explain why these elements generate G. Since $G = \langle B, N\rangle$, it is enough to build every element of N using elements of B and the chosen representatives $\dot{s}$. Let $n \in N$. Since $W = N/T$ is generated by S, we can write

$$nT = s_1 s_2 \cdots s_m$$

with each $s_i \in S$. Then $\dot{s}_1\dot{s}_2 \cdots \dot{s}_m$ represents the same coset of T as n, so

$$n = \dot{s}_1\dot{s}_2 \cdots \dot{s}_m t$$

for some $t \in T$. But $T \leq B$. Therefore every element of N is built from elements of B and the $\dot{s}$'s. Hence G is generated by B and the chosen $\dot{s}$'s. Since $1 \in X$ and X is closed under left multiplication by those generators, every element of G lies in X. Therefore

$$G = \bigcup_{w\in W} D_w.$$

Step 4: the union is disjoint. We use the following elementary fact: two double cosets for the same subgroup B are either equal or disjoint. So, to prove disjointness, it is enough to show that

$$D_w = D_v \quad \Longrightarrow \quad w = v.$$

Assume $D_w = D_v$. We first show that left multiplication by a generator $s \in S$ preserves this equality, in the sense that

$$D_{sw} = D_{sv}.$$

Since $D_w = D_v$, we have

$$D_s D_w = D_s D_v.$$

By Step 2,

$$D_{sw} \subseteq D_s D_w.$$

Thus

$$D_{sw} \subseteq D_s D_v \subseteq D_v \cup D_{sv} = D_w \cup D_{sv}.$$

Since D_{sw} is itself one double coset, it cannot be partly in two different double cosets. Therefore

$$D_{sw} = D_w \qquad \text{or} \qquad D_{sw} = D_{sv}.$$

Doing the same argument with w and v switched gives

$$D_{sv} = D_w \qquad \text{or} \qquad D_{sv} = D_{sw}.$$

In every case, $D_{sw} = D_{sv}$: either this is stated directly, or both double cosets are equal to D_w.
Now repeat this argument. If $x = s_1 \cdots s_m$ is any product of elements of S, then

$$D_{xw} = D_{xv}.$$

Since S generates W, this holds for every $x \in W$. Choose $x = v^{-1}$. Then

$$D_{v^{-1}w} = D_{v^{-1}v} = D_1 = B.$$

Let $y = v^{-1}w$. Choose a representative $\dot{y} \in N$. Since

$$D_y = B\dot{y}B = B,$$

we have $\dot{y} \in B$. But $\dot{y} \in N$ as well, so $\dot{y} \in B \cap N = T$. Hence

$$y = \dot{y}T = T,$$

which is the identity element of W. Therefore $v^{-1}w = 1$, so $w = v$.
Thus different elements of W give different double cosets, and since double cosets are either equal or disjoint, the union is disjoint. We have proved

$$G = \coprod_{w \in W} B\dot{w}B.$$

Since $B\dot{w}B$ is independent of the chosen representative, we may write this as

$$G = \coprod_{w \in W} BwB,$$

where BwB means $B\dot{w}B$ for any $\dot{w} \in N$ representing w. $\square$

5 Parabolic Subgroups

Setup and Complete Flags

- Throughout, $G = \mathrm{GL}(n, F)$ for some field F and some $n \in \mathbb{N}$, with $V_n(F)$ the space of column vectors of length n over F.
- Elements of G are viewed as matrices, with respect to the standard basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$, of invertible linear transformations of $V_n(F)$.
- A *complete flag* on $V_n(F)$ is a sequence of subspaces

$$0 \subset V_1 \subset V_2 \subset \cdots \subset V_{n-1} \subset V_n = V_n(F),$$

denoted $(V_1, \ldots, V_n)$. Since $\subset$ is proper containment, $\dim_F V_i = i$.

- The *standard flag* is given by $V_i = F\mathbf{v}_1 \oplus \cdots \oplus F\mathbf{v}_i = V_{i-1} \oplus F\mathbf{v}_i$ (with $V_0 = 0$).

The G-Action on Complete Flags

- G acts on complete flags by $g(V_1, \ldots, V_n) = (gV_1, \ldots, gV_n)$.
- This action is transitive: any complete flag $(W_1, \ldots, W_n)$ is obtained from the standard flag by the matrix whose ith column is $\mathbf{w}_i \in W_i - W_{i-1}$.
- The stabilizer of the standard flag is precisely the standard Borel subgroup B, so by Lemma 3.2 the Borel subgroups of G are exactly the stabilizers of complete flags on $V_n(F)$.

Upper Unitriangular Matrices and the Decomposition of B

- An upper triangular matrix is *upper unitriangular* if all its main-diagonal entries equal 1, equivalently if it is upper triangular with only eigenvalue 1.
- The set U of all invertible upper unitriangular matrices is a subgroup of B (imitate the proof of Proposition 4.3).
- Let T be the set of diagonal matrices in G. Then $T \leq B$ and $U \cap T = 1$.

Proposition 5.1. $B = U \rtimes T$.

Proof. Let $\varphi\colon B \to T$ be the map which sends an upper triangular matrix to the diagonal matrix with the same main diagonal. If $b = (b_{ij})$ and $c = (c_{ij})$ lie in B, then the (i,i)-entry of bc is $b_{ii}c_{ii}$, since all entries below the main diagonal are zero. Hence

$$\varphi(bc) = \varphi(b)\varphi(c),$$

so φ is a homomorphism. Its kernel consists exactly of those upper triangular matrices whose diagonal entries are all 1, so $\ker\varphi = U$. Therefore $U \trianglelefteq B$.
Now let $b \in B$. Since $\varphi(b) \in T$ and φ restricts to the identity map on T, we have

$$\varphi(b\varphi(b)^{-1}) = \varphi(b)\varphi(b)^{-1} = I.$$

Thus $b\varphi(b)^{-1} \in U$, and so

$$b = (b\varphi(b)^{-1})\varphi(b) \in UT.$$

Therefore $B = UT$. Finally, $U \cap T = 1$, because a diagonal matrix whose diagonal entries are all 1 is the identity matrix. Hence $B = U \rtimes T$. □

Action of U and T on the Standard Flag

- An element of U sends each $\mathbf{v}_i$ to $\mathbf{v}_i$ plus some element of V_{i-1}; conversely, any matrix with this property lies in U.
- Hence U consists of the matrices stabilizing each V_i and inducing the identity on each quotient V_i/V_{i-1}.
- T consists exactly of the matrices stabilizing each of $F\mathbf{v}_1, \ldots, F\mathbf{v}_n$; we say T is the *common stabilizer* of the $F\mathbf{v}_i$.

Unipotent Elements and Subgroups

- An element of G is *unipotent* if its characteristic polynomial is $(X-1)^n$.
- By Jordan form, any unipotent element of G is conjugate to an element of U.
- A subgroup of G is *unipotent* if all its elements are unipotent; U is unipotent and comprises all unipotent elements of B.

Theorem (Kolchin's Theorem)**.** *Any unipotent subgroup of G is conjugate with a subgroup of U.*

Proof. Let H be a unipotent subgroup of G. We first observe that it is enough to show that H stabilizes some complete flag on $V_n(F)$. Indeed, if H stabilizes a complete flag $\mathcal{F}$, then $\mathcal{F} = g\mathcal{F}_0$ for some $g \in G$, where $\mathcal{F}_0$ is the standard flag. By Lemma 3.2, the stabilizer of $\mathcal{F}$ is gBg^{-1}, and hence $g^{-1}Hg \le B$. Since conjugation preserves characteristic polynomials, every element of $g^{-1}Hg$ is still unipotent. The unipotent elements of B all lie in U, so $g^{-1}Hg \le U$. Thus H is conjugate with a subgroup of U.
It remains to show that H stabilizes some complete flag. We argue by induction on n. The case $n = 1$ is immediate. Assume $n > 1$. By Proposition 13.28, there exists some $0 \ne \mathbf{v} \in V_n(F)$ such that $x\mathbf{v} = \mathbf{v}$ for every $x \in H$. Let $W = F\mathbf{v}$. Then H stabilizes W, and hence acts on the quotient $V_n(F)/W$. For each $x \in H$, the induced transformation on $V_n(F)/W$ is again unipotent: with respect to a basis beginning with $\mathbf{v}$, the matrix of x has the block form

$$\begin{pmatrix} 1 & * \\ 0 & \overline{x} \end{pmatrix},$$

so the characteristic polynomial of $\overline{x}$ is $(X-1)^{n-1}$. Therefore the induced subgroup on $V_n(F)/W$ is unipotent. By the induction hypothesis, it stabilizes a complete flag on $V_n(F)/W$. Pulling this flag back to $V_n(F)$ and placing W at the beginning gives a complete flag on $V_n(F)$ stabilized by H. This completes the induction, and hence proves the theorem. □

General Flags and Parabolic Subgroups

- A *flag* on $V_n(F)$ is a nested sequence of non-zero subspaces of $V_n(F)$ which terminates in $V_n(F)$ and has no repeated terms; that is, a sequence $(W_1, \ldots, W_r)$ with

$$0 \subset W_1 \subset W_2 \subset \cdots \subset W_{r-1} \subset W_r = V_n(F).$$

  Since $\subset$ is proper, $r \le n$; a complete flag is the case $r = n$.
- The G-set of all flags is not transitive: two flags $(W_1, \ldots, W_r)$ and $(W'_1, \ldots, W'_s)$ lie in the same orbit iff $r = s$ and $\dim_F W_i = \dim_F W'_i$ for all i (same *dimension sequence*).
- A *parabolic subgroup* of G is the stabilizer of some flag on $V_n(F)$.
- Given a flag $(W_1, \ldots, W_r)$ with parabolic stabilizer P, choose subspaces Y_i with $W_i = W_{i-1} \oplus Y_i$. The *unipotent radical* of P is the subgroup $U_P \le P$ of matrices inducing the identity on each W_i/W_{i-1}; e.g. $U_B = U$.
- A *Levi complement* of U_P is the subgroup $L_P \le P$ which is the common stabilizer of the Y_i.
- A Levi complement is isomorphic with $\mathrm{GL}(y_1, F) \times \cdots \times \mathrm{GL}(y_r, F)$, where $y_i = \dim_F Y_i = \dim_F W_i - \dim_F W_{i-1}$. (In particular, any two Levi complements of U_P are isomorphic.)
- Taking $Y_i = F\mathbf{v}_i$ gives $L_B = T$, isomorphic with a direct product of n copies of $\mathrm{GL}(1, F) \cong F^\times$.

Proposition 5.2. *If P is a parabolic subgroup of G, then we have $P = U_P \rtimes L_P$, where U_P is the unipotent radical of P and L_P is a Levi complement of U_P. Furthermore, $P = N_G(U_P)$.*

Proof. Let P be the stabilizer of the flag $(W_1, \ldots, W_r)$, and put $W_0 = 0$. Choose subspaces Y_i such that $W_i = W_{i-1} \oplus Y_i$ for each i. Then

$$V_n(F) = Y_1 \oplus \cdots \oplus Y_r.$$

With respect to this direct-sum decomposition, elements of P are block upper triangular. This means that if $p \in P$ and $y_i \in Y_i$, then $p(y_i)$ may have a part in the current block Y_i and earlier blocks

$Y_1, \ldots, Y_{i-1}$, but it cannot have a part in any later block. Equivalently, there are maps $a_i \colon Y_i \to Y_i$ such that

$$p(y_i) = w_{i-1} + a_i(y_i)$$

with $w_{i-1} \in W_{i-1}$. The map a_i is the action induced by p on the quotient W_i/W_{i-1}, so it is invertible. We now split p into its diagonal-block part and its upper-block-unipotent part. Define $l \in G$ by letting l act as a_i on each Y_i and by making l preserve every Y_i. Then $l \in L_P$, because l is exactly the block diagonal part of p. Set

$$u = pl^{-1}.$$

We claim that $u \in U_P$. Let $y_i \in Y_i$, and write $z_i = l^{-1}(y_i) \in Y_i$. Since l acts as a_i on Y_i, we have $a_i(z_i) = y_i$. Hence

$$u(y_i) = p(z_i) = y_i + w_{i-1}$$

for some $w_{i-1} \in W_{i-1}$. Therefore, modulo W_{i-1}, the vector $u(y_i)$ is the same as y_i. Since the Y_i build the flag one block at a time, this says exactly that u stabilizes every W_i and induces the identity transformation on each quotient W_i/W_{i-1}. Thus $u \in U_P$, and

$$p = ul \in U_P L_P.$$

Since $p \in P$ was arbitrary, $P = U_P L_P$.
This product is a semidirect product. First, U_P is normal in P, since it is the kernel of the homomorphism

$$P \to \mathrm{GL}(W_1/W_0) \times \cdots \times \mathrm{GL}(W_r/W_{r-1})$$

which records the induced actions on the successive quotients. Second, $U_P \cap L_P = 1$. Indeed, if $x \in U_P \cap L_P$, then x preserves each Y_i, but it also acts as the identity on W_i/W_{i-1}. Therefore x is the identity on every block Y_i, and hence $x = 1$. We have proved

$$P = U_P \rtimes L_P.$$

It remains to prove that $P = N_G(U_P)$. Since $U_P \trianglelefteq P$, every element of P normalizes U_P, so

$$P \leq N_G(U_P).$$

Conversely, suppose that $g \in N_G(U_P)$. We shall prove that g stabilizes every space in the flag, and hence that $g \in P$.
First we recover W_1 from U_P alone. We claim that W_1 is exactly the common fixed subspace of U_P, meaning

$$W_1 = \{v \in V_n(F) \mid uv = v \text{ for every } u \in U_P\}.$$

Every vector in W_1 is fixed because elements of U_P act as the identity on $W_1/W_0 = W_1$. Conversely, take a vector

$$v = y_1 + \cdots + y_r$$

with $y_i \in Y_i$. If some $y_j \neq 0$ for $j > 1$, choose a non-zero vector $x \in Y_1$. Extend y_j to a basis of Y_j, and define a block-unitriangular element u by

$$u(y_j) = y_j + x,$$

while fixing the remaining basis vectors of Y_j and fixing all other blocks. This element lies in U_P: it only adds an earlier-block vector to a vector in the jth block, so it acts as the identity on every quotient W_i/W_{i-1}. But then

$$uv = v + x \neq v.$$

Thus any vector with a non-zero component outside W_1 is moved by some element of U_P. Hence the common fixed subspace of U_P is exactly W_1.
Now use the assumption that g normalizes U_P. We show that $gW_1 = W_1$. Let $w \in W_1$. To prove $gw \in W_1$, it is enough to show that gw is fixed by every element of U_P. Let $u \in U_P$. Since $g \in N_G(U_P)$, we have $g^{-1}ug \in U_P$. Therefore

$$u(gw) = g(g^{-1}ug)w.$$

The element $g^{-1}ug$ lies in U_P, and $w \in W_1$ is fixed by every element of U_P. Hence $(g^{-1}ug)w = w$, so

$$u(gw) = gw.$$

This holds for every $u \in U_P$, so gw belongs to the common fixed subspace of U_P, which is W_1. Thus $gW_1 \le W_1$, and equality follows because g is invertible. Therefore $gW_1 = W_1$.
Now pass to the quotient $V_n(F)/W_1$. On this quotient, the group U_P induces the unipotent radical corresponding to the shorter flag

$$W_2/W_1 \subset \cdots \subset W_r/W_1.$$

Because g normalizes U_P and stabilizes W_1, the induced map of g on $V_n(F)/W_1$ normalizes this induced unipotent radical. The same common-fixed-subspace argument on the quotient shows that the first fixed layer of the quotient, namely W_2/W_1, is stabilized by g. Hence $gW_2 = W_2$. Repeating this argument one layer at a time gives

$$gW_i = W_i$$

for every i. Thus g stabilizes the whole flag, so $g \in P$. Therefore $N_G(U_P) \le P$. Combining both inclusions gives $P = N_G(U_P)$. $\square$

Subflags and Staircase Groups

- A *subflag* of the standard flag $(V_1, \ldots, V_n)$ is a flag $(W_1, \ldots, W_r)$ in which each $W_i = V_j$ for some j; there are 2^{n-1} such subflags.

- A *staircase group* is a parabolic subgroup of G that stabilizes some subflag of the standard flag.

- Example ($n = 6$, subflag $0 \subset V_2 \subset V_3 \subset V_6$): the staircase group consists of all matrices in $G = \mathrm{GL}(6, F)$ of the form

$$\begin{pmatrix} * & * & * & * & * & * \\ * & * & * & * & * & * \\ 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \end{pmatrix}.$$

  Imagine a “staircase” separating the zero entries from the arbitrary entries.

- Its unipotent radical consists of all matrices in G of the form

$$\begin{pmatrix} 1 & 0 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

  and the Levi complement (with canonical choices $Y_2 = F\mathbf{v}_3$ and $Y_3 = F\mathbf{v}_4 \oplus F\mathbf{v}_5 \oplus F\mathbf{v}_6$) consists of all matrices in G of the form

$$\begin{pmatrix} * & * & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 & 0 \\ 0 & 0 & * & 0 & 0 & 0 \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \end{pmatrix},$$

  isomorphic with $\mathrm{GL}(2, F) \times \mathrm{GL}(1, F) \times \mathrm{GL}(3, F)$.

- In general, the unipotent radical of any staircase group is contained in B.

- The G-set of all flags is partitioned into the orbits of the subflags of the standard flag, since any flag has the same dimension sequence as some subflag of the standard flag. By Lemma 3.2, the parabolic subgroups of G are exactly the conjugates of the staircase groups.

Subgroups Containing B

Lemma 5.3. *The only non-zero subspaces of $V_n(F)$ left invariant by B are the subspaces $V_1, \dots, V_n$ that appear in the standard flag.*

Proof. First observe that each V_i is stabilized by B: if $j \le i$ and $b \in B$, then $b\mathbf{v}_j$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_j$, hence lies in V_i. Therefore $bV_i \subseteq V_i$, and equality follows since b is invertible.
Conversely, let Y be a non-zero subspace of $V_n(F)$ stabilized by B. Choose k minimal such that $Y \subseteq V_k$. Then $Y \not\subseteq V_{k-1}$, so there is some

$$y = \alpha_1\mathbf{v}_1 + \cdots + \alpha_k\mathbf{v}_k \in Y$$

with $\alpha_k \neq 0$. Let $b \in B$ be the matrix which is the identity except that its kth column is y:

$$b = \begin{pmatrix} & \alpha_1 & \\ I_{k-1} & \vdots & 0 \\ & \alpha_{k-1} & \\ 0 & \alpha_k & 0 \\ 0 & 0 & I_{n-k} \end{pmatrix}.$$

This matrix is upper triangular and invertible, since its kth diagonal entry is $\alpha_k \neq 0$. Also $b\mathbf{v}_k = y$. Since Y is stabilized by B and $b^{-1} \in B$, we have

$$\mathbf{v}_k = b^{-1}y \in Y.$$

Now let

$$z = \beta_1\mathbf{v}_1 + \cdots + \beta_k\mathbf{v}_k \in V_k - V_{k-1}.$$

Then $\beta_k \neq 0$, and the same matrix construction gives some $c \in B$ with $c\mathbf{v}_k = z$. Since $\mathbf{v}_k \in Y$ and Y is stabilized by B, it follows that $z \in Y$. Thus every vector in $V_k - V_{k-1}$ lies in Y. In particular, if $w \in V_{k-1}$, then $w + \mathbf{v}_k \in Y$, and since $\mathbf{v}_k \in Y$, we get $w \in Y$. Hence $V_{k-1} \subseteq Y$, and together with $\mathbf{v}_k \in Y$ this gives $V_k \subseteq Y$. Since $Y \subseteq V_k$ by choice of k, we conclude that $Y = V_k$. □

Theorem 5.4. *The only subgroups of G that contain B are the staircase groups.*

Proof. First observe that every staircase group contains B, because B stabilizes every subspace in the standard flag and therefore stabilizes every subflag of the standard flag. We prove the converse. Let H be a subgroup with

$$B \le H \le G.$$

If a subspace of $V_n(F)$ is stabilized by H, then it is also stabilized by B. Hence Lemma 5.3 shows that the only possible non-zero H-invariant subspaces are among

$$V_1, \dots, V_n.$$

List the ones which are actually stabilized by H as

$$V_{a_1}, \dots, V_{a_r}, \qquad 1 \le a_1 < \cdots < a_r = n,$$

and put $a_0 = 0$. Let P be the staircase group stabilizing the subflag

$$0 \subset V_{a_1} \subset \cdots \subset V_{a_r} = V_n(F).$$

Then $H \le P$. We shall prove that $H = P$.
First we prove the basic case in which the only H-invariant subspaces are 0 and $V_n(F)$. In this case we claim that $H = G$. If $n = 1$ this is immediate, so assume $n > 1$. The subspace

$$Y = \langle h\mathbf{v}_1 \mid h \in H \rangle$$

is non-zero and H-invariant, so by assumption $Y = V_n(F)$. Thus some $h \in H$ sends $\mathbf{v}_1$ to a vector

$$h\mathbf{v}_1 = \alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n$$

with $\alpha_n \neq 0$. By Bruhat Decomposition Theorem, write $h = b_1wb_2$ with $b_1, b_2 \in B$ and $w \in W$. Since $B \le H$ and $h \in H$, it follows that $w \in H$. By Corollary 4.10, the fact that the last non-zero term in

$h\mathbf{v}_1$ is the $\mathbf{v}_n$-term implies that $w(1) = n$. Let $j = w^{-1}(1)$. Then $j > 1$, so $X_{1j} \le B \le H$. Using Lemma 4.6(v), we get

$$wX_{1j}(\alpha)w^{-1} = X_{n1}(\alpha)$$

for every $\alpha \in F$. Hence $X_{n1} \le H$.

We now get all the remaining root subgroups from this one lower root subgroup. The upper root subgroups X_{ij} with $i < j$ already lie in B, and hence in H. If $n = 2$, this already gives every root subgroup. If $n > 2$, then for $1 < i < n$ and $\alpha \in F$, Lemma 4.6(iv) gives

$$X_{ni}(\alpha) = [X_{n1}(\alpha), X_{1i}(1)] \in H$$

and

$$X_{i1}(\alpha) = [X_{in}(\alpha), X_{n1}(1)] \in H.$$

For distinct $1 < i, j < n$, the same commutator formula gives

$$X_{ij}(\alpha) = [X_{i1}(\alpha), X_{1j}(1)] \in H.$$

Thus H contains every root subgroup X_{ij}, and it also contains every invertible diagonal matrix because all diagonal matrices lie in B. By Theorem 4.12, $H = G$. This proves the basic case.

We return to the general subgroup H. By Proposition 5.2, the staircase group P decomposes as

$$P = U_P \rtimes L_P.$$

Since P is a staircase group, its unipotent radical U_P is contained in B, so $U_P \le H$. Therefore it is enough to show that a Levi complement L_P is contained in H. We choose the standard Levi complement whose blocks are

$$F\mathbf{v}_{a_{k-1}+1} \oplus \cdots \oplus F\mathbf{v}_{a_k}, \qquad 1 \le k \le r.$$

Fix one block, and write

$$s = a_{k-1} + 1, \qquad t = a_k.$$

If $s = t$, there is nothing to prove for this block. Assume $s < t$. Because there is no H-invariant standard flag subspace strictly between $V_{a_{k-1}}$ and V_{a_k}, the subspace

$$V_{a_{k-1}} + \langle h\mathbf{v}_s \mid h \in H \rangle$$

must be all of V_t. Hence there is some $h \in H$ such that the coefficient of $\mathbf{v}_t$ in $h\mathbf{v}_s$ is non-zero. Write $h = b_1wb_2$ by Bruhat Decomposition Theorem. Again $w \in H$. Also $w \in P$, because $b_1, b_2 \in B \le P$ and $h \in P$. Thus w preserves each set of indices

$$\{1, \ldots, a_\ell\}$$

which comes from the subflag stabilized by P. In particular, w sends the block $\{s, \ldots, t\}$ to itself. The row-pivot description in Corollary 4.10 applied to the sth column says that $w(s) = t$, because the first $s - 1$ columns use only rows $1, \ldots, s - 1$, while the sth column has no entries below row t and has a non-zero entry in row t. Let $q = w^{-1}(s)$. Since w preserves the block and $w(s) = t$, we have $s < q \le t$. Therefore $X_{sq} \le B \le H$, and Lemma 4.6(v) gives

$$wX_{sq}(\alpha)w^{-1} = X_{ts}(\alpha)$$

for every $\alpha \in F$. Hence $X_{ts} \le H$.

Now the same commutator ladder used in the basic case works inside this block. All upper root subgroups X_{ij} with $s \le i < j \le t$ lie in B, hence in H. Together with $X_{ts} \le H$, this gives every root subgroup inside the block: for $s < i < t$,

$$X_{ti}(\alpha) = [X_{ts}(\alpha), X_{si}(1)] \in H$$

and

$$X_{is}(\alpha) = [X_{it}(\alpha), X_{ts}(1)] \in H,$$

while for distinct $s < i, j < t$,

$$X_{ij}(\alpha) = [X_{is}(\alpha), X_{sj}(1)] \in H.$$

Thus, for each block of L_P, the subgroup H contains all root subgroups supported inside that block. Since H also contains all diagonal matrices, Theorem 4.12 applied block-by-block shows that $L_P \le H$. Hence

$$P = U_P \rtimes L_P \le H.$$

We already had $H \le P$, so $H = P$. Therefore every subgroup of G containing B is a staircase group. $\square$

Exercises

Let $G = \mathrm{GL}(n, F)$, where $n \geq 2$.

Exercise 5.1. Suppose that $g \in G$ fixes some $0 \neq \mathbf{v} \in V_n(F)$ and induces the identity transformation on $V_n(F)/F\mathbf{v}$. Show that g is conjugate with an element of the root subgroup X_{12}.

Solution. Since g induces the identity transformation on $V_n(F)/F\mathbf{v}$, for every $\mathbf{x} \in V_n(F)$ we have

$$g\mathbf{x} + F\mathbf{v} = \mathbf{x} + F\mathbf{v}.$$

Equivalently,

$$g\mathbf{x} - \mathbf{x} \in F\mathbf{v}.$$

Thus there is a scalar $\lambda(\mathbf{x}) \in F$ such that

$$g\mathbf{x} = \mathbf{x} + \lambda(\mathbf{x})\mathbf{v}.$$

The scalar $\lambda(\mathbf{x})$ is unique because $\mathbf{v} \neq 0$. The assignment $\lambda\colon V_n(F) \to F$ is linear: for example, comparing

$$g(\mathbf{x} + \mathbf{y}) = (\mathbf{x} + \mathbf{y}) + \lambda(\mathbf{x} + \mathbf{y})\mathbf{v}$$

with

$$g\mathbf{x} + g\mathbf{y} = \mathbf{x} + \lambda(\mathbf{x})\mathbf{v} + \mathbf{y} + \lambda(\mathbf{y})\mathbf{v}$$

gives $\lambda(\mathbf{x} + \mathbf{y}) = \lambda(\mathbf{x}) + \lambda(\mathbf{y})$, and scalar multiplication is similar. Also $g\mathbf{v} = \mathbf{v}$, so $\lambda(\mathbf{v}) = 0$.
If $\lambda = 0$, then $g\mathbf{x} = \mathbf{x}$ for every $\mathbf{x}$, so $g = I = X_{12}(0)$, and there is nothing to prove. Assume now that $\lambda \neq 0$. Choose $\mathbf{w}_1 = \mathbf{v}$. Since $\lambda \neq 0$, there exists some vector $\mathbf{w}_2$ with

$$\lambda(\mathbf{w}_2) = \alpha \neq 0.$$

Then

$$g\mathbf{w}_2 = \mathbf{w}_2 + \alpha\mathbf{w}_1.$$

The kernel of λ has dimension $n - 1$. Since $\lambda(\mathbf{w}_1) = 0$, we have $\mathbf{w}_1 \in \ker\lambda$. Choose vectors $\mathbf{w}_3, \ldots, \mathbf{w}_n$ such that

$$\mathbf{w}_1, \mathbf{w}_3, \ldots, \mathbf{w}_n$$

is a basis of $\ker\lambda$. Since $\mathbf{w}_2 \notin \ker\lambda$, the list $\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n$ is a basis of $V_n(F)$. In this basis, g sends

$$\mathbf{w}_1 \mapsto \mathbf{w}_1, \qquad \mathbf{w}_2 \mapsto \mathbf{w}_2 + \alpha\mathbf{w}_1, \qquad \mathbf{w}_i \mapsto \mathbf{w}_i \quad (i \geq 3).$$

Therefore the matrix of g in the basis $\mathbf{w}_1, \ldots, \mathbf{w}_n$ is $X_{12}(\alpha)$. If $p \in G$ is the change-of-basis matrix whose columns are $\mathbf{w}_1, \ldots, \mathbf{w}_n$, then

$$p^{-1}gp = X_{12}(\alpha).$$

Hence g is conjugate with an element of the root subgroup X_{12}. □

Exercise 5.2. We say a basis $\mathcal{B}$ of $V_n(F)$ *belongs* to a complete flag $(V_1, \ldots, V_n)$ if each V_i contains exactly one element of $\mathcal{B}$ that is not in V_{i-1}.

(a) Show that if $(V_1, \ldots, V_n)$ and $(W_1, \ldots, W_n)$ are complete flags on $V_n(F)$, then there is a basis of $V_n(F)$ that belongs to both flags.

(b) Use part (a) to give a different proof of the Bruhat Decomposition Theorem.

Solution.

(a) Put $V_0 = W_0 = 0$. We build the required basis one vector at a time from intersections of the two flags. Fix $i \in \{1, \ldots, n\}$. Since $W_n = V_n(F)$, we have

$$V_i \cap W_n = V_i,$$

and this is not contained in V_{i-1}. Therefore there is a smallest integer j_i such that

$$V_i \cap W_{j_i} \not\leq V_{i-1}.$$

Choose
$$\mathbf{b}_i \in (V_i \cap W_{j_i}) - V_{i-1}.$$
This means that $\mathbf{b}_i$ enters the V-flag at the ith step, and it enters the W-flag no later than the j_ith step.
First we check that these vectors form a basis belonging to the V-flag. Since $\mathbf{b}_1 \in V_1$ and $\mathbf{b}_1 \neq 0$, the first vector works. Suppose that $\mathbf{b}_1, \dots, \mathbf{b}_{i-1}$ are independent and span V_{i-1}. Since $\mathbf{b}_i \notin V_{i-1}$, the vector $\mathbf{b}_i$ is not in the span of the previous vectors. Hence $\mathbf{b}_1, \dots, \mathbf{b}_i$ are independent. They are i independent vectors inside the i-dimensional space V_i, so they span V_i. By induction,
$$V_i = \langle \mathbf{b}_1, \dots, \mathbf{b}_i \rangle$$
for every i. Thus the basis belongs to the V-flag.
It remains to show that the same basis also belongs to the W-flag. Fix $m \in \{1, \dots, n\}$. For each i, the space
$$V_{i-1} \cap W_m$$
is contained in $V_i \cap W_m$, and the dimension can increase by at most 1 because V_i/V_{i-1} is one-dimensional. By the definition of j_i, this increase happens exactly when $j_i \leq m$. Therefore
$$\#\{i \mid j_i \leq m\} = \sum_{i=1}^{n} \big(\dim(V_i \cap W_m) - \dim(V_{i-1} \cap W_m)\big).$$
The sum telescopes:
$$\sum_{i=1}^{n} \big(\dim(V_i \cap W_m) - \dim(V_{i-1} \cap W_m)\big) = \dim(V_n \cap W_m) - \dim(V_0 \cap W_m) = \dim W_m = m.$$
Also, by the minimality of j_i, the vector $\mathbf{b}_i$ lies in W_m exactly when $j_i \leq m$. Thus exactly m of the vectors $\mathbf{b}_1, \dots, \mathbf{b}_n$ lie in W_m. These m vectors are independent and lie inside the m-dimensional space W_m, so they form a basis of W_m. Hence, when we pass from W_{m-1} to W_m, exactly one new basis vector appears. Therefore the same basis belongs to the W-flag as well.

(b) Let
$$\mathcal{F}_0 = (V_1, \dots, V_n)$$
be the standard complete flag, and let $g \in G$. Then
$$g\mathcal{F}_0 = (gV_1, \dots, gV_n)$$
is also a complete flag. By part (a), there is a basis
$$\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$$
which belongs to both $\mathcal{F}_0$ and $g\mathcal{F}_0$. Order the basis so that
$$\mathbf{b}_i \in V_i - V_{i-1}$$
for every i. Let a be the matrix whose ith column is $\mathbf{b}_i$. Since the ith column of a lies in V_i, the matrix a is upper triangular. Since its columns form a basis, a is invertible. Hence
$$a \in B.$$
Now use the fact that the same basis also belongs to $g\mathcal{F}_0$. For each r, exactly one basis vector lies in
$$gV_r - gV_{r-1}.$$
Thus there is a permutation σ of $\{1, \dots, n\}$ such that
$$\mathbf{b}_{\sigma(r)} \in gV_r - gV_{r-1}$$
for every r. Let $w \in W$ be the permutation matrix which reorders the columns of a so that the rth column of aw is $\mathbf{b}_{\sigma(r)}$. Then
$$g^{-1}\mathbf{b}_{\sigma(r)} \in V_r - V_{r-1}$$

for every r. Therefore the rth column of $g^{-1}aw$ lies in V_r. It follows that $g^{-1}aw$ is upper triangular. It is also invertible, so

$$g^{-1}aw \in B.$$

Write

$$c = g^{-1}aw \in B.$$

Then

$$aw = gc,$$

and therefore

$$g = awc^{-1}.$$

Since $a \in B$, $w \in W$, and $c^{-1} \in B$, we have $g \in BWB$. As $g \in G$ was arbitrary, this gives

$$G \subseteq BWB.$$

The reverse inclusion $BWB \subseteq G$ is immediate from $B \leq G$ and $W \leq G$. Hence

$$G = BWB,$$

which is the Bruhat Decomposition Theorem.

□

Exercise 5.3. Let P and Q be distinct parabolic subgroups of G containing the standard Borel subgroup B of G (so that, by Theorem 5.4, P and Q are staircase groups).

(a) Show that P and Q are not conjugate in G.

(b) If L_P and L_Q are corresponding Levi complements, find conditions on P and Q that determine when L_P and L_Q will be conjugate in G.

Solution.

(a) By Theorem 5.4, both P and Q are staircase groups. Thus there are subflags of the standard flag

$$0 \subset V_{a_1} \subset \cdots \subset V_{a_r} = V_n(F)$$

and

$$0 \subset V_{b_1} \subset \cdots \subset V_{b_s} = V_n(F)$$

such that P is the stabilizer of the first subflag and Q is the stabilizer of the second. If the two lists of indices were the same, then the two stabilized standard subflags would be the same, and hence their stabilizers would be the same. Since $P \neq Q$, we know that

$$\{a_1, \ldots, a_r\} \neq \{b_1, \ldots, b_s\}.$$

We first record the invariant-subspace fact we need. Let R be the staircase group stabilizing

$$0 \subset V_{c_1} \subset \cdots \subset V_{c_t} = V_n(F),$$

and put $c_0 = 0$. We claim that the only non-zero subspaces of $V_n(F)$ stabilized by R are

$$V_{c_1}, \ldots, V_{c_t}.$$

This is the block version of Lemma 5.3. Let Y be a non-zero R-invariant subspace. Choose k minimal such that

$$Y \leq V_{c_k}.$$

Then $Y \not\leq V_{c_{k-1}}$, so the image of Y in the quotient

$$V_{c_k}/V_{c_{k-1}}$$

is non-zero. By Proposition 5.2 and the block description of a staircase group, the standard Levi complement of R acts as the full general linear group on the block

$$F\mathbf{v}_{c_{k-1}+1} \oplus \cdots \oplus F\mathbf{v}_{c_k}.$$

Hence it acts on $V_{c_k}/V_{c_{k-1}}$ as the full group $\mathrm{GL}(c_k - c_{k-1}, F)$. Since Y is R-invariant, the image of Y in this quotient is invariant under this full general linear group. But the only non-zero subspace invariant under the full general linear group is the whole quotient, because an invertible linear map can send any non-zero vector to any other non-zero vector. Therefore

$$Y + V_{c_{k-1}} = V_{c_k}.$$

We next show that $V_{c_{k-1}} \leq Y$. If $k = 1$, this is immediate because $V_{c_0} = 0$. Assume $k > 1$. Since $Y + V_{c_{k-1}} = V_{c_k}$, there is some vector $y \in Y$ whose component in the kth block is non-zero. Choose an index m in the kth block such that the coefficient of $\mathbf{v}_m$ in y is a non-zero scalar β. Now let $x \in V_{c_{k-1}}$. Write

$$x = \gamma_1 \mathbf{v}_1 + \cdots + \gamma_{c_{k-1}} \mathbf{v}_{c_{k-1}}.$$

For each $\ell \leq c_{k-1}$, the root subgroup element $X_{\ell m}(\alpha)$ sends $\mathbf{v}_m$ to $\mathbf{v}_m + \alpha \mathbf{v}_\ell$. Therefore, if we choose

$$u = \prod_{\ell=1}^{c_{k-1}} X_{\ell m}(\gamma_\ell/\beta),$$

then u sends y to $y + x$. These are upper root subgroup elements because $\ell < m$, so they lie in B. Since $B \leq R$, we have $u \in R$. Since Y is R-invariant, both y and $uy = y + x$ lie in Y, and hence

$$x = (y + x) - y \in Y.$$

Since $x \in V_{c_{k-1}}$ was arbitrary, $V_{c_{k-1}} \leq Y$. Together with $Y + V_{c_{k-1}} = V_{c_k}$, this gives $V_{c_k} \leq Y$. Since $Y \leq V_{c_k}$ by the choice of k, we get $Y = V_{c_k}$. This proves the claim.
Now suppose, for a contradiction, that P and Q are conjugate in G. Then there exists some $g \in G$ such that

$$Q = gPg^{-1}.$$

For each i, the subspace V_{a_i} is stabilized by P. We claim that gV_{a_i} is stabilized by Q. Indeed, if $q \in Q$, then $q = gpg^{-1}$ for some $p \in P$. If $gv \in gV_{a_i}$ with $v \in V_{a_i}$, then

$$q(gv) = gpg^{-1}gv = gpv.$$

Since p stabilizes V_{a_i}, we have $pv \in V_{a_i}$, and hence $gpv \in gV_{a_i}$. Thus gV_{a_i} is Q-invariant. By the claim applied to Q, the subspace gV_{a_i} must be one of the subspaces

$$V_{b_1}, \ldots, V_{b_s}.$$

Taking dimensions gives

$$a_i = \dim gV_{a_i} = b_j$$

for some j. Therefore

$$\{a_1, \ldots, a_r\} \subseteq \{b_1, \ldots, b_s\}.$$

Applying the same argument to $P = g^{-1}Qg$ gives the reverse inclusion. Hence

$$\{a_1, \ldots, a_r\} = \{b_1, \ldots, b_s\},$$

contradicting the fact that these two sets are different. Therefore no such g exists, and P and Q are not conjugate in G.

(b) Keep the notation from part (a), and put $a_0 = b_0 = 0$. The standard Levi complement of P is the subgroup which preserves each block

$$Y_i = F\mathbf{v}_{a_{i-1}+1} \oplus \cdots \oplus F\mathbf{v}_{a_i},$$

and acts on that block as the full group $\mathrm{GL}(Y_i)$. Thus its block sizes are

$$d_i = \dim Y_i = a_i - a_{i-1} \qquad (1 \leq i \leq r).$$

Similarly, the standard Levi complement of Q has blocks

$$Z_j = F\mathbf{v}_{b_{j-1}+1} \oplus \cdots \oplus F\mathbf{v}_{b_j}$$

of sizes

$$e_j = \dim Z_j = b_j - b_{j-1} \qquad (1 \le j \le s).$$

We claim that L_P and L_Q are conjugate in G exactly when the two lists

$$d_1, \dots, d_r \qquad \text{and} \qquad e_1, \dots, e_s$$

agree up to permutation.

First suppose that the block sizes agree up to permutation. Then $r = s$, and after reordering the Q-blocks there is a bijection σ such that

$$d_i = e_{\sigma(i)}$$

for every i. Choose a permutation matrix w which sends the standard basis vectors in Y_i onto the standard basis vectors in $Z_{\sigma(i)}$. Conjugating by w sends an arbitrary invertible linear transformation on Y_i to an arbitrary invertible linear transformation on $Z_{\sigma(i)}$. Therefore

$$wL_P w^{-1} = L_Q.$$

Hence L_P and L_Q are conjugate.

Conversely, suppose that $L_Q = gL_P g^{-1}$ for some $g \in G$. Conjugation by g sends the block decomposition

$$V_n(F) = Y_1 \oplus \cdots \oplus Y_r$$

to

$$V_n(F) = gY_1 \oplus \cdots \oplus gY_r.$$

On each gY_i, the conjugate subgroup $gL_P g^{-1} = L_Q$ contains a factor acting as the full group $\mathrm{GL}(gY_i)$. Thus the spaces gY_i are the same kind of full general-linear blocks for L_Q as the standard blocks Z_j. The standard block decomposition of L_Q has block dimensions $e_1, \dots, e_s$, so the dimensions

$$\dim gY_i = \dim Y_i = d_i$$

must be exactly the same dimensions as the e_j, possibly in a different order. Hence the block sizes of P and Q agree up to permutation.

Therefore the condition is:

$$L_P \text{ and } L_Q \text{ are conjugate in } G \quad \Longleftrightarrow \quad \{a_i - a_{i-1} \mid 1 \le i \le r\} = \{b_j - b_{j-1} \mid 1 \le j \le s\}$$

as multisets.

□

Exercise 5.4. Prove Proposition 5.2.

Solution. This is exactly Proposition 5.2, proved above. □

Exercise 5.5. Prove the following generalization of Lemma 5.3: If P is a parabolic subgroup of G which is the stabilizer of the flag $(W_1, \dots, W_r)$, then the W_i are the only subspaces of $V_n(F)$ left invariant by P.

Solution. Let

$$0 \subset W_1 \subset \cdots \subset W_r = V_n(F)$$

be the flag stabilized by P, and put $c_i = \dim_F W_i$. By the transitivity of the G-action on flags with a fixed dimension sequence, choose $g \in G$ such that

$$gW_i = V_{c_i}$$

for every i. Hence gPg^{-1} is the staircase group stabilizing the standard subflag

$$0 \subset V_{c_1} \subset \cdots \subset V_{c_r} = V_n(F).$$

Now let Y be a non-zero subspace of $V_n(F)$ stabilized by P. Then gY is stabilized by gPg^{-1}: if $q = gpg^{-1} \in gPg^{-1}$ and $gy \in gY$, with $p \in P$ and $y \in Y$, then

$$q(gy) = gpy \in gY.$$

By the staircase-group claim proved in Exercise 5.3(a), the only non-zero subspaces stabilized by gPg^{-1} are

$$V_{c_1}, \dots, V_{c_r}.$$

Therefore $gY = V_{c_i}$ for some i. Applying g^{-1} gives

$$Y = g^{-1}V_{c_i} = W_i.$$

Thus the only non-zero subspaces stabilized by P are $W_1, \dots, W_r$. Each W_i is stabilized by P by the definition of P as the stabilizer of the flag, so these are exactly the non-zero P-invariant subspaces. □

Exercise 5.6. (cont.) Using Exercise 5.5, show that any parabolic subgroup of G is self-normalizing.

Solution. Let P be the stabilizer of the flag

$$0 \subset W_1 \subset \cdots \subset W_r = V_n(F).$$

We prove that $N_G(P) = P$. The inclusion

$$P \leq N_G(P)$$

is automatic, because every element of a subgroup normalizes that subgroup. It remains to prove the reverse inclusion.
Let $g \in N_G(P)$. We want to show that $g \in P$, i.e. that g stabilizes every W_i. Fix i. We first show that gW_i is stabilized by P. Let $p \in P$ and let $gw \in gW_i$, where $w \in W_i$. Since $g \in N_G(P)$, we have

$$g^{-1}pg \in P.$$

Since P stabilizes W_i, this gives

$$(g^{-1}pg)w \in W_i.$$

Applying g to both sides, we get

$$p(gw) = g(g^{-1}pg)w \in gW_i.$$

Thus gW_i is a non-zero subspace stabilized by P. By Exercise 5.5, the only non-zero subspaces stabilized by P are exactly the spaces in its flag, so

$$gW_i = W_j$$

for some j. But g preserves dimension, so

$$\dim_F W_j = \dim_F(gW_i) = \dim_F W_i.$$

The dimensions in a flag are strictly increasing, so $j = i$. Hence

$$gW_i = W_i$$

for every i. Therefore g stabilizes the whole flag, and so $g \in P$. This proves

$$N_G(P) \leq P.$$

Combining the two inclusions gives $N_G(P) = P$, so every parabolic subgroup of G is self-normalizing. □

Exercise 5.7. Complete the outlined proof of Theorem 5.4.

Solution. Let H be a subgroup of G with $B \leq H \leq G$. If a subspace is stabilized by H, then it is also stabilized by B. Hence Lemma 5.3 says that the only possible non-zero subspaces stabilized by H are among the standard flag subspaces. List the ones actually stabilized by H as

$$V_{a_1}, \ldots, V_{a_r}, \qquad 1 \leq a_1 < \cdots < a_r = n,$$

and put $a_0 = 0$. Let P be the staircase group stabilizing the subflag

$$0 \subset V_{a_1} \subset \cdots \subset V_{a_r} = V_n(F).$$

Then $H \leq P$, because every element of H stabilizes each of the spaces in this subflag. We prove the reverse inclusion.
Step 1: reduce to the Levi complement. By Proposition 5.2, we may write

$$P = U_P \rtimes L_P.$$

Since P is a staircase group, its unipotent radical U_P is contained in B. Hence

$$U_P \leq B \leq H.$$

Therefore, to prove $P \leq H$, it remains only to prove

$$L_P \leq H.$$

Choose the standard Levi complement whose blocks are

$$F\mathbf{v}_{a_{k-1}+1} \oplus \cdots \oplus F\mathbf{v}_{a_k}, \qquad 1 \leq k \leq r.$$

We will show that H contains the full general linear group on each one of these blocks.
Step 2: find one lower root subgroup in a fixed block. Fix a block, and write

$$s = a_{k-1} + 1, \qquad t = a_k.$$

If $s = t$, then the block is one-dimensional, and there are no root subgroups to find in it. Assume $s < t$. Since there is no H-invariant standard flag subspace strictly between $V_{a_{k-1}}$ and V_{a_k}, the H-invariant subspace

$$V_{a_{k-1}} + \langle h\mathbf{v}_s \mid h \in H \rangle$$

is stabilized by H. It contains $\mathbf{v}_s$, so it is larger than $V_{a_{k-1}}$; it is contained in V_t, because $H \leq P$ and P stabilizes V_t. Therefore it must equal V_t. Thus there is some $h \in H$ such that the coefficient of $\mathbf{v}_t$ in $h\mathbf{v}_s$ is non-zero.
Write $h = b_1 w b_2$ by the Bruhat Decomposition Theorem, with $b_1, b_2 \in B$ and $w \in W$. Since $B \leq H$ and $h \in H$, we also have $w \in H$. Since $H \leq P$, the element w lies in P, so it permutes the indices inside each block of the staircase. The row-pivot description in Corollary 4.10, applied to the sth column, gives

$$w(s) = t.$$

Indeed, the first $s - 1$ columns of h lie in V_{s-1}, while the sth column lies in V_t and has a non-zero $\mathbf{v}_t$-coefficient. Thus the row chosen by the Bruhat pivot construction for column s is row t.
Let $q = w^{-1}(s)$. Because w preserves the block $\{s, \ldots, t\}$ and sends s to t, we have

$$s < q \leq t.$$

Hence $X_{sq} \leq B \leq H$. By Lemma 4.6(v),

$$wX_{sq}(\alpha)w^{-1} = X_{ts}(\alpha)$$

for every $\alpha \in F$. Therefore

$$X_{ts} \leq H.$$

Step 3: use commutators to get all root subgroups in the block. The upper root subgroups X_{ij} with $s \leq i < j \leq t$ already lie in B, and hence in H. Together with $X_{ts} \leq H$, the commutator formula in Lemma 4.6(iv) gives the remaining root subgroups inside the block. For $s < i < t$ and $\alpha \in F$,

$$X_{ti}(\alpha) = [X_{ts}(\alpha), X_{si}(1)] \in H$$

and
$$X_{is}(\alpha) = [X_{it}(\alpha), X_{ts}(1)] \in H.$$
For distinct $s < i, j < t$,
$$X_{ij}(\alpha) = [X_{is}(\alpha), X_{sj}(1)] \in H.$$
Thus H contains every root subgroup X_{ij} supported inside this block. Since H also contains all diagonal matrices, Theorem 4.12 applied to this block shows that H contains the full general linear group on the block.

Step 4: finish block by block. Repeating Steps 2 and 3 for every block of the staircase shows that every block factor of L_P is contained in H. Hence
$$L_P \leq H.$$
Since $U_P \leq H$ as well, we have
$$P = U_P \rtimes L_P \leq H.$$
We already had $H \leq P$, so $H = P$. Therefore every subgroup of G containing B is a staircase group, which completes the general case of Theorem 5.4. □

Exercise 5.8. Assume that the case $r = 1$ of Theorem 5.4 holds; we sketch an alternate proof of the general case of Theorem 5.4. We use induction on n, where $G = \mathrm{GL}(n, F)$. Let H be a subgroup of G which contains B, and suppose that H stabilizes V_m, where $1 \leq m < n$. Let S be the stabilizer of V_m; observe that $H \leq S$. We can write $S = U \rtimes (K_1 \times K_2)$, where the elements of $K_1 \cong \mathrm{GL}(m, F)$ have their only non-zero entries in the first m rows and columns, and where K_2 is a direct product of general linear groups whose elements have their non-zero entries concentrated in the last $n - m$ rows and columns. Since $U \leq B \leq H$, by Exercise 2.10 we have $H = U \rtimes Q$ for some $Q \leq K_1 \times K_2$. Use Goursat's theorem (Exercise 2.8), Exercise 5.6, and the induction hypothesis to analyze Q and thereby conclude that H is a staircase group.

Solution. We prove the general case of Theorem 5.4 by strong induction on n, assuming that the case $r = 1$ is already known. Here the case $r = 1$ means the following: if $B \leq H \leq \mathrm{GL}(n, F)$ and H stabilizes no non-zero proper standard flag subspace, then $H = \mathrm{GL}(n, F)$. The case $n = 1$ is immediate, so assume $n > 1$ and that the result is known for all smaller dimensions.

Let H be a subgroup with $B \leq H \leq G$. If H stabilizes no proper standard flag subspace V_m with $1 \leq m < n$, then the assumed $r = 1$ case gives $H = G$, which is a staircase group. Thus we may assume that H stabilizes some V_m, where $1 \leq m < n$. Let S be the stabilizer of V_m. Then $H \leq S$. With respect to the decomposition
$$V_n(F) = V_m \oplus \langle \mathbf{v}_{m+1}, \ldots, \mathbf{v}_n \rangle,$$
the elements of S have block form
$$\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}.$$
Let U be the subgroup with identity diagonal blocks and arbitrary upper-right block, let $K_1 \cong \mathrm{GL}(m, F)$ be the upper-left diagonal block group, and let $K_2 \cong \mathrm{GL}(n - m, F)$ be the lower-right diagonal block group. Then
$$S = U \rtimes (K_1 \times K_2).$$
Also $U \leq B \leq H$. By Exercise 2.10, if a subgroup of a semidirect product contains the normal factor, then it is the semidirect product of that normal factor with its intersection with the complement. Applying this to $U \leq H \leq S$, we get
$$H = U \rtimes Q, \qquad Q = H \cap (K_1 \times K_2).$$
Thus the problem has been reduced to understanding the subgroup Q of the block diagonal group $K_1 \times K_2$.

Step 1: Q contains the two block Borels. Let
$$B_1 = B \cap K_1, \qquad B_2 = B \cap K_2.$$
These are the standard Borel subgroups inside K_1 and K_2. Since every block diagonal element of $B_1 \times B_2$ is also an element of B, we have
$$B_1 \times B_2 \leq B \leq H.$$

But $B_1 \times B_2 \le K_1 \times K_2$, so
$$B_1 \times B_2 \le H \cap (K_1 \times K_2) = Q.$$

Step 2: apply Goursat's theorem to Q. By Exercise 2.8, Goursat's theorem says that a subgroup of a direct product is controlled by its two projections, its two coordinate kernels, and an isomorphism between the corresponding quotient groups. In our case, define

$$P_1 = \{x \in K_1 \mid (x, y) \in Q \text{ for some } y \in K_2\}, \qquad I_1 = \{x \in K_1 \mid (x, 1) \in Q\},$$

and define P_2 and I_2 similarly. Then Exercise 2.8 gives

$$I_i \trianglelefteq P_i \le K_i \qquad (i = 1, 2),$$

together with an isomorphism
$$P_1/I_1 \cong P_2/I_2.$$

Moreover, Q is recovered from these data by

$$Q = \{(x, y) \in P_1 \times P_2 \mid \varphi(xI_1) = yI_2\}.$$

The inclusion $B_1 \times B_2 \le Q$ gives more than just $B_i \le P_i$. Indeed, if $b_1 \in B_1$, then $(b_1, 1) \in Q$, so $b_1 \in I_1$. Similarly, if $b_2 \in B_2$, then $(1, b_2) \in Q$, so $b_2 \in I_2$. Hence

$$B_i \le I_i \le P_i \le K_i \qquad (i = 1, 2).$$

Step 3: use induction and self-normalizing parabolics. Since $K_1 \cong \mathrm{GL}(m, F)$ and $K_2 \cong \mathrm{GL}(n - m, F)$ have smaller dimension than G, the induction hypothesis applies inside both blocks. Because P_i and I_i contain the standard Borel subgroup B_i, the induction hypothesis tells us that P_i and I_i are staircase groups inside K_i. In particular, I_i is a parabolic subgroup of K_i.
Now $I_i \trianglelefteq P_i$. Therefore every element of P_i normalizes I_i, so

$$P_i \le N_{K_i}(I_i).$$

By Exercise 5.6, parabolic subgroups are self-normalizing. Applying this inside K_i, we get

$$N_{K_i}(I_i) = I_i.$$

Therefore
$$P_i \le I_i.$$

Since $I_i \le P_i$ already, we conclude that

$$P_i = I_i \qquad (i = 1, 2).$$

Step 4: the Goursat coupling disappears. Since $P_i = I_i$, both quotient groups in Goursat's theorem are trivial:
$$P_1/I_1 = 1, \qquad P_2/I_2 = 1.$$

Thus the condition $\varphi(xI_1) = yI_2$ in the description of Q is automatic. Hence

$$Q = P_1 \times P_2.$$

The induction hypothesis also tells us that P_1 and P_2 are staircase groups inside their respective blocks.
Step 5: combine the two block staircases. Suppose P_1 stabilizes a standard subflag inside V_m, say

$$0 \subset V_{a_1} \subset \cdots \subset V_{a_r} = V_m.$$

For the lower block, write

$$W_j = \langle \mathbf{v}_{m+1}, \ldots, \mathbf{v}_{m+j} \rangle \qquad (1 \le j \le n - m).$$

Thus $W_1 \subset \cdots \subset W_{n-m}$ is the standard flag inside the lower block. Suppose P_2 stabilizes the lower-block subflag
$$0 \subset W_{b_1} \subset \cdots \subset W_{b_s} = W_{n-m}.$$

When these lower-block spaces are placed back inside $V_n(F)$, we must add the first block V_m. Thus

$$V_m \oplus W_{b_j} = V_{m+b_j} \qquad (1 \leq j \leq s).$$

Thus the two block flags combine to the standard subflag

$$0 \subset V_{a_1} \subset \cdots \subset V_m \subset V_{m+b_1} \subset \cdots \subset V_n(F).$$

The stabilizer of this combined subflag is exactly the group of block upper triangular matrices whose upper-left diagonal block lies in P_1 and whose lower-right diagonal block lies in P_2:

$$U \rtimes (P_1 \times P_2).$$

But this group is H, since

$$H = U \rtimes Q = U \rtimes (P_1 \times P_2).$$

Therefore H is a staircase group. This completes the induction step, and hence proves the general case of Theorem 5.4 from the assumed $r = 1$ case. $\square$

6 The Special Linear Group

Definition

- The *special linear group* is the subgroup $\mathrm{SL}(n,F)$ of $\mathrm{GL}(n,F)$ consisting of all matrices having determinant 1.
- In other words, $\mathrm{SL}(n,F)$ is the kernel of the homomorphism $\det\colon \mathrm{GL}(n,F) \to F^\times$, and hence $\mathrm{SL}(n,F) \trianglelefteq \mathrm{GL}(n,F)$.

Proposition 6.1. *Let $n \in \mathbb{N}$, and let q be a prime power. Then*

$$|\mathrm{SL}(n,q)| = \prod_{k=1}^{n-1}(q^{n+1}-q^k) = q^{\frac{n(n-1)}{2}}(q^n-1)\cdots(q^2-1).$$

Proof. Let F be the field with q elements. The determinant map

$$\det\colon\ \mathrm{GL}(n,F) \to F^\times$$

is surjective: if $\lambda \in F^\times$, then the diagonal matrix with first diagonal entry λ and all other diagonal entries equal to 1 has determinant λ. Its kernel is precisely $\mathrm{SL}(n,F)$. Hence, by the fundamental theorem on homomorphisms,

$$\mathrm{GL}(n,F)/\,\mathrm{SL}(n,F) \cong F^\times.$$

In particular,

$$|\mathrm{GL}(n,q) : \mathrm{SL}(n,q)| = |F^\times| = q-1.$$

Proposition 4.2 gives

$$|\mathrm{GL}(n,q)| = q^{\frac{n(n-1)}{2}}(q^n-1)(q^{n-1}-1)\cdots(q-1).$$

Dividing by the index $q-1$ cancels the final factor, so

$$|\mathrm{SL}(n,q)| = \frac{|\mathrm{GL}(n,q)|}{q-1} = q^{\frac{n(n-1)}{2}}(q^n-1)\cdots(q^2-1).$$

Finally,

$$\prod_{k=1}^{n-1}(q^{n+1}-q^k) = \prod_{k=1}^{n-1} q^k(q^{n+1-k}-1) = q^{1+\cdots+(n-1)}(q^n-1)\cdots(q^2-1),$$

which is the same expression. □

Generation by root subgroups

- We showed in Theorem 4.12 that $\mathrm{GL}(n,F)$ is generated by the root subgroups and the diagonal subgroup. We now establish the analogous result for $\mathrm{SL}(n,F)$.

Proposition 6.2. $\mathrm{SL}(n,F)$ *is generated by the root subgroups X_{ij}.*

Proof. If $n=1$, then $\mathrm{SL}(1,F)=1$, so there is nothing to prove. We assume now that $n \geq 2$. Let H be the subgroup of $\mathrm{GL}(n,F)$ generated by all root subgroups X_{ij}. By Lemma 4.6(i), every transvection has determinant 1, so $H \leq \mathrm{SL}(n,F)$. We prove the reverse containment.
Let $A \in \mathrm{SL}(n,F)$. We shall premultiply A by products of transvections until it becomes the identity. Since the inverse of a transvection is again a transvection by Lemma 4.6(iii), this will imply that $A \in H$.
First we make A upper triangular. By Lemma 4.7, there is a product b of upper triangular transvections such that, for each $1 \leq i \leq n$, the matrix bA has exactly one row whose entries in the first $i-1$ columns are zero and whose entry in the ith column is non-zero. In other words, the rows of bA already have the correct pivot pattern for an upper triangular matrix, but the pivot rows may be in the wrong order.
For distinct i and j, put

$$s_{ij} = X_{ji}(1)X_{ij}(-1)X_{ji}(1).$$

On the two rows indexed by i and j, premultiplication by s_{ij} has the same effect as the matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix};$$

that is, it sends row i to the negative of row j and sends row j to row i. Thus s_{ij} swaps the two rows up to a harmless sign. Using such signed row swaps, we can move the row with the first pivot into the first row, the row with the second pivot into the second row, and so on. Therefore there is a product c of transvections such that

$$cbA$$

is upper triangular.

Next we make the diagonal entries equal to 1. Let M be an upper triangular element of $\mathrm{SL}(n, F)$, with diagonal entries $\lambda_1, \ldots, \lambda_n$. We use the following two-by-two calculation. If $a, c \in F^\times$ and $b' \in F$, then

$$X_{21}(-1)X_{12}(1-c^{-1})X_{21}(c) = \begin{pmatrix} c & 1-c^{-1} \\ 0 & c^{-1} \end{pmatrix},$$

and hence

$$X_{21}(-1)X_{12}(1-c^{-1})X_{21}(c)\begin{pmatrix} a & b' \\ 0 & c \end{pmatrix} = \begin{pmatrix} ac & cb'+c-1 \\ 0 & 1 \end{pmatrix},$$

so the only point is that the two diagonal entries a, c have become $ac, 1$. Applying this calculation to the final two diagonal positions of M changes the final two diagonal entries from λ_{n-1}, λ_n to $\lambda_{n-1}\lambda_n, 1$, while keeping the matrix upper triangular. Applying the same calculation next to positions $n-2, n-1$, then to positions $n-3, n-2$, and continuing leftward, we obtain an upper triangular matrix whose diagonal entries are

$$\lambda_1\lambda_2\cdots\lambda_n, 1, \ldots, 1.$$

Since $M \in \mathrm{SL}(n, F)$, the product of the diagonal entries of M is its determinant, and hence $\lambda_1 \cdots \lambda_n = 1$. Thus a product of transvections premultiplies M to an upper unitriangular matrix. Applying this to $M = cbA$, we find a product d of transvections such that $dcbA$ is upper unitriangular.

Finally, by Lemma 4.11, an upper triangular matrix can be premultiplied by a product of transvections to become the diagonal matrix with the same main diagonal. Since $dcbA$ is upper unitriangular, this diagonal matrix is the identity. Hence there is a product e of transvections such that

$$edcbA = I.$$

It follows that

$$A = b^{-1}c^{-1}d^{-1}e^{-1}.$$

Each factor on the right lies in H, so $A \in H$. Therefore $\mathrm{SL}(n, F) \leq H$, and hence $\mathrm{SL}(n, F) = H$. □

Proposition 6.3. *The subgroups X_{ij} are conjugate in* $\mathrm{SL}(n, F)$.

Proof. If $n = 1$, there are no root subgroups, so there is nothing to prove. Let X_{ij} and $X_{i'j'}$ be two root subgroups. Choose a permutation matrix w such that $w(i) = i'$ and $w(j) = j'$. By Lemma 4.6(v), for every $\alpha \in F$ we have

$$wX_{ij}(\alpha)w^{-1} = X_{i'j'}(\alpha).$$

If $\det w = 1$, then $w \in \mathrm{SL}(n, F)$, and hence $wX_{ij}w^{-1} = X_{i'j'}$.

It remains to handle the case $\det w = -1$. Let d be the diagonal matrix whose first diagonal entry is -1 and whose remaining diagonal entries are 1. Then $\det d = -1$, so $\det(dw) = 1$, and hence $dw \in \mathrm{SL}(n, F)$. For each $\alpha \in F$,

$$(dw)X_{ij}(\alpha)(dw)^{-1} = dX_{i'j'}(\alpha)d^{-1}.$$

Conjugating by d can only change the parameter α by a sign, depending on whether one of i' or j' is equal to 1. Thus

$$dX_{i'j'}(\alpha)d^{-1} = X_{i'j'}(\pm\alpha).$$

As α ranges over F, so does $\pm\alpha$. Therefore

$$(dw)X_{ij}(dw)^{-1} = X_{i'j'}.$$

Thus any two root subgroups are conjugate in $\mathrm{SL}(n, F)$. □

Scalar matrices and the center

- Let Z be the subgroup of $\mathrm{GL}(n,F)$ consisting of the multiples of the identity matrix by elements of $F^\times$. The elements of Z are called the (non-zero) *scalar* matrices. We clearly have $Z \cong F^\times$.

Proposition 6.4. *The center of* $\mathrm{GL}(n,F)$ *is* Z*, and the center of* $\mathrm{SL}(n,F)$ *is* $Z \cap \mathrm{SL}(n,F)$*.*

Proof. If $n=1$, then $\mathrm{GL}(1,F)=F^\times=Z$ and $\mathrm{SL}(1,F)=1=Z\cap\mathrm{SL}(1,F)$, so the result is immediate. We assume now that $n>1$. Let G denote either $\mathrm{GL}(n,F)$ or $\mathrm{SL}(n,F)$. Since scalar matrices commute with every matrix, we have

$$Z \cap G \leq Z(G).$$

It remains to prove the reverse containment. It is enough to show that if an element of G commutes with every element of $\mathrm{SL}(n,F)$, then it is scalar.
Let $M=(m_{rs})$ be such an element of G. Fix distinct indices i and j. By Lemma 4.6(i), the transvection $X_{ij}(1)$ lies in $\mathrm{SL}(n,F)$, so

$$MX_{ij}(1) = X_{ij}(1)M.$$

Writing $X_{ij}(1)=I+E_{ij}$, this equality is equivalent to

$$ME_{ij} = E_{ij}M.$$

Compare entries. The (i,i)-entry of ME_{ij} is 0, while the (i,i)-entry of $E_{ij}M$ is m_{ji}, so $m_{ji}=0$. The (i,j)-entry of ME_{ij} is m_{ii}, while the (i,j)-entry of $E_{ij}M$ is m_{jj}, so $m_{ii}=m_{jj}$.
Since i and j were arbitrary distinct indices, every off-diagonal entry of M is zero, and all diagonal entries are equal. Hence $M=m_{11}I$ is a scalar matrix. Thus $M\in Z\cap G$. Therefore $Z(G)=Z\cap G$. Taking $G=\mathrm{GL}(n,F)$ gives $Z(\mathrm{GL}(n,F))=Z$, and taking $G=\mathrm{SL}(n,F)$ gives $Z(\mathrm{SL}(n,F))=Z\cap\mathrm{SL}(n,F)$. □

Projective groups

- The group $\mathrm{GL}(n,F)/Z$ is called the *projective general linear group*, and the group $\mathrm{SL}(n,F)/Z\cap\mathrm{SL}(n,F)$ is called the *projective special linear group.*
- We denote these groups by $\mathrm{PGL}(n,F)$ and $\mathrm{PSL}(n,F)$, respectively. (Observe that the first isomorphism theorem implies that $\mathrm{PSL}(n,F)\cong Z\,\mathrm{SL}(n,F)/Z \leq \mathrm{GL}(n,F)/Z=\mathrm{PGL}(n,F)$.)
- As $Z\cong F^\times$, we have $|\mathrm{PGL}(n,q)|=|\mathrm{SL}(n,q)|$ for any prime power q; however, to determine $|\mathrm{PSL}(n,q)|$ we must be able to count the number of nth roots of unity in the field of q elements.

Proposition 6.5. *Let q be a prime power. Then*

$$|\mathrm{PSL}(2,q)| = \begin{cases} q^3-q & \text{if } 2 \mid q,\\ \frac{1}{2}(q^3-q) & \text{if } 2\nmid q.\end{cases}$$

Proof. Let F be the field with q elements. By Proposition 6.4,

$$Z(\mathrm{SL}(2,F)) = Z\cap \mathrm{SL}(2,F),$$

and hence

$$|\mathrm{PSL}(2,q)| = \frac{|\mathrm{SL}(2,q)|}{|Z\cap\mathrm{SL}(2,q)|}.$$

By Proposition 6.1, we have

$$|\mathrm{SL}(2,q)| = q(q^2-1) = q^3-q.$$

It remains to count the scalar matrices in $\mathrm{SL}(2,q)$. Such a matrix has the form αI, with $\alpha\in F^\times$, and it lies in $\mathrm{SL}(2,F)$ precisely when

$$\det(\alpha I) = \alpha^2 = 1.$$

Thus α is a root of X^2-1. If $2\nmid q$, then F has characteristic not equal to 2, so

$$X^2-1=(X-1)(X+1)$$

has the two distinct roots 1 and -1. Hence $|Z\cap\mathrm{SL}(2,q)|=2$ in this case. If $2\mid q$, then F has characteristic 2, so $-1=1$ and

$$X^2-1=(X-1)^2$$

has only the single root 1. Hence $|Z\cap\mathrm{SL}(2,q)|=1$ in this case. Dividing the order of $\mathrm{SL}(2,q)$ by these two possible central orders gives the stated formula. □

Simplicity of $\mathrm{PSL}(n, F)$

- Our main objective in this section is to show that $\mathrm{PSL}(n, F)$ is simple whenever $n \geq 2$, except when $n = 2$ and $|F| = 2$ or 3.
- This result dates back to L. E. Dickson, another of the early architects of the mathematical tradition of the University of Chicago, who established it in his 1896 Ph.D. thesis for F a finite field; in 1893 E. H. Moore had established the result for F a finite field and $n = 2$, while in 1870 Camille Jordan had established the result for F a finite field of prime order and n arbitrary.
- We first need two lemmas.

Lemma 6.6. *If* $n \geq 2$*, then every transvection* $X_{ij}(\alpha)$ *is a commutator of elements of* $\mathrm{SL}(n, F)$*, except when* $n = 2$ *and* $|F| = 2$ *or* 3.

Proof. First suppose that $n > 2$. Given distinct i and j, choose k distinct from both of them. By Lemma 4.6(iv),

$$[X_{ik}(\alpha), X_{kj}(1)] = X_{ij}(\alpha).$$

Both $X_{ik}(\alpha)$ and $X_{kj}(1)$ lie in $\mathrm{SL}(n, F)$ by Lemma 4.6(i), so $X_{ij}(\alpha)$ is a commutator of elements of $\mathrm{SL}(n, F)$.
It remains to consider the case $n = 2$. Assume that F is not one of the fields with 2 or 3 elements. Then there is some $\beta \in F^\times$ such that $\beta^2 \neq 1$; otherwise every element of $F^\times$ would be a root of $X^2 - 1$, forcing $|F^\times| \leq 2$. Put

$$d = \begin{pmatrix} \beta & 0 \\ 0 & \beta^{-1} \end{pmatrix}.$$

Then $d \in \mathrm{SL}(2, F)$. For any $\gamma \in F$, a direct calculation gives

$$dX_{12}(\gamma)d^{-1} = X_{12}(\beta^2\gamma),$$

and hence

$$[d, X_{12}(\gamma)] = X_{12}(\beta^2\gamma)X_{12}(-\gamma) = X_{12}((\beta^2 - 1)\gamma).$$

Since $\beta^2 - 1 \neq 0$, choosing $\gamma = \alpha(\beta^2 - 1)^{-1}$ gives

$$X_{12}(\alpha) = [d, X_{12}(\gamma)].$$

Similarly,

$$dX_{21}(\gamma)d^{-1} = X_{21}(\beta^{-2}\gamma),$$

so

$$[d, X_{21}(\gamma)] = X_{21}((\beta^{-2} - 1)\gamma).$$

Since $\beta^{-2} - 1 \neq 0$, choosing $\gamma = \alpha(\beta^{-2} - 1)^{-1}$ gives $X_{21}(\alpha)$ as a commutator of elements of $\mathrm{SL}(2, F)$. This proves the assertion in all non-exceptional cases. □

Consequence

- Observe that Lemma 6.6 and Proposition 6.2 together imply that, except when $n = 2$ and $|F| = 2$ or 3, $\mathrm{SL}(n, F)$ is its own derived group and is consequently also the derived group of $\mathrm{GL}(n, F)$.

Lemma 6.7. *If* $n \geq 2$*, then the action of* $\mathrm{SL}(n, F)$ *on the one-dimensional subspaces of* $V_n(F)$ *is doubly transitive.*

Proof. Let U_1, U_2, W_1, W_2 be one-dimensional subspaces of $V_n(F)$, with $U_1 \neq U_2$ and $W_1 \neq W_2$. Choose non-zero vectors

$$\mathbf{c}_1 \in U_1, \qquad \mathbf{c}_2 \in U_2, \qquad \mathbf{d}_1 \in W_1, \qquad \mathbf{d}_2 \in W_2.$$

Since U_1 and U_2 are distinct one-dimensional subspaces, the vectors $\mathbf{c}_1$ and $\mathbf{c}_2$ are linearly independent. Similarly, $\mathbf{d}_1$ and $\mathbf{d}_2$ are linearly independent. Extend these pairs to bases

$$\mathbf{c}_1, \ldots, \mathbf{c}_n \qquad \text{and} \qquad \mathbf{d}_1, \ldots, \mathbf{d}_n$$

of $V_n(F)$. Let C and D be the matrices whose ith columns are $\mathbf{c}_i$ and $\mathbf{d}_i$, respectively. Then $C, D \in \mathrm{GL}(n, F)$.
Put

$$\varepsilon = \frac{\det D}{\det C},$$

and let E be the diagonal matrix with first diagonal entry ε and all other diagonal entries equal to 1. Then

$$\det(DE^{-1}C^{-1}) = (\det D)\varepsilon^{-1}(\det C)^{-1} = 1,$$

so

$$s = DE^{-1}C^{-1} \in \mathrm{SL}(n, F).$$

Now $C^{-1}\mathbf{c}_1 = \mathbf{v}_1$ and $C^{-1}\mathbf{c}_2 = \mathbf{v}_2$, while $E^{-1}\mathbf{v}_1 = \varepsilon^{-1}\mathbf{v}_1$ and $E^{-1}\mathbf{v}_2 = \mathbf{v}_2$. Hence

$$s\mathbf{c}_1 = \varepsilon^{-1}\mathbf{d}_1, \qquad s\mathbf{c}_2 = \mathbf{d}_2.$$

Thus $sU_1 = W_1$ and $sU_2 = W_2$. Therefore $\mathrm{SL}(n, F)$ sends any ordered pair of distinct one-dimensional subspaces to any other such ordered pair, so the action is doubly transitive. □

Theorem 6.8. *If $n \geq 2$, then $\mathrm{PSL}(n, F)$ is simple, except when $n = 2$ and $|F| = 2$ or 3.*

Proof. Assume that we are not in the exceptional cases. Put

$$S = \mathrm{SL}(n, F), \qquad L = F\mathbf{v}_1, \qquad Z_S = Z \cap S.$$

It is enough to prove that if $N \trianglelefteq S$, then either $N \leq Z_S$ or $N = S$. Indeed, the normal subgroups of

$$\mathrm{PSL}(n, F) = S/Z_S$$

correspond by Correspondence Theorem to the normal subgroups $N \trianglelefteq S$ with $Z_S \leq N \leq S$, via

$$N \longmapsto N/Z_S.$$

If every normal subgroup of S is either contained in Z_S or equal to S, then any normal subgroup N/Z_S of S/Z_S has only two possibilities. In the first case, $Z_S \leq N \leq Z_S$, so $N = Z_S$, and hence

$$N/Z_S = Z_S/Z_S,$$

which is the trivial subgroup of S/Z_S. In the second case, $N = S$, and hence

$$N/Z_S = S/Z_S,$$

which is the whole quotient group. Thus proving the stated assertion about normal subgroups of S proves that $\mathrm{PSL}(n, F)$ is simple.
Let P be the stabilizer of the line L under the action of S on the one-dimensional subspaces of $V_n(F)$. By Lemma 6.7 and Proposition 3.8, the subgroup P is maximal in S. Now let

$$K = \left\{ I + \sum_{j=2}^{n} \alpha_j E_{1j} \;\middle|\; \alpha_j \in F \right\}.$$

Here E_{1j} denotes the matrix with 1 in the $(1, j)$-entry and 0 everywhere else. Equivalently, K is the subgroup of upper unitriangular matrices whose only possible non-zero entries away from the main diagonal lie in the first row. Since $E_{1j}E_{1k} = 0$ for all $j, k \geq 2$, multiplication in K just adds the parameters α_j, and hence K is abelian. Also, K is exactly the subgroup of P acting trivially on L and on the quotient space $V_n(F)/L$; this description is invariant under conjugation by elements of P, because P stabilizes L. Hence $K \trianglelefteq P$.
Let $N \trianglelefteq S$. First suppose that $N \leq P$. Then N stabilizes L. If $s \in S$, then $sNs^{-1} = N$, and hence N stabilizes sL. Since the action of S on one-dimensional subspaces is transitive by Lemma 6.7, it follows that N stabilizes every one-dimensional subspace of $V_n(F)$. In particular, every element of N stabilizes each standard line $F\mathbf{v}_i$, so every element of N is diagonal. If such an element has diagonal entries $\lambda_1, \ldots, \lambda_n$, then stabilizing each line $F(\mathbf{v}_i + \mathbf{v}_j)$ forces $\lambda_i = \lambda_j$ for all $i \neq j$. Thus every element of N is scalar, and hence $N \leq Z_S$.

Now suppose that $N \not\leq P$. Since $N \trianglelefteq S$, the product PN is a subgroup of S. It properly contains P, so the maximality of P gives

$$PN = S.$$

Let $\eta\colon S \to S/N$ be the natural map. Since $K \trianglelefteq P$, we have $\eta(K) \trianglelefteq \eta(P)$. But

$$\eta(P) = PN/N = S/N,$$

so $\eta(K) \trianglelefteq S/N$. By Correspondence Theorem, its inverse image

$$\eta^{-1}(\eta(K)) = KN$$

is normal in S. Since K is generated by the root subgroups $X_{12}, \dots, X_{1n}$, and since KN is normal in S, the subgroup KN contains all S-conjugates of these root subgroups. By Proposition 6.3, all root subgroups are conjugate in S, so KN contains every X_{ij}. Proposition 6.2 then gives

$$KN = S.$$

By First Isomorphism Theorem,

$$S/N = KN/N \cong K/(K \cap N).$$

Since K is abelian, the quotient S/N is abelian. Therefore Proposition 2.6 gives $S' \leq N$. By Lemma 6.6, every transvection $X_{ij}(\alpha)$ is a commutator of elements of S, and hence every transvection lies in N. Since the transvections generate S by Proposition 6.2, we conclude that $N = S$.
Thus every normal subgroup of S is either contained in Z_S or equal to S. Therefore $S/Z_S = \mathrm{PSL}(n, F)$ is simple. □

Exercises

Exercise 6.1. Let B be the standard Borel subgroup of $\mathrm{GL}(n, F)$. Determine all subgroups of $\mathrm{SL}(n, F)$ which contain $B \cap \mathrm{SL}(n, F)$.

Solution. Let $S = \mathrm{SL}(n, F)$ and put $B_S = B \cap S$. We claim that the required subgroups are precisely

$$P \cap S,$$

where P ranges over the staircase groups of $\mathrm{GL}(n, F)$. Indeed, if P is a staircase group, then $B \leq P$, and hence $B_S \leq P \cap S \leq S$.
Conversely, suppose that

$$B_S \leq H \leq S.$$

If $n = 1$, then $S = 1$, so the assertion is immediate. Assume now that $n \geq 2$. We first note that the only non-zero subspaces of $V_n(F)$ stabilized by B_S are the standard flag subspaces $V_1, \dots, V_n$. The proof is the same as Lemma 5.3, with one determinant correction. If

$$y = \alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k \in V_k - V_{k-1},$$

then $\alpha_k \neq 0$, and because $n \geq 2$ we can choose an upper triangular matrix of determinant 1 whose kth column is y by adjusting one diagonal entry different from the kth one. Thus the argument of Lemma 5.3 applies with B_S in place of B.
List the standard flag subspaces stabilized by H as

$$V_{a_1}, \dots, V_{a_r}, \qquad 1 \leq a_1 < \cdots < a_r = n.$$

Let P be the staircase group stabilizing the subflag

$$0 \subset V_{a_1} \subset \cdots \subset V_{a_r} = V_n(F).$$

Then $H \leq P \cap S$. It remains to prove the reverse containment.
We use the same block argument as in Theorem 5.4. Fix a block

$$F\mathbf{v}_s \oplus \cdots \oplus F\mathbf{v}_t, \qquad s = a_{k-1} + 1, \quad t = a_k.$$

If $s = t$, there is nothing to prove for this block. Assume $s < t$. Since there is no H-invariant standard flag subspace strictly between V_{s-1} and V_t, there exists $h \in H$ such that the coefficient of $\mathbf{v}_t$ in $h\mathbf{v}_s$ is non-zero. Write $h = b_1 w b_2$ by Bruhat Decomposition Theorem, with $b_1, b_2 \in B$ and w a permutation matrix. Since $h \in S$, this Bruhat decomposition may be rewritten as

$$h = b_1' \dot{w} b_2',$$

where $b_1', b_2' \in B_S$ and $\dot{w} \in S$ is a diagonal multiple of w. Indeed, choose a diagonal matrix d with $\det(dw) = 1$ and put $\dot{w} = dw$. If $\delta = \det(b_2)$, choose $t_0 \in T$ with $\det(t_0) = \delta^{-1}$. Then

$$h = (b_1 d^{-1} \dot{w} t_0^{-1} \dot{w}^{-1}) \dot{w} (t_0 b_2),$$

and both outside factors lie in B_S. Since $h, b_1', b_2' \in H$, we get $\dot{w} \in H$.
The diagonal factor in $\dot{w}$ does not change the permutation pattern, so the same argument used in Theorem 5.4 gives $w(s) = t$. Put $q = w^{-1}(s)$. Then $s < q \leq t$, so the upper root subgroup X_{sq} lies in $B_S \leq H$. Conjugating by $\dot{w}$ sends this root subgroup onto X_{ts}, up to a non-zero scalar change in the parameter. Hence $X_{ts} \leq H$. Since all upper root subgroups inside the block already lie in B_S, the commutator identities of Lemma 4.6(iv) give every root subgroup X_{ij} with $s \leq i \neq j \leq t$.
Therefore, in each block, H contains all root subgroups supported inside that block. By Proposition 6.2, it contains the corresponding embedded special linear group on each block. It also contains every determinant-one diagonal matrix, since $T \cap S \leq B_S$. Hence H contains the full group $L_P \cap S$, where L_P is the standard Levi complement of P: any element of $L_P \cap S$ is block diagonal of total determinant 1, and after multiplying by a suitable element of $T \cap S$ its blocks all have determinant 1. Finally, the unipotent radical U_P of P lies in B_S, so Proposition 5.2 gives

$$P \cap S = U_P(L_P \cap S) \leq H.$$

Together with $H \leq P \cap S$, this proves $H = P \cap S$. □

Exercise 6.2. Show that $\mathrm{PSL}(2,2) \cong S_3$ and $\mathrm{PSL}(2,3) \cong A_4$, and that these groups are not simple.

Solution. By Proposition 6.5,

$$|\mathrm{PSL}(2,2)| = 2^3 - 2 = 6$$

and

$$|\mathrm{PSL}(2,3)| = \frac{3^3 - 3}{2} = 12.$$

First let F be the field with two elements. The group $\mathrm{SL}(2,F)$ acts on the three non-zero vectors of $V_2(F)$. If a matrix fixes all three non-zero vectors, then in particular it fixes the basis vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, so it is the identity. Hence this action gives an embedding

$$\mathrm{PSL}(2,2) = \mathrm{SL}(2,F) \hookrightarrow S_3.$$

Since both groups have order 6, we get $\mathrm{PSL}(2,2) \cong S_3$.
Now let F be the field with three elements. The group $\mathrm{SL}(2,F)$ acts on the four one-dimensional subspaces of $V_2(F)$, namely

$$F\begin{pmatrix}1\\0\end{pmatrix}, \quad F\begin{pmatrix}0\\1\end{pmatrix}, \quad F\begin{pmatrix}1\\1\end{pmatrix}, \quad F\begin{pmatrix}1\\-1\end{pmatrix}.$$

The kernel of this action consists of the scalar matrices in $\mathrm{SL}(2,F)$: indeed, if a matrix fixes the first two lines, it is diagonal, and if it also fixes the line $F(\mathbf{v}_1 + \mathbf{v}_2)$, then its two diagonal entries are equal. Thus the kernel is $\{I, -I\}$, and the action induces an embedding

$$\mathrm{PSL}(2,3) \hookrightarrow S_4.$$

We show that the image lies in A_4. By Proposition 6.2, $\mathrm{SL}(2,F)$ is generated by X_{12} and X_{21}. Write the four lines as

$$L_0 = F\begin{pmatrix}1\\0\end{pmatrix}, \quad L_1 = F\begin{pmatrix}1\\1\end{pmatrix}, \quad L_{-1} = F\begin{pmatrix}1\\-1\end{pmatrix}, \quad M = F\begin{pmatrix}0\\1\end{pmatrix}.$$

Then $X_{12}(1)$ acts as the cycle

$$(L_1 \; L_{-1} \; M)$$

and $X_{21}(1)$ acts as the cycle

$$(L_0\ L_1\ L_{-1}).$$

These are even permutations, and their powers give the images of the whole root subgroups X_{12} and X_{21}. Therefore the image of $\mathrm{PSL}(2,3)$ in S_4 is contained in A_4. Since this image has order 12 and $|A_4| = 12$, it follows that $\mathrm{PSL}(2,3) \cong A_4$.

Finally, these groups are not simple: $A_3 \trianglelefteq S_3$ is a non-trivial proper normal subgroup, and the Klein four-subgroup is a non-trivial proper normal subgroup of A_4. □

Exercise 6.3. Show that $\mathrm{PSL}(2,4)$ and $\mathrm{PSL}(2,5)$ are both isomorphic with A_5.

Solution. By Proposition 6.5,

$$|\mathrm{PSL}(2,4)| = 4^3 - 4 = 60$$

and

$$|\mathrm{PSL}(2,5)| = \frac{5^3-5}{2} = 60.$$

First let F be the field with four elements. The group $\mathrm{SL}(2,F)$ acts on the five one-dimensional subspaces of $V_2(F)$. The kernel of this action consists of the scalar matrices in $\mathrm{SL}(2,F)$, and since F has characteristic 2, the only scalar matrix in $\mathrm{SL}(2,F)$ is I. Thus this action gives an embedding

$$\mathrm{PSL}(2,4) = \mathrm{SL}(2,F) \hookrightarrow S_5.$$

We show that the image lies in A_5. Write the five lines as

$$L_a = F\begin{pmatrix}1\\a\end{pmatrix} \quad (a \in F), \qquad M = F\begin{pmatrix}0\\1\end{pmatrix}.$$

For $\alpha \neq 0$, the transvection $X_{21}(\alpha)$ fixes M and sends

$$L_a \mapsto L_{a+\alpha}.$$

Since the additive group of F has characteristic 2, the translation $a \mapsto a+\alpha$ on the four elements of F is a product of two disjoint transpositions. Hence $X_{21}(\alpha)$ acts as an even permutation on the five lines. Similarly, $X_{12}(\alpha)$ fixes L_0 and, for $\alpha \neq 0$, fixes no other line: if

$$X_{12}(\alpha)\begin{pmatrix}1\\a\end{pmatrix} = \begin{pmatrix}1+\alpha a\\a\end{pmatrix}$$

spans the same line as $\binom{1}{a}$ and $a \neq 0$, then the second coordinate forces the scalar to be 1, and hence $\alpha a = 0$, a contradiction. Also M is sent to $L_{\alpha^{-1}}$. Thus $X_{12}(\alpha)$ is an involution fixing exactly one of the five lines, so it is again a product of two disjoint transpositions. By Proposition 6.2, the image of $\mathrm{SL}(2,F)$ in S_5 is generated by these even permutations, and hence lies in A_5. Since the image has order $60 = |A_5|$, we conclude that

$$\mathrm{PSL}(2,4) \cong A_5.$$

Now let F be the field with five elements, and put $G = \mathrm{PSL}(2,5)$. We shall find a subgroup of G of index 5. In $\mathrm{SL}(2,F)$, let

$$C = \begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}, \qquad D = \begin{pmatrix}2 & 0\\0 & -2\end{pmatrix}, \qquad R = \begin{pmatrix}1 & 1\\2 & 3\end{pmatrix}.$$

Let c, d, r be their images in G. Since

$$C^2 = D^2 = (CD)^2 = -I$$

and $CD = -DC$, the subgroup

$$V = \{1, c, d, cd\}$$

is a Klein four-subgroup of G. Also $R^3 = I$, so r has order 3. A direct calculation, modulo the scalar subgroup $\{I, -I\}$, gives

$$rcr^{-1} = cd, \qquad r(cd)r^{-1} = d, \qquad rdr^{-1} = c.$$

Thus r normalizes V and cyclically permutes its three non-identity elements. Therefore

$$H = V\langle r\rangle$$

is a subgroup of G of order 12. Since $|G| = 60$, the index of H in G is 5.
Let G act on the five left cosets of H. This gives a homomorphism

$$G \to S_5.$$

The kernel is normal in G. By Theorem 6.8, the group $G = \mathrm{PSL}(2,5)$ is simple, since this is not one of the exceptional cases. The action is not trivial because H is a proper subgroup, so the kernel cannot be all of G. Hence the kernel is trivial, and G embeds in S_5. Finally, composing this embedding with the sign homomorphism $S_5 \to \{\pm 1\}$ gives a homomorphism from the simple group G to a group of order 2. This homomorphism must be trivial, since a non-trivial one would have a non-trivial proper kernel. Therefore the image of G lies in A_5. Since both G and A_5 have order 60, the image is all of A_5, and hence

$$\mathrm{PSL}(2,5) \cong A_5.$$

□

Exercise 6.4. Show that $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$.

Solution. Let F be the field with seven elements, and put $G = \mathrm{PSL}(2,7)$. By Proposition 6.5,

$$|G| = \frac{7^3 - 7}{2} = 168.$$

Also

$$|\mathrm{GL}(3,2)| = (2^3 - 1)(2^3 - 2)(2^3 - 2^2) = 7 \cdot 6 \cdot 4 = 168.$$

We shall construct the same faithful permutation group of degree 7 from G and from $\mathrm{GL}(3,2)$.
First we construct a subgroup of G of index 7. In $\mathrm{SL}(2,F)$, let

$$C = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 3 \\ 3 & -2 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}.$$

Let c, d, r, e be their images in G. A direct calculation gives

$$C^2 = D^2 = (CD)^2 = -I, \qquad CD = -DC.$$

Thus

$$V = \{1, c, d, cd\}$$

is a Klein four-subgroup of G. Also $R^3 = -I$ and $E^2 = -I$, so r has order 3 and e has order 2 in G. Conjugation gives

$$rcr^{-1} = d, \qquad rdr^{-1} = cd, \qquad r(cd)r^{-1} = c,$$

and

$$ece^{-1} = cd, \qquad ede^{-1} = d, \qquad e(cd)e^{-1} = c.$$

Moreover $ere^{-1} = r^{-1}$, so $\langle r, e\rangle \cong S_3$. Hence r and e act on the three non-identity elements of V as a 3-cycle and a transposition. This action is faithful, while every element of V centralizes V, so $V \cap \langle r, e\rangle = 1$. Therefore the subgroup

$$H = V\langle r, e\rangle$$

has order $4 \cdot 6 = 24$. Since $|G| = 168$, the subgroup H has index 7 in G.
Let G act on the seven left cosets of H. The kernel is normal in G. By Theorem 6.8, the group $G = \mathrm{PSL}(2,7)$ is simple. The action is not trivial because H is proper, so the kernel is not all of G. Hence the kernel is trivial, and this coset action embeds G in S_7.
Now take

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The images of S and T generate G: indeed, $T = X_{12}(1)$, and conjugating powers of T by S gives the lower root subgroup X_{21}, so Proposition 6.2 applies. We now make the coset calculation explicit. Choose the following representatives g_i for the seven left cosets of H in G, where each displayed matrix denotes its image in G:

i	g_i
1	$\left(\begin{smallmatrix} 2 & -3 \\ 2 & 1 \end{smallmatrix}\right)$
2	$\left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$
3	$\left(\begin{smallmatrix} 2 & 3 \\ -2 & 1 \end{smallmatrix}\right)$
4	$\left(\begin{smallmatrix} 3 & 0 \\ 2 & -2 \end{smallmatrix}\right)$
5	$\left(\begin{smallmatrix} 3 & -2 \\ 0 & -2 \end{smallmatrix}\right)$
6	$\left(\begin{smallmatrix} 3 & 2 \\ 0 & -2 \end{smallmatrix}\right)$
7	$\left(\begin{smallmatrix} 1 & -1 \\ 2 & -1 \end{smallmatrix}\right)$.

For $X \in \{S, T\}$, the equality $Xg_iH = g_jH$ is equivalent to

$$g_j^{-1} X g_i \in H.$$

Checking this membership for the displayed representatives gives

i	1	2	3	4	5	6	7
Sg_iH	g_1H	g_5H	g_4H	g_3H	g_2H	g_6H	g_7H
Tg_iH	g_3H	g_5H	g_6H	g_7H	g_4H	g_2H	g_1H.

For example,

$$g_5^{-1} S g_2 = \begin{pmatrix} 1 & -2 \\ -3 & 0 \end{pmatrix} = R^{-1} \in H,$$

so $Sg_2H = g_5H$. Reading the two rows of the table as permutations of the labels $1, \ldots, 7$, we get

$$S \mapsto (2\ 5)(3\ 4), \qquad T \mapsto (1\ 3\ 6\ 2\ 5\ 4\ 7).$$

Now consider the natural action of $\mathrm{GL}(3, 2)$ on the seven non-zero vectors of $\mathbb{F}_2^3$. Label these vectors by

label	1	2	3	4	5	6	7
vector	e_1	e_2	$e_1 + e_2$	e_3	$e_1 + e_3$	$e_2 + e_3$	$e_1 + e_2 + e_3$.

With respect to the basis e_1, e_2, e_3, the matrices

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

are invertible over $\mathbb{F}_2$. A direct calculation on the seven labelled vectors gives

$$A \mapsto (2\ 5)(3\ 4), \qquad B \mapsto (1\ 3\ 6\ 2\ 5\ 4\ 7).$$

Thus the two generators of the image of G in S_7 lie in the image of the natural embedding

$$\mathrm{GL}(3, 2) \hookrightarrow S_7.$$

Since S and T generate G, the whole image of G lies in the image of $\mathrm{GL}(3, 2)$. Both images have order 168: the coset action of G is faithful, and the natural action of $\mathrm{GL}(3, 2)$ on the non-zero vectors is faithful. Therefore the two images in S_7 are equal, and hence

$$\mathrm{PSL}(2, 7) \cong \mathrm{GL}(3, 2).$$

□

Exercise 6.5. Show that $\mathrm{PSL}(2, 9) \cong A_6$.

Solution. Let $F = \mathbb{F}_9 = \mathbb{F}_3(i)$, where $i^2 = -1$, and put $G = \mathrm{PSL}(2, 9)$. By Proposition 6.5,

$$|G| = \frac{9^3 - 9}{2} = 360.$$

We first construct a subgroup of G of order 60. In $\mathrm{SL}(2, F)$, let

$$X = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} i & i \\ 1 & 1-i \end{pmatrix}.$$

Both matrices have determinant 1. Let x and y be their images in G. A direct calculation gives

$$X^2 = -I, \qquad Y^3 = -I, \qquad (XY)^5 = -I.$$

Moreover X and Y are not scalar, Y^2 is not scalar, and none of $XY, (XY)^2, (XY)^3, (XY)^4$ is scalar. Hence x has order 2, y has order 3, and xy has order 5 in G.
We use the standard presentation

$$A_5 \cong \langle a, b \mid a^2 = 1,\ b^3 = 1,\ (ab)^5 = 1 \rangle.$$

Therefore there is a homomorphism

$$A_5 \to \langle x, y \rangle$$

sending a to x and b to y. This homomorphism is surjective by definition of $\langle x, y \rangle$. Its image is non-trivial, since x has order 2. Also A_5 is simple by Exercise 6.3 and Theorem 6.8. Hence the kernel is not all of A_5, so the kernel is trivial. Thus

$$H = \langle x, y \rangle \cong A_5,$$

and in particular $|H| = 60$. Since $|G| = 360$, the subgroup H has index 6 in G.
Let G act on the six left cosets of H. This gives a homomorphism

$$G \to S_6.$$

The kernel is normal in G. By Theorem 6.8, the group $G = \mathrm{PSL}(2, 9)$ is simple. The action is not trivial because H is a proper subgroup, so the kernel cannot be all of G. Hence the kernel is trivial, and G embeds in S_6. Composing with the sign homomorphism $S_6 \to \{\pm 1\}$ gives a homomorphism from the simple group G to a group of order 2. This homomorphism must be trivial, since a non-trivial one would have a non-trivial proper kernel. Therefore the image of G lies in A_6. Since both groups have order 360, the image is all of A_6, and hence

$$\mathrm{PSL}(2, 9) \cong A_6.$$

□

Exercise 6.6. By Exercise 6.3 and Theorem 6.8, the group A_5 is simple. Attempt to prove this fact directly by mimicking the proof of Theorem 6.8, with A_5 in place of S, A_4 in place of P, and the Klein four-group in place of K.

Solution. Put $S = A_5$, acting naturally on $\{1, 2, 3, 4, 5\}$, and let

$$P = \{s \in S \mid s(5) = 5\}.$$

Then $P \cong A_4$, since its elements are exactly the even permutations of $\{1, 2, 3, 4\}$. In particular $|P| = 12$, so P has index 5 in S and is maximal. Let

$$K = \{1,\ (12)(34),\ (13)(24),\ (14)(23)\}.$$

Then K is the Klein four-subgroup of P. It is normal in P, because it is the identity together with all three double transpositions in A_4, and conjugation inside A_4 preserves this set.
Let $N \trianglelefteq S$. First suppose that $N \leq P$. Since N is normal in S, for every $s \in S$ we have

$$N = sNs^{-1} \leq sPs^{-1}.$$

The subgroup sPs^{-1} is the stabilizer of the point $s(5)$. Since S acts transitively on $\{1, 2, 3, 4, 5\}$, it follows that N is contained in the stabilizer of every point. Thus every element of N fixes every point, and hence $N = 1$.
Now suppose that $N \not\leq P$. Since P is maximal, the subgroup PN properly contains P, and therefore

$$PN = S.$$

Let $\eta\colon S \to S/N$ be the natural map. Since $K \trianglelefteq P$, we have $\eta(K) \trianglelefteq \eta(P)$. But

$$\eta(P) = PN/N = S/N,$$

so $\eta(K) \trianglelefteq S/N$. By Correspondence Theorem, its inverse image

$$\eta^{-1}(\eta(K)) = KN$$

is normal in S. Since KN contains K and is normal in S, it contains every S-conjugate of every element of K. The non-identity elements of K are double transpositions, and their S-conjugates are all the double transpositions in A_5.
The double transpositions generate A_5. Indeed, if $(a\ b\ c)$ is a 3-cycle and d, e are the remaining two letters, then

$$(a\ b\ c) = (b\ c)(d\ e)(a\ c)(d\ e),$$

which is a product of two double transpositions. Since A_5 is generated by its 3-cycles, it follows that

$$KN = S.$$

Hence

$$S/N = KN/N \cong K/(K \cap N)$$

by First Isomorphism Theorem. Since K is abelian, the quotient S/N is abelian.
It remains to observe that S has no non-trivial abelian quotient. Let $\varphi\colon S \to A$ be a homomorphism to an abelian group. If $\tau = (a\ b\ c)$ is a 3-cycle and d, e are the remaining two letters, then conjugation by the double transposition $(a\ b)(d\ e) \in S$ sends τ to τ^{-1}. Since A is abelian, conjugate elements have the same image, so

$$\varphi(\tau) = \varphi(\tau^{-1}) = \varphi(\tau)^{-1}.$$

Also $\tau^3 = 1$, so $\varphi(\tau)^3 = 1$. Hence $\varphi(\tau) = 1$. Since the 3-cycles generate S, the homomorphism φ is trivial. Applying this to the natural map $S \to S/N$, we get $S/N = 1$, and hence $N = S$.
Thus every normal subgroup of $S = A_5$ is either 1 or S, so A_5 is simple. $\square$

Further Exercises

Let F be a field, and let $V = F^2$. We define an equivalence relation on $V - \{0\}$ by $\mathbf{v} \sim \mathbf{w}$ iff $\mathbf{v} = \alpha\mathbf{w}$ for some $\alpha \in F^\times$. The set of equivalence classes under this relation is called the *one-dimensional projective space* (or *projective line*) over F and is denoted by $\mathbb{P}^1(F)$. We write elements of V as column vectors, and the element of $\mathbb{P}^1(F)$ that is the equivalence class of $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ shall be written as $\left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right]$.

Exercise 6.7. Show that there is a well-defined action of $\mathrm{PSL}(2, F)$ on $\mathbb{P}^1(F)$ given by

$$\overline{\begin{pmatrix} a & b \\ c & d \end{pmatrix}} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

and that this action is doubly transitive. (Here $\overline{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)}$ denotes the image in $\mathrm{PSL}(2, F)$ of $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2, F)$.)

Solution. Let $A \in \mathrm{SL}(2, F)$ and let $0 \neq \mathbf{v} \in V$. Since A is invertible, $A\mathbf{v} \neq 0$, so the displayed formula sends a point of $\mathbb{P}^1(F)$ to another point of $\mathbb{P}^1(F)$. We first check that this point is independent of the chosen vector representative. If $\mathbf{v}' = \lambda\mathbf{v}$ for some $\lambda \in F^\times$, then

$$A\mathbf{v}' = \lambda A\mathbf{v},$$

so $A\mathbf{v}'$ and $A\mathbf{v}$ span the same one-dimensional subspace. Hence the formula is well-defined on projective points.
Next we check that the formula is well-defined on $\mathrm{PSL}(2, F)$. If A' represents the same element of $\mathrm{PSL}(2, F)$ as A, then

$$A' = A(\mu I)$$

for some scalar matrix $\mu I \in Z \cap \mathrm{SL}(2, F)$. In particular $\mu \in F^\times$, and hence

$$A'\mathbf{v} = \mu A\mathbf{v}.$$

Thus $A'\mathbf{v}$ and $A\mathbf{v}$ define the same point of $\mathbb{P}^1(F)$. Therefore the formula depends only on the image $\overline{A} \in \mathrm{PSL}(2, F)$, not on the particular representative $A \in \mathrm{SL}(2, F)$.
The identity element acts as the identity, and for $A, B \in \mathrm{SL}(2, F)$ we have

$$A(B\mathbf{v}) = (AB)\mathbf{v}.$$

Passing to images in $\mathrm{PSL}(2, F)$, this proves that the formula defines an action of $\mathrm{PSL}(2, F)$ on $\mathbb{P}^1(F)$.
It remains to show that the action is doubly transitive. Let $p_1, p_2, q_1, q_2 \in \mathbb{P}^1(F)$ with $p_1 \neq p_2$ and $q_1 \neq q_2$. Choose non-zero vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{w}_1, \mathbf{w}_2 \in V$ such that

$$p_i = [\mathbf{u}_i], \qquad q_i = [\mathbf{w}_i] \quad (i = 1, 2).$$

Since $p_1 \neq p_2$, the vectors $\mathbf{u}_1, \mathbf{u}_2$ are linearly independent; similarly, $\mathbf{w}_1, \mathbf{w}_2$ are linearly independent. Let $\det(\mathbf{u}_1, \mathbf{u}_2)$ denote the determinant of the matrix with columns $\mathbf{u}_1, \mathbf{u}_2$, and similarly for $\mathbf{w}_1, \mathbf{w}_2$. Put

$$\delta = \frac{\det(\mathbf{u}_1, \mathbf{u}_2)}{\det(\mathbf{w}_1, \mathbf{w}_2)} \in F^\times.$$

There is a unique linear map $A \colon V \to V$ satisfying

$$A\mathbf{u}_1 = \delta\mathbf{w}_1, \qquad A\mathbf{u}_2 = \mathbf{w}_2,$$

because $\mathbf{u}_1, \mathbf{u}_2$ form a basis of V. Its determinant is

$$\det A = \frac{\det(\delta\mathbf{w}_1, \mathbf{w}_2)}{\det(\mathbf{u}_1, \mathbf{u}_2)} = \frac{\delta \det(\mathbf{w}_1, \mathbf{w}_2)}{\det(\mathbf{u}_1, \mathbf{u}_2)} = 1.$$

Thus $A \in \mathrm{SL}(2, F)$. Since scalar multiples represent the same projective point, the image $\overline{A} \in \mathrm{PSL}(2, F)$ sends

$$p_1 \mapsto q_1, \qquad p_2 \mapsto q_2.$$

Therefore the action is doubly transitive. $\square$

Exercise 6.8. (cont.) Determine the proper definition of n-dimensional projective space $\mathbb{P}^n(F)$ for arbitrary n, and show that $\mathrm{PSL}(n+1, F)$ acts doubly transitively on $\mathbb{P}^n(F)$.

Solution. Let $V = F^{n+1}$. We define

$$\mathbb{P}^n(F) = (V - \{0\})/\sim,$$

where

$$\mathbf{v} \sim \mathbf{w} \quad \Longleftrightarrow \quad \mathbf{w} = \lambda\mathbf{v} \quad \text{for some } \lambda \in F^\times.$$

Equivalently, $\mathbb{P}^n(F)$ is the set of one-dimensional subspaces of V. We write the equivalence class of a non-zero vector $\mathbf{v} \in V$ as $[\mathbf{v}]$.
The action is defined by

$$\overline{A}[\mathbf{v}] = [A\mathbf{v}], \qquad A \in \mathrm{SL}(n+1, F).$$

This is well-defined for the same reasons as in Exercise 6.7. If $\mathbf{v}' = \lambda\mathbf{v}$ for some $\lambda \in F^\times$, then

$$A\mathbf{v}' = \lambda A\mathbf{v},$$

so $A\mathbf{v}'$ and $A\mathbf{v}$ define the same projective point. Also, if A' represents the same element of $\mathrm{PSL}(n+1, F)$ as A, then $A' = A(\mu I)$ for some scalar matrix $\mu I \in Z \cap \mathrm{SL}(n+1, F)$, and hence

$$A'\mathbf{v} = \mu A\mathbf{v},$$

which again defines the same projective point. Finally, the identity acts as the identity, and

$$A(B\mathbf{v}) = (AB)\mathbf{v}$$

for $A, B \in \mathrm{SL}(n+1, F)$, so the displayed formula defines an action of $\mathrm{PSL}(n+1, F)$ on $\mathbb{P}^n(F)$.
We now prove double transitivity. If $n = 0$, then $\mathbb{P}^0(F)$ has one point, so there are no ordered pairs of distinct points to consider. Assume $n \geq 1$. Let

$$p_1, p_2, q_1, q_2 \in \mathbb{P}^n(F), \qquad p_1 \neq p_2, \quad q_1 \neq q_2.$$

Choose non-zero vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{w}_1, \mathbf{w}_2 \in V$ such that

$$p_i = [\mathbf{u}_i], \qquad q_i = [\mathbf{w}_i] \quad (i = 1, 2).$$

Since $p_1 \neq p_2$, the vectors $\mathbf{u}_1, \mathbf{u}_2$ are linearly independent. Similarly, $\mathbf{w}_1, \mathbf{w}_2$ are linearly independent. Extend these pairs to bases

$$\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \ldots, \mathbf{u}_{n+1} \quad \text{and} \quad \mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \ldots, \mathbf{w}_{n+1}$$

of V. Put

$$\delta = \frac{\det(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{n+1})}{\det(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_{n+1})} \in F^\times.$$

There is a unique linear map $A\colon V \to V$ satisfying

$$A\mathbf{u}_1 = \delta \mathbf{w}_1, \qquad A\mathbf{u}_i = \mathbf{w}_i \quad (2 \leq i \leq n+1).$$

Its determinant is

$$\begin{aligned} \det A &= \frac{\det(\delta\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_{n+1})}{\det(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{n+1})} \\ &= \frac{\delta \det(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_{n+1})}{\det(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{n+1})} = 1. \end{aligned}$$

Thus $A \in \mathrm{SL}(n+1, F)$. Since scalar multiples represent the same projective point, the image $\overline{A} \in \mathrm{PSL}(n+1, F)$ sends

$$p_1 \mapsto q_1, \qquad p_2 \mapsto q_2.$$

Therefore the action is doubly transitive. □

Exercise 6.9. Let F be the field of 7 elements. For $0 \leq i \leq 6$, let i represent $\left[\begin{smallmatrix}1\\ i\end{smallmatrix}\right] \in \mathbb{P}^1(F)$; denote the remaining element $\left[\begin{smallmatrix}0\\ 1\end{smallmatrix}\right]$ of $\mathbb{P}^1(F)$ by ∞. View S_8 as being the group of permutations of $\{\infty, 0, 1, 2, 3, 4, 5, 6\}$, and let $\varphi\colon \mathrm{PSL}(2,7) \to S_8$ be the monomorphism arising (as in Proposition 3.1) from the action of $\mathrm{PSL}(2,7)$ on $\mathbb{P}^1(F)$. Show that the image of φ is the subgroup of A_8 generated by $(0\ 1\ 2\ 3\ 4\ 5\ 6)$, $(1\ 4\ 2)(3\ 5\ 6)$, and $(\infty\ 0)(1\ 3)(2\ 5)(4\ 6)$. (Hint: Show first that $\mathrm{PSL}(2,7)$ is generated by the images under the natural map of the elements $\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}4&0\\0&2\end{smallmatrix}\right)$, and $\left(\begin{smallmatrix}0&3\\2&0\end{smallmatrix}\right)$ of $\mathrm{SL}(2,7)$.) For the relevance of this exercise, see Exercises 7.13–7.17.

Solution. Let

$$A = \begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}, \qquad B = \begin{pmatrix}4 & 0\\ 0 & 2\end{pmatrix}, \qquad C = \begin{pmatrix}0 & 3\\ 2 & 0\end{pmatrix}$$

over F. These are all in $\mathrm{SL}(2,7)$, since

$$\det A = 1, \qquad \det B = 8 = 1, \qquad \det C = -6 = 1$$

in F. Let $\overline{A}, \overline{B}, \overline{C}$ be their images in $\mathrm{PSL}(2,7)$. We first show that these three elements generate $\mathrm{PSL}(2,7)$. The element $A = X_{21}(1)$, so its powers give every element of the lower root subgroup X_{21}. Also a direct calculation gives

$$C X_{21}(\alpha) C^{-1} = X_{12}(5\alpha) \qquad (\alpha \in F).$$

Thus the subgroup generated by $\overline{A}, \overline{B}, \overline{C}$ contains the images of both root subgroups X_{21} and X_{12}. By Proposition 6.2, the root subgroups generate $\mathrm{SL}(2,7)$, so their images generate $\mathrm{PSL}(2,7)$.
It remains to compute the corresponding permutations of $\mathbb{P}^1(F)$. For $0 \leq i \leq 6$, we identify i with $\left[\begin{smallmatrix}1\\ i\end{smallmatrix}\right]$ and identify ∞ with $\left[\begin{smallmatrix}0\\ 1\end{smallmatrix}\right]$. First,

$$A\begin{pmatrix}1\\ i\end{pmatrix} = \begin{pmatrix}1\\ i+1\end{pmatrix}, \qquad A\begin{pmatrix}0\\ 1\end{pmatrix} = \begin{pmatrix}0\\ 1\end{pmatrix}.$$

Hence $\varphi(\overline{A})$ fixes ∞ and sends i to $i+1$ modulo 7, so

$$\varphi(\overline{A}) = (0\ 1\ 2\ 3\ 4\ 5\ 6).$$

Next,

$$B\begin{pmatrix}1\\ i\end{pmatrix}=\begin{pmatrix}4\\ 2i\end{pmatrix}=\begin{pmatrix}1\\ 4i\end{pmatrix},\qquad B\begin{pmatrix}0\\ 1\end{pmatrix}=\begin{pmatrix}0\\ 2\end{pmatrix}.$$

Thus $\varphi(\overline{B})$ fixes 0 and ∞ and sends i to $4i$ on $F^\times$. Since multiplication by 4 on $F^\times$ has cycles

$$1\mapsto 4\mapsto 2\mapsto 1,\qquad 3\mapsto 5\mapsto 6\mapsto 3,$$

we get

$$\varphi(\overline{B})=(1\ 4\ 2)(3\ 5\ 6).$$

Finally,

$$C\begin{pmatrix}0\\ 1\end{pmatrix}=\begin{pmatrix}3\\ 0\end{pmatrix},\qquad C\begin{pmatrix}1\\ 0\end{pmatrix}=\begin{pmatrix}0\\ 2\end{pmatrix}.$$

So $\varphi(\overline{C})$ interchanges ∞ and 0. If $i\neq 0$, then

$$C\begin{pmatrix}1\\ i\end{pmatrix}=\begin{pmatrix}3i\\ 2\end{pmatrix}=\begin{pmatrix}1\\ 3i^{-1}\end{pmatrix},$$

because $(3i)^{-1}2=3i^{-1}$ in F. Therefore

$$1\leftrightarrow 3,\qquad 2\leftrightarrow 5,\qquad 4\leftrightarrow 6,$$

and hence

$$\varphi(\overline{C})=(\infty\ 0)(1\ 3)(2\ 5)(4\ 6).$$

Since $\overline{A},\overline{B},\overline{C}$ generate $\mathrm{PSL}(2,7)$, their images generate $\operatorname{im}\varphi$. Thus

$$\operatorname{im}\varphi=\langle(0\ 1\ 2\ 3\ 4\ 5\ 6),\ (1\ 4\ 2)(3\ 5\ 6),\ (\infty\ 0)(1\ 3)(2\ 5)(4\ 6)\rangle\,.$$

Each of these generators is an even permutation, so this subgroup lies in A_8. □

Recall the definition of a BN-pair from the further exercises to Section 4.

Exercise 6.10. Let $G=\mathrm{GL}(n,F)$. Let B be the standard Borel subgroup of G, let N be the subgroup of G of monomial matrices, and let $T=B\cap N$ be the subgroup of diagonal matrices. Let $B_0=B\cap\mathrm{SL}(n,F)$, let $N_0=N\cap\mathrm{SL}(n,F)$, and let $T_0=T\cap\mathrm{SL}(n,F)=B_0\cap N_0$. Show that B_0 and N_0 form a BN-pair of $\mathrm{SL}(n,F)$, with the associated Weyl group $W_0=N_0/T_0$ being isomorphic with S_n.

Solution. Put

$$G_0=\mathrm{SL}(n,F).$$

We verify the defining conditions for a BN-pair. Thus we must show that $G_0=\langle B_0,N_0\rangle$, that $T_0=B_0\cap N_0$ is normal in N_0, that $W_0=N_0/T_0$ is generated by a finite set of involutions, and that the two double-coset conditions hold.

We first record a determinant-one version of the usual Bruhat double cosets. Suppose that $n_0\in N_0$ and that

$$x\in G_0\cap Bn_0B.$$

Write $x=b_1n_0b_2$ with $b_1,b_2\in B$. Since x and n_0 have determinant 1, we have

$$\det(b_1)\det(b_2)=1.$$

Choose a diagonal matrix $d\in T$ with $\det(d)=\det(b_1)^{-1}$. Then

$$x=(b_1d)n_0(n_0^{-1}d^{-1}n_0b_2).$$

The left factor b_1d lies in B_0. Also $n_0^{-1}d^{-1}n_0\in T$, because monomial matrices normalize the diagonal subgroup. Hence the right factor lies in B. Its determinant is

$$\det(d)^{-1}\det(b_2)=\det(b_1)\det(b_2)=1,$$

so it lies in B_0. Therefore

$$G_0\cap Bn_0B\subseteq B_0n_0B_0.$$

The reverse inclusion is immediate from $B_0, n_0 \le G_0$ and $B_0 \le B$, so

$$G_0 \cap B n_0 B = B_0 n_0 B_0 \qquad (n_0 \in N_0).$$

We now prove that G_0 is generated by B_0 and N_0. Let $g \in G_0$. By the Bruhat Decomposition Theorem in $\mathrm{GL}(n, F)$, we can write

$$g = b_1 p b_2,$$

where $b_1, b_2 \in B$ and p is a permutation matrix. Choose a diagonal matrix $e \in T$ with $\det(e) = \det(p)^{-1}$. Then

$$n_0 = ep$$

lies in N_0, and $g = (b_1 e^{-1}) n_0 b_2 \in B n_0 B$. Since $g \in G_0$, the determinant-one double-coset observation gives $g \in B_0 n_0 B_0$. Thus

$$G_0 = B_0 N_0 B_0,$$

and in particular $G_0 = \langle B_0, N_0 \rangle$.
Next,

$$B_0 \cap N_0 = (B \cap \mathrm{SL}(n, F)) \cap (N \cap \mathrm{SL}(n, F)) = T \cap \mathrm{SL}(n, F) = T_0.$$

Also $T_0 \trianglelefteq N_0$, because conjugating a diagonal matrix by a monomial matrix gives another diagonal matrix, and determinant 1 is preserved under conjugation.
We identify the Weyl group. Define

$$\pi \colon N_0 \to S_n$$

by sending a monomial matrix to its underlying permutation of the standard basis lines. Equivalently, if

$$m\mathbf{v}_j = \lambda_j \mathbf{v}_{\sigma(j)} \qquad (\lambda_j \in F^\times),$$

then $\pi(m) = \sigma$. This is a homomorphism. Its kernel consists exactly of the determinant-one monomial matrices whose underlying permutation is the identity; these are precisely the determinant-one diagonal matrices, so

$$\ker \pi = T_0.$$

The map π is surjective. Indeed, if $\sigma \in S_n$ and p_σ is its permutation matrix, choose $d \in T$ with $\det(d) = \det(p_\sigma)^{-1}$. Then $dp_\sigma \in N_0$ and $\pi(dp_\sigma) = \sigma$. Hence, by the Fundamental Theorem on Homomorphisms,

$$W_0 = N_0/T_0 \cong S_n.$$

For $1 \le i \le n-1$, let $\dot{s}_i \in N_0$ be the monomial matrix which is the identity away from rows and columns $i, i+1$ and has block

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

on those rows and columns. This block has determinant 1, so $\dot{s}_i \in N_0$. Let $s_i = \dot{s}_i T_0 \in W_0$. Under the isomorphism $W_0 \cong S_n$, the element s_i corresponds to the adjacent transposition $(i\ i+1)$. Also $\dot{s}_i^2 \in T_0$, so $s_i^2 = 1$ in W_0. Thus

$$S_0 = \{s_1, \ldots, s_{n-1}\}$$

is a finite set of involutions generating W_0, because the adjacent transpositions generate S_n.
It remains to check the two double-coset conditions. Let $s_i \in S_0$, let $w \in W_0$, and choose a representative $\dot{w} \in N_0$. From the BN-pair for $\mathrm{GL}(n, F)$ proved in Exercise 4.8, we have

$$\dot{s}_i B \dot{w} \subseteq B\dot{w}B \cup B\dot{s}_i\dot{w}B.$$

Here we may use the determinant-one representatives $\dot{s}_i$ and $\dot{w}$, because changing a representative by an element of T does not change the corresponding B-double coset. Now take $x \in \dot{s}_i B_0 \dot{w}$. Then $x \in G_0$, and the displayed inclusion puts x in either $B\dot{w}B$ or $B\dot{s}_i\dot{w}B$. Since both $\dot{w}$ and $\dot{s}_i\dot{w}$ lie in N_0, the determinant-one double-coset observation gives

$$x \in B_0\dot{w}B_0 \quad \text{or} \quad x \in B_0\dot{s}_i\dot{w}B_0.$$

Therefore

$$\dot{s}_i B_0 \dot{w} \subseteq B_0\dot{w}B_0 \cup B_0\dot{s}_i\dot{w}B_0.$$

Finally, we show that
$$\dot{s}_i B_0 \dot{s}_i \not\subseteq B_0.$$
The transvection $X_{i,i+1}(1)$ lies in B_0. In the $i, i+1$ rows and columns, the product
$$\dot{s}_i X_{i,i+1}(1) \dot{s}_i$$
has block
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix}.$$
This matrix has a non-zero entry below the diagonal, so it is not in B_0. Hence $\dot{s}_i B_0 \dot{s}_i \not\subseteq B_0$. All the defining conditions are satisfied, so B_0 and N_0 form a BN-pair of $\mathrm{SL}(n, F)$, and the associated Weyl group is $W_0 \cong S_n$. □

Exercise 6.11. Let $P = \mathrm{PSL}(n, F)$, and let $\overline{B_0}$, $\overline{N_0}$, and $\overline{T_0}$ be the images in P of the subgroups B_0, N_0, and T_0 of $\mathrm{SL}(n, F)$. Show that $\overline{B_0}$ and $\overline{N_0}$ form a BN-pair of P (with again the Weyl group $\overline{W_0} = \overline{N_0}/\overline{T_0}$ being isomorphic with S_n).

Solution. Let
$$q\colon\ \mathrm{SL}(n, F) \to P = \mathrm{PSL}(n, F)$$
be the natural map, and let
$$K = \ker q.$$
Then K consists of the scalar matrices in $\mathrm{SL}(n, F)$. In particular $K \leq T_0$, since scalar matrices are diagonal. We write $\overline{x} = q(x)$ for $x \in \mathrm{SL}(n, F)$.
By Exercise 6.10, the subgroups B_0 and N_0 form a BN-pair of $\mathrm{SL}(n, F)$. We check that the defining conditions pass to the quotient. First,
$$P = q(\mathrm{SL}(n, F)) = q(\langle B_0, N_0 \rangle) = \langle \overline{B_0}, \overline{N_0} \rangle.$$
Next we compute the intersection. Certainly $\overline{T_0} \leq \overline{B_0} \cap \overline{N_0}$. Conversely, suppose that
$$y \in \overline{B_0} \cap \overline{N_0}.$$
Then $y = \overline{b} = \overline{n}$ for some $b \in B_0$ and $n \in N_0$. Thus
$$bn^{-1} \in K \leq T_0 \leq N_0.$$
Since $n \in N_0$, it follows that $b \in N_0$. But also $b \in B_0$, so
$$b \in B_0 \cap N_0 = T_0.$$
Hence $y = \overline{b} \in \overline{T_0}$, and therefore
$$\overline{B_0} \cap \overline{N_0} = \overline{T_0}.$$
Also $\overline{T_0} \trianglelefteq \overline{N_0}$, because $T_0 \trianglelefteq N_0$.
Now we identify the Weyl group. Define
$$\theta\colon N_0/T_0 \to \overline{N_0}/\overline{T_0}, \qquad \theta(nT_0) = \overline{n}\,\overline{T_0}.$$
This map is well-defined because replacing n by nt with $t \in T_0$ replaces $\overline{n}$ by $\overline{n}\,\overline{t}$, which gives the same coset modulo $\overline{T_0}$. It is also surjective. If $\theta(nT_0) = \overline{T_0}$, then $\overline{n} \in \overline{T_0}$, so $\overline{n} = \overline{t}$ for some $t \in T_0$. Hence $nt^{-1} \in K \leq T_0$, and therefore $n \in T_0$. Thus $\ker\theta = 1$, and θ is an isomorphism. Since Exercise 6.10 gives $N_0/T_0 \cong S_n$, we get
$$\overline{W_0} = \overline{N_0}/\overline{T_0} \cong S_n.$$
In particular $\overline{W_0}$ is generated by the images of the same adjacent-transposition generators used in Exercise 6.10. If $\dot{s}_i \in N_0$ is the determinant-one monomial representative with block
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$
in rows and columns $i, i+1$, then $\overline{\dot{s}_i}\,\overline{T_0}$ is an involution in $\overline{W_0}$, and these elements generate $\overline{W_0}$.
It remains to verify the two double-coset conditions. Let s_i be one of the above generators, and let

$w \in \overline{W_0}$. Choose representatives $\dot{s}_i \in N_0$ and $\dot{w} \in N_0$ whose images in P represent s_i and w. In $\mathrm{SL}(n, F)$, Exercise 6.10 gives

$$\dot{s}_i B_0 \dot{w} \subseteq B_0 \dot{w} B_0 \cup B_0 \dot{s}_i \dot{w} B_0.$$

Applying q gives

$$\overline{\dot{s}_i}\,\overline{B_0}\,\overline{\dot{w}} \subseteq \overline{B_0}\,\overline{\dot{w}}\,\overline{B_0} \cup \overline{B_0}\,\overline{\dot{s}_i}\,\overline{\dot{w}}\,\overline{B_0}.$$

This is the first double-coset condition for P.

Finally, we prove the non-containment condition. In Exercise 6.10, the element

$$y = \dot{s}_i X_{i,i+1}(1) \dot{s}_i$$

lies in $\dot{s}_i B_0 \dot{s}_i$ but does not lie in B_0. We claim that $\overline{y} \notin \overline{B_0}$. Indeed, if $\overline{y} \in \overline{B_0}$, then $\overline{y} = \overline{b}$ for some $b \in B_0$. Hence $yb^{-1} \in K \leq T_0 \leq B_0$, so $y \in B_0$, a contradiction. Therefore

$$\overline{\dot{s}_i}\,\overline{B_0}\,\overline{\dot{s}_i} \not\subseteq \overline{B_0}.$$

All the defining conditions are satisfied, so $\overline{B_0}$ and $\overline{N_0}$ form a BN-pair of $P = \mathrm{PSL}(n, F)$, and the Weyl group is again isomorphic with S_n. $\square$

Chapter 3

Local Structure

7 Sylow's Theorem

Setup and notation

- Throughout the section, G is a finite group and p a prime divisor of $|G|$.
- $|G|_p$ denotes the highest power of p dividing $|G|$, so $|G|_p = p^n$ with $p^n \mid |G|$ but $p^{n+1} \nmid |G|$.

p-elements, p-groups, p-subgroups

- $g \in G$ is a *p-element* if its order is a power of p.
- G is a *p-group* if $|G|$ is a power of p; $H \le G$ is a *p-subgroup* if $|H|$ is a power of p.
- Every element of a p-group is a p-element. (For infinite groups, p-group means every element is a p-element.)
- $H \le G$ is a *Sylow p-subgroup* of G if H is a p-subgroup of order $|G|_p$, the maximal possible order of a p-subgroup of G.
- A subgroup of G is a *Sylow subgroup* if it is a Sylow p-subgroup for some prime divisor p of $|G|$.

Motivating example: $\mathrm{GL}(n,p)$

- For $G = \mathrm{GL}(n,p)$, Proposition 4.2 gives $|G| = p^{\frac{n(n-1)}{2}}(p^n-1)\cdots(p-1)$, so $|G|_p = p^{\frac{n(n-1)}{2}}$.
- The subgroup U of upper uni-triangular matrices satisfies $|U| = p^{n-1}p^{n-2}\cdots p^2p = p^{\frac{n(n-1)}{2}} = |G|_p$, hence U is a Sylow p-subgroup.
- Every p-element of $\mathrm{GL}(n,p)$ is unipotent (its characteristic polynomial is $(X-1)^n$), and by Kolchin's Theorem any p-subgroup is conjugate with a subgroup of U.
- In particular: $\mathrm{GL}(n,p)$ has a Sylow p-subgroup, all are conjugate, and every p-subgroup is contained in one. Sylow's Theorem extends this to arbitrary finite G.

Theorem (Sylow's Theorem)**.** *(i) G has at least one Sylow p-subgroup.*
(ii) All the Sylow p-subgroups of G are conjugate.
(iii) Any p-subgroup of G is contained in a Sylow p-subgroup.
(iv) The number of Sylow p-subgroups of G is congruent to 1 modulo p.

Proof. Let $|G| = p^n m$, where $p^n = |G|_p$ and $p \nmid m$. Let X be the set of all subsets of G having p^n elements. Then G acts on X by left multiplication.
We first examine the orbits of this action whose sizes are not divisible by p. Suppose that $\mathcal{O}$ is such an orbit. Choose some $A \in \mathcal{O}$. Since A is non-empty, there is some $a \in A$, and then $a^{-1}A$ lies in the same orbit and contains 1. Replacing A by $a^{-1}A$, we may assume that $1 \in A$. Let P be the stabilizer of A. If $x \in P$, then $xA = A$, so $x = x1 \in A$. Hence $P \subseteq A$, and therefore

$$|P| \le |A| = p^n.$$

By Corollary 3.5, we have

$$|\mathcal{O}| = |G : P|,$$

and hence $|G| = |P|\,|\mathcal{O}|$. Since $p \nmid |\mathcal{O}|$, the full p-part p^n of $|G|$ must divide $|P|$. Together with $|P| \leq p^n$, this gives $|P| = p^n$. Thus P is a Sylow p-subgroup of G. Since $P \subseteq A$ and $|P| = |A|$, we have $A = P$, so $\mathcal{O}$ is the coset space G/P.

Conversely, if P is a Sylow p-subgroup of G, then the set of left cosets G/P is contained in X, and it is exactly the orbit of the subset P under left multiplication. This orbit has size

$$|G : P| = m,$$

which is not divisible by p. Therefore the Sylow p-subgroups of G are in bijective correspondence with the orbits in X whose sizes are not divisible by p.

Let X' be the union of all orbits in X whose sizes are not divisible by p, and let r be the number of Sylow p-subgroups of G. Every orbit in $X - X'$ has size divisible by p, so

$$|X| \equiv |X'| \pmod{p}.$$

Each orbit contained in X' has size m, and there are r such orbits. Hence

$$|X'| = rm,$$

so

$$rm \equiv |X| = \binom{p^n m}{p^n} \pmod{p}.$$

This congruence depends only on the order of the group. Now compare with the cyclic group of order $p^n m$. By Theorem 1.4, this cyclic group has exactly one subgroup of order p^n, so the same congruence gives

$$m \equiv \binom{p^n m}{p^n} \pmod{p}.$$

Since $p \nmid m$, we may cancel m modulo p, obtaining

$$r \equiv 1 \pmod{p}.$$

This proves (iv), and in particular $r > 0$, so G has at least one Sylow p-subgroup. This proves (i).

It remains to prove conjugacy and containment. Let P be a Sylow p-subgroup of G, and let Q be an arbitrary p-subgroup of G. Let Y be the set of conjugates of P in G. The group Q acts on Y by conjugation. Each orbit of this action has size equal to the index of a subgroup of the p-group Q, and hence has size a power of p.

By Proposition 3.14,

$$|Y| = |G : N_G(P)|.$$

Since $P \leq N_G(P)$, the index $|G : N_G(P)|$ divides

$$|G : P| = m.$$

Thus $p \nmid |Y|$. Therefore at least one orbit of the action of Q on Y has size 1. Let this orbit be $\{P_1\}$, where P_1 is a conjugate of P. Then $xP_1x^{-1} = P_1$ for every $x \in Q$, so $QP_1 = P_1Q$. By Proposition 1.3, $QP_1 \leq G$. Moreover, by Proposition 3.12,

$$|QP_1| = \frac{|Q||P_1|}{|Q \cap P_1|},$$

so QP_1 is a p-subgroup of G. Since P_1 is already a Sylow p-subgroup and $P_1 \leq QP_1$, we must have $QP_1 = P_1$. Hence $Q \leq P_1$. This proves that every p-subgroup of G is contained in a Sylow p-subgroup, proving (iii).

Finally, if Q itself is a Sylow p-subgroup of G, then $Q \leq P_1$ and $|Q| = |P_1|$, so $Q = P_1$. Since P_1 is a conjugate of P, every Sylow p-subgroup of G is conjugate to P. This proves (ii). $\square$

Corollary 7.1. *The number of Sylow p-subgroups of G divides $|G|/|G|_p$.*

Proof. Let r be the number of Sylow p-subgroups of G, and let P be one of them. By part (ii) of Sylow's Theorem, every Sylow p-subgroup of G is conjugate to P, so r is exactly the number of conjugates of P in G. By Proposition 3.14, this number is

$$r = |G : N_G(P)|.$$

Since $P \leq N_G(P) \leq G$, we have

$$|G : P| = |G : N_G(P)|\,|N_G(P) : P|.$$

Therefore $r = |G : N_G(P)|$ divides

$$|G : P| = \frac{|G|}{|P|} = \frac{|G|}{|G|_p},$$

as required. □

Theorem (Cauchy's Theorem)**.** *G has an element of order p.*

Proof. By part (i) of Sylow's Theorem, G has a Sylow p-subgroup P. Since p divides $|G|$, this subgroup is non-trivial. Choose some $1 \neq x \in P$. Then $o(x)$ is a power of p, say $o(x) = p^k$ with $k \geq 1$. Therefore

$$o\left(x^{p^{k-1}}\right) = p.$$

Thus G has an element of order p. □

(Other proofs of Sylow's Theorem often assume that Cauchy's Theorem is already known, as was the case historically.)

Sylow subgroups of normal subgroups and quotients

- Sylow p-subgroups behave well with respect to a normal subgroup $N \trianglelefteq G$: both PN/N and $P \cap N$ are Sylow p-subgroups (of G/N and N respectively).
- For a non-normal subgroup $H \leq G$, $P \cap H$ need not be Sylow in H, but some conjugate $gPg^{-1} \cap H$ is.

Proposition 7.2. *Let $N \trianglelefteq G$, and let P be a Sylow p-subgroup of G. Then PN/N is a Sylow p-subgroup of G/N, and $P \cap N$ is a Sylow p-subgroup of N.*

Proof. By Proposition 1.7, PN is a subgroup of G. The Correspondence Theorem applied to the natural map $G \to G/N$ gives

$$|G/N : PN/N| = |G : PN|.$$

Since $P \leq PN \leq G$, the index $|G : PN|$ divides $|G : P|$. But P is a Sylow p-subgroup of G, so

$$|G : P| = \frac{|G|}{|G|_p}$$

is not divisible by p. Hence $p \nmid |G/N : PN/N|$. On the other hand, by the First Isomorphism Theorem,

$$PN/N \cong P/(P \cap N),$$

so PN/N is a p-group. Thus PN/N is a p-subgroup of G/N whose index in G/N is not divisible by p, and therefore PN/N is a Sylow p-subgroup of G/N.
It remains to prove that $P \cap N$ is Sylow in N. Clearly $P \cap N$ is a p-subgroup. By Proposition 3.12,

$$|PN| = \frac{|P||N|}{|P \cap N|},$$

and hence

$$|N : P \cap N| = \frac{|N|}{|P \cap N|} = \frac{|PN|}{|P|} = |PN : P|.$$

Since $P \leq PN \leq G$, the index $|PN : P|$ divides $|G : P|$, and again $p \nmid |G : P|$. Therefore $p \nmid |N : P \cap N|$. Hence $P \cap N$ is a p-subgroup of N whose index in N is not divisible by p, so $P \cap N$ is a Sylow p-subgroup of N. □

Applications: groups of order pq and simplicity of A_5

- Sylow's theorem classifies groups of order pq (distinct primes $p > q$).
- It also gives a clean count of p-elements via the orbit count of Sylow p-subgroups, used to prove A_5 is simple.

Proposition 7.3. *Let p and q be distinct primes, with $p > q$. If $p \not\equiv 1 \pmod q$, then any group of order pq is isomorphic with $\mathbb{Z}_{pq}$. If $p \equiv 1 \pmod q$, then any abelian group of order pq is isomorphic with $\mathbb{Z}_{pq}$, and there is exactly one isomorphism class of non-abelian groups of order pq.*

Proof. Let G be a group of order pq. Let P be a Sylow p-subgroup of G, and let Q be a Sylow q-subgroup of G. Then $|P| = p$ and $|Q| = q$, so $P \cong \mathbb{Z}_p$ and $Q \cong \mathbb{Z}_q$. Also, $P \cap Q = 1$, and Proposition 3.12 gives

$$|PQ| = \frac{|P||Q|}{|P \cap Q|} = pq = |G|,$$

so $G = PQ$.
We first count the Sylow subgroups. By Sylow's Theorem and Corollary 7.1, the number of Sylow p-subgroups divides q and is congruent to 1 modulo p. Since $p > q$, we have $q \not\equiv 1 \pmod p$, so there is exactly one Sylow p-subgroup. Hence $P \trianglelefteq G$. Similarly, the number of Sylow q-subgroups divides p and is congruent to 1 modulo q. Therefore this number is either 1 or p, and the second case can occur only when $p \equiv 1 \pmod q$.
If $Q \trianglelefteq G$, then both P and Q are normal in G. Since $G = PQ$ and $P \cap Q = 1$, Lemma 2.7 gives

$$G \cong P \times Q \cong \mathbb{Z}_p \times \mathbb{Z}_q \cong \mathbb{Z}_{pq},$$

where the final isomorphism follows from Lemma 2.8. Thus G is cyclic whenever Q is normal. This proves the first assertion when $p \not\equiv 1 \pmod q$, and it also proves that every abelian group of order pq is cyclic, since then Q is automatically normal.
It remains to classify the non-abelian case, so assume $p \equiv 1 \pmod q$ and that G is non-abelian. Then Q is not normal, while $P \trianglelefteq G$, $G = PQ$, and $P \cap Q = 1$. Hence $G = P \rtimes Q$. Let

$$\varphi \colon Q \to \mathrm{Aut}(P)$$

be the conjugation homomorphism. If $\ker \varphi \neq 1$, then $\ker \varphi = Q$, because Q has prime order. In that case the action is trivial, so $G = P \times Q$, contradicting that G is non-abelian. Therefore φ is injective. Thus every non-abelian group of order pq arises from a monomorphism

$$\mathbb{Z}_q \to \mathrm{Aut}(\mathbb{Z}_p).$$

Conversely, any such monomorphism gives a non-abelian semidirect product $\mathbb{Z}_p \rtimes_\varphi \mathbb{Z}_q$ of order pq.
By Proposition 2.2, we have $\mathrm{Aut}(\mathbb{Z}_p) \cong \mathbb{Z}_{p-1}$. Since $q \mid (p-1)$, Theorem 1.4 shows that $\mathrm{Aut}(\mathbb{Z}_p)$ has a unique subgroup K of order q. Hence a monomorphism $\varphi \colon \mathbb{Z}_q \to \mathrm{Aut}(\mathbb{Z}_p)$ exists, and every such monomorphism has image K. If $\psi \colon \mathbb{Z}_q \to \mathrm{Aut}(\mathbb{Z}_p)$ is another monomorphism, then $\varphi(\mathbb{Z}_q) = K = \psi(\mathbb{Z}_q)$, so Proposition 2.11 gives

$$\mathbb{Z}_p \rtimes_\varphi \mathbb{Z}_q \cong \mathbb{Z}_p \rtimes_\psi \mathbb{Z}_q.$$

Therefore, when $p \equiv 1 \pmod q$, there is exactly one isomorphism class of non-abelian groups of order pq. □

Theorem 7.4. *A_5 is simple.*

Proof. We have

$$|A_5| = \frac{5!}{2} = 60 = 2^2 \cdot 3 \cdot 5.$$

We first count the elements of prime-power order that will be needed below. A 5-cycle is even, and the number of 5-cycles in S_5 is $4! = 24$. Each Sylow 5-subgroup of A_5 has order 5 and hence contains 4 non-identity elements, and two distinct subgroups of order 5 intersect trivially. Thus A_5 has

$$\frac{24}{4} = 6$$

Sylow 5-subgroups, and these account for all 24 elements of order 5. Similarly, the 3-cycles are even, and there are

$$\binom{5}{3} \cdot 2 = 20$$

of them. Each Sylow 3-subgroup has two non-identity elements, so A_5 has 10 Sylow 3-subgroups, accounting for all 20 elements of order 3.

Now consider the Sylow 2-subgroups. For $1 \leq i \leq 5$, let $\{a, b, c, d\}$ be the complement of $\{i\}$ in $\{1, 2, 3, 4, 5\}$, and put

$$V_i = \{1, (ab)(cd), (ac)(bd), (ad)(bc)\}.$$

Then V_i is a subgroup of A_5 of order 4, hence a Sylow 2-subgroup. If $i \neq j$, then $V_i \cap V_j = 1$, since a non-identity element of V_i fixes exactly the point i. Moreover, for $\sigma \in A_5$ we have

$$\sigma V_i \sigma^{-1} = V_{\sigma(i)}.$$

Since A_5 acts transitively on $\{1, 2, 3, 4, 5\}$, the five subgroups $V_1, \ldots, V_5$ are conjugate. By part (ii) of Sylow's Theorem, these are all the Sylow 2-subgroups of A_5. Thus A_5 has $5(4-1) = 15$ elements of order 2. Also, every element of order 2 in A_5 has a distinct conjugate in A_5: for instance, if $x = (ab)(cd)$, with i the remaining point, then conjugation by the 3-cycle (acb) sends x to $(ac)(bd) \neq x$.

Now let $N \trianglelefteq A_5$ be a proper normal subgroup, and put $n = |N|$. By Lagrange's Theorem, n divides 60, and since N is proper we have $n \leq 30$. Suppose first that 5 divides n. Then by Cauchy's Theorem, N contains an element of order 5, and hence contains a Sylow 5-subgroup of A_5. Since N is normal in A_5, it contains every conjugate of this Sylow 5-subgroup, so it contains all 24 elements of order 5. This forces $n = 30$. But then 3 divides n, so the same argument with Sylow 3-subgroups shows that N contains all 20 elements of order 3. These are disjoint from the 24 elements of order 5, and with the identity this gives more than 30 elements in N, a contradiction. Therefore $5 \nmid n$.

If 3 divides n, then again by Cauchy's Theorem, N contains an element of order 3, and normality forces N to contain all 20 elements of order 3. But $5 \nmid n$ and $n \mid 60$, so $n \leq 12$, a contradiction. Hence $3 \nmid n$. Therefore n is one of $1, 2, 4$. If $n = 4$, then N is a Sylow 2-subgroup of A_5, but a Sylow 2-subgroup has five conjugates, contradicting the normality of N. If $n = 2$, then $N = \{1, x\}$ for some element x of order 2, but we have just observed that x has a conjugate different from itself, again contradicting normality. Hence $n = 1$.

Thus every proper normal subgroup of A_5 is trivial, and so A_5 is simple. □

Theorem 7.5. *Any simple group of order* 60 *is isomorphic with* A_5.

Proof. Let G be a simple group of order 60. We first observe that G has no proper subgroup of index less than 5. Indeed, if $H < G$ has index r, then the action of G on G/H by left multiplication gives a homomorphism

$$G \to S_r$$

by Proposition 3.1. Since this action is transitive and $r > 1$, it is non-trivial. The kernel is therefore a proper normal subgroup of the simple group G, and hence the kernel is trivial. Thus G embeds in S_r. But if $r < 5$, then

$$|S_r| = r! < 60 = |G|,$$

which is impossible.

We next show that G has a subgroup of index 5. Suppose, for a contradiction, that it does not. Let S be a Sylow 2-subgroup of G. Then $|S| = 4$. By Corollary 7.1, the number of Sylow 2-subgroups divides $60/4 = 15$, and by Proposition 3.14 this number is the index $|G : N_G(S)|$. This number is not 1, since otherwise S would be normal in G, contradicting simplicity. Since G has no proper subgroup of index less than 5 and, by assumption, none of index 5, this number must be 15.

Let S_1 and S_2 be distinct Sylow 2-subgroups. Suppose that there is some $1 \neq t \in S_1 \cap S_2$. Since every group of order 4 is abelian, both S_1 and S_2 lie in $C_G(t)$. Thus $|C_G(t)| > 4$, and since $S_1 \leq C_G(t)$, the order $|C_G(t)|$ is divisible by 4. Hence

$$|G : C_G(t)| \leq 5.$$

The first paragraph and our assumption that G has no subgroup of index 5 force $C_G(t) = G$. Thus $t \in Z(G)$, so $\langle t \rangle$ is a non-trivial proper normal subgroup of G, contradicting simplicity. Therefore distinct Sylow 2-subgroups intersect trivially. Since there are 15 Sylow 2-subgroups, G has

$$15(4-1) = 45$$

non-identity elements lying in Sylow 2-subgroups.
Since G is simple, its Sylow 5-subgroup is not normal. By Sylow's Theorem and Corollary 7.1, the number of Sylow 5-subgroups divides $60/5 = 12$ and is congruent to 1 modulo 5, so this number must be 6. These subgroups intersect trivially away from the identity, and hence contribute

$$6(5-1) = 24$$

elements of order 5. This is impossible, since $45 + 24 > 60$. Therefore G has a subgroup of index 5.
Let H be a subgroup of index 5. The action of G on G/H gives a homomorphism $G \to S_5$. As above, simplicity forces this homomorphism to be injective, so we identify G with its image in S_5. Then $|G| = 60$, and hence $|S_5 : G| = 2$. By Proposition 1.8, $G \trianglelefteq S_5$.
We claim that $G = A_5$. Suppose not. Since $A_5 \trianglelefteq S_5$, Proposition 1.7 gives $GA_5 \leq S_5$. The subgroup GA_5 contains A_5 properly, because $G \neq A_5$ and both groups have order 60. Hence $|GA_5| > 60$, so $GA_5 = S_5$. By Proposition 3.12,

$$|G \cap A_5| = \frac{|G||A_5|}{|GA_5|} = \frac{60 \cdot 60}{120} = 30.$$

Therefore $G \cap A_5$ has index 2 in G, so Proposition 1.8 gives $G \cap A_5 \trianglelefteq G$. This is a non-trivial proper normal subgroup of G, contradicting simplicity. Thus $G = A_5$, and the original group is isomorphic with A_5. □

Corollary 7.6. $\mathrm{PSL}(2,4) \cong \mathrm{PSL}(2,5) \cong A_5$.

Proof. By Proposition 6.5, we have

$$|\mathrm{PSL}(2,4)| = 4^3 - 4 = 60$$

and

$$|\mathrm{PSL}(2,5)| = \frac{1}{2}(5^3 - 5) = 60.$$

By Theorem 6.8, both groups are simple, since the exceptional groups in degree 2 occur only over fields of order 2 and 3. Therefore Theorem 7.5 gives

$$\mathrm{PSL}(2,4) \cong A_5 \qquad \text{and} \qquad \mathrm{PSL}(2,5) \cong A_5.$$

Hence $\mathrm{PSL}(2,4) \cong \mathrm{PSL}(2,5) \cong A_5$. □

It is a relatively easy exercise to show that the only simple groups of order less than 60 are the cyclic groups of prime order, and hence that A_5 is the non-abelian finite simple group having smallest order. The non-abelian finite simple group of next smallest order is $\mathrm{PSL}(2,7)$, which has order 168.

Exercises

Throughout these exercises, G is a finite group and p is a prime divisor of $|G|$.

Exercise 7.1. Let $H \leq G$ and let P be a Sylow p-subgroup of G. Prove, without reference to Sylow's Theorem, that there is some conjugate of P whose intersection with H is a Sylow p-subgroup of H.

Solution. Let H act on the left cosets G/P by left multiplication. Since P is a Sylow p-subgroup of G, the index $|G : P|$ is not divisible by p. The H-orbits on G/P partition G/P, so at least one such orbit has size not divisible by p. Choose a coset gP in such an orbit.
The stabilizer of gP in H is

$$\mathrm{Stab}_H(gP) = \{h \in H : hgP = gP\}.$$

Now

$$hgP = gP \iff g^{-1}hg \in P \iff h \in gPg^{-1},$$

and hence

$$\mathrm{Stab}_H(gP) = H \cap gPg^{-1}.$$

By Corollary 3.5, the size of the orbit of gP is

$$|H : \mathrm{Stab}_H(gP)| = |H : H \cap gPg^{-1}|.$$

This index is not divisible by p. Also, $H \cap gPg^{-1}$ is a p-subgroup of H, because it is contained in the p-group gPg^{-1}. Therefore $H \cap gPg^{-1}$ is a p-subgroup of H whose index in H is not divisible by p, so its order is the full p-part of $|H|$. Thus $H \cap gPg^{-1}$ is a Sylow p-subgroup of H. □

Exercise 7.2. (cont.) Use Exercise 7.1 to give an alternate proof of part (i) of Sylow's Theorem.

Solution. Let K be a finite group with $p \mid |K|$, and put $n = |K|$. By Cayley's Theorem, we may regard K as a subgroup of S_n. Embedding S_n into $\mathrm{GL}(n,p)$ by permutation matrices, we may therefore regard

$$K \leq \mathrm{GL}(n,p).$$

Let U be the subgroup of upper uni-triangular matrices in $\mathrm{GL}(n,p)$. As computed at the beginning of this section,

$$|U| = p^{\frac{n(n-1)}{2}} = |\mathrm{GL}(n,p)|_p,$$

so U is a Sylow p-subgroup of $\mathrm{GL}(n,p)$ by definition. Applying Exercise 7.1 to the subgroup $K \leq \mathrm{GL}(n,p)$ and the Sylow p-subgroup U, there is some $x \in \mathrm{GL}(n,p)$ such that

$$K \cap xUx^{-1}$$

is a Sylow p-subgroup of K. Hence K has a Sylow p-subgroup. Since K was arbitrary, this proves part (i) of Sylow's Theorem. □

Exercise 7.3. Give an alternate proof of parts (ii) and (iii) of Sylow's Theorem by considering the action of an arbitrary p-subgroup Q of G on the coset space G/P, where P is a Sylow p-subgroup of G.

Solution. Let Q be a p-subgroup of G, and let Q act on G/P by left multiplication:

$$q \cdot gP = qgP.$$

Since P is a Sylow p-subgroup of G, the index $|G : P|$ is not divisible by p. On the other hand, every orbit of the p-group Q on G/P has size a power of p, because its size is the index of a stabilizer in Q. Since the orbits partition G/P, at least one orbit must have size not divisible by p, and hence must have size 1. Choose a fixed coset gP.
For every $q \in Q$, we have

$$qgP = gP.$$

Equivalently,

$$g^{-1}qgP = P,$$

so $g^{-1}qg \in P$, and therefore $q \in gPg^{-1}$. Thus

$$Q \leq gPg^{-1}.$$

Since gPg^{-1} is a conjugate of P, it is a Sylow p-subgroup of G. This proves part (iii): every p-subgroup of G is contained in a Sylow p-subgroup.
Now let Q itself be a Sylow p-subgroup of G. The containment just proved gives

$$Q \leq gPg^{-1}.$$

But Q and gPg^{-1} have the same order, namely $|G|_p$. Hence $Q = gPg^{-1}$, so every Sylow p-subgroup of G is conjugate to P. This proves part (ii). □

Exercise 7.4. Prove the following generalization of part (iv) of Sylow's Theorem: If $|G|$ is divisible by p^b, and $H \leq G$ has order p^a where $a \leq b$, then the number of subgroups of G that both contain H and have order p^b is congruent to 1 modulo p.

Solution. We first record a one-step counting fact. Suppose that $K \leq G$ has order p^c and that p^{c+1} divides $|G|$. We claim that the number of subgroups $L \leq G$ such that

$$K \leq L, \qquad |L| = p^{c+1}$$

is congruent to 1 modulo p.
To see this, first note that if M is any finite group whose order is divisible by p, then the number of subgroups of M of order p is congruent to 1 modulo p. Indeed, let

$$X = \{(x_1, \ldots, x_p) \in M^p : x_1 x_2 \cdots x_p = 1\}.$$

Then $|X| = |M|^{p-1}$, so $p \mid |X|$. The cyclic group of order p acts on X by cyclically permuting the entries. The fixed points are exactly the tuples $(x, \dots, x)$ with $x^p = 1$. Hence the number of elements $x \in M$ satisfying $x^p = 1$ is congruent to 0 modulo p. If r is the number of subgroups of M of order p, then these elements are the identity together with the $p-1$ non-identity elements in each such subgroup, so

$$1 + (p-1)r \equiv 0 \pmod{p}.$$

Thus $r \equiv 1 \pmod{p}$.

Now choose a Sylow p-subgroup P of G containing K, using part (iii) of Sylow's Theorem. Since p^{c+1} divides $|G|$, we have $K < P$. The p-group K acts by left multiplication on the coset space P/K. The total number of cosets is divisible by p, and the coset K is fixed. Since all non-trivial orbits have size divisible by p, there must be some other fixed coset xK. The condition that xK is fixed says that $kxK = xK$ for every $k \in K$, or equivalently $x^{-1}kx \in K$ for every $k \in K$. Thus $x \in N_P(K) - K$, so p divides $|N_G(K) : K|$.

Subgroups $L \leq G$ with $K \leq L$ and $|L| = p^{c+1}$ are now in bijection with subgroups of $N_G(K)/K$ of order p. Indeed, if $K \leq L$ and $|L : K| = p$, then $K \trianglelefteq L$: the action of L on the left cosets of K gives a transitive homomorphism $L \to S_p$, whose image is a p-subgroup of S_p and hence has order p, so the kernel has the same order as K and is contained in K. Hence $L \leq N_G(K)$ and L/K has order p. Conversely, each subgroup of $N_G(K)/K$ of order p lifts to such an L. Since p divides $|N_G(K)/K|$, the preceding paragraph applied to $M = N_G(K)/K$ proves the one-step claim.

We now prove the exercise by induction on $b - a$. If $b = a$, the only subgroup of order p^b containing H is H itself, so the number is 1. Assume $b > a$, and for $a \leq c \leq b$ put

$$\mathcal{A}_c = \{K \leq G : H \leq K,\ |K| = p^c\}.$$

By induction, $|\mathcal{A}_{b-1}| \equiv 1 \pmod{p}$. Count the pairs

$$(K, L) \quad \text{with} \quad K \in \mathcal{A}_{b-1},\ L \in \mathcal{A}_b,\ K \leq L.$$

For each fixed $K \in \mathcal{A}_{b-1}$, the one-step claim shows that the number of possible L is congruent to 1 modulo p. Hence the total number of such pairs is congruent to $|\mathcal{A}_{b-1}|$, and therefore to 1 modulo p.

On the other hand, fix $L \in \mathcal{A}_b$. The number of possible $K \in \mathcal{A}_{b-1}$ with $K \leq L$ is the number of subgroups of the p-group L that contain H and have order p^{b-1}. By the induction hypothesis applied inside L, this number is congruent to 1 modulo p. Therefore the same total number of pairs is congruent to $|\mathcal{A}_b|$ modulo p. Combining the two counts gives

$$|\mathcal{A}_b| \equiv 1 \pmod{p}.$$

This is exactly the desired congruence. □

Exercise 7.5. Show that if P is a Sylow p-subgroup of G, then $N_G(P)$ is a self-normalizing subgroup of G, meaning that $N_G(N_G(P)) = N_G(P)$. (Any subgroup of G that can be written as $N_G(P)$ for some non-trivial p-subgroup P of G is said to be a *p-local subgroup* of G.)

Solution. Put $H = N_G(P)$. Since $P \leq H \leq G$ and P has the full p-part of $|G|$, the subgroup P also has the full p-part of $|H|$. Hence P is a Sylow p-subgroup of H. Moreover, by the definition of normalizer, $P \trianglelefteq H$. Thus P is the unique Sylow p-subgroup of H: if Q is any Sylow p-subgroup of H, then part (ii) of Sylow's Theorem, applied inside H, gives $Q = hPh^{-1}$ for some $h \in H$, and this equals P because $P \trianglelefteq H$.

Now let $g \in N_G(H)$. Then $gHg^{-1} = H$. Since $P \leq H$, we have

$$gPg^{-1} \leq H.$$

Also gPg^{-1} has the same order as P, so it is a Sylow p-subgroup of H. By the uniqueness just proved, $gPg^{-1} = P$. Therefore $g \in N_G(P) = H$, and so

$$N_G(H) \leq H.$$

The reverse inclusion is automatic, since every subgroup normalizes itself. Hence $H = N_G(H)$, which is exactly

$$N_G(N_G(P)) = N_G(P).$$

□

Exercise 7.6. Given a prime p, find an example of a finite group having exactly $1+p$ Sylow p-subgroups. Can this be done for $1+2p$? $1+3p$?

Solution. First take

$$G = \mathrm{GL}(2,p)$$

and let

$$U = \left\{ \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} : a \in \mathbb{Z}_p \right\}.$$

Then $|U| = p$, while

$$|\,\mathrm{GL}(2,p)| = (p^2-1)(p^2-p) = p(p-1)^2(p+1),$$

so U is a Sylow p-subgroup of G. The normalizer of U is the upper triangular subgroup

$$B = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, d \in \mathbb{Z}_p^\times,\ b \in \mathbb{Z}_p \right\}.$$

Indeed, U fixes the line $\langle e_1 \rangle$, and this is its unique fixed line in $\mathbb{Z}_p^2$; hence any element normalizing U must preserve $\langle e_1 \rangle$. Thus $N_G(U) = B$, and

$$|B| = p(p-1)^2.$$

Therefore the number of Sylow p-subgroups of $\mathrm{GL}(2,p)$ is

$$|G : N_G(U)| = \frac{p(p-1)^2(p+1)}{p(p-1)^2} = p+1.$$

So $1+p$ can be done for every prime p.
For the next two questions, there is a useful general construction. Suppose that q is a prime power with $q \equiv 1 \pmod{p}$. Let F be the finite field of order q, and choose an element $\lambda \in F^\times$ of order p, which is possible because $F^\times$ is cyclic of order $q-1$. Let

$$G = F^+ \rtimes \langle \lambda \rangle,$$

where λ acts on the additive group F^+ by multiplication. Then $|G| = qp$, and $A = \langle \lambda \rangle$ is a Sylow p-subgroup of G. We claim that $N_G(A) = A$. In the affine notation, A consists of the maps $x \mapsto \lambda^i x$. If t_b is the translation $x \mapsto x+b$, then

$$t_b(x \mapsto \lambda x)t_b^{-1}$$

is the map

$$x \mapsto \lambda x + (1-\lambda)b.$$

This lies again in A only when $b=0$. Hence no non-trivial translation normalizes A, and so $N_G(A) = A$. Therefore

$$n_p(G) = |G : N_G(A)| = q.$$

Thus $1+2p$ can be done whenever $1+2p$ is a prime power, and $1+3p$ can be done whenever $1+3p$ is a prime power. For example, $p=2$ gives $1+3p=7$, and $p=5$ gives $1+3p=16$. Also, for $p=3$, the group A_5 has exactly $10 = 1+3p$ Sylow 3-subgroups, since A_5 has 20 elements of order 3 and each subgroup of order 3 has two non-identity elements.
These constructions do not work for every prime. In fact, a theorem of M. Hall, Jr. on possible Sylow numbers (“On the Number of Sylow Subgroups in a Finite Group”, *J. Algebra* 7 (1967), Theorems 3.1–3.2) shows that $1+2p$ occurs exactly when $1+2p$ is a prime power, and that $1+3p$ occurs only for $p = 2, 3, 5$. In particular, there is no finite group with 15 Sylow 7-subgroups, and no finite group with $1+3p$ Sylow p-subgroups for $p \geq 7$. □

Exercise 7.7. Show that if G has exactly $1+kp$ Sylow p-subgroups for some $k \in \mathbb{N}$, then there is a subgroup of S_{1+kp} having exactly $1+kp$ Sylow p-subgroups.

Solution. Let Ω be the set of Sylow p-subgroups of G. By assumption,

$$|\Omega| = 1+kp.$$

The group G acts on Ω by conjugation:

$$g \cdot P = gPg^{-1}.$$

This gives a homomorphism

$$\theta\colon G \to \mathrm{Sym}(\Omega) \cong S_{1+kp}.$$

Put $\overline{G} = \theta(G)$. Then $\overline{G}$ is a subgroup of S_{1+kp}. We claim that $\overline{G}$ has exactly $1 + kp$ Sylow p-subgroups. Let $K = \ker\theta$. Since $K \trianglelefteq G$, Proposition 7.2 applied to the quotient G/K shows that, if P is a Sylow p-subgroup of G, then

$$\theta(P) \cong PK/K$$

is a Sylow p-subgroup of $\overline{G} \cong G/K$. Moreover, every Sylow p-subgroup of $\overline{G}$ arises in this way from some Sylow p-subgroup of G.

It remains only to check that distinct Sylow p-subgroups of G have distinct images in $\overline{G}$. Suppose that P and Q are Sylow p-subgroups of G and that

$$\theta(P) = \theta(Q).$$

Equivalently, $PK = QK$. Every element of K fixes every point of Ω, so in particular $K \leq N_G(P)$. Hence PK is a subgroup of G in which P is normal. Since P is a Sylow p-subgroup of G, it is also a Sylow p-subgroup of PK, and normality makes it the unique Sylow p-subgroup of PK. But $Q \leq QK = PK$, so Q is also a Sylow p-subgroup of PK. Therefore $Q = P$.

Thus the map

$$P \mapsto \theta(P)$$

gives a bijection from the Sylow p-subgroups of G to the Sylow p-subgroups of $\overline{G}$. Hence $\overline{G} \leq S_{1+kp}$ has exactly $1 + kp$ Sylow p-subgroups. □

Exercise 7.8. Let $n \geq 6$, and assume by induction that A_{n-1} is simple. Let N be a non-trivial proper normal subgroup of A_n.

(a) Show that $A_n = N \rtimes A_{n-1}$.

(b) Let $N^\#$ be the set of non-identity elements of N, and consider the action of A_{n-1} on $N^\#$ given by the conjugation homomorphism from A_{n-1} to $\mathrm{Aut}(N)$. Show that $N^\#$ is isomorphic as an A_{n-1}-set to $\{1, \ldots, n-1\}$.

(c) Show that A_{n-1} acts triply transitively on $N^\#$. (An action of a group G on a set X is *triply transitive* if for every pair of triples (x_1, x_2, x_3) and (y_1, y_2, y_3), where $x_i, y_i \in X$ for all i and $x_i \neq x_j$ (resp., $y_i \neq y_j$) when $i \neq j$, there is some $g \in G$ such that $gx_i = y_i$ for all i.)

(d) Derive a contradiction, and conclude that A_n is simple.

Solution.

(a) Let $H = A_{n-1}$ be the stabilizer of n in the natural action of A_n on $\{1, \ldots, n\}$. First observe that this action is doubly transitive. Indeed, given two ordered pairs of distinct points, choose a permutation in S_n sending the first pair to the second. If this permutation is odd, compose it on the right with a transposition of two points outside the first pair; this is possible because $n \geq 6$. The resulting permutation is even and still sends the first pair to the second. Thus Proposition 3.8 shows that H is a maximal subgroup of A_n.

Now $N \cap H \trianglelefteq H$, because $N \trianglelefteq A_n$. By the induction hypothesis, $H \cong A_{n-1}$ is simple, so $N \cap H$ is either 1 or H. If $N \cap H = H$, then $H \leq N$. Since H is maximal and N is proper, this would force $N = H$. But H is not normal in A_n: its conjugates are the stabilizers of the other points. This contradicts $N \trianglelefteq A_n$. Hence

$$N \cap H = 1.$$

Since N is non-trivial, we have $N \not\leq H$. Therefore NH is a subgroup of A_n properly containing H. By maximality of H, we get

$$NH = A_n.$$

Since $N \trianglelefteq A_n$ and $N \cap H = 1$, this gives

$$A_n = N \rtimes H = N \rtimes A_{n-1}.$$

(b) From part (a), we have
$$|N| = |A_n : H| = n.$$
The subgroup N acts on $\{1, \dots, n\}$ by restriction of the natural action of A_n. The stabilizer in N of the point n is
$$N \cap H = 1,$$
so the orbit of n under N has size $|N| = n$. Hence the map
$$N \to \{1, \dots, n\}, \qquad x \mapsto x(n),$$
is a bijection. It sends $1 \in N$ to n, and therefore restricts to a bijection
$$N^\# \to \{1, \dots, n-1\}.$$
This bijection is H-equivariant. Indeed, if $h \in H$ and $x \in N$, then h fixes n, so
$$(hxh^{-1})(n) = hx(h^{-1}(n)) = hx(n).$$
Thus conjugating x by h corresponds exactly to applying h to the point $x(n)$. Therefore $N^\#$ is isomorphic as an A_{n-1}-set to $\{1, \dots, n-1\}$.

(c) By part (b), it is enough to prove that the natural action of A_{n-1} on $\{1, \dots, n-1\}$ is triply transitive. Let
$$(a_1, a_2, a_3), \qquad (b_1, b_2, b_3)$$
be ordered triples of distinct elements of $\{1, \dots, n-1\}$. Choose a permutation $\sigma \in S_{n-1}$ such that $\sigma(a_i) = b_i$ for $i = 1, 2, 3$. If σ is even, then $\sigma \in A_{n-1}$ and we are done. If σ is odd, then because $n - 1 \geq 5$, there are at least two points of $\{1, \dots, n-1\}$ outside $\{a_1, a_2, a_3\}$. Let τ be the transposition of two such points. Then $\sigma\tau$ is even and still sends each a_i to b_i. Hence $\sigma\tau \in A_{n-1}$ gives the required element. Therefore A_{n-1} acts triply transitively on $N^\#$.

(d) We now derive a contradiction from the triple transitivity of the conjugation action of $H = A_{n-1}$ on $N^\#$. Since $|N| = n \geq 6$, the set $N^\#$ has at least five elements. Choose $x \in N^\#$, and choose
$$y \in N^\# - \{x, x^{-1}\}.$$
Then xy is non-identity and is distinct from both x and y. Choose
$$z \in N^\# - \{x, y, xy\}.$$
By triple transitivity, there is some $h \in H$ such that
$$hxh^{-1} = x, \qquad hyh^{-1} = y, \qquad h(xy)h^{-1} = z.$$
But conjugation by h is an automorphism of N, so
$$h(xy)h^{-1} = (hxh^{-1})(hyh^{-1}) = xy.$$
This contradicts $z \neq xy$. Therefore no non-trivial proper normal subgroup N of A_n exists. Hence A_n is simple.

□

The remaining exercises develop a proof that if G is a finite simple group with $|G| \leq 200$, then either $G \cong \mathbb{Z}_p$ for some prime p, or $G \cong A_5$, or $G \cong \mathrm{PSL}(2, 7)$. This was established by Hölder in 1892. We will need to assume two facts, which will be proved in later sections:

A. If $|G| = p^n$ where $n > 1$, then G is not simple (Section 8).

B. If G is a p-group and $H < G$, then $H < N_G(H)$ (Section 11).

Exercise 7.9. Show that G is not simple whenever one of the following statements is true, where p is an odd prime:

(a) $|G| = 2^m p^n$, where $2^k \not\equiv 1 \pmod p$ for any $1 \leq k \leq m$.

(b) $|G| = p^n q$, where $q \neq p$ is prime and $q \not\equiv 1 \pmod p$.

(c) $|G| = ap$, where $p \nmid a$ and $kp + 1 \nmid a$ for any $k \in \mathbb{N}$.

Solution. We use the same Sylow-counting argument in each case. Let n_p denote the number of Sylow p-subgroups of G. By Sylow's Theorem and Corollary 7.1,

$$n_p \equiv 1 \pmod p \quad \text{and} \quad n_p \mid |G : P|,$$

where P is a Sylow p-subgroup of G. If $n_p = 1$, then P is the unique Sylow p-subgroup of G, and hence $P \trianglelefteq G$. This gives a non-trivial proper normal subgroup of G, so G is not simple.

(a) Suppose $|G| = 2^m p^n$. Then a Sylow p-subgroup has order p^n, so

$$n_p \mid 2^m.$$

Hence $n_p = 2^k$ for some $0 \leq k \leq m$. If $k \geq 1$, then the Sylow congruence gives

$$2^k = n_p \equiv 1 \pmod p,$$

contradicting the hypothesis. Therefore $k = 0$, and so $n_p = 1$. Hence the Sylow p-subgroup is normal, and G is not simple.

(b) Suppose $|G| = p^n q$, where $q \neq p$ is prime. Then

$$n_p \mid q \quad \text{and} \quad n_p \equiv 1 \pmod p.$$

Since q is prime, we have $n_p = 1$ or $n_p = q$. The second possibility would force

$$q \equiv 1 \pmod p,$$

contrary to the hypothesis. Thus $n_p = 1$, so the Sylow p-subgroup is normal and G is not simple.

(c) Suppose $|G| = ap$, where $p \nmid a$. Then a Sylow p-subgroup has order p, so

$$n_p \mid a \quad \text{and} \quad n_p \equiv 1 \pmod p.$$

If $n_p > 1$, then $n_p = kp + 1$ for some $k \in \mathbb{N}$. But this is impossible, because the hypothesis says that no such number $kp + 1$ divides a. Hence $n_p = 1$. Therefore the Sylow p-subgroup is normal, and G is not simple.

□

Exercise 7.10. Show that G is not simple whenever one of the following statments is true:

(a) $|G| = p^a(p + 1)$, where $a > 1$.

(b) $|G| = p^a(p + 3)$, where $a > 3$ if $p = 2$ and $a > 1$ if $p > 3$.

(c) $|G| = p^a(p^2 - 1)$, where $a > 1$ and p is odd.

Solution. We again write n_p for the number of Sylow p-subgroups of G. If $n_p = 1$, then the Sylow p-subgroup is normal, so G is not simple. We may therefore assume, in each case, that $n_p > 1$ and derive a contradiction.

(a) Let $|G| = p^a(p + 1)$ with $a > 1$, and let P be a Sylow p-subgroup of G. By Sylow's Theorem and Corollary 7.1,

$$n_p \equiv 1 \pmod p \quad \text{and} \quad n_p \mid p + 1.$$

Thus $n_p = 1$ or $n_p = p + 1$. If G were simple, then $n_p \neq 1$, so $n_p = p + 1$. The conjugation action of G on its Sylow p-subgroups gives a homomorphism

$$G \to S_{p+1}.$$

Its kernel is normal in G, and the action is non-trivial because there is more than one Sylow p-subgroup. Hence, if G is simple, the kernel is trivial and G embeds in S_{p+1}. Therefore

$$|G| \mid (p + 1)!.$$

But $p^2 \mid |G|$ because $a > 1$, whereas $p^2 \nmid (p + 1)!$, since among the integers $1, \ldots, p + 1$ there is only one multiple of p. This contradiction shows that G is not simple.

(b) First suppose $p = 2$. Then $|G| = 2^a \cdot 5$ with $a > 3$. The number n_2 of Sylow 2-subgroups satisfies

$$n_2 \mid 5 \quad \text{and} \quad n_2 \equiv 1 \pmod 2.$$

Thus $n_2 = 1$ or $n_2 = 5$. If G were simple, then $n_2 = 5$, and the conjugation action on the five Sylow 2-subgroups would embed G in S_5. Hence $2^a \cdot 5$ would divide $5! = 120 = 2^3 \cdot 3 \cdot 5$, contradicting $a > 3$. Therefore G is not simple in this case.

Now suppose $p > 3$. Then $p \nmid p + 3$, so a Sylow p-subgroup has order p^a, and

$$n_p \mid p + 3 \quad \text{and} \quad n_p \equiv 1 \pmod p.$$

Since $n_p \leq p+3$, the only values congruent to 1 modulo p are 1 and $p+1$. But $p+1 \nmid p+3$. Hence $n_p = 1$, so the Sylow p-subgroup is normal and G is not simple.

(c) Let $|G| = p^a(p^2 - 1)$, where p is odd and $a > 1$. Then

$$n_p \mid p^2 - 1 \quad \text{and} \quad n_p \equiv 1 \pmod p.$$

Write $n_p = kp + 1$. Since $n_p \mid p^2 - 1$, we have $0 \leq k \leq p - 1$. Also $kp \equiv -1 \pmod{n_p}$, so

$$k^2p^2 \equiv 1 \pmod{n_p}.$$

As $n_p \mid p^2 - 1$, it follows that

$$n_p \mid k^2(p^2 - 1).$$

Combining this with the congruence above gives

$$n_p \mid k^2 - 1.$$

If $k \geq 2$, then $n_p = kp + 1 > k^2 - 1$, because $k \leq p - 1$. This is impossible. Hence $k = 0$ or $k = 1$, and therefore

$$n_p = 1 \quad \text{or} \quad n_p = p + 1.$$

If G were simple, then $n_p \neq 1$, so $n_p = p + 1$. As in part (a), the conjugation action on the Sylow p-subgroups would embed G in S_{p+1}. Thus $|G| \mid (p + 1)!$. But $p^2 \mid |G|$ and $p^2 \nmid (p + 1)!$, contradicting $a > 1$. Therefore G is not simple.

□

Exercise 7.11. Show that if $|G| \in \{30, 56, 105, 132\}$, then G is not simple.

Solution. We argue by contradiction in each case, assuming that G is simple.

1. If $|G| = 30 = 2 \cdot 3 \cdot 5$, then

$$n_5 \mid 6, \qquad n_5 \equiv 1 \pmod 5,$$

so $n_5 = 1$ or 6. Simplicity forces $n_5 = 6$. These six Sylow 5-subgroups are distinct subgroups of order 5, and hence intersect pairwise only in the identity. They therefore contribute $6(5 - 1) = 24$ non-identity elements. Similarly,

$$n_3 \mid 10, \qquad n_3 \equiv 1 \pmod 3,$$

so $n_3 = 1$ or 10. Simplicity forces $n_3 = 10$, and these Sylow 3-subgroups contribute $10(3 - 1) = 20$ non-identity elements. Elements of order 3 and elements of order 5 are disjoint, so this gives at least $24 + 20 = 44$ non-identity elements, impossible in a group of order 30.

2. If $|G| = 56 = 2^3 \cdot 7$, then

$$n_7 \mid 8, \qquad n_7 \equiv 1 \pmod 7,$$

so $n_7 = 1$ or 8. Simplicity forces $n_7 = 8$. These eight Sylow 7-subgroups contribute $8(7 - 1) = 48$ non-identity elements. Hence only $56 - 1 - 48 = 7$ non-identity elements remain outside the Sylow 7-subgroups. On the other hand,

$$n_2 \mid 7, \qquad n_2 \equiv 1 \pmod 2,$$

so $n_2 = 1$ or 7. Simplicity forces $n_2 = 7$. Each Sylow 2-subgroup has order 8, so its seven non-identity elements must be exactly the seven remaining non-identity elements. Thus all Sylow 2-subgroups are equal, contradicting $n_2 = 7$.

3. If $|G| = 105 = 3 \cdot 5 \cdot 7$, then
$$n_7 \mid 15, \qquad n_7 \equiv 1 \pmod 7,$$
so $n_7 = 1$ or 15. Simplicity forces $n_7 = 15$, giving $15(7-1) = 90$ non-identity elements of order 7. Also
$$n_5 \mid 21, \qquad n_5 \equiv 1 \pmod 5,$$
so $n_5 = 1$ or 21. Simplicity forces $n_5 = 21$, giving $21(5-1) = 84$ non-identity elements of order 5. These are disjoint from the elements of order 7, giving more elements than G has. This is impossible.

4. If $|G| = 132 = 2^2 \cdot 3 \cdot 11$, then
$$n_{11} \mid 12, \qquad n_{11} \equiv 1 \pmod{11},$$
so $n_{11} = 1$ or 12. Simplicity forces $n_{11} = 12$, and these Sylow 11-subgroups contribute $12(11-1) = 120$ non-identity elements. Now
$$n_3 \mid 44, \qquad n_3 \equiv 1 \pmod 3,$$
so $n_3 \in \{1, 4, 22\}$. Simplicity gives $n_3 \neq 1$. If $n_3 = 22$, then the Sylow 3-subgroups contribute $22(3-1) = 44$ non-identity elements, disjoint from the elements of order 11, which is impossible. Hence $n_3 = 4$. The conjugation action of G on its four Sylow 3-subgroups gives a non-trivial homomorphism
$$G \to S_4.$$
Since G is simple, the kernel must be trivial, so G embeds in S_4. This is impossible because $132 \nmid 24 = |S_4|$.

□

Exercise 7.12. Show that G is not simple in the following cases:

(a) $|G| = 90$. (Hint: Show that if Q is a Sylow 3-subgroup of G and $1 \neq x \in Q$, then $C_G(x) = Q$.)

(b) $|G| = 112$. (Hint: Use fact B above to show that any two Sylow 2-subgroups of G intersect trivially.)

(c) $|G| = 120$. (Hint: Show that G imbeds in S_6, and contradict Exercise 3.8.)

(d) $|G| = 144$. (Hint: Argue as in part (i) above.)

(e) $|G| = 180$. (Hint: Show that if Q is a Sylow 3-subgroup of G and $1 \neq x \in Q$, then $|C_G(x) : Q| \leq 2$.)

Solution. We argue by contradiction in each case, assuming that G is simple.

(a) Let $|G| = 90 = 2 \cdot 3^2 \cdot 5$. Then
$$n_3 \mid 10, \qquad n_3 \equiv 1 \pmod 3,$$
so simplicity forces $n_3 = 10$. Also
$$n_5 \mid 18, \qquad n_5 \equiv 1 \pmod 5,$$
so simplicity forces $n_5 = 6$. Let Q be a Sylow 3-subgroup and let $1 \neq x \in Q$. Since $|Q| = 9$, the group Q is abelian, so $Q \leq C_G(x)$. Hence $|C_G(x)|$ is divisible by 9. If $|C_G(x)| = 90$, then $\langle x \rangle$ is a non-trivial proper normal subgroup of G. If $|C_G(x)| = 45$, then $C_G(x)$ has index 2 and is normal in G. If $|C_G(x)| = 18$, then the action of G on the five cosets of $C_G(x)$ gives a non-trivial homomorphism $G \to S_5$, which must be injective by simplicity; this is impossible because $90 \nmid 5!$. Therefore $|C_G(x)| = 9$, and so
$$C_G(x) = Q.$$
It follows that distinct Sylow 3-subgroups intersect trivially: if two of them contain the same non-identity element x, then both are contained in $C_G(x)$ and hence are equal. Thus the Sylow 3-subgroups contribute $10(9-1) = 80$ non-identity elements. The Sylow 5-subgroups contribute $6(5-1) = 24$ further non-identity elements, disjoint from these. This is impossible in a group of order 90.

(b) Let $|G| = 112 = 2^4 \cdot 7$. Then
$$n_2 \mid 7, \qquad n_2 \equiv 1 \pmod 2,$$
and
$$n_7 \mid 16, \qquad n_7 \equiv 1 \pmod 7.$$
Simplicity forces $n_2 = 7$ and $n_7 = 8$. We first show that distinct Sylow 2-subgroups intersect trivially. Suppose otherwise, and choose distinct Sylow 2-subgroups S and T such that
$$D = S \cap T \neq 1$$
has maximal possible order among intersections of distinct Sylow 2-subgroups. By fact B, applied inside the 2-groups S and T, we have
$$D < N_S(D) \quad \text{and} \quad D < N_T(D).$$
Put $N = N_G(D)$. We claim that N has more than one Sylow 2-subgroup. If U were the unique Sylow 2-subgroup of N, then U would contain both $N_S(D)$ and $N_T(D)$. Let R be a Sylow 2-subgroup of G containing U. Since
$$N_S(D) \leq S \cap R \quad \text{and} \quad N_S(D) > D,$$
the maximality of D forces $R = S$. The same argument with T forces $R = T$, a contradiction. Hence N has more than one Sylow 2-subgroup. The number of Sylow 2-subgroups of N is odd and greater than 1, so 7 divides $|N|$. Also N contains the 2-subgroup $N_S(D)$, which has order at least 4. Hence 28 divides $|N|$. Thus $|G : N|$ is 1, 2, or 4. If $|G : N| = 1$, then $D \trianglelefteq G$; if $|G : N| = 2$, then $N \trianglelefteq G$; and if $|G : N| = 4$, then the coset action gives an embedding $G \hookrightarrow S_4$, impossible because $112 \nmid 24$. This proves that distinct Sylow 2-subgroups intersect trivially.
Therefore the seven Sylow 2-subgroups contribute $7(16-1) = 105$ non-identity elements. The eight Sylow 7-subgroups contribute $8(7-1) = 48$ non-identity elements, disjoint from the elements in the Sylow 2-subgroups. This is impossible in a group of order 112.

(c) Let $|G| = 120 = 2^3 \cdot 3 \cdot 5$. Then
$$n_5 \mid 24, \qquad n_5 \equiv 1 \pmod 5,$$
so simplicity forces $n_5 = 6$. The conjugation action of G on its six Sylow 5-subgroups gives a non-trivial homomorphism
$$G \to S_6.$$
Since G is simple, this homomorphism is injective. Let H be its image. The sign map on H must be trivial, since otherwise $H \cap A_6$ would be a proper normal subgroup of the simple group H. Hence $H \leq A_6$. But then
$$|A_6 : H| = \frac{360}{120} = 3.$$
By Exercise 7.8, the group A_6 is simple. Its action on the three cosets of H would therefore give an embedding $A_6 \hookrightarrow S_3$, impossible because $360 \nmid 6$. Thus G is not simple.

(d) Let $|G| = 144 = 2^4 \cdot 3^2$. Then
$$n_3 \mid 16, \qquad n_3 \equiv 1 \pmod 3,$$
so $n_3 \in \{1, 4, 16\}$. Simplicity gives $n_3 \neq 1$. If $n_3 = 4$, then the conjugation action on the four Sylow 3-subgroups embeds G in S_4, impossible because $144 \nmid 24$. Hence $n_3 = 16$.
We claim that distinct Sylow 3-subgroups intersect trivially. Suppose Q and R are distinct Sylow 3-subgroups with $D = Q \cap R \neq 1$. Since $|Q| = |R| = 9$, both Q and R are abelian, and hence both lie in $N_G(D)$. Thus $N_G(D)$ has more than one Sylow 3-subgroup. The number of its Sylow 3-subgroups divides 16 and is congruent to 1 modulo 3, so it is at least 4. Hence 36 divides $|N_G(D)|$. Therefore $|G : N_G(D)|$ is 1, 2, or 4. These cases are impossible respectively because they would make D normal in G, make $N_G(D)$ normal in G, or embed G in S_4. Thus distinct Sylow 3-subgroups intersect trivially.
The sixteen Sylow 3-subgroups therefore contribute $16(9-1) = 128$ non-identity elements. Only $144 - 1 - 128 = 15$ non-identity elements remain. But a Sylow 2-subgroup has order 16, and hence has exactly 15 non-identity elements, all outside the Sylow 3-subgroups. Therefore every Sylow 2-subgroup is the same subgroup, making it normal. This contradicts simplicity.

(e) Let $|G| = 180 = 2^2 \cdot 3^2 \cdot 5$. Let Q be a Sylow 3-subgroup and let $1 \neq x \in Q$. Since $|Q| = 9$, the group Q is abelian, so $Q \leq C_G(x)$. Thus $|C_G(x)|$ is divisible by 9. The possibilities are 9, 18, 36, 45, 90, and 180. If $|C_G(x)| = 180$, then $\langle x \rangle \trianglelefteq G$. If $|C_G(x)| = 90$, then $C_G(x)$ has index 2 and is normal in G. If $|C_G(x)| = 45$, then the coset action gives an embedding $G \hookrightarrow S_4$, impossible because $180 \nmid 24$. If $|C_G(x)| = 36$, then the coset action gives an embedding $G \hookrightarrow S_5$, impossible because $180 \nmid 120$. Hence

$$|C_G(x) : Q| \leq 2.$$

In particular, Q is the unique Sylow 3-subgroup of $C_G(x)$. Therefore distinct Sylow 3-subgroups intersect trivially. Now

$$n_3 \mid 20, \qquad n_3 \equiv 1 \pmod 3,$$

so $n_3 \in \{1, 4, 10\}$, and simplicity gives $n_3 \neq 1$. If $n_3 = 4$, then the non-identity elements in the Sylow 3-subgroups number $4(9-1) = 32$. But each conjugacy class of such an element has size $|G : C_G(x)|$, which is either 20 or 10, contradicting that 32 is not a sum of numbers of this form. Hence $n_3 = 10$, and there are $10(9-1) = 80$ non-identity elements lying in Sylow 3-subgroups. Finally,

$$n_5 \mid 36, \qquad n_5 \equiv 1 \pmod 5,$$

so $n_5 \in \{1, 6, 36\}$. Simplicity gives $n_5 \neq 1$. If $n_5 = 36$, then the Sylow 5-subgroups contribute $36(5-1) = 144$ non-identity elements, disjoint from the 80 elements already counted in Sylow 3-subgroups, impossible. Hence $n_5 = 6$. The conjugation action on the six Sylow 5-subgroups embeds G in S_6. Its image lies in A_6 by the sign-map argument used in part (c), and has order 180, hence index 2 in A_6. This contradicts the simplicity of A_6.

□

From fact A above and Exercises 9–12, we can conclude that if G is a simple group with $|G| \leq 200$, then $|G|$ either is prime or is equal to either 60 or 168. (Verify this.) In light of Theorem 7.5 and Theorem 6.8, all we need now show is that any two simple groups of order $168 = 2^3 \cdot 3 \cdot 7$ are isomorphic. We accomplish this in Exercises 13–17 below by showing that any simple group of order 168 is isomorphic with a specific subgroup of A_8. (Compare with Exercise 6.9.) Let G be a simple group of order 168, let $P = \langle x \rangle$ be a Sylow 7-subgroup of G, and let $H = N_G(P)$.

Exercise 7.13. Show that $H = \{g \in G \mid gxg^{-1} \in P\}$ is a non-abelian group of order 21 which is generated by x and y, where y has order 3 and $yxy^{-1} = x^4$.

Solution. Since $P = \langle x \rangle$, an element $g \in G$ normalizes P if and only if

$$gPg^{-1} = P.$$

If g normalizes P, then certainly $gxg^{-1} \in P$. Conversely, if $gxg^{-1} \in P$, then gxg^{-1} has order 7, so it is a non-identity element of the cyclic group P of order 7. Hence it generates P, and therefore

$$gPg^{-1} = \langle gxg^{-1} \rangle = P.$$

Thus

$$H = N_G(P) = \{g \in G \mid gxg^{-1} \in P\}.$$

Next,

$$n_7 \mid 24, \qquad n_7 \equiv 1 \pmod 7,$$

so $n_7 = 1$ or 8. Since G is simple, $n_7 \neq 1$, and hence $n_7 = 8$. The Sylow 7-subgroups are the conjugates of P, so

$$8 = n_7 = |G : N_G(P)| = |G : H|.$$

Therefore

$$|H| = \frac{168}{8} = 21.$$

We now show that H is non-abelian. Suppose, for a contradiction, that H is abelian. By Cauchy's Theorem, H contains a subgroup Y of order 3. Since H is abelian, $Y \trianglelefteq H$, so $H \leq N_G(Y)$. Thus $|N_G(Y)|$ is divisible by 21, and hence

$$|N_G(Y)| \in \{21, 42, 84, 168\}.$$

If $|N_G(Y)| = 168$, then $Y \trianglelefteq G$. If $|N_G(Y)| = 84$, then $N_G(Y)$ has index 2 in G and is normal. If $|N_G(Y)| = 42$, then the action of G on the four cosets of $N_G(Y)$ gives an embedding $G \hookrightarrow S_4$, impossible because $168 \nmid 24$. Finally, if $|N_G(Y)| = 21$, then the number of Sylow 3-subgroups of G is

$$|G : N_G(Y)| = 8,$$

contradicting the congruence $n_3 \equiv 1 \pmod 3$. Hence H is non-abelian.
Choose an element $y \in H$ of order 3, again by Cauchy's Theorem. Since $P \trianglelefteq H$, the subgroup $P\langle y\rangle$ is a subgroup of H of order

$$\frac{|P|\,|\langle y\rangle|}{|P \cap \langle y\rangle|} = 7 \cdot 3 = 21.$$

Thus

$$H = P\langle y\rangle = \langle x, y\rangle.$$

Since $y \in H = N_G(P)$, conjugation by y restricts to an automorphism of P. Therefore

$$yxy^{-1} = x^a$$

for some $a \in \{1, 2, 3, 4, 5, 6\}$. Because y has order 3, this automorphism has order dividing 3, so

$$a^3 \equiv 1 \pmod 7.$$

Since H is non-abelian, the action is not trivial, so $a \neq 1$. The non-trivial solutions are $a = 2$ and $a = 4$. If necessary, replacing y by y^2 changes a to a^2, so we may choose y so that

$$yxy^{-1} = x^4.$$

□

Exercise 7.14. (cont.) Show that $N_G(\langle y\rangle)$ is isomorphic with S_3 and is generated by y and z, where z has order 2 and $yzy = z$.

Solution. Put $Y = \langle y\rangle$ and $N = N_G(Y)$. We first compute the number of Sylow 3-subgroups of G. Inside H, the subgroup $P = \langle x\rangle$ is normal. If Y were also normal in H, then P and Y would be normal subgroups of coprime orders, so they would commute elementwise and $H = PY$ would be abelian. This contradicts Exercise 7.13. Hence Y is not normal in H, and the number of Sylow 3-subgroups of H is 7. Now the Sylow congruence in G gives

$$n_3 \mid 56, \qquad n_3 \equiv 1 \pmod 3,$$

so $n_3 \in \{1, 4, 7, 28\}$. Simplicity gives $n_3 \neq 1$. If $n_3 = 4$, then the conjugation action on the four Sylow 3-subgroups gives an embedding $G \hookrightarrow S_4$, impossible because $168 \nmid 24$. We also cannot have $n_3 = 7$. Indeed, every conjugate of H contains seven Sylow 3-subgroups, so if G had only seven Sylow 3-subgroups, then every conjugate of H would contain all of them. But there are eight conjugates of H, one for each Sylow 7-subgroup, and two distinct conjugates of H would then have intersection containing the identity and all fourteen non-identity elements in the Sylow 3-subgroups. This is impossible, since the intersection of two subgroups of order 21 has order dividing 21. Therefore

$$n_3 = 28.$$

Since the Sylow 3-subgroups are the conjugates of Y, we get

$$|G : N_G(Y)| = 28.$$

Hence

$$|N| = |N_G(Y)| = \frac{168}{28} = 6.$$

We claim that N is not cyclic. Suppose otherwise. Then, for each Sylow 3-subgroup Y' of G, the group $N_G(Y')$ is cyclic of order 6, and hence contains exactly two elements of order 6. Distinct Sylow 3-subgroups give disjoint such elements of order 6, because an element of order 6 determines its unique subgroup of order 3. Thus G would contain

$$28 \cdot 2 = 56$$

elements of order 6. The Sylow 3-subgroups contribute $28 \cdot 2 = 56$ elements of order 3, and the Sylow 7-subgroups contribute $8 \cdot 6 = 48$ elements of order 7. Altogether this accounts for

$$56 + 56 + 48 = 160$$

non-identity elements. Only seven non-identity elements remain. But a Sylow 2-subgroup has order 8, so its seven non-identity elements must be exactly these seven remaining elements. Hence the Sylow 2-subgroup is unique and normal, contradicting the simplicity of G. Therefore N is non-cyclic.
A non-cyclic group of order 6 is isomorphic with S_3. By Cauchy's Theorem, choose an element $z \in N$ of order 2. Then $N = \langle y, z\rangle$, since this subgroup has order divisible by both 2 and 3 and is contained in the group N of order 6. Since $z \in N_G(Y)$, conjugation by z sends y either to y or to y^{-1}. It cannot send y to itself, since then y and z would commute and $N = \langle y, z\rangle$ would be cyclic. Therefore

$$zyz^{-1} = y^{-1}.$$

As z has order 2, this is equivalent to

$$yzy = z.$$

Thus $N_G(\langle y\rangle) = \langle y, z\rangle \cong S_3$, with z of order 2 and $yzy = z$. □

Exercise 7.15. (cont.) Show that $G/H = \{H, zH, xzH, \ldots, x^6zH\}$. Conclude that $G = \langle x, y, z\rangle$.

Solution. We know from Exercise 7.13 that $|H| = 21$, so

$$|G : H| = \frac{168}{21} = 8.$$

Thus it is enough to show that the eight displayed left cosets are distinct. First, $zH \neq H$, because otherwise $z \in H$, impossible since z has order 2 while $|H| = 21$. Similarly, $x^izH \neq H$ for every $0 \leq i \leq 6$, because $x^i \in H$ and $x^iz \in H$ would force $z \in H$.
Now suppose that

$$x^izH = x^jzH$$

for some $0 \leq i, j \leq 6$. Then

$$zx^{i-j}z \in H.$$

If $i \neq j$, then x^{i-j} is a non-identity element of P, and hence generates P. Since $zx^{i-j}z$ is a non-identity element of order 7 lying in H, it lies in the unique Sylow 7-subgroup P of H. Therefore

$$zPz^{-1} = P,$$

so $z \in N_G(P) = H$, again impossible. Hence $i = j$. The eight cosets

$$H,\ zH,\ xzH, \ldots, x^6zH$$

are therefore distinct, and since there are exactly eight left cosets of H in G, they are all of them.
Finally, Exercise 7.13 gives $H = \langle x, y\rangle$. Let $K = \langle x, y, z\rangle$. Then $H \leq K$, and each coset representative in the displayed list lies in K. Hence every element of G lies in K, so

$$G = K = \langle x, y, z\rangle.$$

□

Let $\varphi\colon G \to S_8$ be the monomorphism corresponding to the action of G on the coset space G/H. Observe that $\varphi(G) = \varphi(G)' \leq S_8' \leq A_8$. View S_8 as the group of permutations of the set $\{\infty, 0, 1, 2, 3, 4, 5, 6\}$; for each $0 \leq i \leq 6$, associate i with the coset x^izH, and associate ∞ with the coset H.

Exercise 7.16. (cont.) Show that $\varphi(x) = (0\,1\,2\,3\,4\,5\,6)$, that $\varphi(y) = (1\,4\,2)(3\,5\,6)$, and that $\varphi(z) = z_i$ for some i, where $z_1 = (\infty\,0)(1\,3)(2\,5)(4\,6)$, $z_2 = (\infty\,0)(1\,5)(2\,6)(3\,4)$, and $z_3 = (\infty\,0)(1\,6)(2\,3)(4\,5)$.

Solution. The action is left multiplication on the eight left cosets of H. With the labelling above, ∞ denotes the coset H, while i denotes the coset x^izH for $0 \leq i \leq 6$. Since $x \in H$, the element x fixes H. Also, for $0 \leq i \leq 6$,

$$x \cdot x^izH = x^{i+1}zH,$$

where the exponent is read modulo 7. Hence

$$\varphi(x) = (0\,1\,2\,3\,4\,5\,6).$$

Similarly, $y \in H$, so y fixes H. From Exercise 7.13, we have $yxy^{-1} = x^4$, and hence

$$yx^i = x^{4i}y$$

for every i. From Exercise 7.14, we also have $yzy = z$, so $yz = zy^{-1}$. Since $y^{-1} \in H$, this gives $yzH = zH$. Therefore

$$y \cdot x^i zH = yx^i zH = x^{4i}yzH = x^{4i}zH.$$

Thus $\varphi(y)$ sends i to $4i$ modulo 7, while fixing ∞ and 0. Hence

$$\varphi(y) = (1\,4\,2)(3\,5\,6).$$

It remains to determine the possible form of $\varphi(z)$. Write $t = \varphi(z)$ and $\sigma = \varphi(y)$. Since $z^2 = 1$, we have $t^2 = 1$. Also,

$$z \cdot H = zH \quad \text{and} \quad z \cdot zH = H,$$

so t contains the transposition $(\infty\,0)$. Since $\varphi(G) \leq A_8$, the permutation t is even. Finally, Exercise 7.14 gives $zyz^{-1} = y^{-1}$, so

$$t\sigma t^{-1} = \sigma^{-1}.$$

The two non-trivial cycles of σ have supports $\{1, 4, 2\}$ and $\{3, 5, 6\}$. The last conjugation relation forces t either to preserve these two supports or to interchange them. It cannot preserve both supports: on each support it would have to reverse a 3-cycle, giving one transposition on each support, and together with $(\infty\,0)$ this would make t odd. Hence t interchanges the two supports.
Thus $t(1)$ is one of $3, 5, 6$. Since $t\sigma = \sigma^{-1}t$, the image of 1 determines the rest:

$t(1)$	$t(4)$	$t(2)$	t
3	6	5	$(\infty\,0)(1\,3)(2\,5)(4\,6)$
5	3	6	$(\infty\,0)(1\,5)(2\,6)(3\,4)$
6	5	3	$(\infty\,0)(1\,6)(2\,3)(4\,5)$.

Therefore $\varphi(z) = z_i$ for some $i \in \{1, 2, 3\}$. □

Exercise 7.17. (cont.) Let $L_i = \langle \varphi(x), \varphi(y), z_i \rangle \leq A_8$ for $i = 1, 2, 3$. Show that $L_1 = L_2 = L_3$. Conclude that G is uniquely determined up to isomorphism.

Solution. Put

$$a = \varphi(x) = (0\,1\,2\,3\,4\,5\,6) \quad \text{and} \quad b = \varphi(y) = (1\,4\,2)(3\,5\,6).$$

Then $L_i = \langle a, b, z_i \rangle$. We first compare the three possible choices of z_i. Since b fixes ∞ and 0, direct conjugation gives

$$bz_1b^{-1} = (\infty\,0)(4\,5)(1\,6)(2\,3) = z_3$$

and

$$b^{-1}z_1b = (\infty\,0)(2\,6)(4\,3)(1\,5) = z_2.$$

Thus $z_2, z_3 \in L_1$, and hence

$$L_2 = \langle a, b, z_2 \rangle \leq L_1, \qquad L_3 = \langle a, b, z_3 \rangle \leq L_1.$$

Conversely, the same two identities give

$$z_1 = bz_2b^{-1} \quad \text{and} \quad z_1 = b^{-1}z_3b.$$

Hence $z_1 \in L_2$ and $z_1 \in L_3$, so $L_1 \leq L_2$ and $L_1 \leq L_3$. Therefore

$$L_1 = L_2 = L_3.$$

It remains to connect this with the original group G. By Exercise 7.15, we have $G = \langle x, y, z \rangle$. Applying φ gives

$$\varphi(G) = \langle \varphi(x), \varphi(y), \varphi(z) \rangle.$$

By Exercise 7.16, the element $\varphi(z)$ is one of z_1, z_2, z_3. Hence

$$\varphi(G) = L_i$$

for some $i \in \{1, 2, 3\}$. Since $L_1 = L_2 = L_3$, the image $\varphi(G)$ is the fixed subgroup $L_1 \leq A_8$, independent of the possible choice of $\varphi(z)$. Finally, φ is a monomorphism, so

$$G \cong \varphi(G) = L_1.$$

Thus G is uniquely determined up to isomorphism. $\square$

8 Finite p-groups

We have seen that Sylow's Theorem allows us to study finite groups by concentrating on their subgroups of prime-power order. This motivates the study of finite p-groups, a rich subject whose surface we shall only scratch in this section. Our first result is the most basic structural fact about finite p-groups.

Theorem 8.1. *If P is a non-trivial finite p-group, then $Z(P)$ is also non-trivial. Moreover, if $1 \neq N \trianglelefteq P$, then $N \cap Z(P) \neq 1$.*

Proof. Let $K_1 = \{1\}, K_2, \ldots, K_r$ be the conjugacy classes of P. Suppose first that $Z(P) = 1$. If some K_i with $i > 1$ were a singleton, say $K_i = \{x\}$, then $gxg^{-1} = x$ for every $g \in P$, so $x \in Z(P) = 1$, contradicting $i > 1$. Hence $|K_i| > 1$ for every $i > 1$. Each K_i is an orbit for the conjugation action of P on itself, as in Proposition 3.13, and so $|K_i|$ divides $|P|$ by Corollary 3.5. Since P is a p-group, it follows that p divides $|K_i|$ for every $i > 1$. Therefore

$$|P| = 1 + |K_2| + \cdots + |K_r| \equiv 1 \pmod{p},$$

which is impossible because P is a non-trivial p-group. Thus $Z(P) \neq 1$.
Now let $1 \neq N \trianglelefteq P$. Since N is normal in P, it is a union of conjugacy classes of P, and since $1 \in N$, this union contains $K_1 = \{1\}$. If $N \cap Z(P) = 1$, then every conjugacy class of P contained in N other than K_1 has size greater than 1, and hence has size divisible by p by the same orbit-size argument. Therefore $|N| \equiv 1 \pmod{p}$, contradicting the fact that N is a non-trivial subgroup of the p-group P. Hence $N \cap Z(P) \neq 1$. □

In particular, the only finite simple p-groups are the groups of prime order. Indeed, if P is a non-trivial simple p-group, then Theorem 8.1 gives $Z(P) \neq 1$. Since $Z(P) \trianglelefteq P$, simplicity forces $Z(P) = P$, so P is abelian; hence P has prime order.

Lemma 8.2. *If G is a non-abelian group, then $G/Z(G)$ cannot be cyclic. In particular, if P is a finite p-group, then $|P : Z(P)| \neq p$.*

Proof. Suppose, for a contradiction, that $G/Z(G)$ is cyclic, say $G/Z(G) = \langle xZ(G)\rangle$ for some $x \in G$. Then every element of G has the form $x^m z$ for some $m \in \mathbb{Z}$ and some $z \in Z(G)$. If $g = x^m z$ and $h = x^n w$, where $z, w \in Z(G)$, then

$$gh = x^m z x^n w = x^{m+n} zw = x^{n+m} wz = x^n w x^m z = hg.$$

Thus any two elements of G commute, so G is abelian, contradicting the hypothesis. Therefore $G/Z(G)$ cannot be cyclic.
Now let P be a finite p-group. If P is abelian, then $Z(P) = P$, so $|P : Z(P)| = 1$. If P is non-abelian and $|P : Z(P)| = p$, then $P/Z(P)$ has prime order and is therefore cyclic, contradicting the first part. Hence $|P : Z(P)| \neq p$. □

Recall that a proper subgroup H of a group G is said to be *maximal* if there is no proper subgroup of G properly containing H. If G is non-trivial and finite, then G has maximal subgroups, and every proper subgroup of G is contained in one. We now show that maximal subgroups of finite p-groups are especially well-behaved.

Proposition 8.3. *If P is a finite p-group, then every maximal subgroup of P is normal in P.*

Proof. We argue by induction on $|P|$. If $|P| = 1$, then P has no maximal subgroups, and if $|P| = p$, then the only maximal subgroup is 1, which is normal in P. Thus we may assume that $|P| > p$ and that the result holds for all finite p-groups of smaller order. Let M be a maximal subgroup of P, and put $Z = Z(P)$. Since Z is central, MZ is a subgroup of P containing M. By maximality, either $MZ = P$ or $MZ = M$.
If $MZ = P$, then every element of P has the form mz with $m \in M$ and $z \in Z$. For any $a \in M$, we have

$$(mz)a(mz)^{-1} = mzaz^{-1}m^{-1} = mam^{-1} \in M,$$

so $M \trianglelefteq P$.
It remains to consider the case $MZ = M$. Then $Z \leq M$. By Theorem 8.1, $Z \neq 1$, so $|P/Z| < |P|$. Moreover, by the Correspondence Theorem, M/Z is a maximal subgroup of the finite p-group P/Z. The induction hypothesis gives $M/Z \trianglelefteq P/Z$, and applying the Correspondence Theorem once more gives $M \trianglelefteq P$. □

Corollary 8.4. *Any maximal subgroup of a finite p-group is of index p.*

Proof. Let M be a maximal subgroup of the finite p-group P. By Proposition 8.3, we have $M \trianglelefteq P$, so P/M is a non-trivial finite p-group. By Cauchy's Theorem, P/M has a subgroup of order p. By the Correspondence Theorem, this subgroup has the form L/M for some subgroup L satisfying $M < L \leq P$. Since M is maximal in P, we must have $L = P$. Therefore

$$|P : M| = |P/M| = |L/M| = p.$$

□

Our next task is to classify the finite abelian p-groups. The following lemma isolates the elementary fact about cyclic p-groups that will be used in the induction.

Lemma 8.5. *Every non-generator of a cyclic p-group P is a pth power in P.*

Proof. If $P = 1$, then there is nothing to prove. Otherwise, write $P = \langle x \rangle$ with $|P| = p^a$ for some $a \geq 1$. By Theorem 1.4, the subgroup

$$Q = \langle x^p \rangle$$

is the unique subgroup of P of index p. Every element of Q is a pth power in P, since

$$(x^p)^n = (x^n)^p$$

for every $n \in \mathbb{Z}$.
We claim that Q is precisely the set of non-generators of P. Since $Q < P$, no element of Q can generate P. Conversely, if $y \in P$ does not generate P, then $\langle y \rangle$ is a proper subgroup of P, and hence is contained in some maximal subgroup M of P. By Corollary 8.4, M has index p, so by the uniqueness of Q we have $M = Q$. Thus $y \in Q$, and therefore y is a pth power in P. □

Theorem 8.6. *A finite abelian p-group is a direct product of cyclic p-groups.*

Proof. We argue by induction on $|P|$. If $P = 1$, there is nothing to prove, and if $|P| = p$, then P is cyclic. Thus assume that $|P| > p$ and that the result holds for abelian p-groups of smaller order. Let Q be a maximal subgroup of P. By Corollary 8.4, we have $|P : Q| = p$, and by induction we may write

$$Q = Q_1 \times \cdots \times Q_s,$$

where each Q_i is cyclic of order p^{a_i}. Reordering if necessary, assume that $a_1 \geq \cdots \geq a_s \geq 1$.
Choose $x \in P - Q$. Since P/Q has order p, we have $x^p \in Q$. Write

$$x^p = y_1 \cdots y_s,$$

where $y_i \in Q_i$ for each i. If some y_i is a non-generator of Q_i, then Lemma 8.5 gives $y_i = z_i^p$ for some $z_i \in Q_i$. Replacing x by xz_i^{-1} keeps x outside Q, and changes the ith coordinate of x^p to 1. Repeating this process, we may assume that each y_i is either 1 or a generator of Q_i.
If $x^p = 1$, then $\langle x \rangle \cap Q = 1$, because xQ has order p in P/Q. Hence $\langle x \rangle Q$ is a direct product of order $p|Q| = |P|$, so $P = \langle x \rangle \times Q$, and the result follows. We may therefore assume that $x^p \neq 1$. Let j be minimal such that $y_j \neq 1$. Then y_j is a generator of Q_j, so it has order p^{a_j}; since $a_j \geq a_i$ for all $i \geq j$, the element $x^p = y_j \cdots y_s$ has order p^{a_j}. Therefore $\langle x \rangle$ has order p^{a_j+1}.
Let R be the direct product of all the Q_i except Q_j. We claim that $\langle x \rangle \cap R = 1$. Suppose $x^t \in R$, with $0 \leq t < |\langle x \rangle|$. Since xQ has order p in P/Q, we must have $p \mid t$, say $t = mp$ with $0 \leq m < p^{a_j}$. Then

$$x^t = (x^p)^m = y_j^m \cdots y_s^m.$$

The Q_j-coordinate of this element is y_j^m. Since y_j has order p^{a_j} and $0 \leq m < p^{a_j}$, this coordinate is the identity only when $m = 0$. Thus $x^t \in R$ forces $t = 0$, and so $\langle x \rangle \cap R = 1$. It follows that $\langle x \rangle R$ is a direct product of order

$$|\langle x \rangle||R| = p^{a_j+1} \frac{|Q|}{p^{a_j}} = p|Q| = |P|.$$

Therefore $P = \langle x \rangle \times R$, which is a direct product of cyclic p-groups. □

We now consider how finite p-groups can be used to study arbitrary finite groups. The next result is a standard tool in finite group theory: it turns conjugacy of Sylow subgroups inside a normal subgroup into a factorization of the whole group.

Theorem (Frattini Argument)**.** *Let G be a finite group, let $N \trianglelefteq G$, and let P be a Sylow subgroup of N. Then $G = N_G(P)N$.*

Proof. Let $g \in G$. Since $N \trianglelefteq G$, we have $gNg^{-1} = N$, and hence $gPg^{-1} \le N$. Moreover, gPg^{-1} has the same order as P, so gPg^{-1} is also a Sylow subgroup of N. By Sylow's Theorem, applied inside N, there exists $n \in N$ such that

$$n(gPg^{-1})n^{-1} = P.$$

Equivalently, $(ng)P(ng)^{-1} = P$, so $ng \in N_G(P)$. Thus $g = n^{-1}(ng) \in NN_G(P)$. Since $N \trianglelefteq G$, we have $NN_G(P) = N_G(P)N$, and hence $g \in N_G(P)N$. As g was arbitrary, $G \le N_G(P)N$. The reverse inclusion is immediate, since both $N_G(P)$ and N are subgroups of G. Therefore $G = N_G(P)N$. □

Theorem 8.7. *The following statements about a finite group G are equivalent:*

(1) Every Sylow subgroup of G is normal in G.

(2) G is the direct product of its Sylow subgroups.

(3) Every maximal subgroup of G is normal in G.

A finite group satisfying these equivalent conditions is said to be *nilpotent.* Thus finite abelian groups and finite p-groups are nilpotent. A different but equivalent definition, one that extends beyond finite groups, will be discussed in Section 11.

Proof. We prove the implications in a cycle. First assume that every Sylow subgroup of G is normal in G. Let $p_1, \ldots, p_n$ be the distinct prime divisors of $|G|$, and let P_i be the Sylow p_i-subgroup of G. By hypothesis, each P_i is normal in G. We claim by induction that

$$P_1 \cdots P_i$$

is a normal subgroup of G of order $|P_1| \cdots |P_i|$ for every i. This is clear for $i = 1$. If it holds for i, then Proposition 1.7 gives $P_1 \cdots P_i P_{i+1} \trianglelefteq G$, and

$$(P_1 \cdots P_i) \cap P_{i+1} = 1$$

because its order divides both $|P_1| \cdots |P_i|$ and $|P_{i+1}|$, which are relatively prime. Hence Proposition 3.12 gives

$$|P_1 \cdots P_i P_{i+1}| = |P_1 \cdots P_i||P_{i+1}| = |P_1| \cdots |P_{i+1}|.$$

The claim follows. Taking $i = n$, we get $|P_1 \cdots P_n| = |G|$, so $G = P_1 \cdots P_n$. Moreover, for each i, the subgroup $P_i \cap P_1 \cdots P_{i-1} P_{i+1} \cdots P_n$ has order dividing $|P_i|$ and also dividing the product of the other Sylow orders, so it is trivial. Therefore G is the internal direct product of its Sylow subgroups. This proves (1) ⇒ (2).

Now assume that $G = P_1 \times \cdots \times P_n$, where the P_i are the Sylow subgroups of G. Let M be a maximal subgroup of G. Since the orders $|P_i|$ are pairwise relatively prime, Proposition 2.10 gives

$$M = (M \cap P_1) \times \cdots \times (M \cap P_n).$$

At least one factor $M \cap P_j$ is proper in P_j, since otherwise $M = G$. Let j be any index for which $M \cap P_j < P_j$. If $M \cap P_j$ were not maximal in P_j, then replacing it by an intermediate subgroup of P_j would give a subgroup strictly between M and G, contradicting the maximality of M. Hence each proper factor $M \cap P_j$ is maximal in P_j. There cannot be two such proper factors: if $M \cap P_j < P_j$ and $M \cap P_k < P_k$ with $j \ne k$, then replacing $M \cap P_j$ by P_j while leaving the kth factor proper again gives a subgroup strictly between M and G. Thus there is exactly one index j for which $M \cap P_j < P_j$, and this subgroup is maximal in the finite p_j-group P_j. By Proposition 8.3, $M \cap P_j \trianglelefteq P_j$. The other factors are the whole P_i, so they are normal in P_i. It follows directly from the direct product multiplication that $M \trianglelefteq G$. This proves (2) ⇒ (3).

Finally assume that every maximal subgroup of G is normal in G. Let P be a Sylow subgroup of G.

Suppose, for a contradiction, that P is not normal in G. Then $N_G(P) < G$, so choose a maximal subgroup M of G with

$$P \leq N_G(P) \leq M < G.$$

By hypothesis, $M \trianglelefteq G$. Also P is a Sylow subgroup of M, since $P \leq M$ and P already has the full p-part possible in G. Applying the Frattini Argument to the normal subgroup $M \trianglelefteq G$ and its Sylow subgroup P, we obtain

$$G = N_G(P)M.$$

But $N_G(P) \leq M$, so the right hand side is just M, contradicting $M < G$. Therefore every Sylow subgroup of G is normal in G, proving (3) $\Rightarrow$ (1). ☐

As a consequence of this result and our previous classification of finite abelian p-groups, we obtain the usual basis theorem for finite abelian groups.

Corollary 8.8. *A finite abelian group is a direct product of cyclic p-groups.*

Proof. Let G be a finite abelian group. Since every subgroup of an abelian group is normal, every Sylow subgroup of G is normal in G. By Theorem 8.7, G is the direct product of its Sylow subgroups. Each Sylow subgroup is a finite abelian p-group, and hence is a direct product of cyclic p-groups by Theorem 8.6. Combining these direct-product decompositions gives the result. ☐

We close this section with a discussion of p-groups of small order. Any group of order p is isomorphic with $\mathbb{Z}_p$. By Theorem 8.1 and Lemma 8.2, any group of order p^2 is abelian, and hence is isomorphic with either $\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$ by Theorem 8.6. The same theorem shows that the abelian groups of order p^3 are precisely $\mathbb{Z}_{p^3}$, $\mathbb{Z}_{p^2} \times \mathbb{Z}_p$, and $\mathbb{Z}_p \times \mathbb{Z}_p \times \mathbb{Z}_p$. We therefore turn to the non-abelian groups of order p^3. For odd primes p, the following commutator identities provide the technical input for the classification.

Lemma 8.9. *Let G be a group and let $x, y \in G$. If $[x, y] \in Z(G)$, then $[x^n, y] = [x, y]^n$ and $x^n y^n = (xy)^n [x, y]^{\frac{n(n-1)}{2}}$ for any $n \in \mathbb{N}$.*

Proof. Put $c = [x, y]$. Since $c \in Z(G)$, the relation $xy = cyx$ allows us to move y past powers of x. We first prove by induction on n that

$$x^n y = c^n y x^n.$$

The case $n = 1$ is exactly $xy = cyx$. If $x^n y = c^n y x^n$, then

$$x^{n+1} y = x(c^n y x^n) = c^n x y x^n = c^{n+1} y x^{n+1},$$

because c is central and $xy = cyx$. Therefore $x^n y = c^n y x^n$ for all $n \in \mathbb{N}$, and hence

$$[x^n, y] = x^n y x^{-n} y^{-1} = c^n = [x, y]^n.$$

It remains to prove the second identity. We again argue by induction on n. The case $n = 1$ is immediate. Suppose that

$$x^n y^n = (xy)^n c^{\frac{n(n-1)}{2}}.$$

Using the identity already proved, we have $x^n y = [x^n, y] y x^n = c^n y x^n$. Thus

$$x^{n+1} y^{n+1} = x(x^n y) y^n = x c^n y x^n y^n = c^n x y x^n y^n.$$

By the induction hypothesis, this becomes

$$x^{n+1} y^{n+1} = c^n (xy)(xy)^n c^{\frac{n(n-1)}{2}} = (xy)^{n+1} c^{\frac{n(n-1)}{2} + n} = (xy)^{n+1} c^{\frac{(n+1)n}{2}}.$$

Substituting back $c = [x, y]$ gives the desired formula. ☐

Proposition 8.10. *If p is an odd prime, then there is exactly one isomorphism class of non-abelian groups of order p^3 having elements of order p^2.*

Proof. Let P be a non-abelian group of order p^3 having an element of order p^2. We first show that P can be written as a semidirect product of $\mathbb{Z}_{p^2}$ by $\mathbb{Z}_p$. Choose $y \in P$ with $o(y) = p^2$, and choose $x \in P - \langle y \rangle$. Since $\langle y \rangle$ has index p, it is maximal in P, and hence $\langle y \rangle \trianglelefteq P$ by Proposition 8.3.
We may arrange that x has order p. If $x^p = 1$, there is nothing to change, so suppose that $x^p \neq 1$. Since P is non-abelian of order p^3, Theorem 8.1 and Lemma 8.2 force $|Z(P)| = p$. Also $\langle y \rangle \cap Z(P) \neq 1$ by Theorem 8.1, so the unique subgroup of order p in the cyclic group $\langle y \rangle$ gives

$$Z(P) = \langle y^p \rangle.$$

The quotient $P/Z(P)$ has order p^2, and it is not cyclic by Lemma 8.2; hence every element of $P/Z(P)$ has order dividing p. Thus $x^p \in Z(P)$. Since $x^p \neq 1$, we have $x^p = y^{kp}$ for some $1 \leq k < p$. Replacing y by y^{-k}, which is still a generator of $\langle y \rangle$, we may assume that

$$x^p = y^{-p}.$$

Now $P/Z(P)$ is abelian, so Proposition 2.6 gives $P' \leq Z(P)$. In particular, $[x, y] \in Z(P)$. By Lemma 8.9,

$$x^p y^p = (xy)^p [x, y]^{\frac{p(p-1)}{2}}.$$

The left hand side is 1, and $[x, y]^{\frac{p(p-1)}{2}} = 1$, since $[x, y] \in Z(P)$ has order dividing p and p divides $\frac{p(p-1)}{2}$ for odd p. Therefore $(xy)^p = 1$. Also $xy \notin \langle y \rangle$, because $x \notin \langle y \rangle$. Replacing x by xy, we have $o(x) = p$, $o(y) = p^2$, and $x \notin \langle y \rangle$.
It follows that $\langle x \rangle \cap \langle y \rangle = 1$, and hence

$$|\langle y \rangle \langle x \rangle| = p^2 p = p^3.$$

Thus $P = \langle y \rangle \langle x \rangle$, with $\langle y \rangle \trianglelefteq P$, so P is a semidirect product of $\langle y \rangle \cong \mathbb{Z}_{p^2}$ by $\langle x \rangle \cong \mathbb{Z}_p$. Since P is non-abelian, the corresponding homomorphism $\varphi \colon \mathbb{Z}_p \to \mathrm{Aut}(\mathbb{Z}_{p^2})$ is non-trivial, and hence is a monomorphism.
It remains to see that this semidirect product is unique up to isomorphism. By Proposition 2.1, $\mathrm{Aut}(\mathbb{Z}_{p^2}) \cong (\mathbb{Z}/p^2\mathbb{Z})^\times$, an abelian group of order $p(p-1)$. By Sylow's Theorem, $\mathrm{Aut}(\mathbb{Z}_{p^2})$ has a subgroup of order p, and because $\mathrm{Aut}(\mathbb{Z}_{p^2})$ is abelian, this Sylow p-subgroup is unique. Therefore every monomorphism $\mathbb{Z}_p \to \mathrm{Aut}(\mathbb{Z}_{p^2})$ has the same image. Proposition 2.11 then implies that all non-trivial semidirect products of $\mathbb{Z}_{p^2}$ by $\mathbb{Z}_p$ are isomorphic. Such a semidirect product exists by choosing an isomorphism from $\mathbb{Z}_p$ onto the unique subgroup of order p in $\mathrm{Aut}(\mathbb{Z}_{p^2})$, and it is non-abelian because the action is non-trivial. Hence there is exactly one isomorphism class of non-abelian groups of order p^3 having an element of order p^2. $\square$

Proposition 8.11. *If p is an odd prime, then there is exactly one isomorphism class of non-abelian groups of order p^3 having no elements of order p^2.*

Proof. Let P be a non-abelian group of order p^3 having no elements of order p^2. Choose a maximal subgroup M of P. By Corollary 8.4, we have $|P : M| = p$, and by Proposition 8.3, we have $M \trianglelefteq P$. Thus $|M| = p^2$. By Lemma 8.2, every group of order p^2 is abelian; since P has no elements of order p^2, the subgroup M is not cyclic, and hence Theorem 8.6 gives $M \cong \mathbb{Z}_p \times \mathbb{Z}_p$. Choose $x \in P - M$. Again using the hypothesis, x has order p. Also $\langle x \rangle \cap M = 1$, and so

$$|M\langle x \rangle| = |M||\langle x \rangle| = p^3.$$

Therefore $P = M\langle x \rangle$, and P is a semidirect product of $M \cong \mathbb{Z}_p \times \mathbb{Z}_p$ by $\langle x \rangle \cong \mathbb{Z}_p$. Since P is non-abelian, the corresponding homomorphism $\varphi \colon \mathbb{Z}_p \to \mathrm{Aut}(\mathbb{Z}_p \times \mathbb{Z}_p)$ is non-trivial, and hence is a monomorphism. We now show that all such semidirect products are isomorphic. Put $E = \mathbb{Z}_p \times \mathbb{Z}_p$. By Proposition 4.1, $\mathrm{Aut}(E) \cong \mathrm{GL}(2, p)$, and by Proposition 4.2,

$$|\mathrm{Aut}(E)| = |\mathrm{GL}(2, p)| = (p^2 - 1)(p^2 - p) = p(p - 1)^2(p + 1).$$

Thus a Sylow p-subgroup of $\mathrm{Aut}(E)$ has order p, so all subgroups of order p in $\mathrm{Aut}(E)$ are conjugate by Sylow's Theorem. If $\varphi, \psi \colon \mathbb{Z}_p \to \mathrm{Aut}(E)$ are monomorphisms, then there is some $f \in \mathrm{Aut}(E)$ such that

$$\psi(\mathbb{Z}_p) = f\varphi(\mathbb{Z}_p)f^{-1}.$$

Equivalently, if $\widehat{f}$ denotes conjugation by f on $\mathrm{Aut}(E)$, then $\psi(\mathbb{Z}_p) = (\widehat{f} \circ \varphi)(\mathbb{Z}_p)$. By Proposition 2.12,

$$E \rtimes_{\widehat{f}\circ\varphi} \mathbb{Z}_p \cong E \rtimes_{\varphi} \mathbb{Z}_p,$$

while Proposition 2.11 gives

$$E \rtimes_{\widehat{f}\circ\varphi} \mathbb{Z}_p \cong E \rtimes_{\psi} \mathbb{Z}_p,$$

because $\mathbb{Z}_p$ is cyclic and the two actions have the same image. Hence all non-trivial semidirect products of $\mathbb{Z}_p \times \mathbb{Z}_p$ by $\mathbb{Z}_p$ are isomorphic.
It remains only to check that this isomorphism class actually occurs among groups with no elements of order p^2. Let E be written additively as a two-dimensional vector space over $\mathbb{F}_p$, and let a generator t of $\mathbb{Z}_p$ act on E by

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Writing $A = I + N$, we have $N^2 = 0$, so $A^p = I + pN = I$ and $A \neq I$. Thus this defines a non-trivial semidirect product $E \rtimes \langle t \rangle$. We claim that every element has order dividing p. An element has the form (v, t^i). If $i = 0$, then $(v, 1)^p = (0, 1)$ because E has exponent p. If $i \neq 0$, then

$$(v, t^i)^p = \left(\sum_{j=0}^{p-1} A^{ij} v, 1 \right) = \left(\sum_{j=0}^{p-1} (I + ijN) v, 1 \right) = \left(\left(pI + i\frac{p(p-1)}{2} N \right) v, 1 \right) = (0, 1),$$

since we are working over $\mathbb{F}_p$ and p is odd. Therefore this semidirect product is non-abelian of order p^3 and has no elements of order p^2. Combined with the uniqueness just proved, this gives exactly one isomorphism class. □

It remains to mention the case $p = 2$. A non-abelian group of order 8 has an element of order 4 by Exercise 1.2. If it also has an element of order 2 outside the cyclic subgroup of order 4, then it is the non-trivial semidirect product $\mathbb{Z}_4 \rtimes \mathbb{Z}_2$, namely the dihedral group D_8. This will happen unless the group has a unique element t of order 2. In that remaining case, choose an element y of order 4, put $Q = \langle y \rangle$, and choose x so that $P/Q = \langle xQ \rangle$. Since x cannot have order 2, we must have $x^2 = t = y^2$. Every element can then be written uniquely as $x^a y^b$, where $0 \leq a \leq 3$ and $0 \leq b \leq 1$, and the multiplication is forced by the relation $yx = x^3 y$. Thus, if this case exists, it is unique up to isomorphism.
To see that it exists, work in $\mathrm{SL}(2, 3)$ over the field $\mathbb{F}_3$ and take

$$x = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad y = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}.$$

The subgroup generated by these two matrices is non-abelian of order 8, and it has the unique element

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

of order 2. This group is the quaternion group Q_8, which appeared in Exercise 2.11 and cannot be written non-trivially as a semidirect product.

Exercises

Exercise 8.1. Show that if P is a non-cyclic finite p-group, then P has a normal subgroup N such that $P/N \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

Solution. Suppose first that P has only one maximal subgroup, say M. If $x \in P - M$, then $\langle x \rangle$ is not contained in M. Since every proper subgroup of the finite group P is contained in a maximal subgroup, $\langle x \rangle$ cannot be proper. Hence $\langle x \rangle = P$, so P is cyclic. Therefore, since P is non-cyclic, it has at least two distinct maximal subgroups. Choose two of them, say M and K.
By Proposition 8.3, both M and K are normal in P, and by Corollary 8.4 both have index p. Put

$$N = M \cap K.$$

Then $N \trianglelefteq P$. Since $M \neq K$, we have $K < MK \leq P$, and the maximality of K gives $MK = P$. Hence

$$|P| = |MK| = \frac{|M||K|}{|M \cap K|} = \frac{(|P|/p)(|P|/p)}{|N|},$$

so $|P : N| = p^2$. Thus P/N has order p^2. Moreover, M/N and K/N are distinct subgroups of P/N of order p. A cyclic group of order p^2 has a unique subgroup of order p, so P/N is not cyclic. By Lemma 8.2, every group of order p^2 is abelian, and then Theorem 8.6 gives $P/N \cong \mathbb{Z}_p \times \mathbb{Z}_p$. □

Exercise 8.2. Let P be a group of order p^n. Show that P has a normal subgroup N_a of order p^a for every $0 \le a \le n$, and that these subgroups can be chosen so that N_a is contained in N_b whenever $a \le b$.

Solution. We argue by induction on n. If $n = 0$, then $P = 1$, and we take $N_0 = 1$. Now assume that $n \ge 1$ and that the result is known for groups of order p^{n-1}. By Theorem 8.1, the center $Z(P)$ is non-trivial. Hence, by Cauchy's Theorem, there is a subgroup

$$C \le Z(P)$$

of order p. Since C is central, it is normal in P.
Let $\pi\colon P \to P/C$ be the quotient map. The group P/C has order p^{n-1}, so by the induction hypothesis there are normal subgroups

$$\overline{N}_0 \le \overline{N}_1 \le \cdots \le \overline{N}_{n-1}$$

of P/C, with $|\overline{N}_a| = p^a$ for each $0 \le a \le n-1$. Define

$$N_0 = 1, \qquad N_a = \pi^{-1}(\overline{N}_{a-1}) \quad \text{for } 1 \le a \le n.$$

Each N_a is normal in P, because it is the preimage of a normal subgroup under the homomorphism π. Also $C \le N_a$ for $a \ge 1$, and

$$N_a/C \cong \overline{N}_{a-1}.$$

Therefore

$$|N_a| = |C|\,|\overline{N}_{a-1}| = p \cdot p^{a-1} = p^a$$

for every $1 \le a \le n$, while $|N_0| = 1 = p^0$. Finally, the inclusions among the $\overline{N}_a$ give

$$N_0 \le N_1 \le \cdots \le N_n,$$

so the subgroups may be chosen with $N_a \le N_b$ whenever $a \le b$. □

Exercise 8.3. Let $G = \mathrm{GL}(n,p)$, and let P be a Sylow p-subgroup of G. What is the order of $Z(P)$? What is the order of $Z(P/Z(P))$? If we let $Z_2(P) \le P$ be such that $Z_2(P)/Z(P) = Z(P/Z(P))$ and continue in this way, what happens?

Solution. Let U be the subgroup of upper unitriangular matrices in $\mathrm{GL}(n,p)$. Then $|U| = p^{n(n-1)/2}$, which is the p-part of $|\mathrm{GL}(n,p)|$, so we may take $P = U$. If $n = 1$, then $P = 1$ and there is nothing to prove, so assume $n \ge 2$. For $i < j$, put

$$X_{ij} = \{I + aE_{ij} : a \in \mathbb{F}_p\},$$

where E_{ij} is the matrix with 1 in the (i,j)-position and 0 elsewhere. These are the root subgroups of U. The basic matrix-unit calculation shows that a commutator between X_{ij} and X_{kl} can be non-trivial only when $j = k$ or $l = i$.
The subgroup X_{1n} commutes with every root subgroup: the conditions $n = k$ and $l = 1$ are impossible for $k < l$ inside U. Conversely, in an arbitrary element of U, any non-zero X_{ij}-coordinate with $(i,j) \ne (1,n)$ is detected by an adjacent root subgroup: either $i > 1$ and $X_{i-1,i}$ detects it, or $j < n$ and $X_{j,j+1}$ detects it. Thus centrality forces all root coordinates except possibly the X_{1n}-coordinate to vanish, and so

$$Z(P) = X_{1n}$$

and $|Z(P)| = p$.
If $n = 2$, then P is already abelian, so the quotient case is trivial. Assume now that $n \ge 3$. Let $Z_2(P) \le P$ be defined by $Z_2(P)/Z(P) = Z(P/Z(P))$. Modulo $Z(P) = X_{1n}$, a root subgroup becomes central exactly when all of its possible non-trivial commutators land in X_{1n}. This happens precisely for

$$X_{1,n-1}, \qquad X_{1n}, \qquad X_{2n}.$$

Hence

$$Z_2(P) = X_{1,n-1}X_{1n}X_{2n},$$

so $|Z_2(P)| = p^3$, and therefore

$$|Z(P/Z(P))| = |Z_2(P)/Z(P)| = \frac{p^3}{p} = p^2.$$

Continuing in this way gives the upper central series of P. If $Z_1(P) = Z(P)$, then for $1 \leq r \leq n-1$,

$$Z_r(P) = \langle X_{ij} : j - i \geq n - r \rangle.$$

Equivalently, $Z_r(P)$ consists of the upper unitriangular matrices whose non-zero off-diagonal entries can occur only on the r outer superdiagonals nearest the top-right corner. Thus each step adds the next outer superdiagonal, and

$$|Z_r(P)| = p^{1+2+\cdots+r} = p^{r(r+1)/2}$$

for $1 \leq r \leq n-1$. After $n-1$ steps, we reach $Z_{n-1}(P) = P$. □

Exercise 8.4. Let U be the subgroup of $\mathrm{GL}(n,p)$ consisting of the upper unitriangular matrices, and let Q be the subgroup of U consisting of matrices whose (i,j)-entry is zero whenever $1 < i < j < n$. Determine $Z(Q)$, and show that $Q/Z(Q)$ is abelian.

Solution. For $i < j$, write

$$X_{ij} = \{I + aE_{ij} : a \in \mathbb{F}_p\}.$$

The subgroup Q consists of the upper unitriangular matrices whose possible non-zero off-diagonal entries lie either in the first row or in the last column. Equivalently,

$$Q = \langle X_{1j}, X_{in} : 2 \leq j \leq n,\ 1 \leq i < n \rangle.$$

The only possible non-trivial commutators among these root subgroups occur when a first-row root meets the corresponding last-column root:

$$[X_{1j}, X_{jn}] \leq X_{1n} \qquad (2 \leq j \leq n-1).$$

All other commutators between the displayed root subgroups are trivial. In particular, $X_{1n} \leq Z(Q)$.
We now show that there are no other central root directions. Let $g \in Q$. If the X_{1j}-coordinate of g is non-zero for some $2 \leq j \leq n-1$, then commuting g with an element of X_{jn} gives a non-trivial element of X_{1n}, so g is not central. Similarly, if the X_{jn}-coordinate of g is non-zero for some $2 \leq j \leq n-1$, then commuting g with an element of X_{1j} gives a non-trivial element of X_{1n}. Thus a central element of Q can have only its X_{1n}-coordinate non-zero, and hence

$$Z(Q) = X_{1n}.$$

Finally, since every commutator in Q lies in $X_{1n} = Z(Q)$, all commutators become trivial in the quotient $Q/Z(Q)$. Therefore $Q/Z(Q)$ is abelian. □

Exercise 8.5. Show that subgroups and quotient groups of finite nilpotent groups are nilpotent, and that direct products of finite nilpotent groups are nilpotent.

Solution. Let G be a finite nilpotent group. By Theorem 8.7, we may write

$$G = P_1 \times \cdots \times P_r,$$

where P_i is the Sylow p_i-subgroup of G.
First let $H \leq G$. We claim that

$$H = (H \cap P_1) \times \cdots \times (H \cap P_r).$$

Indeed, if $h \in H$, write $h = x_1 \cdots x_r$ with $x_i \in P_i$. Since the factors have pairwise relatively prime orders, each x_i is a power of h, and hence $x_i \in H$. Thus $x_i \in H \cap P_i$, proving the claim. Each $H \cap P_i$ is a normal subgroup of H, because $P_i \trianglelefteq G$. These intersections are precisely the Sylow subgroups of H which are non-trivial. Hence every Sylow subgroup of H is normal in H, so H is nilpotent by Theorem 8.7.
Now let $N \trianglelefteq G$. For each i, the subgroup

$$P_iN/N$$

is a normal p_i-subgroup of G/N, and

$$G/N = (P_1N/N)\cdots(P_rN/N).$$

These factors have pairwise relatively prime orders and commute with one another, since the P_i commute in G. Therefore the Sylow subgroups of G/N are normal, and G/N is nilpotent.
Finally, let $G_1, \ldots, G_m$ be finite nilpotent groups. If $S_{i,p}$ denotes the Sylow p-subgroup of G_i, with $S_{i,p} = 1$ when $p \nmid |G_i|$, then

$$S_{1,p} \times \cdots \times S_{m,p}$$

is the Sylow p-subgroup of $G_1 \times \cdots \times G_m$. Each $S_{i,p}$ is normal in G_i, so this product is normal in $G_1 \times \cdots \times G_m$. Thus every Sylow subgroup of the direct product is normal, and the direct product is nilpotent. □

Exercise 8.6. Let U be the subgroup of $\mathrm{GL}(3, p)$ consisting of the upper unitriangular matrices. Show that if p is an odd prime, then U is a non-abelian group of order p^3 having no elements of order p^2. If $p = 2$, with which group of order 8 is U isomorphic?

Solution. Write an element of U as

$$u(a,b,c) = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad a,b,c \in \mathbb{F}_p.$$

There are p choices for each of a, b, c, so $|U| = p^3$. A direct multiplication gives

$$u(a,b,c)u(a',b',c') = u(a+a', b+b', c+c'+ab').$$

Thus U is not abelian, since

$$u(1,0,0)u(0,1,0) = u(1,1,1), \qquad u(0,1,0)u(1,0,0) = u(1,1,0).$$

Now let

$$N = aE_{12} + bE_{23} + cE_{13}.$$

Then $u(a,b,c) = I + N$, while $N^2 = abE_{13}$ and $N^3 = 0$. Hence, for $m \geq 1$,

$$u(a,b,c)^m = (I+N)^m = I + mN + \binom{m}{2}N^2 = u\left(ma, mb, mc + \binom{m}{2}ab\right).$$

If p is odd, then in $\mathbb{F}_p$ we have $pa = pb = pc = 0$ and

$$\binom{p}{2} = \frac{p(p-1)}{2} \equiv 0 \pmod{p}.$$

Therefore $u(a,b,c)^p = I$ for every element of U. Thus every element has order dividing p, so no element can have order p^2.

It remains to consider $p = 2$. Put

$$r = u(1,1,0), \qquad s = u(1,0,0).$$

Then $r^2 = u(0,0,1)$, so r has order 4, while s has order 2. Also

$$srs^{-1} = srs = u(1,1,1) = r^{-1}.$$

Since $s \notin \langle r \rangle$, the elements r and s generate all of U. Thus U has the usual presentation of the dihedral group of order 8, and so

$$U \cong D_8.$$

□

Further Exercises

Exercise 8.7. Show that a finite group G has a largest nilpotent normal subgroup, in the sense that it contains all nilpotent normal subgroups of G. (This subgroup is called the *Fitting subgroup* of G.)

Solution. We first prove that the product of two nilpotent normal subgroups of G is again nilpotent. Let $A, B \trianglelefteq G$ be nilpotent. By Theorem 8.7, write

$$A = \prod_p A_p, \qquad B = \prod_p B_p,$$

where A_p and B_p are the Sylow p-subgroups of A and B, with the trivial subgroup inserted when p does not divide the relevant order. Since A and B are nilpotent, these Sylow subgroups are unique in A and B, respectively. Hence each A_p is characteristic in A and each B_p is characteristic in B. Since $A, B \trianglelefteq G$, it follows that $A_p, B_p \trianglelefteq G$.

If $p \neq q$, then

$$[A_p, B_q] \leq A_p \cap B_q = 1,$$

because the commutator subgroup lies in both normal subgroups A_p and B_q, while $A_p \cap B_q$ is both a p-group and a q-group. Thus different-prime factors commute. For each prime p, the product

$$A_pB_p$$

is a normal p-subgroup of AB. Moreover,

$$AB = \prod_p A_pB_p,$$

and the factors for distinct primes commute and intersect trivially. Therefore the Sylow subgroups of AB are precisely the subgroups A_pB_p, and they are normal in AB. By Theorem 8.7, AB is nilpotent.

Now let $\mathcal{N}$ be the set of nilpotent normal subgroups of G, and define

$$F = \prod_{N \in \mathcal{N}} N.$$

Since G is finite, it has only finitely many subgroups, so this is a finite product. Repeatedly applying the previous paragraph shows that F is nilpotent. Also, a product of normal subgroups is normal, so $F \trianglelefteq G$. Finally, by construction, every nilpotent normal subgroup N of G is one of the factors in this product, and hence $N \leq F$. Thus F is the largest nilpotent normal subgroup of G. □

The intersection of all maximal subgroups of a finite group G is called the *Frattini subgroup* of G and is denoted by $\Phi(G)$.

Exercise 8.8. Show that $\Phi(G)$ is a nilpotent normal subgroup of G.

Solution. If $G = 1$, then the result is clear, so assume $G \neq 1$. First we show that $\Phi(G) \trianglelefteq G$. Every automorphism of G sends maximal subgroups to maximal subgroups. Indeed, if $\alpha \in \mathrm{Aut}(G)$ and M is maximal in G, then any subgroup strictly between $\alpha(M)$ and G would pull back under α^{-1} to a subgroup strictly between M and G. Thus automorphisms permute the maximal subgroups of G, and so preserve their intersection. Hence $\Phi(G)$ is characteristic in G, and therefore $\Phi(G) \trianglelefteq G$.

Now let P be a Sylow p-subgroup of $\Phi(G)$. Since $\Phi(G) \trianglelefteq G$, the Frattini Argument gives

$$G = N_G(P)\Phi(G).$$

If $N_G(P) < G$, then, since G is finite, there is a maximal subgroup M of G with

$$N_G(P) \leq M < G.$$

But $\Phi(G)$ is contained in every maximal subgroup of G, so $\Phi(G) \leq M$. Hence

$$G = N_G(P)\Phi(G) \leq M,$$

a contradiction. Therefore $N_G(P) = G$, so $P \trianglelefteq G$, and in particular $P \trianglelefteq \Phi(G)$.

Thus every Sylow subgroup of $\Phi(G)$ is normal in $\Phi(G)$. By Theorem 8.7, $\Phi(G)$ is nilpotent. □

Exercise 8.9. Show that $g \in \Phi(G)$ iff whenever $G = \langle S \rangle$ and $g \in S$, then $G = \langle S - \{g\} \rangle$.

Solution. First suppose that $g \in \Phi(G)$. Let S be a subset of G such that $G = \langle S \rangle$ and $g \in S$. Put

$$H = \langle S - \{g\} \rangle.$$

We claim that $H = G$. If not, then H is contained in some maximal subgroup M of G. Since $g \in \Phi(G)$, and $\Phi(G)$ is contained in every maximal subgroup of G, we have $g \in M$. Also $S - \{g\} \subseteq H \leq M$. Hence every element of S lies in M, so

$$G = \langle S \rangle \leq M < G,$$

a contradiction. Therefore $G = \langle S - \{g\} \rangle$.

Conversely, suppose that $g \notin \Phi(G)$. Since $\Phi(G)$ is the intersection of all maximal subgroups of G, there is a maximal subgroup M of G such that $g \notin M$. Let

$$S = M \cup \{g\}.$$

Then $\langle S \rangle = \langle M, g \rangle$. This subgroup contains M properly, because $g \notin M$, and so maximality of M forces

$$\langle S \rangle = G.$$

However,

$$\langle S - \{g\} \rangle = \langle M \rangle = M \neq G.$$

Thus g does not have the stated removable-generator property. This proves the equivalence. □

Exercise 8.10. Show that if P is a finite p-group, then $P/\Phi(P)$ is an elementary abelian p-group.

Solution. If $P = 1$, then the result is clear. Assume therefore that $P \neq 1$. Let M be a maximal subgroup of P. By Proposition 8.3 and Corollary 8.4, we have $M \trianglelefteq P$ and $|P : M| = p$. Hence P/M is cyclic of order p.

Now let $x, y \in P$. Since P/M has order p, the coset xM has order dividing p, so

$$x^p \in M.$$

Also P/M is abelian, so

$$[x, y] \in M.$$

These conclusions hold for every maximal subgroup M of P. Therefore

$$x^p \in \Phi(P) \qquad \text{and} \qquad [x, y] \in \Phi(P)$$

for all $x, y \in P$.

It follows that every element of $P/\Phi(P)$ has order dividing p, since

$$(x\Phi(P))^p = x^p\Phi(P) = \Phi(P),$$

and all commutators are trivial in $P/\Phi(P)$, since

$$[x\Phi(P), y\Phi(P)] = [x, y]\Phi(P) = \Phi(P).$$

Thus $P/\Phi(P)$ is an abelian p-group of exponent p. By Theorem 8.6, it is a direct product of cyclic p-groups; since every element has order dividing p, each non-trivial cyclic factor has order p. Therefore $P/\Phi(P)$ is elementary abelian. □

9 The Schur-Zassenhaus Theorem

Setting and terminology

- This section is devoted to another of the basic results of finite group theory, Schur–Zassenhaus Theorem.
- A *complement* to a normal subgroup N of a group G is a subgroup H of G such that $G = N \rtimes H$.
- A subgroup H of a finite group G is called a *Hall subgroup* if $|H|$ and $|G : H|$ are coprime.

Theorem (Schur–Zassenhaus Theorem)**.** *Any normal Hall subgroup of a finite group has a complement.*

Proof. Let N be a normal Hall subgroup of a finite group G, and put $n = |G : N|$. We first observe that it suffices to find a subgroup of G of order n. Indeed, if $K \leq G$ and $|K| = n$, then $|N \cap K|$ divides both $|N|$ and n, so $N \cap K = 1$ by Lagrange's Theorem. Hence Proposition 3.12 gives

$$|NK| = \frac{|N||K|}{|N \cap K|} = |N|n = |G|,$$

and therefore $NK = G$. Thus K is a complement to N.
We argue by induction on $|G|$. If $N = 1$, then G itself is a complement to N, so assume $N \neq 1$. Let P be a Sylow p-subgroup of N for some prime p dividing $|N|$. By Frattini Argument, we have

$$G = N_G(P)N.$$

Applying First Isomorphism Theorem to the map $N_G(P) \to G/N$ given by $x \mapsto xN$, we obtain

$$G/N \cong N_G(P)/(N_G(P) \cap N) = N_G(P)/N_N(P).$$

Thus $|N_G(P) : N_N(P)| = n$. Moreover, $N_N(P) = N_G(P) \cap N$ is normal in $N_G(P)$, and $|N_N(P)|$ divides $|N|$. Hence $N_N(P)$ is a normal Hall subgroup of $N_G(P)$. If $N_G(P) < G$, then the induction hypothesis gives a subgroup of $N_G(P)$ of order n, and hence a subgroup of G of order n. We may therefore assume that $N_G(P) = G$, or equivalently that $P \trianglelefteq G$.
Suppose first that $P < N$. Then N/P is a normal Hall subgroup of G/P, and

$$|G/P : N/P| = |G : N| = n.$$

Since $|G/P| < |G|$, the induction hypothesis gives a subgroup of G/P of order n. By the Correspondence Theorem, this subgroup has the form L/P for some subgroup L of G containing P, and $|L/P| = n$. Hence $|L| = n|P|$. Now $|L \cap N|$ divides both $|L| = n|P|$ and $|N|$, and n is coprime to $|N|$, so $|L \cap N|$ divides $|P|$. Since $P \leq L \cap N$, it follows that $L \cap N = P$. In particular $L < G$, because $P < N$. Inside L, the subgroup P is a normal Hall subgroup with $|L : P| = n$, so the induction hypothesis gives a subgroup of L, and hence of G, of order n. We may therefore assume that $N = P$, so N is a p-group.
Suppose next that N is non-abelian. Put $Z = Z(N)$. By Theorem 8.1, we have $Z \neq 1$, and since N is non-abelian, we also have $Z < N$. The center $Z(N)$ is characteristic in N, and $N \trianglelefteq G$, so $Z \trianglelefteq G$. Thus N/Z is a normal Hall subgroup of G/Z, and

$$|G/Z : N/Z| = |G : N| = n.$$

Since $|G/Z| < |G|$, the induction hypothesis gives a subgroup L/Z of G/Z of order n. Then $|L| = n|Z|$, and the same divisibility argument as above gives $L \cap N = Z$. Hence $L < G$, because $Z < N$. Inside L, the subgroup Z is a normal Hall subgroup with $|L : Z| = n$, so induction gives a subgroup of L, and hence of G, of order n. We may therefore assume that the normal Hall subgroup is abelian; write it as A.
It remains to prove the theorem when A is an abelian normal Hall subgroup of G. We use additive notation inside A, while retaining multiplicative notation in G, and put $H = G/A$. If $h \in H$ and $x \in A$, define

$${}^h x = txt^{-1},$$

where t is any element of the coset h. This is well-defined: if u is another element of h, then $u = ta$ for some $a \in A$, and since A is abelian,

$$uxu^{-1} = taxa^{-1}t^{-1} = txt^{-1}.$$

Conjugation by t is an automorphism of A, and the definition is compatible with multiplication in H, so H acts on A by automorphisms. Choose, for each $h \in H$, an element $t_h \in h$. Since $t_{h_1}t_{h_2}$ and $t_{h_1h_2}$ lie in the same coset of A, there is a unique element $f(h_1, h_2) \in A$ such that

$$t_{h_1}t_{h_2} = f(h_1, h_2)t_{h_1h_2}.$$

Associativity in G gives $t_{h_1}(t_{h_2}t_{h_3}) = (t_{h_1}t_{h_2})t_{h_3}$, and substituting the definition of f gives

$${}^{h_1}f(h_2, h_3) + f(h_1, h_2h_3) = f(h_1, h_2) + f(h_1h_2, h_3)$$

for all $h_1, h_2, h_3 \in H$.
We now seek to modify the representatives t_h by elements of A. Suppose there is a function $c\colon H \to A$ satisfying

$$c(h_1h_2) = c(h_1) + {}^{h_1}c(h_2) + f(h_1, h_2)$$

for all $h_1, h_2 \in H$. Then the map $\varphi\colon H \to G$ defined by

$$\varphi(h) = c(h)t_h$$

is a homomorphism, since

$$\varphi(h_1)\varphi(h_2) = c(h_1)t_{h_1}c(h_2)t_{h_2} = \big(c(h_1) + {}^{h_1}c(h_2) + f(h_1, h_2)\big)t_{h_1h_2} = \varphi(h_1h_2).$$

Moreover, if $h \neq 1$, then $\varphi(h)$ lies in the non-identity coset h, and so $\varphi(h) \neq 1$. Hence φ is injective, and its image is a subgroup of G of order $|H| = n$. It remains only to construct such a function c.
For $h \in H$, define

$$e(h) = \sum_{k \in H} f(h, k).$$

Using the cocycle identity and summing over $k \in H$, we obtain

$${}^{h_1}e(h_2) + e(h_1) = nf(h_1, h_2) + e(h_1h_2),$$

and hence

$$nf(h_1, h_2) = -e(h_1h_2) + e(h_1) + {}^{h_1}e(h_2).$$

Since A is finite abelian and $\gcd(n, |A|) = 1$, multiplication by n is an automorphism of A. Let $\eta\colon A \to A$ denote its inverse, so that $n\eta(x) = x$ for every $x \in A$. The action of H on A commutes with multiplication by n, and hence also with η. Define

$$c(h) = -\eta(e(h)).$$

Applying η to the preceding identity gives

$$f(h_1, h_2) = -\eta(e(h_1h_2)) + \eta(e(h_1)) + {}^{h_1}\eta(e(h_2)),$$

and therefore

$$c(h_1h_2) = c(h_1) + {}^{h_1}c(h_2) + f(h_1, h_2).$$

As shown above, this gives a subgroup of G of order n, and hence a complement to A. This completes the induction and the proof. □

Cohomological framework

- We now wish to fit some of the ideas developed in the proof of Schur–Zassenhaus Theorem into a more general framework.
- Let H be an arbitrary group, and let A be an abelian group written additively. Suppose that we have an action of H as automorphisms of A, or equivalently a homomorphism from H to $\mathrm{Aut}(A)$.
- A function $f : H \times H \to A$ satisfying the cocycle identity given in the above proof is called a 2-*cocycle* of the pair (H, A).
- The set of all 2-cocycles of (H, A) is denoted by $Z^2(H, A)$, and if we define $(f + g)(h_1, h_2) = f(h_1, h_2) + g(h_1, h_2)$ for $f, g \in Z^2(H, A)$ and $h_1, h_2 \in H$, then $Z^2(H, A)$ becomes an abelian group.

- A 2-cocycle is called a 2-*coboundary* if there is a function $c : H \to A$ such that $f(h_1, h_2) = c(h_1) + {}^{h_1}c(h_2) - c(h_1h_2)$ for all $h_1, h_2 \in H$.
- The set of 2-coboundaries of (H, A) is denoted by $B^2(H, A)$ and is a subgroup of $Z^2(H, A)$.
- The quotient group $Z^2(H, A)/B^2(H, A)$ is called the *second cohomology group* of (H, A) and is denoted by $H^2(H, A)$. (This terminology comes from algebraic topology. We shall define the nth cohomology group of (H, A) for any $n \in \mathbb{N}$ in the further exercises to Section 12.)
- Now suppose that A is an abelian normal Hall subgroup of a finite group G. We showed during the proof of Schur–Zassenhaus Theorem that any 2-cocycle f of $(G/A, A)$ was a 2-coboundary, or equivalently that $H^2(G/A, A) = 0$. (The difference of sign between the definition of 2-coboundary given above and what was established in the proof is irrelevant.) Furthermore, we showed that this fact implied the existence of a complement to A in G. We attempt to make this connection between the second cohomology group of $(G/A, A)$ and the existence of complements to A in G more transparent in the further exercises below.
- A stronger version of Schur–Zassenhaus Theorem asserts that if N is a normal Hall subgroup of a finite group G, then not only does N have a complement in G, but all such complements are conjugate in G. To prove this, one needs the concept of solvable groups, which we shall introduce in Section 11; see [22, pp. 246–8] for further details.

Exercises

Exercise 9.1. Suppose that G is a finite group such that $G = N \rtimes H$, where H is abelian. Show that if $|N|$ and $|H|$ are relatively coprime, then all complements to N in G are conjugate.

Solution. Let K be another complement to N in G. We show that K is conjugate to H. Let $\pi\colon G \to G/N$ be the natural map, and identify G/N with H. Since $K \cap N = 1$ and $|K| = |G : N| = |H|$, the restriction $\pi|_K\colon K \to G/N$ is an isomorphism. Hence $K \cong H$, and in particular K is abelian.
We argue by induction on $|G|$. If $|H| = 1$, there is nothing to prove. If $|H| = p$ is prime, then H and K are Sylow p-subgroups of G, because $p \nmid |N|$. Hence they are conjugate by part (ii) of Sylow's Theorem.
Assume now that $|H|$ is composite. Choose a subgroup $L \leq H$ of prime order. Put $K_L = K \cap NL$. By Second Isomorphism Theorem, we have $NL/N \cong L/(L \cap N) \cong L$. Under the isomorphism $\pi|_K$, the subgroup K_L is the inverse image of NL/N, so $|K_L| = |L|$. Also $K_L \cap N = 1$ and $NK_L = NL$, so K_L is a complement to N in $NL = N \rtimes L$. Since $L < H$, we have $|NL| < |G|$. By induction inside NL, the complements K_L and L to N are conjugate in NL. Replacing K by a conjugate in G, we may therefore assume that $L \leq K$.
Since both H and K are abelian and contain L, we have $H, K \leq C_G(L)$. We claim that

$$C_G(L) = C_N(L) \rtimes H.$$

Indeed, let $g \in C_G(L)$ and write $g = nh$ with $n \in N$ and $h \in H$. Since H is abelian, h centralizes L. Hence $n = gh^{-1}$ also centralizes L, and so $n \in C_N(L)$. This gives $C_G(L) \leq C_N(L)H$, and the reverse containment is clear. Moreover, $C_N(L) \trianglelefteq C_G(L)$ and $C_N(L) \cap H = 1$, so the displayed product is a semidirect product. Since $K \leq C_G(L)$, $K \cap C_N(L) = 1$, and $|K| = |H| = |C_G(L) : C_N(L)|$, the subgroup K is also a complement to $C_N(L)$ in $C_G(L)$.
If $C_N(L) < N$, then $|C_G(L)| < |G|$. Applying the induction hypothesis inside $C_G(L)$ to the normal Hall subgroup $C_N(L)$ and its abelian complement H, we conclude that the two complements K and H are conjugate in $C_G(L)$, and hence in G.
It remains to consider the case $C_N(L) = N$. Then L centralizes N, and since H is abelian, L also centralizes H. Thus $L \leq Z(G)$. In the quotient G/L, the subgroup NL/L is a normal Hall subgroup and

$$G/L = (NL/L) \rtimes (H/L),$$

where H/L is abelian. Also K/L is a complement to NL/L, because $K \cap NL = L$ and $NK = G$. Since $|G/L| < |G|$, induction gives some $g \in G$ such that

$$gKg^{-1}L/L = H/L.$$

Both gKg^{-1} and H contain L, so this equality in the quotient implies $gKg^{-1} = H$. Therefore every complement to N in G is conjugate to H. □

Exercise 9.2. Repeat Exercise 9.1 under the assumption that H is nilpotent rather than abelian.

Solution. Let K be another complement to N in G. As in Exercise 9.1, the quotient map restricts to an isomorphism $K \cong G/N \cong H$, so K is nilpotent. We argue by induction on $|G|$. The cases $|H| = 1$ and $|H|$ prime are handled exactly as in Exercise 9.1.
Assume that $|H|$ is composite. Since H is finite nilpotent, $Z(H) \neq 1$, so choose a subgroup $L \leq Z(H)$ of prime order. Put $K_L = K \cap NL$. By the same quotient-map argument as in Exercise 9.1, using Second Isomorphism Theorem to identify NL/N with L, the subgroup K_L is the subgroup of K corresponding to L. Since $L \leq Z(H)$ and $K \cong H$, this gives $K_L \leq Z(K)$. As before, K_L is a complement to N in $NL = N \rtimes L$. Since $|NL| < |G|$, induction inside NL conjugates K_L to L. Replacing K by this conjugate, we may assume that $L \leq Z(K)$.
Now H and K both centralize L, so $H, K \leq C_G(L)$. Since $L \leq Z(H)$, the same calculation as in Exercise 9.1 gives

$$C_G(L) = C_N(L) \rtimes H,$$

and K is also a complement to $C_N(L)$ in $C_G(L)$. If $C_N(L) < N$, induction inside $C_G(L)$ gives that K and H are conjugate. If $C_N(L) = N$, then L centralizes both N and H, hence $L \leq Z(G)$. Passing to G/L, induction gives that K/L and H/L are conjugate complements to NL/L. Lifting this conjugacy back to G gives that K and H are conjugate. □

Further Exercises

These exercises are a continuation of the further exercises to Section 2. Let N and H be groups. A *factor pair* of N by H is a pair (f, φ) of set maps $f : H \times H \to N$ and $\varphi : H \to \mathrm{Aut}(N)$ satisfying properties **1**, **2**, and **3** listed on page 27. Let $\mathcal{E}$ be the set of extensions of N by H, and let $\mathcal{F}$ be the set of factor pairs of N by H. In what follows, we will always use "extension" to mean an element of $\mathcal{E}$, and "factor pair" to mean an element of $\mathcal{F}$.

Exercise 9.3. Let (f, φ) be a factor pair, and define

$$(x, \alpha) \cdot (y, \beta) = (x\varphi(\alpha)(y)f(\alpha, \beta), \alpha\beta)$$

for $(x, \alpha), (y, \beta) \in N \times H$. Show that this gives a group structure on $N \times H$; call this group $E_{f,\varphi}$. Show further that $(E_{f,\varphi}, i, \pi)$ is an extension, where $i(x) = (x, 1)$ and $\pi(x, \alpha) = \alpha$, and that (f, φ) is the factor pair arising from some normalized section of $E_{f,\varphi}$. (Observe that this construction generalizes the notion of external semidirect product.)

Solution. We first prove associativity. Let $(x, \alpha), (y, \beta), (z, \gamma) \in N \times H$. Then

$$((x, \alpha)(y, \beta))(z, \gamma) = (x\varphi(\alpha)(y)f(\alpha, \beta)\varphi(\alpha\beta)(z)f(\alpha\beta, \gamma), \alpha\beta\gamma).$$

On the other hand,

$$(x, \alpha)((y, \beta)(z, \gamma)) = (x\varphi(\alpha)(y)\varphi(\alpha)\varphi(\beta)(z)\varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma), \alpha\beta\gamma).$$

Using property **2** of factor pairs, we have

$$\varphi(\alpha)\varphi(\beta)(z) = f(\alpha, \beta)\varphi(\alpha\beta)(z)f(\alpha, \beta)^{-1}.$$

Using property **3**, we have

$$f(\alpha, \beta)^{-1}\varphi(\alpha)(f(\beta, \gamma))f(\alpha, \beta\gamma) = f(\alpha\beta, \gamma).$$

Substituting these two identities into the second product gives the first product. Hence the operation is associative.
By property **1**, the element $(1, 1)$ is an identity:

$$(1, 1)(x, \alpha) = (x, \alpha) = (x, \alpha)(1, 1).$$

For $(x, \alpha) \in N \times H$, put

$$u = \varphi(\alpha)^{-1}\big(x^{-1}f(\alpha, \alpha^{-1})^{-1}\big), \qquad v = f(\alpha^{-1}, \alpha)^{-1}\varphi(\alpha^{-1})(x^{-1}).$$

Then $(x, \alpha)(u, \alpha^{-1}) = (1, 1)$ and $(v, \alpha^{-1})(x, \alpha) = (1, 1)$. Since the operation is associative, the left and right inverses agree, so every element has an inverse. Thus $E_{f,\varphi}$ is a group.
Define $i\colon N \to E_{f,\varphi}$ by $i(x) = (x, 1)$ and $\pi\colon E_{f,\varphi} \to H$ by $\pi(x, \alpha) = \alpha$. Property **1** gives

$$i(x)i(y) = (xy, 1) = i(xy),$$

and i is visibly injective, so i is a monomorphism. Also

$$\pi((x, \alpha)(y, \beta)) = \alpha\beta = \pi(x, \alpha)\pi(y, \beta),$$

and π is surjective since $\pi(1, \alpha) = \alpha$, so π is an epimorphism. Its kernel is

$$\ker \pi = \{(x, 1) \mid x \in N\} = i(N).$$

Therefore $(E_{f,\varphi}, i, \pi)$ is an extension of N by H.
Finally define a normalized section $t\colon H \to E_{f,\varphi}$ by $t(\alpha) = (1, \alpha)$. Then $t(1) = (1, 1)$ and $\pi(t(\alpha)) = \alpha$. Moreover,

$$t(\alpha)t(\beta) = (f(\alpha, \beta), \alpha\beta) = i(f(\alpha, \beta))t(\alpha\beta),$$

so the factor-set component arising from t is f. Also

$$t(\alpha)i(x) = (\varphi(\alpha)(x), \alpha) = i(\varphi(\alpha)(x))t(\alpha),$$

so conjugation by $t(\alpha)$ restricts to $\varphi(\alpha)$ on $i(N)$. Hence the factor pair arising from the normalized section t is precisely (f, φ). □

We have seen in Exercise 2.13 that an extension gives rise to a factor pair via a choice of normalized section, and we have just given an explicit construction of an extension from a given factor pair. We view these processes as giving maps between $\mathcal{E}$ and $\mathcal{F}$, and we now investigate the relationship between these maps. We must first consider the relation between factor pairs arising from different normalized sections of the same extension.

Exercise 9.4. (cont.) Suppose that t and u are normalized sections of an extension E, and let (f, φ) and (g, ρ) be the factor pairs arising from t and u, respectively. Let $c : H \to N$ be the set map such that $u(\alpha) = c(\alpha)t(\alpha)$ for every $\alpha \in H$. Show that the following properties hold:

4 $\rho(\alpha) = \Psi(c(\alpha))\varphi(\alpha)$ for $\alpha \in H$, where $\Psi(c(\alpha))$ is the inner automorphism of N corresponding to $c(\alpha)$.

5 $g(\alpha, \beta) = c(\alpha)\varphi(\alpha)(c(\beta))f(\alpha, \beta)c(\alpha\beta)^{-1}$ for $\alpha, \beta \in H$.

Solution. By definition of the factor pairs arising from t and u, we have

$$\varphi(\alpha) = \Psi(t(\alpha)), \qquad \rho(\alpha) = \Psi(u(\alpha)).$$

Since $u(\alpha) = c(\alpha)t(\alpha)$ and Ψ is a homomorphism into the automorphism group, it follows that

$$\rho(\alpha) = \Psi(c(\alpha)t(\alpha)) = \Psi(c(\alpha))\Psi(t(\alpha)) = \Psi(c(\alpha))\varphi(\alpha).$$

This proves property **4**.
For property **5**, use the definition of the factor-set component:

$$g(\alpha, \beta) = u(\alpha)u(\beta)u(\alpha\beta)^{-1}.$$

Substituting $u(\delta) = c(\delta)t(\delta)$ gives

$$g(\alpha, \beta) = c(\alpha)t(\alpha)c(\beta)t(\beta)t(\alpha\beta)^{-1}c(\alpha\beta)^{-1}.$$

Now $t(\alpha)c(\beta)t(\alpha)^{-1} = \varphi(\alpha)(c(\beta))$, and

$$t(\alpha)t(\beta)t(\alpha\beta)^{-1} = f(\alpha, \beta).$$

Therefore

$$g(\alpha, \beta) = c(\alpha)\varphi(\alpha)(c(\beta))f(\alpha, \beta)c(\alpha\beta)^{-1},$$

as required. □

The above exercise motivates the following definition: We say that two factor pairs (f, φ) and (g, ρ) are *equivalent* if there is a map $c : H \to N$ such that properties **4** and **5** hold. (Verify that this is an equivalence relation on $\mathcal{F}$.) Let $\overline{\mathcal{F}}$ denote the set of equivalence classes of factor pairs; we will use $[f, \varphi]$ to denote the class of the factor pair (f, φ). We have a well-defined map from $\mathcal{E}$ to $\overline{\mathcal{F}}$ which sends an extension to the class of a factor pair arising from any normalized section. We must now consider what happens when we pass from $\overline{\mathcal{F}}$ back to $\mathcal{E}$ via the construction in Exercise 9.3. Here we will need to recall the exact definition of an extension.

Exercise 9.5. (cont.) Let (f, φ) and (g, ρ) be factor pairs, with $[f, \varphi] = [g, \rho]$. Let $i : N \to E_{f,\varphi}$ and $j : N \to E_{g,\rho}$ be the natural inclusions (of N into the underlying set $N \times H$), and let $\pi : E_{f,\varphi} \to H$ and $\tau : E_{g,\rho} \to H$ be the natural projections (of the underlying set $N \times H$ onto H). Show that there is an isomorphism $\xi : E_{f,\varphi} \to E_{g,\rho}$ such that $\xi \circ i = j$ and $\tau \circ \xi = \pi$.

Solution. Let $c\colon H \to N$ be a map witnessing the equivalence of (f, φ) and (g, ρ), so properties **4** and **5** hold. Taking $\alpha = \beta = 1$ in property **5**, and using property **1** for factor pairs, gives $c(1) = 1$. Define

$$\xi\colon E_{f,\varphi} \to E_{g,\rho}, \qquad \xi(x, \alpha) = (xc(\alpha)^{-1}, \alpha).$$

This map is bijective, with inverse $(x, \alpha) \mapsto (xc(\alpha), \alpha)$. We check that it is a homomorphism. In $E_{g,\rho}$, we have

$$\xi(x, \alpha)\xi(y, \beta) = (xc(\alpha)^{-1}\rho(\alpha)(yc(\beta)^{-1})g(\alpha, \beta), \alpha\beta).$$

By property **4**, $\rho(\alpha) = \Psi(c(\alpha))\varphi(\alpha)$, and hence

$$\rho(\alpha)(yc(\beta)^{-1}) = c(\alpha)\varphi(\alpha)(y)\varphi(\alpha)(c(\beta)^{-1})c(\alpha)^{-1}.$$

Substituting this and then using property **5**, the first coordinate becomes

$$\begin{aligned} &xc(\alpha)^{-1}c(\alpha)\varphi(\alpha)(y)\varphi(\alpha)(c(\beta)^{-1})c(\alpha)^{-1}c(\alpha)\varphi(\alpha)(c(\beta))f(\alpha, \beta)c(\alpha\beta)^{-1} \\ &\quad = x\varphi(\alpha)(y)f(\alpha, \beta)c(\alpha\beta)^{-1}. \end{aligned}$$

Therefore

$$\xi(x, \alpha)\xi(y, \beta) = (x\varphi(\alpha)(y)f(\alpha, \beta)c(\alpha\beta)^{-1}, \alpha\beta) = \xi\big((x, \alpha)(y, \beta)\big),$$

so ξ is a group isomorphism. Finally, for $x \in N$ we have

$$\xi(i(x)) = \xi(x, 1) = (xc(1)^{-1}, 1) = (x, 1) = j(x),$$

so $\xi \circ i = j$. Also

$$(\tau \circ \xi)(x, \alpha) = \tau(xc(\alpha)^{-1}, \alpha) = \alpha = \pi(x, \alpha),$$

so $\tau \circ \xi = \pi$. □

Motivated by the above exercise, we say that two extensions (E, i, π) and (F, j, τ) are *equivalent* if there is an isomorphism $\xi : E \to F$ such that $\xi \circ i = j$ and $\tau \circ \xi = \pi$. (Verify that this gives an equivalence relation on $\mathcal{E}$.) We let $\overline{\mathcal{E}}$ denote the set of equivalence classes of extensions, and we let $[E]$ denote the class of an extension E. In this context, Exercise 9.5 asserts that there is a well-defined map from $\overline{\mathcal{F}}$ to $\overline{\mathcal{E}}$, sending $[f, \varphi]$ to $[E_{f,\varphi}]$.

Exercise 9.6. (cont.) If p is an odd prime, show that $\mathbb{Z}_{p^2}$ can be realized in $p - 1$ nonequivalent ways as an extension of $\mathbb{Z}_p$ by $\mathbb{Z}_p$.

Solution. Write all cyclic groups additively, and put $E = \mathbb{Z}_{p^2}$. Define

$$i\colon \mathbb{Z}_p \to E, \qquad i(\overline{x}) = \overline{px}.$$

For each $a \in \{1, \ldots, p - 1\}$, define

$$\pi_a\colon E \to \mathbb{Z}_p, \qquad \pi_a(\overline{x}) = \overline{ax}.$$

Then π_a is an epimorphism, and

$$\ker \pi_a = \{\overline{x} \in \mathbb{Z}_{p^2} \mid p \mid x\} = i(\mathbb{Z}_p).$$

Thus (E, i, π_a) is an extension of $\mathbb{Z}_p$ by $\mathbb{Z}_p$ for each $a = 1, \ldots, p - 1$.
We show that these $p - 1$ extensions are pairwise nonequivalent. Suppose that (E, i, π_a) and (E, i, π_b)

are equivalent. Then there is an isomorphism $\xi\colon E \to E$ such that $\xi \circ i = i$ and $\pi_b \circ \xi = \pi_a$. Since E is cyclic, ξ is multiplication by some unit u modulo p^2. The condition $\xi \circ i = i$ gives

$$\overline{up} = \overline{p} \quad \text{in } \mathbb{Z}_{p^2},$$

so $u \equiv 1 \pmod p$. Applying $\pi_b \circ \xi = \pi_a$ to $\overline{1} \in E$, we get

$$\overline{bu} = \overline{a} \quad \text{in } \mathbb{Z}_p.$$

Since $u \equiv 1 \pmod p$, this gives $b = a$. Hence distinct choices of a give nonequivalent extensions, so $\mathbb{Z}_{p^2}$ can be realized in $p-1$ nonequivalent ways as an extension of $\mathbb{Z}_p$ by $\mathbb{Z}_p$. □

Exercise 9.7. (cont.) We have already obtained a map from $\mathcal{E}$ to $\overline{\mathcal{F}}$, sending an extension E to the class of the factor pair arising from any normalized section of E. Show that this map induces a map from $\overline{\mathcal{E}}$ to $\overline{\mathcal{F}}$.

Solution. Let (E,i,π) and (F,j,τ) be equivalent extensions, and let $\xi\colon E \to F$ be an isomorphism such that $\xi \circ i = j$ and $\tau \circ \xi = \pi$. Choose a normalized section $t\colon H \to E$, and let (f,φ) be the factor pair arising from t. Define

$$u\colon H \to F, \qquad u(\alpha) = \xi(t(\alpha)).$$

Then $u(1) = 1$ and

$$\tau(u(\alpha)) = \tau(\xi(t(\alpha))) = \pi(t(\alpha)) = \alpha,$$

so u is a normalized section of F. Let (g,ρ) be the factor pair arising from u. Since $t(\alpha)t(\beta) = i(f(\alpha,\beta))t(\alpha\beta)$, applying ξ gives

$$u(\alpha)u(\beta) = j(f(\alpha,\beta))u(\alpha\beta).$$

Hence $g(\alpha,\beta) = f(\alpha,\beta)$. Similarly, for $x \in N$ we have

$$u(\alpha)j(x)u(\alpha)^{-1} = \xi\big(t(\alpha)i(x)t(\alpha)^{-1}\big) = j(\varphi(\alpha)(x)),$$

so $\rho(\alpha) = \varphi(\alpha)$. Thus the factor pair obtained from F using the section u is the same as the factor pair obtained from E using t. Therefore equivalent extensions determine the same class in $\overline{\mathcal{F}}$, and the map $\mathcal{E} \to \overline{\mathcal{F}}$ induces a well-defined map $\overline{\mathcal{E}} \to \overline{\mathcal{F}}$. □

Exercise 9.8. (cont.) Show that the map from $\overline{\mathcal{E}}$ to $\overline{\mathcal{F}}$ obtained in Exercise 9.7 is inverse to the map from $\overline{\mathcal{F}}$ to $\overline{\mathcal{E}}$ sending $[f,\varphi]$ to $[E_{f,\varphi}]$. Conclude that there is a bijective correspondence between the set of equivalence classes of extensions and the set of equivalence classes of factor pairs.

Solution. First start with a class $[f,\varphi] \in \overline{\mathcal{F}}$. By Exercise 9.3, the extension $E_{f,\varphi}$ has a normalized section $t(\alpha) = (1,\alpha)$ whose factor pair is precisely (f,φ). Hence the composite

$$\overline{\mathcal{F}} \to \overline{\mathcal{E}} \to \overline{\mathcal{F}}$$

sends $[f,\varphi]$ back to $[f,\varphi]$.
Conversely, start with an extension (E,i,π), choose a normalized section $t\colon H \to E$, and let (f,φ) be the factor pair arising from t. We show that E is equivalent to $E_{f,\varphi}$. Let $i_{f,\varphi}\colon N \to E_{f,\varphi}$ and $\pi_{f,\varphi}\colon E_{f,\varphi} \to H$ denote the natural inclusion and projection. Define

$$\theta\colon E_{f,\varphi} \to E, \qquad \theta(x,\alpha) = i(x)t(\alpha).$$

This map is bijective. Indeed, if $e \in E$ and $\alpha = \pi(e)$, then $et(\alpha)^{-1} \in \ker\pi = i(N)$, so $e = i(x)t(\alpha)$ for some $x \in N$. Also, if $i(x)t(\alpha) = i(y)t(\beta)$, then applying π gives $\alpha = \beta$, and hence $i(x) = i(y)$, so $x = y$. We check that θ is a homomorphism. Using the defining equations for the factor pair arising from t, we have

$$\begin{aligned}
\theta(x,\alpha)\theta(y,\beta) &= i(x)t(\alpha)i(y)t(\beta)\\
&= i(x)i(\varphi(\alpha)(y))t(\alpha)t(\beta)\\
&= i(x\varphi(\alpha)(y)f(\alpha,\beta))t(\alpha\beta)\\
&= \theta\big(x\varphi(\alpha)(y)f(\alpha,\beta),\alpha\beta\big)\\
&= \theta\big((x,\alpha)(y,\beta)\big).
\end{aligned}$$

Thus θ is an isomorphism. Finally,

$$\theta(i_{f,\varphi}(x)) = \theta(x,1) = i(x)t(1) = i(x),$$

so $\theta \circ i_{f,\varphi} = i$, and

$$(\pi \circ \theta)(x,\alpha) = \pi(i(x)t(\alpha)) = \alpha = \pi_{f,\varphi}(x,\alpha).$$

Therefore $E_{f,\varphi}$ is equivalent to E, so the composite

$$\overline{\mathcal{E}} \to \overline{\mathcal{F}} \to \overline{\mathcal{E}}$$

sends $[E]$ back to $[E]$. The two maps are inverse to each other, and hence they give a bijective correspondence between equivalence classes of extensions and equivalence classes of factor pairs. $\square$

Exercise 9.8 implies that in order to study extensions up to equivalence, it suffices to study equivalence classes of factor pairs. The next two exercises give a slight refinement of the correspondence just obtained.

Exercise 9.9. (cont.) Let $\eta : \mathrm{Aut}(N) \to \mathrm{Out}(N)$ be the natural map. Show that if (f,φ) and (g,ρ) are any two factor pairs arising from an extension E, then $\eta \circ \varphi = \eta \circ \rho$, and this map from H to $\mathrm{Out}(N)$ (which we denote by ψ_E) is a homomorphism. Conclude that there is a well-defined map from $\mathcal{E}$ to the set of homomorphisms from H to $\mathrm{Out}(N)$, sending E to ψ_E.

Solution. Let (f,φ) and (g,ρ) be factor pairs arising from two normalized sections of the same extension E. By Exercise 9.4, there is a map $c\colon H \to N$ such that

$$\rho(\alpha) = \Psi(c(\alpha))\varphi(\alpha)$$

for every $\alpha \in H$. Since $\Psi(c(\alpha))$ is an inner automorphism of N, it lies in $\ker \eta$. Hence

$$\eta(\rho(\alpha)) = \eta(\Psi(c(\alpha)))\eta(\varphi(\alpha)) = \eta(\varphi(\alpha)).$$

Thus $\eta \circ \rho = \eta \circ \varphi$, so the map $\eta \circ \varphi$ depends only on the extension E, not on the chosen normalized section. We denote it by ψ_E.
It remains to check that ψ_E is a homomorphism. Let (f,φ) be any factor pair arising from E. Property **2** of factor pairs says that

$$\varphi(\alpha)\varphi(\beta) = \Psi(f(\alpha,\beta))\varphi(\alpha\beta),$$

where $\Psi(f(\alpha,\beta))$ is inner. Applying η gives

$$\psi_E(\alpha)\psi_E(\beta) = \eta(\varphi(\alpha))\eta(\varphi(\beta)) = \eta(\varphi(\alpha\beta)) = \psi_E(\alpha\beta).$$

Therefore $\psi_E\colon H \to \mathrm{Out}(N)$ is a homomorphism. Since the construction is independent of the normalized section, we obtain a well-defined map from $\mathcal{E}$ to the set of homomorphisms $H \to \mathrm{Out}(N)$, sending E to ψ_E. $\square$

Exercise 9.10. (cont.) Let $\psi : H \to \mathrm{Out}(N)$ be a given homomorphism. Show that there is a bijective correspondence between the set of classes $[E]$ of $\overline{\mathcal{E}}$ for which $\psi_E = \psi$ and the set of classes $[f,\varphi]$ of $\overline{\mathcal{F}}$ for which $\eta \circ \varphi = \psi$, where $\eta : \mathrm{Aut}(N) \to \mathrm{Out}(N)$ is the natural map.

Solution. First note that the condition $\eta \circ \varphi = \psi$ is independent of the representative of the factor-pair class. Indeed, if (f,φ) and (g,ρ) are equivalent factor pairs, then property **4** gives

$$\rho(\alpha) = \Psi(c(\alpha))\varphi(\alpha)$$

for some map $c\colon H \to N$, and hence $\eta(\rho(\alpha)) = \eta(\varphi(\alpha))$ for every $\alpha \in H$. Similarly, by Exercise 9.7, equivalent extensions determine the same class of factor pairs, so the condition $\psi_E = \psi$ is well-defined on $\overline{\mathcal{E}}$.
Now let

$$\Phi\colon \overline{\mathcal{E}} \to \overline{\mathcal{F}}$$

be the bijection from Exercise 9.8, sending an extension class to the class of a factor pair arising from any normalized section. If $\Phi([E]) = [f,\varphi]$, then by Exercise 9.9 we have

$$\psi_E = \eta \circ \varphi.$$

Therefore Φ sends the classes $[E]$ satisfying $\psi_E = \psi$ into the classes $[f, \varphi]$ satisfying $\eta \circ \varphi = \psi$.
Conversely, suppose that $[f, \varphi] \in \overline{\mathcal{F}}$ satisfies $\eta \circ \varphi = \psi$. Under the inverse bijection from Exercise 9.8, this class maps to $[E_{f,\varphi}]$. By Exercise 9.3, the natural normalized section $t(\alpha) = (1, \alpha)$ of $E_{f,\varphi}$ gives back the factor pair (f, φ), so

$$\psi_{E_{f,\varphi}} = \eta \circ \varphi = \psi.$$

Thus the inverse bijection sends the classes $[f, \varphi]$ satisfying $\eta \circ \varphi = \psi$ into the classes $[E]$ satisfying $\psi_E = \psi$. Hence the bijection of Exercise 9.8 restricts to the required bijective correspondence. $\square$

We now consider the case where the group N is abelian; we write A instead of N, and we will use additive notation for A. Observe that $\mathrm{Out}(A) = \mathrm{Aut}(A)$. We fix a homomorphism $\varphi : H \to \mathrm{Aut}(A)$, and we write ${}^x a$ in lieu of $\varphi(x)(a)$ for $x \in H$ and $a \in A$. We would like to study those equivalence classes $[E]$ of extensions of N by H for which $\psi_E = \varphi$; we say that such extensions *respect* the action of H on A. By Exercise 9.10, it suffices to study equivalence classes $[f, \varphi]$ of factor pairs of A by H. We suppress φ in our notation, so that we are studying functions $f : H \times H \to A$ such that $f(x, 1) = f(1, x) = 0$ for all $x \in H$ and which in addition satisfy

$$f(x, y) + f(xy, z) = {}^x f(y, z) + f(x, yz)$$

for all $x, y, z \in H$, with two such functions f and g being equivalent if there is a map $c : H \to A$ such that $c(1) = 0$ and

$$g(x, y) = f(x, y) + c(x) + {}^x c(y) - c(xy)$$

for all $x, y \in H$. As discussed in the section, the set $H^2(H, A)$ of equivalence classes of such functions forms an abelian group that is called the second cohomology group of (H, A). It now follows from Exercise 9.10 that there is a bijective correspondence between the group $H^2(H, A)$ and the set of equivalence classes of extensions of A by H which respect the action of H on A. In particular, if $H^2(H, A) = 0$, then every extension of A by H is split.

Exercise 9.11. (cont.) Suppose that H and A are both finite. Show that the order of each element of $H^2(H, A)$ divides both $|H|$ and the exponent of A. (This implies that $H^2(G/A, A) = 0$ when A is an abelian normal Hall subgroup of a finite group G, which we established in proving Schur–Zassenhaus Theorem.)

Solution. Let $[f] \in H^2(H, A)$, where $f \colon H \times H \to A$ is a normalized 2-cocycle. First let m be the exponent of A. Since $mf(x, y) = 0$ for all $x, y \in H$, we have $m[f] = [0]$. Hence the order of $[f]$ divides m.
It remains to show that $|H|[f] = [0]$. Define a map $c \colon H \to A$ by

$$c(x) = \sum_{z \in H} f(x, z).$$

Since $f(1, z) = 0$ for all $z \in H$, we have $c(1) = 0$. We claim that the coboundary determined by c is $|H|f$. Indeed, for $x, y \in H$, the cocycle identity gives

$${}^x f(y, z) - f(xy, z) = f(x, y) - f(x, yz).$$

Therefore

$$\begin{aligned} c(x) + {}^x c(y) - c(xy) &= \sum_{z \in H} \big(f(x, z) + {}^x f(y, z) - f(xy, z) \big) \\ &= \sum_{z \in H} \big(f(x, z) + f(x, y) - f(x, yz) \big) \\ &= |H| f(x, y). \end{aligned}$$

The last equality holds because $z \mapsto yz$ permutes the elements of H. Hence $|H|f$ is a 2-coboundary, so $|H|[f] = [0]$. Thus the order of $[f]$ divides both $|H|$ and the exponent of A.
Finally suppose that A is an abelian normal Hall subgroup of a finite group G, and put $H = G/A$. Then $\gcd(|H|, |A|) = 1$, while m divides $|A|$. Hence $\gcd(|H|, m) = 1$. Since every element of $H^2(H, A)$ has order dividing both of these relatively prime numbers, every element is trivial. Therefore $H^2(G/A, A) = 0$. $\square$

Chapter 4

Normal Structure

10 Composition Series

Composition Series

- A series of subgroups $G = G_0 > G_1 > \ldots > G_r = 1$ of a group G is a *composition series* of G if $G_{i+1} \trianglelefteq G_i$ for every i and if each *successive quotient* G_i/G_{i+1} is simple.
- The composition series above is said to have *length* r.
- The successive quotients of a composition series are called the *composition factors* of the series.
- More generally, a group is said to be a *composition factor* of G if it is isomorphic with one of the composition factors in some composition series of G.
- Example: $S_5 > A_5 > 1$ is the only composition series of S_5 (since A_5 is simple by Theorem 7.4 and is the only non-trivial proper normal subgroup of S_5 by Theorem 7.5), with composition factors $\mathbb{Z}_2$ and A_5.

Maximal Normal Subgroups

- N is a *maximal normal subgroup* of G if $N \trianglelefteq G$ and there is no proper normal subgroup of G that properly contains N.
- By Correspondence Theorem, N is a maximal normal subgroup of G iff G/N is simple.
- A maximal subgroup that is also normal is clearly a maximal normal subgroup (and must be of prime index), but a maximal normal subgroup need not be a maximal subgroup, for it could be properly contained in a proper subgroup that is not normal.
- For example, $A_5 \times \mathbb{Z}_2$ has $1 \times \mathbb{Z}_2$ as a maximal normal subgroup that is not maximal.

Proposition 10.1. *Finite groups have composition series.*

Proof. We use induction on $|G|$. If $G = 1$, then the series $G = 1$ is a composition series of length 0. If G is non-trivial and simple, then $G > 1$ is a composition series of G. Suppose now that G is not simple. Since G is finite, the set of proper normal subgroups of G has a maximal element; call it G_1. Then G_1 is a maximal normal subgroup of G, so G/G_1 is simple by the Correspondence Theorem. Also $|G_1| < |G|$, so the induction hypothesis gives a composition series

$$G_1 > G_2 > \cdots > G_r = 1$$

of G_1. Therefore

$$G = G_0 > G_1 > G_2 > \cdots > G_r = 1$$

is a composition series of G: the first quotient G/G_1 is simple, and the remaining successive quotients are simple by the composition series of G_1. $\square$

- Infinite groups need not have composition series. For example, using Theorem 1.5 we see that every non-trivial subgroup of the infinite cyclic group $\mathbb{Z}$ is isomorphic with $\mathbb{Z}$; as $\mathbb{Z}$ is not simple, $\mathbb{Z}$ has no simple subgroups, and hence we cannot construct a composition series of $\mathbb{Z}$, since the last non-trivial term of such a series must be a simple subgroup of $\mathbb{Z}$.

Lemma 10.2. *Let G be a group that has a composition series, and let $N \trianglelefteq G$. Then N has a composition series.*

Proof. Let
$$G = G_0 > G_1 > \cdots > G_r = 1$$
be a composition series of G, and put $N_i = N \cap G_i$ for each i. Then
$$N = N_0 \geq N_1 \geq \cdots \geq N_r = 1$$
is a series of subgroups of N. For each i, we have $N_{i+1} \trianglelefteq N_i$, because $G_{i+1} \trianglelefteq G_i$ and $N_i \leq G_i$. Moreover, let $\psi_i \colon G_i \to G_i/G_{i+1}$ be the natural map. Then the restriction of ψ_i to N_i has kernel $N_i \cap G_{i+1} = N_{i+1}$, so by First Isomorphism Theorem we have
$$N_i/N_{i+1} \cong \psi_i(N_i) = (N \cap G_i)G_{i+1}/G_{i+1}.$$
The subgroup $(N \cap G_i)G_{i+1}/G_{i+1}$ is normal in G_i/G_{i+1}, since $N \trianglelefteq G$ gives $N \cap G_i \trianglelefteq G_i$. Since G_i/G_{i+1} is simple, it follows that either N_i/N_{i+1} is trivial or $N_i/N_{i+1} \cong G_i/G_{i+1}$ is simple. Deleting any repeated terms from
$$N = N_0 \geq N_1 \geq \cdots \geq N_r = 1$$
therefore gives a composition series of N. □

Equivalence of Composition Series

- Let $G = G_0 > G_1 > \ldots > G_r = 1$ be a composition series, and suppose $G = H_0 > H_1 > \ldots > H_r = 1$ is another composition series of the same length r. These series are *equivalent* if there is some $\rho \in S_r$ such that $G_{i-1}/G_i \cong H_{\rho(i)-1}/H_{\rho(i)}$ for every i.
- Example: $G = \langle x \rangle \cong \mathbb{Z}_6$ with $G_1 = \langle x^2 \rangle$, $H_1 = \langle x^3 \rangle$ gives equivalent composition series $G > G_1 > 1$ and $G > H_1 > 1$, since $G/G_1 \cong H_1/1 \cong \mathbb{Z}_2$ and $G_1/1 \cong G/H_1 \cong \mathbb{Z}_3$. (Here $\rho = (1\ 2) \in S_2$.)
- Up to equivalence, a group has at most one composition series, so a group having a composition series has a well-defined collection of composition factors.
- Since composition factors are simple groups, classifying finite simple groups (completed in 1980) gives comprehensive understanding of finite groups.

Theorem (Jordan–Hölder Theorem)**.** *Suppose that G is a group that has a composition series. Then any two composition series of G have the same length and are equivalent.*

Proof. Let
$$G = G_0 > G_1 > \cdots > G_r = 1$$
and
$$G = H_0 > H_1 > \cdots > H_s = 1$$
be two composition series of G. We use induction on r. If $r = 0$, then $G = 1$, and there is nothing to prove. If $r = 1$, then G is simple, so $G > 1$ is the only composition series of G. Thus assume $r > 1$, and assume the result is known for every group having a composition series of length less than r.
If $G_1 = H_1$, then the two tails
$$G_1 > G_2 > \cdots > G_r = 1$$
and
$$H_1 > H_2 > \cdots > H_s = 1$$
are composition series of G_1. By induction they have the same length and are equivalent, so the original two composition series of G also have the same length and are equivalent.
We may therefore assume that $G_1 \neq H_1$. Since $G_1 \trianglelefteq G$ and $H_1 \trianglelefteq G$, Proposition 1.7 gives $G_1H_1 \trianglelefteq G$. Also $G_1H_1 \neq G_1$, for otherwise $H_1 \leq G_1$, contradicting the maximal normality of H_1 in G. Since G/G_1

is simple and $G_1 < G_1H_1 \trianglelefteq G$, it follows that $G_1H_1 = G$. Put $K = G_1 \cap H_1$. By First Isomorphism Theorem, applied to the natural maps into G/G_1 and G/H_1, we have

$$H_1/K \cong G/G_1 \qquad \text{and} \qquad G_1/K \cong G/H_1.$$

In particular, H_1/K and G_1/K are simple. Since $K \trianglelefteq G$, Lemma 10.2 gives a composition series

$$K = K_0 > K_1 > \cdots > K_t = 1$$

of K, with $t = 0$ if $K = 1$.
Now compare two composition series of G_1:

$$G_1 > G_2 > \cdots > G_r = 1$$

and

$$G_1 > K > K_1 > \cdots > K_t = 1.$$

By induction these have the same length and are equivalent. Hence $t + 1 = r - 1$, so $t = r - 2$. Similarly, compare the two composition series of H_1:

$$H_1 > H_2 > \cdots > H_s = 1$$

and

$$H_1 > K > K_1 > \cdots > K_{r-2} = 1.$$

Induction gives $s - 1 = r - 1$, so $s = r$, and these two series are equivalent. Finally, the two intermediate composition series

$$G > G_1 > K > K_1 > \cdots > K_{r-2} = 1$$

and

$$G > H_1 > K > K_1 > \cdots > K_{r-2} = 1$$

are equivalent, because $G/G_1 \cong H_1/K$, $G/H_1 \cong G_1/K$, and all remaining factors are the same. Combining these equivalences shows that the original two composition series of G are equivalent and have the same length. □

Subnormal, Normal, and Chief Series

- A series of subgroups $G = G_0 \geq G_1 \geq \ldots \geq G_r = 1$ of a group G is called a *subnormal series* of G if $G_{i+1} \trianglelefteq G_i$ for every i.
- A subnormal series is called a *normal series* if $G_i \trianglelefteq G$ for every i.
- Some authors use "normal series" for what we call a subnormal series, and "invariant series" for what we call a normal series.
- Composition series are examples of subnormal series, but subnormal series can have successive quotients that are trivial or that are non-trivial but not simple.
- Two subnormal series of the same length are said to be *equivalent* if they satisfy the condition stated as the definition of equivalence of composition series.
- A *refinement* of a subnormal series is one obtained by interposing additional terms; it is *proper* if at least one of the interposed terms was not already present.
- A composition series is a subnormal series that has no repeated terms and does not admit a proper refinement.
- A *chief series* of G is a normal series of G with no repeated terms and with the additional property that no normal subgroup of G is contained properly between any two terms of the series.
- Analogy: A composition series is a subnormal series having no subnormal series as a proper refinement, and a chief series is a normal series having no normal series as a proper refinement.
- The successive quotients of a chief series are its *chief factors*; a group is a *chief factor* of G if it is isomorphic with a chief factor in some chief series of G.
- The analogue of the Jordan–Hölder Theorem for chief series holds—namely, any two chief series of a group have the same length and are equivalent, with a proof virtually identical to that for composition series. This is a special case of the Jordan–Hölder theorem for groups with operators (see [26, Section 2.3]).

Minimal Normal Subgroups

- N is a *minimal normal subgroup* of G if $1 \neq N \trianglelefteq G$ and there is no non-trivial normal subgroup of G that is properly contained in N.
- Any non-trivial finite group has minimal normal subgroups.
- A simple group has a unique minimal normal subgroup, namely itself.

Proposition 10.3. *Finite groups have chief series.*

Proof. We use induction on $|G|$. If $G = 1$, then the series $G = 1$ is a chief series of length 0. If G is non-trivial and simple, then $G > 1$ is a chief series of G. Suppose now that G is not simple. Since G is finite, it has a proper minimal normal subgroup N. By induction, the finite group G/N has a chief series. By the Correspondence Theorem, this chief series has the form

$$G/N = G_0/N > G_1/N > \cdots > G_r/N = 1,$$

where $G_i \trianglelefteq G$ for every i, $G_0 = G$, and $G_r = N$. Moreover, no normal subgroup of G lies properly between G_{i-1} and G_i, for such a subgroup would correspond to a normal subgroup of G/N lying properly between G_{i-1}/N and G_i/N. Since N is minimal normal in G, no non-trivial normal subgroup of G lies properly between N and 1. Therefore

$$G = G_0 > G_1 > \cdots > G_r = N > 1$$

is a chief series of G. □

- By definition, the composition factors of finite groups are simple groups. We now turn to determining the nature of chief factors of finite groups; the first step is to rephrase the problem in different terms.

Lemma 10.4. *Let G be a group having a chief series. Then every chief factor of G is a minimal normal subgroup of a quotient group of G.*

Proof. Let

$$G = G_0 > G_1 > \cdots > G_r = 1$$

be a chief series of G. A chief factor of this series has the form G_i/G_{i+1} for some i. Since $G_i \trianglelefteq G$ and $G_{i+1} \trianglelefteq G$, the quotient G_i/G_{i+1} is a normal subgroup of G/G_{i+1}. If there were a non-trivial normal subgroup of G/G_{i+1} properly contained in G_i/G_{i+1}, then by the Correspondence Theorem it would have the form M/G_{i+1} for some normal subgroup $M \trianglelefteq G$ satisfying

$$G_{i+1} < M < G_i.$$

This contradicts the definition of a chief series. Hence G_i/G_{i+1} is a minimal normal subgroup of the quotient group G/G_{i+1}. Therefore every chief factor of G is a minimal normal subgroup of a quotient group of G. □

Theorem 10.5. *A minimal normal subgroup of a finite group is a direct product of mutually isomorphic simple groups.*

Proof. Let G be a finite group, and let N be a minimal normal subgroup of G. Choose a maximal normal subgroup N_1 of N, so N/N_1 is simple. Let $N_1, N_2, \ldots, N_r$ be the distinct conjugates of N_1 in G. Since $N \trianglelefteq G$, each N_i is a maximal normal subgroup of N, and conjugation by an element of G gives an isomorphism $N/N_1 \cong N/N_i$. Hence all quotients N/N_i are mutually isomorphic simple groups.
Conjugation by elements of G permutes the set $\{N_1, \ldots, N_r\}$, so

$$I = N_1 \cap N_2 \cap \cdots \cap N_r$$

is normal in G. Since $I \leq N_1 < N$, the minimality of N gives $I = 1$.
For $1 \leq i \leq r$, put

$$A_i = N_1 \cap \cdots \cap N_i.$$

We prove by induction on i that N/A_i is a direct product of groups isomorphic with N/N_1. For $i = 1$, this is immediate. Suppose $i > 1$ and the claim holds for $i - 1$. If $A_{i-1} \leq N_i$, then $A_i = A_{i-1}$, and there is nothing new to prove. Otherwise $N_i < A_{i-1}N_i \trianglelefteq N$. Since N_i is maximal normal in N, we have $A_{i-1}N_i = N$. By Lemma 2.7,

$$N/A_i \cong A_{i-1}/A_i \times N_i/A_i.$$

Now the natural map $A_{i-1} \to N/N_i$ has kernel $A_{i-1} \cap N_i = A_i$ and image $A_{i-1}N_i/N_i = N/N_i$, so by First Isomorphism Theorem,

$$A_{i-1}/A_i \cong N/N_i \cong N/N_1.$$

Similarly, the natural map $N_i \to N/A_{i-1}$ has kernel $N_i \cap A_{i-1} = A_i$ and image $A_{i-1}N_i/A_{i-1} = N/A_{i-1}$, so by First Isomorphism Theorem,

$$N_i/A_i \cong N/A_{i-1}.$$

By the induction hypothesis, N/A_{i-1} is a direct product of groups isomorphic with N/N_1. Therefore N/A_i is also a direct product of groups isomorphic with N/N_1. Taking $i = r$ gives $A_r = I = 1$, and hence N itself is a direct product of mutually isomorphic simple groups. □

Corollary 10.6. *A chief factor of a finite group is a direct product of mutually isomorphic simple groups.*

Proof. Let G_i/G_{i+1} be a chief factor of a finite group G. By Lemma 10.4, this chief factor is a minimal normal subgroup of the quotient group G/G_{i+1}. Since G/G_{i+1} is finite, Theorem 10.5 applies to this quotient group. Therefore G_i/G_{i+1} is a direct product of mutually isomorphic simple groups. □

Exercises

Exercise 10.1. Show that an abelian group has a composition series iff it is finite.

Solution. If G is finite, then Proposition 10.1 gives a composition series of G. Conversely, suppose that G is abelian and has a composition series

$$G = G_0 > G_1 > \cdots > G_r = 1.$$

Each factor G_i/G_{i+1} is simple by definition, and it is abelian because $G_i \leq G$ is abelian. We first note that a simple abelian group has prime order. Indeed, if A is simple and abelian, and if $1 \neq a \in A$, then $\langle a \rangle \trianglelefteq A$; hence $\langle a \rangle = A$. Thus A is cyclic. If A were infinite cyclic, then it would have a proper non-trivial subgroup; if A were finite cyclic of composite order, then it would also have a proper non-trivial subgroup. Therefore A has prime order. Applying this to each G_i/G_{i+1}, we see that each factor is finite. Since

$$|G| = \prod_{i=0}^{r-1} |G_i : G_{i+1}|,$$

it follows that G is finite. □

Exercise 10.2. Show that $\mathrm{GL}(n, F)$ has a composition series iff F is finite.

Solution. If F is finite, then $\mathrm{GL}(n, F)$ is finite, so Proposition 10.1 gives a composition series of $\mathrm{GL}(n, F)$. Conversely, suppose that $\mathrm{GL}(n, F)$ has a composition series. Let

$$Z = \{aI_n \mid a \in F^\times\}$$

be the subgroup of non-zero scalar matrices. This subgroup is central, hence normal, in $\mathrm{GL}(n, F)$, and $Z \cong F^\times$. By Lemma 10.2, the subgroup Z has a composition series. Since Z is abelian, Exercise 10.1 shows that Z is finite. Therefore $F^\times$ is finite. Since

$$F = F^\times \cup \{0\},$$

it follows that F is finite. □

Exercise 10.3. Using Third Isomorphism Theorem, prove the *Schreier refinement theorem*: Any two subnormal series of a given group have equivalent refinements.

Solution. Let
$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1$$
and
$$G = H_0 \geq H_1 \geq \cdots \geq H_s = 1$$
be two subnormal series of G. For each $0 \leq i < r$ and $0 \leq j \leq s$, put
$$G_{i,j} = G_{i+1}(G_i \cap H_j).$$
Then
$$G_i = G_{i,0} \geq G_{i,1} \geq \cdots \geq G_{i,s} = G_{i+1},$$
so these subgroups refine the first series. The normality condition holds because $G_{i+1} \trianglelefteq G_i$ and $H_{j+1} \trianglelefteq H_j$: if $x \in G_{i,j}$ and $y \in G_{i,j+1}$, writing $x = ab$ with $a \in G_{i+1}$ and $b \in G_i \cap H_j$, and writing $y = cd$ with $c \in G_{i+1}$ and $d \in G_i \cap H_{j+1}$, a direct conjugation calculation gives $xyx^{-1} \in G_{i+1}(G_i \cap H_{j+1}) = G_{i,j+1}$. Hence
$$G = G_{0,0} \geq G_{0,1} \geq \cdots \geq G_{0,s} = G_{1,0} \geq \cdots \geq G_{r-1,s} = 1$$
is a subnormal refinement of the first series.
Similarly, for each $0 \leq j < s$ and $0 \leq i \leq r$, put
$$H_{j,i} = H_{j+1}(H_j \cap G_i).$$
Then
$$H_j = H_{j,0} \geq H_{j,1} \geq \cdots \geq H_{j,r} = H_{j+1},$$
and the same normality argument shows that these subgroups give a subnormal refinement of the second series.
It remains to compare the factors. Fix i and j. Applying Third Isomorphism Theorem with
$$H_1 = G_i, \qquad K_1 = G_{i+1}, \qquad H_2 = H_j, \qquad K_2 = H_{j+1},$$
we obtain
$$G_{i+1}(G_i \cap H_j)/G_{i+1}(G_i \cap H_{j+1}) \cong H_{j+1}(H_j \cap G_i)/H_{j+1}(H_j \cap G_{i+1}).$$
That is,
$$G_{i,j}/G_{i,j+1} \cong H_{j,i}/H_{j,i+1}.$$
Thus the factors of the refinement of the first series are precisely the factors of the refinement of the second series, after reindexing the pair (i,j) as (j,i). Therefore the two refinements are equivalent. $\square$

Exercise 10.4. (cont.) Use the Schreier refinement theorem to give an alternate proof of Jordan–Hölder Theorem.

Solution. Let
$$G = G_0 > G_1 > \cdots > G_r = 1$$
and
$$G = H_0 > H_1 > \cdots > H_s = 1$$
be two composition series of G. By Exercise 10.3, these two subnormal series have equivalent refinements. Since each factor G_i/G_{i+1} is simple, no subgroup can be inserted properly between G_i and G_{i+1} in a subnormal refinement; any inserted term must repeat either G_i or G_{i+1}. Thus the non-trivial factors of any refinement of the first composition series are precisely the factors G_i/G_{i+1}. Similarly, the non-trivial factors of any refinement of the second composition series are precisely the factors H_j/H_{j+1}.
The equivalent refinements therefore have the same non-trivial factors up to reordering, because their trivial factors correspond only to trivial factors. Hence $r = s$, and there is some permutation $\rho \in S_r$ such that
$$G_{i-1}/G_i \cong H_{\rho(i)-1}/H_{\rho(i)}$$
for every i. Therefore the two original composition series are equivalent. This proves Jordan–Hölder Theorem. $\square$

Exercise 10.5. Determine all composition series and all chief series of S_n for $n \geq 2$. (Hint: Use Exercise 3.8.)

Solution. For $n = 2$, the group S_2 has prime order, so the only composition series and chief series is

$$S_2 > 1.$$

For $n = 3$, the only non-trivial proper normal subgroup of S_3 is A_3: indeed, a normal subgroup is a union of conjugacy classes, and the conjugacy classes in S_3 have sizes 1, 2, and 3. Hence the only composition series and chief series is

$$S_3 > A_3 > 1.$$

Now let $n = 4$, and put

$$K = \{1,\ (1\,2)(3\,4),\ (1\,3)(2\,4),\ (1\,4)(2\,3)\}.$$

The conjugacy classes in S_4 have sizes 1, 3, 8, 6, and 6, corresponding respectively to the identity, double transpositions, 3-cycles, transpositions, and 4-cycles. Since a normal subgroup is a union of conjugacy classes and its order must divide 24, the normal subgroups of S_4 are precisely 1, K, A_4, and S_4. Thus the only chief series of S_4 is

$$S_4 > A_4 > K > 1.$$

The composition series of S_4 are obtained by refining the final factor $K/1$. Since $K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, its three subgroups of order 2 give the three composition series

$$S_4 > A_4 > K > L > 1,$$

where

$$L \in \{\langle(1\,2)(3\,4)\rangle,\ \langle(1\,3)(2\,4)\rangle,\ \langle(1\,4)(2\,3)\rangle\}.$$

Finally, suppose $n \geq 5$. We first record that the only non-trivial proper normal subgroup of S_n is A_n. Indeed, if $N \trianglelefteq S_n$, then $N \cap A_n \trianglelefteq A_n$. By the simplicity of A_n (Theorem 7.4 for $n = 5$ and Exercise 7.8 for $n \geq 6$), if $N \leq A_n$, then $N = 1$ or $N = A_n$. If $N \not\leq A_n$, then $NA_n = S_n$. If $N \cap A_n = A_n$, then $N = S_n$; while if $N \cap A_n = 1$, then $N \cong S_n/A_n \cong \mathbb{Z}_2$, which would force the non-identity element of N to be central in S_n, impossible for $n \geq 3$. Thus the only non-trivial proper normal subgroup of S_n is A_n. Hence the only composition series and the only chief series of S_n is

$$S_n > A_n > 1.$$

□

Exercise 10.6. Show that any group having a composition series has a chief series.

Solution. Let

$$G = G_0 > G_1 > \cdots > G_r = 1$$

be a composition series of G. We first note that the lengths of normal series of G with no repeated terms are bounded above by r. Indeed, any normal series is a subnormal series, so by Exercise 10.3 it has an equivalent refinement with a refinement of the composition series above. Since a composition series admits no proper non-trivial refinement, the other series cannot have more than r non-trivial factors. Hence among all normal series of G with no repeated terms, choose one of maximal length:

$$G = N_0 > N_1 > \cdots > N_s = 1.$$

We claim that this is a chief series. If not, then for some i there is a normal subgroup $M \trianglelefteq G$ such that

$$N_i > M > N_{i+1}.$$

But then inserting M gives a longer normal series

$$G = N_0 > \cdots > N_i > M > N_{i+1} > \cdots > N_s = 1,$$

still with no repeated terms. This contradicts the maximality of the chosen normal series. Therefore no normal subgroup of G lies properly between two consecutive terms, and so the series is a chief series of G. □

Exercise 10.7. (cont.) Show that any chief factor of a group having a composition series is a direct product of mutually isomorphic simple groups.

Solution. Let A/B be a chief factor of G. By Exercise 10.6, G has a chief series, so Lemma 10.4 shows that A/B is a minimal normal subgroup of the quotient group G/B. Also, since G has a composition series, the quotient group G/B has a composition series: take the images of a composition series of G under the natural map $G \to G/B$ and delete repeated terms. Therefore Lemma 10.2 shows that A/B has a composition series.

We now prove the following claim. If N is a minimal normal subgroup of a group X, and if N has a composition series, then N is a direct product of mutually isomorphic simple groups. Choose a maximal normal subgroup N_1 of N, so N/N_1 is simple. Let $\mathcal{C}$ be the set of X-conjugates of N_1. Since $N \trianglelefteq X$, every member of $\mathcal{C}$ is again a maximal normal subgroup of N, and all quotients N/C, with $C \in \mathcal{C}$, are isomorphic to N/N_1.

Consider finite intersections of members of $\mathcal{C}$. Since N has a composition series, the same bounded-length argument used in Exercise 10.6 shows that descending chains of normal subgroups of N cannot be strictly decreasing indefinitely. Hence there is a finite intersection

$$I = N_1 \cap N_2 \cap \cdots \cap N_r$$

which is minimal among all finite intersections of members of $\mathcal{C}$. This subgroup is normal in X: for any $x \in X$, the intersection $I \cap xIx^{-1}$ is again a finite intersection of members of $\mathcal{C}$ contained in I, so minimality gives $I \cap xIx^{-1} = I$; applying the same argument to x^{-1} gives equality. Since $I \leq N_1 < N$ and N is minimal normal in X, we get $I = 1$.

For $1 \leq i \leq r$, put

$$A_i = N_1 \cap \cdots \cap N_i.$$

We prove by induction on i that N/A_i is a direct product of groups isomorphic with N/N_1. The case $i = 1$ is immediate. Suppose $i > 1$ and the claim is known for $i-1$. If $A_{i-1} \leq N_i$, then $A_i = A_{i-1}$, and there is nothing to prove. Otherwise $N_i < A_{i-1}N_i \trianglelefteq N$, so maximal normality of N_i in N gives $A_{i-1}N_i = N$. By Lemma 2.7,

$$N/A_i \cong A_{i-1}/A_i \times N_i/A_i.$$

The first factor is isomorphic to $N/N_i \cong N/N_1$, while the second factor is isomorphic to N/A_{i-1} by First Isomorphism Theorem. Hence, by induction, N/A_i is a direct product of groups isomorphic with N/N_1. Taking $i = r$ and using $A_r = I = 1$, we conclude that N itself is a direct product of mutually isomorphic simple groups.

Applying this claim to the minimal normal subgroup A/B of G/B proves that the chief factor A/B is a direct product of mutually isomorphic simple groups. □

Exercise 10.8. Show that any finite group that has no proper non-trivial characteristic subgroups is a direct product of mutually isomorphic simple groups. Use this to give a new proof of Theorem 10.5.

Solution. We prove the first assertion by induction on $|G|$. If $G = 1$, there is nothing to prove, and if G is simple, then G is itself a direct product with one simple factor. Suppose then that G is not simple, and choose a minimal normal subgroup N of G. Since G is not simple, we may choose $N < G$.

We first observe that N has no proper non-trivial characteristic subgroups. Indeed, if C is characteristic in N, then $C \trianglelefteq G$, because $N \trianglelefteq G$ and conjugation by each element of G restricts to an automorphism of N. Since N is minimal normal in G, this gives $C = 1$ or $C = N$. Thus, by induction,

$$N \cong S \times \cdots \times S$$

for some simple group S.

Now let $N_1, \ldots, N_t$ be the distinct images of N under automorphisms of G. Each N_i is again a minimal normal subgroup of G, and each N_i is isomorphic with N. The product

$$K = N_1 N_2 \cdots N_t$$

is characteristic in G, since automorphisms of G merely permute the subgroups N_i. It is non-trivial, so our hypothesis on G gives $K = G$. Choose a subcollection, relabelled as $N_1, \ldots, N_r$, minimal with respect to the property that

$$G = N_1 N_2 \cdots N_r.$$

We claim that this product is direct. Put $P_{i-1} = N_1 \cdots N_{i-1}$. Since P_{i-1} and N_i are normal in G, their intersection $P_{i-1} \cap N_i$ is normal in G and is contained in the minimal normal subgroup N_i. Hence $P_{i-1} \cap N_i$ is either 1 or N_i. The second possibility would give $N_i \leq P_{i-1}$, contradicting the minimality of the chosen subcollection. Therefore $P_{i-1} \cap N_i = 1$. Also $[P_{i-1}, N_i] \leq P_{i-1} \cap N_i = 1$, so $P_i = P_{i-1}N_i$ is the direct product $P_{i-1} \times N_i$. Inducting on i gives

$$G = N_1 \times \cdots \times N_r.$$

Since each N_i is isomorphic with N, and N is a direct product of copies of the same simple group S, it follows that G is a direct product of mutually isomorphic simple groups.

We now use this to prove Theorem 10.5. Let N be a minimal normal subgroup of a finite group G. If C is characteristic in N, then, as above, $C \trianglelefteq G$. By the minimal normality of N, we have $C = 1$ or $C = N$. Thus N is a finite group with no proper non-trivial characteristic subgroups. By the first part, N is a direct product of mutually isomorphic simple groups. $\square$

11 Solvable Groups

The derived series and solvability

- We define a series $G^{(k)}$ of subgroups of a group G by setting $G^{(0)} = G$ and taking $G^{(k)}$ to be the derived group of $G^{(k-1)}$ for $k \in \mathbb{N}$. This series is called the *derived series* of G.
- Since $G^{(k+1)}$ is a characteristic subgroup of $G^{(k)}$ for each k by Lemma 2.5, we see via Lemma 2.4 that the derived series is a normal series of G. Moreover, we see from Proposition 2.6 that each successive quotient $G^{(k)}/G^{(k+1)}$ of the derived series is abelian.
- We say that a group is *solvable* if its derived series terminates in the identity. (Authors from British Commonwealth countries often write "soluble" in lieu of "solvable" and "insoluble" in lieu of "non-solvable.")
- Since a group is abelian iff its derived group is trivial, we see that abelian groups are solvable. However, not all groups are solvable; for example, A_5 is non-abelian and simple by Theorem 7.4, so $A_5^{(k)} = A_5$ for all k.
- A group whose derived group is itself is said to be *perfect*; any non-abelian simple group is perfect, and in particular is not solvable. However, a perfect group need not be simple: $\mathrm{SL}(n, F)$ is perfect but has non-trivial center, except when $n = 2$ and $|F| = 3$.
- Solvable groups are to be thought of as the opposite of simple groups: simple groups have very few normal subgroups while solvable groups are rife with them.

Proposition 11.1. *A simple solvable group has prime order.*

Proof. Let G be a simple solvable group. Since G is solvable, the derived series of G eventually reaches 1, so we cannot have $G' = G$. But G' is characteristic in G by Lemma 2.5, and hence $G' \trianglelefteq G$. Since G is simple, it follows that $G' = 1$. Thus G is abelian.
Let $g \in G$ be non-trivial. Since G is abelian, every subgroup of G is normal in G, so $\langle g \rangle \trianglelefteq G$. By simplicity, $\langle g \rangle = G$. Hence G is cyclic. If G were infinite cyclic, then $\langle g^2 \rangle$ would be a proper non-trivial normal subgroup of G. If G had finite composite order, then again G would have a proper non-trivial subgroup, contradicting simplicity. Therefore G has prime order. $\square$

Proposition 11.2. *The following statements about a group G are equivalent:*

(1) G is solvable.

(2) G has a normal series in which every successive quotient is abelian.

(3) G has a subnormal series in which every successive quotient is abelian.

Proof. We first show that (1) ⇒ (2). If G is solvable, then its derived series terminates in the identity. As observed above, the derived series is a normal series of G, and each successive quotient is abelian. Thus G has a normal series with abelian successive quotients. Since every normal series is in particular a subnormal series, (2) ⇒ (3).
It remains to show that (3) ⇒ (1). Suppose that

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1$$

is a subnormal series in which each quotient G_i/G_{i+1} is abelian. We claim that $G^{(i)} \leq G_i$ for every i. For $i = 0$ this is immediate. Now suppose that $i \geq 1$ and that $G^{(i-1)} \leq G_{i-1}$. Then

$$G^{(i)} = (G^{(i-1)})' \leq G'_{i-1}.$$

Since G_{i-1}/G_i is abelian, Proposition 2.6 gives $G'_{i-1} \leq G_i$. Hence $G^{(i)} \leq G_i$, proving the claim by induction. Taking $i = r$, we obtain $G^{(r)} \leq G_r = 1$, so the derived series terminates in the identity. Therefore G is solvable. $\square$

Proposition 11.3. *(i) If G is solvable and $H \leq G$, then H is solvable.*

(ii) If G is solvable and $N \trianglelefteq G$, then G/N is solvable.

(iii) If $N \trianglelefteq G$ and both N and G/N are solvable, then G is solvable.

(iv) If G and H are solvable, then $G \times H$ is solvable.

Proof. For (i), suppose that G is solvable and let $H \leq G$. We claim that $H^{(k)} \leq G^{(k)}$ for every k. This is immediate when $k = 0$. If $H^{(k-1)} \leq G^{(k-1)}$, then

$$H^{(k)} = (H^{(k-1)})' \leq (G^{(k-1)})' = G^{(k)}.$$

Since the derived series of G terminates in the identity, the derived series of H also terminates in the identity. Thus H is solvable.
For (ii), suppose that G is solvable and that $N \trianglelefteq G$. By Proposition 11.2, there is a normal series

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1$$

whose successive quotients are abelian. Then

$$G/N = G_0N/N \geq G_1N/N \geq \cdots \geq G_rN/N = 1$$

is a normal series of G/N. Moreover, each quotient

$$(G_iN/N)/(G_{i+1}N/N)$$

is isomorphic with $G_iN/G_{i+1}N$ by the Third Isomorphism Theorem, and by First Isomorphism Theorem this is a quotient of G_i/G_{i+1}. Hence each successive quotient is abelian. Therefore G/N is solvable by Proposition 11.2.
For (iii), suppose that $N \trianglelefteq G$ and that both N and G/N are solvable. By Proposition 11.2, there are subnormal series

$$G/N = G_0/N \geq G_1/N \geq \cdots \geq G_s/N = 1$$

and

$$N = N_0 \geq N_1 \geq \cdots \geq N_r = 1$$

whose successive quotients are abelian. By the Third Isomorphism Theorem, the quotients

$$(G_i/N)/(G_{i+1}/N)$$

are isomorphic with G_i/G_{i+1}. Hence

$$G = G_0 \geq G_1 \geq \cdots \geq G_s = N = N_0 \geq N_1 \geq \cdots \geq N_r = 1$$

is a subnormal series of G whose successive quotients are abelian. Therefore G is solvable by Proposition 11.2.
For (iv), suppose that G and H are solvable. The subgroup $1 \times H$ is normal in $G \times H$ and is isomorphic with H, so it is solvable. Also, by First Isomorphism Theorem,

$$(G \times H)/(1 \times H) \cong G,$$

so the quotient is solvable. It follows from part (iii) that $G \times H$ is solvable. □

Proposition 11.4. *A group having a composition series is solvable iff all of its composition factors have prime order. (In particular, such groups are finite.)*

Proof. Suppose first that all composition factors of G have prime order. Then every composition factor is abelian, so any composition series of G is a subnormal series whose successive quotients are abelian. Hence G is solvable by Proposition 11.2.
Conversely, suppose that G is solvable, and let H/K be a composition factor of G, where $K \trianglelefteq H \leq G$ and H/K is simple. By Proposition 11.3, the subgroup H is solvable, and then the quotient H/K is solvable. Thus H/K is a simple solvable group, so Proposition 11.1 shows that H/K has prime order.
Under either equivalent condition, each factor in a composition series of G has prime order. Starting from the final term 1 and moving upward through the series, each term is a finite extension of a finite group, and hence is finite. Therefore G is finite. □

Examples and consequences

- There are groups, such as S_5, which are neither simple nor solvable, and there are non-abelian groups, such as S_3 (which has the composition series $S_3 > A_3 > 1$), which are solvable.
- Proposition 11.4 also implies that an infinite abelian group, being solvable, cannot possess a composition series (which was Exercise 10.1).

Corollary 11.5. *Finite p-groups are solvable.*

Proof. Let P be a finite p-group. By Proposition 10.1, P has a composition series. Each composition factor of P is a finite simple p-group. If S is such a factor, then Theorem 8.1 gives $Z(S) \neq 1$, and simplicity forces $Z(S) = S$. Thus S is abelian, and a simple abelian group has prime order. Therefore all composition factors of P have prime order, so P is solvable by Proposition 11.4. □

Proposition 11.6. *A group having a composition series is solvable iff all of its chief factors are elementary abelian.*

(Any group having a composition series also has a chief series by Exercise 10.6. Alternately, a solvable group having a composition series is finite by Proposition 11.4 and hence has a chief series by Proposition 10.3.)

Proof. Suppose first that all chief factors of G are elementary abelian. Since G has a composition series, Exercise 10.6 gives a chief series

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1.$$

Each quotient G_i/G_{i+1} is elementary abelian, hence abelian. Therefore this chief series is a normal series with abelian successive quotients, so G is solvable by Proposition 11.2.

Conversely, suppose that G is solvable, and let A/B be a chief factor of G. Since G has a composition series, Proposition 11.4 implies that G is finite. By Corollary 10.6, the chief factor A/B is a direct product of mutually isomorphic simple groups; say

$$A/B \cong S \times \cdots \times S.$$

Since $A \leq G$, Proposition 11.3 shows that A is solvable, and then that A/B is solvable. Hence each direct factor S is solvable, again by Proposition 11.3. Thus S is a simple solvable group, so Proposition 11.1 gives that S has prime order. Therefore A/B is a direct product of copies of a cyclic group of prime order, and hence is elementary abelian. □

Corollary 11.7. *A group having a composition series is solvable iff it has a normal series in which every successive quotient is a p-group.*

Proof. Suppose first that G is solvable. Since G has a composition series, Exercise 10.6 gives a chief series

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1.$$

By Proposition 11.6, each chief factor G_i/G_{i+1} is elementary abelian, and hence is a p-group. Thus G has a normal series whose successive quotients are p-groups.

Conversely, suppose that G has a normal series

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1$$

in which every successive quotient is a p-group. We prove that G is solvable by induction on r. If $r = 0$, then $G = 1$, so there is nothing to prove. If $r \geq 1$, put $N = G_1$. Then

$$N = G_1 \geq G_2 \geq \cdots \geq G_r = 1$$

is a normal series of N whose successive quotients are p-groups, so N is solvable by induction. Also, $G/N = G_0/G_1$ is a finite p-group, and hence is solvable by Corollary 11.5. Therefore G is solvable by Proposition 11.3. □

Supersolvable groups and Hall's theorems

- Those finite groups whose chief factors all have prime order are said to be *supersolvable*; finite supersolvable groups are solvable by Proposition 11.6.

- It follows from Exercise 8.2 that finite p-groups are supersolvable. However, not all finite solvable groups are supersolvable. The series $S_4 > A_4 > K > 1$, where K is the Klein four-group, is a chief series of S_4, and we have $S_4/A_4 \cong \mathbb{Z}_2$, $A_4/K \cong \mathbb{Z}_3$, and $K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, which by Proposition 11.6 shows that S_4 is solvable but not supersolvable.

- There is an equivalent definition of supersolvability which extends to infinite groups; see Exercise 11.6.

- Solvable groups possess a number of properties beyond those which hold for arbitrary groups. The following theorem, due to Philip Hall, is a generalization for finite solvable groups of part (i) of Sylow's Theorem.

Theorem 11.8. *Let G be a finite solvable group of order mn, where m and n are coprime. Then G has a subgroup of order m.*

(In other words, this theorem asserts that a finite solvable group possesses Hall subgroups of all possible orders.)

Proof. We argue by induction on $|G|$. If $G = 1$, there is nothing to prove. Suppose then that $G \neq 1$, and choose a minimal normal subgroup N of G. Since $N/1$ is a chief factor of the finite solvable group G, Proposition 11.6 shows that N is elementary abelian, say $|N| = p^a$ for some prime p. Since $\gcd(m, n) = 1$, either $|N|$ divides m or $|N|$ divides n.
First suppose that $|N|$ divides m. Then

$$|G/N| = \frac{m}{|N|}n,$$

and the two factors on the right are coprime. The quotient G/N is solvable by Proposition 11.3, so the induction hypothesis gives a subgroup H/N of G/N of order $m/|N|$. Hence $|H| = m$, and H is the required subgroup of G.
Now suppose that $|N|$ divides n. Then

$$|G/N| = m\frac{n}{|N|},$$

again with coprime factors. By induction, G/N has a subgroup K/N of order m. Thus $|K| = m|N|$. Inside K, the subgroup N is a normal Hall subgroup, since $\gcd(m, |N|) = 1$. By Schur–Zassenhaus Theorem, N has a complement in K. This complement has order $|K : N| = m$, and hence is the required subgroup of G. □

More on solvability criteria

- Theorem 11.8 does not hold for all finite groups; for instance, we observed in the proof of Theorem 7.5 that the group A_5 of order $60 = 20 \cdot 3$ has no subgroup of order 20.

- P. Hall also proved the following generalization for finite solvable groups of parts (ii) and (iii) of Sylow's Theorem, which the reader is asked to prove in the exercises.

Theorem 11.9. *Let G be a finite solvable group of order mn, where m and n are coprime. Then any two subgroups of G of order m are conjugate, and any subgroup of G whose order divides m is contained in a subgroup of order m.*

Proof. We prove both assertions simultaneously by induction on $|G|$. If $G = 1$, there is nothing to prove. We first record a special conjugacy fact which will be needed in the induction. Suppose that A is an abelian normal Hall subgroup of a finite group E, and that $E = A \rtimes H$. If K is another complement to A in E, then K is conjugate to H in E. Indeed, write A additively. The quotient map $E \to E/A$ identifies both H and K with E/A, so for each $h \in H$ there is a unique element $c(h) \in A$ such that $c(h)h \in K$. Since K is a subgroup, we have

$$c(h_1h_2) = c(h_1) + {}^{h_1}c(h_2)$$

for all $h_1, h_2 \in H$. Put $b = \sum_{h \in H} c(h)$. Then, for $h_0 \in H$,

$$^{h_0}b = \sum_{h \in H} {}^{h_0}c(h) = \sum_{h \in H} \big(c(h_0h) - c(h_0)\big) = b - |H|c(h_0).$$

Since $\gcd(|H|, |A|) = 1$, multiplication by $|H|$ is an automorphism of A. Choose $a \in A$ such that $|H|a = b$. Applying h_0 to this equality and using the displayed equation gives

$$|H|^{h_0}a = |H|a - |H|c(h_0),$$

so $^{h_0}a = a - c(h_0)$, and hence $c(h_0) = a - {}^{h_0}a$. Therefore

$$ah_0a^{-1} = c(h_0)h_0 \in K$$

for every $h_0 \in H$, and it follows that $aHa^{-1} = K$.

Now let N be a minimal normal subgroup of G. Since G is finite and solvable, Proposition 11.6 shows that N is elementary abelian, say $|N| = p^r$. Since $\gcd(m, n) = 1$, either $|N|$ divides m or $|N|$ divides n. First suppose that $|N|$ divides m. Let H and K be subgroups of G of order m. If P is a Sylow p-subgroup of H, then P is also a Sylow p-subgroup of G, and since N is a normal p-subgroup of G, we have $N \le P \le H$. Similarly, $N \le K$. By induction in G/N, the subgroups H/N and K/N are conjugate, and hence H and K are conjugate in G. Now let $L \le G$ have order dividing m. The subgroup LN has order dividing m, because its order divides $|G|$ and all of its prime divisors divide m. Hence LN/N has order dividing $m/|N|$. By induction in G/N, there is a subgroup M/N of G/N of order $m/|N|$ containing LN/N. Then M has order m and contains L.

It remains to consider the case where $|N|$ divides n. Let H and K be subgroups of G of order m. Then $H \cap N = K \cap N = 1$, so HN/N and KN/N are subgroups of G/N of order m. By induction in G/N, these two subgroups are conjugate. Replacing H by a conjugate in G, we may assume that $HN = KN$. Inside HN, the subgroup N is an abelian normal Hall subgroup, and H and K are complements to N. By the special conjugacy fact proved above, H and K are conjugate in HN, and hence in G.

Finally let $L \le G$ have order dividing m. Since $|N|$ divides n, we have $L \cap N = 1$, and so LN/N has order $|L|$, which divides m. By induction in G/N, there is a subgroup M/N of G/N of order m containing LN/N. Then $|M| = m|N|$, and N is a normal Hall subgroup of M. By Schur–Zassenhaus Theorem, choose a complement H to N in M, so $|H| = m$. Under the quotient isomorphism $H \to M/N$, let L_H be the subgroup of H corresponding to LN/N. Then $NL_H = NL$, and both L and L_H are complements to N in NL. Applying the special conjugacy fact inside NL, there is some $x \in NL$ such that $xL_Hx^{-1} = L$. Therefore $L \le xHx^{-1}$, and xHx^{-1} is a subgroup of G of order m. □

Classical theorems on solvability

- We now state, without proof, a number of well-known theorems giving conditions under which a finite group is solvable. We start with a classical result that generalizes Corollary 11.5.

Theorem (Burnside's Theorem). *If p and q are primes, then any group of order p^aq^b is solvable.*

Proof. We defer the proof to the Appendix, where it is proved as Burnside's Theorem, using the character theory developed in Chapter 6. □

- Burnside's Theorem is the best possible result, in the sense that a finite group whose order has exactly three prime divisors need not be solvable, with A_5 providing a counterexample.
- Burnside proved this theorem in 1904 using the methods of character theory, which will be discussed in Chapter 6; we present a character-theoretic proof of Burnside's theorem in the Appendix (Burnside's Theorem). Until the 1960s, there was no proof of this theorem that did not involve character theory.

Theorem (Feit–Thompson Theorem). *All finite groups of odd order are solvable.*

Proof. □

- This result, also known as the "odd order theorem," had first been conjectured by Burnside in 1911. As a corollary, we see that the only simple groups of odd order are cyclic groups of prime order.

- Once it had finally been established that all non-abelian finite simple groups are of even order, the movement toward a classification of all finite simple groups gathered steam. Feit and Thompson completed their proof of this theorem during a special "Finite Group Theory Year," held at the University of Chicago in the school year 1960–61. Their original proof [12] was 255 pages long, and even at that length the proof requires too much background knowledge for it to be readily comprehensible to anyone but the specialists at the time at which the proof was published. In recent years there has been a program, of which [8] represents a major part, to produce a more accessible proof of this fundamental result.

- The following result is a converse to Theorem 11.8 and is also due to P. Hall. While this theorem generalizes Burnside's Theorem, its proof relies on the fact that groups of order $p^a q^b$ are solvable.

Theorem 11.10. *Let G be a finite group. If G has a subgroup of order m whenever m and n are coprime numbers such that $|G| = mn$, then G is solvable.*

Proof. □

- This next result was conjectured by P. Hall in the same paper [14] in which he proved Theorem 11.10; it was later proved by Thompson in [27].

Theorem 11.11. *A finite group G is not solvable iff there exist non-trivial elements x, y, z of G of pairwise coprime orders a, b, c such that $xy = z$.*

Proof. □

- For example, in A_5 this criterion for non-solvability is satisfied by taking $a = 5$, $b = 2$, $c = 3$ and $x = (1\,2\,3\,4\,5)$, $y = (1\,2)(3\,4)$, $z = (1\,3\,5)$. (Compare with Exercise 1.5.)

Galois's theorem and the affine group

- The concept of solvable groups originated with Galois around 1830 in his work on the solution by radicals of polynomial equations; this in fact is the origin of the term "solvable" for this class of groups.

- Let V be a vector space over a field F. For $v \in V$, let T_v be the self-map of V corresponding to translation by v, and let $\mathcal{T}(V)$ be the set of all such translations; this is a group isomorphic with the abelian group V, and if we regard $\mathcal{T}(V)$ and $\mathrm{GL}(V)$ as subgroups of the group of all invertible self-maps of V, then $\mathrm{GL}(V) \cap \mathcal{T}(V)$ is trivial.

- We easily verify that $ST_vS^{-1} = T_{S(v)}$ for any $S \in \mathrm{GL}(V)$ and $v \in V$. This gives rise to a homomorphism from $\mathrm{GL}(V)$ to $\mathrm{Aut}(\mathcal{T}(V))$, and hence we have an (internal) semidirect product $\mathcal{T}(V) \rtimes \mathrm{GL}(V)$. This group is called the *affine group* of V and is denoted $\mathrm{Aff}(V)$; its elements are called the *affine transformations* of V.

Theorem 11.12. *Let G be a finite, solvable, primitive permutation group on a set X. Then X can be given the structure of a vector space over the field $\mathbb{Z}/p\mathbb{Z}$ for some prime p in such a way that G is isomorphic with a subgroup of $\mathrm{Aff}(X)$ that contains $\mathcal{T}(X)$.*

Proof. If X has one element, then the result is immediate, taking X to be the zero vector space over any field $\mathbb{Z}/p\mathbb{Z}$. We may therefore assume that $G \neq 1$. Let N be a minimal normal subgroup of G. Since G is finite and solvable, Proposition 11.6 shows that N is elementary abelian, say $N \cong \mathbb{Z}_p^r$ for some prime p. We regard N as a vector space over $\mathbb{Z}_p$ in the usual way.
Choose $x_0 \in X$, and let $H = G_{x_0}$. Since X is primitive, Corollary 3.10 shows that H is a maximal subgroup of G. We claim that $N \not\leq H$. Indeed, if $N \leq H$, then N fixes x_0. Since $N \trianglelefteq G$, Lemma 3.2 gives

$$N = gNg^{-1} \leq gHg^{-1} = G_{gx_0}$$

for every $g \in G$. Since X is transitive, this would mean that N fixes every point of X. But G is a permutation group on X, so the action is faithful, and hence $N = 1$, a contradiction. Thus $H < NH \leq G$, and maximality of H gives $NH = G$.
Now $N \cap H$ is normal in both N and H: normality in N follows because N is abelian, and normality in H follows from $N \trianglelefteq G$. Hence $N \cap H \trianglelefteq NH = G$. Since N is a minimal normal subgroup of G and

$N \nleq H$, we get $N \cap H = 1$. Therefore $G = N \rtimes H$.
By Proposition 3.4, the map

$$G/H \to X, \qquad gH \mapsto gx_0$$

is an isomorphism of G-sets. Since $G = NH$ and $N \cap H = 1$, every coset of H has the form nH for a unique $n \in N$. Thus every element of X has the form nx_0 for a unique $n \in N$. We transfer the vector-space structure of N to X by declaring

$$nx_0 + n'x_0 = (n + n')x_0, \qquad \lambda(nx_0) = (\lambda n)x_0$$

for $n, n' \in N$ and $\lambda \in \mathbb{Z}_p$. This makes X a vector space over $\mathbb{Z}_p$.
Under this vector-space structure, the elements of N act as translations. Indeed, if $n_0, n \in N$, then

$$n_0(nx_0) = (n_0 + n)x_0 = nx_0 + n_0x_0.$$

Since every vector of X has the form n_0x_0 for a unique $n_0 \in N$, the image of N in the permutation group of X is exactly $\mathcal{T}(X)$.
It remains to see that the stabilizer H acts linearly. Let $h \in H$ and let $x = nx_0 \in X$. Since h fixes x_0, we have

$$hx = h(nx_0) = (hnh^{-1})x_0.$$

Conjugation by h is an automorphism of the elementary abelian group N, and hence is a linear automorphism of the vector space N over $\mathbb{Z}_p$. Therefore the action of h on X is linear. Since every element of G has the form nh with $n \in N$ and $h \in H$, every element of G acts on X as an affine transformation. Thus G is isomorphic with a subgroup of $\mathrm{Aff}(X)$, and this subgroup contains $\mathcal{T}(X)$. □

Nilpotent groups

- A normal series $G = G_0 \geq G_1 \geq \ldots \geq G_r = 1$ of a group G is said to be a *central series* of G if, for each i, G_i/G_{i+1} is contained in the center of G/G_{i+1}.
- A group G is said to be *nilpotent* if it has a central series. An abelian group G has the central series $G > 1$, and hence abelian groups are nilpotent.

Proposition 11.13. *Nilpotent groups are solvable.*

Proof. Let

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1$$

be a central series for G. For each i, the quotient G_i/G_{i+1} is contained in $Z(G/G_{i+1})$, and hence is abelian. Therefore this central series is a normal series whose successive quotients are abelian. By Proposition 11.2, G is solvable. □

- There are solvable groups that are not nilpotent. For example, the group S_3 cannot have a central series, as the penultimate term of such a series would have to be a non-trivial subgroup of $Z(S_3) = 1$.

Lemma 11.14. *Finite p-groups are nilpotent.*

Proof. Let P be a finite p-group. We argue by induction on $|P|$. If $P = 1$, then the result is immediate. Suppose now that $P \neq 1$. By Theorem 8.1, the center $Z(P)$ is non-trivial. If $Z(P) = P$, then P is abelian, so

$$P \geq 1$$

is a central series. We may therefore assume that $1 < Z(P) < P$.
The quotient $P/Z(P)$ is a smaller finite p-group, so by induction it has a central series

$$P/Z(P) = \overline{P}_0 \geq \overline{P}_1 \geq \cdots \geq \overline{P}_r = 1.$$

Let $\pi \colon P \to P/Z(P)$ be the quotient map, and put $P_i = \pi^{-1}(\overline{P}_i)$. Then

$$P = P_0 \geq P_1 \geq \cdots \geq P_r = Z(P) \geq 1.$$

For each $i < r$, the centrality of $\overline{P}_i/\overline{P}_{i+1}$ in $(P/Z(P))/\overline{P}_{i+1}$ says precisely that P_i/P_{i+1} is contained in $Z(P/P_{i+1})$. Finally, $Z(P)/1$ is contained in $Z(P/1)$. Hence the displayed series is a central series for P, and P is nilpotent. □

- If H and K are subgroups of a group G, we define a new subgroup $[H, K]$ of G by $[H, K] = \langle\{[h, k] \mid h \in H, k \in K\}\rangle$. Observe that $G' = [G, G]$.

- We now reconcile the above definition of nilpotence with that made in Section 8 for finite groups. Recall that a subgroup H of a group G is said to be *self-normalizing* if $N_G(H) = H$.

Theorem 11.15. *The following statements about a finite group G are equivalent:*

(1) G is nilpotent.

(2) G has no proper self-normalizing subgroups.

(3) Every Sylow subgroup of G is normal in G.

(4) G is the direct product of its Sylow subgroups.

(5) Every maximal subgroup of G is normal in G.

Proof. If $G = 1$, then all five conditions are immediate. We may therefore assume that $G \neq 1$. We first prove (1) $\Rightarrow$ (2). Let

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1$$

be a central series for G, and let $H < G$. Choose i minimal such that $G_i \leq H$. Since $G_0 = G \nleq H$ and $G_r = 1 \leq H$, we have $i \geq 1$. Choose $x \in G_{i-1} - H$. For $h \in H$, the containment

$$G_{i-1}/G_i \leq Z(G/G_i)$$

gives $xhG_i = hxG_i$, so $xhx^{-1}h^{-1} \in G_i \leq H$. Hence $xhx^{-1} \in H$. Thus $xHx^{-1} \leq H$, and since G is finite, this inclusion is equality. Therefore $x \in N_G(H) - H$, so H is not self-normalizing.
We next prove (2) $\Rightarrow$ (3). Let P be a Sylow p-subgroup of G, and put $K = N_G(P)$. We claim that K is self-normalizing. Indeed, $P \trianglelefteq K$, so P is the unique Sylow p-subgroup of K. If $g \in N_G(K)$, then $gPg^{-1} \leq K$ and gPg^{-1} is again a Sylow p-subgroup of K. By uniqueness, $gPg^{-1} = P$, so $g \in N_G(P) = K$. Thus $N_G(K) \leq K$, and the reverse inclusion is automatic, so $N_G(K) = K$. If P were not normal in G, then $K < G$, giving a proper self-normalizing subgroup of G, contrary to (2). Therefore every Sylow subgroup of G is normal in G.
The implications (3) $\Rightarrow$ (4) and (4) $\Rightarrow$ (5) are exactly the corresponding implications in Theorem 8.7. It remains to prove (5) $\Rightarrow$ (1). By Theorem 8.7, condition (5) implies that

$$G = P_1 \times \cdots \times P_n,$$

where the P_i are the Sylow subgroups of G. By Lemma 11.14, each P_i is nilpotent. Choose a central series

$$P_i = P_{i,0} \geq P_{i,1} \geq \cdots \geq P_{i,r_i} = 1$$

for each i, and extend the notation by setting $P_{i,j} = 1$ for $j > r_i$. If $r = \max\{r_1, \ldots, r_n\}$, define

$$G_j = P_{1,j} \times \cdots \times P_{n,j} \qquad (0 \leq j \leq r).$$

Then

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1.$$

If $0 \leq j < r$, $x = (x_1, \ldots, x_n) \in G_j$, and $g = (g_1, \ldots, g_n) \in G$, then the centrality of each $P_{i,j}/P_{i,j+1}$ gives

$$x_i g_i x_i^{-1} g_i^{-1} \in P_{i,j+1}$$

for every i. Hence $xgx^{-1}g^{-1} \in G_{j+1}$, so $G_j/G_{j+1} \leq Z(G/G_{j+1})$. Therefore the displayed series is a central series for G, and G is nilpotent. $\square$

Exercises

Exercise 11.1. Show that any finite group has a largest solvable normal subgroup.

Solution. Let A and B be solvable normal subgroups of G. Then

$$gABg^{-1} = (gAg^{-1})(gBg^{-1}) = AB$$

for every $g \in G$, so $AB \trianglelefteq G$. Also, $A \trianglelefteq AB$, and by the Second Isomorphism Theorem we have

$$AB/A \cong B/(A \cap B).$$

Since B is solvable, Proposition 11.3 shows that $B/(A \cap B)$ is solvable. Hence AB/A is solvable. Since A is solvable, another application of Proposition 11.3 gives that AB is solvable. Thus the product of two solvable normal subgroups of G is again a solvable normal subgroup of G.
Because G is finite, it has only finitely many subgroups. List the solvable normal subgroups of G as

$$N_1, \dots, N_r.$$

Put $R = N_1 \cdots N_r$. Repeatedly applying the previous paragraph shows that R is a solvable normal subgroup of G. Moreover, each N_i is contained in R. Therefore every solvable normal subgroup of G is contained in R, so R is the largest solvable normal subgroup of G. □

Exercise 11.2. Let N be a solvable normal Hall subgroup of a finite group G. Show that any two complements to N in G are conjugate. (The strong version of Schur–Zassenhaus Theorem asserts that this is true for any normal Hall subgroup N, whether solvable or not. The proof uses Feit–Thompson Theorem to argue that if N is not solvable, then G/N must be solvable.)

Solution. We argue by induction on $|G|$. If $N = 1$, then any complement to N is equal to G, so there is nothing to prove. Suppose then that $N \neq 1$, and let H and K be complements to N in G. Since N is solvable, its derived series has a last non-trivial term, say $N^{(r)}$. Then $N^{(r)}$ is abelian. Since $N^{(r)}$ is characteristic in N and $N \trianglelefteq G$, we have $N^{(r)} \trianglelefteq G$. Choose a minimal non-trivial normal subgroup A of G with $A \leq N^{(r)}$. Then A is abelian, $A \trianglelefteq G$, and $A \leq N$.
In G/A, the subgroup N/A is a solvable normal Hall subgroup, and HA/A and KA/A are complements to N/A. By induction, these two complements are conjugate in G/A. Replacing H by a conjugate in G, we may assume that

$$HA = KA.$$

Put $E = HA = KA$. Then A is an abelian normal Hall subgroup of E, and H and K are complements to A in E. It remains to show that two complements to an abelian normal Hall subgroup are conjugate. Write A additively. Since both H and K map isomorphically onto E/A, for each $h \in H$ there is a unique element $c(h) \in A$ such that $c(h)h \in K$. Because K is a subgroup, we have

$$c(h_1h_2) = c(h_1) + {}^{h_1}c(h_2)$$

for all $h_1, h_2 \in H$. Put $b = \sum_{h \in H} c(h)$. For any $h_0 \in H$, we get

$${}^{h_0}b = \sum_{h \in H} {}^{h_0}c(h) = \sum_{h \in H} (c(h_0h) - c(h_0)) = b - |H|c(h_0).$$

Since A is a Hall subgroup of E, $\gcd(|A|, |H|) = 1$, so multiplication by $|H|$ is an automorphism of A. Choose $a \in A$ such that $|H|a = b$. Applying h_0 to this equality and using the displayed equation gives

$$|H|^{h_0}a = |H|a - |H|c(h_0),$$

whence ${}^{h_0}a = a - c(h_0)$. Thus $c(h_0) = a - {}^{h_0}a$, and so

$$ah_0a^{-1} = c(h_0)h_0 \in K$$

for every $h_0 \in H$. Therefore $aHa^{-1} = K$. Hence H and K are conjugate in G. □

Exercise 11.3. Complete the following sketch, which gives a proof of Theorem 11.8 that is independent of Schur–Zassenhaus Theorem. Here G is a finite solvable group of order mn, where m and n are coprime, and we are to show that G has a subgroup of order m.

(a) Using induction, reduce to the case where G has a unique minimal normal subgroup N of order n.

(b) Let M/N be a minimal normal subgroup of G/N; since G/N is solvable by Proposition 11.3, it follows from Proposition 11.6 that $|M/N|$ is a power of some prime divisor p of m. If P is a Sylow p-subgroup of M, show that $|N_G(P)| = m$.

Solution.

(a) We prove the theorem by induction on $|G|$ and show that all cases reduce to the stated one. Let A be a minimal normal subgroup of G. Since G is finite and solvable, Proposition 11.6 shows that A is elementary abelian. Hence $|A|$ divides either m or n.
If $|A|$ divides m, then
$$|G/A| = \frac{m}{|A|}n,$$
with coprime factors. By induction, G/A has a subgroup of order $m/|A|$. Its preimage in G has order m, and we are done.
Suppose next that $|A|$ divides n and $|A| < n$. Then
$$|G/A| = m\frac{n}{|A|}.$$
By induction, G/A has a subgroup K/A of order m. Thus $|K| = m|A|$. Since K is a subgroup of the solvable group G, Proposition 11.3 shows that K is solvable. Applying induction to K, whose order is smaller than $|G|$, gives a subgroup of K of order m.
Thus the only remaining case is the case in which every minimal normal subgroup of G has order n. In that case such a subgroup is unique. Indeed, if A and B were distinct minimal normal subgroups of order n, then $A \cap B = 1$, so $|AB| = n^2$ would divide $|G| = mn$, contradicting $\gcd(m, n) = 1$ unless $n = 1$. If $n = 1$, the result is trivial. Hence, in the non-trivial remaining case, G has a unique minimal normal subgroup N of order n.

(b) We now work in the reduced case from part (a). Let M/N be a minimal normal subgroup of G/N. Since G/N is solvable by Proposition 11.3, Proposition 11.6 shows that M/N is elementary abelian, say of order p^a, where p divides m. Let P be a Sylow p-subgroup of M. Since $|N| = n$ is coprime to p, we have $|P| = p^a = |M/N|$, and hence $M = NP$ with $N \cap P = 1$.
Since $M \trianglelefteq G$, the Frattini argument gives
$$G = MN_G(P).$$
We claim first that $N_M(P) = P$. Put
$$C = N \cap N_G(P).$$
Since N is the unique minimal normal subgroup of the solvable group G, it is elementary abelian, so N normalizes C. Also, $N_G(P)$ normalizes both N and $N_G(P)$, and hence normalizes C. Since $G = MN_G(P) = NPN_G(P)$, it follows that $C \trianglelefteq G$. If $C \neq 1$, then the minimality and uniqueness of N force $C = N$. Thus $N \leq N_G(P)$, and so $M = NP \leq N_G(P)$. Together with $G = MN_G(P)$, this would imply that $P \trianglelefteq G$. Then G would have a non-trivial minimal normal subgroup contained in P, contradicting the uniqueness of N, since p does not divide $|N|$. Therefore $C = 1$.
Now every element of $N_M(P)$ has the form np with $n \in N$ and $p \in P$. Since both np and p normalize P, the element n also normalizes P, so $n \in C = 1$. Hence $N_M(P) = P$. Using $G = MN_G(P)$, we obtain
$$|G| = \frac{|M||N_G(P)|}{|M \cap N_G(P)|} = \frac{|M||N_G(P)|}{|N_M(P)|} = \frac{n|P||N_G(P)|}{|P|} = n|N_G(P)|.$$
Since $|G| = mn$, it follows that $|N_G(P)| = m$. Therefore $N_G(P)$ is the required subgroup of order m.

□

Exercise 11.4. Prove Theorem 11.9. If your proof involves Schur–Zassenhaus Theorem, then construct a second proof that is independent of Schur-Zassenhaus by mimicking Exercise 11.3 above.

Solution. We give the Schur–Zassenhaus-free proof directly. We prove both assertions simultaneously by induction on $|G|$. If $G = 1$, there is nothing to prove. Let N be a minimal normal subgroup of G. Since G is finite and solvable, Proposition 11.6 shows that N is elementary abelian, say $|N| = p^r$. Since $\gcd(m, n) = 1$, either $|N|$ divides m or $|N|$ divides n.
First suppose that $|N|$ divides m. Let H and K be subgroups of G of order m. If P is a Sylow p-subgroup of H, then P is also a Sylow p-subgroup of G. Since N is a normal p-subgroup of G, we have $N \leq P \leq H$. Similarly, $N \leq K$. In the quotient G/N, the subgroups H/N and K/N have order $m/|N|$. By induction in G/N, they are conjugate. Hence H and K are conjugate in G.
Now let $L \leq G$ have order dividing m. Then LN has order dividing m: it is a subgroup of G, and all of its prime divisors divide m. Hence LN/N has order dividing $m/|N|$. By induction in G/N, there is a subgroup M/N of G/N of order $m/|N|$ containing LN/N. Then M has order m and contains L. This proves both assertions in the case $|N| \mid m$.
It remains to consider the case where $|N|$ divides n. Let H and K be subgroups of G of order m. Then $H \cap N = K \cap N = 1$, so HN/N and KN/N are subgroups of G/N of order m. By induction in G/N, they are conjugate. Replacing H by a conjugate in G, we may assume that

$$HN = KN.$$

Inside HN, the subgroup N is a solvable normal Hall subgroup, and H and K are complements to N. By Exercise 11.2, the complements H and K are conjugate in HN, and hence in G.
Finally let $L \leq G$ have order dividing m. Since $|N|$ divides n, we have $L \cap N = 1$, and so LN/N has order $|L|$, which divides m. By induction in G/N, there is a subgroup M/N of G/N of order m containing LN/N. Then $|M| = m|N|$, and M is solvable by Proposition 11.3. By the Schur–Zassenhaus-free proof of Theorem 11.8 from Exercise 11.3, the group M has a subgroup H of order m. Since $N \trianglelefteq M$ and $\gcd(|H|, |N|) = 1$, we have $H \cap N = 1$ and $HN = M$. Thus H is a complement to N in M.
Under the quotient isomorphism $H \to M/N$, let L_H be the subgroup of H corresponding to LN/N. Then $NL_H = NL$. Also, both L and L_H are complements to N in NL. Applying Exercise 11.2 inside NL, there is some $x \in NL$ such that

$$xL_Hx^{-1} = L.$$

Therefore $L \leq xHx^{-1}$, and xHx^{-1} is a subgroup of G of order m. This proves the containment assertion, and completes the proof of Theorem 11.9. □

Exercise 11.5. Let V be a finite-dimensional vector space over the field of p elements. Suppose that G is a solvable subgroup of $\mathrm{Aff}(V)$ which contains $\mathcal{T}(V)$ and that the stabilizer of the origin under the action of G on V does not leave any non-zero proper subspace of V invariant. Show that G is a primitive permutation group on V.

Solution. Let $H = G_0$ be the stabilizer of the origin. Since G contains the full translation group $\mathcal{T}(V)$, the action of G on V is transitive: for any $v \in V$, the translation T_v sends 0 to v. By Corollary 3.10, it remains to show that H is a maximal subgroup of G.
Let K be a subgroup of G such that $H < K \leq G$. We will show that $K = G$. Choose $g \in K - H$, and write $g(0) = v$. Then $v \neq 0$. Since $g \in \mathrm{Aff}(V)$, we may write $g = T_vS$ for some $S \in \mathrm{GL}(V)$. Moreover,

$$S = T_{-v}g \in G,$$

because $T_{-v} \in \mathcal{T}(V) \leq G$. Since S fixes the origin, we have $S \in H \leq K$. Hence $T_v = gS^{-1} \in K$.
Now define

$$W = \{w \in V \mid T_w \in K\}.$$

The subgroup $K \cap \mathcal{T}(V)$ of the translation group corresponds to the additive subgroup W of V, so W is a subspace of V over the field of p elements. Since $T_v \in K$ with $v \neq 0$, the subspace W is non-zero. Also, if $h \in H$ and $w \in W$, then

$$hT_wh^{-1} = T_{h(w)}$$

lies in K, because $h, T_w \in K$. Thus $h(w) \in W$, and so W is invariant under H. By hypothesis, H leaves no non-zero proper subspace of V invariant. Therefore $W = V$.
Hence K contains every translation T_w. Since every element of G has the form T_wh with $w \in V$ and $h \in H$, it follows that $K = G$. Thus there is no subgroup strictly between H and G, so H is maximal in G. By Corollary 3.10, the action of G on V is primitive. Therefore G is a primitive permutation group on V. □

Exercise 11.6. Show that a finite group is supersolvable iff it has a normal series having cyclic successive quotients. (An arbitrary group is called supersolvable if this latter condition holds.)

Solution. Suppose first that G is finite supersolvable. By Proposition 10.3, G has a chief series

$$G = G_0 > G_1 > \cdots > G_r = 1.$$

By definition of supersolvability, each chief factor G_i/G_{i+1} has prime order, and hence is cyclic. Thus this chief series is a normal series with cyclic successive quotients.
Conversely, suppose that G has a normal series

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1$$

whose successive quotients are cyclic. Removing repeated terms if necessary, we may assume the inclusions are strict. Since G is finite, each cyclic quotient G_i/G_{i+1} is finite. We refine each such quotient into cyclic groups of prime order. Indeed, if $|G_i/G_{i+1}| = p_1 \cdots p_s$, with primes repeated according to multiplicity, then Theorem 1.4 gives a chain of subgroups in G_i/G_{i+1} whose successive quotients have orders $p_1, \ldots, p_s$. These subgroups are characteristic in the cyclic group G_i/G_{i+1}, since they are the unique subgroups of their respective orders. Because $G_i/G_{i+1} \trianglelefteq G/G_{i+1}$, Exercise 2.4 and the Correspondence Theorem show that the corresponding intermediate subgroups are normal in G.
Refining every factor in this way gives a normal series of G whose successive quotients have prime order. No normal subgroup of G can be inserted properly between two consecutive terms of this refined series, because a group of prime order has no non-trivial proper subgroup. Hence the refined series is a chief series. Therefore G has a chief series whose factors all have prime order, and so G is supersolvable. □

Exercise 11.7. Show that finite nilpotent groups are supersolvable. (If we define supersolvability for arbitrary groups as in Exercise 11.6, is an infinite nilpotent group necessarily supersolvable?)

Solution. We first prove that finite nilpotent groups are supersolvable. By Exercise 11.6, it is enough to show that a finite nilpotent group has a normal series with cyclic successive quotients. We argue by induction on $|G|$. If $G = 1$, there is nothing to prove. Suppose $G \neq 1$, and let

$$G = G_0 \geq G_1 \geq \cdots \geq G_r = 1$$

be a central series for G. The last non-trivial term of this series is contained in $Z(G)$, so $Z(G) \neq 1$. Since G is finite, Cauchy's Theorem gives a subgroup $C \leq Z(G)$ of prime order. Then $C \trianglelefteq G$, and G/C is nilpotent, since the image of a central series for G in G/C is a central series after repeated terms are deleted.
By induction, G/C has a normal series with cyclic successive quotients. Lifting this series to G by the Correspondence Theorem and then appending $C > 1$ gives a normal series of G with cyclic successive quotients. Hence G is supersolvable by Exercise 11.6.
The corresponding statement for infinite nilpotent groups is false. The additive group $(\mathbb{Q}, +)$ is abelian, and hence nilpotent, but it is not supersolvable in the sense of Exercise 11.6. Indeed, any group with a finite normal series whose successive quotients are cyclic is finitely generated, by lifting one generator from each cyclic quotient. The group $(\mathbb{Q}, +)$ is not finitely generated: if $q_1, \ldots, q_m \in \mathbb{Q}$ and D is a common denominator for them, then $\langle q_1, \ldots, q_m \rangle \leq D^{-1}\mathbb{Z}$, so these elements cannot generate all of $\mathbb{Q}$. □

Exercise 11.8. (cont.) Give an example of a finite supersolvable group that is not nilpotent. (Hence we have a proper ascending chain of classes of finite groups, starting with cyclic groups, then abelian, then nilpotent, then supersolvable, and ending with solvable.)

Solution. The group S_3 is such an example. The series

$$S_3 > A_3 > 1$$

is a normal series with cyclic successive quotients, since $S_3/A_3 \cong \mathbb{Z}_2$ and $A_3 \cong \mathbb{Z}_3$. Thus S_3 is supersolvable by Exercise 11.6.
However, S_3 is not nilpotent. Its Sylow 2-subgroups are the three subgroups generated by the transpositions, so no Sylow 2-subgroup is normal in S_3. By Theorem 11.15, a finite nilpotent group has all Sylow subgroups normal. Therefore S_3 is supersolvable but not nilpotent. □

Further Exercises

Let G be a group. Let $\Gamma_1 = G$, and for $n \in \mathbb{N}$ let $\Gamma_{n+1} = [\Gamma_n, G] \leq G$. Let $\mathcal{Z}_0 = 1$, and for $n \in \mathbb{N}$ let $\mathcal{Z}_n$ be the unique subgroup of G such that $\mathcal{Z}_n/\mathcal{Z}_{n-1} = Z(G/\mathcal{Z}_{n-1})$. (Observe that $\Gamma_2 = G'$ and $\mathcal{Z}_1 = Z(G)$.) We call the series $G = \Gamma_1 \geq \Gamma_2 \geq \Gamma_3 \geq \dots$ the *lower central series* of G and the series $1 = \mathcal{Z}_0 \leq \mathcal{Z}_1 \leq \mathcal{Z}_2 \leq \dots$ the *upper central series* of G.

Exercise 11.9. Show that each Γ_n and $\mathcal{Z}_n$ is a characteristic subgroup of G.

Solution. We first consider the lower central series. Clearly $\Gamma_1 = G$ is characteristic in G. Suppose that Γ_n is characteristic in G, and let $\varphi \in \mathrm{Aut}(G)$. Then

$$\varphi(\Gamma_{n+1}) = \varphi([\Gamma_n, G]) = [\varphi(\Gamma_n), \varphi(G)] = [\Gamma_n, G] = \Gamma_{n+1}.$$

Hence Γ_{n+1} is characteristic in G, and induction proves that every Γ_n is characteristic.
We next consider the upper central series. Clearly $\mathcal{Z}_0 = 1$ is characteristic in G. Suppose that $\mathcal{Z}_{n-1}$ is characteristic in G, and let $\varphi \in \mathrm{Aut}(G)$. Then φ induces an automorphism

$$\overline{\varphi}\colon G/\mathcal{Z}_{n-1} \to G/\mathcal{Z}_{n-1}.$$

The center of a group is characteristic, so $\overline{\varphi}$ fixes

$$Z(G/\mathcal{Z}_{n-1}) = \mathcal{Z}_n/\mathcal{Z}_{n-1}.$$

Therefore φ fixes the full preimage of this subgroup in G, namely $\mathcal{Z}_n$. Hence $\mathcal{Z}_n$ is characteristic in G, and induction proves that every $\mathcal{Z}_n$ is characteristic. □

Exercise 11.10. (cont.) Show that G is nilpotent iff $\Gamma_r = 1$ for some r iff $\mathcal{Z}_s = G$ for some s.

Solution. We use the following translation of the centrality condition: if $K \trianglelefteq G$ and $K \leq H \leq G$, then

$$H/K \leq Z(G/K) \quad \text{iff} \quad [H, G] \leq K.$$

Indeed, hK commutes with every gK exactly when $[h, g] \in K$.
Suppose first that $\Gamma_r = 1$ for some r. Then

$$G = \Gamma_1 \geq \Gamma_2 \geq \dots \geq \Gamma_r = 1$$

is a normal series by Exercise 11.9. Since $\Gamma_{i+1} = [\Gamma_i, G]$, the observation above gives

$$\Gamma_i/\Gamma_{i+1} \leq Z(G/\Gamma_{i+1})$$

for every i. Thus this is a central series, and G is nilpotent. Conversely, suppose that G is nilpotent, and let

$$G = G_0 \geq G_1 \geq \dots \geq G_t = 1$$

be a central series. We claim that $\Gamma_{i+1} \leq G_i$ for $0 \leq i \leq t$. This is clear for $i = 0$. If $\Gamma_i \leq G_{i-1}$, then

$$\Gamma_{i+1} = [\Gamma_i, G] \leq [G_{i-1}, G] \leq G_i,$$

where the last inclusion follows from the centrality of G_{i-1}/G_i in G/G_i. Hence $\Gamma_{t+1} \leq G_t = 1$, so $\Gamma_{t+1} = 1$.
Now suppose that $\mathcal{Z}_s = G$ for some s. By Exercise 11.9, the series

$$G = \mathcal{Z}_s \geq \mathcal{Z}_{s-1} \geq \dots \geq \mathcal{Z}_0 = 1$$

is a normal series, and it is central by the defining relation

$$\mathcal{Z}_i/\mathcal{Z}_{i-1} = Z(G/\mathcal{Z}_{i-1}).$$

Hence G is nilpotent. Conversely, suppose again that

$$G = G_0 \geq G_1 \geq \dots \geq G_t = 1$$

is a central series. We claim that $G_{t-i} \leq \mathcal{Z}_i$ for $0 \leq i \leq t$. This is clear for $i = 0$, since $G_t = 1 = \mathcal{Z}_0$. If $G_{t-i} \leq \mathcal{Z}_i$, then centrality gives

$$[G_{t-i-1}, G] \leq G_{t-i} \leq \mathcal{Z}_i.$$

Therefore $G_{t-i-1}/\mathcal{Z}_i \leq Z(G/\mathcal{Z}_i)$, and so $G_{t-i-1} \leq \mathcal{Z}_{i+1}$. Taking $i = t$ gives $G = G_0 \leq \mathcal{Z}_t$, hence $\mathcal{Z}_t = G$. This proves the equivalence of all three conditions. □

Exercise 11.11. (cont.) Suppose that G is nilpotent. Show that the lower and upper central series of G have the same length, say c, and that no central series of G can have length less than c. (This number c is called the *nilpotency class* of G. The groups of nilpotency class 1 are exactly the abelian groups.)

Solution. Let

$$c_L = \min\{c \geq 0 \mid \Gamma_{c+1} = 1\}, \qquad c_U = \min\{c \geq 0 \mid \mathcal{Z}_c = G\}.$$

These integers exist by Exercise 11.10. The lower central series has length c_L, and the upper central series has length c_U.
Since

$$G = \mathcal{Z}_{c_U} \geq \mathcal{Z}_{c_U-1} \geq \cdots \geq \mathcal{Z}_0 = 1$$

is a central series, the argument in Exercise 11.10 gives $\Gamma_{c_U+1} = 1$. Hence $c_L \leq c_U$. Similarly,

$$G = \Gamma_1 \geq \Gamma_2 \geq \cdots \geq \Gamma_{c_L+1} = 1$$

is a central series, so the upper-series argument in Exercise 11.10 gives $\mathcal{Z}_{c_L} = G$. Hence $c_U \leq c_L$. Therefore $c_L = c_U$; call this common value c.
Finally, let

$$G = G_0 \geq G_1 \geq \cdots \geq G_t = 1$$

be any central series of G. The lower-series argument from Exercise 11.10 gives $\Gamma_{t+1} = 1$, so by the minimality of $c = c_L$ we have $c \leq t$. Thus no central series of G can have length less than c. For a non-trivial group, $c = 1$ iff $\Gamma_2 = G' = 1$, which is equivalent to G being abelian. $\square$

Chapter 5

Semisimple Algebras

12 Modules and Representations

From Group Actions to Linear Actions

- In Section 3 we studied actions of groups on sets; if a group acts on a set carrying additional algebraic structure, it is the structure-respecting actions that are most interesting.
- Here we focus on actions of groups on vector spaces that respect the vector space structure.
- Let F be a field and let G be a group acting on an F-vector space V. The action is *linear* if
  - $g(v + w) = gv + gw$ for all $g \in G$ and $v, w \in V$;
  - $g(\alpha v) = \alpha(gv)$ for all $g \in G$, $\alpha \in F$, and $v \in V$.
- Proposition 3.1 gave the bijection between group actions on a set and homomorphisms to symmetric groups; we now deduce its analogue for linear actions.

Proposition 12.1. *There is a bijective correspondence between the set of linear actions of a group G on an F-vector space V and the set of homomorphisms from G to* $\mathrm{GL}(V)$.

Proof. Suppose first that $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism. Define an action of G on V by setting $gv = \rho(g)(v)$ for $g \in G$ and $v \in V$. Since $\rho(1)$ is the identity map on V, we have $1v = v$, and since $\rho(g_1g_2) = \rho(g_1)\rho(g_2)$, we have

$$(g_1g_2)v = \rho(g_1g_2)(v) = \rho(g_1)(\rho(g_2)(v)) = g_1(g_2v).$$

Thus this is an action of G on V. Moreover, the action is linear because each $\rho(g)$ is an element of $\mathrm{GL}(V)$.
Conversely, suppose that G acts linearly on V. For each $g \in G$, define $\rho(g)\colon V \to V$ by $\rho(g)(v) = gv$. The linearity of the action says precisely that $\rho(g)$ is an F-linear map, and $\rho(g^{-1})$ is its inverse, so $\rho(g) \in \mathrm{GL}(V)$. The action law gives

$$\rho(g_1g_2)(v) = (g_1g_2)v = g_1(g_2v) = \rho(g_1)(\rho(g_2)(v))$$

for every $v \in V$, whence $\rho(g_1g_2) = \rho(g_1)\rho(g_2)$. Therefore $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism. These two constructions clearly undo one another, so the correspondence is bijective. □

Linear Representations and Modules

- A homomorphism $\rho\colon G \to \mathrm{GL}(V)$, where G is a group and V a vector space, is called a *linear representation* of G in V.
- By Proposition 12.1, the study of linear representations is equivalent to the study of linear actions; with emphasis on finite groups and finite-dimensional vector spaces, this area originated in the late nineteenth century.

- The modern approach to the representation theory of finite groups involves yet another equivalent concept, that of finitely generated modules over group algebras. It is therefore necessary at this juncture to review some elementary module theory.
- As in Section 1, we will omit most proofs on the assumption that the reader will have seen this material previously; however, our account requires only a mild knowledge of rings, fields, and vector spaces.
- Let R be a ring with unit, and let M be an abelian group written additively. Then M is a *left R-module* if there is a map $R \times M \to M$, $(r, m) \mapsto rm$, satisfying:
  - $1m = m$ for all $m \in M$;
  - $r(m + n) = rm + rn$ for all $r \in R$ and $m, n \in M$;
  - $(r + s)m = rm + sm$ for all $r, s \in R$ and $m \in M$;
  - $r(sm) = (rs)m$ for all $r, s \in R$ and $m \in M$.
- If $R = \mathbb{Z}$, then $\mathbb{Z}$-modules are exactly abelian groups; if F is a field, then F-modules are precisely F-vector spaces. A module is the natural generalisation of a vector space to an arbitrary ring.
- One similarly defines right R-modules; if R is commutative, every left module is canonically a right module. In general, a left R-module is a right module over the *opposite ring* R^{op}.
- Let S be another ring with unit. An abelian group M that is both a left R-module and a right S-module is an *(R, S)-bimodule* provided $r(ms) = (rm)s$ for all $r \in R$, $m \in M$, $s \in S$. Any left R-module is an $(R, \mathbb{Z})$-bimodule, and R itself is an (R, R)-bimodule.
- An R-module M is *finitely generated* if every element is an R-linear combination of elements from some finite subset of M. Unlike vector spaces, minimal generating sets need not have a common size.
- If N is a subgroup of M closed under the R-action, N is an *R-submodule*. The submodules of the regular module R are exactly the left ideals. A nonzero module with no submodules other than 0 and itself is *simple*.
- Quotient modules M/N are defined by $r(m + N) = rm + N$.
- Sums, intersections, internal and external direct sums of submodules are defined as for groups; N is a *direct summand* of M if $M = N \oplus N'$ for some submodule N'. We write nM or M^n for $\underbrace{M \oplus \cdots \oplus M}_{n}$.
- A *composition series* of M is a descending series of submodules ending at 0 in which each successive quotient is simple. Modules need not have one (any infinite $\mathbb{Z}$-module is an example), but when they do, Jordan–Hölder Theorem holds.
- Every composition factor of any submodule or quotient module of an R-module M must also be a composition factor of M, as any composition series of a submodule (resp., quotient module) can be extended (resp., lifted) to give a composition series of M.
- An *R-module homomorphism* $\varphi\colon M \to N$ is a group homomorphism with $\varphi(rm) = r\varphi(m)$ for all $r \in R$, $m \in M$. Mono-, epi-, iso-, endo-, automorphisms are defined as for groups; the kernel is a submodule, the image is a submodule, and Fundamental Theorem on Homomorphisms, Correspondence Theorem, and the two isomorphism theorems (First Isomorphism Theorem and Second Isomorphism Theorem) all carry over.

Lemma (Schur's Lemma)**.** *Any non-zero homomorphism between simple R-modules is an isomorphism.*

Proof. Let M and N be simple R-modules, and let $\varphi\colon M \to N$ be a non-zero R-module homomorphism. Since $\ker\varphi$ is a submodule of M, simplicity of M gives either $\ker\varphi = M$ or $\ker\varphi = 0$. The first case would force $\varphi = 0$, contrary to assumption, so $\ker\varphi = 0$. Thus φ is injective. Similarly, $\operatorname{im}\varphi$ is a submodule of N, so simplicity of N gives either $\operatorname{im}\varphi = 0$ or $\operatorname{im}\varphi = N$. Since φ is non-zero, its image is non-zero, and hence $\operatorname{im}\varphi = N$. Thus φ is surjective. Therefore φ is an isomorphism. □

Hom-Modules, Balanced Maps, and Tensor Products

- Given R-modules M and N, we write $\mathrm{Hom}_R(M,N)$ for the set of R-module homomorphisms $M \to N$, and $\mathrm{End}_R(M)$ for $\mathrm{Hom}_R(M,M)$. Pointwise addition makes $\mathrm{Hom}_R(M,N)$ an abelian group.

- If S is another ring, M an (R,S)-bimodule, and N an R-module, then $\mathrm{Hom}_R(M,N)$ is an S-module via $(s\varphi)(m) = \varphi(ms)$. In particular, when F is a field and U, V are F-vector spaces, $\mathrm{Hom}_F(U,V)$ is an F-vector space.

- The map $\mathrm{Hom}_R(R,M) \to M$ sending φ to $\varphi(1)$ is an R-module isomorphism.

- Let M be an (R,S)-bimodule and N a left S-module. A map $f\colon M \times N \to U$ to an R-module U is *balanced* if

  - $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$;
  - $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$;
  - $f(ms, n) = f(m, sn)$;
  - $f(rm, n) = rf(m, n)$.

- The *tensor product* $M \otimes_S N$ is the R-module equipped with a balanced map $\eta\colon M \times N \to M \otimes_S N$ such that any balanced $f\colon M \times N \to U$ factors uniquely through η: there is a unique R-module homomorphism $\alpha\colon M \otimes_S N \to U$ with $f = \alpha \circ \eta$. We write $m \otimes n = \eta(m,n)$.

- Concretely, $M \otimes_S N$ is generated by the symbols $\{m \otimes n \mid m \in M, n \in N\}$ subject to the relations

  - $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$;
  - $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$;
  - $(ms) \otimes n = m \otimes (sn)$;
  - $(rm) \otimes n = r(m \otimes n)$.

  A general element of $M \otimes_S N$ is a *sum* of elementary tensors, not just one.

- Special case: if F is a field and U, V are finite-dimensional with bases $\{u_1, \ldots, u_r\}$ and $\{v_1, \ldots, v_s\}$, then $U \otimes_F V$ has basis $\{u_i \otimes v_j\}$ and dimension rs. For $u = \sum_i a_i u_i$ and $v = \sum_j b_j v_j$, $u \otimes v = \sum_{i,j} a_i b_j (u_i \otimes v_j)$.

- (An understanding of this special case will, strictly speaking, suffice in reading the remainder of the book; however, an understanding of the general case will be beneficial in Section 16.)

Proposition 12.2. *Let R be a ring with unit, and let M be an R-module. Then M and $R \otimes_R M$ are isomorphic R-modules.*

Proof. Define $f\colon R \times M \to M$ by $f(r,m) = rm$. This map is balanced, so it induces an R-module homomorphism
$$\alpha\colon R \otimes_R M \to M, \qquad \alpha(r \otimes m) = rm.$$
We define a second R-module homomorphism
$$\beta\colon M \to R \otimes_R M, \qquad \beta(m) = 1 \otimes m.$$
This is R-linear because $\beta(rm) = 1 \otimes rm = r \otimes m = r(1 \otimes m) = r\beta(m)$. Now $\alpha\beta(m) = \alpha(1 \otimes m) = m$ for every $m \in M$, while
$$\beta\alpha(r \otimes m) = \beta(rm) = 1 \otimes rm = r \otimes m$$
for every elementary tensor $r \otimes m$. Since elementary tensors generate $R \otimes_R M$, the maps α and β are inverse isomorphisms. $\square$

Proposition 12.3. *Let R and S be rings with unit, let $M_1, \ldots, M_r$ be (R,S)-bimodules, and let N be an S-module. Then $(\bigoplus_i M_i) \otimes_S N$ and $\bigoplus_i M_i \otimes_S N$ are isomorphic R-modules.*

Proof. Define

$$f\colon \Big(\bigoplus_i M_i\Big) \times N \to \bigoplus_i (M_i \otimes_S N)$$

by

$$f((m_1, \ldots, m_r), n) = (m_1 \otimes n, \ldots, m_r \otimes n).$$

This map is balanced, so it induces an R-module homomorphism

$$\alpha\colon \Big(\bigoplus_i M_i\Big) \otimes_S N \to \bigoplus_i (M_i \otimes_S N).$$

Conversely, for each i, let $\iota_i\colon M_i \to \bigoplus_i M_i$ be the natural inclusion. The map $M_i \times N \to (\bigoplus_i M_i) \otimes_S N$ given by $(m, n) \mapsto \iota_i(m) \otimes n$ is balanced, and hence induces an R-module homomorphism

$$\beta_i\colon M_i \otimes_S N \to \Big(\bigoplus_i M_i\Big) \otimes_S N.$$

Combining these maps gives an R-module homomorphism

$$\beta\colon \bigoplus_i (M_i \otimes_S N) \to \Big(\bigoplus_i M_i\Big) \otimes_S N, \qquad \beta(z_1, \ldots, z_r) = \sum_i \beta_i(z_i).$$

On elementary tensors, $\alpha\beta$ sends an element $m \otimes n$ in the ith summand to itself in the ith summand, so $\alpha\beta$ is the identity. Also,

$$\beta\alpha((m_1, \ldots, m_r) \otimes n) = \sum_i \iota_i(m_i) \otimes n = (m_1, \ldots, m_r) \otimes n.$$

Since elementary tensors generate $(\bigoplus_i M_i) \otimes_S N$, $\beta\alpha$ is also the identity. Thus α and β are inverse isomorphisms. □

Proposition 12.4. *Let F be a field, and let U and V be F-vector spaces, with $\dim_F(U) < \infty$. Let $U^* = \mathrm{Hom}_F(U, F)$. Then the map $\Gamma\colon U^* \otimes_F V \to \mathrm{Hom}_F(U, V)$ defined by setting $\Gamma(\varphi \otimes v)(u) = \varphi(u)v$ and extending linearly is an isomorphism of F-vector spaces.*

Proof. Let $\{u_1, \ldots, u_n\}$ be a basis for U, and let $\{\varepsilon_1, \ldots, \varepsilon_n\}$ be the dual basis for U^*, so that $\varepsilon_i(u_j) = \delta_{ij}$. By the preceding description of tensor products over a field, every element of $U^* \otimes_F V$ can be written uniquely in the form $\sum_i \varepsilon_i \otimes v_i$ with $v_i \in V$. If

$$\Gamma\Big(\sum_i \varepsilon_i \otimes v_i\Big) = 0,$$

then evaluating at u_j gives

$$0 = \Gamma\Big(\sum_i \varepsilon_i \otimes v_i\Big)(u_j) = \sum_i \varepsilon_i(u_j) v_i = v_j.$$

Hence all v_j are zero, so Γ is injective. To show surjectivity, let $T \in \mathrm{Hom}_F(U, V)$. We claim that

$$\Gamma\Big(\sum_i \varepsilon_i \otimes T(u_i)\Big) = T.$$

Indeed, if $u = \sum_i a_i u_i$, then

$$\Gamma\Big(\sum_i \varepsilon_i \otimes T(u_i)\Big)(u) = \sum_i \varepsilon_i(u) T(u_i) = \sum_i a_i T(u_i) = T(u).$$

Thus Γ is surjective, and therefore an isomorphism. □

Group Rings and Group Algebras

- Let R be a ring and G a group. The *group ring* of G over R, denoted RG, consists of all finite formal R-linear combinations of elements of G, with the obvious addition and with multiplication extending that in G:

$$\Big(\sum_{x\in G}\alpha_x x\Big)\Big(\sum_{y\in G}\beta_y y\Big)=\sum_{x\in G}\sum_{y\in G}\alpha_x\beta_y\,xy=\sum_{x\in G}\Big(\sum_{g\in G}\alpha_g\beta_{g^{-1}x}\Big)x.$$

- The unit element is the identity of G.
- When $R=F$ is a field and G is finite, FG is also an F-vector space with basis G, hence has finite dimension $|G|$. In this case FG is called the *group algebra*.
- An *algebra* over F (or *F-algebra*) is a set A carrying both a ring structure and an F-vector space structure that share the same addition, with $(\lambda a)b=a(\lambda b)=\lambda(ab)$ for $\lambda\in F$, $a,b\in A$. We do *not* assume an algebra has a ring unit.
- An algebra is *finite-dimensional* if it has finite dimension as an F-vector space; the matrix ring $\mathcal{M}_n(F)$ is a finite-dimensional F-algebra, and FG is finite-dimensional whenever G is finite.
- A homomorphism of F-algebras is a ring homomorphism that is also F-linear.

Lemma 12.5. *If F is a field and G a finite group, then an FG-module is finitely generated iff it has finite dimension as an F-vector space.*

Proof. Suppose first that V is generated as an FG-module by $v_1,\ldots,v_t$. Every element of V is then a sum of elements of the form $\alpha_i v_i$ with $\alpha_i\in FG$. Writing $\alpha_i=\sum_{g\in G}a_g g$, we have

$$\alpha_i v_i=\sum_{g\in G}a_g(gv_i).$$

Hence V is generated as an F-vector space by the finite set $\{gv_i\mid g\in G,\ 1\le i\le t\}$, so $\dim_F(V)<\infty$. Conversely, if V is finite-dimensional over F, then any finite F-basis for V also generates V as an FG-module, since F acts through the scalar elements $\lambda 1\in FG$. Thus V is finitely generated as an FG-module. $\square$

Proposition 12.6. *If F is a field and G a finite group, then there is a bijective correspondence between finitely generated FG-modules and linear actions of G on finite-dimensional F-vector spaces.*

Proof. Let V be a finitely generated FG-module. By Lemma 12.5, V is finite-dimensional as an F-vector space. Restricting the module structure map $FG\times V\to V$ to $G\times V$ gives an action of G on V, and this action is linear because, for each $g\in G$, multiplication by g is F-linear on V. Conversely, suppose that V is a finite-dimensional F-vector space on which G acts linearly. Define an action of FG on V by

$$\Big(\sum_{g\in G}a_g g\Big)v=\sum_{g\in G}a_g(gv).$$

The linearity of the G-action and the group action law show that this makes V an FG-module. These two constructions undo one another: restricting an FG-module to the basis elements $g\in G$ recovers the corresponding linear action, and extending a linear action F-linearly recovers the corresponding FG-module structure. Therefore the correspondence is bijective. $\square$

Examples and Constructions of FG-Modules

- In what follows G is a finite group, F a field, and all F-vector spaces are finite-dimensional; all FG-modules will be finitely generated and hence finite-dimensional by Lemma 12.5.
- It is enough to specify the action of G on V; the FG-action then extends linearly.
- *Trivial module*: F regarded as an FG-module via $g\lambda=\lambda$.
- *Permutation module*: if G acts on a finite set $X=\{x_1,\ldots,x_n\}$, the F-vector space FX with basis X is an FG-module via $g\big(\sum_i c_i x_i\big)=\sum_i c_i(gx_i)$.

- Direct sums of FG-modules: $g(u, v) = (gu, gv)$.
- *Tensor product* $U \otimes_F V$ with the diagonal action $g(u \otimes v) = gu \otimes gv$.
- $\mathrm{Hom}_F(U, V)$ with action $(g\varphi)(u) = g(\varphi(g^{-1}u))$; one checks $(g_1g_2)\varphi = g_1(g_2\varphi)$.
- Taking $V = F$ trivial gives the *dual module* $U^* = \mathrm{Hom}_F(U, F)$ with $(g\varphi)(u) = \varphi(g^{-1}u)$.

Proposition 12.7. *Let U and V be FG-modules. Then $U^* \otimes_F V$ and $\mathrm{Hom}_F(U, V)$ are isomorphic FG-modules.*

Proof. By Proposition 12.4, the map

$$\Gamma\colon U^* \otimes_F V \to \mathrm{Hom}_F(U, V), \qquad \Gamma(\varphi \otimes v)(u) = \varphi(u)v$$

is an isomorphism of F-vector spaces. It remains only to check that Γ respects the G-action. For $g \in G$, $\varphi \in U^*$, $v \in V$, and $u \in U$, we have

$$\Gamma(g(\varphi \otimes v))(u) = \Gamma((g\varphi) \otimes gv)(u) = (g\varphi)(u)gv = \varphi(g^{-1}u)gv.$$

On the other hand,

$$(g\Gamma(\varphi \otimes v))(u) = g\big(\Gamma(\varphi \otimes v)(g^{-1}u)\big) = g\big(\varphi(g^{-1}u)v\big) = \varphi(g^{-1}u)gv.$$

Thus $\Gamma(g(\varphi \otimes v)) = g\Gamma(\varphi \otimes v)$ on elementary tensors, and hence by linearity on all of $U^* \otimes_F V$. Therefore Γ is an FG-module isomorphism. □

Maschke's Theorem and Semisimplicity

- We now reach the most basic result of the representation theory of finite groups, discovered in 1898 by Heinrich Maschke at the University of Chicago.

Theorem (Maschke's Theorem)**.** *Let G be a group, and suppose that the characteristic of F is either zero or coprime to $|G|$. If U is an FG-module and V is an FG-submodule of U, then V is a direct summand of U as FG-modules.*

Proof. Since V is an F-subspace of U, choose an F-subspace W of U such that $U = V \oplus W$ as F-vector spaces. Let $\pi\colon U \to V$ be the projection onto V along W. This map is F-linear and restricts to the identity on V, but it need not be an FG-module homomorphism. We average it over G by defining

$$\pi'(u) = \frac{1}{|G|} \sum_{g \in G} g\pi(g^{-1}u)$$

for $u \in U$. The scalar $1/|G|$ exists in F because the characteristic of F does not divide $|G|$. Since V is an FG-submodule of U, each term $g\pi(g^{-1}u)$ lies in V, so π' maps U into V. If $v \in V$, then $g^{-1}v \in V$ and $\pi(g^{-1}v) = g^{-1}v$, so

$$\pi'(v) = \frac{1}{|G|} \sum_{g \in G} gg^{-1}v = v.$$

Thus π' is still a projection onto V. We now show that π' is an FG-module homomorphism. It is enough to check the action of elements $x \in G$, since the FG-action is obtained by F-linear extension. For $x \in G$ and $u \in U$,

$$\pi'(xu) = \frac{1}{|G|} \sum_{g \in G} g\pi(g^{-1}xu).$$

Put $g = xy$. As g runs through G, so does y, and hence

$$\pi'(xu) = \frac{1}{|G|} \sum_{y \in G} xy\pi(y^{-1}u) = x\pi'(u).$$

Therefore π' is an FG-module homomorphism. Its kernel is then an FG-submodule of U, and because π' is a projection onto V, we have

$$U = V \oplus \ker \pi'.$$

Thus V is a direct summand of U as an FG-module. □

- We summarise the hypothesis by saying that the characteristic of F *does not divide* $|G|$, even though this is a slight abuse.
- A module is *semisimple* if it is a direct sum of simple modules.

Corollary 12.8. *Let G be a group, and let F be a field whose characteristic does not divide $|G|$. Then every non-zero finitely generated FG-module is semisimple.*

Proof. Let U be a non-zero finitely generated FG-module. By Lemma 12.5, U is finite-dimensional over F, so we argue by induction on $\dim_F(U)$. If U is simple, then U is already semisimple. Otherwise, U has a non-zero proper FG-submodule V. By Maschke's Theorem, there is an FG-submodule W of U such that

$$U = V \oplus W.$$

Both V and W have smaller F-dimension than U, so by induction they are semisimple. Their direct sum is therefore semisimple, and hence U is semisimple. □

- In Chapter 6 we shall be concerned only with the case $F = \mathbb{C}$, called *ordinary* representation theory; since $\mathbb{C}$ has characteristic zero, by Corollary 12.8 every non-zero finitely generated $\mathbb{C}G$-module is semisimple.
- The next section concentrates on algebras with this property. The case where the characteristic of F divides $|G|$, in which FG-modules need not be semisimple, is the subject of *modular* representation theory (originated by Dickson in 1907; mostly developed by Brauer from the 1930s onward).

Exercises

Throughout these exercises, let F be a field and let G be a finite group.

Exercise 12.1. Let U and V be FG-modules having the same dimension n, and let $\rho\colon G \to \mathrm{GL}(U)$ and $\tau\colon G \to \mathrm{GL}(V)$ be the corresponding representations. By fixing F-bases for U and V, we consider ρ and τ as homomorphisms from G to $\mathrm{GL}(n, F)$. Show that $U \cong V$ iff there exists some $M \in \mathrm{GL}(n, F)$ such that $\rho(g)M = M\tau(g)$ for every $g \in G$. (Two representations are said to be *equivalent* if this latter condition holds; hence two representations are equivalent iff their corresponding modules over the group algebra are isomorphic.)

Solution. Fix the chosen F-bases for U and V, and write vectors in coordinates with respect to these bases.
($\Rightarrow$) Suppose that $U \cong V$ as FG-modules. Then there is an FG-module isomorphism $T\colon V \to U$. Let M be the matrix of T with respect to the chosen bases. Since T is an F-vector space isomorphism, $M \in \mathrm{GL}(n, F)$. For every $v \in V$ and $g \in G$, the FG-linearity of T gives

$$T(gv) = gT(v).$$

In coordinates, the left hand side is represented by $M\tau(g)v$, while the right hand side is represented by $\rho(g)Mv$. Hence $M\tau(g)v = \rho(g)Mv$ for every coordinate vector v, and therefore $\rho(g)M = M\tau(g)$ for every $g \in G$.
($\Leftarrow$) Conversely, suppose that there exists $M \in \mathrm{GL}(n, F)$ such that $\rho(g)M = M\tau(g)$ for every $g \in G$. Let $T\colon V \to U$ be the F-linear map represented by M. Since M is invertible, T is an F-vector space isomorphism. For $v \in V$ and $g \in G$, the given matrix identity says exactly that

$$gT(v) = T(gv).$$

Thus T commutes with the action of every $g \in G$. Since the FG-action is obtained by extending the G-action F-linearly, T is an FG-module homomorphism. Therefore T is an FG-module isomorphism, and so $U \cong V$. □

Exercise 12.2. Let $\sigma = \sum_{g \in G} g \in FG$. Show that the subspace $F\sigma$ of FG is the unique submodule of FG that is isomorphic with the trivial module.

Solution. For any $x \in G$, left multiplication gives

$$x\sigma = \sum_{g\in G} xg = \sum_{h\in G} h = \sigma,$$

since $g \mapsto xg$ is a permutation of G. Thus every element of G acts trivially on $F\sigma$, so $F\sigma$ is a submodule isomorphic with the trivial module.
Conversely, let W be a submodule of FG isomorphic with the trivial module. Choose a non-zero element

$$u = \sum_{g\in G} a_g g \in W.$$

Since G acts trivially on W, we have $xu = u$ for every $x \in G$. Comparing the coefficient of $h \in G$ in this equality gives

$$a_{x^{-1}h} = a_h.$$

Given any $g, h \in G$, choose $x = hg^{-1}$. Then $x^{-1}h = g$, so $a_g = a_h$. Hence all coefficients of u are equal, say $a_g = c$ for every $g \in G$. Since $u \neq 0$, we have $c \neq 0$, and therefore $u = c\sigma$. Thus $W = Fu = F\sigma$, proving uniqueness. □

Exercise 12.3. (cont.) Let $\epsilon\colon FG \to F$ be the FG-module epimorphism defined by $\epsilon(g) = 1$ for all $g \in G$, and let $\Delta = \ker \epsilon$. (We call Δ the *augmentation ideal* of FG.) Show that Δ is the unique submodule of FG whose quotient is isomorphic with the trivial module.

Solution. Write

$$\epsilon\Big(\sum_{g\in G} a_g g\Big) = \sum_{g\in G} a_g.$$

This is an FG-module homomorphism: it is F-linear, and for $x \in G$,

$$\epsilon\Big(x\sum_{g\in G} a_g g\Big) = \epsilon\Big(\sum_{g\in G} a_g xg\Big) = \sum_{g\in G} a_g = x\epsilon\Big(\sum_{g\in G} a_g g\Big),$$

where the last equality uses the trivial action of G on F. Since $\epsilon(1) = 1$, the map is onto, and by Fundamental Theorem on Homomorphisms we have

$$FG/\Delta \cong F.$$

Thus Δ is a submodule whose quotient is isomorphic with the trivial module.
Conversely, suppose that N is a submodule of FG and that $FG/N \cong F$ as FG-modules. Compose the quotient map $FG \to FG/N$ with such an isomorphism to obtain an FG-module epimorphism $q\colon FG \to F$ with kernel N. Put $c = q(1)$. For every $g \in G$, we have

$$q(g) = q(g \cdot 1) = gq(1) = c,$$

again because F is the trivial module. Hence, for $\sum_{g\in G} a_g g \in FG$,

$$q\Big(\sum_{g\in G} a_g g\Big) = \sum_{g\in G} a_g c = c\epsilon\Big(\sum_{g\in G} a_g g\Big).$$

Since q is onto, $c \neq 0$. Therefore $\ker q = \ker \epsilon = \Delta$, so $N = \Delta$. This proves the desired uniqueness. □

Exercise 12.4. (cont.) Suppose that the characteristic of F divides $|G|$. Show that $F\sigma \subseteq \Delta$. Conclude that Δ is not a direct summand of FG and hence that the FG-module FG is not semisimple. (This shows that the converse of Corollary 12.8 is true.)

Solution. Since $\operatorname{char}(F)$ divides $|G|$, we have $|G| \cdot 1_F = 0$ in F. Therefore

$$\epsilon(\sigma) = \epsilon\Big(\sum_{g\in G} g\Big) = \sum_{g\in G} 1 = |G| \cdot 1_F = 0.$$

Hence $\sigma \in \Delta$, and so $F\sigma \subseteq \Delta$.
We now show that Δ cannot be a direct summand of FG. Suppose, for contradiction, that

$$FG = \Delta \oplus W$$

for some FG-submodule W. Then
$$W \cong FG/\Delta \cong F,$$
so W is a submodule isomorphic with the trivial module. By Exercise 12.2, we must have $W = F\sigma$. But $F\sigma \subseteq \Delta$, so
$$W \cap \Delta = F\sigma \neq 0,$$
contradicting the directness of the sum. Thus Δ is not a direct summand of FG. Since every submodule of a semisimple module is a direct summand, it follows that FG is not semisimple. □

Further Exercises

In Section 9 we defined the second cohomology group of a pair (H, A), where H is a group and A an abelian group, with respect to a given homomorphism from H to $\mathrm{Aut}(A)$. It is easy to verify that, for an abelian group A, specifying a homomorphism from a group H to $\mathrm{Aut}(A)$ is exactly the same as specifying a $\mathbb{Z}H$-module structure on A. Therefore, it is customary to talk about the second cohomology group of a pair (H, A) where H is a group and A a $\mathbb{Z}H$-module. (We anticipated this development by writing xa instead of $\varphi(x)(a)$ on page 86.) We will now generalize the second cohomology group.

Let H be a group, let H^n be the set of n-tuples of elements of H for some $n \in \mathbb{N}$, and let A be a $\mathbb{Z}H$-module. A *normalized n-cochain* of (H, A) is a set function $f\colon H^n \to A$ such that $f(h_1, \ldots, h_n) = 0$ whenever $h_i = 1$ for some i; the set of normalized n-cochains of (H, A) is denoted $C^n(H, A)$. We give $C^n(H, A)$ an abelian group structure by defining
$$(f + g)(h_1, \ldots, h_n) = f(h_1, \ldots, h_n) + g(h_1, \ldots, h_n).$$
We define a homomorphism $\delta^n\colon C^{n-1}(H, A) \to C^n(H, A)$ by
$$(\delta^n f)(h_1, \ldots, h_n) = h_1 f(h_2, \ldots, h_n) + \sum_{i=1}^{n-1} (-1)^i f(h_1, \ldots, h_i h_{i+1}, \ldots, h_n) + (-1)^n f(h_1, \ldots, h_{n-1}).$$
The image of δ^n is called the set of *n-coboundaries* and is denoted $B^n(H, A)$; the kernel of δ^{n+1} is called the set of *n-cocycles* and is denoted $Z^n(H, A)$. For example, $\ker \delta^3$ consists of all normalized 2-cochains f satisfying $h_1 f(h_2, h_3) - f(h_1 h_2, h_3) + f(h_1, h_2 h_3) - f(h_1, h_2) = 0$ for all $h_1, h_2, h_3 \in H$, which agrees with our definition of a 2-cocycle in Section 9. Both $B^n(H, A)$ and $Z^n(H, A)$ are abelian subgroups of $C^n(H, A)$. We find that $\delta^{n+1} \circ \delta^n$ is the zero map, or equivalently that $B^n(H, A) \subseteq Z^n(H, A)$. (Verify this.) The quotient group $Z^n(H, A)/B^n(H, A)$ is denoted $H^n(H, A)$ and is called the *nth cohomology group* of (H, A).

While $H^3(H, A)$ plays some role in the theory of extensions of a non-abelian group N by H, where $Z(N) = A$ (see [20, pp. 124–131]), the groups $H^n(H, A)$ for $n > 3$ have no known group-theoretic meaning. Nonetheless, higher cohomology groups are studied for their own sake, and in recent years connections between the cohomology of groups and the representation theory of groups have attracted much attention. (As the title of one of the first author's papers [6] proclaims, "Cohomology *is* representation theory.") We developed the group-theoretic meaning of $H^2(H, A)$ in the further exercises to Section 9, and in what follows we develop the meaning of $H^1(H, A)$.

Exercise 12.5. (cont.) Define a homomorphism $\varphi\colon H \to \mathrm{Aut}(A)$ by $\varphi(x)(a) = xa$. Show that there is a bijective correspondence between the set of splitting maps of $A \rtimes_\varphi H$ and $Z^1(H, A)$. (Recall that a homomorphism $t\colon H \to A \rtimes_\varphi H$ is called a splitting map if $\eta \circ t$ is the identity map on H, where $\eta\colon A \rtimes_\varphi H \to H$ is the natural map. The natural inclusion of H into $A \rtimes_\varphi H$ is a splitting map, but there may be others.)

Solution. Since A is a $\mathbb{Z}H$-module, each $x \in H$ acts on A by an automorphism of the underlying abelian group, and the action law gives $\varphi(xy) = \varphi(x)\varphi(y)$. Thus $\varphi\colon H \to \mathrm{Aut}(A)$ is a homomorphism. We use additive notation for A, so multiplication in $A \rtimes_\varphi H$ is given by
$$(a, x)(b, y) = (a + xb, xy).$$
Let $t\colon H \to A \rtimes_\varphi H$ be a splitting map. Since $\eta \circ t$ is the identity on H, each $t(x)$ has the form
$$t(x) = (f(x), x)$$

for a unique function $f\colon H \to A$. Since t is a homomorphism, $t(1) = (0,1)$, so $f(1) = 0$. Also, for $x, y \in H$,

$$(f(xy), xy) = t(xy) = t(x)t(y) = (f(x), x)(f(y), y) = (f(x) + xf(y), xy).$$

Hence

$$f(xy) = f(x) + xf(y).$$

This is precisely the condition $\delta^2 f = 0$, so $f \in Z^1(H, A)$.
Conversely, suppose that $f \in Z^1(H, A)$. Define $t_f\colon H \to A \rtimes_\varphi H$ by

$$t_f(x) = (f(x), x).$$

The normalization of f gives $t_f(1) = (0,1)$, and the cocycle identity gives

$$t_f(x)t_f(y) = (f(x) + xf(y), xy) = (f(xy), xy) = t_f(xy).$$

Therefore t_f is a homomorphism. Moreover, $\eta(t_f(x)) = x$, so t_f is a splitting map. These two constructions are inverse to one another, and hence the splitting maps of $A \rtimes_\varphi H$ are in bijective correspondence with $Z^1(H, A)$. □

Exercise 12.6. (cont.) We say that two splitting maps t and u of $A \rtimes_\varphi H$ are *conjugate* if there is some $a \in A$ such that $u(x) = at(x)a^{-1}$ for all $x \in H$. (Verify that this is an equivalence relation on the set of splitting maps.) Show that there is a bijective correspondence between the set of conjugacy classes of splitting maps of $A \rtimes_\varphi H$ and $H^1(H, A)$.

Solution. First observe that conjugacy of splitting maps is an equivalence relation. Reflexivity is obtained by conjugating by $0 \in A$; symmetry follows by conjugating by $-a$ if the original conjugating element is a; and transitivity follows because two successive conjugations by elements of A are the same as conjugation by their sum.
By Exercise 12.5, a splitting map t corresponds to a 1-cocycle $f \in Z^1(H, A)$ via

$$t(x) = (f(x), x).$$

Let u be another splitting map, corresponding to $k \in Z^1(H, A)$, so that

$$u(x) = (k(x), x).$$

We identify $a \in A$ with $(a, 1) \in A \rtimes_\varphi H$. Then

$$(a, 1)(f(x), x)(-a, 1) = (f(x) + a - xa, x).$$

Therefore $u(x) = at(x)a^{-1}$ for every $x \in H$ iff

$$k(x) = f(x) + a - xa$$

for every $x \in H$. Identifying $C^0(H, A)$ with A, the coboundary of a is the 1-cochain

$$(\delta^1 a)(x) = xa - a.$$

Thus the preceding identity says

$$k - f = -\delta^1 a = \delta^1(-a).$$

In particular, conjugate splitting maps determine 1-cocycles in the same coset of $B^1(H, A)$.
Conversely, suppose that two 1-cocycles $f, k \in Z^1(H, A)$ differ by a 1-coboundary. Say

$$k - f = \delta^1 c,$$

so $k(x) = f(x) + xc - c$ for every $x \in H$. Then

$$k(x) = f(x) + (-c) - x(-c),$$

and the calculation above shows that the corresponding splitting maps satisfy

$$u(x) = (-c)t(x)(-c)^{-1}$$

for every $x \in H$. Hence they are conjugate. It follows that the bijection from Exercise 12.5 descends exactly to a bijection between conjugacy classes of splitting maps and cosets in $Z^1(H, A)/B^1(H, A) = H^1(H, A)$. □

13 Wedderburn Theory

Setting and Goal

- Maschke (Corollary 12.8) shows that if $\mathrm{char}(F) \nmid |G|$ then every finitely generated FG-module decomposes as a direct sum of simple modules.
- This section identifies the finite-dimensional F-algebras with this property and uses the resulting structure theorems to finish Kolchin's theorem on unipotent subgroups.
- Throughout, all algebras are finite-dimensional F-algebras (with unit unless stated), and all modules are finitely generated, equivalently finite-dimensional as F-vector spaces (Lemma 12.5).
- The classification is due to J. H. M. Wedderburn.

Semisimplicity for Modules

- Three equivalent formulations of semisimplicity for an A-module: every submodule is a direct summand; the module is a direct sum of simple submodules; the module is a sum of simple submodules.
- Semisimplicity is preserved under submodules and quotient modules.
- An algebra A is *semisimple* if every non-zero A-module is semisimple; equivalently, A is semisimple as an A-module over itself.

Lemma 13.1. *The following statements about an A-module M are equivalent:*

1. *Any submodule of M is a direct summand of M.*
2. *M is semisimple.*
3. *M is a sum (but not, a priori, a direct sum) of simple submodules.*

Proof. We first prove that (1) implies (2). If $M = 0$, then there is nothing to prove. Otherwise, choose a non-zero submodule $S \leq M$ of minimal F-dimension. Then S is simple. By (1), we may write $M = S \oplus M_1$ for some submodule M_1. The same direct-summand condition passes to M_1: if $L \leq M_1$, then L is a direct summand of M, say $M = L \oplus W$, and hence

$$M_1 = L \oplus (M_1 \cap W).$$

Since $\dim_F(M_1) < \dim_F(M)$, induction on $\dim_F(M)$ gives that M_1 is a direct sum of simple submodules. Hence M is also a direct sum of simple submodules, so M is semisimple.
The implication (2) implies (3) is immediate from the definition of semisimplicity. It remains to prove that (3) implies (1). Assume that M is a sum of simple submodules, and let $N \leq M$. Choose a submodule $V \leq M$ maximal among those satisfying $N \cap V = 0$; such a V exists because M is finite-dimensional. We claim that $N + V = M$. Suppose not. Since M is a sum of simple submodules, some simple submodule $S \leq M$ is not contained in $N + V$. Then $S \cap (N + V)$ is a proper submodule of the simple module S, so $S \cap (N + V) = 0$. In particular, $S \cap V = 0$, and hence $V \subsetneq V + S$. If $n \in N \cap (V + S)$, write $n = v + s$ with $v \in V$ and $s \in S$. Then $s = n - v \in S \cap (N + V) = 0$, so $n = v \in N \cap V = 0$. Thus $N \cap (V + S) = 0$, contradicting the maximality of V. Therefore $N + V = M$, and since $N \cap V = 0$, we have $M = N \oplus V$. Thus every submodule of M is a direct summand. □

Lemma 13.2. *Submodules and quotient modules of semisimple modules are semisimple.*

Proof. Let M be a semisimple A-module. We first show that quotient modules of M are semisimple. Let $N \leq M$, and let $\pi\colon M \to M/N$ be the natural epimorphism. Since M is semisimple, write $M = S_1 + \cdots + S_t$ as a sum of simple submodules. Then

$$M/N = \pi(M) = \pi(S_1) + \cdots + \pi(S_t).$$

Each $\pi(S_i)$ is isomorphic to a quotient of S_i, so it is either zero or simple. Hence M/N is a sum of simple submodules, and Lemma 13.1 shows that M/N is semisimple.
Now let $L \leq M$. By Lemma 13.1, L is a direct summand of M, so $M = L \oplus K$ for some submodule K. The projection $M \to L$ is an epimorphism, and hence L is isomorphic to a quotient of M. Since quotient modules of M are semisimple by the preceding paragraph, L is semisimple. Thus both submodules and quotient modules of semisimple modules are semisimple. □

Lemma 13.3. *The algebra A is semisimple iff the A-module A is semisimple.*

Proof. Suppose first that the A-module A is semisimple, and let M be a non-zero A-module generated by $m_1, \dots, m_r$. The direct sum A^r is semisimple, since it is a finite direct sum of semisimple modules. Define

$$\varphi\colon A^r \to M, \qquad \varphi(a_1, \dots, a_r) = a_1 m_1 + \cdots + a_r m_r.$$

This is an A-module epimorphism, so M is isomorphic to a quotient module of the semisimple module A^r. By Lemma 13.2, M is semisimple. Therefore every non-zero A-module is semisimple, and hence A is a semisimple algebra.
Conversely, if A is a semisimple algebra, then every non-zero A-module is semisimple by definition. In particular, the regular A-module A is semisimple. □

Counting Simple Summands

- Decomposing $A \cong S_1 \oplus \cdots \oplus S_r$ into simple summands captures all isomorphism types of simple A-modules.
- Multiplicities of simple summands in a semisimple module are determined uniquely (Jordan–Hölder Theorem).

Proposition 13.4. *Let A be a semisimple algebra, and suppose that as A-modules we have $A \cong S_1 \oplus \cdots \oplus S_r$, where the S_i are simple submodules of A. Then any simple A-module is isomorphic with some S_i.*

Proof. Let $\theta\colon S_1 \oplus \cdots \oplus S_r \to A$ be an A-module isomorphism. Let S be a simple A-module. Choose $0 \neq s \in S$, and define

$$\varphi\colon A \to S, \qquad \varphi(a) = as.$$

This is an A-module homomorphism, since $\varphi(ba) = (ba)s = b(as) = b\varphi(a)$. Since S is simple and $\varphi(1) = s \neq 0$, the image of φ is a non-zero submodule of S, so φ is surjective. For each i, let $\iota_i\colon S_i \to S_1 \oplus \cdots \oplus S_r$ be the natural inclusion and set $\varphi_i = \varphi\theta\iota_i$. If every φ_i were zero, then $\varphi\theta$ would vanish on $S_1 \oplus \cdots \oplus S_r$, contrary to the fact that φ is non-zero and θ is surjective. Hence φ_i is non-zero for some i. By Schur's Lemma, this non-zero homomorphism between simple modules is an isomorphism. Therefore $S \cong S_i$ for some i. □

Proposition 13.5. *Suppose that A is a semisimple algebra, and let $S_1, \dots, S_r$ be a collection of simple A-modules such that every simple A-module is isomorphic with exactly one S_i. Let M be an A-module, and write $M \cong n_1 S_1 \oplus \cdots \oplus n_r S_r$ for some non-negative integers n_i. Then the n_i are uniquely determined.*

Proof. The point is that the Jordan–Hölder theorem does not only record the length of a composition series; it records the simple factors, up to reordering and isomorphism. In the module

$$n_1 S_1 \oplus \cdots \oplus n_r S_r,$$

we can build a composition series by removing one simple summand at a time. This gives a composition series in which S_i appears exactly n_i times as a composition factor. Now suppose that the same module could also be written as

$$M \cong m_1 S_1 \oplus \cdots \oplus m_r S_r.$$

This second decomposition gives another composition series, this time with S_i appearing exactly m_i times. By Jordan–Hölder Theorem, the two composition series have the same composition factors up to order and isomorphism. Since each simple module is isomorphic with exactly one of the S_i, the number of factors isomorphic to S_i must be the same in both series. Thus $n_i = m_i$ for every i, and the multiplicities are uniquely determined. □

Matrix Algebras over Division Algebras

- For a finite-dimensional F-algebra D, the matrix algebra $\mathcal{M}_n(D)$ has dimension $n^2 \dim_F D$.
- $E_{ij}(\alpha)$ denotes the matrix with sole non-zero entry α in position (i, j).
- D^n is the natural left $\mathcal{M}_n(D)$-module of column vectors.

- A *division algebra* is an algebra whose non-zero elements form a group under multiplication; field extensions of F are division algebras, but non-commutative examples also exist.

Theorem 13.6. *Let D be a division algebra, and let $n \in \mathbb{N}$. Then any simple $\mathcal{M}_n(D)$-module is isomorphic with D^n, and $\mathcal{M}_n(D)$ is isomorphic as $\mathcal{M}_n(D)$-modules with the direct sum of n copies of D^n. In particular, $\mathcal{M}_n(D)$ is a semisimple algebra.*

Proof. First we show that D^n is simple as a left $\mathcal{M}_n(D)$-module. Let $0 \neq U \leq D^n$, and choose a non-zero vector $u \in U$. Some entry of u is non-zero; say the jth entry is $x \in D^\times$. Multiplying u by $E_{jj}(x^{-1})$ gives the jth standard basis vector e_j, so $e_j \in U$. For every i, multiplying e_j by $E_{ij}(1)$ gives e_i, so every standard basis vector lies in U. Finally, multiplying e_i by $E_{ii}(\alpha)$ gives the vector with α in the ith position and zero elsewhere. Since any vector in D^n is the sum of its coordinate vectors, this shows that U contains every vector in D^n. Hence $U = D^n$, so D^n is simple.
Now let C_k be the submodule of $\mathcal{M}_n(D)$ consisting of matrices whose only possibly non-zero entries are in the kth column. Left multiplication by matrices preserves columns, so each C_k is an $\mathcal{M}_n(D)$-submodule, and

$$\mathcal{M}_n(D) = C_1 \oplus \cdots \oplus C_n$$

as $\mathcal{M}_n(D)$-modules. The map $C_k \to D^n$ sending a matrix to its kth column is an $\mathcal{M}_n(D)$-module isomorphism: left multiplication by a matrix sends the kth column to the same result as the usual action on that column. Hence $\mathcal{M}_n(D)$ is a direct sum of n copies of the simple module D^n. By Lemma 13.3, $\mathcal{M}_n(D)$ is a semisimple algebra. Finally, Proposition 13.4 shows that every simple $\mathcal{M}_n(D)$-module is isomorphic with one of these simple summands, and hence is isomorphic with D^n. $\square$

Simple Algebras

- An algebra is *simple* if its only two-sided ideals are itself and the zero ideal.
- Simple algebras are semisimple, and matrix algebras over division algebras are simple.

Lemma 13.7. *Simple algebras are semisimple.*

Proof. Let Σ be the sum of all simple submodules of the regular left A-module A. This sum is non-zero, because A is finite-dimensional and therefore contains a non-zero submodule of minimal dimension; such a submodule is simple, since any proper non-zero submodule would have smaller dimension. We claim that Σ is a two-sided ideal. It is already a left ideal because it is an A-submodule of A. Now let S be a simple submodule of A, and let $a \in A$. The map

$$S \to A, \qquad s \mapsto sa$$

is an A-module homomorphism, since $(bs)a = b(sa)$ for all $b \in A$. The image of a simple module under a homomorphism is isomorphic to a quotient of that simple module, so Sa is either zero or simple. In either case $Sa \subseteq \Sigma$. Since every element of Σ is a finite sum of elements from simple submodules, multiplying such a sum on the right by a still lands in Σ. Hence $\Sigma a \subseteq \Sigma$ for every $a \in A$. Thus Σ is also a right ideal, and hence a two-sided ideal.
Because A is simple and $\Sigma \neq 0$, we must have $\Sigma = A$. Thus A, as a left A-module, is a sum of simple submodules. By Lemma 13.1, the A-module A is semisimple, and then Lemma 13.3 shows that A is a semisimple algebra. $\square$

Theorem 13.8. *Let D be a division algebra, and let $n \in \mathbb{N}$. Then $\mathcal{M}_n(D)$ is a simple algebra.*

Proof. Let $0 \neq J$ be a two-sided ideal of $\mathcal{M}_n(D)$. We show that $J = \mathcal{M}_n(D)$. Choose a non-zero matrix $M \in J$, and let x be a non-zero entry of M, say in position (r, s). Since D is a division algebra, x^{-1} exists. Because J is a two-sided ideal,

$$E_{sr}(x^{-1})ME_{ss}(1) \in J.$$

This product is the matrix $E_{ss}(1)$: the right multiplication by $E_{ss}(1)$ keeps only column s, and the left multiplication by $E_{sr}(x^{-1})$ moves row r into row s while multiplying the surviving entry x by x^{-1}. Hence $E_{ss}(1) \in J$.
For any i, j, we have

$$E_{ij}(1) = E_{is}(1)E_{ss}(1)E_{sj}(1),$$

so $E_{ij}(1) \in J$. Since J is a right ideal, $E_{ij}(\alpha) = E_{ij}(1)E_{jj}(\alpha)$ also lies in J for every $\alpha \in D$. The matrices $E_{ij}(\alpha)$ span $\mathcal{M}_n(D)$ as an F-vector space, so $J = \mathcal{M}_n(D)$. Thus the only two-sided ideals are 0 and the whole algebra, and $\mathcal{M}_n(D)$ is simple. $\square$

Direct Sums of Algebras

- The *external direct sum* $B = B_1 \oplus \cdots \oplus B_r$ has the Cartesian product of underlying sets with componentwise operations; each B_i embeds as an ideal.
- A B-module M becomes a B_i-module via $b_i m = (0, \dots, b_i, \dots, 0)m$, and is simple/semisimple as a B_i-module iff it is simple/semisimple as a B-module.
- An *internal direct sum* of ideals $B_1, \dots, B_r$ (vector-space direct sum) is isomorphic to the external direct sum.
- Two-sided ideals of a finite direct sum decompose componentwise.

Lemma 13.9. *Let $B = B_1 \oplus \cdots \oplus B_n$ be a direct sum of algebras. Then the two-sided ideals of B are exactly the sets of the form $J_1 \oplus \cdots \oplus J_n$, where J_i is a two-sided ideal of B_i for each i.*

Proof. Let J be a two-sided ideal of B, and set $J_i = J \cap B_i$ for each i. Each J_i is a two-sided ideal of B_i: if $x \in J_i$ and $c \in B_i$, then cx and xc lie in J because J is a two-sided ideal of B, and they lie in B_i because multiplication is componentwise. Clearly $J_1 \oplus \cdots \oplus J_n \subseteq J$. Conversely, take $b \in J$, and write

$$b = b_1 + \cdots + b_n$$

with $b_i \in B_i$. Let e_i be the element of B whose ith component is the identity of B_i and whose other components are zero. This element acts as a projection onto the ith component. Since J is a right ideal, $b_i = be_i$ lies in J, and therefore $b_i \in J_i$. Hence $b \in J_1 \oplus \cdots \oplus J_n$, proving that $J = J_1 \oplus \cdots \oplus J_n$.
The converse is direct: if each J_i is a two-sided ideal of B_i, then $J_1 \oplus \cdots \oplus J_n$ is closed under left and right multiplication by elements of B, since multiplication in B is componentwise. Thus these are exactly the two-sided ideals of B. □

Theorem 13.10. *Let $r \in \mathbb{N}$. For each $1 \le i \le r$, let D_i be a division algebra over F, let $n_i \in \mathbb{N}$, and let $B_i = \mathcal{M}_{n_i}(D_i)$. Let B be the external direct sum of the B_i. Then B is a semisimple algebra having exactly r isomorphism classes of simple modules and exactly 2^r two-sided ideals, namely every sum of the form $\bigoplus_{j \in J} B_j$, where J is a subset of $\{1, \dots, r\}$.*

Proof. By Theorem 13.6, each B_i is a direct sum

$$B_i \cong G_{i1} \oplus \cdots \oplus G_{in_i},$$

where the G_{ij} are simple B_i-modules and are all isomorphic to the natural module $D_i^{n_i}$. We regard each G_{ij} as a B-module by letting B act through its ith component. Explicitly, $(b_1, \dots, b_r)$ acts on G_{ij} as b_i acts. This does not introduce any new submodules, so each G_{ij} is still simple as a B-module. Since $B = B_1 \oplus \cdots \oplus B_r$, we obtain a decomposition of B as a direct sum of simple B-modules. Therefore B is semisimple by Lemma 13.3.
Proposition 13.4 now implies that every simple B-module is isomorphic with some G_{ij}. If $i = k$, then $G_{ij} \cong G_{k\ell}$ by Theorem 13.6. If $i \ne k$, they cannot be isomorphic as B-modules. Indeed, let $e_i = (0, \dots, 1_i, \dots, 0)$. This element acts as the identity on G_{ij} but as zero on $G_{k\ell}$. If $f\colon G_{ij} \to G_{k\ell}$ were a B-module isomorphism, then for $v \ne 0$ we would have

$$f(v) = f(e_i v) = e_i f(v) = 0,$$

a contradiction. Thus there are exactly r isomorphism classes of simple B-modules.
Finally, by Theorem 13.8, each B_i has only the two two-sided ideals 0 and B_i. Lemma 13.9 therefore shows that every two-sided ideal of B is obtained by choosing, for each i, either 0 or B_i. Hence the two-sided ideals are precisely the sums $\bigoplus_{j \in J} B_j$ with $J \subseteq \{1, \dots, r\}$, and there are 2^r of them. □

Endomorphism and Opposite Algebras

- For an A-module M, $\mathrm{End}_A(M)$ is an F-algebra under composition; we call it the *endomorphism algebra* of M.
- The *opposite algebra* B^{op} has the same underlying additive structure but reversed multiplication: $a \cdot b := ba$. Note $(B^{op})^{op} = B$, and the opposite of a division algebra is again a division algebra.

- These two notions are linked by $B^{op} \cong \mathrm{End}_B(B)$: every B-endomorphism of B is right multiplication by some element.
- Endomorphism algebras decompose along A-module direct sums into block form, and over a single simple module they yield a matrix algebra.

Lemma 13.11. *Let B be an algebra. Then $B^{op} \cong \mathrm{End}_B(B)$.*

Proof. For $a \in B$, let $P_a : B \to B$ be right multiplication by a, so $P_a(x) = xa$. This is a left B-module homomorphism because

$$P_a(bx) = (bx)a = b(xa) = bP_a(x).$$

Conversely, if $\varphi \in \mathrm{End}_B(B)$, then for every $b \in B$ we have

$$\varphi(b) = \varphi(b \cdot 1) = b\varphi(1).$$

Thus $\varphi = P_{\varphi(1)}$, so every B-endomorphism of the regular module is right multiplication by a unique element of B.
It remains to check the multiplication. In B^{op} the product of a and b is $a \cdot b = ba$. Hence, for $x \in B$,

$$(P_a P_b)(x) = P_a(xb) = xba = P_{ba}(x) = P_{a \cdot b}(x).$$

Therefore $a \mapsto P_a$ is an algebra isomorphism $B^{op} \cong \mathrm{End}_B(B)$. □

Lemma 13.12. *Let $S_1, \dots, S_r$ be the distinct simple A-modules; for each i, let U_i be a direct sum of copies of S_i, and let $U = U_1 \oplus \cdots \oplus U_r$. Then we have $\mathrm{End}_A(U) \cong \mathrm{End}_A(U_1) \oplus \cdots \oplus \mathrm{End}_A(U_r)$.*

Proof. Let $\varphi \in \mathrm{End}_A(U)$. We first show that φ preserves each isotypic summand U_i. The image $\varphi(U_i)$ is isomorphic to a quotient of U_i, so every composition factor of $\varphi(U_i)$ is isomorphic to S_i. If $\varphi(U_i)$ were not contained in U_i, then its image in U/U_i would be non-zero. That image would have a composition factor isomorphic to S_i. But U/U_i is a direct sum of copies of the S_j with $j \neq i$, so none of its composition factors is isomorphic to S_i. This contradiction shows that $\varphi(U_i) \subseteq U_i$ for every i.
Thus restriction gives endomorphisms $\varphi_i = \varphi|_{U_i} \in \mathrm{End}_A(U_i)$, and we obtain a homomorphism

$$\Gamma : \mathrm{End}_A(U) \to \mathrm{End}_A(U_1) \oplus \cdots \oplus \mathrm{End}_A(U_r), \qquad \Gamma(\varphi) = (\varphi_1, \dots, \varphi_r).$$

This map is injective because an endomorphism of U is determined by its restrictions to the direct summands U_i. It is also surjective: given $(\psi_1, \dots, \psi_r)$ with $\psi_i \in \mathrm{End}_A(U_i)$, define $\psi \in \mathrm{End}_A(U)$ by writing $u = u_1 + \cdots + u_r$ and setting

$$\psi(u) = \psi_1(u_1) + \cdots + \psi_r(u_r).$$

This gives $\Gamma(\psi) = (\psi_1, \dots, \psi_r)$. Since composition is componentwise under this decomposition, Γ is an algebra isomorphism. □

Lemma 13.13. *If S is a simple A-module, then for any $n \in \mathbb{N}$ we have $\mathrm{End}_A(nS) \cong \mathcal{M}_n(\mathrm{End}_A(S))$.*

Proof. View nS as the module of column vectors with entries in S. Given a matrix $\Phi = (\varphi_{ij}) \in \mathcal{M}_n(\mathrm{End}_A(S))$, define an endomorphism $\Gamma(\Phi)$ of nS by the usual matrix rule:

$$\Gamma(\Phi) \begin{pmatrix} s_1 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} \varphi_{11}(s_1) + \cdots + \varphi_{1n}(s_n) \\ \vdots \\ \varphi_{n1}(s_1) + \cdots + \varphi_{nn}(s_n) \end{pmatrix}.$$

This is A-linear because each φ_{ij} is A-linear. The assignment Γ respects addition directly. For multiplication, if $\Phi = (\varphi_{ij})$ and $\Psi = (\psi_{ij})$, then the ith component of $\Gamma(\Phi)\Gamma(\Psi)(s_1, \dots, s_n)^t$ is

$$\sum_k \varphi_{ik}\Big(\sum_j \psi_{kj}(s_j)\Big) = \sum_j \Big(\sum_k \varphi_{ik}\psi_{kj}\Big)(s_j),$$

which is exactly the ith component coming from the matrix product $\Phi\Psi$. Thus Γ is an algebra homomorphism.
If $\Gamma(\Phi) = 0$, apply it to a vector whose only non-zero entry is an arbitrary $s \in S$ in the jth position.

The ith component of the result is $\varphi_{ij}(s)$, so every φ_{ij} is zero. Thus Γ is injective. Conversely, let $\psi \in \mathrm{End}_A(nS)$. Let $\iota_j : S \to nS$ insert an element into the jth coordinate, and let $\pi_i : nS \to S$ be projection onto the ith coordinate. Then

$$\psi_{ij} = \pi_i \psi \iota_j$$

lies in $\mathrm{End}_A(S)$, and the matrix (ψ_{ij}) maps under Γ to ψ. Hence Γ is surjective, and therefore $\mathrm{End}_A(nS) \cong \mathcal{M}_n(\mathrm{End}_A(S))$. ☐

Lemma 13.14. *Suppose that F is algebraically closed, and let S be a simple A-module. Then $\mathrm{End}_A(S) \cong F$.*

Proof. By Schur's Lemma, $\mathrm{End}_A(S)$ is a division algebra: every non-zero endomorphism of the simple module S is invertible. Let $\varphi \in \mathrm{End}_A(S)$. If $\varphi = 0$, then it is scalar multiplication by 0. Otherwise, since S is a finite-dimensional vector space over the algebraically closed field F, the F-linear map φ has an eigenvalue $\lambda \in F$. Thus $\varphi - \lambda I$ has non-zero kernel, so it is not invertible. But $\varphi - \lambda I$ is still an element of the division algebra $\mathrm{End}_A(S)$; the only non-invertible element of a division algebra is 0. Therefore $\varphi = \lambda I$.
Hence every A-endomorphism of S is scalar multiplication by an element of F. The map $F \to \mathrm{End}_A(S)$ given by $\lambda \mapsto \lambda I$ is therefore an algebra isomorphism. ☐

Lemma 13.15. *Let B be an algebra. Then $\mathcal{M}_n(B)^{op} \cong \mathcal{M}_n(B^{op})$ for any $n \in \mathbb{N}$.*

Proof. Define

$$\Psi : \mathcal{M}_n(B)^{op} \to \mathcal{M}_n(B^{op})$$

by $\Psi(X) = X^t$. This map is clearly bijective and F-linear. We check multiplication carefully because both the transpose and the opposite multiplication reverse order. Let $X = (x_{ij})$ and $Y = (y_{ij})$ be elements of $\mathcal{M}_n(B)^{op}$. In the domain, $X \cdot Y$ means YX using the ordinary multiplication in $\mathcal{M}_n(B)$. In the codomain, entries multiply in B^{op}, so $u \cdot v = vu$. Therefore

$$(\Psi(X)\Psi(Y))_{ij} = \sum_k x_{ki} \cdot y_{jk} = \sum_k y_{jk} x_{ki} = (YX)_{ji} = \Psi(X \cdot Y)_{ij}.$$

Thus $\Psi(X \cdot Y) = \Psi(X)\Psi(Y)$, and Ψ is an algebra isomorphism. ☐

Wedderburn's Structure Theorems

- Wedderburn's main structure theorem (Theorem 13.16): a finite-dimensional algebra is semisimple iff it is isomorphic to a direct sum of matrix algebras over division algebras.
- Equivalently, semisimple iff a direct sum of simple algebras, confirming the consistency of terminology.
- An algebra is simple iff it is isomorphic to a single matrix algebra over a division algebra (Theorem 13.17).
- Over an algebraically closed field, every semisimple algebra is a direct sum of full matrix algebras over F itself (Theorem 13.18).

Theorem 13.16. *The algebra A is semisimple iff it is isomorphic with a direct sum of matrix algebras over division algebras.*

Proof. Suppose first that A is semisimple. As a left A-module, decompose A into its simple isotypic parts:

$$A = U_1 \oplus \cdots \oplus U_r,$$

where $U_i \cong n_i S_i$, the S_i are pairwise non-isomorphic simple A-modules, and each n_i is positive. By Lemma 13.11,

$$A^{op} \cong \mathrm{End}_A(A).$$

By Lemma 13.12, the endomorphism algebra of A splits across the isotypic parts:

$$\mathrm{End}_A(A) \cong \mathrm{End}_A(U_1) \oplus \cdots \oplus \mathrm{End}_A(U_r).$$

Since $U_i \cong n_i S_i$, Lemma 13.13 gives

$$\mathrm{End}_A(U_i) \cong \mathcal{M}_{n_i}(\mathrm{End}_A(S_i)).$$

Combining these isomorphisms,

$$A^{op} \cong \mathcal{M}_{n_1}(\mathrm{End}_A(S_1)) \oplus \cdots \oplus \mathcal{M}_{n_r}(\mathrm{End}_A(S_r)).$$

Taking opposite algebras and using Lemma 13.15, we obtain

$$A \cong \mathcal{M}_{n_1}(\mathrm{End}_A(S_1)^{op}) \oplus \cdots \oplus \mathcal{M}_{n_r}(\mathrm{End}_A(S_r)^{op}).$$

By Schur's Lemma, each $\mathrm{End}_A(S_i)$ is a division algebra, and hence so is its opposite algebra. Thus A is isomorphic with a direct sum of matrix algebras over division algebras.
Conversely, if A is isomorphic with a direct sum of matrix algebras over division algebras, then Theorem 13.10 shows that A is semisimple. □

Theorem 13.17. *The algebra A is simple iff it is isomorphic with a matrix algebra over a division algebra.*

Proof. Suppose first that A is simple. By Lemma 13.7, A is semisimple, so Theorem 13.16 writes A as a direct sum of r matrix algebras over division algebras. By Theorem 13.10, such a direct sum has exactly 2^r two-sided ideals. Since A is simple, it has exactly two two-sided ideals, namely 0 and A. Hence $2^r = 2$, so $r = 1$. Therefore A is isomorphic with a single matrix algebra over a division algebra.
Conversely, every matrix algebra over a division algebra is simple by Theorem 13.8. □

Theorem 13.18. *Suppose that the field F is algebraically closed. Then any semisimple algebra is isomorphic with a direct sum of matrix algebras over F.*

Proof. In the proof of Theorem 13.16, the division algebras that occur are the opposite algebras of the endomorphism algebras $\mathrm{End}_A(S_i)$, where the S_i are simple A-modules. If F is algebraically closed, Lemma 13.14 gives $\mathrm{End}_A(S_i) \cong F$ for each i. Since $F^{op} = F$, the decomposition in Theorem 13.16 becomes a direct sum of matrix algebras over F. □

Nilpotent Ideals and the Radical

- In this part, A is an algebra possibly without unit. An element $x \in A$ is *nilpotent* if $x^n = 0$ for some $n \in \mathbb{N}$.
- An ideal I is *nilpotent* if $I^n = 0$ for some n, where I^n is the ideal spanned by all products of n elements; an algebra is nilpotent if some such n exists for $I = A$.
- A nilpotent algebra cannot have a unit.
- The sum of two nilpotent ideals is nilpotent, and (using finite-dimensionality) a unique largest nilpotent ideal exists.
- Submodules with semisimple quotient are closed under finite intersection, so a smallest such submodule of A exists.
- For an A-module M, $\mathrm{Ann}(M) = \{a \in A \mid aM = 0\}$ is the *annihilator* of M, an ideal of A; the *common annihilator* of a family $\{M_t\}_{t\in\mathcal{T}}$ is $\bigcap_{t\in\mathcal{T}} \mathrm{Ann}(M_t)$.
- Wedderburn (Theorem 13.23): the largest nilpotent ideal, the common annihilator of all simple A-modules, and the smallest submodule of A with semisimple quotient all coincide. This common object is the *radical* $\mathrm{rad}(A)$, also called the Jacobson radical $J(A)$.

Lemma 13.19. *Let I and J be nilpotent ideals of A. Then $I + J$ is also a nilpotent ideal of A.*

Proof. Choose $m, n \in \mathbb{N}$ such that $I^m = 0$ and $J^n = 0$. Consider a product of $m + n$ elements of $I + J$. Expanding each factor as a sum of one element from I and one element from J, the product becomes a sum of terms

$$z_1 z_2 \cdots z_{m+n},$$

where each z_k lies in either I or J. In any such term, either at least m of the factors lie in I, or at least n of the factors lie in J. If at least m factors lie in I, group the product around those I-factors. The pieces between them stay inside A, and because I is a two-sided ideal, multiplying an element of I on either side by elements of A still leaves an element of I. Thus the whole term lies in $I^m = 0$. The case with at least n factors in J is the same. Hence every such term is zero, and therefore $(I + J)^{m+n} = 0$. Thus $I + J$ is nilpotent. □

Corollary 13.20. *A has a largest nilpotent ideal.*

Proof. Let K be the sum of all nilpotent ideals of A. Since A is finite-dimensional, this sum is already the sum of finitely many nilpotent ideals: choose finitely many elements spanning K, and each one lies in a finite sum of nilpotent ideals. Thus

$$K = I_1 + \cdots + I_t.$$

By repeated application of Lemma 13.19, this finite sum is nilpotent. By construction, K contains every nilpotent ideal of A. Hence K is the largest nilpotent ideal of A. □

Lemma 13.21. *Let M and N be submodules of A such that A/M and A/N are semisimple A-modules. Then $A/(M \cap N)$ is a semisimple A-module.*

Proof. Define

$$\Phi\colon A/(M \cap N) \to A/M \oplus A/N$$

by

$$\Phi(a + M \cap N) = (a + M, a + N).$$

This is a well-defined A-module homomorphism. Its kernel consists of those $a + M \cap N$ such that $a \in M$ and $a \in N$, which is exactly the zero coset. Hence Φ is injective. Thus $A/(M \cap N)$ is isomorphic to a submodule of $A/M \oplus A/N$. The direct sum $A/M \oplus A/N$ is semisimple, and Lemma 13.2 says that submodules of semisimple modules are semisimple. Therefore $A/(M \cap N)$ is semisimple. □

Corollary 13.22. *A has a smallest submodule having semisimple quotient.*

Proof. Intersect all submodules $M \leq A$ for which A/M is semisimple. Since A is finite-dimensional, this intersection is already the intersection of finitely many such submodules: repeatedly intersecting with a new submodule strictly lowers dimension only finitely many times. Say these finitely many submodules are $M_1, \ldots, M_t$. By Lemma 13.21 and induction on t, the quotient

$$A/(M_1 \cap \cdots \cap M_t)$$

is semisimple. Therefore this finite intersection itself has semisimple quotient. It is contained in every submodule of A with semisimple quotient by construction, so it is the smallest such submodule. □

Theorem 13.23. *Let A be an algebra with unit. Then the following are equal:*

- *The largest nilpotent ideal of A.*
- *The common annihilator of all simple A-modules.*
- *The smallest submodule of A having semisimple quotient.*

Proof. Let I be the largest nilpotent ideal of A, let Λ be the common annihilator of all simple A-modules, and let M be the smallest submodule of the regular module A having semisimple quotient. We prove

$$I = \Lambda = M.$$

First let J be any nilpotent ideal of A, and let S be a simple A-module. The subspace JS is a submodule of S, so either $JS = 0$ or $JS = S$. If $JS = S$, then $J^k S = S$ for every $k \geq 1$ by induction. This contradicts nilpotence of J, since $J^n = 0$ for some n. Therefore $JS = 0$. Thus every nilpotent ideal annihilates every simple module, so $I \subseteq \Lambda$.

Conversely, take a composition series of the regular A-module:

$$A = A_0 \supset A_1 \supset \cdots \supset A_r = 0.$$

Each quotient A_k/A_{k+1} is simple, so Λ annihilates it. Equivalently, $\Lambda A_k \subseteq A_{k+1}$ for every k. It follows inductively that $\Lambda^k A \subseteq A_k$. In particular, $\Lambda^r A = 0$. Since A has a unit, every element $x \in \Lambda^r$ satisfies $x = x1$, so $\Lambda^r \subseteq \Lambda^r A = 0$. Hence Λ is nilpotent, and therefore $\Lambda \subseteq I$. Thus $I = \Lambda$.

Next, since Λ annihilates every simple module, it annihilates every semisimple module, because a semisimple module is a direct sum of simple modules. The quotient A/M is semisimple, so $\Lambda(A/M) = 0$. In particular, for $\lambda \in \Lambda$ we have $\lambda(1 + M) = \lambda + M = 0$, and hence $\lambda \in M$. Thus $\Lambda \subseteq M$.

It remains to prove $M \subseteq \Lambda$. Suppose, for contradiction, that $\Lambda \subsetneq M$. Then some element of M does not annihilate all simple modules, so there is a simple A-module S with $MS \neq 0$. Since S is simple, $MS = S$. Hence there is some $s \in S$ with $Ms \neq 0$; the set Ms is a submodule of S, so simplicity gives $Ms = S$. We may therefore choose $m \in M$ such that $ms = -s$. Then $(m + 1)s = 0$, so $m + 1$ lies in the annihilator of s:

$$\mathrm{Ann}(s) = \{a \in A \mid as = 0\}.$$

This annihilator is a proper submodule of the regular module A, because $1s = s \neq 0$. Since A is finite-dimensional, $\mathrm{Ann}(s)$ is contained in some maximal submodule N of A. The quotient A/N is simple, hence semisimple. By the defining minimality of M, we have $M \subseteq N$. Thus $m \in N$, and also $m+1 \in N$. Hence $1 = (m + 1) - m \in N$, forcing $N = A$, a contradiction. Therefore $M \subseteq \Lambda$. We conclude that $I = \Lambda = M$. □

Corollary 13.24. *Let A be an algebra with unit. Then A is semisimple iff A has no non-zero nilpotent ideals.*

Proof. By Lemma 13.3, A is semisimple iff the regular A-module A is semisimple. This is equivalent to saying that the smallest submodule of A having semisimple quotient is 0. By Theorem 13.23, that smallest submodule is the radical, which is also the largest nilpotent ideal of A. Hence A is semisimple iff the largest nilpotent ideal is 0, equivalently iff A has no non-zero nilpotent ideals. □

Algebras Without Unit; Nilpotence

- Any algebra (possibly without unit) embeds as an ideal in an algebra with unit, with quotient F.
- Consequence: an algebra without unit having radical zero cannot exist—no such algebras occur.
- Wedderburn: an algebra generated as an F-vector space by nilpotent elements is itself nilpotent. This finishes Kolchin's Theorem.

Lemma 13.25. *Let A be an algebra, possibly without unit. Then there exists an algebra B with unit and an ideal I of B such that, as algebras, we have $I \cong A$ and $B/I \cong F$.*

Proof. Define B to be the vector space $F \oplus A$, with multiplication

$$(\lambda, a)(\mu, b) = (\lambda\mu, \lambda b + \mu a + ab).$$

This multiplication is the usual multiplication on A plus the missing scalar identity terms; expanding both $((\lambda, a)(\mu, b))(\nu, c)$ and $(\lambda, a)((\mu, b)(\nu, c))$ gives the same expression by associativity and distributivity in A. Thus B is an algebra, and the element $(1, 0)$ is a unit for B. Let

$$I = \{(0, a) \mid a \in A\}.$$

Then I is an ideal of B, since multiplying $(0, a)$ on either side by an element of B again gives an element whose first coordinate is zero. The map $A \to I$ given by $a \mapsto (0, a)$ is an algebra isomorphism. Finally, the projection $B \to F$ onto the first coordinate is an algebra epimorphism with kernel I, so $B/I \cong F$. □

Theorem 13.26. *Let A be a non-zero algebra, possibly without unit. If A has no non-zero nilpotent ideals, then A is an algebra with unit.*

Proof. Let B and I be as in Lemma 13.25, so $I \cong A$ and $B/I \cong F$. Let J be a nilpotent ideal of B. Then $J \cap I$ is a nilpotent ideal of I, and hence corresponds to a nilpotent ideal of A. By hypothesis, A has no non-zero nilpotent ideals, so $J \cap I = 0$.

The natural map $B \to B/I$ restricts to an injection on J, and its image is $(I + J)/I$. Equivalently,

$$J \cong J/(J \cap I) \cong (I + J)/I.$$

This image is a nilpotent ideal of B/I. But $B/I \cong F$, and a field has no non-zero nilpotent ideals; the whole field is not nilpotent because it has a unit. Hence $(I+J)/I = 0$, so $J = 0$. Therefore B has no non-zero nilpotent ideals. Since B has a unit, Corollary 13.24 shows that B is semisimple.
By Theorem 13.16, B is a direct sum of matrix algebras over division algebras. Theorem 13.10 describes the ideals of such a direct sum: every ideal is a direct sum of some of the matrix-algebra summands. In particular, every non-zero ideal of B has a unit, namely the sum of the identity elements in the selected summands. Since $I \cong A$ is a non-zero ideal of B, I has a unit. Transporting this unit across the isomorphism $A \cong I$ gives a unit for A. □

Theorem 13.27. *Suppose that the algebra A can be generated as an F-vector space by a set consisting of nilpotent elements. Then A is nilpotent.*

Proof. We first reduce to the case where F is algebraically closed. Let $\overline{F}$ be an algebraic closure of F, and form the scalar extension
$$\overline{A} = \overline{F} \otimes_F A.$$
with multiplication $(\alpha \otimes a)(\beta \otimes b) = \alpha\beta \otimes ab$. Choosing an F-basis of A shows that A embeds in $\overline{A}$ by $a \mapsto 1 \otimes a$, and that $\dim_{\overline{F}} \overline{A} = \dim_F A$. If $x \in A$ is nilpotent, then $1 \otimes x$ is nilpotent in $\overline{A}$. Thus, if a set of nilpotent elements spans A over F, the corresponding elements $1 \otimes x$ span $\overline{A}$ over $\overline{F}$ and are still nilpotent. If $\overline{A}$ is nilpotent, then its embedded subalgebra A is nilpotent. Hence it suffices to prove the theorem after replacing F by $\overline{F}$; from now on assume that F is algebraically closed.
We argue by induction on $\dim_F A$. The zero algebra is nilpotent, and if $\dim_F A = 1$, then A is spanned by a nilpotent element, so a sufficiently long product of elements of A is zero. Now assume $\dim_F A > 1$. If A has a non-zero nilpotent ideal I, then A/I is generated by the images of the original nilpotent generators, so by induction A/I is nilpotent. Say $(A/I)^m = 0$, so $A^m \subseteq I$. Since I is nilpotent, say $I^n = 0$, we get $A^{mn} \subseteq I^n = 0$. Hence A is nilpotent.
It remains to rule out the case where A has no non-zero nilpotent ideals. By Theorem 13.26, A has a unit, and by Corollary 13.24, A is semisimple. Since F is algebraically closed, Theorem 13.18 gives
$$A \cong \mathcal{M}_{n_1}(F) \oplus \cdots \oplus \mathcal{M}_{n_r}(F).$$
Projecting the nilpotent generating set of A onto any summand shows that each summand $\mathcal{M}_{n_i}(F)$ is spanned by nilpotent matrices. But over an algebraically closed field, a nilpotent matrix has only the eigenvalue 0, so its trace is zero. Thus all nilpotent matrices lie in the proper subspace $\ker(\mathrm{tr})$ of $\mathcal{M}_{n_i}(F)$, and they cannot span the whole matrix algebra. This contradiction shows that the no-non-zero-nilpotent-ideal case cannot occur. Therefore A is nilpotent. □

Proposition 13.28. *Let $V_n(F)$ be the space of column vectors of length $n \in \mathbb{N}$ with entries from a field F, and let H be a unipotent subgroup of* $\mathrm{GL}(n, F)$*. Then there exists some $0 \neq \mathbf{v} \in V_n(F)$ such that $x\mathbf{v} = \mathbf{v}$ for all $x \in H$.*

Proof. Let I denote the identity matrix, and let W be the F-subspace of $\mathcal{M}_n(F)$ spanned by
$$\{x - I \mid x \in H\}.$$
If $x \in H$, then x is unipotent, so $x - I$ is nilpotent. Also, for $x, y \in H$,
$$(x - I)(y - I) = (xy - I) - (x - I) - (y - I).$$
Since $xy \in H$, the right-hand side lies in W. Hence W is closed under multiplication, so W is an algebra, possibly without unit, generated as an F-vector space by nilpotent elements. By Theorem 13.27, W is nilpotent.
If $W = 0$, then $x - I = 0$ for every $x \in H$, so every non-zero vector is fixed by H. Otherwise, choose $t \in \mathbb{N}$ minimal such that every product
$$(x_1 - I) \cdots (x_t - I)$$
with $x_1, \ldots, x_t \in H$ is zero. Such a t exists because W is nilpotent and the elements $x - I$ span W. By minimality, there exist $y_1, \ldots, y_{t-1} \in H$ such that
$$P = (y_1 - I) \cdots (y_{t-1} - I)$$

is not the zero matrix. Choose a vector $\mathbf{v}'$ such that $\mathbf{v} = P\mathbf{v}' \neq 0$. For any $x \in H$, the product

$$(x - I)(y_1 - I) \cdots (y_{t-1} - I)$$

has t factors from W, and hence is zero. Therefore

$$(x - I)\mathbf{v} = 0$$

for every $x \in H$, which is exactly $x\mathbf{v} = \mathbf{v}$ for every $x \in H$. □

Exercises

Throughout these exercises, A denotes a finite-dimensional algebra with unit over a field F, and all A-modules are finitely generated.

Exercise 13.1. Let $n \in \mathbb{N}$. Let V be the n-dimensional subspace of A^n spanned by the identity elements of the summands. Show that if M is an A-module and $T\colon V \to M$ is a linear transformation, then there is a unique A-module homomorphism from A^n to M which extends T.

Solution. Let e_i denote the element of A^n whose ith coordinate is 1 and whose other coordinates are zero. Then V is the F-span of $e_1, \ldots, e_n$, and every element of A^n has a unique expression

$$(a_1, \ldots, a_n) = a_1 e_1 + \cdots + a_n e_n.$$

Define

$$\widetilde{T}(a_1, \ldots, a_n) = a_1 T(e_1) + \cdots + a_n T(e_n).$$

This map is F-linear, and for $b \in A$ we have

$$\widetilde{T}(b(a_1, \ldots, a_n)) = \widetilde{T}(ba_1, \ldots, ba_n) = ba_1 T(e_1) + \cdots + ba_n T(e_n) = b\widetilde{T}(a_1, \ldots, a_n).$$

Hence $\widetilde{T}$ is an A-module homomorphism. Also $\widetilde{T}(e_i) = T(e_i)$ for each i, so $\widetilde{T}$ extends T. Finally, if $\varphi\colon A^n \to M$ is any A-module homomorphism extending T, then

$$\varphi(a_1, \ldots, a_n) = a_1\varphi(e_1) + \cdots + a_n\varphi(e_n) = a_1 T(e_1) + \cdots + a_n T(e_n).$$

Thus $\varphi = \widetilde{T}$, proving uniqueness. □

Exercise 13.2. (cont.) Suppose that B is an A-module having an n-dimensional subspace U such that whenever M is an A-module and $T\colon U \to M$ is a linear transformation, there is a unique extension of T to an A-module homomorphism from B to M. Show that B and A^n are isomorphic A-modules. (Modules of the form A^n for some $n \in \mathbb{N}$ are called *free* A-modules; hence this exercise provides an alternate characterization of freeness, which we shall generalize for group algebras in the exercises to Section 16.)

Solution. Choose a basis $u_1, \ldots, u_n$ of U, and let $e_1, \ldots, e_n$ be the standard elements of the subspace $V \leq A^n$ from Exercise 13.1. Define linear maps

$$\alpha_0\colon V \to B, \qquad \alpha_0(e_i) = u_i,$$

and

$$\beta_0\colon U \to A^n, \qquad \beta_0(u_i) = e_i.$$

By Exercise 13.1, α_0 extends uniquely to an A-module homomorphism $\alpha\colon A^n \to B$. By the assumed property of B, β_0 extends uniquely to an A-module homomorphism $\beta\colon B \to A^n$.
The composite $\beta\alpha\colon A^n \to A^n$ extends the identity map on V, because $\beta\alpha(e_i) = \beta(u_i) = e_i$. The identity map on A^n is also an A-module homomorphism extending the identity map on V. By the uniqueness in Exercise 13.1, we have $\beta\alpha = 1_{A^n}$. Similarly, $\alpha\beta\colon B \to B$ extends the identity map on U, and so does 1_B. By the uniqueness property of B, we have $\alpha\beta = 1_B$. Therefore α and β are inverse isomorphisms, and $B \cong A^n$ as A-modules. □

Exercise 13.3. Let $n \in \mathbb{N}$, and let $T_n(F)$ be the algebra of upper triangular $n \times n$ matrices. Show that the set $V_n(F)$ of column vectors over F of length n is a $T_n(F)$-module that has a unique composition series in which every simple $T_n(F)$-module appears exactly once as a composition factor.

Solution. The usual matrix action makes $V_n(F)$ a $T_n(F)$-module. Let

$$W_k = \langle e_1, \ldots, e_k \rangle_F$$

for $0 \le k \le n$, where $W_0 = 0$ and $e_1, \ldots, e_n$ are the standard column vectors. Since an upper triangular matrix sends e_j into $\langle e_1, \ldots, e_j \rangle_F$, each W_k is a $T_n(F)$-submodule.
We claim that these are the only submodules. Let $0 \ne W \le V_n(F)$, and choose the largest index k such that some vector in W has non-zero kth coordinate. Then $W \subseteq W_k$. For such a vector v, the matrix unit $E_{kk}(1)$ gives a non-zero scalar multiple of e_k, so $e_k \in W$. Applying $E_{ik}(1)$ for $1 \le i \le k$ gives $e_i \in W$. Hence $W_k \subseteq W$, and so $W = W_k$. Thus the submodule lattice is exactly

$$0 = W_0 < W_1 < \cdots < W_n = V_n(F).$$

Therefore this is the unique composition series of $V_n(F)$. The quotient W_k/W_{k-1} is one-dimensional, and a matrix $(a_{ij}) \in T_n(F)$ acts on it by the scalar a_{kk}.
It remains to see that every simple module occurs in this list. Let N be the ideal of strictly upper triangular matrices. Then $N^n = 0$. If S is a simple $T_n(F)$-module, then NS is a submodule of S, so either $NS = 0$ or $NS = S$. The second possibility would imply $N^m S = S$ for every $m \ge 1$, contradicting $N^n = 0$. Hence $NS = 0$, and S is a simple module for

$$T_n(F)/N \cong F \oplus \cdots \oplus F.$$

The simple modules for this direct sum are the coordinate modules. Indeed, the diagonal idempotents split any module into the sum of their images, and a simple module can have only one non-zero such image. Thus one diagonal copy of F acts by scalars and the other copies act as zero. These are exactly the modules W_k/W_{k-1}. Hence every simple $T_n(F)$-module appears exactly once as a composition factor in the displayed series. □

Exercise 13.4. (cont.) Show that the $T_n(F)$-module $T_n(F)$ is isomorphic with the direct sum of all non-zero submodules of $V_n(F)$.

Solution. By Exercise 13.3, the non-zero submodules of $V_n(F)$ are

$$W_k = \langle e_1, \ldots, e_k \rangle_F, \qquad 1 \le k \le n.$$

For each k, let C_k be the subspace of $T_n(F)$ consisting of matrices whose only possibly non-zero entries lie in the kth column. Since left multiplication by an upper triangular matrix acts on each column separately, C_k is a left $T_n(F)$-submodule of the regular module $T_n(F)$. Also every upper triangular matrix is the sum of its columns, so

$$T_n(F) = C_1 \oplus \cdots \oplus C_n$$

as $T_n(F)$-modules.
The map $C_k \to W_k$ sending a matrix to its kth column is a $T_n(F)$-module isomorphism. Indeed, the kth column of XM is X times the kth column of M, and the possible kth columns of upper triangular matrices are exactly the vectors in W_k. Therefore

$$T_n(F) \cong W_1 \oplus \cdots \oplus W_n,$$

which is the direct sum of all non-zero submodules of $V_n(F)$. □

Exercise 13.5. (cont.) Find the largest nilpotent ideal of $T_n(F)$, and show without reference to Theorem 13.23 that it is also the common annihilator of all simple $T_n(F)$-modules and the smallest submodule of $T_n(F)$ having semisimple quotient.

Solution. Let N be the ideal of strictly upper triangular matrices. Then $N^n = 0$, so N is nilpotent. We first show that N is the largest nilpotent ideal. Let I be a nilpotent ideal of $T_n(F)$. If some $X \in I$ had a non-zero diagonal entry x_{ii}, then

$$E_{ii}(1) X E_{ii}(1) = x_{ii} E_{ii}(1)$$

would lie in I. Hence $E_{ii}(1) \in I$. But $E_{ii}(1)$ is idempotent and non-zero, so it is not nilpotent, contradicting the nilpotence of I. Thus every element of I has zero diagonal, and $I \subseteq N$. Hence N is

the largest nilpotent ideal.
Next we compute the common annihilator of the simple modules. From Exercise 13.3, the simple modules are the one-dimensional quotients W_i/W_{i-1}. A strictly upper triangular matrix sends W_i into W_{i-1}, so N annihilates every W_i/W_{i-1}. Conversely, if $X = (x_{ij})$ is not in N, then $x_{ii} \neq 0$ for some i, and X acts on W_i/W_{i-1} by the non-zero scalar x_{ii}. Hence X does not annihilate all simple modules. Therefore N is exactly the common annihilator of all simple $T_n(F)$-modules.
Finally, $T_n(F)/N$ is the diagonal algebra, isomorphic to $F \oplus \cdots \oplus F$, so it is semisimple as a $T_n(F)$-module. Thus N is a submodule of $T_n(F)$ with semisimple quotient. If $L \leq T_n(F)$ is any submodule such that $T_n(F)/L$ is semisimple, then the common annihilator of all simple modules annihilates $T_n(F)/L$. In particular, for $x \in N$,

$$x(1 + L) = x + L = 0,$$

so $x \in L$. Hence $N \subseteq L$. Therefore N is the smallest submodule of $T_n(F)$ having semisimple quotient. □

Exercise 13.6. Let U be an A-module, let $n \in \mathbb{N}$, and let U^n be the set of column vectors of length n with entries from U, considered in the obvious way as an $\mathcal{M}_n(A)$-module. Show that U is a simple A-module iff U^n is a simple $\mathcal{M}_n(A)$-module.

Solution. Suppose first that U is simple, and let $0 \neq W \leq U^n$ be an $\mathcal{M}_n(A)$-submodule. Choose a vector $w = (u_1, \ldots, u_n)^t \in W$ with some $u_j \neq 0$. Applying $E_{jj}(1)$ to w shows that the vector with u_j in the jth coordinate and zero elsewhere lies in W. Since U is simple, $Au_j = U$. Applying $E_{jj}(a)$ for $a \in A$ shows that every vector with an arbitrary element of U in the jth coordinate lies in W. Applying $E_{ij}(1)$ for $1 \leq i \leq n$ then shows that every coordinate copy of U lies in W. Hence $W = U^n$, proving that U^n is simple.
Conversely, suppose that U^n is simple as an $\mathcal{M}_n(A)$-module. If $0 \neq L < U$ were an A-submodule, then L^n would be a non-zero proper $\mathcal{M}_n(A)$-submodule of U^n, contradicting simplicity. Therefore U is simple as an A-module. □

Exercise 13.7. (cont.) Show that $\mathrm{Hom}_A(U, V) \cong \mathrm{Hom}_{\mathcal{M}_n(A)}(U^n, V^n)$ for any A-modules U and V.

Solution. Given $\varphi \in \mathrm{Hom}_A(U, V)$, define $\varphi^n \colon U^n \to V^n$ by applying φ to each coordinate. This is an $\mathcal{M}_n(A)$-module homomorphism, so we have a linear map

$$\Gamma\colon \mathrm{Hom}_A(U, V) \to \mathrm{Hom}_{\mathcal{M}_n(A)}(U^n, V^n), \qquad \Gamma(\varphi) = \varphi^n.$$

If $\varphi^n = 0$, then $\varphi = 0$, so Γ is injective.
To prove surjectivity, let $\Psi\colon U^n \to V^n$ be an $\mathcal{M}_n(A)$-module homomorphism. Let ι_i denote insertion into the ith coordinate and π_i projection onto the ith coordinate. Since $E_{11}(1)\iota_1(u) = \iota_1(u)$, we have

$$\Psi\iota_1(u) = \Psi E_{11}(1)\iota_1(u) = E_{11}(1)\Psi\iota_1(u),$$

so $\Psi\iota_1(u)$ has only its first coordinate possibly non-zero. Define $\varphi\colon U \to V$ by

$$\Psi\iota_1(u) = \iota_1(\varphi(u)).$$

For $a \in A$,

$$\iota_1(\varphi(au)) = \Psi E_{11}(a)\iota_1(u) = E_{11}(a)\Psi\iota_1(u) = \iota_1(a\varphi(u)),$$

so φ is A-linear. For each j,

$$\Psi\iota_j(u) = \Psi E_{j1}(1)\iota_1(u) = E_{j1}(1)\Psi\iota_1(u) = \iota_j(\varphi(u)).$$

By additivity, Ψ applies φ to each coordinate. Thus $\Psi = \varphi^n$, and Γ is surjective. Therefore

$$\mathrm{Hom}_A(U, V) \cong \mathrm{Hom}_{\mathcal{M}_n(A)}(U^n, V^n).$$

□

Exercise 13.8. (cont.) Show that every $\mathcal{M}_n(A)$-module is isomorphic with a module of the form U^n for some A-module U. (What Exercises 13.6–13.8 show is that the module theory of the algebras A and $\mathcal{M}_n(A)$ is, in a sense that can be made precise, the same. In the language of category theory, we say that the categories of A-modules and $\mathcal{M}_n(A)$-modules are *Morita equivalent*.)

Solution. Let W be an $\mathcal{M}_n(A)$-module, and put $e_i = E_{ii}(1)$. The idempotents $e_1, \dots, e_n$ are orthogonal and satisfy $e_1 + \cdots + e_n = 1$, so

$$W = e_1W \oplus \cdots \oplus e_nW.$$

Let $U = e_1W$, made into an A-module by

$$a \cdot u = E_{11}(a)u.$$

For each i, the maps

$$U \to e_iW, \qquad u \mapsto E_{i1}(1)u,$$

and

$$e_iW \to U, \qquad w \mapsto E_{1i}(1)w,$$

are inverse F-linear isomorphisms. Define

$$\Phi \colon U^n \to W, \qquad \Phi((u_1, \dots, u_n)^t) = \sum_{i=1}^{n} E_{i1}(1)u_i.$$

The inverse sends $w \in W$ to

$$(E_{11}(1)w, E_{12}(1)w, \dots, E_{1n}(1)w)^t.$$

Hence Φ is an F-linear isomorphism. It remains only to check the action. If $X = (a_{ij}) \in \mathcal{M}_n(A)$, then

$$X\Phi((u_1, \dots, u_n)^t) = \sum_{i,j} E_{ij}(a_{ij})E_{j1}(1)u_j = \sum_i E_{i1}(1)\Big(\sum_j E_{11}(a_{ij})u_j\Big).$$

This is exactly $\Phi(X(u_1, \dots, u_n)^t)$, using the A-module structure on U. Thus Φ is an $\mathcal{M}_n(A)$-module isomorphism, and $W \cong U^n$. □

Exercise 13.9. Show that A is a simple A-module iff A is a division algebra.

Solution. Suppose first that A is a division algebra. If $0 \neq I \leq A$ is a left ideal, choose $0 \neq x \in I$. Then x is invertible, so $1 = x^{-1}x \in I$, and hence $I = A$. Thus the regular left A-module A has no non-zero proper submodules, so it is simple.
Conversely, suppose that A is simple as a left A-module. Let $0 \neq a \in A$. The left ideal Aa is non-zero, so $Aa = A$. Hence there exists $b \in A$ such that $ba = 1$. Right multiplication by a is then surjective, since every $x \in A$ can be written as

$$x = (xb)a.$$

Since A is finite-dimensional over F, this right multiplication map is also injective. Now

$$(ab - 1)a = a(ba) - a = 0,$$

so injectivity gives $ab - 1 = 0$. Thus $ab = ba = 1$, and a is invertible. Every non-zero element of A is invertible, so A is a division algebra. □

Exercise 13.10. We can clearly regard an $A/\operatorname{rad} A$-module as being an A-module; on the other hand, a simple A-module is annihilated by $\operatorname{rad}(A)$ and hence can be considered as an $A/\operatorname{rad} A$-module. Show that an A-module is simple iff it is a simple $A/\operatorname{rad}(A)$-module. (Since $A/\operatorname{rad}(A)$ is a semisimple algebra, this implies that the determination of simple modules over arbitrary algebras reduces to the case of semisimple algebras.)

Solution. Let $R = \operatorname{rad}(A)$. If M is a simple A-module, then Theorem 13.23 shows that R annihilates every simple A-module. Hence $RM = 0$, so the action of A on M factors through A/R. The A-submodules of M are then exactly the A/R-submodules of M, because the two actions differ only by the quotient map $A \to A/R$. Therefore M is simple as an A/R-module.
Conversely, suppose that M is simple as an A/R-module. Regard M as an A-module through the quotient map. Then R acts as zero, and again the A-submodules and the A/R-submodules of M are the same. Since there are no non-zero proper A/R-submodules, there are no non-zero proper A-submodules. Hence M is simple as an A-module. □

Exercise 13.11. Let B be a finite-dimensional F-algebra, possibly without unit. Show that if B is a simple algebra whose multiplication is not identically zero, then B is isomorphic with a matrix algebra over a division algebra.

Solution. Since the multiplication on B is not identically zero, we have $B^2 \neq 0$. The subspace B^2, spanned by all products of two elements of B, is a two-sided ideal of B. Since B is simple, this forces $B^2 = B$.
We claim that B has no non-zero nilpotent ideals. If I were a non-zero nilpotent ideal, then simplicity would give $I = B$, so B itself would be nilpotent. But $B^2 = B$ implies $B^m = B$ for every $m \geq 1$, contradicting nilpotence. Thus no such ideal exists. By Theorem 13.26, the algebra B has a unit. Now Theorem 13.17 applies to the simple algebra with unit B, and gives

$$B \cong \mathcal{M}_n(D)$$

for some $n \in \mathbb{N}$ and some division algebra D. □

Further Exercises

Let A be as above, and let M be a finitely generated A-module.

Exercise 13.12. Show that the following objects (exist and) are equal: the intersection of all maximal submodules of M; the smallest submodule of M having semisimple quotient; and $\mathrm{rad}(A)M$. This submodule of M is called the *radical* of M and is denoted $\mathrm{rad}(M)$. (Observe that our two definitions of the symbol $\mathrm{rad}(A)$ agree.)

Solution. Put $R = \mathrm{rad}(A)$. We first show that RM is the smallest submodule of M having semisimple quotient. Since R annihilates M/RM, the quotient M/RM is naturally an A/R-module. The algebra A/R is semisimple by Theorem 13.23, so M/RM is semisimple. Conversely, if $N \leq M$ and M/N is semisimple, then R annihilates every simple summand of M/N, and therefore annihilates all of M/N. Hence $RM \subseteq N$. Thus RM is exactly the smallest submodule of M with semisimple quotient.
It remains to identify this submodule with the intersection of all maximal submodules. If $M = 0$, all three objects are zero. Assume $M \neq 0$. Every maximal submodule L of M has simple quotient M/L, so the preceding paragraph gives $RM \subseteq L$. Hence RM is contained in the intersection of all maximal submodules.
For the reverse inclusion, consider the semisimple module M/RM. In a finite-dimensional semisimple module, the intersection of all maximal submodules is zero: if $0 \neq \overline{m}$, choose a decomposition into simple summands and project onto a simple summand where $\overline{m}$ has non-zero component; the kernel of that projection is a maximal submodule not containing $\overline{m}$. Pulling this statement back along $M \to M/RM$ shows that the intersection of the maximal submodules of M containing RM is exactly RM. Since every maximal submodule contains RM, the intersection of all maximal submodules of M is RM. Therefore all three objects are equal. □

We define a descending series $\mathrm{rad}^n(M)$ of submodules of M by setting $\mathrm{rad}^0(M) = M$ and $\mathrm{rad}^n(M) = \mathrm{rad}(\mathrm{rad}^{n-1}(M))$ for $n \in \mathbb{N}$. It follows from Exercise 13.12 that $\mathrm{rad}^n(M) = (\mathrm{rad}(A))^n M$ for any $n \in \mathbb{N}$. This series is called the *radical series* of M. Observe that $\mathrm{rad}^n(M) = 0$ for some n, since $\mathrm{rad}(A)$ is a nilpotent ideal, and that each successive quotient of the radical series is a semisimple module, with M being semisimple iff $\mathrm{rad}(M) = 0$.

Exercise 13.13. (cont.) Show that M has a unique largest semisimple submodule and that this submodule is equal to $\{m \in M \mid \mathrm{rad}(A)m = 0\}$. We call this submodule the *socle* of M, and we denote it by $\mathrm{soc}(M)$.

Solution. Put $R = \mathrm{rad}(A)$, and set

$$K = \{m \in M \mid Rm = 0\}.$$

This is an A-submodule of M: if $a \in A$, $r \in R$, and $m \in K$, then

$$r(am) = (ra)m = 0$$

because $ra \in R$. Since R annihilates K, the module K is naturally an A/R-module. The algebra A/R is semisimple, so K is semisimple.
Conversely, every simple A-module is annihilated by R, by Theorem 13.23. Hence every simple submodule of M lies in K, and therefore every semisimple submodule of M lies in K. Since K itself is semisimple, it is the unique largest semisimple submodule of M. Thus

$$\mathrm{soc}(M) = K = \{m \in M \mid \mathrm{rad}(A)m = 0\}.$$

□

Exercise 13.14. (cont.) We define an ascending series $\operatorname{soc}^n(M)$ of submodules of M as follows: We let $\operatorname{soc}^0(M) = 0$ and $\operatorname{soc}^1(M) = \operatorname{soc}(M)$, and for $n > 1$ we let $\operatorname{soc}^n(M)$ be the submodule of M such that $\operatorname{soc}^n(M)/\operatorname{soc}^{n-1}(M) = \operatorname{soc}(M/\operatorname{soc}^{n-1}(M))$. Show that we have $\operatorname{soc}^n(M) = \{m \in M \mid \operatorname{rad}(A)^n m = 0\}$ for any $n \in \mathbb{N}$. (This series is called the *socle series* of M. As with the radical series, we observe that $\operatorname{soc}^n(M) = M$ for some n, since $\operatorname{rad}(A)$ is nilpotent, and that each successive quotient of the socle series is semisimple, with M being semisimple iff $\operatorname{soc}(M) = M$.)

Solution. Put $R = \operatorname{rad}(A)$. We prove the formula by induction on n. For $n = 0$, the formula says

$$\operatorname{soc}^0(M) = \{m \in M \mid Am = 0\},$$

and both sides are zero because A has a unit. The case $n = 1$ is exactly Exercise 13.13.
Assume now that $n > 1$ and that

$$\operatorname{soc}^{n-1}(M) = \{m \in M \mid R^{n-1}m = 0\}.$$

By definition, $m \in \operatorname{soc}^n(M)$ iff the coset $m + \operatorname{soc}^{n-1}(M)$ lies in

$$\operatorname{soc}(M/\operatorname{soc}^{n-1}(M)).$$

By Exercise 13.13, this is equivalent to

$$Rm \subseteq \operatorname{soc}^{n-1}(M).$$

Using the induction hypothesis, the last condition is equivalent to $R^{n-1}(Rm) = R^n m = 0$. Therefore

$$\operatorname{soc}^n(M) = \{m \in M \mid R^n m = 0\}$$

for every $n \in \mathbb{N}$. □

Exercise 13.15. (cont.) Show that $\operatorname{rad}^n(M) = 0$ iff $\operatorname{soc}^n(M) = M$; conclude that the radical and socle series of M have the same finite length. If this common length is r, then show that $\operatorname{rad}^n(M) \subseteq \operatorname{soc}^{r-n}(M)$ for every $0 \le n \le r$.

Solution. Put $R = \operatorname{rad}(A)$. By Exercise 13.12, the radical series satisfies

$$\operatorname{rad}^n(M) = R^n M$$

for every $n \in \mathbb{N}$. By Exercise 13.14, the socle series satisfies

$$\operatorname{soc}^n(M) = \{m \in M \mid R^n m = 0\}.$$

Hence

$$\operatorname{rad}^n(M) = 0 \iff R^n M = 0 \iff \text{every } m \in M \text{ satisfies } R^n m = 0 \iff \operatorname{soc}^n(M) = M.$$

Therefore the least n for which the radical series reaches zero is the same as the least n for which the socle series reaches M. Since R is nilpotent, this common length is finite.
Let this common length be r. For $0 \le n \le r$, take $x \in \operatorname{rad}^n(M) = R^n M$. Then

$$R^{r-n}x \subseteq R^{r-n}R^n M = R^r M = \operatorname{rad}^r(M) = 0.$$

By the formula for the socle series, this means $x \in \operatorname{soc}^{r-n}(M)$. Hence

$$\operatorname{rad}^n(M) \subseteq \operatorname{soc}^{r-n}(M)$$

for every $0 \le n \le r$. □

Chapter 6

Group Representations

14 Characters

Standing assumptions

- In this chapter G denotes a finite group.
- All $\mathbb{C}G$-modules are finitely generated, equivalently, finite-dimensional as $\mathbb{C}$-vector spaces.
- $\mathbb{C}$ is the field of complex numbers.

Wedderburn structure of $\mathbb{C}G$

- Since $\mathbb{C}$ is algebraically closed of characteristic zero, Wedderburn theory pins down the structure of $\mathbb{C}G$ precisely.
- The simple summands $\mathcal{M}_{f_i}(\mathbb{C})$ correspond bijectively to isomorphism classes of simple $\mathbb{C}G$-modules S_i, with $\dim_{\mathbb{C}} S_i = f_i$.

Theorem 14.1. *There is some $r \in \mathbb{N}$ and some $f_1, \ldots, f_r \in \mathbb{N}$ such that $\mathbb{C}G \cong \mathcal{M}_{f_1}(\mathbb{C}) \oplus \ldots \oplus \mathcal{M}_{f_r}(\mathbb{C})$ as $\mathbb{C}$-algebras. Furthermore, there are exactly r isomorphism classes of simple $\mathbb{C}G$-modules, and if we let $S_1, \ldots, S_r$ be representatives of these r classes, then we can order the S_i so that $\mathbb{C}G \cong f_1 S_1 \oplus \ldots \oplus f_r S_r$ as $\mathbb{C}G$-modules, where $\dim_{\mathbb{C}} S_i = f_i$ for each i. Any $\mathbb{C}G$-module can be written uniquely in the form $a_1 S_1 \oplus \ldots \oplus a_r S_r$, where the a_i are non-negative integers.*

Proof. Since $\operatorname{char}(\mathbb{C}) = 0$, Corollary 12.8 shows that every finitely generated $\mathbb{C}G$-module is semisimple. In particular, the regular module $\mathbb{C}G$ is semisimple, and hence $\mathbb{C}G$ is a semisimple algebra by Lemma 13.3. Since $\mathbb{C}$ is algebraically closed, Theorem 13.18 gives a $\mathbb{C}$-algebra isomorphism

$$\mathbb{C}G \cong \mathcal{M}_{f_1}(\mathbb{C}) \oplus \cdots \oplus \mathcal{M}_{f_r}(\mathbb{C})$$

for some positive integers $f_1, \ldots, f_r$.
For the algebra on the right, Theorem 13.10 says that there are exactly r isomorphism classes of simple modules, one coming from each matrix-algebra summand. More concretely, by Theorem 13.6, the simple module attached to the summand $\mathcal{M}_{f_i}(\mathbb{C})$ is the natural column module $\mathbb{C}^{f_i}$, and the regular module $\mathcal{M}_{f_i}(\mathbb{C})$ is the direct sum of f_i copies of this module. Transporting these modules across the displayed algebra isomorphism gives simple $\mathbb{C}G$-modules $S_1, \ldots, S_r$ with $\dim_{\mathbb{C}} S_i = f_i$, and as $\mathbb{C}G$-modules we obtain

$$\mathbb{C}G \cong f_1 S_1 \oplus \cdots \oplus f_r S_r.$$

Finally, every finitely generated $\mathbb{C}G$-module is semisimple, so it is a direct sum of simple modules. Since $S_1, \ldots, S_r$ represent all simple isomorphism classes, every such module has the form $a_1 S_1 \oplus \cdots \oplus a_r S_r$ with $a_i \geq 0$. The multiplicities a_i are uniquely determined by Proposition 13.5. $\square$

Degrees of G

- The $\mathbb{C}$-dimensions $f_1, \ldots, f_r$ of the r simple $\mathbb{C}G$-modules are called the *degrees* of G.
- The trivial $\mathbb{C}G$-module $\mathbb{C}$ is one-dimensional and hence simple, so G always has at least one degree equal to 1; by convention we set $f_1 = 1$.

Corollary 14.2. *We have* $\sum_{i=1}^{r} f_i^2 = |G|$.

Proof. Taking $\mathbb{C}$-dimensions in Theorem 14.1 gives

$$|G| = \dim_{\mathbb{C}} \mathbb{C}G = \dim_{\mathbb{C}} (\mathcal{M}_{f_1}(\mathbb{C}) \oplus \cdots \oplus \mathcal{M}_{f_r}(\mathbb{C})) = \sum_{i=1}^{r} \dim_{\mathbb{C}} \mathcal{M}_{f_i}(\mathbb{C}) = \sum_{i=1}^{r} f_i^2.$$

□

Degrees and conjugacy classes

- In fact the degrees of G must divide $|G|$; we shall not use this fact, but we provide a proof of it in the Appendix.
- The next theorem links the number of simple $\mathbb{C}G$-modules to the conjugacy structure of G via the center of $\mathbb{C}G$.

Theorem 14.3. *The number r of simple $\mathbb{C}G$-modules is equal to the number of conjugacy classes of G.*

Proof. Let Z be the center of the algebra $\mathbb{C}G$. By Theorem 14.1, $\mathbb{C}G$ is isomorphic with

$$\mathcal{M}_{f_1}(\mathbb{C}) \oplus \cdots \oplus \mathcal{M}_{f_r}(\mathbb{C}).$$

The center of a direct sum is the direct sum of the centers, and the center of $\mathcal{M}_{f_i}(\mathbb{C})$ consists of the scalar matrices: commuting with the matrix units forces all off-diagonal entries to be zero and all diagonal entries to be equal. Hence $Z \cong \mathbb{C}^r$, so $\dim_{\mathbb{C}} Z = r$.
We now compute the same dimension using the group basis of $\mathbb{C}G$. Write an arbitrary element of $\mathbb{C}G$ as

$$a = \sum_{g \in G} \lambda_g g.$$

If $a \in Z$, then $hah^{-1} = a$ for every $h \in G$. Comparing the coefficients of the basis elements of G, we get

$$\lambda_g = \lambda_{hgh^{-1}}$$

for all $g, h \in G$. Thus the coefficients of a central element are constant on conjugacy classes. Conversely, if the coefficients are constant on conjugacy classes, then $hah^{-1} = a$ for every $h \in G$, so a commutes with every element of G, and hence with every element of $\mathbb{C}G$. Therefore Z has as a basis the class sums

$$\sum_{g \in K} g,$$

where K runs through the conjugacy classes of G. These class sums have disjoint supports in the basis G, so they are linearly independent. It follows that $\dim_{\mathbb{C}} Z$ is the number of conjugacy classes of G. Since we already proved that $\dim_{\mathbb{C}} Z = r$, the result follows. □

Defining characters

- In general there is no natural bijective correspondence between the conjugacy classes of G and the simple $\mathbb{C}G$-modules. However, if G is a symmetric group, then there is such a correspondence, although we shall not develop it here; for an overview, see [13, Section 4.1].
- If U is a $\mathbb{C}G$-module, then each $g \in G$ defines an invertible linear transformation of U that sends $u \in U$ to gu. The *character* of U is the function $\chi_U : G \to \mathbb{C}$, where $\chi_U(g)$ is the trace of this linear transformation of U defined by g.

- For example, $\chi_U(1) = \dim_{\mathbb{C}} U$, since the identity element of G induces the identity transformation on U.
- If $\rho\colon G \to \mathrm{GL}(U)$ is the representation corresponding to U, then $\chi_U(g)$ is just the trace of the map $\rho(g)$.
- Isomorphic $\mathbb{C}G$-modules have equal characters.
- For any $g, h \in G$ the linear transformations of U defined by g and hgh^{-1} are similar and hence have the same trace; so any character of G is constant on each conjugacy class of G.
- For example, let $U = \mathbb{C}G$ and let $g \in G$. With respect to the basis G of $\mathbb{C}G$, $\chi_U(g)$ equals the number of elements $x \in G$ for which $gx = x$. Therefore $\chi_U(1) = |G|$ and $\chi_U(g) = 0$ for every $1 \neq g \in G$. This character is called the *regular character* of G.
- The theory of characters was developed by Frobenius and others, starting in 1896. At first nothing was known about linear representations, let alone modules over the group algebra; Frobenius defined characters as being functions from G to $\mathbb{C}$ satisfying certain properties, but it turned out that his characters were exactly the trace functions of finitely generated $\mathbb{C}G$-modules.
- We will denote the characters of the r simple $\mathbb{C}G$-modules by $\chi_1, \ldots, \chi_r$; these characters will be referred to as the *irreducible characters* of G. Whenever we say that $S_1, \ldots, S_r$ are the distinct simple $\mathbb{C}G$-modules, we implicitly order them so that $\chi_{S_i} = \chi_i$ for each i.
- In keeping with the convention that f_1 denotes the degree of the trivial representation, we let χ_1 be the character of the trivial representation; we call χ_1 the *principal character* of G, and we have $\chi_1(g) = 1$ for all $g \in G$.
- A character of a one-dimensional $\mathbb{C}G$-module is called a *linear character*. Since one-dimensional modules are simple, all linear characters are irreducible. If χ arises from a one-dimensional $\mathbb{C}G$-module U, then for $g, h \in G$ and $u \in U$ we have $gu = \chi(g)u$ and $hu = \chi(h)u$, so $\chi(gh)u = (gh)u = \chi(g)\chi(h)u$; hence χ is a homomorphism from G to $\mathbb{C}^\times$. Conversely, given a homomorphism $\varphi\colon G \to \mathbb{C}^\times$, we can define a one-dimensional $\mathbb{C}G$-module U by $gu = \varphi(g)u$ for $g \in G$ and $u \in U$, and we then have $\chi_U = \varphi$. Therefore, linear characters of G are exactly the same as group homomorphisms from G to $\mathbb{C}^\times$.

Proposition 14.4. *Let U be a $\mathbb{C}G$-module, let $\rho\colon G \to \mathrm{GL}(U)$ be the representation corresponding to U, and let $g \in G$ be of order n. Then:*

(i) *$\rho(g)$ is diagonalizable.*

(ii) *$\chi_U(g)$ equals the sum (with multiplicities) of the eigenvalues of $\rho(g)$.*

(iii) *$\chi_U(g)$ is a sum of $\chi_U(1)$ nth roots of unity.*

(iv) *$\chi_U(g^{-1}) = \overline{\chi_U(g)}$. (Here $\overline{z}$ denotes the complex conjugate of $z \in \mathbb{C}$.)*

(v) *$|\chi_U(g)| \leq \chi_U(1)$.*

(vi) *$\{x \in G \mid \chi_U(x) = \chi_U(1)\}$ is a normal subgroup of G.*

Proof. Since $g^n = 1$, we have $\rho(g)^n = I$, so the minimal polynomial of $\rho(g)$ divides $x^n - 1$. Over $\mathbb{C}$, the polynomial $x^n - 1$ splits into distinct linear factors. Hence the minimal polynomial of $\rho(g)$ also splits into distinct linear factors, and therefore $\rho(g)$ is diagonalizable. This proves (i).
Once $\rho(g)$ is diagonalizable, we may choose a basis of U consisting of eigenvectors for $\rho(g)$. In such a basis the matrix of $\rho(g)$ is diagonal, so its trace is the sum of its eigenvalues, counted with multiplicity. This proves (ii). Since every eigenvalue λ satisfies $\lambda^n = 1$, each eigenvalue is an nth root of unity. There are $\dim_{\mathbb{C}} U = \chi_U(1)$ eigenvalues with multiplicity, so $\chi_U(g)$ is a sum of $\chi_U(1)$ nth roots of unity. This proves (iii).
The eigenvalues of $\rho(g^{-1}) = \rho(g)^{-1}$ are the inverses of the eigenvalues of $\rho(g)$. If λ is a root of unity, then $\lambda^{-1} = \overline{\lambda}$. Hence the eigenvalues of $\rho(g^{-1})$ are the complex conjugates of the eigenvalues of $\rho(g)$, and taking traces gives $\chi_U(g^{-1}) = \overline{\chi_U(g)}$. This proves (iv).
Part (iii) writes $\chi_U(g) = \lambda_1 + \cdots + \lambda_m$, where $m = \chi_U(1)$ and each $|\lambda_i| = 1$. Therefore

$$|\chi_U(g)| \leq |\lambda_1| + \cdots + |\lambda_m| = m = \chi_U(1),$$

proving (v).
It remains to prove (vi). Let

$$K = \{x \in G \mid \chi_U(x) = \chi_U(1)\}.$$

For $x \in G$, the preceding parts write $\chi_U(x)$ as a sum of $\chi_U(1)$ roots of unity. Such a sum can equal the positive real number $\chi_U(1)$ only when every summand is 1. Since $\rho(x)$ is diagonalizable, this happens exactly when $\rho(x) = I$. Thus $K = \ker\rho$, and hence $K \trianglelefteq G$. □

Algebra of characters

- If χ and ψ are characters of G, we can define new functions $\chi + \psi$ and $\chi\psi$ from G to $\mathbb{C}$ by $(\chi + \psi)(g) = \chi(g) + \psi(g)$ and $(\chi\psi)(g) = \chi(g)\psi(g)$ for $g \in G$.
- These new functions obtained from characters are not, a priori, characters themselves.
- Given a scalar $\lambda \in \mathbb{C}$, we can also define a new function $\lambda\chi\colon G \to \mathbb{C}$ by $(\lambda\chi)(g) = \lambda\chi(g)$, so we view characters of G as elements of a $\mathbb{C}$-vector space of functions from G to $\mathbb{C}$.

Proposition 14.5. *The irreducible characters of G are, as functions from G to $\mathbb{C}$, linearly independent over $\mathbb{C}$.*

Proof. For each irreducible character χ_i, extend χ_i linearly from the basis G of $\mathbb{C}G$ to a linear map $\mathbb{C}G \to \mathbb{C}$. Equivalently, $\chi_i(a)$ is the trace of the linear transformation of S_i defined by $a \in \mathbb{C}G$.
Using the algebra decomposition from Theorem 14.1, let e_i be the element that is the identity matrix in the ith summand $\mathcal{M}_{f_i}(\mathbb{C})$ and zero in all other summands. Then e_i acts as the identity on S_i, so $\chi_i(e_i) = \dim_{\mathbb{C}} S_i = f_i$. If $j \neq i$, then e_i acts as zero on S_j, so $\chi_j(e_i) = 0$.
Suppose that $\lambda_1, \ldots, \lambda_r \in \mathbb{C}$ satisfy

$$\lambda_1\chi_1 + \cdots + \lambda_r\chi_r = 0$$

as functions on G. Since G is a basis of $\mathbb{C}G$, the same equality holds after extending the characters linearly to $\mathbb{C}G$. Evaluating at e_i gives

$$0 = \sum_{j=1}^{r} \lambda_j \chi_j(e_i) = \lambda_i f_i.$$

Since $f_i > 0$, we get $\lambda_i = 0$. This holds for every i, so the irreducible characters are linearly independent. □

Lemma 14.6. $\chi_{U\oplus V} = \chi_U + \chi_V$ *for any $\mathbb{C}G$-modules U and V.*

Proof. Fix $g \in G$. Choose a basis of $U \oplus V$ whose first vectors form a basis of $U \oplus 0$ and whose remaining vectors form a basis of $0 \oplus V$. With respect to this basis, the linear transformation induced by g has block diagonal form

$$\begin{pmatrix} \rho_U(g) & 0 \\ 0 & \rho_V(g) \end{pmatrix}.$$

The trace of a block diagonal matrix is the sum of the traces of its diagonal blocks. Therefore

$$\chi_{U\oplus V}(g) = \chi_U(g) + \chi_V(g).$$

Since this holds for every $g \in G$, we have $\chi_{U\oplus V} = \chi_U + \chi_V$. □

Characters distinguish modules

- Characters of $\mathbb{C}G$-modules suffice to distinguish $\mathbb{C}G$-modules up to isomorphism.

Theorem 14.7. *If $S_1, \ldots, S_r$ are the distinct simple $\mathbb{C}G$-modules, then the character of the $\mathbb{C}G$-module $a_1S_1 \oplus \ldots \oplus a_rS_r$ (where the a_i are non-negative integers) is $a_1\chi_1 + \ldots + a_r\chi_r$. Consequently, two $\mathbb{C}G$-modules are isomorphic iff their characters are equal.*

Proof. Repeatedly applying Lemma 14.6 gives

$$\chi_{a_1S_1\oplus\cdots\oplus a_rS_r} = a_1\chi_1 + \cdots + a_r\chi_r.$$

Thus the displayed formula for the character of a decomposed module holds.
Isomorphic modules have equal characters, so only the converse needs proof. Suppose that U and V are $\mathbb{C}G$-modules with $\chi_U = \chi_V$. By Theorem 14.1, write

$$U \cong a_1S_1 \oplus \cdots \oplus a_rS_r, \qquad V \cong b_1S_1 \oplus \cdots \oplus b_rS_r.$$

Taking characters and using the first part of the proof gives

$$0 = \chi_U - \chi_V = (a_1 - b_1)\chi_1 + \cdots + (a_r - b_r)\chi_r.$$

Proposition 14.5 says that the χ_i are linearly independent, so $a_i - b_i = 0$ for every i. Hence $a_i = b_i$ for every i, and therefore $U \cong V$. □

Tensor products, duals, and Hom

- Although each character determines a $\mathbb{C}G$-module up to isomorphism by Theorem 14.7, there is no generic way to construct a module from its corresponding character, and so in some sense information is lost by studying characters in lieu of modules.
- Characters turn out to be an efficient means of translating information about the ordinary representation theory of G to information about G itself; from now on we focus on characters rather than modules.
- Recall from Section 12 that if U and V are $\mathbb{C}G$-modules, then $U \otimes_{\mathbb{C}} V$ and $\mathrm{Hom}_{\mathbb{C}}(U, V)$, written here as $U \otimes V$ and $\mathrm{Hom}(U, V)$, admit natural $\mathbb{C}G$-module structures.

Proposition 14.8. *Let U and V be $\mathbb{C}G$-modules. Then:*

(i) $\chi_{U\otimes V} = \chi_U\chi_V$.

(ii) $\chi_{U^*} = \overline{\chi_U}$. *(Recall that $U^* = \mathrm{Hom}_{\mathbb{C}}(U, \mathbb{C})$.)*

(iii) $\chi_{\mathrm{Hom}(U,V)} = \overline{\chi_U}\chi_V$.

Proof. We prove the three assertions separately.
(i) Fix $g \in G$. By Proposition 14.4, the transformations induced by g on U and V are diagonalizable. Choose eigenbases $u_1, \ldots, u_m$ of U and $v_1, \ldots, v_n$ of V, with

$$gu_i = \lambda_iu_i, \qquad gv_j = \mu_jv_j.$$

Then $u_i \otimes v_j$, with $1 \le i \le m$ and $1 \le j \le n$, is a basis of $U \otimes V$, and

$$g(u_i \otimes v_j) = gu_i \otimes gv_j = \lambda_i\mu_j(u_i \otimes v_j).$$

Hence the eigenvalues of g on $U \otimes V$ are the products $\lambda_i\mu_j$, and so

$$\chi_{U\otimes V}(g) = \sum_{i,j}\lambda_i\mu_j = \left(\sum_i\lambda_i\right)\left(\sum_j\mu_j\right) = \chi_U(g)\chi_V(g).$$

This proves (i).
(ii) Again fix $g \in G$, and choose an eigenbasis $u_1, \ldots, u_m$ of U with $gu_i = \lambda_iu_i$. Let $\varphi_1, \ldots, \varphi_m$ be the dual basis of U^*. Since each λ_i is a root of unity, $\lambda_i^{-1} = \overline{\lambda_i}$. For each i, j we have

$$(g\varphi_i)(u_j) = \varphi_i(g^{-1}u_j) = \varphi_i(\lambda_j^{-1}u_j) = \overline{\lambda_j}\delta_{ij}.$$

Thus $g\varphi_i = \overline{\lambda_i}\varphi_i$. Therefore the eigenvalues of g on U^* are $\overline{\lambda_1}, \ldots, \overline{\lambda_m}$, and

$$\chi_{U^*}(g) = \overline{\lambda_1} + \cdots + \overline{\lambda_m} = \overline{\chi_U(g)}.$$

This proves (ii).
(iii) Proposition 12.7 gives an isomorphism of $\mathbb{C}G$-modules

$$\mathrm{Hom}(U, V) \cong U^* \otimes V.$$

Isomorphic modules have equal characters, so by (i) and (ii),

$$\chi_{\mathrm{Hom}(U,V)} = \chi_{U^*\otimes V} = \chi_{U^*}\chi_V = \overline{\chi_U}\chi_V.$$

□

Virtual characters and class functions

- A *virtual character* of G is a $\mathbb{Z}$-linear combination of the irreducible characters of G. (Some authors prefer the term "generalized character.") Characters are virtual characters by Theorem 14.7.

Corollary 14.9. *The virtual characters of G form a ring.*

Proof. Addition and additive inverses are immediate from the definition of virtual characters as $\mathbb{Z}$-linear combinations of the irreducible characters. The only point to check is closure under multiplication.
If χ and ψ are ordinary characters, then Proposition 14.8 shows that $\chi\psi$ is the character of a tensor product module. Hence $\chi\psi$ is again an ordinary character, and by Theorem 14.7 it is a non-negative integer combination of the irreducible characters. Now if

$$\alpha = \sum_i a_i\chi_i, \qquad \beta = \sum_j b_j\chi_j$$

are virtual characters, then

$$\alpha\beta = \sum_{i,j} a_i b_j(\chi_i\chi_j),$$

and each product $\chi_i\chi_j$ is an integer combination of irreducible characters by the preceding paragraph. Thus $\alpha\beta$ is again a virtual character. The principal character is the multiplicative identity, so the virtual characters form a ring. □

Class functions

- A *class function* on G is a function from G to $\mathbb{C}$ whose value within any conjugacy class is constant.
- Characters of $\mathbb{C}G$-modules are class functions.
- The set of class functions on G forms a $\mathbb{C}$-vector space of dimension r, where r is the number of conjugacy classes of G; an obvious basis is the set of functions taking the value 1 on a single conjugacy class and 0 on all other classes.

Proposition 14.10. *The irreducible characters of G form a basis for the space of class functions on G.*

Proof. Every irreducible character is a character, and hence is a class function. By Proposition 14.5, the irreducible characters $\chi_1, \dots, \chi_r$ are linearly independent. By Theorem 14.3, the number r is also the number of conjugacy classes of G, which is the dimension of the space of class functions on G. Therefore we have r linearly independent vectors in an r-dimensional vector space, so the irreducible characters form a basis. □

Inner product on class functions

- If α and β are class functions on G, then their *inner product* is the complex number

$$(\alpha, \beta) = \frac{1}{|G|}\sum_{g\in G}\alpha(g)\overline{\beta(g)}.$$

- This function $(\,\cdot\,,\,\cdot\,)$ is, as the name suggests, an inner product on the space of class functions:
  - $(\alpha, \alpha) \geq 0$ for all α, and $(\alpha, \alpha) = 0$ iff $\alpha = 0$.
  - $(\alpha, \beta) = \overline{(\beta, \alpha)}$ for all α, β.
  - $(\lambda\alpha, \beta) = \lambda(\alpha, \beta)$ for all α, β and all $\lambda \in \mathbb{C}$.
  - $(\alpha_1 + \alpha_2, \beta) = (\alpha_1, \beta) + (\alpha_2, \beta)$ for all $\alpha_1, \alpha_2, \beta$.
- The following are easy consequences of the above properties:
  - $(\alpha, \lambda\beta) = \overline{\lambda}(\alpha, \beta)$ for all α, β and all $\lambda \in \mathbb{C}$.
  - $(\alpha, \beta_1 + \beta_2) = (\alpha, \beta_1) + (\alpha, \beta_2)$ for all α, β_1, β_2.
- We need a small piece of notation: if U is a $\mathbb{C}G$-module, the set of elements of U on which G acts trivially is a $\mathbb{C}G$-submodule, denoted $U^G = \{u \in U \mid gu = u \text{ for all } g \in G\}$.

Lemma 14.11. *If U is a $\mathbb{C}G$-module, then* $\dim_{\mathbb{C}} U^G = \frac{1}{|G|}\sum_{g\in G}\chi_U(g)$.

Proof. Let
$$a = \frac{1}{|G|}\sum_{g\in G} g \in \mathbb{C}G,$$
and let T be the linear transformation of U defined by the action of a. For any $h \in G$, we have $ha = a$, because left multiplication by h merely permutes the elements of G inside the sum. Also,
$$a^2 = \frac{1}{|G|^2}\sum_{g,k\in G} gk = \frac{1}{|G|}\sum_{y\in G} y = a,$$
since each $y \in G$ occurs as gk exactly $|G|$ times. Hence $T^2 = T$. The polynomial $x^2 - x = x(x-1)$ has distinct roots, so T is diagonalizable with eigenvalues only 0 and 1.
Let U_1 be the eigenspace of T with eigenvalue 1. If $u \in U_1$, then for every $h \in G$ we have
$$hu = hTu = (ha)u = au = Tu = u,$$
so $u \in U^G$. Conversely, if $u \in U^G$, then
$$Tu = au = \frac{1}{|G|}\sum_{g\in G} gu = \frac{1}{|G|}\sum_{g\in G} u = u,$$
so $u \in U_1$. Thus $U^G = U_1$.
Since T is diagonalizable with eigenvalues 0 and 1, its trace is the dimension of its 1-eigenspace, namely $\dim_{\mathbb{C}} U^G$. On the other hand, by linearity of trace,
$$\mathrm{tr}(T) = \frac{1}{|G|}\sum_{g\in G}\chi_U(g).$$
Therefore
$$\dim_{\mathbb{C}} U^G = \frac{1}{|G|}\sum_{g\in G}\chi_U(g).$$
□

Theorem 14.12. *We have $(\chi_U, \chi_V) = \dim_{\mathbb{C}} \mathrm{Hom}_{\mathbb{C}G}(U, V)$ for any $\mathbb{C}G$-modules U and V.*

Proof. First we identify a fixed-point space. Inside the $\mathbb{C}G$-module $\mathrm{Hom}(U, V)$, the $\mathbb{C}G$-homomorphisms are exactly the G-fixed maps. Recall that the G-action is given by
$$(g\varphi)(u) = g\varphi(g^{-1}u).$$
If $\varphi \in \mathrm{Hom}_{\mathbb{C}G}(U, V)$, then
$$(g\varphi)(u) = g\varphi(g^{-1}u) = gg^{-1}\varphi(u) = \varphi(u),$$
so φ is fixed by every $g \in G$. Conversely, if $g\varphi = \varphi$ for every $g \in G$, then applying this equality to gu gives
$$\varphi(gu) = (g\varphi)(gu) = g\varphi(u),$$
so φ is G-linear. Hence
$$\mathrm{Hom}_{\mathbb{C}G}(U, V) = \mathrm{Hom}(U, V)^G.$$
By Lemma 14.11 and Proposition 14.8, we now get
$$\dim_{\mathbb{C}} \mathrm{Hom}_{\mathbb{C}G}(U, V) = \frac{1}{|G|}\sum_{g\in G}\chi_{\mathrm{Hom}(U,V)}(g) = \frac{1}{|G|}\sum_{g\in G}\overline{\chi_U(g)}\chi_V(g) = (\chi_V, \chi_U).$$
The right-hand side is equal to a dimension, so it is a real number. By conjugate symmetry of the inner product,
$$(\chi_U, \chi_V) = \overline{(\chi_V, \chi_U)} = (\chi_V, \chi_U).$$
Therefore $(\chi_U, \chi_V) = \dim_{\mathbb{C}} \mathrm{Hom}_{\mathbb{C}G}(U, V)$. □

Exercises

Throughout these exercises, G denotes a finite group, and all modules are finitely generated.

Exercise 14.1. Let A be a semisimple $\mathbb{C}$-algebra. Let $[A, A]$ be the subspace of A spanned by the set $\{ab - ba \mid a, b \in A\}$. Show that the codimension of $[A, A]$ in A (which is just the dimension of $A/[A, A]$) is equal to the number of isomorphism classes of simple A-modules.

Solution. □

Exercise 14.2. (cont.) Let $A = \mathbb{C}G$, and let $g_1, \ldots, g_r$ be representatives of the r conjugacy classes of G. Show that $\{g_i + [A, A] \mid 1 \leq i \leq r\}$ is a basis of $A/[A, A]$.

Solution. □

Exercise 14.3. Show that if S is a simple $\mathbb{C}G$-module and U is a one-dimensional $\mathbb{C}G$-module, then $S \otimes U$ is simple.

Solution. □

Exercise 14.4. Show that the set of isomorphism classes of the one-dimensional $\mathbb{C}G$-modules forms a group under the taking of tensor products.

Solution. □

Exercise 14.5. Let χ be an irreducible character of G. Show that if λ is any $|G|$th root of unity, then $\{x \in G \mid \chi(x) = \lambda\chi(1)\} \trianglelefteq G$.

Solution. □

Exercise 14.6. Show that a $\mathbb{C}G$-module U is simple iff its dual U^* is simple. Conclude that the complex conjugate of an irreducible character is again an irreducible character.

Solution. □

If F is any field and U is any FG-module, then we define the character of U to be an F-valued function on U whose value at $g \in G$ is the trace of the F-endomorphism of U induced by g, exactly as was done in this section in the case $F = \mathbb{C}$.

Exercise 14.7. Suppose that F has characteristic $p > 0$. Give an example to demonstrate that two FG-modules that have the same character need not be isomorphic.

Solution. □

Exercise 14.8. Suppose that F has characteristic $p > 0$. Show that if U is an FG-module then we have $\chi_U(g) = \chi_U(g_{p'})$ for any $g \in G$, where $g_{p'}$ is the p'-part of g as defined in Exercise 1.4.

Solution. □

15 The Character Table

Definition of the Character Table

- Any character of G is a $\mathbb{Z}$-linear combination of the r irreducible characters $\chi_1, \ldots, \chi_r$, where r is the number of conjugacy classes of G (Theorem 14.3).
- Each irreducible character is determined by its values on the r conjugacy classes, so the irreducible characters are completely determined by an $r \times r$ array.
- The *character table* of G is this $r \times r$ array $\mathcal{X} = (\chi_i(g_j))_{1 \le i,j \le r}$, where $g_1, \ldots, g_r$ are conjugacy class representatives.
- The character table is well-defined only up to reordering of rows and columns.
- By convention, $g_1 = 1$, so the first column consists of the degrees of G.
- We write $\mathcal{X}$ in the form

$$\begin{array}{c|cccc} & 1 & k_2 & \cdots & k_r \\ & 1 & g_2 & \cdots & g_r \\ \hline \chi_1 & 1 & 1 & \cdots & 1 \\ \chi_2 & f_2 & \chi_2(g_2) & \cdots & \chi_2(g_r) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \chi_r & f_r & \chi_r(g_2) & \cdots & \chi_r(g_r) \end{array}$$

  where the f_i are the degrees of G, and $k_i = |G : C_G(g_i)|$ is the order of the conjugacy class of g_i (Proposition 3.13).
- By convention χ_1 is always the principal character.

Orthogonality of Rows

Theorem (Row Orthogonality Theorem)**.** *$(\chi_i, \chi_j) = \delta_{ij}$ for any i and j.*

Proof. □

In other words, this theorem asserts that

$$\delta_{ij} = \frac{1}{|G|} \sum_{g \in G} \chi_i(g)\overline{\chi_j(g)} = \frac{1}{|G|} \sum_{t=1}^{r} k_t \chi_i(g_t)\overline{\chi_j(g_t)}$$

for any i and j, where the g_t are conjugacy class representatives and the k_t are the orders of the conjugacy classes. The rows of the character table, considered as vectors in $\mathbb{C}^r$, are orthogonal with respect to an inner product that differs slightly from the standard one.

Corollary 15.1. *The irreducible characters of G form an orthonormal basis for the vector space of class functions on G.*

Proof. □

Corollary 15.2. *If $\alpha = \sum_i a_i \chi_i$ and $\beta = \sum_j b_j \chi_j$ are virtual characters of G, then $(\alpha, \beta) = \sum_i a_i b_i$.*

Proof. □

Corollary 15.3. *If α is a character of G and $n \in \{1, 2, 3\}$, then $(\alpha, \alpha) = n$ iff α is a sum of n irreducible characters.*

Proof. □

Corollary 15.4. *If α is a virtual character of G, then each χ_j appears with coefficient (α, χ_j) in the unique expression of α as a $\mathbb{Z}$-linear combination of the irreducible characters of G.*

Proof. □

Proposition 15.5. *If α is a linear character of G and χ is an irreducible character of G, then $\alpha\chi$ is an irreducible character of G.*

Proof. □

Orthogonality of Columns

Theorem (Column Orthogonality Theorem). *If $g_1, \ldots, g_r$ are a set of conjugacy class representatives of G, and $k_1, \ldots, k_r$ are the orders of the conjugacy classes, then for any $1 \le i, j \le r$ we have*

$$\sum_{t=1}^{r} \chi_t(g_i)\overline{\chi_t(g_j)} = \frac{|G|}{k_i}\delta_{ij} = |C_G(g_i)|\delta_{ij}.$$

Proof. □

Characters and Quotient Groups

Lemma 15.6. *Let $N \trianglelefteq G$, and let U be a $\mathbb{C}(G/N)$-module. Then U admits a canonical $\mathbb{C}G$-module structure, with a subspace of U being a $\mathbb{C}G$-submodule iff it is a $\mathbb{C}(G/N)$-submodule. If ψ is the character of the $\mathbb{C}(G/N)$-module U, then the character of the $\mathbb{C}G$-module U is $\psi \circ \eta$, where $\eta\colon G \to G/N$ is the natural map.*

Proof. □

We define $K_\chi = \{x \in G \mid \chi(x) = \chi(1)\}$ for any character χ of G; we have $K_\chi \trianglelefteq G$ by part (vi) of Proposition 14.4. We call K_χ the *kernel* of χ, as it is the kernel of the corresponding representation. We write K_i instead of K_{χ_i}.

Proposition 15.7. *The normal subgroups of G are exactly the sets of the form $\bigcap_{i \in I} K_i$ for some $I \subseteq \{1, \ldots, r\}$.*

Proof. □

Corollary 15.8. *G is simple iff the only irreducible character χ_i for which $\chi_i(g) = \chi_i(1)$ for some $1 \ne g \in G$ is the principal character χ_1.*

Proof. □

Corollary 15.9. *The character table of G can be used to determine whether or not G is solvable.*

Proof. □

We define $Z_\chi = \{x \in G \mid |\chi(x)| = \chi(1)\}$ for any character χ of G. Again we write Z_i instead of Z_{χ_i}.

Lemma 15.10. *$Z_\chi \le G$ for any character χ of G, and if in addition χ is irreducible, then $Z_\chi/K_\chi = Z(G/K_\chi)$.*

Proof. □

Corollary 15.11. *If G is non-abelian and simple, then $Z_i = 1$ for all $i > 1$.*

Proof. □

Proposition 15.12. $Z(G) = \bigcap_{i=1}^{r} Z_i$.

Proof. □

Lemma 15.13. *If $N \trianglelefteq G$, then the irreducible characters of G/N can be determined from those of G.*

Proof. □

Corollary 15.14. *The character table of G can be used to determine whether or not G is nilpotent.*

Proof. □

Observe that in Corollary 15.9 we assumed that we knew both the irreducible characters of G and the orders of the conjugacy classes, whereas in Corollary 15.14 we did not need to know the orders of the classes. As evidenced by Lemma 15.13, there may be circumstances in which we can construct a character table without knowing the orders of the conjugacy classes. There is a theorem of G. Higman (see [15, p. 136]) which asserts that if we know the irreducible characters of G, but not the orders of the conjugacy classes, then we can at least determine the sets of prime divisors of the orders of the conjugacy classes.

Examples: Constructing Character Tables

The remainder of this section consists of a series of examples in which we develop methods of finding characters and use these methods to construct the character tables of various groups.

Example 15.1. Let $G = \langle g \rangle \cong \mathbb{Z}_n$ for some $n \in \mathbb{N}$, and let λ be a primitive nth root of unity. For each $1 \leq i \leq n$, let V_i be a one-dimensional $\mathbb{C}$-vector space, and let g act on V_i by multiplication by λ^{i-1}; since G is cyclic, this definition completely determines a $\mathbb{C}G$-module structure on V_i. Each V_i, being one-dimensional, is a simple $\mathbb{C}G$-module. If χ_i is the character of V_i, then $\chi_i(g) = \lambda^{i-1}$, and hence $\chi_i(g^a) = \lambda^{a(i-1)}$ for any a. The characters $\chi_1, \ldots, \chi_n$ are n distinct linear characters of G, and since G can have at most $|G| = n$ irreducible characters, we see that the χ_i are precisely the irreducible characters of G. Thus, the character table of G is

$$\begin{array}{c|cccccc} & 1 & 1 & 1 & \cdots & 1 \\ & 1 & g & g^2 & \cdots & g^{n-1} \\ \hline \chi_1 & 1 & 1 & 1 & \cdots & 1 \\ \chi_2 & 1 & \lambda & \lambda^2 & \cdots & \lambda^{n-1} \\ \chi_3 & 1 & \lambda^2 & \lambda^4 & \cdots & \lambda^{n-2} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \chi_n & 1 & \lambda^{n-1} & \lambda^{n-2} & \cdots & \lambda \end{array}$$

Observe that the set $\{\chi_1, \ldots, \chi_n\}$ is a cyclic group under multiplication of characters, with generator χ_2; we have $\chi_i = \chi_2^{i-1}$ for any i.

Example 15.2. Let G and H be groups with respective irreducible characters $\chi_1, \ldots, \chi_r$ and $\psi_1, \ldots, \psi_s$. We wish to determine the irreducible characters of $G \times H$. We see that two elements (x, y) and (x', y') of $G \times H$ are conjugate iff x and x' are conjugate in G and y and y' are conjugate in H. Since by Theorem 14.3, G and H have r and s conjugacy classes, respectively, we conclude that $G \times H$ has rs conjugacy classes, with each class of $G \times H$ being the product of a class of G and a class of H; hence by Theorem 14.3, $G \times H$ has rs irreducible characters.

Let $S_1, \ldots, S_r$ and $T_1, \ldots, T_s$ be the distinct simple $\mathbb{C}G$- and $\mathbb{C}H$-modules, respectively. For each i and j, we give $S_i \otimes T_j$ the structure of a $\mathbb{C}(G \times H)$-module by $(g, h)(s \otimes t) = gs \otimes ht$ and linear extension. Let τ_{ij} be the character of $S_i \otimes T_j$ for each i and j. By imitating the proof of part (i) of Proposition 14.8, we find that $\tau_{ij}(g, h) = \chi_i(g)\psi_j(h)$; we adopt the notation $\tau_{ij} = \chi_i \times \psi_j$. We note that the τ_{ij} are distinct and that we can recover the χ_i and ψ_j from the τ_{ij} by appropriate restriction. Now for any i, i', j, j' we have

$$\begin{aligned}(\tau_{ij}, \tau_{i'j'}) &= \frac{1}{|G \times H|} \sum_{(g,h) \in G \times H} \tau_{ij}(g, h)\overline{\tau_{i'j'}(g, h)} \\ &= \frac{1}{|G||H|} \sum_{g \in G} \sum_{h \in H} \chi_i(g)\psi_j(h)\overline{\chi_{i'}(g)\psi_{j'}(h)} \\ &= \left(\frac{1}{|G|} \sum_{g \in G} \chi_i(g)\overline{\chi_{i'}(g)}\right)\left(\frac{1}{|H|} \sum_{h \in H} \psi_j(h)\overline{\psi_{j'}(h)}\right) \\ &= (\chi_i, \chi_{i'})(\psi_j, \psi_{j'}) = \delta_{ii'}\delta_{jj'}\end{aligned}$$

by Row Orthogonality Theorem. In particular, we have $(\tau_{ij}, \tau_{ij}) = 1$ for each i and j, which by Corollary 15.3 implies that τ_{ij} is an irreducible character. Therefore the τ_{ij} are the rs irreducible characters of $G \times H$; that is, the set of irreducible characters of $G \times H$ is $\{\chi_i \times \psi_j\}_{i,j}$.

Example 15.3. Suppose that G is abelian. Then every conjugacy class of G contains exactly one element, and consequently G has $|G|$ irreducible characters by Theorem 14.3. But $\sum_{i=1}^{|G|} f_i^2 = |G|$ by Corollary 14.2, so we must have $f_i = 1$ for each i. Therefore all irreducible characters of G are linear. By Corollary 8.8, G is a direct product of cyclic p-groups; let these cyclic p-groups be of orders $p_1^{a_1}, \ldots, p_t^{a_t}$ with respective generators $g_1, \ldots, g_t$. We could determine the character table of G from those of the cyclic p-groups using Examples 15.1 and 15.2, but there is an alternate method, which we now describe. A linear character χ of G is just a homomorphism from G to $\mathbb{C}^\times$, and hence to define such a χ it suffices to specify $\chi(g_i)$ for each i, with the only restriction on $\chi(g_i)$ being that it is a $p_i^{a_i}$-th root of unity.

Therefore, there is a bijective correspondence between irreducible characters of G and ordered t-tuples $(\lambda_1, \ldots, \lambda_t)$ in which each λ_i is a $p_i^{a_i}$-th root of unity.

As a simple example, let $G = \langle a, b \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Both a and b have order 2, so by the above paragraph we see that the four irreducible characters of G correspond to the ordered pairs $(1,1), (1,-1), (-1,1)$, and $(-1,-1)$. Therefore, the character table for G is

	1	1	1	1
	1	a	b	ab
χ_1	1	1	1	1
χ_2	1	1	-1	-1
χ_3	1	-1	1	-1
χ_4	1	-1	-1	1

Here the second and third columns correspond to the ordered pairs given above, and the fourth column is wholly determined by the previous two columns.

Example 15.4. Let X be a finite G-set. We saw in Section 12 that the vector space $\mathbb{C}X$ having basis X is a $\mathbb{C}G$-module. We wish to determine the character π of $\mathbb{C}X$. Consider the matrix, with respect to the basis X, of the linear transformation of $\mathbb{C}X$ defined by $g \in G$. Since $gx \in X$ for every $x \in X$, this matrix is a permutation matrix; but $\pi(g)$ is the sum of the diagonal entries of this matrix, so we conclude that $\pi(g)$ equals the number of elements of X which are fixed by g.

Let χ_1 be the principal character of G. We have

$$\begin{aligned}
(\pi, \chi_1) &= \frac{1}{|G|} \sum_{g \in G} \pi(g)\chi_1(g) = \frac{1}{|G|} \sum_{g \in G} \pi(g) \\
&= \frac{1}{|G|} \sum_{g \in G} |\{x \in X \mid gx = x\}| \\
&= \frac{1}{|G|} |\{(g, x) \in G \times X \mid gx = x\}| \\
&= \frac{1}{|G|} \sum_{x \in X} |\{g \in G \mid gx = x\}|.
\end{aligned}$$

For each $x \in X$, let G_x be the stabilizer of x; using Corollary 3.5, we now have

$$(\pi, \chi_1) = \sum_{x \in X} \frac{|G_x|}{|G|} = \sum_{\text{orbits } \mathcal{O}} \sum_{x \in \mathcal{O}} \frac{1}{|G : G_x|} = \sum_{\text{orbits } \mathcal{O}} \sum_{x \in \mathcal{O}} \frac{1}{|\mathcal{O}|} = \sum_{\text{orbits } \mathcal{O}} |\mathcal{O}| \cdot \frac{1}{|\mathcal{O}|} = \sum_{\text{orbits } \mathcal{O}} 1,$$

which allows us to conclude that (π, χ_1) is equal to the number of orbits of X under the action of G. (This result is often attributed to Burnside, although as argued in [21] it would be more accurate to call this result the Cauchy-Frobenius lemma.) In particular, if G acts transitively on X, then we see via Corollary 15.4 that the unique expression of the character $\pi - \chi_1$ as a linear combination of the irreducible characters of G does not involve χ_1.

Example 15.5. The group G/G' is abelian by Proposition 2.6, and hence (as seen in Example 15.3) all of its irreducible characters are linear. It follows from Lemma 15.6 that each linear character of G/G' can be lifted to a linear character of G, and that the lifts of distinct characters are distinct. Now let χ be a linear character of G, so that $\chi\colon G \to \mathbb{C}^\times$ is a homomorphism whose kernel is K_χ. Then G/K_χ is abelian, being isomorphic with X, a subgroup of $\mathbb{C}^\times$, and hence $G' \leq K_\chi$ by Proposition 2.6. We can now define a linear character ψ of G/G' whose lift is χ by $\psi(gG') = \chi(g)$. We conclude that every linear character of G is the lift of a linear character of G/G'.

Example 15.6. Consider S_3. The conjugacy classes of S_3 are

$$\{1\}, \qquad \{(1\,2), (1\,3), (2\,3)\}, \qquad \{(1\,2\,3), (1\,3\,2)\}.$$

This follows from Proposition 1.10. We have $S_3/A_3 \cong \mathbb{Z}_2$, so S_3 has two linear characters, which are lifted from the characters of $\mathbb{Z}_2$ as in Lemma 15.6. Letting these characters be χ_1 and χ_2, we have

	1	3	2
	1	$(1\,2)$	$(1\,2\,3)$
χ_1	1	1	1
χ_2	1	-1	1
χ_3			

Since $6 = |S_3| = f_1^2 + f_2^2 + f_3^2 = 2 + f_3^2$ by Corollary 14.2, we have $f_3 = 2$. Column Orthogonality Theorem now gives

$$0 = \sum_{i=1}^{3} f_i \chi_i((1\,2)) = 1 \cdot 1 + 1 \cdot (-1) + 2\chi_3((1\,2))$$

and

$$0 = \sum_{i=1}^{3} f_i \chi_i((1\,2\,3)) = 1 \cdot 1 + 1 \cdot 1 + 2\chi_3((1\,2)),$$

from which we see that the character table of S_3 is

	1	3	2
	1	$(1\,2)$	$(1\,2\,3)$
χ_1	1	1	1
χ_2	1	-1	1
χ_3	2	0	-1

Example 15.7. Consider S_4. Recall that S_4 has a normal subgroup of order 4, namely

$$K = \{1,\ (1\,2)(3\,4),\ (1\,3)(2\,4),\ (1\,4)(2\,3)\}.$$

The subgroup S_3 of S_4 intersects K trivially, and $|S_4| = |S_3||K|$; therefore $S_4 = K \rtimes S_3$. In particular, we have $S_4/K \cong S_3$, and thus we can obtain irreducible characters of S_4 from the irreducible characters of S_3 as in Lemma 15.6.

By Proposition 1.10, each conjugacy class of S_4 consists of all elements having a given cycle structure. Hence S_4 has 5 classes, with representatives 1, $(1\,2)(3\,4)$, $(1\,2\,3)$, $(1\,2)$, and $(1\,2\,3\,4)$; the orders of the classes are 1, 3, 8, 6, and 6, respectively.

Let χ_1, χ_2, and χ_3 be the irreducible characters of S_4 obtained from those of S_3. To determine these characters, we need information about the images in S_4/K of the class representatives. We easily see that the images of 1 and $(1\,2)(3\,4)$ are trivial, that the image of $(1\,2\,3)$ has order 3, and that the images of $(1\,2)$ and $(1\,2\,3\,4)$ have order 2.

Using this observation and the character table for S_3 obtained in Example 15.6, we obtain the following partial character table for S_4:

	1	3	8	6	6
	1	$(1\,2)(3\,4)$	$(1\,2\,3)$	$(1\,2)$	$(1\,2\,3\,4)$
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	2	2	-1	0	0
χ_4					
χ_5					

Now S_4 acts transitively on the set $X = \{1, 2, 3, 4\}$; let π be the character of the $\mathbb{C}G$-module $\mathbb{C}X$. Since $\pi(g)$ equals the number of fixed points under the action of g on X by Example 15.4, we have

	1	3	8	6	6
	1	$(1\,2)(3\,4)$	$(1\,2\,3)$	$(1\,2)$	$(1\,2\,3\,4)$
π	4	0	1	2	0
$\pi - \chi_1$	3	-1	0	1	-1

We also have

$$(\pi - \chi_1, \pi - \chi_1) = \frac{1}{|S_4|} \sum_{i=1}^{5} k_i [(\pi - \chi_1)(g_i)]^2 = \frac{1}{24}(1 \cdot 9 + 3 \cdot 1 + 8 \cdot 0 + 6 \cdot 1 + 6 \cdot 1) = 1,$$

and hence $\pi - \chi_1$ is irreducible by Corollary 15.3. Let $\chi_4 = \pi - \chi_1$. Now $\chi_2\chi_4 \neq \chi_4$, and $\chi_2\chi_4$ is irreducible by Proposition 15.5; therefore $\chi_5 = \chi_2\chi_4$, and hence the character table of S_4 is

	1	3	8	6	6
	1	$(1\,2)(3\,4)$	$(1\,2\,3)$	$(1\,2)$	$(1\,2\,3\,4)$
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	2	2	-1	0	0
χ_4	3	-1	0	1	-1
χ_5	3	-1	0	-1	1

Example 15.8. Let G be a non-abelian group of order 8. We will show that this information completely determines the character table of G. Since there are two isomorphism classes of non-abelian groups of order 8 (see page 79), this will imply that the character table does not specify a group up to isomorphism.

We have $|Z(G)| = 2$ by Theorem 8.1 and Lemma 8.2, and hence $Z(G) \cong \mathbb{Z}_2$. Since G is non-abelian and $G/Z(G)$ is abelian (being a group of order 4), this forces $G' = Z(G)$ by Proposition 2.6. Now $|G/G'| = 4$, and by Lemma 8.2 G/G' cannot be cyclic since G is non-abelian and $G' = Z(G)$, so we must have $G/G' \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. By Examples 15.3 and 15.5, we now know that G has exactly 4 linear characters. By Corollary 14.2, we have $\sum_{i=1}^{r} f_i^2 = 8$, where $f_i = 1$ for $1 \le i \le 4$ and $f_i > 1$ for $i > 4$; this forces $r = 5$ and $f_5 = 2$.

Let x be the generator of G'. As $G' = Z(G)$, we see that 1 and x are the only elements of G lying in single-element conjugacy classes, and hence each of the other three classes must have order 2. Let a, b, and c be representatives of the multi-element classes. Since G/G' also has four conjugacy classes, we see that the images of a, b, and c in G/G' must lie in distinct conjugacy classes. Using the character table for $\mathbb{Z}_2 \times \mathbb{Z}_2$ determined in Example 15.3, we can now obtain a partial character table for G by lifting:

	1	1	2	2	2
	1	x	a	b	c
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2				

We see via Column Orthogonality Theorem that $\chi_5(a) = \chi_5(b) = \chi_5(c) = 0$ and that $\chi_5(x) = -2$, completing the character table.

Example 15.9. Consider the group A_5. Using Proposition 1.10, we find that the elements of A_5 form 4 conjugacy classes in S_5, with representatives 1, $(1\,2)(3\,4)$, $(1\,2\,3)$, and $(1\,2\,3\,4\,5)$ and respective orders 1, 15, 20, and 24. However, elements of A_5 that are conjugate in S_5 may not be conjugate in A_5.

Let $x \in A_5$, and let $\mathcal{K}$ be the conjugacy class in S_5 of x. We see from Proposition 3.13 that $|\mathcal{K}| = |S_5 : C_{S_5}(x)|$ and that the conjugacy class in A_5 of x is contained in $\mathcal{K}$ and has order $|A_5 : C_{A_5}(x)|$. Suppose first that $C_{S_5}(x) \not\leq A_5$. This forces $A_5 C_{S_5}(x) = S_5$ since A_5 is a maximal subgroup of S_5. The first isomorphism theorem gives $S_5/A_5 = A_5C_{S_5}(x)/A_5 \cong C_{S_5}(x)/C_{S_5}(x) \cap A_5 = C_{S_5}(x)/C_{A_5}(x)$, and hence

$$\begin{aligned} |A_5 : C_{A_5}(x)| &= \frac{|S_5 : C_{A_5}(x)|}{|S_5 : A_5|} = \frac{|S_5 : C_{A_5}(x)|}{|C_{S_5}(x) : C_{A_5}(x)|} = |S_5 : C_{S_5}(x)| \\ &= |\mathcal{K}|. \end{aligned}$$

Thus $\mathcal{K}$ is the conjugacy class in A_5 of x. Now suppose $C_{S_5}(x) \le A_5$. Then we have $C_{A_5}(x) = C_{S_5}(x) \cap A_5 = C_{S_5}(x)$, and hence

$$|A_5 : C_{A_5}(x)| = |A_5 : C_{S_5}(x)| = \tfrac{1}{2}|S_5 : C_{S_5}(x)| = \tfrac{1}{2}|\mathcal{K}|,$$

from which we see that $\mathcal{K}$ splits into two conjugacy classes in A_5 of equal cardinality.

We easily see that each of the elements 1, $(1\,2)(3\,4)$, and $(1\,2\,3)$ commutes with some odd permutation in S_5; for example, $(4\,5)$ and $(1\,2\,3)$ commute. Hence by the above paragraph, the conjugacy classes in A_5 of these elements are the respective classes in S_5. Since $24 \nmid 60$, we can now conclude that the conjugacy class in S_5 of $(1\,2\,3\,4\,5)$ splits into two classes in A_5, and hence that the conjugacy classes of A_5 have orders 1, 15, 20, 12, and 12. (Using this fact and the remark made after Proposition 3.13, we could now give a slick proof of the simplicity of A_5.) We will now show that $x = (1\,2\,3\,4\,5)$ and $x^2 = (1\,3\,5\,2\,4)$ are not conjugate in A_5, thereby establishing that the conjugacy class in A_5 of x splits in A_5 into the class containing x and the class containing x^2.

Suppose that $g \in A_5$ is such that $gxg^{-1} = x^2$. Then we have $g\langle x\rangle g^{-1} = \langle x^2\rangle = \langle x\rangle$, and hence $g \in N_{A_5}(\langle x\rangle)$. Now $\langle x\rangle$ is a Sylow 5-subgroup of A_5, and we observed in the proof of Theorem 7.4 that A_5 has 6 Sylow 5-subgroups. Thus $|N_{A_5}(\langle x\rangle)| = 10$, and we find that $N_{A_5}(\langle x\rangle) = \langle x\rangle \rtimes \langle t\rangle \cong D_{10}$, where $t = (2\,5)(3\,4)$. But $txt^{-1} = x^{-1} \neq x^2$; from this, we see that there is no element $g \in N_{A_5}(\langle x\rangle)$ such that $gxg^{-1} = x^2$. Contradiction; therefore x and x^2 are not conjugate in A_5.

We now see that A_5 has 5 conjugacy classes, with representatives 1, $(1\,2)(3\,4)$, $(1\,2\,3)$, $(1\,2\,3\,4\,5)$, and $(1\,3\,5\,2\,4)$ and respective orders 1, 15, 20, 12, and 12. Since A_5 is simple, we cannot produce irreducible characters of A_5 by lifting irreducible characters of quotient groups, as we have done in previous examples. However, A_5 is a group of permutations, so we have as a starting point the characters of certain obvious permutation modules.

Let $X = \{1,2,3,4,5\}$; A_5 acts transitively on X in the obvious way. Let π be the character of the $\mathbb{C}A_5$-module $\mathbb{C}X$. By Example 15.4, we see that $\pi(g)$ equals the number of fixed points of X under the action of $g \in A_5$, and hence we have

	1	15	20	12	12
	1	(1 2)(3 4)	(1 2 3)	(1 2 3 4 5)	(1 3 5 2 4)
π	5	1	2	0	0
$\pi - \chi_1$	4	0	1	-1	-1

We also have

$$(\pi - \chi_1, \pi - \chi_1) = \frac{1}{|A_5|}\sum_{i=1}^{5} k_i[(\pi - \chi_1)(g_i)]^2 = \frac{1}{60}(1\cdot 16 + 15\cdot 0 + 20\cdot 1 + 12\cdot 1 + 12\cdot 1) = 1,$$

and hence $\pi - \chi_1$ is irreducible by Corollary 15.3. Let $\chi_2 = \pi - \chi_1$.

Let Y be the set of two-element subsets of X; this is a transitive A_5-set under the obvious action. Let ψ be the character of the $\mathbb{C}A_5$-module $\mathbb{C}Y$. We have

	1	15	20	12	12
	1	(1 2)(3 4)	(1 2 3)	(1 2 3 4 5)	(1 3 5 2 4)
ψ	10	2	1	0	0
$\psi - \chi_1$	9	1	0	-1	-1

Now

$$(\psi - \chi_1, \psi - \chi_1) = \frac{1}{|A_5|}\sum_{i=1}^{5} k_i[(\psi - \chi_1)(g_i)]^2 = \frac{1}{60}(1\cdot 81 + 15\cdot 1 + 20\cdot 0 + 12\cdot 1 + 12\cdot 1) = 2,$$

so by Corollary 15.3, $\psi - \chi_1$ is the sum of two irreducible characters; as noted in Example 15.4, neither of these two characters is χ_1, since Y is transitive. We have

$$(\psi - \chi_1, \chi_2) = \frac{1}{|A_5|}\sum_{i=1}^{5} k_i(\psi - \chi_1)(g)\chi_2(g) = \frac{1}{60}(1\cdot 9\cdot 4 + 15\cdot 1\cdot 0 + 20\cdot 0\cdot 1 + 24\cdot(-1)(-1)) = 1,$$

and therefore by Corollary 15.4 we see that $\psi - \chi_1 - \chi_2$ is an irreducible character that is neither χ_1 nor χ_2. Let $\chi_3 = \psi - \chi_1 - \chi_2$, and observe that $f_3 = 5$.

By Corollary 14.2 we have $\sum_{i=1}^{5} f_i^2 = |A_5| = 60$, which gives $f_4^2 + f_5^2 = 18$ and hence $f_4 = f_5 = 3$. Thus we have the following partial character table:

	1	15	20	12	12
	1	(1 2)(3 4)	(1 2 3)	(1 2 3 4 5)	(1 3 5 2 4)
χ_1	1	1	1	1	1
χ_2	4	0	1	-1	-1
χ_3	5	1	-1	0	0
χ_4	3				
χ_5	3				

Let $a = \chi_4((1\,2)(3\,4))$ and $b = \chi_5((1\,2)(3\,4))$. By Column Orthogonality Theorem, we have

$$0 = \sum_{i=1}^{5} f_i\chi_i((1\,2)(3\,4)) = 1\cdot 1 + 4\cdot 0 + 5\cdot 1 + 3a + 3b,$$

and hence $a + b = -2$. By part (iii) of Proposition 14.4, we see that each of a and b is a sum of three square roots of unity; therefore $a, b \in \{-3,-1,1,3\}$. But by Column Orthogonality Theorem we have

$$4 = \frac{60}{15} = \frac{|A_5|}{k_2} = \sum_{i=1}^{5} |\chi_i((1\,2)(3\,4))|^2 = 1 + 0 + 1 + a^2 + b^2,$$

and hence $a^2 + b^2 = 2$. It now follows that $a = b = -1$.

By Column Orthogonality Theorem, we have

$$3 = \frac{60}{20} = \frac{|A_5|}{k_3} = \sum_{i=1}^{5} |\chi_i((1\,2\,3))|^2 = 1 + 1 + 1 + |\chi_4((1\,2\,3))|^2 + |\chi_5((1\,2\,3))|^2;$$

thus $|\chi_4((1\,2\,3))|^2 + |\chi_5((1\,2\,3))|^2 = 0$, and therefore we conclude that $\chi_4((1\,2\,3)) = \chi_5((1\,2\,3)) = 0$. We now have the following partial character table:

	1 1	15 $(1\,2)(3\,4)$	20 $(1\,2\,3)$	12 $(1\,2\,3\,4\,5)$	12 $(1\,3\,5\,2\,4)$
χ_1	1	1	1	1	1
χ_2	4	0	1	-1	-1
χ_3	5	1	-1	0	0
χ_4	3	-1	0		
χ_5	3	-1	0		

Let $x = (1\,2\,3\,4\,5)$. Recall that in proving that x and x^2 are not conjugate in A_5, we found that x and x^{-1} are conjugate in A_5; hence $\chi_4(x) = \chi_4(x^{-1})$, and thus $\chi_4(x)$ is real-valued by part (iv) of Proposition 14.4. The same argument shows that $\chi_4(x^2)$, $\chi_5(x)$, and $\chi_5(x^2)$ are also real-valued. Let $c = \chi_4(x)$ and $d = \chi_4(x^2)$. Then by Row Orthogonality Theorem, we have

$$0 = \sum_{i=1}^{5} k_i \chi_4(g_i) = 1 \cdot 3 + 15 \cdot (-1) + 20 \cdot 0 + 12c + 12d,$$

which gives $c + d = 1$, and also (since χ_4 is real-valued)

$$60 = \sum_{i=1}^{5} k_i \chi_4(g_i)^2 = 1 \cdot 9 + 15 \cdot 1 + 20 \cdot 0 + 12c^2 + 12d^2,$$

which gives $c^2 + d^2 = 3$. Hence $c^2 - c - 1 = 0$, so without loss of generality we have $c = \frac{1+\sqrt{5}}{2}$ and $d = 1 - c = \frac{1-\sqrt{5}}{2}$. We can similarly show that we must have $\chi_5(x), \chi_5(x^2) \in \{\frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}\}$; we leave it to the reader to verify that $\chi_5(x) = \frac{1-\sqrt{5}}{2}$ and $\chi_5(x^2) = \frac{1+\sqrt{5}}{2}$, which completes the character table of A_5.

Exercises

Throughout these exercises, G denotes a finite group, and $\mathcal{X}$ denotes the character table of G.

Exercise 15.1. What is the determinant of $\mathcal{X}$?

Solution. □

Exercise 15.2. Show that the sum of the entries in any row of $\mathcal{X}$ is a non-negative integer.

Solution. □

Exercise 15.3. Compute the character of the G-set G, where G acts via conjugation, and determine the multiplicities in this character of the irreducible characters.

Solution. □

Exercise 15.4. Observe that the complex conjugate of an irreducible character is an irreducible character. (This is either Exercise 14.6 or an easy consequence of part (ii) of Proposition 14.8 and Corollary 15.3.) Let P be the $r \times r$ permutation matrix such that $P\mathcal{X}$ is the matrix obtained from $\mathcal{X}$ by interchanging two rows if their corresponding characters are conjugate. Let Q be the $r \times r$ permutation matrix such that $\mathcal{X}Q$ is the matrix obtained from $\mathcal{X}$ by interchanging two columns if their corresponding conjugacy classes are inverse, in the sense that one class consists of the inverses of the elements of the other class. Show that $P\mathcal{X} = \mathcal{X}Q$.

Solution. □

Exercise 15.5. Suppose that $|G|$ is odd. Show that the only irreducible character that is real-valued is the principal character.

Solution. □

Exercise 15.6. Let χ be an irreducible character of G, and let $e \in \mathbb{C}G$ be the identity element of the corresponding matrix summand of $\mathbb{C}G$. Show that

$$e = \frac{\chi(1)}{|G|} \sum_{g \in G} \chi(g^{-1})g.$$

Solution. □

Exercise 15.7. If $N \trianglelefteq G$ and $g \in G$, show that $|C_G(g)| \geq |C_{G/N}(gN)|$.

Solution. □

Exercise 15.8. Show that if G acts doubly transitively on X, then the character of $\mathbb{C}X$ is the sum of the principal character and exactly one other irreducible character.

Solution. □

Exercise 15.9. Determine the character table of D_{2n} for any $n \in \mathbb{N}$.

Solution. □

Exercise 15.10. Determine the character table of S_5.

Solution. □

16 Induction

Overview

- Induced modules play a critical role in the representation theory of finite groups.
- Two equivalent definitions of induction are presented: one via tensor products over arbitrary rings (slick but heavier), the other using only tensor products over fields (clumsier but more elementary). A third definition is developed in Exercise 16.1.
- Let $H \leq G$ and let V be an FH-module. Since the group algebra FG is an (FG, FH)-bimodule, we form the FG-module $FG \otimes_{FH} V$, the *induced* FG-module of V (or the *induction* of V to G), denoted $\mathrm{Ind}_H^G V$. Other notations: V^G, $V \uparrow G$, and $V \uparrow_H^G$.
- For an FG-module U, regarding U as an FH-module yields the *restriction* $\mathrm{Res}_H^G U$. Other notations: U_H, $U \downarrow H$, and $U \downarrow_H^G$.

Lemma 16.1. *We have* $\dim_F \mathrm{Ind}_H^G V = |G : H| \dim_F V$ *for* V *an* FH*-module. Moreover, as* F*-vector spaces we have*

$$\mathrm{Ind}_H^G V = \bigoplus_{t \in T} t \otimes V,$$

where T *is a transversal for* H *in* G *and* $t \otimes V = \{t \otimes v \mid v \in V\}$.

Proof. □

Theorem (Frobenius Reciprocity Theorem)**.** *Let* U *be an* FG*-module, let* $H \leq G$*, and let* V *be an* FH*-module. Then as* F*-vector spaces we have* $\mathrm{Hom}_{FH}(V, \mathrm{Res}_H^G U) \cong \mathrm{Hom}_{FG}(\mathrm{Ind}_H^G V, U)$.

Proof. □

Induced Characters

- Restrict attention to $F = \mathbb{C}$. If V is a $\mathbb{C}H$-module with character φ, the character of $\mathrm{Ind}_H^G V$ is denoted φ^G and called an *induced character*; we have $\varphi^G(1) = |G : H|\varphi(1)$.
- If U is a $\mathbb{C}G$-module with character χ, the character of $\mathrm{Res}_H^G U$ is denoted $\chi|_H$, and $\chi|_H(1) = \chi(1)$.
- The form of "Frobenius reciprocity" actually known to Frobenius (Theorem 16.2) follows from Frobenius Reciprocity Theorem and Theorem 14.12.

Theorem 16.2. *Let* $H \leq G$*, let* U *be a* $\mathbb{C}G$*-module having character* χ*, and let* V *be a* $\mathbb{C}H$*-module having character* φ*. Then* $(\varphi, \chi|_H)_H = (\varphi^G, \chi)_G$.

Proof. □

Induction–Restriction Tables

- If $H \leq G$ and $\varphi_1, \ldots, \varphi_r$ and $\chi_1, \ldots, \chi_s$ are the irreducible characters of H and G respectively, the *induction-restriction table* of (G, H) is the $r \times s$ matrix whose (i, j)-entry gives the common value of $(\varphi_i^G, \chi_j)_G$ and $(\varphi_i, \chi_j|_H)_H$.
- By Corollary 15.4, the ith row gives the multiplicities with which the χ_j appear in the decomposition of φ_i^G, while the jth column gives the multiplicities with which the φ_i appear in the decomposition of $\chi_j|_H$.
- One can compute the table by calculating the restrictions of the χ_j and reading off the expressions for the φ_i^G in terms of the χ_j.
- For example, with $G = S_3$ and $H = S_2 \leq G$, inspection gives $\chi_1|_H = \varphi_1$, $\chi_2|_H = \varphi_2$, $\chi_3|_H = \varphi_1 + \varphi_2$, so the induction-restriction table yields $\varphi_1^G = \chi_1 + \chi_3$ and $\varphi_2^G = \chi_2 + \chi_3$.

Proposition 16.3. *Let $H \leq G$, and let V be a $\mathbb{C}H$-module having character χ. If T is a transversal for H in G, then for any $g \in G$ we have*

$$\chi^G(g) = \sum_{\substack{t \in T, \\ t^{-1}gt \in H}} \chi(t^{-1}gt).$$

Proof. □

Corollary 16.4. *Let χ be a character of $H \leq G$, and let $g \in G$. Then*

$$\chi^G(g) = \frac{1}{|H|} \sum_{\substack{x \in G, \\ x^{-1}gx \in H}} \chi(x^{-1}gx).$$

Proof. □

Proposition 16.5. *Let χ be a character of $H \leq G$, and let $g \in G$, and let s be the number of conjugacy classes of H whose members are conjugate in G to g. If $s = 0$, then $\chi^G(g) = 0$. Otherwise, if we let $h_1, \ldots, h_s$ be representatives of these s conjugacy classes of H, then*

$$\chi^G(g) = \sum_{i=1}^{s} \frac{|C_G(g)|}{|C_H(h_i)|} \chi(h_i).$$

Proof. □

Corollary 16.6. *Let χ be a character of $H \leq G$. Let $g \in G$, and suppose that the number s of conjugacy classes of H whose members are conjugate in G to g is positive. Let ℓ be the order of the conjugacy class in G of g; let $h_1, \ldots, h_s$ be representatives of these s conjugacy classes of H, and let $k_1, \ldots, k_s$ be the orders of these classes. Then*

$$\chi^G(g) = \sum_{i=1}^{s} |G : H| \frac{k_i}{\ell} \chi(h_i).$$

Proof. □

Frobenius Groups

- We end the section by presenting an important theorem on finite groups which, quite remarkably, has no known proof that does not use character theory.
- A *Frobenius group* is a group G having a subgroup H such that $H \cap gHg^{-1} = 1$ for every $g \in G - H$. We call H a *Frobenius complement* of G.
- If G is a finite Frobenius group, applying Frobenius' Theorem yields a normal subgroup N, called the *Frobenius kernel* of G. One has $G = N \rtimes H$, $N \cap H = 1$, and $|N||H| = |G|$.
- For $G = N \rtimes H$ a finite Frobenius group with Frobenius kernel N and complement H, the conjugation map $\varphi : H \to \mathrm{Aut}(N)$ has the property that for each $1 \neq h \in H$, $\varphi(h)$ is a *fixed-point-free automorphism* of N: $\varphi(h)(n) = n$ forces $n = 1$. In particular, φ is injective.
- Conversely, given finite groups N and H and a homomorphism $\varphi : H \to \mathrm{Aut}(N)$ for which every $\varphi(h)$, $h \neq 1$, is fixed-point-free, $N \rtimes_\varphi H$ is a Frobenius group.
- Example: if p, q are distinct primes with $p \equiv 1 \pmod q$, then any monomorphism $\varphi : \mathbb{Z}_q \to \mathrm{Aut}(\mathbb{Z}_p)$ is fixed-point-free, and hence the unique non-abelian group of order pq is a Frobenius group.
- A theorem of Thompson (his 1959 Chicago Ph.D. thesis) states that any finite group having a fixed-point-free automorphism of prime order is nilpotent. Hence the Frobenius kernel of a finite Frobenius group is nilpotent.

Theorem (Frobenius' Theorem)**.** *Let G be a transitive permutation group on a finite set X, and suppose that each non-identity element of G fixes at most one element of X. Then the union of the identity element and the set of elements of G that have no fixed points is a normal subgroup of G.*

Proof. □

Exercises

Throughout these exercises, G is a finite group, H is a subgroup of G, F is a field, and all FG-modules are finitely generated.

Exercise 16.1. Let V be an FH-module, and let W be an FG-module containing V. Suppose that W has the property that for any FG-module U and any $\varphi \in \mathrm{Hom}_{FH}(V, U)$, there is a unique FG-module homomorphism from W to U which extends V. (In this case, we say that W is relatively H-free with respect to V; compare with Exercises 13.1–13.2.) Show that $W \cong \mathrm{Ind}_H^G V$.

Solution. □

Exercise 16.2. Let W be an FG-module that is generated by an FH-submodule V, and suppose that $\dim_F W = |G : H| \dim_F V$. Show that we must have $W \cong \mathrm{Ind}_H^G V$.

Solution. □

Exercise 16.3. Show that $\mathrm{Hom}_{FG}(U, \mathrm{Ind}_H^G V) \cong \mathrm{Hom}_{FH}(\mathrm{Res}_H^G U, V)$ as F-vector spaces for any FG-module U and FH-module V. (Hint: Show first that if $\varphi \in \mathrm{Hom}_{FH}(\mathrm{Res}_H^G U, V)$, then the map sending $u \in U$ to $\sum_{t \in T} t \otimes \varphi(t^{-1}u)$, where T is a transversal for H in G, lies in $\mathrm{Hom}_{FG}(U, \mathrm{Ind}_H^G V)$.)

Solution. □

Exercise 16.4. Show that if X is a transitive G-set, then $FX \cong \mathrm{Ind}_{G_x}^G F$ for any $x \in X$.

Solution. □

Exercise 16.5. Compute the induction-restriction table of (A_5, A_4).

Solution. □

Exercise 16.6. Compute the character table of the unique non-abelian group of order pq, where p and q are distinct primes and $p \equiv 1 \pmod{q}$.

Solution. □

Exercise 16.7. Suppose that G is a Frobenius group having Frobenius kernel N. Show that the irreducible characters of G either are lifts of irreducible characters of G/N or are induced from the non-principal irreducible characters of N. When is it true that the inductions to G of two distinct non-principal irreducible characters of N are equal?

Solution. □

Further Exercises

In this set of exercises, we develop the character table of $G = \mathrm{GL}(2, q)$, where q is any prime power. Recall from Proposition 4.2 that we have $|G| = q(q-1)^2(q+1)$. Our first task is to determine the conjugacy classes of G. Here we sketch a comparatively "hands-on" approach, which uses only some standard topics from linear algebra. (A slicker approach might, for instance, make use of the invariant factor formulation of rational canonical form.) We denote by $m_g(X)$ the minimal polynomial of an element $g \in G$. Our strategy is to exploit the following:

Exercise 16.8. Show that conjugate elements of G have the same minimal polynomial.

Solution. □

Since $m_g(X)$ is a factor of the characteristic polynomial of g, which is the quadratic polynomial $\det(XI - g)$, we see that $m_g(X)$ is either linear, or a product of two like or unlike linear factors, or an irreducible monic quadratic.

Exercise 16.9. (cont.) Show that if $m_g(X) = X - a$ for some $a \in \mathbb{F}_q^\times$, then $g = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in Z(G)$.

Solution. □

Exercise 16.10. (cont.) Show that if $m_g(X) = (X - a)^2$ for some $a \in \mathbb{F}_q^\times$, then g is conjugate with $\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$, and the conjugacy class of g has order $q^2 - 1$.

Solution. □

Exercise 16.11. (cont.) Show that if $m_g(X) = (X - a)(X - b)$ for some distinct $a, b \in \mathbb{F}_q^\times$, then g is conjugate with $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, and the conjugacy class of g has order $q^2 + q$.

Solution. □

Exercise 16.12. (cont.) Show that if $r, s \in \mathbb{F}_q$ are such that $X^2 - rX - s$ is irreducible, then the conjugacy class of $\begin{pmatrix} 0 & 1 \\ s & r \end{pmatrix}$ has order $q^2 - q$.

Solution. □

(Hint for Exercises 16.10–16.12: Compute centralizers and use Proposition 3.13.)

Exercise 16.13. (cont.) Verify that what follows is a complete list of the conjugacy classes of G: $q - 1$ classes of 1 element each, indexed by elements of $\mathbb{F}_q^\times$; $q - 1$ classes of $q^2 - 1$ elements each, indexed by elements of $\mathbb{F}_q^\times$; $(q - 1)(q - 2)/2$ classes of $q^2 + q$ elements each, indexed by unordered pairs of distinct elements of $\mathbb{F}_q^\times$; and $q(q - 1)/2$ classes of $q^2 - q$ elements each, indexed by irreducible monic quadratic polynomials with coefficients in $\mathbb{F}_q$.

Solution. □

Having determined the classes of G, we now turn to finding irreducible characters. Recall that we have an epimorphism $\det : G \to \mathbb{F}_q^\times$. Consequently, we can lift characters of $\mathbb{F}_q^\times$ to characters of G, as in Lemma 15.6. Now $\mathbb{F}_q^\times$ is an abelian group of order $q - 1$, and hence it has $q - 1$ linear characters, which we denote by $\tau_1, \dots, \tau_{q-1}$. (In fact, it is true (see [17, p. 132]) that $\mathbb{F}_q^\times \cong \mathbb{Z}_{q-1}$, and so we can, upon fixing a generator of $\mathbb{F}_q^\times$, completely specify the τ_i.) For each $1 \le i \le q - 1$, let $\chi_i = \tau_i \circ \det$. We have

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ s & r \end{pmatrix}$
χ_i	$\tau_i(a)^2$	$\tau_i(a)^2$	$\tau_i(a)\tau_i(b)$	$\tau_i(-s)$

The characters χ_i are linear, and χ_1 is indeed the principal character of G.
As in Chapter 2, let B, U, and T be the subgroups of G consisting, respectively, of all upper triangular, all upper unitriangular, and all diagonal elements of G. We have $T \cong \mathbb{F}_q^\times \times \mathbb{F}_q^\times$, and so we see from Example 15.2 that the irreducible characters of T have the form $\tau_i \times \tau_j$ (after making appropriate identifications). Now $B = U \rtimes T$ by Proposition 5.1, so we can lift each character $\tau_i \times \tau_j$ of $T \cong B/U$ to obtain a character θ_{ij} of B. The epimorphism from B to T sends $\begin{pmatrix} x & y \\ 0 & z \end{pmatrix}$ to $\begin{pmatrix} x & 0 \\ 0 & z \end{pmatrix}$, and consequently we have $\theta_{ij} \begin{pmatrix} x & y \\ 0 & z \end{pmatrix} = (\tau_i \times \tau_j) \begin{pmatrix} x & 0 \\ 0 & z \end{pmatrix} = \tau_i(x)\tau_j(z)$. We now wish to consider the induced characters θ_{ij}^G.

Exercise 16.14. (cont.) Show that

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ s & r \end{pmatrix}$
θ_{ij}^G	$(q+1)\tau_i(a)\tau_j(a)$	$\tau_i(a)\tau_j(a)$	$\tau_i(a)\tau_j(b) + \tau_i(b)\tau_j(a)$	0

Observe that $\theta_{ij}^G = \theta_{ji}^G$.

Solution. □

Exercise 16.15. (cont.) Show that $(\theta_{ij}^G, \theta_{ij}^G) = 1 + \delta_{ij}$ and that $(\theta_{ii}^G, \chi_i) = 1$. Conclude that $\theta_{ii}^G - \chi_i$ is an irreducible character of G for each i and that θ_{ij}^G is an irreducible character of G whenever i and j are distinct.

Solution. □

At this point, we have constructed $q-1$ distinct irreducible characters of degree 1 (the χ_i), $q-1$ distinct irreducible characters of degree q (the $\theta_{ii}^G - \chi_i$), and $(q-1)(q-2)/2$ distinct irreducible characters of degree $q+1$ (the θ_{ij}^G for $i < j$). Our partial character table of G is now

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ s & r \end{pmatrix}$
χ_i	$\tau_i(a)^2$	$\tau_i(a)^2$	$\tau_i(a)\tau_i(b)$	$\tau_i(-s)$
$\theta_{ii}^G - \chi_i$	$q\tau_i(a)^2$	0	$\tau_i(a)\tau_i(b)$	$-\tau_i(-s)$
θ_{ij}^G	$(q+1)\tau_i(a)\tau_j(a)$	$\tau_i(a)\tau_j(a)$	$\tau_i(a)\tau_j(b) + \tau_i(b)\tau_j(a)$	0

To complete the character table, it will be convenient to work with a different set of representatives for the conjugacy classes of those elements whose minimal polynomials are irreducible quadratics. (We shall refer to these classes as being the "fourth-column" classes, owing to their placement in our partial character table above.) By Exercise 4.4, G has a subgroup C which is isomorphic with $\mathbb{F}_{q^2}^\times$, where $\mathbb{F}_{q^2}$ is the field of q^2 elements. As we remarked before, we have $C \cong \mathbb{Z}_{q^2-1}$. It is true, although we are not in a position to prove it, that any subgroup of G that is isomorphic with $\mathbb{F}_{q^2}^\times$ is a conjugate of C. (The proof of this fact requires some familiarity with field theory, but the essential point is that all fields of order q^2 are isomorphic; see [19, Corollary XIV.2.7].) It is also true that C must contain $Z(G) \cong \mathbb{F}_q^\times$, and so we will consider $\mathbb{F}_q^\times$ as being a subgroup of C.

Exercise 16.16. (cont.) Show that the elements of $C - \mathbb{F}_q^\times$ form a double set of representatives of the fourth-column classes, with $x \in C - \mathbb{F}_q^\times$ being conjugate with (but unequal to) x^q. (HINT: Argue that each element of G whose minimal polynomial is an irreducible quadratic determines a subgroup of G that is isomorphic with $\mathbb{F}_{q^2}^\times$.)

Solution. □

We now take $x \in C - \mathbb{F}_q^\times$ as our arbitrary fourth-column class representative. We have $\chi_i(x) = \tau_i(\det x)$, $(\theta_{ii}^G - \chi_i)(x) = -\tau_i(\det x)$, and $\theta_{ij}^G(x) = 0$.
Let $\beta_1, \ldots, \beta_{q^2-1}$ be the irreducible characters of C. (By choosing a generator for C, we can specify the β_i.) We will consider the induced characters β_k^G.

Exercise 16.17. (cont.) Show

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$x \in C - \mathbb{F}_q^\times$
β_k^G	$(q^2-q)\beta_k(a)$	0	0	$\beta_k(x) + \beta_k(x)^q$

Observe that $(\beta_k^q)^G = \beta_k^G$, where by β_k^q we mean the linear character of C defined by $\beta_k^q(x) = \beta_k(x)^q$.

Solution. □

It is not hard to show that there are exactly $q-1$ values of k for which $\beta_k^q = \beta_k$. (This follows, for instance, from the observation made in Example 15.1 that the linear characters of a cyclic group of order n themselves form a cyclic group of order n.)
Choose some k such that $\beta_k^q \neq \beta_k$, and let i_k be such that the restriction of β_k to $\mathbb{F}_q^\times \subseteq C$ is τ_{i_k}. Let $\psi_k = (\theta_{11}^G - \chi_1)\theta_{i_k 1}^G - \theta_{i_k 1}^G - \beta_k^G$; this is a virtual character.

Exercise 16.18. (cont.) Show that

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$x \in C - \mathbb{F}_q^\times$
ψ_k	$(q-1)\tau_{i_k}(a)$	$-\tau_{i_k}(a)$	0	$-\beta_k(x) - \beta_k(x)^q$

Solution. □

Exercise 16.19. (cont.) If k is such that $\beta_k^q \neq \beta_k$, show that $(\psi_k, \psi_k) = 1$. Conclude that we have constructed $q(q-1)/2$ distinct irreducible characters of degree $q-1$.

Solution. □

The number of irreducible characters that we have constructed is now equal to the number of conjugacy classes of G, and so by Theorem 14.3 we are done.

Appendix A

Algebraic Integers and Characters

17 Algebraic Integers and Characters

Setting and motivation

- This appendix continues character theory with a different flavour from Chapter 6, leaning on Galois theory and a higher level of algebraic background.
- Logically the appendix follows Section 15.
- Throughout, G is a finite group.
- We say that $x \in \mathbb{C}$ is an *algebraic integer* if there is some monic polynomial $f \in \mathbb{Z}[X]$ such that $f(x) = 0$.

Lemma 17.1. *The set of rational numbers that are also algebraic integers is exactly the set of ordinary integers.*

Proof. □

Algebraic integers form a ring

- The next standard result is used freely; its proof is sketched in the exercises.

Proposition 17.2. *The algebraic integers form a subring of* $\mathbb{C}$.

Proof. □

Characters take algebraic-integer values

- The relevance of algebraic integers to representation theory of finite groups is established by the following basic fact.

Proposition 17.3. *Let* χ *be a character of* G*. Then* $\chi(g)$ *is an algebraic integer for any* $g \in G$.

Proof. □

Class-sum integrality

- The next result is necessary for both intended applications.

Lemma 17.4. *If* χ *is an irreducible character of* G *and* $g \in G$*, then* $|G : C_G(g)|\chi(g)/\chi(1)$ *is an algebraic integer.*

Proof. □

Degrees of irreducible characters divide $|G|$

- This proves the first main result of the appendix, a fact already mentioned in Section 14.

Proposition 17.5. *Let χ be an irreducible character of G. Then $\chi(1)$ divides $|G|$.*

Proof. □

Toward Burnside's $p^a q^b$ theorem

- The goal of the remainder of the appendix is to prove Burnside's Theorem on the solvability of groups of order $p^a q^b$, mentioned in Section 11.

Lemma 17.6. *Let χ be a character of G, let $g \in G$, and define $\gamma = \chi(g)/\chi(1)$. If γ is a non-zero algebraic integer, then $|\gamma| = 1$.*

Proof. □

A classical theorem of Burnside

- We now prove a classical theorem due to Burnside.

Theorem 17.7. *If G has a conjugacy class of non-trivial prime power order, then G is not simple.*

Proof. □

Burnside's famous $p^a q^b$ theorem

- Finally, we have Burnside's famed $p^a q^b$ theorem.

Theorem (Burnside's Theorem)**.** *If $|G| = p^a q^b$, where p and q are primes, then G is solvable.*

Proof. We argue by induction on $|G|$. If $|G|$ is a prime power, then G is solvable by Corollary 11.5. We may therefore assume that p and q are distinct and both divide $|G|$.
Suppose first that G is not simple. Choose a normal subgroup N of G with $1 < N < G$. Both N and G/N have orders of the form $p^r q^s$, so they are solvable by induction. Hence G is solvable by Proposition 11.3.
It remains to consider the case where G is simple. If G is abelian, then G has prime order and is solvable.
Suppose then that G is non-abelian simple. In particular, $Z(G) = 1$. If every non-identity conjugacy class of G had order divisible by pq, then the class equation would give

$$|G| = 1 + \sum_i |C_i|$$

with each $|C_i|$ divisible by pq, a contradiction since pq divides $|G|$. Therefore G has a non-trivial conjugacy class whose order is not divisible by both p and q. Since every conjugacy class size divides $|G|$, this class has prime power order. Theorem 17.7 then implies that G is not simple, contradicting our assumption. Thus the non-abelian simple case cannot occur, and G is solvable. □

Beyond two primes

- It was observed on page 100 that there are non-solvable groups of order $p^a q^b r^c$, where p, q, and r are distinct primes.

- In fact, up to isomorphism there are eight such groups.

- Six of these groups are projective special linear groups. (Recall that A_5 and A_6 are, by Exercises 6.2 and 6.5, isomorphic with projective special linear groups.)

- The remaining two groups are members of a family of groups called the projective special unitary groups (see [24, p. 245]).

Exercises

Exercise 17.1. For $x \in \mathbb{C}$, let $\mathbb{Z}[x] = \{f(x) \mid f(X) \in \mathbb{Z}[X]\}$; this is an abelian subgroup of $\mathbb{C}$. Show that x is an algebraic integer iff $\mathbb{Z}[x]$ is finitely generated.

Solution. □

Exercise 17.2. (cont.) Show that the set of algebraic integers is a subring of $\mathbb{C}$. (Hint: Use Exercise 17.1 and the well-known fact (see [1, p. 145]) that any subgroup of a finitely generated abelian group is finitely generated.)

Solution. □

Exercise 17.3. Let G be a finite group, and define a function $\psi\colon G \to \mathbb{C}$ by setting $\psi(g) = |\{(x, y) \in G \times G \mid [x, y] = g\}|$ for $g \in G$. Show that

$$\psi = \sum_{i=1}^{r} \frac{|G|}{\chi_i(1)} \chi_i,$$

which in light of Proposition 17.5 shows that ψ is a character. (Compare with Exercise 3.6.)

Solution. □